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United States Patent	12398154
Kind Code	B2
Date of Patent	August 26, 2025
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Azaquinazoline pan-KRas inhibitors

Abstract

The present invention relates to compounds that inhibit at least one of KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and KRas Q61H, pharmaceutical compositions comprising the compounds and methods of use therefor.

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Appl. No.: 17/300930

Filed: December 15, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230072276 A1	Mar. 09, 2023

Related U.S. Application Data

us-provisional-application US 63164338 20210322
us-provisional-application US 63159868 20210311
us-provisional-application US 63125776 20201215

Publication Classification

Int. Cl.: C07D519/00 (20060101); C07D471/04 (20060101)

U.S. Cl.:

CPC C07D519/00 (20130101); C07D471/04 (20130101);

Field of Classification Search

CPC: C07D (519/00); C07D (471/04)

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to compounds that inhibit multiple mutated forms of KRas, i.e., pan-KRas inhibitors. In particular, the present invention relates to pan-KRas compounds, pharmaceutical compositions comprising the compounds and methods of use therefor.

BACKGROUND OF THE INVENTION

(2) Kirsten Rat Sarcoma 2 Viral Oncogene Homolog (“KRas”) is a small GTPase and a member of the Ras family of oncogenes. KRas serves as a molecular switch cycling between inactive (GDP-bound) and active (GTP-bound) states to transduce upstream cellular signals received from multiple tyrosine kinases to downstream effectors to regulate a wide variety of processes, including cellular proliferation (e.g., see Alamgeer et al., (2013) Current Opin Pharmacol. 13:394-401).

(3) The role of activated KRas in malignancy was observed over thirty years ago (e.g., see Santos et al., (1984) Science 223:661-664). Aberrant expression of KRas accounts for up to 20% of all cancers and oncogenic KRas mutations that stabilize GTP binding and lead to constitutive activation of KRas. KRas mutations at codons 12, 13, 61 and other positions of the KRas primary amino acid sequence are present in 88% of all pancreatic adenocarcinoma patients, 50% of all colon/rectal adenocarcinoma patients, and 32% lung adenocarcinoma patients (e.g., see Prior et al., (2020) Cancer Res 80:2969-74). A recent publication also suggested wild type Kras inhibition could be a viable therapeutic strategy to treat KRas.sup.WT dependent cancers (e.g., see Bery et al., (2020) Nat. Commun. 11: 3233).

(4) The well-known role of KRas in malignancy and the discovery of these frequent mutations in KRas in various tumor types made KRas a highly attractive target of the pharmaceutical industry for cancer therapy. Notwithstanding thirty years of large-scale discovery efforts to develop inhibitors of KRas for treating cancer, no KRas inhibitor has yet demonstrated sufficient safety and/or efficacy to obtain regulatory approval (e.g., see McCormick (2015) Clin Cancer Res. 21 (8):1797-1801).

(5) Compounds that inhibit KRas activity are still highly desirable and under investigation, including those that disrupt effectors such as guanine nucleotide exchange factors (e.g., see Sun et al., (2012) Agnew Chem Int Ed Engl. 51(25):6140-6143 doi: 10.1002/anie.201201358) as well recent advances in the covalent targeting of an allosteric pocket of KRas G12C (e.g., see Ostrem et al., (2013) Nature 503:548-551 and Fell et al., (2018) ACS Med. Chem. Lett. 9:1230-1234). Clearly there remains a continued interest and effort to develop inhibitors of KRas, particularly inhibitors of activating KRas mutants-.

(6) Thus, there is a need to develop new pan-KRas inhibitors that demonstrate sufficient efficacy for treating KRas-mediated cancers.

SUMMARY OF THE INVENTION

(7) In one aspect of the invention, compounds are provided that inhibit KRas activity. In certain embodiments, the compounds are represented by Formula (I):

(8) ##STR00001##

or a pharmaceutically acceptable salt thereof, wherein:

(9) W is:

(10) ##STR00002##

(11) A is aryl or heteroaryl, wherein the aryl or the heteroaryl is optionally substituted with 1-4 R.sup.1;

(12) B is selected from:

(13) ##STR00003##

(14) Y.sup.1 is hydrogen, L-hydroxy optionally substituted with 1-4 R.sup.8, L-alkoxy optionally

substituted with 1-4 R^{sup.8}, halogen, L-C3-C6 cycloalkyl optionally substituted with 1-4 R^{sup.9}, L-heteroaryl optionally substituted with 1-4 R^{sup.8}, L-aryl optionally substituted with 1-4 R^{sup.8}, L-C(O)—NH_{sub.2}, and L-heterocycle substituted with 1-2 oxo (=O) or oxo-containing substituent, and optionally further substituted with 1-2 R^{sup.8};

(15) Y^{sup.2} is hydrogen or C1-C4 alkyl;

(16) or Y^{sup.1} and Y^{sup.2} join to form:

(17) ##STR00004##

(18) where X is selected from: a bond, —S—, —O—, —N< bound to a fused ring, —CH_{sub.2}—, —CH_{sub.2}—N—, —CH_{sub.2}—N—CH_{sub.2}—, —CH_{sub.2}—CH_{sub.2}—CH_{sub.2}—, —CH_{sub.2}—CH_{sub.2}—, —O—CH_{sub.2}— and —S—CH_{sub.2}—;

(19) Y^{sup.3} is hydrogen or C1-C4 alkyl;

(20) Y^{sup.4} is hydrogen or C1-C4 alkyl;

(21) or Y^{sup.3} and Y^{sup.4} join to form:

(22) ##STR00005##

where J is selected from: a bond, —O—, —NH—, —CH_{sub.2}—, —C(C1-C3 alkyl)_{sub.2}—, —CH(C1-C3alkyl)- and —N(C1-C3 alkyl)-;

(23) each R^{sup.1} is independently halogen, cyano, hydroxy, C1-C4 alkyl, —S—C1-C3 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C2-C4 hydroxyalkynyl, C1-C3 cyanoalkyl, triazolyl, C1-C3 haloalkyl, —O—C1-C3 haloalkyl, —S—C1-C3 haloalkyl, C1-C3 alkoxy, hydroxyC1-C3 alkyl, —CH_{sub.2}C(=O)N(R^{sup.5})_{sub.2}, —C3-C4 alkynyl(NR^{sup.5})_{sub.2}, —N(R^{sup.5})_{sub.2}, deuterioC2-C4 alkynyl, (C1-C3 alkoxy)haloC1-C3 alkyl-, or C3-C6 cycloalkyl wherein said C3-C6 cycloalkyl is optionally substituted with halogen or C1-C3 alkyl;

(24) each R^{sup.2} is independently hydrogen, hydroxy, halogen, cyano, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R^{sup.5})_{sub.2}, —CO_{sub.2}R^{sup.5}, or —CO_{sub.2}N(R^{sup.5})_{sub.2};

(25) each R^{sup.3} is independently hydrogen, hydroxy, halogen, cyano, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R^{sup.5})_{sub.2}, —CO_{sub.2}R, or —CO_{sub.2}N(R^{sup.5})_{sub.2};

(26) each R^{sup.4} is independently hydrogen, halogen or C1-C3 alkyl;

(27) each R^{sup.5} is independently hydrogen or C1-C3 alkyl, or two R^{sup.5} join to form cycloalkyl or heterocycle;

(28) each R^{sup.6} is independently hydrogen, hydroxy, C1-C4 hydroxyalkyl or heteroaryl,

(29) or two R^{sup.6} join to form C3-C6 cycloalkyl or heterocycle;

(30) each R^{sup.7} is independently hydrogen, C1-C3 alkyl, hydroxy, halogen, C1-C3 haloalkyl, -L-NH_{sub.2}, —NH(C1-C3 alkyl), —N(C1-C3 alkyl)_{sub.2}, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R^{sup.10})_{sub.2}, —NHC(O)H, —CN, aryl, —(CH_{sub.2})_{sub.1-2}S(O)_{sub.2}N(R^{sup.10})_{sub.2}, —NH—S(O)_{sub.2}N(R^{sup.10})_{sub.2}, —O—S(O)_{sub.2}N(R^{sup.10})_{sub.2}, S(O)_{sub.2}R^{sup.10}, or heteroaryl or heterocycle optionally independently substituted with 1-2 substituents independently selected from C1-C3 alkyl, —CN and C(O)NH_{sub.2},

(31) two R^{sup.7} on the same atom optionally join to form a spirocyclic ring selected from C3-C6 cycloalkyl and heterocycle, where said spirocyclic ring is optionally substituted with 1-4 substituents independently selected from oxo (=O), halogen, hydroxy, C1-C3 alkyl and —O—(C1-C3 alkyl),

(32) two R^{sup.7} on adjacent atoms optionally join to form a bond or a fused ring selected from C3-C6 cycloalkyl optionally substituted with 1-4 R^{sup.8}, heteroaryl optionally substituted with 1-4 R^{sup.8}, aryl optionally substituted with 1-4 R^{sup.8}, and heterocycle optionally substituted with 1-4 R^{sup.8}, and

(33) two R^{sup.7} on non-adjacent atoms optionally join to form a 1-2 carbon bridge;

(34) each R^{sup.8} is independently C1-C3 alkyl, hydroxy, halogen, —N(R¹⁰)_{sub.2}, —

$\text{N(R}_{10}\text{)C(O)R}_{10}$, oxo ($=\text{O}$), $-\text{O}-(\text{C}1-\text{C}3 \text{ alkyl})$, $-(\text{C}1-\text{C}3 \text{ alkyl})-\text{OH}$, $-\text{C(O)OH}$, $-\text{C(O)O}(\text{C}1-\text{C}3 \text{ alkyl})$, $-\text{C(O)N(R}_{\text{sup.10}}\text{)}_{\text{sub.2}}$, heteroaryl or $-\text{CN}$;
 (35) each $\text{R}_{\text{sup.9}}$ is independently $\text{C}1-\text{C}3$ alkyl, hydroxy, halogen, oxo ($=\text{O}$), $-\text{O}-(\text{C}1-\text{C}3 \text{ alkyl})$, $-(\text{C}1-\text{C}3 \text{ alkyl})-\text{OH}$, $-\text{C(O)OH}$, $-\text{C(O)O}(\text{C}1-\text{C}3 \text{ alkyl})$, $-\text{C(O)NH}_{\text{sub.2}}$, $-\text{C(O)NH}(\text{C}1-\text{C}3 \text{ alkyl})$, $-\text{C(O)N}(\text{C}1-\text{C}3 \text{ alkyl})_{\text{sub.2}}$ or $-\text{CN}$;
 (36) each $\text{R}_{\text{sup.10}}$ is independently hydrogen, halogen, $\text{C}1-\text{C}3$ alkyl, or two $\text{R}_{\text{sup.10}}$ join to form cycloalkyl or heterocycle optionally substituted with 1-2 $\text{C}1-\text{C}3$ alkyl;
 (37) each L is independently a bond, $-\text{C}1-\text{C}4 \text{ alkyl}-$, $-\text{C}1-\text{C}4 \text{ alkyl}-\text{NH}-$, $-\text{NH}-$, $-\text{N}(\text{C}1-\text{C}3 \text{ alkyl})-$ or cyclopropyl- $\text{CH}_{\text{sub.2}}-$;
 (38) each n is 0-3;
 (39) o is 1-6; and
 (40) p is 1-8.
 (41) In another aspect of the invention, pharmaceutical compositions are provided comprising a therapeutically effective amount of a compound of the present invention or a pharmaceutically acceptable salt thereof and a pharmaceutically acceptable excipient.
 (42) In yet another aspect of the invention, methods for inhibiting the activity of cells containing wild type KRas or one or more KRas mutations, for instance the KRas mutations G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H, in a cell, comprising contacting the cell with a compound of Formula (I), or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof as defined herein. In one embodiment, the contacting is in vitro. In one embodiment, the contacting is in vivo.
 (43) Also provided herein is a method of inhibiting cell proliferation, in vitro or in vivo, the method comprising contacting a cell with an effective amount of a compound of Formula (I), or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof as defined herein.
 (44) Also provided are methods for treating cancer in a patient comprising administering a therapeutically effective amount of a compound or pharmaceutical composition of the present invention or a pharmaceutically acceptable salt thereof to a patient in need thereof.
 (45) Also provided herein is a method of treating a KRas wild type, KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated disease or disorder in a patient in need of such treatment, the method comprising administering to the patient a therapeutically effective amount of a compound of Formula (1), or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof as defined herein.
 (46) Also provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof as defined herein for use in therapy.
 (47) Also provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof or a pharmaceutical composition thereof as defined herein for use in the treatment of cancer.
 (48) Also provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof for use in the inhibition of KRas wild type or multiple types of KRas mutations, for instance KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutations.
 (49) Also provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof or a pharmaceutical composition thereof as defined herein, for use in the treatment of a KRas wild type associated disease or disorder or a KRas mutation G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated disease or disorder.
 (50) Also provided herein is the use of a compound of Formula (I), or a pharmaceutically acceptable salt thereof, as defined herein in the manufacture of a medicament for the treatment of cancer.
 (51) Also provided herein is a use of a compound of Formula (I), or a pharmaceutically acceptable salt thereof, as defined herein in the manufacture of a medicament for the inhibition of activity of

the wild type form of KRas or mutated forms of KRas, including the mutations: G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H.

(52) Also provided herein is the use of a compound of Formula (I), or a pharmaceutically acceptable salt thereof, as defined herein, in the manufacture of a medicament for the treatment of a KRas wild type associated disease or disorder or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated disease or disorder.

(53) Also provided herein is a method for treating cancer in a patient in need thereof, the method comprising (a) determining that the cancer is associated with KRas wild type or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation (i.e., a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated cancer); and (b) administering to the patient a therapeutically effective amount of a compound of Formula (I), or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof.

(54) One potential utility of the herein-described pan-KRas inhibitors, including pan-KRas inhibitors such as (R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (Example 5 herein), is for the treatment of cancers that develop resistance following long-term treatment with KRas G12C inhibitors. Thus, embodiments of the invention include those wherein a patient suffering from cancer is treated with a herein-described pan-KRas inhibitor such as Example 5 after treatment with a G12C inhibitor becomes ineffective or less effective due to the emergence of resistance-imparting mutations.

(55) Treatment of KRas G12C mutant cancers with covalent KRas G12C inhibitors such as adagrasib (MRTX849) or sotorasib (AMG510) may result in the incorporation of additional mutations that confer resistance to adagrasib. These mutations could confer resistance through numerous mechanisms.

(56) Mutations that change the mutant cysteine at codon 12 to another amino acid would render the current covalent KRas G12C inhibitors ineffective since current inhibitors make a covalent bond with the mutant cysteine amino acid side chain. Likewise, in patients that have one wild type KRas allele in addition to the KRas G12C-mutant allele, mutations in the wild type codon 12 glycine to another codon would allow bypass signaling in these tumors through the novel mutant protein. The repertoire of codon 12 mutations that can occur with a single nucleotide substitution in the wild type gene (glycine codon) includes mutations commonly observed in cancer such as G12S, G12V, G12R, G12C. The repertoire of codon 12 mutations that can occur with single nucleotide base substitutions of the cysteine codon 12 include mutations not frequently observed in cancer, G12Y, G12F and G12W, in addition to G12S and G12R.

(57) Second-site mutations may also occur in another location in the KRas G12C mutant gene that confers resistance to KRas G12C inhibitor treatment. These mutations may confer resistance through different mechanisms. RAS proteins are small GTPases that normally cycle between an active, GTP-bound state and an inactive, GDP-bound state. RAS proteins are loaded with GTP through guanine nucleotide exchange factors (GEFs; e.g., SOS1) which are activated by upstream receptor tyrosine kinases, triggering subsequent interaction with effector proteins that activate RAS-dependent signaling. RAS proteins hydrolyze GTP to GDP through their intrinsic GTPase activity which is dramatically enhanced by GTPase-activating proteins (GAPs). Mutations at codons 12 and 13 in RAS proteins impair GAP-stimulated GTP hydrolysis leaving RAS predominantly in the GTP-bound, active state. Covalent KRas G12C inhibitors in current clinical development only bind GDP-bound KRas G12C. Mutations such as Q61 codon mutations, which may or may not occur on the same allele as the G12C mutation, reduce the intrinsic GTPase activity of KRas and may represent a mechanism of resistance to KRas G12C inhibitor treatment by shifting KRas into the GTP-loaded state where it is not susceptible to covalent inhibition. Co-mutations such as R68, H95 and Y96 may be present along with the KRas G12C mutation and may diminish the binding affinity of KRas G12C inhibitors to the Switch II binding pocket.

(58) The herein-described pan-KRas inhibitors may demonstrate activity against common as well as uncommon codon 12 mutations or mutations that occur in the KRas protein that diminish binding of KRas G12C inhibitors to the KRas protein.

(59) Also provided herein is a process for preparing a compound of Formula (I), or a pharmaceutically acceptable salt thereof.

(60) Also provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof obtained by a process of preparing the compound as defined herein.

Description

DETAILED DESCRIPTION OF THE INVENTION

(1) The present invention relates to inhibitors of KRas wild type and/or multiple mutated forms of KRas, for instance KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutations. In particular, the present invention relates to compounds that inhibit the activity of KRas wild type and/or KRas mutations such as G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H, pharmaceutical compositions comprising a therapeutically effective amount of the compounds and methods of use therefor.

Definitions

(2) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this invention belongs. All patents, patent applications, and publications referred to herein are incorporated by reference.

(3) As used herein, “wild type KRas” refers to a non-mutant form of a mammalian KRas protein. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “wild type KRas inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of wild type KRas G12A. A “wild type KRas-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having wild type KRas. A non-limiting example of a wild type KRas-associated disease or disorder is a wild type KRas-associated cancer.

(4) As used herein, “KRas G12A” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of an alanine for a glycine at amino acid position 12. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G12A inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G12A. A “KRas G12A-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G12A mutation. A non-limiting example of a KRas G12A-associated disease or disorder is a KRas G12A-associated cancer.

(5) As used herein, “KRas G12C” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of a cysteine for a glycine at amino acid position 12. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G12C inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G12C. A “KRas G12C-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G12C mutation. A non-limiting example of a KRas G12C-associated disease or

disorder is a KRas G12CD-associated cancer.

(6) As used herein, “KRas G12D” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of an aspartic acid for a glycine at amino acid position 12. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G12D inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G12D. A “KRas G12D-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G12D mutation. A non-limiting example of a KRas G12D-associated disease or disorder is a KRas G12D-associated cancer.

(7) As used herein, “KRas G12R” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of an arginine for a glycine at amino acid position 12. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G12R inhibitor” refers to compounds of the present invention that are represented by Formula (1), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G12R. A “KRas G12R-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G12R mutation. A non-limiting example of a KRas G12R-associated disease or disorder is a KRas G12R-associated cancer.

(8) As used herein, “KRas G12S” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of a serine for a glycine at amino acid position 12. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G12S inhibitor” refers to compounds of the present invention that are represented by Formula (1), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G12S. A “KRas G12S-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G12S mutation. A non-limiting example of a KRas G12S-associated disease or disorder is a KRas G12S-associated cancer.

(9) As used herein, “KRas G12V” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of a valine for a glycine at amino acid position 12. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G12V inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G12V. A “KRas G12V-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G12V mutation. A non-limiting example of a KRas G12V-associated disease or disorder is a KRas G12V-associated cancer.

(10) As used herein, “KRas G13D” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of an aspartic acid for a glycine at amino acid position 13. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas G13D inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas G13D. A “KRas G13D-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas G13D mutation. A non-limiting example of a KRas G13D-associated disease or

disorder is a KRas G13D-associated cancer.

(11) As used herein, “KRas Q61H” refers to a mutant form of a mammalian KRas protein that contains an amino acid substitution of a histidine for a glutamine at amino acid position 61. The assignment of amino acid codon and residue positions for human KRas is based on the amino acid sequence identified by UniProtKB/Swiss-Prot P01116: Variantp.Gly12Asp. As used herein, a “KRas Q61H inhibitor” refers to compounds of the present invention that are represented by Formula (I), as described herein. These compounds are capable of negatively modulating or inhibiting all or a portion of the enzymatic activity of KRas Q61H. A “KRas Q61H-associated disease or disorder” as used herein refers to diseases or disorders associated with or mediated by or having a KRas Q61H mutation. A non-limiting example of a KRas Q61H-associated disease or disorder is a KRas Q61H-associated cancer.

(12) As used herein, the term “subject,” “individual,” or “patient,” used interchangeably, refers to any animal, including mammals such as mice, rats, other rodents, rabbits, dogs, cats, swine, cattle, sheep, horses, primates, and humans. In some embodiments, the patient is a human. In some embodiments, the subject has experienced and/or exhibited at least one symptom of the disease or disorder to be treated and/or prevented. In some embodiments, the subject has been identified or diagnosed as having a cancer having wild type KRas or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation (e.g., as determined using a regulatory agency-approved, e.g., FDA-approved, assay or kit). In some embodiments, the subject has a tumor that is positive for wild type KRas or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation (e.g., as determined using a regulatory agency-approved assay or kit). The subject can be a subject with a tumor(s) that is positive for wild type KRas or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation (e.g., identified as positive using a regulatory agency-approved, e.g., FDA-approved, assay or kit). The subject can be a subject whose tumors have wild type KRas or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation (e.g., where the tumor is identified as such using a regulatory agency-approved, e.g., FDA-approved, kit or assay). In some embodiments, the subject is suspected of having wild type KRas or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H gene-associated cancer. In some embodiments, the subject has a clinical record indicating that the subject has a tumor that has wild type KRas or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation (and optionally the clinical record indicates that the subject should be treated with any of the compositions provided herein).

(13) In some embodiments of any of the methods or uses described herein, an assay is used to determine whether the patient has wild type KRas or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H mutation using a sample (e.g., a biological sample or a biopsy sample (e.g., a paraffin-embedded biopsy sample) from a patient (e.g., a patient suspected of having wild type KRas-associated or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated cancer, a patient having one or more symptoms of wild type KRas-associated or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated cancer, and/or a patient that has an increased risk of developing wild type KRas-associated or a KRas G12A, G12C, G12D, G12R, G12S, G12V, G13D and/or Q61H-associated cancer) can include, for example, next generation sequencing, immunohistochemistry, fluorescence microscopy, break apart FISH analysis, Southern blotting, Western blotting, FACS analysis, Northern blotting, and PCR-based amplification (e.g., RT-PCR and quantitative real-time RT-PCR). As is well-known in the art, the assays are typically performed, e.g., with at least one labelled nucleic acid probe or at least one labelled antibody or antigen-binding fragment thereof.

(14) The term “regulatory agency” is a country's agency for the approval of the medical use of pharmaceutical agents with the country. For example, a non-limiting example of a regulatory agency is the U.S. Food and Drug Administration (FDA).

(15) The term “acyl” refers to —C(O)CH₂sub.3.

(16) The terms “C1-C6 alkyl”, “C1-C4 alkyl” and “C1-C3 alkyl” as employed herein refers to straight and branched chain aliphatic groups having from 1-6 carbon atoms, or 1-4 carbon atoms, or 1-3 carbon atoms, respectively. Examples of alkyl groups include, without limitation, methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, and hexyl.

(17) The terms “C1-C3 haloalkyl” and “C1-C4 haloalkyl” refer to a C1-C3 alkyl chain or C1-C4 alkyl chain, respectively, as defined herein in which one or more hydrogen has been replaced by a halogen. Examples include trifluoromethyl, difluoromethyl and fluoromethyl.

(18) An “C1-C4 alkylene,” group is a C1-C4 alkyl group, as defined hereinabove, that is positioned between and serves to connect two other chemical groups. Exemplary alkylene groups include, without limitation, methylene, ethylene, propylene, and butylene.

(19) The terms “C1-C3 alkoxy” and “C1-C4 alkoxy” refer to —OC1-C3 alkyl and —OC1-C4 alkyl, respectively, wherein the alkyl portion is as defined herein above.

(20) The term “cycloalkyl” as employed herein includes saturated and partially unsaturated cyclic hydrocarbon groups having 3 to 12 carbons, for example 3 to 8 carbons, and as a further example 3 to 6 carbons, wherein the cycloalkyl group additionally is optionally substituted with one or more R_{sup.8} or R_{sup.9} groups as defined herein. Examples of cycloalkyl groups include, without limitation, cyclopropyl, cyclobutyl, cyclopentyl, cyclopentenyl, cyclohexyl, cyclohexenyl, cycloheptyl, and cyclooctyl. The term “cycloalkyl” also includes bridged cycloalkyls, such as bicyclo[1.1.1]pentanyl.

(21) As used herein, the terms “C1-C3 hydroxyalkyl” and “C1-C4 hydroxyalkyl” refer to —C1-C3 alkylene-OH and —C1-C4 alkylene-OH, respectively.

(22) As used herein, the term “C2-C4 hydroxyalkynyl” refers to —C2-C4 alkynylene-OH.

(23) An “aryl” group is a C_{sub.6}-C_{sub.14} aromatic moiety comprising one to three aromatic rings, which is optionally substituted with one or more R_{sup.8} or R_{sup.9} groups as defined herein. As one embodiment, the aryl group is a C_{sub.6}-C_{sub.10} aryl group. Examples of aryl groups include, without limitation, phenyl, naphthyl, anthracenyl, fluorenyl, and dihydrobenzofuranyl. “Aryl” also refers to bicyclic or tricyclic ring systems in which one or two rings, respectively, of said aryl ring system may be saturated or partially saturated, and wherein if said ring system includes two saturated rings, said saturated rings may be fused or spirocyclic. An example of an aryl ring system comprising two saturated rings wherein the rings are spirocyclic includes the following ring system:

(24) ##STR00006##

(25) An “araC1-C6 alkyl” or “arylalkyl” group comprises an aryl group covalently linked to an alkyl group, either of which may independently be optionally substituted or unsubstituted. An example of an aralkyl group is (C_{sub.6}-C_{sub.10})aryl(C_{sub.1}-C_{sub.6})alkyl-, including, without limitation, benzyl, phenethyl, and naphthylmethyl. An example of a substituted araC1-C6 alkyl is wherein the alkyl group is substituted with hydroxyalkyl.

(26) A “heterocyclyl” or “heterocyclic” group is a ring structure having from 3 to 12 atoms, for example 4 to 8 atoms, wherein one or more atoms are selected from the group consisting of N, O, and S wherein the ring N atom may be oxidized to N—O, and the ring S atom may be oxidized to SO or SO_{sub.2}, the remainder of the ring atoms being carbon. The heterocyclyl may be a monocyclic, a bicyclic, a spirocyclic or a bridged ring system. The heterocyclic group is optionally substituted with one or more R_{sup.8} or R_{sup.9} groups on ring carbon or ring nitrogen at one or more positions, wherein R_{sup.6} is as defined for Formula I. The heterocyclic group is also independently optionally substituted on a ring nitrogen atom with alkyl, aralkyl, alkylcarbonyl, or on sulfur with lower alkyl. Examples of heterocyclic groups include, without limitation, epoxy, azetidiny, aziridiny, tetrahydrofuranyl, tetrahydropyranyl, pyrrolidiny, pyrrolidinonyl, piperidiny, piperaziny, imidazolidiny, imidazopyridiny, thiazolidiny, dithianyl, trithianyl, dioxolanyl, oxazolidiny, oxazolidinonyl, decahydroquinolinyl, piperidonyl, 4-piperidinonyl, quinuclidiny, thiomorpholinyl, thiomorpholinyl 1,1 dioxide, morpholinyl, azepanyl, oxazepanyl,

azabicyclohexanyls, azabicycloheptanyl, azabicyclooctanyls, azabicyclononanyls (e.g., octahydroindoliziny), azaspiroheptanyls, dihydro-1H, 3H, 5H-oxazolo[3,4-c]oxazolyl, tetrahydro-1'H, 3'H-spiro[cyclopropane-1,2'-pyrrolizine], hexahydro-1H-pyrroliziny, hexahydro-1H-pyrrolo[2,1-c][1,4]oxazinyl, octahydroindoliziny, oxaazaspiroonanyls, oxaazaspirooctanyls, diazaspiroonanyls, oxaazabicycloheptanyls, hexahydropyrroliziny 4 (1H)-oxide, tetrahydro-2H-thiopyranyl 1-oxide and tetrahydro-2H-thiopyranyl 1,1-dioxide. Specifically excluded from the scope of this term are compounds having adjacent annular O and/or S atoms.

(27) As used herein, the term "heteroaryl" refers to groups having 5 to 14 ring atoms, preferably 5, 6, 9, or 10 ring atoms; having 6, 10, or 14 π electrons shared in a cyclic array; and having, in addition to carbon atoms, from one to three heteroatoms per ring, or from one to three heteroatoms in at least one ring, selected from the group consisting of N, O, and S. Examples of heteroaryl groups include acridinyl, azocinyl, benzimidazolyl, benzofuranyl, benzothiofuranyl, benzothiophenyl, benzoxazolyl, benzthiazolyl, benztriazolyl, benztetrazolyl, benzisoxazolyl, benzisothiazolyl, benzimidazoliny, carbazolyl, 4aH-carbazolyl, carboliny, chromanyl, chromenyl, cinnoliny, 6,7-dihydro-5H-pyrrolo[1,2-a]imidazole, furanyl, furazanyl, imidazoliny, imidazolyl, 1H-indazolyl, indolenyl, indoliny, indoliziny, indolyl, 3H-indolyl, isobenzofuranyl, isochromanyl, isoindazolyl, isoindoliny, isoindolyl, isoquinoliny, isothiazolyl, isoxazolyl, methylenedioxyphenyl, naphthyridiny, octahydroisoquinoliny, oxadiazolyl, 1,2,3-oxadiazolyl, 1,2,4-oxadiazolyl, 1,2,5-oxadiazolyl, 1,3,4-oxadiazolyl, oxazolidiny, oxazolyl, oxazolidiny, pyrimidiny, phenanthridiny, phenanthroliny, phenazinyl, phenothiaziny, phenoxathiiny, phenoxazinyl, phthalazinyl, piperonyl, pteridiny, puriny, pyranyl, pyraziny, pyrazolidiny, pyrazoliny, pyrazolyl, pyridazinyl, pyridooxazole, pyridoimidazole, pyridothiazole, pyridiny, pyridyl, pyrimidiny, pyrroliny, 2H-pyrrolyl, pyrrolyl, quinazoliny, quinoliny, 4H-quinoliziny, quinoxaliny, quinuclidiny, tetrahydroisoquinoliny, tetrahydroquinoliny, tetrazolyl, 6H-1,2,5-thiadiaziny, 1,2,3-thiadiazolyl, 1,2,4-thiadiazolyl, 1,2,5-thiadiazolyl, 1,3,4-thiadiazolyl, thianthrenyl, thiazolyl, thienyl, thienothiazolyl, thienooxazolyl, thienoimidazolyl, thiophenyl, triazinyl, 1,2,3-triazolyl, 1,2,4-triazolyl, 1,2,5-triazolyl, 1,3,4 triazolyl, and xanthenyl. "Heteroaryl" also refers to bicyclic ring systems having, in addition to carbon atoms, from one to three heteroatoms per ring selected from the group consisting of N, O, and S in which one ring system may be saturated or partially saturated.

(28) As used herein, "an effective amount" of a compound is an amount that is sufficient to negatively modulate or inhibit the activity of one or more of wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H. Such amount may be administered as a single dosage or may be administered according to a regimen, whereby it is effective.

(29) As used herein, a "therapeutically effective amount" of a compound is an amount that is sufficient to ameliorate, or in some manner reduce a symptom or stop or reverse progression of a condition, or negatively modulate or inhibit the activity of one or more of wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H. Such amount may be administered as a single dosage or may be administered according to a regimen, whereby it is effective.

(30) As used herein, treatment means any manner in which the symptoms or pathology of a condition, disorder or disease are ameliorated or otherwise beneficially altered. Treatment also encompasses any pharmaceutical use of the compositions herein.

(31) As used herein, amelioration of the symptoms of a particular disorder by administration of a particular pharmaceutical composition refers to any lessening, whether permanent or temporary, lasting or transient that can be attributed to or associated with administration of the composition.

(32) Compounds

(33) In certain embodiments of the invention there are provided compounds of Formula (I):

(34) ##STR00007##

or a pharmaceutically acceptable salt thereof, wherein:

(35) W is:

(36) ##STR00008##

(37) A is aryl or heteroaryl, wherein the aryl or the heteroaryl is optionally substituted with 1-4 R.sup.1;

(38) B is selected from:

(39) ##STR00009##

(40) Y.sup.1 is hydrogen, L-hydroxy optionally substituted with 1-4 R.sup.8, L-alkoxy optionally substituted with 1-4 R.sup.8, halogen, L-C3-C6 cycloalkyl optionally substituted with 1-4 R.sup.9, L-heteroaryl optionally substituted with 1-4 R.sup.8, L-aryl optionally substituted with 1-4 R.sup.8, L-C(O)—NH.sub.2, and L-heterocycle substituted with 1-2 oxo (=O) or oxo-containing substituent, and optionally further substituted with 1-2 R.sup.8;

(41) Y.sup.2 is hydrogen or C1-C4 alkyl;

(42) or Y.sup.1 and Y.sup.2 join to form:

(43) ##STR00010##

where X is selected from: a bond, —S—, —O—, —N< bound to a fused ring, —CH.sub.2—, —CH.sub.2—N—, —CH.sub.2—N—CH.sub.2—, —CH.sub.2—CH.sub.2—CH.sub.2—, —CH.sub.2—CH.sub.2—, —O—CH.sub.2— and —S—CH.sub.2—;

(44) Y.sup.3 is hydrogen or C1-C4 alkyl;

(45) Y.sup.4 is hydrogen or C1-C4 alkyl;

(46) or Y.sup.3 and Y.sup.4 join to form:

(47) ##STR00011##

where J is selected from: a bond, —O—, —NH—, —CH.sub.2—, —C(C1-C3 alkyl).sub.2-, —CH(C1-C3alkyl)- and —N(C1-C3 alkyl)-;

(48) each R.sup.1 is independently halogen, cyano, hydroxy, C1-C4 alkyl, —S—C1-C3 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C2-C4 hydroxyalkynyl, C1-C3 cyanoalkyl, triazolyl, C1-C3 haloalkyl, —O—C1-C3 haloalkyl, —S—C1-C3 haloalkyl, C1-C3 alkoxy, hydroxyC1-C3 alkyl, —CH.sub.2C(=O)N(RV).sub.2, —C3-C4 alkynyl(NR.sup.5).sub.2, —N(R).sub.2, deuterioC2-C4 alkynyl, (C1-C3 alkoxy)haloC1-C3 alkyl-, or C3-C6 cycloalkyl wherein said C3-C6 cycloalkyl is optionally substituted with halogen or C1-C3 alkyl;

(49) each R.sup.2 is independently hydrogen, hydroxy, halogen, cyano, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R.sup.5).sub.2, —CO.sub.2R.sup.5, or —CO.sub.2N(R.sup.5).sub.2;

(50) each R.sup.3 is independently hydrogen, hydroxy, halogen, cyano, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R.sup.5).sub.2, —CO.sub.2R.sup.5, or —CO.sub.2N(R.sup.5).sub.2;

(51) each R.sup.4 is independently hydrogen, halogen or C1-C3 alkyl;

(52) each R.sup.5 is independently hydrogen or C1-C3 alkyl, or two R.sup.5 join to form cycloalkyl or heterocycle;

(53) each R.sup.6 is independently hydrogen, hydroxy, C1-C4 hydroxyalkyl or heteroaryl,

(54) or two R.sup.6 join to form C3-C6 cycloalkyl or heterocycle;

(55) each R.sup.7 is independently hydrogen, C1-C3 alkyl, hydroxy, halogen, C1-C3 haloalkyl, -L-NH.sub.2, —NH(C1-C3 alkyl), —N(C1-C3 alkyl).sub.2, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R.sup.10).sub.2, —NHC(O)H, —CN, aryl, —(CH.sub.2).sub.1-2S(O).sub.2N(R.sup.10).sub.2, —NH—S(O).sub.2N(R.sup.10).sub.2, —O—S(O).sub.2N(R.sup.10).sub.2, S(O).sub.2R.sup.10, or heteroaryl or heterocycle optionally independently substituted with 1-2 substituents independently selected from C1-C3 alkyl, —CN and C(O)NH.sub.2,

(56) two R.sup.7 on the same atom optionally join to form a spirocyclic ring selected from C3-C6 cycloalkyl and heterocycle, where said spirocyclic ring is optionally substituted with 1-4

substituents independently selected from oxo (=O), halogen, hydroxy, C1-C3 alkyl and —O—(C1-C3 alkyl),

(57) two R.sup.7 on adjacent atoms optionally join to form a bond or a fused ring selected from C3-C6 cycloalkyl optionally substituted with 1-4 R.sup.8, heteroaryl optionally substituted with 1-4 R.sup.8, aryl optionally substituted with 1-4 R.sup.8, and heterocycle optionally substituted with 1-4 R.sup.8, and

(58) two R.sup.7 on non-adjacent atoms optionally join to form a 1-2 carbon bridge;

(59) each R.sup.8 is independently C1-C3 alkyl, hydroxy, halogen, —N(R.sup.10).sub.2, —N(R.sup.10)C(O)R.sup.10, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R.sup.10).sub.2, heteroaryl or —CN;

(60) each R.sup.9 is independently C1-C3 alkyl, hydroxy, halogen, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)NH.sub.2, —C(O)NH(C1-C3 alkyl), —C(O)N(C1-C3 alkyl).sub.2 or —CN;

(61) each R.sup.10 is independently hydrogen, halogen, C1-C3 alkyl, or two R.sup.10 join to form cycloalkyl or heterocycle optionally substituted with 1-2 C1-C3 alkyl;

(62) each L is independently a bond, —C1-C4 alkyl-, —C1-C4 alkyl-NH—, —NH—, —N(C1-C3 alkyl)- or cyclopropyl-CH.sub.2—;

(63) each n is 0-3;

(64) o is 1-6; and

(65) p is 1-8.

(66) Embodiments of the invention also include compounds of Formula (I):

(67) ##STR00012##

or a pharmaceutically acceptable salt thereof, wherein:

(68) W is:

(69) ##STR00013##

(70) A is naphthyl optionally substituted with 1-4 R.sup.1;

(71) B is:

(72) ##STR00014##

(73) Y.sup.1 and Y.sup.2 join to form:

(74) ##STR00015##

where X is selected from: —CH.sub.2—, —CH.sub.2—CH.sub.2— and —O—CH.sub.2—;

(75) each R.sup.1 is independently halogen, cyano, hydroxy, C1-C4 alkyl, —S—C1-C3 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C2-C4 hydroxyalkynyl, C1-C3 cyanoalkyl, triazolyl, C1-C3 haloalkyl, —O—C1-C3 haloalkyl, —S—C1-C3 haloalkyl, C1-C3 alkoxy, hydroxyC1-C3 alkyl, —CH.sub.2C(=O)N(R.sup.5).sub.2, —C3-C4 alkynyl(NR.sup.5).sub.2, —N(R.sup.5).sub.2, deuterioC2-C4 alkynyl, (C1-C3 alkoxy)haloC1-C3 alkyl-, or C3-C6 cycloalkyl wherein said C3-C6 cycloalkyl is optionally substituted with halogen or C1-C3 alkyl;

(76) each R.sup.2 is independently hydrogen, hydroxy, halogen, cyano, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R.sup.5).sub.2, —CO.sub.2R, or —CO.sub.2N(R.sup.5).sub.2;

(77) each R.sup.3 is independently hydrogen, hydroxy, halogen, cyano, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R.sup.5).sub.2, —CO.sub.2R.sup.3, or —CO.sub.2N(R.sup.5).sub.2;

(78) each R.sup.4 is independently hydrogen, halogen or C1-C3 alkyl;

(79) each R.sup.5 is independently hydrogen or C1-C3 alkyl, or two R.sup.5 join to form cycloalkyl or heterocycle;

(80) each R.sup.6 is independently hydrogen, hydroxy, C1-C4 hydroxyalkyl or heteroaryl,

(81) or two R.sup.6 join to form C3-C6 cycloalkyl or heterocycle;

(82) each R.sup.7 is independently hydrogen, C1-C3 alkyl, hydroxy, halogen, C1-C3 haloalkyl, -L-NH.sub.2, —NH(C1-C3 alkyl), —N(C1-C3 alkyl).sub.2, oxo (=O), —O—(C1-C3 alkyl), —(C1-

- C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R.sup.10).sub.2, —NHC(O)H, —CN, aryl, —(CH.sub.2).sub.1-2S(O).sub.2N(R.sup.10).sub.2, —NH—S(O).sub.2N(R.sup.10).sub.2, —O—S(O).sub.2N(R.sup.10).sub.2, S(O).sub.2R.sup.10, or heteroaryl or heterocycle optionally independently substituted with 1-2 substituents independently selected from C1-C3 alkyl, —CN and C(O)NH.sub.2,
- (83) two R.sup.7 on the same atom optionally join to form a spirocyclic ring selected from C3-C6 cycloalkyl and heterocycle, where said spirocyclic ring is optionally substituted with 1-4 substituents independently selected from oxo (=O), halogen, hydroxy, C1-C3 alkyl and —O—(C1-C3 alkyl),
- (84) two R.sup.7 on adjacent atoms optionally join to form a bond or a fused ring selected from C3-C6 cycloalkyl optionally substituted with 1-4 R.sup.8, heteroaryl optionally substituted with 1-4 R.sup.8, aryl optionally substituted with 1-4 R.sup.8, and heterocycle optionally substituted with 1-4 R.sup.8, and
- (85) two R.sup.7 on non-adjacent atoms optionally join to form a 1-2 carbon bridge;
- (86) each R.sup.8 is independently C1-C3 alkyl, hydroxy, halogen, —N(R.sup.10).sub.2, —N(R.sup.10)C(O)R.sup.10, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R.sup.10).sub.2, heteroaryl or —CN;
- (87) each R.sup.9 is independently C1-C3 alkyl, hydroxy, halogen, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)—OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)NH.sub.2, —C(O)NH(C1-C3 alkyl), —C(O)N(C1-C3 alkyl).sub.2 or —CN;
- (88) each R.sup.10 is independently hydrogen, halogen, C1-C3 alkyl, or two R.sup.10 join to form cycloalkyl or heterocycle optionally substituted with 1-2 C1-C3 alkyl;
- (89) each L is independently a bond, —C1-C4 alkyl-, —C1-C4 alkyl-NH—, —NH—, —N(C1-C3 alkyl)- or cyclopropyl-CH.sub.2—;
- (90) each n is 0-3;
- (91) o is 1-6; and
- (92) p is 1-8.
- (93) Embodiments also include such compounds or salts wherein each R.sup.1 is independently selected from halogen, hydroxy, C1-C3 alkoxy and C1-C4 alkyl.
- (94) Embodiments also include such compounds or salts wherein each R.sup.2, if present, is selected from hydrogen and halogen, and wherein each R.sup.3, if present, is selected from hydrogen and halogen.
- (95) Embodiments also include such compounds or salts wherein each R.sup.7 is independently selected from hydrogen, C1-C4 alkyl, hydroxy, C1-C3 alkoxy, and wherein two R.sup.7 on non-adjacent atoms optionally join to form a 1-2 carbon bridge.
- (96) Embodiments also include such compounds or salts wherein each R.sup.6 is independently hydrogen or hydroxy.
- (97) Embodiments also include such compounds or salts wherein B is:
- (98) ##STR00016##
- (99) Embodiments also include such compounds or salts wherein B is:
- (100) ##STR00017##
- (101) Embodiments also include such compounds or salts wherein B is:
- (102) ##STR00018##
- (103) Embodiments also include such compounds or salts wherein Y.sup.1 and Y.sup.2 join to form:
- (104) ##STR00019##
- (105) Embodiments also include such compounds or salts wherein Y.sup.1 and Y.sup.2 join to form:
- (106) ##STR00020##
- (107) Embodiments also include such compounds or salts wherein Y.sup.1 and Y.sup.2 join to

form:

(108) ##STR00021##

(109) Embodiments also include such compounds or salts wherein A is naphthyl.

(110) Embodiments also include such compounds or salts wherein A is indazolyl.

(111) Embodiments also include such compounds or salts wherein A is phenyl.

(112) Embodiments also include such compounds or salts wherein A is pyridyl.

(113) In certain embodiments of the invention at least one R^{sup.1} is C1-C4 alkyl.

(114) In certain embodiments of the invention at least one R^{sup.1} is halogen, preferably fluorine or chlorine.

(115) In certain embodiments of the invention at least one R^{sup.1} is hydroxy

(116) In certain embodiments of the invention at least one R^{sup.2} is C1-C4 alkyl.

(117) In certain embodiments of the invention at least one R^{sup.2} is halogen, preferably fluorine or chlorine.

(118) In certain embodiments of the invention at least one R^{sup.2} is hydroxy.

(119) In certain embodiments of the invention at least one R^{sup.3} is C1-C4 alkyl.

(120) In certain embodiments of the invention at least one R^{sup.3} is halogen, preferably fluorine or chlorine.

(121) In certain embodiments of the invention at least one R^{sup.3} is hydroxy.

(122) In certain embodiments of the invention R^{sup.4} is halogen, preferably fluorine.

(123) In certain embodiments of the invention at least one R^{sup.5} is C1-C4 alkyl.

(124) In certain embodiments of the invention at least one R^{sup.5} is hydrogen.

(125) In certain embodiments of the invention at least one R^{sup.6} is C1-C4 alkyl.

(126) In certain embodiments of the invention, two R^{sup.6} join to form C3-C6 cycloalkyl or heterocycle.

(127) In certain embodiments of the invention at least one R^{sup.6} is hydrogen.

(128) In certain embodiments of the invention both R^{sup.6} are C1-C4 alkyl.

(129) In certain embodiments of the invention both R^{sup.6} are hydrogen.

(130) In certain embodiments Y^{sup.1} is L-C3-C6 cycloalkyl, L-heteroaryl, L-aryl, or L-heterocycle. In certain of these embodiments, L is a bond. In certain of these embodiments L is C1-C4 alkyl. In certain of these embodiments L is NH or N(C1-C3) alkyl.

(131) In certain embodiments Y^{sup.1} is L-heteroaryl where the heteroaryl is thietane dioxide, isothiazolidine dioxide, imidazopyrazine, pyridine or pyrimidine.

(132) In certain embodiments Y^{sup.1} is L-C3-C6 cycloalkyl where the cycloalkyl is preferably cyclobutane, cyclopentane, cyclohexane or cycloheptane.

(133) In certain embodiments Y^{sup.1} is L-heterocycle where the heterocycle is preferably pyrrolidinone.

(134) In certain embodiments of the invention Y^{sup.2} is hydrogen.

(135) In certain embodiments of the invention Y^{sup.2} is C1-C4 alkyl;

(136) In certain embodiments of the invention at least one R^{sup.8} is C1-C4 alkyl, preferably methyl.

(137) In certain embodiments of the invention at least one R^{sup.8} is hydroxy or C1-C3 alkyl-hydroxy.

(138) In certain embodiments of the invention one or two R^{sup.8} is oxo (=O).

(139) In certain embodiments of the invention at least one R^{sup.8} is aryl or heteroaryl.

(140) In certain embodiments of the invention at least one R^{sup.8} is C(O)OH.

(141) In certain embodiments of the invention at least one R^{sup.8} is —C(O)NH_{sub.2}, —C(O)NH(C1-C3 alkyl) or —C(O)N(C1-C3 alkyl)_{sub.2}.

(142) In certain embodiments of the invention R^{sup.8} is —NH_{sub.2}, —NH(C1-C3 alkyl); —N(C1-C3 alkyl)_{sub.2}.

(143) In certain embodiments of the invention at least one R^{sup.9} is C1-C4 alkyl, preferably

methyl.

(144) In certain embodiments of the invention at least one R.sup.9 is hydroxy or C1-C3 alkyl-hydroxy.

(145) In certain embodiments of the invention one or two R.sup.9 is oxo (=O).

(146) In certain embodiments of the invention at least one R.sup.9 is aryl or heteroaryl.

(147) In certain embodiments of the invention at least one R.sup.9 is C(O)OH.

(148) In certain embodiments of the invention at least one R.sup.9 is —C(O)NH.sub.2, —C(O)NH(C1-C3 alkyl) or —C(O)N(C1-C3 alkyl).sub.2.

(149) In certain embodiments of the invention Y.sup.1 and Y.sup.2 join to form a piperidine, azepane, azocane, thiazepine, diazepane, oxazepane, azetidine, pyrrolidine, piperazine bound to a fused ring via nitrogen or thiomorpholine.

(150) In certain embodiments of the invention, two R.sup.7 on the same atom join to form a spirocyclic ring selected from C3-C6 cycloalkyl and heterocycle, where said spirocyclic ring is optionally substituted with oxo (=O), halogen, hydroxy, C1-C3 alkyl and —O—(C1-C3 alkyl).

(151) In certain embodiments of the invention, two R.sup.7 on adjacent atoms optionally join to form a bond or a fused ring selected from C3-C6 cycloalkyl optionally substituted with 1-4 R.sup.8; heteroaryl optionally substituted with 1-4 R.sup.8; aryl optionally substituted with 1-4 R.sup.8, and heterocycle optionally substituted with 1-4 R.sup.8.

(152) In certain embodiments of the invention, two R.sup.7 on non-adjacent atoms optionally join to form a 1-2 carbon bridge.

(153) Non-limiting examples of compounds of Formula (I) are selected from the group consisting of the compounds described in the below Examples, and pharmaceutically acceptable salts thereof.

(154) In one embodiment, the compounds of Formula (1) include bis-hydrochloride, tris-hydrochloride, trifluoroacetic acid, bis-trifluoroacetic acid, and tris-trifluoroacetic acid salts of the above compounds. The compounds of Formula (I) or pharmaceutically acceptable salt thereof may be formulated into pharmaceutical compositions.

(155) Pharmaceutical Compositions

(156) In another aspect, the invention provides pharmaceutical compositions comprising a wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H inhibitor according to the invention and a pharmaceutically acceptable carrier, excipient, or diluent. Compounds of the invention may be formulated by any method well known in the art and may be prepared for administration by any route, including, without limitation, parenteral, intraperitoneal, intradermal, intracardiac, intraventricular, intracranial, intracerebrospinal, intrasynovial, intrathecal administration, intramuscular injection, intravitreal injection, intravenous injection, intra-arterial injection, oral, buccal, sublingual, transdermal, topical, intranasal, intratracheal, intrarectal, subcutaneous, and topical administration. In certain embodiments, compounds of the invention are administered intravenously in a hospital setting. In one embodiment, administration may be by the oral route. In some embodiments, the provided pharmaceutical compositions may be administered to a subject in need of treatment by injection systemically, such as by intravenous injection; or by injection or application to the relevant site, such as by direct injection via syringe, or direct application to the site when the site is exposed in surgery; or by topical administration.

(157) Parenteral administration can be by bolus injection or continuous infusion. Pharmaceutical compositions for injection may be presented in unit dosage form, e.g., in ampoules or in multi-dose containers, with an added preservative.

(158) The provided pharmaceutical compositions can also be formulated as a depot preparation. Such long acting formulations may be administered by implantation (for example subcutaneously or intramuscularly) or by intramuscular injection. Thus, for example, the formulations may be modified with suitable polymeric or hydrophobic materials (for example as an emulsion in an acceptable oil) or ion exchange resins, or as sparingly soluble derivatives, for example, as a

sparingly soluble salt.

(159) The pharmaceutical compositions may, if desired, be presented in a vial, pack or a medical device, including but not limited to a dispenser device which may contain one or more unit dosage forms containing the active ingredient. In one embodiment the dispenser device can comprise a syringe having a single dose of the liquid formulation ready for injection. The syringe can be accompanied by instructions for administration.

(160) The characteristics of the carrier will depend on the route of administration. As used herein, the term “pharmaceutically acceptable” means a non-toxic material that is compatible with a biological system such as a cell, cell culture, tissue, or organism, and that does not interfere with the effectiveness of the biological activity of the active ingredient(s). Thus, compositions according to the invention may contain, in addition to the inhibitor, diluents, fillers, salts, buffers, stabilizers, solubilizers, and other materials well known in the art. The preparation of pharmaceutically acceptable formulations is described in, e.g., Remington's Pharmaceutical Sciences, 18th Edition, ed. A. Gennaro, Mack Publishing Co., Easton, Pa., 1990.

(161) As used herein, the term pharmaceutically acceptable salt refers to salts that retain the desired biological activity of the above-identified compounds and exhibit minimal or no undesired toxicological effects. Examples of such salts include but are not limited to acid addition salts formed with inorganic acids (for example, hydrochloric acid, hydrobromic acid, sulfuric acid, phosphoric acid, nitric acid, and the like), and salts formed with organic acids such as acetic acid, oxalic acid, tartaric acid, succinic acid, malic acid, ascorbic acid, benzoic acid, tannic acid, pamoic acid, alginic acid, polyglutamic acid, naphthalenesulfonic acid, naphthalenedisulfonic acid, and polygalacturonic acid. The compounds can also be administered as pharmaceutically acceptable quaternary salts known by those skilled in the art, which specifically include the quaternary ammonium salt of the formula $\text{—NR}^+\text{Z—}$, wherein R is hydrogen, alkyl, or benzyl, and Z is a counterion, including chloride, bromide, iodide, —O—alkyl , toluenesulfonate, methylsulfonate, sulfonate, phosphate, or carboxylate (such as benzoate, succinate, acetate, glycolate, maleate, malate, citrate, tartrate, ascorbate, benzoate, cinnamoate, mandelate, benzyloate, and diphenylacetate).

(162) The active compound is included in the pharmaceutically acceptable carrier or diluent in an amount sufficient to deliver to a patient a therapeutically effective amount without causing serious toxic effects in the patient treated. In one embodiment, a dose of the active compound for all of the above-mentioned conditions is in the range from about 0.01 to 300 mg/kg, for example 0.1 to 100 mg/kg per day, and as a further example 0.5 to about 25 mg per kilogram body weight of the recipient per day. A typical topical dosage will range from 0.01-3% wt/wt in a suitable carrier. The effective dosage range of the pharmaceutically acceptable derivatives can be calculated based on the weight of the parent compound to be delivered. If the derivative exhibits activity in itself, the effective dosage can be estimated as above using the weight of the derivative, or by other means known to those skilled in the art.

(163) The pharmaceutical compositions comprising compounds of the present invention may be used in the methods of use described herein.

(164) Methods of Use

(165) In yet another aspect, the invention provides for methods for inhibiting wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V and/or KRas Q61H activity in a cell, comprising contacting the cell in which inhibition of wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V and/or Q61H activity is desired with an effective amount of a compound of Formula (I), pharmaceutically acceptable salts thereof, or pharmaceutical compositions containing the compound or pharmaceutically acceptable salt thereof. In one embodiment, the contacting is in vitro. In one embodiment, the contacting is in vivo.

(166) As used herein, the term “contacting” refers to the bringing together of indicated moieties in

an in vitro system or an in vivo system. For example, "contacting" wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H with a compound provided herein includes the administration of a compound provided herein to an individual or patient, such as a human, having wild type KRas or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H mutation, as well as, for example, introducing a compound provided herein into a sample containing a cellular or purified preparation containing wild type KRas or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H mutation.

(167) In one embodiment, a cell in which inhibition of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H activity is desired is contacted with an effective amount of a compound of Formula (I) or pharmaceutically acceptable salt thereof to negatively modulate the activity of one or more of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and KRas Q61H.

(168) By negatively modulating the activity of one or more of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and KRas Q61H, the methods described herein are designed to inhibit undesired cellular proliferation resulting from enhanced wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H activity within the cell. The cells may be contacted in a single dose or multiple doses in accordance with a particular treatment regimen to affect the desired negative modulation of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H. The ability of compounds to bind one or more of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and KRas Q61H may be monitored in vitro using well known methods, including those described in Examples A and B below. In addition, the inhibitory activity of exemplary compounds in cells may be monitored, for example, by measuring the inhibition of one or more of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H activity of the amount of phosphorylated ERK, for example using the method described in Example C below.

(169) In another aspect, methods of treating cancer in a patient in need thereof, comprising administering to said patient a therapeutically effective amount of a compound of Formula (I) or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition comprising the compound or pharmaceutically acceptable salt thereof are provided.

(170) The compositions and methods provided herein may be used for the treatment of a wild type KRas-associated or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H-associated cancer in a patient in need thereof, comprising administering to said patient a therapeutically effective amount of a compound of Formula (I), a pharmaceutically acceptable salt thereof, or a pharmaceutical composition comprising the compound or pharmaceutically acceptable salt thereof are provided. In one embodiment, the wild type KRas-associated or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H-associated cancer is lung cancer.

(171) The compositions and methods provided herein may be used for the treatment of a wide variety of cancers including tumors such as lung, prostate, breast, brain, skin, cervical carcinomas, testicular carcinomas, etc. More particularly, cancers that may be treated by the compositions and methods of the invention include but are not limited to tumor types such as astrocytic, breast, cervical, colorectal, endometrial, esophageal, gastric, head and neck, hepatocellular, laryngeal, lung, oral, ovarian, prostate and thyroid carcinomas and sarcomas. More specifically, these compounds can be used to treat: Cardiac: sarcoma (angiosarcoma, fibrosarcoma, rhabdomyosarcoma, liposarcoma), myxoma, rhabdomyoma, fibroma, lipoma and teratoma; Lung: bronchogenic carcinoma (squamous cell, undifferentiated small cell, undifferentiated large cell,

adenocarcinoma), alveolar (bronchiolar) carcinoma, bronchial adenoma, sarcoma, lymphoma, chondromatous hamartoma, mesothelioma; Gastrointestinal: esophagus (squamous cell carcinoma, adenocarcinoma, leiomyosarcoma, lymphoma), stomach (carcinoma, lymphoma, leiomyosarcoma), pancreas (ductal adenocarcinoma, insulinoma, glucagonoma, gastrinoma, carcinoid tumors, vipoma), small bowel (adenocarcinoma, lymphoma, carcinoid tumors, Kaposi's sarcoma, leiomyoma, hemangioma, lipoma, neurofibroma, fibroma), large bowel (adenocarcinoma, tubular adenoma, villous adenoma, hamartoma, leiomyoma); Genitourinary tract: kidney (adenocarcinoma, Wilm's tumor (nephroblastoma), lymphoma, leukemia), bladder and urethra (squamous cell carcinoma, transitional cell carcinoma, adenocarcinoma), prostate (adenocarcinoma, sarcoma), testis (seminoma, teratoma, embryonal carcinoma, teratocarcinoma, choriocarcinoma, sarcoma, interstitial cell carcinoma, fibroma, fibroadenoma, adenomatoid tumors, lipoma); Liver: hepatoma (hepatocellular carcinoma), cholangiocarcinoma, hepatoblastoma, angiosarcoma, hepatocellular adenoma, hemangioma; Biliary tract: gall bladder carcinoma, ampullary carcinoma, cholangiocarcinoma; Bone: osteogenic sarcoma (osteosarcoma), fibrosarcoma, malignant fibrous histiocytoma, chondrosarcoma, Ewing's sarcoma, malignant lymphoma (reticulum cell sarcoma), multiple myeloma, malignant giant cell tumor chordoma, osteochondroma (osteochondrogenous exostoses), benign chondroma, chondroblastoma, chondromyxofibroma, osteoid osteoma and giant cell tumors; Nervous system: skull (osteoma, hemangioma, granuloma, xanthoma, osteitis deformans), meninges (meningioma, meningiosarcoma, gliomatosis), brain (astrocytoma, medulloblastoma, glioma, ependymoma, germinoma (pinealoma), glioblastoma multiform, oligodendroglioma, schwannoma, retinoblastoma, congenital tumors), spinal cord neurofibroma, meningioma, glioma, sarcoma); Gynecological: uterus (endometrial carcinoma), cervix (cervical carcinoma, pre-tumor cervical dysplasia), ovaries (ovarian carcinoma (serous cystadenocarcinoma, mucinous cystadenocarcinoma, unclassified carcinoma), granulosa-thecal cell tumors, Sertoli-Leydig cell tumors, dysgerminoma, malignant teratoma), vulva (squamous cell carcinoma, intraepithelial carcinoma, adenocarcinoma, fibrosarcoma, melanoma), vagina (clear cell carcinoma, squamous cell carcinoma, botryoid sarcoma (embryonal rhabdomyosarcoma), fallopian tubes (carcinoma); Hematologic: blood (myeloid leukemia (acute and chronic), acute lymphoblastic leukemia, chronic lymphocytic leukemia, myeloproliferative diseases, multiple myeloma, myelodysplastic syndrome), Hodgkin's disease, non-Hodgkin's lymphoma (malignant lymphoma); Skin: malignant melanoma, basal cell carcinoma, squamous cell carcinoma, Kaposi's sarcoma, moles dysplastic nevi, lipoma, angioma, dermatofibroma, keloids, psoriasis; and Adrenal glands: neuroblastoma. In certain embodiments, the cancer is non-small cell lung cancer, small cell lung cancer, colorectal cancer, rectal cancer or pancreatic cancer. In certain embodiments, the cancer is non-small cell lung cancer.

(172) The concentration and route of administration to the patient will vary depending on the cancer to be treated. The compounds, pharmaceutically acceptable salts thereof and pharmaceutical compositions comprising such compounds and salts also may be co-administered with other anti-neoplastic compounds, e.g., chemotherapy, or used in combination with other treatments, such as radiation or surgical intervention, either as an adjuvant prior to surgery or post-operatively.

(173) Also provided herein is a compound of Formula (I) or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof as defined herein for use in therapy.

(174) Also provided herein is a compound of Formula (I) or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof as defined herein for use in the treatment of cancer.

(175) Also provided herein is a compound of Formula (I) or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof for use in the inhibition of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H.

(176) Also provided herein is a compound of Formula (I) or a pharmaceutically acceptable salt

thereof, or a pharmaceutical composition thereof as defined herein, for use in the treatment of wild type KRas-associated or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H-associated disease or disorder.

(177) Also provided herein is the use of a compound of Formula (I) or a pharmaceutically acceptable salt thereof, as defined herein in the manufacture of a medicament for the treatment of cancer.

(178) Also provided herein is a use of a compound of Formula (I) or a pharmaceutically acceptable salt thereof, as defined herein in the manufacture of a medicament for the inhibition of activity of wild type KRas or KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H.

(179) Also provided herein is the use of a compound of Formula (I) or a pharmaceutically acceptable salt thereof, as defined herein, in the manufacture of a medicament for the treatment of wild type KRas-associated or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H-associated disease or disorder.

(180) Also provided herein is a method for treating cancer in a patient in need thereof, the method comprising (a) determining that cancer is associated with wild type KRas or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D and/or KRas Q61H mutation (e.g., as determined using a regulatory agency-approved, e.g., FDA-approved, assay or kit); and (b) administering to the patient a therapeutically effective amount of a compound of Formula (1), or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition thereof.

(181) One skilled in the art will recognize that, both in vivo and in vitro trials using suitable, known and generally accepted cell and/or animal models are predictive of the ability of a test compound to treat or prevent a given disorder.

(182) One skilled in the art will further recognize that human clinical trials including first-in-human, dose ranging and efficacy trials, in healthy patients and/or those suffering from a given disorder, may be completed according to methods well known in the clinical and medical arts.

Reaction Schemes and Examples

(183) The compounds of the present invention may be prepared from commercially available reagents using the synthetic methods and reaction schemes described herein, or using other reagents and conventional methods well known to those skilled in the art. For instance, compounds of the present invention may be prepared according to the reaction schemes and examples outlines below.

(184) The compounds of the present invention may have one or more chiral center and may be synthesized as stereoisomeric mixtures, isomers of identical constitution that differ in the arrangement of their atoms in space. The compounds may be used as mixtures or the individual components/isomers may be separated using commercially available reagents and conventional methods for isolation of stereoisomers and enantiomers well-known to those skilled in the art, e.g., using CHIRALPAK® (Sigma-Aldrich) or CHIRALCEL® (Diacel Corp) chiral chromatographic HPLC columns according to the manufacturer's instructions. Alternatively, compounds of the present invention may be synthesized using optically pure, chiral reagents and intermediates to prepare individual isomers or enantiomers. Unless otherwise indicated, all chiral (enantiomeric and diastereomeric) and racemic forms are within the scope of the invention. Unless otherwise indicated, whenever the specification, including the claims, refers to compounds of the invention, the term "compound" is to be understood to encompass all chiral (enantiomeric and diastereomeric) and racemic forms.

(185) The compounds of the present invention may be in anhydrous, solvated or hydrated forms, and all such forms are included within the scope of the invention.

(186) The following Intermediates are intended to illustrate further certain embodiments of the invention and are not intended to limit the scope of the invention.

Intermediate 1

(187) ##STR00022##

2,4-dichloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine

(188) ##STR00023##

(189) Step A. methyl 4-(tert-butoxycarbonylamino)-6-chloro-5-fluoro-pyridine-3-carboxylate. To a solution of 4-((tert-butoxycarbonyl)amino)-6-chloro-5-fluoronicotinic acid (14.3 g, 49.2 mmol, 1 equiv.) in MeOH (70 mL) and toluene (210 mL) was added TMSCHN.sub.2 (2 M in hexane, 44.3 mL, 1.8 equiv.) slowly. After stirring at 15° C. for 2 hours, the mixture was quenched with 2N HCl (100 mL) and layers were separated. The organic phase was washed with saturated aqueous NaHCO.sub.3 (150 mL), followed by brine (150 mL). The organic layer was dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate 10:1 to 1:1) to give methyl 4-(tert-butoxycarbonylamino)-6-chloro-5-fluoro-pyridine-3-carboxylate (15 g, 91%). Colorless oil; R_f=0.50 (3:1 petroleum ether/ethyl acetate); ¹H NMR (400 MHz, CDCl₃): δ 8.85 (br s, 1H), 8.68 (s, 1H), 3.98 (s, 3H), 1.57-1.49 (m, 9H); LCMS [ESI, M+1]: 305.

(190) Step B. methyl 4-amino-6-chloro-5-fluoro-pyridine-3-carboxylate. To a solution of methyl 4-(tert-butoxycarbonylamino)-6-chloro-5-fluoro-pyridine-3-carboxylate (15 g, 49.2 mmol, 1.0 equiv.) in MeCN (150 mL) was added HCl.dioxane (4 M, 290 mL, 23.6 equiv.) at 0° C. The mixture was stirred at 15° C. for 0.5 hour, and the solvent was removed under reduced pressure. The residue was diluted with saturated Na.sub.2CO.sub.3 solution (100 mL) and extracted with ethyl acetate (3×200 mL). The organic layers were dried over Na.sub.2SO.sub.4 and concentrated in vacuum to give methyl 4-amino-6-chloro-5-fluoro-pyridine-3-carboxylate (9.07 g, 89%) which was used directly in the next step without further purification. Orange solid; LCMS [ESI, M+1]: 205.

(191) Step C. methyl 4-amino-6-(8-chloro-1-naphthyl)-5-fluoro-pyridine-3-carboxylate. A mixture of methyl 4-amino-6-chloro-5-fluoro-pyridine-3-carboxylate (6 g, 29.3 mmol, 1.0 equiv.), (8-chloronaphthalen-1-yl)trimethylstannane (21.0 g, 64.5 mmol, 2.2 equiv.), CuI (1.68 g, 8.80 mmol, 0.3 equiv.), Pd(dppf)Cl.sub.2 (2.15 g, 2.93 mmol, 0.1 equiv.), and BINAP (3.65 g, 5.87 mmol, 0.2 equiv.) in toluene (120 mL) was degassed and then heated to 100° C. for 11 hours under N.sub.2. The mixture was filtered and the filtrate was concentrated in vacuum. The residue was diluted with water (50 mL) and extracted with ethyl acetate (2×60 mL). The organic layers were dried over Na.sub.2SO.sub.4 and concentrated in vacuum. The residue was purified by chromatography (Al.sub.2O.sub.3, petroleum ether/ethyl acetate 30/1 to 1/1). The product was triturated with a mixed solution (DMAc/methanol 1/2, 30 mL) at 15° C. for 10 minutes to give methyl 4-amino-6-(8-chloro-1-naphthyl)-5-fluoro-pyridine-3-carboxylate (5.33 g, 54%). Yellow solid; R_f=0.20 (3:1 petroleum ether/ethyl acetate); LCMS [ESI, M+1]: 331.

(192) Step D. methyl 6-(8-chloro-1-naphthyl)-5-fluoro-4-[(2,2,2-trichloroacetyl)carbamoylamino]pyridine-3-carboxylate. To a solution of methyl 4-amino-6-(8-chloro-1-naphthyl)-5-fluoro-pyridine-3-carboxylate (5.5 g, 16.6 mmol, 1.0 equiv.) in THF (82 mL) was added 2,2,2-trichloroacetyl isocyanate (3.45 g, 18.3 mmol, 2.17 mL, 1.1 equiv.) dropwise. The mixture was stirred at 15° C. for 10 minutes, and the mixture was concentrated in vacuum. The residue was triturated with MTBE (20 mL) at 15° C. for 15 minutes to give methyl 6-(8-chloro-1-naphthyl)-5-fluoro-4-[(2,2,2-trichloroacetyl)carbamoylamino]pyridine-3-carboxylate (8 g, crude). Yellow solid; LCMS [ESI, M+1]: 520.

(193) Step E. 7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol. A suspension of methyl 6-(8-chloro-1-naphthyl)-5-fluoro-4-[(2,2,2-trichloroacetyl)carbamoylamino]pyridine-3-carboxylate (8 g, 15.4 mmol, 1.0 equiv.) in NH.sub.3-MeOH (20 mL, 20% purity) was stirred at 15° C. for 0.5 hour, the mixture was concentrated in vacuum. The residue was triturated with MTBE (30 mL) at 15° C. for 15 minutes to give 7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (5.3 g, two steps 93%). Yellow solid; ¹H NMR (400 MHz, DMSO): δ 9.59-8.27 (m, 1H), 8.24-8.13 (m, 1H), 8.11-8.03 (m, 1H), 7.74-7.61 (m, 2H), 7.60-7.52 (m, 2H),

3.59-3.31 (m, 2H); LCMS [ESI, M+1]: 342.

(194) Step F. 2,4-dichloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine. A solution of POCl.sub.3 (1.62 g, 10.6 mmol, 985 µL, 36.2 equiv.) and N-ethyl-N-isopropylpropan-2-amine (189 mg, 1.46 mmol, 255 µL, 5.0 equiv.) was stirred at 0° C., followed by the addition of 7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (0.1 g, 293 µmol, 1.0 equiv.). The suspension was stirred at 110° C. for 1 hour, the mixture was concentrated in vacuum to give 2,4-dichloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine (0.11 g, crude) which was used directly in the next step without further purification. Black oil.

Intermediate 2

(195) ##STR00024##

(R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(196) ##STR00025##

(197) Step A. 3-methylpiperidin-3-ol: To the solution of tert-butyl 3-hydroxy-3-methyl-piperidine-1-carboxylate (2.45 g, 11.4 mmol) in acetonitrile (9 mL) was added HCl.dioxane (4 M, 18 mL) at 0° C., and the mixture was stirred at 0° C. for 0.5 h. After completion, the reaction mixture was concentrated to give 3-methylpiperidin-3-ol (1.75 g, crude) as a yellow oil which was used in the next step without further purification.

(198) Step B. 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the mixture of 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (3.65 g, 14.5 mmol), DIEA (7.47 g, 57.8 mmol) in dichloromethane (40 mL) was added 3-methylpiperidin-3-ol (1.75 g, crude) at -40° C., and the mixture was stirred at -40° C. for 0.5 hour. After completion, the reaction mixture was diluted with water (50 mL) and extracted with dichloromethane (100 mL). The combined organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated to give a residue. The residue was purified by column chromatography (5% to 50% EA/PE) to give 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.05 g, 43% yield). Yellow Solid.

LCMS (ESI, M+1): m/z 331.0.

(199) Step C. (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.05 g) was separated by SFC (column: DAICEL CHIRALCEL OJ (250 mm×30 mm, 10 µm); mobile phase: [0.1% NH.sub.3WATER MeOH]; B %: 20%-20%, 3.7; 1035 min) to give (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (1.00 g, 48%). Yellow Solid; LCMS (ESI, M+1): m/z 331.1.

Intermediate 3

(200) ##STR00026##

(S)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(201) (S)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol was isolated as the other enantiomer in step C of the synthesis of intermediate 2. (1.05 g, 52%, yellow solid)

LCMS (ESI, M+1): m/z 331.1.

Intermediate 4

(202) ##STR00027##

2,7-dichloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]imidine

(203) ##STR00028##

(204) Step A. 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoropyrido[4,3-d]pyrimidine-2,4-diol (2.8 g, 13.0 mmol) in POCl.sub.3 (20 mL) was added DIEA (5.04 g, 39.0 mmol, 6.79 mL) in one portion at 25° C. under N.sub.2. The mixture was heated to 110° C. and stirred for 2 h. The mixture was concentrated in reduced pressure to give a residue. The residue was purified by silica gel chromatography (Silica gel, Petroleum ether/Ethyl acetate=20/1, 3/1) to afford 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (3.1 g, 89% yield) as a yellow solid; LCMS (ESI, M+1): m/z 251.9.

(205) Step B. 2,7-dichloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a mixture of

2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (100 mg, 396 μmol) and DIPEA (76.8 mg, 594 μmol , 103 μL) in DCM (1 mL) was added piperidine (40.5 mg, 475 μmol , 46.9 μL) in portions at -40°C . under N.sub.2. The mixture was stirred at -40°C . for 1 hour. The reaction mixture was filtered and concentrated in vacuum. The residue was purified by silica gel chromatography (Silica gel, Petroleum ether/Ethyl acetate=1/0 to 0/1) affording 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (105 mg, 88% yield) as a yellow solid. LCMS (ESI, M+1): m/z 301.0.

Intermediate 5

(206) ##STR00029##

((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane

(207) ##STR00030##

(208) Step A. 5-(2-(4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione. To a solution of 2-(4-fluorophenyl)acetic acid (500 g, 3.24 mol, 1 equiv.), Meldrum's acid (514 g, 3.57 mol, 1.1 equiv.), DMAP (33.7 g, 275 mmol, 0.085 equiv.) in CH.sub.3CN (1500 mL) was added DIPEA (901 g, 6.97 mol, 1.21 L, 2.15 equiv.) while maintaining the temperature below 45°C ., and then pivaloyl chloride (430 g, 3.57 mol, 439 mL, 1.1 equiv.) was slowly added over 3 hours while maintaining the temperature below 45°C . The resulted solution was stirred at 45°C . for 3 hours. The mixture solution was cooled to 0°C ., then IN HCl (5 L) was slowly added, and the resulted solution was stirred at 0°C . for 2 hours. Lot of solid was generated, and the mixture was filtered to give the crude yellow solid. The crude was washed with CH.sub.3CN/WATER (3 L/12 L) to give 5-(2-(4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (800 g, 88% yield). White Solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =15.35 (s, 1H), 7.40-7.38 (m, 2H), 7.05-7.01 (m, 2H), 4.40 (s, 2H), 1.72 (s, 6H).

(209) Step B. tert-butyl 4-(4-fluorophenyl)-3-oxobutanoate. A solution of 5-(2-(4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (1 kg) in t-BuOH (3 L) was stirred at 90°C . for 2 hours, then the mixture solution was concentrated to give the crude solid, and the crude solid was washed with petroleum ether (350 mL) to give tert-butyl 4-(4-fluorophenyl)-3-oxobutanoate (850 g, 94% yield). Light-yellow Solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =7.27-7.18 (m, 2H), 7.18-7.08 (m, 2H), 3.86 (s, 2H), 3.55 (s, 2H), 1.40 (s, 9H).

(210) Step C. 4-(4-fluorophenyl)-3-oxobutanoic acid. A solution of tert-butyl 4-(4-fluorophenyl)-3-oxobutanoate (800 g, 3.17 mol, 1 equiv.) and TFA (2.46 kg, 21.6 mol, 1.6 L, 6.81 equiv.) in DCM (1.6 L) was stirred at 20°C . for 1 hour. The mixture was concentrated to dryness. The residue was washed with petroleum ether (500 mL) to give 4-(4-fluorophenyl)-3-oxobutanoic acid (516 g, 83% yield). White Solid; .sup.1H NMR (400 MHz, CDCl.sub.3-d) δ =10.01 (s, 1H), 7.20-7.17 (m, 2H), 7.07-7.03 (m, 2H), 3.84 (s, 2H), 3.54-3.52 (m, 2H).

(211) Step D. 7-fluoronaphthalene-1,3-diol. A solution of 4-(4-fluorophenyl)-3-oxobutanoic acid (450 g, 2.29 mol, 1 equiv.) in CF.sub.3SO.sub.3H (8.5 kg, 56 mol, 5 L, 25 equiv.) was stirred at 25°C . for 24 hours, the reaction was cooled to 0°C ., and slowly added to ice-water (15 L). Precipitates were formed, and the mixture was filtered to give the crude product. Then the crude was slurred with petroleum ether (1 L), and filtered to give the 7-fluoronaphthalene-1,3-diol (325 g, 79% yield). Light-yellow Solid.

(212) Step E. 7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol. To the mixture of 7-fluoronaphthalene-1,3-diol (120 g, 673 mmol, 1 equiv.), 2-bromoethynyl(triisopropyl)silane (184 g, 707 mmol, 1.05 equiv.), AcOK (132 g, 1.34 mol, 2 equiv.) in dioxane (800 mL) was added dichlororuthenium; 1-isopropyl-4-methyl-benzene dimer (41.3 g, 67.4 mmol, 0.1 equiv.) under N.sub.2. The mixture was stirred at 110°C . for 2 hours. The mixture was filtered and concentrated to give a residue. Then the residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=1/0 to 5/1) to give 7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol (213 g, 88% yield) was obtained. Black Oil; LCMS [ESI, M+1]: 359.2

(213) Step F. 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl. To the mixture of 7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol (170 g, 474 mmol, 1 equiv.), DIEA (184 g, 1.42 mol, 3 equiv.) and DCM (1700 mL) was added MOMCl (49.8 g, 618 mmol, 1.3 equiv.) at 0° C. The mixture was warmed to 15° C. and stirred for 0.5 hour. The reaction mixture was diluted with ice-water (1000 mL) and extracted with ethyl acetate (500 mL×2). The combined organic phase was washed with brine (1000 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=I/O to 50/1) to give 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-ol (96 g, 50% yield). Yellow Solid; .sup.1H NMR (400 MHz, CDCl₃-d) δ=9.13 (s, 1H), 7.68-7.64 (m, 1H), 7.21-7.16 (m, 1H), 6.97-6.96 (m, 1H), 6.81-6.80 (m, 1H), 5.26 (s, 2H), 3.51 (s, 3H), 1.24-1.17 (m, 21H). LCMS [ESI, M+1]: 403.2.

(214) Step G: 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate. To the solution of 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-ol (80 g, 198 mmol, 1 equiv.), DIEA (77.0 g, 596 mmol, 104 mL, 3 equiv.) in DCM (1200 mL) was added Tf₂O (84.1 g, 298 mmol, 49.2 mL, 1.5 equiv.) at -40° C., and the mixture was stirred at -40° C. for 0.5 hour. The reaction mixture was diluted with ice-water (500 mL), and then extracted with DCM (300 mL). The combined organic phase was dried over Na₂SO₄ and concentrated to dryness. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=I/O to 60/1) to afford 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate (100 g, 94% yield). Yellow oil;

(215) Step H. ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane. To the mixture of 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate (105 g, 196 mmol, 1 equiv.), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (100 g, 393 mmol, 2 equiv.), AcOK (57.8 g, 589 mmol, 3 equiv.) in toluene (1100 mL) was added Pd(dppf)Cl₂ (14.4 g, 20 mmol, 0.1 equiv.). The mixture was degassed and stirred at 130° C. for 3 hours. The reaction mixture was filtered and concentrated to give a residue. To the residue was added EtOAc (1000 mL) and water (800 mL). The organic phase was washed with brine (500 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=100/1 to 3/1) and triturated with MeCN (40 mL) to give ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (41 g, 43% yield). Yellow Solid; .sup.1H NMR (400 MHz, CDCl₃-d) δ=7.69-7.65 (m, 1H), 7.51 (d, J=2.4 Hz, 1H), 7.38 (d, J=2.4 Hz, 1H), 7.25 (t, J=8.8 Hz, 1H), 5.28 (s, 2H), 3.50 (s, 3H), 1.44 (s, 12H), 1.18-1.16 (m, 21H); LCMS [ESI, M+1]: 513.4.

Intermediate 6

(216) ##STR00031##

2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane

(217) ##STR00032##

(218) Step A. 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl pivalate. To the solution of 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-ol (2.00 g, 4.97 mmol, 1.0 equiv.), DMAP (122 mg, 999 μmol, 0.2 equiv.), TEA (1.51 g, 14.9 mmol, 3.0 equiv.) in DCM (20 mL) was added 2,2-dimethylpropanoyl chloride (1.80 g, 14.9 mmol, 3.0 equiv.) dropwise at 0° C., and then the mixture was stirred at 20° C. for 1 hour. After completion, the reaction mixture was diluted with DCM (15 mL) and water (15 mL), and then the aqueous layer was extracted with DCM (10 mL). The combined organic phase was dried over Na₂SO₄ and concentrated. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=I/O to 15/1) to give the title compound (3.00 g, crude). Yellow oil. LCMS [ESI, M+1]:487.2.

(219) Step B. 8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl pivalate. To the solution of 7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl pivalate (3.00 g, 6.16 mmol, 1.0 equiv.) in DMF (50 mL) was added CsF (9.36 g, 61.6 mmol, 10 equiv.), and the mixture was stirred at 20° C. for 0.25 hour. After completion, to the reaction mixture was added water (250 mL), and then the mixture was extracted with ethyl acetate (2×120 mL). The combined organic phase was washed with brine 100 mL, dried over Na.sub.2SO.sub.4 and concentrated to give the title compound (2.20 g, crude). Yellow oil. LCMS [ESI, M+1]:331.1.

(220) Step C. 8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl pivalate. To the solution of 8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl pivalate (2.00 g, 6.05 mmol, 1.0 equiv.) in MeOH (20 mL) was added Pd/C (200 mg, 10% purity) under N.sub.2. The suspension was degassed in vacuum and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (15 psi) at 20° C. for 20 minutes. After completion, the mixture was filtered and concentrated to give the title compound (1.06 g, crude). LCMS [ESI, M+1]:335.1.

(221) Step D. 8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-ol. To the solution of 8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl pivalate (1.00 g, 2.99 mmol, 1.0 equiv.) in MeOH (15 mL) was added KOH (504 mg, 8.98 mmol, 3.0 equiv.), and the mixture was stirred at 20° C. for 0.5 hour. After completion, the reaction solution was adjusted to pH=4 with 0.5 M HCl at 0° C. and extracted with ethyl acetate (80 mL×2), the combined organic phase was washed with brine 50 mL, dried over anhydrous sodium sulfate, filtered and concentrated. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=50/1 to 10/1) to give the title compound (570 mg, four steps 51% yield). Yellow solid. .sup.1H NMR (400 MHz, CDCl.sub.3) δ=7.55-7.43 (m, 1H), 7.18 (t, J=9.2 Hz, 1H), 6.98 (d, J=2.4 Hz, 1H), 6.57 (d, J=2.0 Hz, 1H), 5.32 (s, 1H), 5.25 (s, 2H), 3.52 (s, 3H), 3.40-3.25 (m, 2H), 1.30 (t, J=7.6 Hz, 3H).

(222) Step E. 8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl trifluoromethane sulfonate. To the solution of 8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-ol (520 mg, 2.08 mmol, 1.0 equiv.), DIEA (806 mg, 6.24 mmol, 3.0 equiv.) in DCM (10 mL), trifluoromethylsulfonyl trifluoromethanesulfonate (879 mg, 3.12 mmol, 1.5 equiv.) was added dropwise at -40° C., and then the mixture was stirred at -40° C. for 0.5 hr. After completion, the reaction mixture was quenched with ice-water (15 mL), and then extracted with DCM (2×15 mL). The combined organic phase was dried over Na.sub.2SO.sub.4 and concentrated. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=100/1 to 30/1) to give the title compound (620 mg, 78% yield). Yellow oil. .sup.1H NMR (400 MHz, CDCl.sub.3) δ=7.67-7.59 (m, 1H), 7.43 (d, J=2.4 Hz, 1H), 7.37 (d, J=2.4 Hz, 1H), 7.33-7.27 (m, 1H), 5.29 (s, 2H), 3.53 (s, 3H), 3.33-3.14 (m, 2H), 1.25 (t, J=7.6 Hz, 3H).

(223) Step F. 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane. To the mixture of 8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl trifluoromethane sulfonate (500 mg, 1.31 mmol, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (665 mg, 2.62 mmol, 2.0 equiv.), AcOK (385 mg, 3.92 mmol, 3.0 equiv.) in dioxane (6 mL) was added Pd(dppf)Cl.sub.2 (96.0 mg, 131 μmol, 0.1 equiv.) under N.sub.2. The mixture was degassed and stirred at 100° C. for 1 hour. After completion, the mixture was diluted with ethyl acetate (20 mL) and water (10 mL), and extracted with ethyl acetate (10 mL). The combined organic phases were washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=100/1 to 25/1) to give the title compound (143 mg, 30% yield). Yellow oil. .sup.1H NMR (400 MHz, CDCl.sub.3) δ=7.62-7.53 (m, 1H), 7.44-7.34 (m, 2H), 7.21 (t, J=9.2 Hz, 1H), 5.28 (s, 2H), 3.51 (s, 3H), 3.20-3.06 (m, 2H), 1.45 (s, 12H), 1.30-1.25 (m, 3H).

Intermediate 7

(224) ##STR00033##

((2-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-

yl)ethynyl)triisopropylsilane

(225) ##STR00034##

(226) Step A. 7-fluoronaphthalen-1-ol. To a solution of 7-fluoro-3,4-dihydronaphthalen-1 (2H)-one (75.0 g, 457 mmol, 1.00 equiv.) in acetic acid (1.50 L) and hydrogen bromide in AcOH (33%, 7.50 mL) was added bromine (80.3 g, 503 mmol, 25.9 mL, 1.1 equiv.) in acetic acid (50 mL) at 0° C., and the mixture was stirred at 25° C. for 3 hours. The mixture was diluted with DCM (1.5 L), washed with water (3×500 mL). The organic layer was dried over anhydrous sodium sulfate, filtered and concentrated to afford a brown oil, which was dissolved in DMF (750 mL).

(227) Lithium bromide (67.4 g, 777 mmol, 19.5 mL, 1.70 equiv.), lithium carbonate (57.4 g, 777 mmol, 1.70 equiv.) were added. The reaction mixture was stirred at 160° C. for 3.5 hours. The reaction was diluted with ethyl acetate (1.00 L), washed with brine (2×500 mL), dried with anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate=I/O to 5/1) affording the title compound (61.0 g, 82% yield). Brown solid; ¹H NMR (400 MHz, CDCl₃) δ 7.84-7.77 (m, 2H), 7.44 (d, J=8.0 Hz, 1H), 7.31-7.24 (m, 2H), 6.84 (d, J=7.6 Hz, 1H), 5.39 (s, 1H).

(228) Step B. 7-fluoro-8-(2-triisopropylsilylethynyl)naphthalen-1-ol. To a solution of (bromoethynyl)triisopropylsilane (72.0 g, 275 mmol, 1.20 equiv.) and 7-fluoronaphthalen-1-ol (37.2 g, 230 mmol, 1.0 equiv.) in DCE (500 mL) were added dichlororuthenium; 1-isopropyl-4-methyl-benzene (21.1 g, 34.4 mmol, 0.15 equiv.), K₂CO₃ (31.7 g, 230 mmol, 1.0 equiv.) and NaOAc (3.77 g, 45.9 mmol, 0.20 equiv.). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate=I/O to 50/1) affording the title compound (73.0 g, 93% yield). Yellow oil; ¹H NMR (400 MHz, CDCl₃) δ 9.10 (s, 1H), 7.79 (dd, J=5.6, 8.8 Hz, 1H), 7.41-7.33 (m, 2H), 7.23 (t, J=8.8 Hz, 1H), 7.08-7.00 (m, 1H), 1.24-1.14 (m, 21H); LCMS [ESI, M+1, 2 M+1]: 343.1, 685.3.

(229) Step C. [7-fluoro-8-(2-triisopropylsilylethynyl)-naphthyl]trifluoromethanesulfonate. To a solution of 7-fluoro-8-(2-triisopropylsilylethynyl)naphthalen-1-ol (73.0 g, 213 mmol, 1.00 equiv.) in DCM (600 mL) were added DIEA (55.1 g, 426 mmol, 74.2 mL, 2.00 equiv.) and Tf₂O (90.2 g, 320 mmol, 52.7 mL, 1.50 equiv.) at -40° C. The mixture was stirred at -40° C. for 0.5 hour. The combined reaction mixture was filtered and concentrated under reduced pressure to give a residue. The mixture was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate=I/O to 50/1) affording the title compound (78.0 g, 77% yield).

(230) Yellow solid; ¹H NMR (400 MHz, CDCl₃) δ 7.88-7.79 (m, 2H), 7.59-7.52 (m, 1H), 7.46 (t, J=8.0 Hz, 1H), 7.37 (t, J=8.8 Hz, 1H), 1.32-1.16 (m, 21H).

(231) Step D. ((2-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane. To a solution of [7-fluoro-8-(2-triisopropylsilylethynyl)-1-naphthyl]trifluoromethanesulfonate (20.0 g, 42.1 mmol, 1.00 equiv.) and bis(pinacolato)diboron (16.0 g, 63.2 mmol, 1.50 equiv.) in dioxane (6.00 mL) were added KOAc (8.27 g, 84.3 mmol, 2.0 equiv.) and Pd(dppf)Cl₂ (3.08 g, 4.21 mmol, 0.10 equiv.). The mixture was stirred at 110° C. for 12 hours. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate=I/O to 10/1) affording the title compound (9.0 g, 47% yield). Yellow solid; ¹H NMR (400 MHz, CDCl₃) δ 7.85-7.75 (m, 3H), 7.43 (dd, J=7.2, 8.0 Hz, 1H), 7.30-7.24 (m, 1H), 1.45 (s, 12H), 1.21-1.14 (m, 21H); LCMS [ESI, M+1]: 453.2.

Intermediate 8

(232) ##STR00035##

((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol

(233) ##STR00036##

(234) Step A. Ethyl 2-methylene-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate. To a mixture of ethyl 5-oxopyrrolidine-2-carboxylate (1.50 kg, 9.54 mol, 1.00 equiv.) and 3-chloro-2-

(chloromethyl)prop-1-ene (1.91 kg, 15.3 mol, 1.77 L, 1.60 equiv.) in THE (7.50 L) was added LiHMDS (1 M, 19.1 L, 2.00 equiv.) drop-wise at -40°C . under N.sub.2. The mixture was stirred at 25°C . for 20 hrs. The reaction mixture was poured into HCl (1 M, 2.50 L) and pH was adjusted to 7 with HCl (2 M) at 0°C . The mixture was extracted with EtOAc (4.50 L $\times$ 3). The combined organic layers were washed with brine (4.50 L), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=10/1 to 1/1, Rf=0.40) to afford the title compound (898 g, 3.88 mol, 40.6% yield, 82% purity) as a yellow oil. LCMS: Rt=0.716 min, m/z=210.1 (M+H). .sup.1H NMR: 400 MHz CDCl.sub.3 δ : 5.02-5.07 (m, 2H), 4.28 (m, 1H), 4.16-4.22 (m, 2H), 3.71 (dd, J=15.6, 1.6 Hz, 1H), 3.04 (m, 1H), 2.73-2.80 (m, 1H), 2.57-2.64 (m, 1H), 2.41-2.49 (m, 2H), 2.03-2.17 (m, 2H), 1.24-1.30 (m, 3H).

(235) Step B. ethyl 2,5-dioxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate. To a mixture of ethyl 2-methylene-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate (165 g, 646 mmol, 1.00 equiv.) in DCM (1650 mL) and MeOH (165 mL) was added O₃ (15 psi) at -70°C . under N.sub.2. The solution became pale blue, and then the mixture was purged by N.sub.2 for 30 min. Me.sub.2S (80.4 g, 1.29 mol, 95.0 mL, 2.00 equiv.) was added to the mixture at -70°C . The mixture was stirred at 25°C . for 16 hours. The reaction mixture was concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=10/1 to 1/1, Rf=0.50) to afford the title compound (821 g, 3.62 mol, 93.3% yield, 93.1% purity) as a yellow oil. LCMS: Rt=0.543 min, m/z=212.1 (M+H). .sup.1H NMR: 400 MHz CDCl.sub.3 δ : 4.23 (m, 2H), 4.12 (m, 1H), 3.56 (m, 1H), 2.96-3.01 (m, 2H), 2.77-2.86 (m, 1H), 2.43-2.50 (m, 2H), 2.14-2.22 (m, 1H), 1.28 (m, 1H).

(236) Step C. ethyl 2-hydroxy-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate. To a solution of ethyl 2,5-dioxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate (257 g, 1.22 mol, 1.00 equiv.) in EtOH (1300 mL) was slowly added NaBH.sub.4 (13.8 g, 365 mmol, 0.30 equiv.) at 0°C . under N.sub.2. The mixture was stirred at 0°C . for 10 min. The reaction was quenched with saturated NH.sub.4Cl (65.0 mL) at 5°C . and stirred at 5°C . for 0.5 hr, then the mixture was concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=10/1 to 1/1) to afford the title compound (56.8% yield) as a yellow oil. .sup.1H NMR: 400 MHz CDCl.sub.3 δ : 4.65 (s, 1H), 4.14 (q, J=7.2 Hz, 2H), 3.95 (dd, J=12.8, 6.0 Hz, 1H), 3.10 (d, J=12.8 Hz, 1H), 2.75-2.84 (m, 2H), 2.49-2.49 (m, 2H), 2.39-2.45 (m, 1H), 2.02-2.10 (m, 1H), 1.84 (dd, J=13.6, 6.0 Hz, 1H), 1.30 (t, J=7.2 Hz, 1H).

(237) Step D. ethyl (2S,7aR)-2-fluoro-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate. To a solution of ethyl 2-hydroxy-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate (150 g, 642 mmol, 1.00 equiv.) in DCM (750 mL) was added a solution of DAST (131 g, 813 mmol, 107 mL, 1.50 equiv.) drop-wise at -70°C . under N.sub.2. The reaction mixture was warmed to 25°C . stirred at 25°C . for 16 hours. The reaction mixture was quenched with MeOH (40.0 mL) at 10°C ., then diluted with water (750 mL) and extracted with DCM (750 mL $\times$ 3). The combined organic layers were washed with brine (750 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=I/O to 0/1, Rf=0.30) to afford ethyl 2-fluoro-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate (50.6% yield, 74.7% purity) as a yellow oil. This compound (61 g, 283.43 mmol, 1.00 equiv.) was further purified by HPLC (column: Welch ultimate XB-NH.sub.2 250 $\times$ 50 $\times$ 10 μm ; mobile phase: [Heptane-EtOH (0.1% NH.sub.3WATER)]); B %: 10%-10%, 10 min) to give a yellow oil (49.0 g, 226.08 mmol, 99.3% purity). .sup.1H NMR: 400 MHz CDCl.sub.3 δ : 5.30 (m, 1H), 4.10-4.23 (m, 3H), 3.11-3.14 (m, 1H), 2.67-2.76 (m, 3H), 2.41-2.45 (m, 1H), 2.03-2.12 (m, 2H), 1.23-1.29 (m, 3H). SFC separation (column: DAICEL CHIRALPAK IC (250 mm $\times$ 50 mm, 10 μm); mobile phase: [0.1% NH.sub.3H.sub.2O-IPA]; B %: 40%-40%, 4.7 min; 200 minmin, desired product: Peak 2, Rt=1.959 min) of the racemic material (280 g, 1.22 mol, 1 equiv.) gave the title compound (114 g, 96.0% purity).

(238) Step E. ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol. To a suspension of LiAlH₄.sub.4 (33.1 g, 871 mmol, 1.50 equiv.) in THF (625 mL) was added a solution of ethyl (2S,7aR)-2-fluoro-5-oxotetrahydro-1H-pyrrolizine-7a(5H)-carboxylate (125 g, 581 mmol, 1.00 equiv.) in THF (375 mL) drop-wise at 0° C. under N₂. The reaction mixture was warmed to 70° C. and stirred at 70° C. for 3 hours. The mixture was cooled to 0° C. Then to the mixture was added water (33.0 mL), NaOH (15%, 99.0 mL) and water (99 mL) dropwise in sequence 0° C. After addition, the mixture was stirred at 0° C. stirred for 5 min. The mixture was filtered, and the filtered cake was washed with EtOAc (1000 mL×2). The filtrate was dried with MgSO₄, filtered, and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, DCM: MeOH=100/1 to 10/1) to afford the title compound (180 g, 1.10 mol, 94.7% yield, 97.3% purity) as a yellow oil. ¹H NMR: 400 MHz CDCl₃: 5.12-5.27 (m, 1H), 3.25 (s, 2H), 3.14-3.18 (m, 2H), 3.12-3.13 (m, 1H), 3.02-3.09 (m, 1H), 2.01-2.11 (m, 2H), 1.75-1.86 (m, 4H).

Intermediate 9

(239) ##STR00037##

7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(240) ##STR00038##

(241) Step A. 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine. To a mixture of 8-chloro-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (20 g, 92.8 mmol, 1.00 equiv.) in toluene (100 mL) was added POCl₃.sub.3 (42.7 g, 278 mmol, 25.9 mL, 3.00 equiv.) and N-ethyl-N-isopropylpropan-2-amine (36.0 g, 278 mmol, 48.5 mL, 3.00 equiv.) at 0° C. The mixture was stirred at 110° C. for 3 hours. After completion, the mixture was concentrate under reduced pressure at 40° C. to dryness affording 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (23.4 g, crude) as a black oil.

(242) Step B. 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. To a solution of 2,2,2-trifluoroethanol (11.1 g, 111 mmol, 8.01 mL, 1.20 equiv.) in toluene (200 mL) was added t-BuONa (26.7 g, 278 mmol, 3.00 equiv.) at 0° C. The mixture was first stirred at 10° C. for 0.5 hour. Then the above mixture was added to 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (23.4 g, 92.7 mmol, 1.00 equiv.) in toluene (200 mL) at -10° C. After addition, the mixture was stirred at -10° C.~25° C. for 16 hours. After monitored, a mixture of t-BuONa (1.78 g, 18.5 mmol, 0.2 equiv.) and 2,2,2-trifluoroethanol (1.85 g, 18.5 mmol, 1.33 mL, 0.20 equiv.) in toluene (20.0 mL) was added thereto at 0° C. The mixture was continued to stir at 25° C. for 30 hours. After completion, the mixture was poured onto Silica gel column, purified by column chromatography (Silica gel, petroleum ether/ethyl acetate=30/1 to 10/1), and then further purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) affording 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (16.3 g, 55.6% yield); Yellow solid; LCMS [ESI, M+1]: 316.

(243) Step C. 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. To a mixture of (tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (17.9 g, 126 mmol, 2.00 equiv.), 4 Å MS (15.0 g) and N-ethyl-N-isopropylpropan-2-amine (16.4 g, 126 mmol, 22.0 mL, 2.00 equiv.) in 2-methyltetrahydrofuran (200 mL) was added 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 g, 63.3 mmol, 1.00 equiv.) in 2-methyltetrahydrofuran (200 mL) at 0-5° C. The mixture was stirred at 0-25° C. for 2 hours. After completion, the mixture was filtered and washed with ethyl acetate (100 mL). The filtrate was quenched by saturated NH₄Cl aqueous solution (300 mL), and the organic layer was separated and dried over anhydrous Na₂SO₄. The mixture was filtered, and the filtrate was concentrated under reduced pressure at 40° C. to dryness. The crude product was triturated with CH₃CN (20 mL) at 25° C. for 15 minutes and filtered, the filter cake was dried in vacuum at 40° C. affording the title compound (18.2 g, 64.6% yield). Light yellow solid. ¹H NMR (400 MHz, CDCl₃) δ 8.99 (s, 1H), 5.03 (q, J=8.4 Hz, 2H), 4.32 (s, 2H), 3.23-3.05 (m,

2H), 2.67 (td, J=6.8, 10.4 Hz, 2H), 2.11-1.96 (m, 21H), 1.96-1.85 (m, 4H), 1.74-1.69 (m, 2H); LCMS [ESI, M+1]: 421.

(244) Step D. 7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. To a mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (5.00 g, 11.9 mmol, 1.00 equiv.), (8-chloronaphthalen-1-yl)trimethylstannane (7.73 g, 23.8 mmol, 2.00 equiv.) in toluene (150 mL) was added 4 Å MS (5.00 g) at 25° C. The mixture was stirred at 25° C. for 1 hour. Then CuI (792 mg, 4.16 mmol, 0.35 equiv.), Pd(dppf)Cl.sub.2 (1.30 g, 1.78 mmol, 0.15 equiv.) and BINAP (1.85 g, 2.97 mmol, 0.25 equiv.) were added thereto at 25° C. The mixture was degassed in vacuum and purged with N.sub.2 several times over 30 minutes. Then the mixture was heated to 90° C. and stirred for 2 hours. The mixture was cooled to 25° C., and then (8-chloronaphthalen-1-yl)trimethylstannane (1.93 g, 5.94 mmol, 0.50 equiv.) was added thereto at 25° C. The mixture was heated to 90° C. and stirred for 1 hour. After completion, the mixture was filtered, and the filtrate was concentrated under reduced pressure at 40° C. to dryness. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) affording the title compound (2.3 g, 33.9% yield); Yellow solid. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 9.25 (s, 1H), 8.02 (dd, J=1.2, 8.0 Hz, 1H), 7.89 (dd, J=0.8, 8.0 Hz, 1H), 7.65-7.60 (m, 1H), 7.59-7.53 (m, 2H), 7.46-7.41 (m, 1H), 5.08 (q, J=8.0 Hz, 2H), 4.46 (s, 2H), 3.32 (br d, J=3.8 Hz, 2H), 2.83-2.70 (m, 2H), 2.20-2.09 (m, 2H), 2.03-1.90 (m, 4H), 1.82-1.72 (m, 2H); LCMS [ESI, M+1]: 547.

Intermediate 10

(245) ##STR00039##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(246) ##STR00040##

(247) Step A. 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine. To a solution of 7-chloro-8-fluoropyrido[4,3-d]pyrimidine-2,4(1H,3H)-dione (100 g, 463 mmol, 1.00 equiv.) in toluene (500 mL) were added POCl.sub.3 (213 g, 1.39 mol, 129 mL, 3.00 equiv.) and DIEA (179 g, 1.39 mol, 242 mL, 3.00 equiv.) at 0° C. The mixture was stirred at 110° C. for 5 h. The reaction was distilled in vacuum (80° C., water pump) to give 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (100 g, 396.10 mmol, 85.39% yield) as brown oil.

(248) Step B. 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (100 g, 396 mmol, 1.00 equiv.) and 2,2,2-trifluoroethanol (59.4 g, 594 mmol, 42.7 mL, 1.50 equiv.) in toluene (2 L) was added t-BuONa (152 g, 1.58 mol, 4.00 equiv.) at 25° C. The mixture was stirred at 25° C. for 2 hr. The reaction mixture was filtered through a pad of Celite, washed with brine (3 L×2) and concentrated under reduced pressure to give a residue, which was purified by reversed-phase HPLC (water (0.1% formic acid)-ACN) to give 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (45.0 g, 140 mmol, 35.5% yield, 99.0% purity) as a brown solid. LCMS: M+1, 316.

(249) Step C. 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. A mixture of (tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (35.7 g, 253 mmol, 2.00 equiv.), DIEA (32.7 g, 253 mmol, 44.0 mL, 2.00 equiv.) and 4 Å molecular sieves (40.0 g) in 2-methyltetrahydrofuran (400 mL) was stirred at 25° C. for 1 hr. Then a solution of 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (40.0 g, 126 mmol, 1.00 equiv.) in 2-methyltetrahydrofuran (400 mL) was added and the resulting mixture was stirred at 25° C. for 2 hrs. The reaction mixture was filtered. The filtrate was washed with sat. aq. NH.sub.4Cl solution (1 L×2), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was triturated with acetonitrile (300 mL) at 25° C. for 30 min to give 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (26.0 g, 61.1 mmol, 48.3% yield, 99.0% purity) as a light

yellow solid. LCMS: M+1, 421.

(250) Step D. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (17.0 g, 40.4 mmol, 1.00 equiv.), 2-(8-fluoronaphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (16.4 g, 60.6 mmol, 1.50 equiv.), BrettPhos Pd G3 (4.25 g, 4.69 mmol, 1.16e-1 equiv.), K₂CO₃ (1.5 M, 80.8 mL, 3.00 equiv.) in toluene (170 mL) was degassed and purged with N₂ for 3 times, and then the mixture was stirred at 65° C. for 4 hrs under N₂ atmosphere. The reaction mixture was filtered. The filtrate was extracted with toluene (170 mL×3). The combined organic layers were dried over Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed-phase HPLC (water (0.1% formic acid)-ACN) to give 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (10.85 g, 16.6 mmol, 41.2% yield, 95.8% purity) as a yellow solid. NMR: 5.928 (s, 1H), 8.23 (d, J=8.1 Hz, 1H), 7.97 (d, J=8.1 Hz, 1H), 7.81-7.76 (m, 1H), 7.68 (dd, J=0.9, 7.2 Hz, 1H), 7.61 (dt, J=5.1, 7.9 Hz, 1H), 7.34 (dd, J=7.1, 13.3 Hz, 1H), 5.47-5.37 (m, 2H), 4.77-4.67 (m, 2H), 3.56-3.49 (m, 2H), 3.22 (td, J=6.0, 11.7 Hz, 2H), 2.27-2.00 (m, 8H); LCMS: M+1, 531.

Intermediate 11

(251) ##STR00041##

(2-((tert-butoxycarbonyl)amino)-5,6-dimethylbenzo[d]thiazol-4-yl)boronic Acid

(252) ##STR00042##

(253) Step A. N-((2-bromo-3,4-dimethylphenyl)carbamothiol)benzamide: To a solution of 2-bromo-3,4-dimethylaniline (2.17 g, 10.9 mmol) in acetone (30 mL) was added benzoyl isothiocyanate (1.9 g, 11.6 mmol) in acetone (10 mL) at 25° C. The mixture was stirred at 25° C. for 5 minutes. The reaction mixture was concentrated under reduced pressure to give a residue. The residue was dispersed in petroleum ether/ethyl acetate=20/1 (40 mL) and stirred for 0.5 hour. The mixture was filtered and the solid was dried under reduced pressure to give N-((2-bromo-3,4-dimethylphenyl)carbamothioyl)benzamide (3.56 g, 90% yield) as a light yellow solid; ¹H NMR (400 MHz, CDCl₃) δ=12.45 (br s, 1H), 9.21 (br s, 1H), 7.95-7.92 (m, 2H), 7.79 (d, J=8.0 Hz, 1H), 7.67-7.59 (m, 1H), 7.58-7.56 (m, 2H), 7.19 (d, J=8.0 Hz, 1H), 2.44 (s, 3H), 2.37 (s, 3H).

(254) Step B. 1-(2-bromo-3,4-dimethylphenyl)thiourea: A mixture of N-((2-bromo-3,4-dimethylphenyl)carbamothioyl)benzamide (3.30 g, 9.08 mmol) and NaOH (50 mL, 10% aqueous) was stirred at 80° C. for 3 hours. A white precipitate appeared. After completion, the reaction mixture was cooled to 10° C. The mixture was filtered and filter cake was washed with water until the pH of the filtrate was 8-9. The filter cake was washed with petroleum ether (10 mL) and dried under reduced pressure to give 1-(2-bromo-3,4-dimethylphenyl)thiourea (2.2 g, 90% yield) as a white solid; LCMS (ESI, M+1): m/z 259.0, 261.0.

(255) Step C. 4-bromo-5,6-dimethylbenzo[d]thiazol-2-amine: To a solution of 1-(2-bromo-3,4-dimethylphenyl)thiourea (2.20 g, 8.49 mmol) in CHCl₃ (30 mL) was added drop-wise Br₂ (1.36 g, 8.49 mmol) in CHCl₃ (2 mL) at 0° C. The mixture was stirred at 70° C. for 1 hour. The reaction mixture was concentrated under reduced pressure to give a residue. The residue was diluted with ethyl acetate (200 mL), saturated Na₂SO₃ aqueous (50 mL) and saturated NaHCO₃ aqueous solution (50 mL). The mixture was extracted with ethyl acetate (50 mL×3). The combined organic layers were dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure to give 4-bromo-5,6-dimethylbenzo[d]thiazol-2-amine (2 g, 88% yield) as a white solid; ¹H NMR (400 MHz, DMSO-d₆) δ=7.66 (s, 2H), 7.43 (s, 1H), 2.34 (s, 3H), 2.29 (s, 3H). LCMS (ESI, M+1): m/z 257.0, 259.0.

(256) Step D. tert-butyl (4-bromo-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate: A mixture of 4-bromo-5,6-dimethylbenzo[d]thiazol-2-amine (2 g, 7.78 mmol), (Boc)₂O (2.1 g, 9.62 mmol),

DIPEA (3.04 g, 23.5 mmol) and DMAP (96 mg, 786 μ mol) in THF (40 mL) was stirred at 25° C. for 16 hour. Then DIPEA (1.04 g, 8.04 mmol) and (Boc).sub.2O (570 mg, 2.61 mmol) was added. The mixture was stirred at 25° C. for 4 hours. After completion, the reaction mixture was diluted with water (50 mL) and ethyl acetate (200 mL). The mixture was extracted with ethyl acetate (50 mL $\times$ 3). The combined organic layers were washed with brine (20 mL) and dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=1/19] to give tert-butyl (4-bromo-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate (2.6 g, 91% yield) as a light yellow solid; LCMS (ESI, M-55): m/z 300.9, 302.9.

(257) Step E. (2-((tert-butoxycarbonyl)amino)-5,6-dimethylbenzo[d]thiazol-4-yl)boronic acid: A mixture of tert-butyl (4-bromo-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate (1 g, 2.80 mmol), 2-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-5,5-dimethyl-1,3,2-dioxaborinane (1.58 g, 7.00 mmol) and AcOK (1.00 g, 10.2 mmol) in dioxane (15 mL) was degassed and purged with N.sub.2 for 3 times. Then [2-(2-aminophenyl)phenyl]-chloro-palladium; tricyclohexylphosphane (100 mg, 169 μ mol, 0.06 equiv.) was added. The mixture was stirred at 80° C. for 40 hours under N.sub.2 atmosphere. After completion, the reaction mixture was diluted with water (1 mL) and brine (1 mL), and extracted with ethyl acetate (2 mL $\times$ 4). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=1/9] to give (2-((tert-butoxycarbonyl)amino)-5,6-dimethylbenzo[d]thiazol-4-yl)boronic acid (0.73 g, 65% yield) as a light yellow solid; LCMS (ESI, M+1): m/z 323.1.

Intermediate 12

(258) ##STR00043##

triisopropyl((8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)silane

(259) ##STR00044##

(260) Step A. 8-((triisopropylsilyl)ethynyl)naphthalen-1-ol: To a solution of naphthalen-1-ol (500 mg, 3.47 mmol, 1.25 mL), potassium carbonate (479 mg, 3.47 mmol), dichlororuthenium; 1-isopropyl-4-methyl-benzene (531 mg, 867 μ mol) and sodium acetate (56.9 mg, 694 μ mol) in DCE (20.0 mL) was added 2-bromoethynyl(triisopropyl)silane (1.09 g, 4.16 mmol). The reaction was stirred at 40° C. for 12 hours. The reaction mixture was cooled to 25° C. and filtered. The filtrate was concentrated in vacuum to give a residue. The residue was purified by silica gel chromatography (Petroleum ether/ethyl acetate=I/O to 10/1) to give 8-

((triisopropylsilyl)ethynyl)naphthalen-1-ol (760 mg, 67% yield) as a brown solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =9.22 (s, 1H), 7.81 (dd, J=1.2, 8.4 Hz, 1H), 7.64 (dd, J=1.2, 6.8 Hz, 1H), 7.42-7.34 (m, 3H), 7.01 (dd, J=4.0, 5.6 Hz, 1H), 1.25-1.13 (m, 21H).

(261) Step B. 8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate: To a solution of 8-((triisopropylsilyl)ethynyl)naphthalen-1-ol (760 mg, 2.34 mmol) and DIEA (605 mg, 4.68 mmol, 816 μ L) in DCM (8.00 mL) was added Tf.sub.2O (991 mg, 3.51 mmol, 580 μ L) at -40° C. The reaction was stirred at 25° C. for 0.5 hour. The reaction was quenched with water (10.0 mL). The aqueous phase was extracted with DCM (2 $\times$ 20.0 mL). The combined organic phase was washed with brine (2 $\times$ 20 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate=1/0 to 10/1) to give 8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate (1.00 g, 93% yield) as a brown oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =7.92-7.84 (m, 3H), 7.56-7.47 (m, 3H), 1.26-1.12 (m, 21H).

(262) Step C. triisopropyl((8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)silane: To a solution of 8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate (950 mg, 2.08 mmol) in dioxane (15.0 mL) were added 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (687 mg, 2.70 mmol), Pd(dppf)Cl.sub.2

(152 mg, 208 μ mol) and KOAc (408 mg, 4.16 mmol). The reaction was stirred at 110° C. for 5 hours under nitrogen. The reaction was filtered and concentrated in vacuum to give a residue. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate=I/O to 20/1) to give triisopropyl((8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)silane (340 mg, 38% yield) as a red oil; ¹H NMR (400 MHz, CDCl₃) δ =7.78-7.66 (m, 3H), 7.39-7.26 (m, 3H), 1.36 (s, 12H), 1.12-1.05 (m, 21H).

Intermediate 13

(263) ##STR00045##

1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(264) ##STR00046##

(265) Step A. 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoropyrido[4,3-d]pyrimidine-2,4-diol (2.8 g, 13.0 mmol) in POCl₃ (20 mL) was added DIEA (5.04 g, 39.0 mmol, 6.79 mL) in one portion at 25° C. under N₂. The mixture was heated to 110° C. and stirred for 2 hours. After completion, the mixture was concentrated under reduced pressure at 45° C. The residue was purified by silica gel chromatography (Silica gel, Petroleum ether/Ethyl acetate=20/1, 3/1) to afford 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (3.1 g, 89% yield) as a yellow solid; LCMS (ESI, M+1): m/z 251.9.

(266) Step B. 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (3 g, 11.9 mmol) and DIEA (6.14 g, 47.5 mmol, 8.28 mL) in DCM (10 mL) was added 3-methylpiperidin-3-ol hydrochloride (1.52 g, 13.2 mmol) in portions at -40° C. under N₂. The mixture was stirred at -40° C. for 1 hour. After completion, the mixture was filtered and concentrated in vacuum. The residue was purified by silica gel chromatography (Silica gel, Petroleum ether/Ethyl acetate=1/0, 0/1) affording 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.81 g, 71% yield) as a yellow solid; ¹H NMR (400 MHz, DMSO-d₆) δ 9.18 (s, 1H), 4.73 (s, 1H), 4.46 (d, J=12.4 Hz, 1H), 4.12 (d, J=13.2 Hz, 1H), 3.57 (d, J=13.2 Hz, 1H), 3.31-3.21 (m, 1H), 2.03-1.89 (m, 1H), 1.73-1.60 (m, 3H), 1.16 (s, 3H); LCMS (ESI, M+1): m/z 331.0.

(267) Step C. 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (hexahydro-1H-pyrrolizin-7a-yl)methanol (1.43 g, 10.1 mmol) and 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.8 g, 8.45 mmol) in dioxane (18 mL) were added DIEA (3.28 g, 25.4 mmol, 4.42 mL) and 4 Å molecular sieves (1.5 g, 8.45 mmol) in one portion at 25° C. under N₂. The mixture was heated to 90° C. and stirred for 20 hours. After completion, the mixture was filtered and concentrated in vacuum. The crude product was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) affording 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (1.6 g, 41% yield) as a yellow solid; ¹H NMR (400 MHz, CDCl₃): δ 8.89 (s, 1H), 4.46-4.37 (m, 1H), 4.36-4.29 (m, 1H), 4.26-4.18 (m, 2H), 3.42-3.33 (m, 1H), 3.31-3.24 (m, 1H), 3.14-3.06 (m, 2H), 2.74-2.60 (m, 3H), 2.05-1.97 (m, 3H), 1.92-1.82 (m, 5H), 1.70-1.61 (m, 4H), 1.33 (s, 3H); LCMS (ESI, M+1): m/z 436.2.

Intermediate 14

(268) ##STR00047##

2,4-dichloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidine

(269) ##STR00048##

(270) Step A. 7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol: To a mixture of 7-chloro-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol, 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (14.0 g, 39.0 mmol) and K₃PO₄ (1.5 M, 46.4 mL) in EtOH (140 mL) was added cataCXium-A-

Pd-G3 cataCXium-A-Pd-G3 (1.39 g, 1.90 mmol) under N.sub.2. The mixture was de-gassed and heated to 78° C. for 9.5 hours under N.sub.2. The reaction mixture was concentrated in vacuum. Then the mixture was diluted with ethyl acetate (500 mL) and filtered. The filtrate was diluted with water (100 mL). The organic phase was separated. The aqueous phase was extracted with ethyl acetate (80 mL). The combined organic layers were washed with brine (120 mL) and dried over with anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (2.34 g, 24% yield) as a yellow solid. LCMS [ESI, M+H]: m/z 414.1.

(271) Step B. 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine: A mixture of POCl.sub.3 (278 mg, 1.81 mmol, 169 µL) in toluene (3 mL) were added DIEA (141 mg, 1.09 mmol, 190 µL) and 7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (150 mg, 363 µmol). The reaction mixture was stirred at 110° C. for 25 minutes. After completion, the mixture was concentrated in vacuum and the pH value was adjusted to 8 with ice cold saturated NaHCO.sub.3 solution. Then the mixture was extracted with ethyl acetate (8 mL×2). The combined organic layer was washed with brine (10 mL) and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum to give 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine (164 mg, crude) as a brown oil, which was used in the next step without further purification.

Intermediate 15

(272) ##STR00049##

7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(273) ##STR00050##

(274) Step A. 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a mixture of 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (33.3 g, 1.0 equiv.), DIEA (54.5 g, 4.0 equiv.) and 4 Å molecular sieves (4.0 g) in THF (340 mL) was added ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (20.1 g, 1.2 equiv.). The reaction was stirred at 40° C. for 14 hours. The reaction mixture was diluted with water (20 mL) and was extracted with EtOAc (3×20 mL), the combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated under vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (28.8 g, 62% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ 8.98 (s, 1H), 5.40-5.19 (m, 1H), 5.02 (q, J=8.0 Hz, 1H, 2H), 4.40-4.27 (m, 2H), 3.34-3.12 (m, 3H), 3.05-2.94 (m, 1H), 2.32-2.06 (m, 3H), 2.03-1.84 (m, 3H); LCMS [ESI, M+1]: 439.1.

Intermediate 16

(275) ##STR00051##

7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(276) ##STR00052##

(277) Step A. 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (1.0 g, 1.0 equiv.), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (1.64 g, 2.0 equiv.) and Cs.sub.2CO.sub.3 (1.5 M, 3.0 equiv.) in methoxycyclopentane (15.0 mL) was added CataCXium A Pd G3 (332 mg, 0.20 equiv.). The reaction was stirred at 100° C. for 2 hours. The

mixture was filtered. To the filtrate water (20 mL) was added and the aqueous layer was extracted with ethyl acetate (2×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [C18, mobile phase: [water (0.1% formic acid)/acetonitrile]] to afford the title compound (820 mg, 56% yield) as a yellow solid; LCMS (ESI, M+1): m/z=637.3.

Intermediate 17

(278) ##STR00053##

2-[7,8-difluoro-3-(methoxymethoxy)-1-naphthyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane

(279) ##STR00054## ##STR00055##

(280) Step A. (E)-4-(2,3-difluorophenyl)but-3-enoic acid: To a solution of 2,3-difluorobenzaldehyde (100 g, 1.0 equiv.) and 2-carboxyethyl(triphenyl)phosphonium bromide (321 g, 1.1 equiv.) in THF (1 L) was added t-BuOK (1 M in THF, 1.41 L, 2.0 equiv.) at -70° C. The mixture was stirred at -70° C. for 1 hour. Then the mixture was warmed up to 20° C. and stirred for 1 hour. The reaction mixture was diluted with water (1 L) and concentrated under reduced pressure to remove the THF. Then the mixture was filtered and the filtrate was adjusted to pH~2 with HCl (1 M). The mixture was extracted with ethyl acetate (3×1000 mL). The combined organic layers were washed with brine (1000 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give the residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 10:1 to 2:1) to afford the title compound (75 g, 50% yield) as a pink solid; .sup.1H NMR (400 MHz, chloroform-d) 5-11.50 (br s, 1H), 7.25-7.17 (m, 1H), 7.12-6.96 (m, 2H), 6.67 (d, J=16.0 Hz, 1H), 6.42 (td, J=7.2, 16.0 Hz, 1H), 3.36 (dd, J=1.2, 7.2 Hz, 2H).

(281) Step B. 4-(2,3-difluorophenyl)butanoic acid: To a solution of (E)-4-(2,3-difluorophenyl)but-3-enoic acid (14 g, 1.0 equiv.) in EtOAc (500 mL) was added dry Pd/C (3 g, 10% purity) under N.sub.2. The suspension was degassed and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (15 psi) at 20° C. for 1 hour. The reaction mixture was filtered and the filtrate was concentrated to afford the title compound (13.5 g, 95% yield) as a yellow solid and used to next step without purification.

(282) Step C. 5,6-difluoro-3,4-dihydronaphthalen-1 (2H)-one: To a solution of 4-(2,3-difluorophenyl)butanoic acid (13.5 g, 1.0 equiv.) in DCM (300 mL) was added DMF (246 mg, 0.05 equiv.) and oxalyl chloride (17.1 g, 2.0 equiv.). The mixture was stirred at 20° C. for 0.5 hour. The mixture was concentrated under vacuum and then the residue was dissolved in DCM (300 mL). Then, to the mixture was added AlCl.sub.3 (12.3 g, 1.5 equiv.). The reaction mixture was stirred at 40° C. for 1 hour before being quenched with water (200 mL) and extracted with T)CM (3×300 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1 to 5:1) to afford the title compound (11 g, 98% yield) as a yellow solid; LCMS (ESI, M+1): m/z=183.2.

(283) Step D. 5,6-difluoronaphthalen-1-ol: To a solution of 5,6-difluoro-3,4-dihydronaphthalen-1 (2H)-one (11 g, 1.0 equiv.) and HBr (1.48 g, 0.1 equiv.) in AcOH (240 mL) was added a solution of Br.sub.2 (9.65 g, 1.0 equiv.) in AcOH (40 mL) at 0° C. The mixture was stirred at 25° C. for 1 hour. Then the mixture was diluted with DCM (100 mL) and washed with water (3×100 mL). The organic layer was dried over anhydrous sodium sulfate, filtered and concentrated to afford a brown oil which was dissolved in DMF (260 mL). LiBr (8.91 g, 102 mmol, 2.58 mL, 1.7 equiv.), Li.sub.2CO.sub.3 (7.59 g, 102 mmol, 1.7 equiv.) was added. The reaction mixture was stirred at 160° C. for 1 hour. The reaction mixture was diluted with ethyl acetate (400 mL) and washed with water (3×300 mL). The organic layer was washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 50:1 to 10:1) to afford the title compound (10 g, 90% yield) as a yellow solid; .sup.1H NMR (400 MHz,

chloroform-d) δ =8.06-7.88 (m, 1H), 7.66 (d, J=8.4 Hz, 1H), 7.40 (t, J=8.0 Hz, 1H), 7.32 (dt, J=7.6, 9.6 Hz, 1H), 6.83 (d, J=7.6 Hz, 1H), 5.44 (s, 1H).

(284) Step E. 5,6-difluoronaphthalen-1-yl trifluoromethanesulfonate: To a solution of 5,6-difluoronaphthalen-1-ol (21 g, 1.0 equiv.) in DCM (200 mL) was added DIEA (37.6 g, 2.5 equiv.) and Tf.sub.2O (42.7 g, 1.3 equiv.) at 0° C. The mixture was stirred at 20° C. for 1 hour. The reaction mixture was diluted with water (200 mL) and extracted with DCM (3×200 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 20:1 to 5:1) to afford the title compound (30 g, 82% yield) as a colorless oil; .sup.1H NMR (400 MHz, chloroform-d) δ =8.14 (d, J=8.4 Hz, 1H), 7.91-7.82 (m, 1H), 7.61 (d, J=8.0 Hz, 1H), 7.57-7.46 (m, 2H).

(285) Step F. N-(diphenylmethylene)-5,6-difluoronaphthalen-1-amine: A mixture of 5,6-difluoronaphthalen-1-yl trifluoromethanesulfonate (30 g, 1.0 equiv.), diphenylmethanimine (52.2 g, 3.0 equiv.), Pd.sub.2(dba).sub.3 (8.80 g, 0.1 equiv.), Xantphos (11.1 g, 0.2 equiv.) and Cs.sub.2CO.sub.3 (93.9 g, 3 equiv.) in toluene (500 mL) was degassed and stirred at 90° C. for 12 hours under N.sub.2 atmosphere. The reaction mixture was diluted with water (300 mL) and extracted with ethyl acetate (3×400 mL). The combined organic layers were washed with brine (300 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1 to 5:1) to afford title compound (40 g, 92% yield) as a yellow solid; LCMS (ESI, M+1): m/z=344.0.

(286) Step G. 5,6-difluoronaphthalen-1-amine: A solution of N-(5,6-difluoronaphthalen-1-yl)-1,1-diphenylmethanimine (40 g, 116 mmol, 1.0 equiv.) in HCl-MeOH (4 M, 300 mL, 10.3 equiv.) was stirred at 10° C. for 0.5 hour. The mixture was concentrated under vacuum. The pH of the residue was adjusted to ~8 with saturated aqueous NaHCO.sub.3 solution and extracted with ethyl acetate (3×300 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 20:1 to 3:1) to afford the title compound (14 g, 66% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ =7.61-7.51 (m, 2H), 7.38 (t, J=8.0 Hz, 1H), 7.33-7.27 (m, 1H), 6.79 (d, J=7.2 Hz, 1H), 4.31-4.13 (s, 2H); LCMS (ESI, M+1): m/z=180.2.

(287) Step H. 2,4-dibromo-5,6-difluoronaphthalen-1-amine: To a solution of 5,6-difluoronaphthalen-1-amine (10 g, 1.0 equiv.) in AcOH (200 mL) was added a solution of Br.sub.2 (19.4 g, 2.18 equiv.) in AcOH (100 mL) at 0° C. The mixture was stirred at 70° C. for 1 hour. The reaction mixture was filtered and the filter cake was washed with AcOH (200 mL). Then the residue was diluted with 15% aqueous of NaOH (100 mL). The mixture was stirred for 20 minutes and filtered. The solid was washed with water (200 mL) and dried under vacuum to afford the title compound (16 g, 85% yield) as a yellow solid. LCMS (ESI, M+1): m/z=337.9.

(288) Step I. 5-bromo-6,7-difluoronaphtho[1,2-d][1,2,3]oxadiazole: 2,4-dibromo-5,6-difluoronaphthalen-1-amine (16 g, 47.5 mmol, 1.0 equiv.) was dissolved in AcOH (280 g, 98.2 equiv.) and propionic acid (26.5 g, 7.53 equiv.) and cooled to 0° C. Then NaNO.sub.2 (4.91 g, 1.5 equiv.) was added and the mixture was stirred at 20° C. for 1 hour. The reaction mixture was filtered and the filter cake was washed with water (300 mL) to afford the title compound (11.3 g, 83% yield) as a yellow solid and used to next step without purification; .sup.1H NMR (400 MHz, chloroform-d) 5-7.51-7.40 (m, 1H), 7.24 (s, 1H), 7.08-6.99 (m, 1H).

(289) Step J. 4-bromo-5,6-difluoronaphthalen-2-ol: To a suspension of 5-bromo-6,7-difluoronaphtho[1,2-d][1,2,3]oxadiazole (11.3 g, 1.0 equiv.) in EtOH (150 mL) and THF (50 mL) at 0° C. was added NaBH.sub.4 (3.49 g, 2.33 equiv.). Bubbles evolved immediately. The mixture was stirred at 0° C. for 0.5 hour. The mixture was quenched with water (50 mL) and concentrated under vacuum to remove EtOH. The mixture was extracted with ethyl acetate (3×100 mL). The

combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 20:1 to 3:1) to afford the title compound (6.3 g, 59% yield) as a black solid; LCMS (EST, M-1): m/z=257.1.

(290) Step K. 8-bromo-1,2-difluoro-6-(methoxymethoxy)naphthalene: To a solution of 4-bromo-5,6-difluoronaphthalen-2-ol (6.3 g, 1.0 equiv.) in DCM (120 mL) was added DIEA (7.86 g, 2.5 equiv.) and MOMCl (3.44 g, 1.76 equiv.). The mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was diluted with water (50 mL) and extracted with DCM (3×100 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 20:1 to 5:1) to afford the title compound (5.5 g, 75% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ=7.61 (d, J=2.4 Hz, 1H), 7.52-7.46 (m, 1H), 7.39-7.30 (m, 2H), 5.27 (s, 2H), 3.52 (s, 3H).

(291) Step L. 2-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: A mixture of 8-bromo-1,2-difluoro-6-(methoxymethoxy)naphthalene (3 g, 1.0 equiv.), Pin.sub.2B.sub.2 (6.28 g, 2.5 equiv.), KOAc (2.91 g, 3.0 equiv.), Pd(dppf)Cl.sub.2 (724 mg, 0.1 equiv.) in dioxane (60 mL) was degassed and stirred at 110° C. for 1 hour under N.sub.2 atmosphere. The reaction mixture was diluted with water (50 mL) and extracted with ethyl acetate (3×50 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 50:1 to 5:1) to afford the title compound (2 g, 58% yield) as a white solid; .sup.1H NMR (400 MHz, chloroform-d) δ=7.49-7.44 (m, 1H), 7.42 (d, J=2.4 Hz, 1H), 7.41-7.39 (m, 1H), 7.33-7.27 (m, 1H), 5.29 (s, 2H), 3.51 (s, 3H), 1.45 (s, 12H).

Intermediate 18

(292) ##STR00056##

((6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)ethynyl)naphthalen-2-yl)oxy)triisopropylsilane

(293) ##STR00057##

(294) Step A. 5-(2-(4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione: To a mixture of 2-(4-fluorophenyl)acetic acid (250 g, 1 equiv.) and 2,2-dimethyl-1,3-dioxane-4,6-dione (257 g, 1.1 equiv.) in ACN (1.25 L) was added DMAP (16.9 g, 0.09 equiv.) at 15° C. DIPEA (451 g, 2.1 equiv.) was added dropwise below 30° C. for 1 hour. Pivaloyl chloride (215 g, 1.1 equiv.) was added dropwise below 40° C. for 1 hour. The mixture was stirred at 45° C. for 3 hours. The mixture was cooled to 0° C. 4 N aqueous HCl (5.0 L) was added dropwise to adjust pH to 5 while maintaining the temperature between below 15° C. The mixture was stirred at 0° C. for 1 hour. The mixture was diluted with H.sub.2O (15 L) and the pH of the mixture was adjusted to 2 with 4N HCl. The mixture was filtered. The filter cake was washed with H.sub.2O until the pH of filter cake was 5-6. The solid was dried under reduced pressure to afford the title compound (500 g, crude) as a white solid; .sup.1H NMR (400 MHz, chloroform-d) 5-7.36 (dd, J=5.6, 8.4 Hz, 2H), 7.01 (t, J=8.4 Hz, 2H), 4.38 (s, 2H), 1.72 (s, 6H).

(295) Step B. 7-fluoro-1,3-dihydroxy-2-naphthoic acid: 5-(2-(4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (490 g, 1 equiv., crude) was added into CF.sub.3SO.sub.3H (2.04 kg, 7.8 equiv.) portion wise maintaining the temperature below 30° C. The mixture was stirred at 20° C. for 2 hour. The mixture was poured into ice water (30 L) slowly. The mixture was filtered. The filter cake was washed with water until the pH of the filtrate was 3-4 to afford the title compound (500 g, crude, wet) as a brown solid.

(296) Step C. 7-fluoronaphthalene-1,3-diol: A mixture of 7-fluoro-1,3-dihydroxy-2-naphthoic acid (375 g, crude, wet) in H.sub.2O (1.8 L) and ACN (1.8 L) was stirred at 78° C. for 13 hours. The mixture was concentrated to remove ACN. The mixture was diluted with H.sub.2O (1 L) and

saturated NaHCO₃ aqueous (0.3 L), and then extracted with ethyl acetate (4×0.5 L). The combined organic layers were washed with saturated aqueous NaHCO₃ (0.5 L), water (0.5 L) and brine (0.5 L), dried over anhydrous Na₂SO₄ and concentrated. The residue was treated with n-heptanes (0.8 L) for 1 hour. The mixture filtered and the solid was dried under reduced pressure to afford the title compound (145 g, 60% yield over three steps) as light a red solid; ¹H NMR (400 MHz, DMSO-d₆) δ=10.18 (s, 1H), 9.48 (s, 1H), 7.65-7.56 (m, 2H), 7.23 (dt, J=2.8, 8.8 Hz, 1H), 6.64 (d, J=1.6 Hz, 1H), 6.56 (d, J=1.6 Hz, 1H).

(297) Step D. 7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol: A mixture of 7-fluoronaphthalene-1,3-diol (173 g, 1 equiv.), (bromoethynyl)triisopropylsilane (266 g, 1.05 equiv.), AcOK (191 g, 2 equiv.) and dichloro(p-cymene)ruthenium(II) dimer (17.8 g, 0.03 equiv.) in dioxane (1.5 L) was degassed and stirred at 100° C. for 3.5 hours. The reaction mixture was filtered through a pad of Celite. The filter cake was washed with ethyl acetate (4×500 mL). The filtrate was concentrated under reduced pressure to give a residue. The residue was dissolved in ethyl acetate (3 L). The solution was washed with saturated NaHCO₃ aqueous (0.5 L) and brine (0.2 L), dried over anhydrous Na₂SO₄, and concentrated to give a residue. The residue was purified by column chromatography [SiO₂, petroleum ether/ethyl acetate 15:1 to 10:1] to afford a crude product. The crude product was dispersed in n-heptanes (0.5 L) and stirred for 1 hour. The mixture was filtered and the filter cake was washed with n-heptanes (0.5 L). The solid was dried under reduced pressure to afford the title compound (204 g, 56% yield) as a light yellow solid; ¹H NMR (400 MHz, DMSO-d₆) δ=10.04 (s, 1H), 9.58 (s, 1H), 7.63 (dd, J=5.6, 9.2 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 6.63 (d, J=2.4 Hz, 1H), 6.58 (d, J=2.0 Hz, 1H), 1.13 (s, 21H).

(298) Step E. 7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyloxy)naphthalen-1-ol: To a solution of 7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol (197 g, 1 equiv.) and DIPEA (142 g, 2.0 equiv.) in DCM (1 L) was added TIPSCl (122 g, 1.2 equiv.) dropwise between 0 and 10° C. for 1 hour. The mixture was stirred 25° C. for 1 hour. The mixture was poured H₂O (2 L). The DCM phase was separated and washed with brine (3×1 L). The DCM phase was dried over anhydrous Na₂SO₄ and filtered to afford the title compound (282 g, in DCM) as a red liquid, which was used in next step directly.

(299) Step F. 7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yltrifluoromethanesulfonate: To a solution of 7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-ol (282 g, 1 equiv.) and DIPEA (248 g, 3.5 equiv.) in DCM (1.7 L) was added Tf₂O (263 g, 1.7 equiv.) dropwise at -40° C. for 3 hours. The mixture was stirred at -40° C. for 0.5 hour. The mixture was poured into water (1 L). The DCM layer was separated and washed with water (3×3 L), 0.001N HCl (3×2 L), H₂O (1.5 L), brine (2×1 L), dried over anhydrous Na₂SO₄ and concentrated to give a residue. The residue was purified by column chromatography [SiO₂, petroleum ether] to afford the title compound (321 g, crude) as a light red oil; ¹H NMR (400 MHz, chloroform-d) δ=7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.34-7.25 (m, 3H), 1.34-1.14 (m, 42H); ¹⁹F NMR (376 MHz, chloroform-d) δ=-79, -105.

(300) Step G. ((6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)ethynyl)naphthalen-2-yl)oxy)triisopropylsilane: A mixture of 7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yltrifluoromethanesulfonate (229 g, 1 equiv.), TEA (144 g, 4.0 equiv.), 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (182 g, 4.0 equiv.) and Pd(dppf)Cl₂ (16 g, 0.06 equiv.) in MeCN (1.5 L) was degassed stirred at 78° C. for 4 hours under N₂ atmosphere. The mixture was slowly quenched with of MeOH (0.5 L) maintaining the temperature below 25° C. producing a precipitate. The mixture was filtered and the filter cake was washed with MeOH (1 L). The solid was dispersed in MeOH (0.5 L) and stirred for 0.5 hour. The mixture was filtered. The solid was dried under reduced pressure to afford the title compound (170 g, 69% yield over three steps, crude) as a white solid; ¹H NMR (400 MHz, CHLOROFORM-d) δ=7.60 (dd, J=5.6, 9.2 Hz, 1H), 7.43 (d, J=2.4 Hz, 1H), 7.22-7.17 (m, 2H),

1.43 (s, 12H), 1.32-1.12 (m, 42H); LCMS (ESI, M+1): m/z=625.6.

Intermediate 19

(301) ##STR00058##

((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane

(302) ##STR00059##

(303) Step A. 5-ethynyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol:

To a solution of ((6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)ethynyl)naphthalen-2-yl)oxy)triisopropylsilane (1.50 kg, 1.0 equiv.) in DMSO (15.0 L) was added CsF (2.19 kg, 6.0 equiv.) in one portion at 25° C. under N.sub.2. The reaction mixture was stirred at 25° C. for 12 hrs. The mixture was diluted with EtOAc (5.00 L) and water (20.0 L) and the layers were separated. The aqueous phase was extracted with EtOAc (5.00 L×3). The combined organic phase was washed with brine (5.00 L×5), dried with anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The crude product was triturated with n-heptanes (4.5 L) at 50° C. for 6 hrs. The mixture was cooled to room temperature and filtered to afford the title compound (600 g, 1.86 mol, 80.1% yield) as a yellow solid. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.48-7.46 (m, 1H), 7.30-7.29 (d, J=2.4, 1H), 7.12-7.08 (m, 1H), 7.00-6.98 (m, 1H), 5.56 (s, 1H), 3.61 (s, 1H), 1.37 (s, 12H).

(304) Step B. 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol: To a solution of 5-ethynyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (40.0 g, 1.00 equiv.) in THF (400 mL) was added Pd/C (4.00 g, 10.0% purity). The mixture was purged with H.sub.2 three times and stirred at 25° C. under H.sub.2 (15 Psi) for 5 hrs. The mixture was filtered through a diatomite powder pad and washed with THF (200 mL×2), the filtrate was concentrated in vacuum. The residue was purified by silica gel chromatography (petroleum ether/EtOAc 50:1 to 3:1) to afford the title compound (74.0 g, 88.5% yield) as a yellow solid. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.57-7.55 (m, 1H), 7.41-7.37 (m, 1H), 7.23-7.18 (m, 2H), 5.28 (s, 1H), 3.15-3.12 (m, 2H), 1.45 (s, 12H), 1.29-1.26 (m, 3H).

(305) Step C. ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane: To a solution of 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (73.0 g, 1.0 equiv.) in DMF (750 mL) was added imidazole (47.2 g, 3.0 equiv.) and TIPSCl (89.0 g, 2.0 equiv.). The mixture was stirred at 25° C. for 12 hours. The mixture was diluted with H.sub.2O (500 mL) and extracted with MTBE (3×300 mL). The combined organic phase was washed with brine (3×300 mL), dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by silica gel chromatography [petroleum ether/ethyl acetate 100:1 to 20:1] to afford the title compound (102 g, 94% yield) as a yellow solid; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ=7.43-7.40 (m, 1H), 7.22-7.21 (m, 1H), 7.13-7.09 (m, 2H), 3.07-3.01 (m, 2H), 1.36 (s, 12H), 1.23-1.18 (m, 6H), 1.06-0.97 (m, 18H).

Intermediate 20

(306) ##STR00060##

7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(307) ##STR00061##

(308) Step A. 7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.00 equiv.), 2-(8-ethyl-1-naphthyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (201 mg, 1.50 equiv.), CataCXium A Pd G3 (34.6 mg, 0.10 equiv.), K.sub.3PO.sub.4 (1.5 M, 3.0 equiv.) in THF (2 mL) was degassed and stirred at 60° C. for 12 hours under N.sub.2 atmosphere. The reaction mixture was diluted with water (5 mL) and extracted with EtOAc (15 mL×3). The combined organic layers were washed with brine (5 mL×2), dried over Na.sub.2SO.sub.4, concentrated and purified by

reversed-phase HPLC [column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 25% Y-55%, 10 min] to afford the title compound (20.0 mg, 7.8% yield) as an off-white solid; LCMS (ESI, M+1): m/z=541.3.

Intermediate 21

(309) ##STR00062##

((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol

(310) ##STR00063##

(311) Step A: (2R,4S)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxy)pyrrolidine-1,2-dicarboxylate: To a solution of (2R,4S)-1-tert-butyl 2-methyl 4-hydroxypyrrolidine-1,2-dicarboxylate (20.0 g, 1.0 equiv.) and imidazole (11.1 g, 2.0 equiv.) in DCM (200 mL) was added TBDPSCl (33.6 g, 31.4 mL, 1.50 equiv.). The mixture was stirred at 25° C. for 1 hour. After reaction completion, the reaction mixture was added water (150 mL), separated, the aqueous layer was extracted with EtOAc (2×100 mL), the combined organic layer was dried over Na.sub.2SO.sub.4, filtered and concentrated. The crude product was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1-30:1) to afford the title compound (39 g, 93% yield) as a white solid; LCMS (ESI, M-100): m/z=384.3.

(312) Step B. (4S)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxyl-2-(3-chloropropyl)pyrrolidine-1,2-dicarboxylate: To a solution of (2R,4S)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxy)pyrrolidine-1,2-dicarboxylate (37.0 g, 1.0 equiv.) in THF (400 mL) was added LDA (2 M, 49.7 mL, 1.30 equiv.) at -70° C. slowly. The mixture was stirred at -70° C. for 1 hour. To the mixture was added 1-bromo-3-chloro-propane (60.2 g, 37.6 mL, 5.0 equiv.) at -70° C. The mixture was stirred at -70-20° C. for 12 hours. Upon completion, the reaction mixture was diluted with water (300 mL), separated, the aqueous layer was extracted with EtOAc (2×100 mL), the combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified by reversed phase flash chromatography (C18, 0.1% formic acid condition) and column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1-20:1) to afford the title compound (8.70 g, 20% yield) as a colorless oil; LCMS (ESI, M-55, M-100): 504.3, 461.2.

(313) Step C. (4S)-methyl 4-((tert-butyldiphenylsilyl)oxy)-2-(3-chloropropyl)pyrrolidine-2-carboxylate: To a solution of (4S)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxy)-2-(3-chloropropyl)pyrrolidine-1,2-dicarboxylate (8.60 g, 1.0 equiv.) in MeCN (40 mL) was added HCl.dioxane (4 M, 40 mL, 10.4 equiv.). The mixture was stirred at 20° C. for 0.5 hour. Upon completion, the reaction mixture was concentrated to afford the title compound (7.8 g, crude, HCl salt) as a yellow solid.

(314) Step D. (2S)-methyl 2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizine-7a-carboxylate (90122-E): To a solution of (4S)-methyl 4-((tert-butyldiphenylsilyl)oxy)-2-(3-chloropropyl)pyrrolidine-2-carboxylate (7.80 g, 1.0 equiv., HCl) in ACN (80 mL) was added NaHCO.sub.3 (7.12 g, 3.30 mL) and KI (281 mg). The mixture was stirred at 50° C. for 12 hours. Upon completion, the reaction mixture was diluted with water (100 mL) and extracted with ethyl acetate (2×50 mL). The organic layer was dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified by prep-HPLC (column: Phenomenex luna C18 250*80 mm*10 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 75%-100%, 20 min) to afford the title compound (1.3 g, two steps 18% yield) as a colorless oil.

(315) Step E. ((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol: To a solution of (2S)-methyl 2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizine-7a-carboxylate (1.30 g, 1.0 equiv.) in THF (15 mL) was added LiAlH.sub.4 (349 mg, 3.0 equiv.) at -40° C. The mixture was stirred at -40° C. for 1 hour. Upon completion, the reaction mixture was quenched with saturated aqueous Na.sub.2SO.sub.4 solution (1 mL), filtered and concentrated to afford the title compound (1.1 g, 91% yield) as a yellow oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.70-7.60 (m, 4H), 7.48-7.35 (m, 6H), 4.49-4.30 (m, 1H), 3.12 (d, J=2.4 Hz, 2H), 3.06-2.98 (m, 2H), 2.95 (dd, J=4.8, 11.2 Hz, 1H), 2.73 (dd, J=4.4, 11.2 Hz, 1H), 2.07-1.89 (m,

3H), 1.81-1.65 (m, 3H), 1.06 (s, 9H); LCMS (ESI, M+1): m/z=396.7.

Intermediate 22

(316) ##STR00064##

((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol

Intermediate 23

(317) ##STR00065##

((3S,7aS)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol

Intermediate 24

(318) ##STR00066##

((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol

Intermediate 25

(319) ##STR00067##

((3R,7aS)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol

(320) ##STR00068## ##STR00069##

(321) Step A. benzyl 2-methyl 2-(but-3-en-1-yl)pyrrolidine-1,2-dicarboxylate: To a solution of 1-benzyl 2-methyl (S)-pyrrolidine-1,2-dicarboxylate (220 g, 1 equiv.) in THF (2.2 L) was added LiHMDS (1 M, 1.00 L, 1.2 equiv.) at -65° C. The mixture was stirred at -65° C. for 1 hour. Then 4-bromobut-1-ene (225.61 g, 2 equiv.) was added to the mixture at -65° C. The mixture was stirred at 25° C. for 12 hours. The mixture was quenched with saturated aqueous NH₄Cl (500 mL) and extracted with ethyl acetate (3×500 mL). The combined organic layer was dried over anhydrous sodium sulfate. The mixture was filtered and the filtrate was concentrated under reduced pressure to dryness. The residue was purified by column chromatography [SiO₂, petroleum ether/ethyl acetate 10:1 to 5:1] to afford the title compound (220 g, 74.6% yield) as a yellow oil; ¹H NMR (400 MHz, chloroform-d) δ=7.25 (s, 5H), 5.88-5.63 (m, 1H), 5.17-5.05 (m, 2H), 5.04-4.88 (m, 2H), 3.83-3.62 (m, 3H), 3.54-3.42 (m, 2H), 2.48-2.17 (m, 1H), 2.15-2.03 (m, 3H), 2.02-1.77 (m, 4H); LCMS[ESI, M+1]: m/z=318.2.

(322) Step B. benzyl 2-methyl 2-(2-(oxiran-2-yl)ethyl)pyrrolidine-1,2-dicarboxylate: To a solution of benzyl 2-methyl 2-(but-3-en-1-yl)pyrrolidine-1,2-dicarboxylate (242 g, 1 equiv.) in DCM (2.40 L) was added m-CPBA (309 g, 85% purity, 2 equiv.) in portions at 0° C. The mixture was stirred at 25° C. for 5 hours. The mixture was quenched by addition of saturated aqueous Na₂SO₃ solution (500 mL), extracted with dichloromethane (3×500 mL). The combined organic layers were washed with saturated brine (2×200 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and the filtrate was concentrated under reduced pressure to give a residue. The residue was purified by column chromatography [SiO₂, petroleum ether/ethyl acetate 10:1 to 5:1] to afford the title compound (200 g, 62.9% h yield) as a yellow oil. ¹H NMR (400 MHz, chloroform-d) δ=7.43-7.10 (m, 5H), 5.20-4.93 (m, 2H), 3.78-3.56 (m, 3H), 3.48-3.34 (m, 2H), 2.89-2.53 (m, 2H), 2.46-2.10 (m, 2H), 2.06-1.97 (m, 3H), 1.91-1.71 (m, 2H), 1.59-1.31 (m, 2H); LCMS[ESI, M+1]: m/z=334.1.

(323) Step C. methyl 3-(hydroxymethyl)hexahydro-1H-pyrrolizine-7a-carboxylate: To a suspension of Pd/C (16.0 g, 10% purity) in MeOH (1.0 L) was added benzyl 2-methyl 2-(2-(oxiran-2-yl)ethyl)pyrrolidine-1,2-dicarboxylate (130 g, 1 equiv.) under N₂ atmosphere. The suspension was degassed under vacuum and purged with H₂ (50 psi) several times. The mixture was stirred at 25° C. under H₂ (50 psi) for 1 hour. The mixture was filtered and the filtrate was concentrated to afford the title compound (75 g, 96.5% yield) as a yellow oil; ¹H NMR (400 MHz, chloroform-d) δ (ppm)=3.88-3.73 (m, 1H), 3.73-3.66 (m, 1H), 3.61-3.54 (m, 1H), 3.42-3.35 (m, 1H), 3.09-3.03 (m, 1H), 2.99-2.88 (m, 1H), 2.74-2.65 (m, 1H), 2.55-2.48 (m, 1H), 2.31 (td, J=4.7, 12.4 Hz, 1H), 2.14 (br d, J=1.6 Hz, 1H), 1.89-1.74 (m, 5H), 1.73-1.45 (m, 2H).

(324) Steps D and E. methyl 3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizine-7a-carboxylate: To a solution of methyl 3-(hydroxymethyl)-1,2,3,5,6,7-hexahydropyrrolizine-8-carboxylate (165 g, 1 equiv.) and imidazole (169 g, 3 equiv.), DMAP (10.1 g, 0.1 equiv.) in DCM

(1.4 L) was added TBDPSCl (296 g, 1.3 equiv.) drop-wise at 0° C. The mixture was stirred at 25° C. for 2 hours. The reaction mixture was washed with H.sub.2O (2×500 mL), the combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduce pressure to give a residue. The residue was purified by column chromatography (SiO.sub.2, 0.1% NH.sub.3.H.sub.2O, Petroleum ether/Ethyl acetate 20:1 to 1:1) to afford peak 1 (330 g, crude), lower polarity, as a yellow oil and peak 2 (166 g, 44.9% yield), higher polarity as a yellow oil; Peak 1: LCMS[ESI, M+1]: m/z=438.3; Peak 2 LCMS[ESI, M+1]: m/z=438.3.

(325) Step F1: rac-((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol: To a mixture of rac-(3R,7aR)-methyl 3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizine-7a-carboxylate (160 g, 70% purity, 1 equiv.) in THF (2 L) was added LiAlH.sub.4 (12.63 g, 1.3 equiv.) portion wise at -40° C. The mixture was stirred at -40° C. for 3 hours. The reaction mixture was quenched with H.sub.2O (13 mL), 15% aqueous NaOH (13 mL), H.sub.2O (39 mL), dried over anhydrous Na.sub.2SO.sub.4 at 0° C., filtered and concentrated. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)-ACN) to afford the title compound (85 g, 64.87% yield) as a yellow oil. LCMS [ESI, M+1]: m/z=410.2.

(326) Step G1. ((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol and ((3S,7aS)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol: The stereoisomers of rac-((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (175 g) were separated by SFC (column: Phenomenex-Cellulose-2 (250 mm×50 mm, 10 um); mobile phase: [0.1% NH.sub.3—H.sub.2O IPA]; B %: 40%-40%, 4.9 min) to afford title compounds ((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (54 g, 40% yield) as a yellow oil .sup.1H NMR (400 MHz, CHLOROFORM-d) δ=7.71 (br t, J=7.4 Hz, 4H), 7.47-7.34 (m, 6H), 3.77-3.57 (m, 2H), 3.35-3.21 (m, 2H), 3.01-2.78 (m, 3H), 1.98-1.84 (m, 2H), 1.83-1.64 (m, 4H), 1.64-1.51 (m, 3H), 1.13-1.01 (m, 10H), -0.89--0.90 (m, 1H); LCMS (ESI, M+1): m/z=410.3; SFC: 100% ee.

(327) and ((3S,7aS)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (62 g, 48% yield) as a yellow oil. .sup.1H NMR (400 MHz, CHLOROFORM-d) δ=7.77-7.64 (m, 4H), 7.48-7.35 (m, 6H), 3.96-3.62 (m, 1H), 3.57-3.30 (m, 2H), 3.18-2.85 (m, 3H), 2.01-1.90 (m, 2H), 1.89-1.70 (m, 4H), 1.69-1.56 (m, 2H), 1.11-1.03 (m, 10H); LCMS (ESI, M+1): m/z=410.3; SFC: 99.3% ee.

(328) Step F2: rac-((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol: To a solution of methyl 3-[[tert-butyl(diphenyl)silyl]oxymethyl]-1,2,3,5,6,7-hexahydropyrrolizine-8-carboxylate (83.0 g, 1 equiv.) in THF (830 mL) was added LAH (8.05 g, 1.3 equiv.) portion wise at -40° C. The mixture was stirred at -40° C. for 2 hours. The reaction mixture was quenched with H.sub.2O (8.00 mL), NaOH (15%, 8.00 mL) and H.sub.2O (24.0 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated to afford the title compound (72.0 g, crude) as yellow oil. LCMS [ESI, M+1]: m/z=410.2.

(329) Step G2: ((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol and ((3R,7aS)-3-(((tert-butyldiphenylsilyl)oxy)methyl)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol: The stereoisomers of rac-((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (110 g) were separated by SFC (column: REGIS (s,s) WHELK-O1 (250 mm×50 mm, 10 um); mobile phase: [0.1% NH.sub.3—H.sub.2O ETOH]; B %: 40%-40%, 3.7 min) to afford title compounds ((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)tetrahydro-1H-pyrrolizin-7a (5H)-yl)methanol (45 g, 40% yield) as yellow oil .sup.1H NMR (400 MHz, chloroform-d) δ=7.77-7.63 (m, 4H), 7.52-7.31 (m, 6H), 3.96-3.86 (m, 1H), 3.82-3.68 (m, 1H), 3.33-3.22 (m, 2H), 3.22-3.13 (m, 1H), 2.86-2.79 (m, 1H), 2.76-2.64 (m, 1H), 2.00-1.92 (m, 1H), 1.82-1.48 (m, 7H), 1.31-1.21 (m, 1H), 1.07 (s, 9H) LCMS[ESI, M+1]: m/z=410.3.

(330) and ((3R,7aS)-3-(((tert-butyldiphenylsilyl)oxy)methyl)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (90466-H2B) (45 g, 40% yield) as a yellow oil. ¹H NMR (400 MHz, chloroform-d) δ=7.75-7.64 (m, 4H), 7.35 (br s, 6H), 3.96-3.88 (m, 1H), 3.78-3.72 (m, 1H), 3.35-3.23 (m, 2H), 3.22-3.13 (m, 1H), 2.87-2.79 (m, 1H), 2.74-2.65 (m, 1H), 2.00-1.92 (m, 1H), 1.83-1.63 (m, 4H), 1.63-1.43 (m, 3H), 1.28-1.23 (m, 1H), 1.10-1.04 (m, 9H); LCMS[ESI, M+1]: m/z=410.3.

Intermediate 26

(331) ##STR00070##

((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methanol

(332) ##STR00071## ##STR00072##

(333) Step A. (2S,4R)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxy)pyrrolidine-1,2-dicarboxylate: To a solution of (2S,4R)-1-tert-butyl 2-methyl 4-hydroxypyrrolidine-1,2-dicarboxylate (20.0 g, 1.0 equiv.) in DCM (250 mL) was added imidazole (11.1 g, 2.0 equiv.) and TBDPSCl (26.9 g, 1.20 equiv.) at 0° C. The mixture was stirred at 25° C. for 1 hour. The reaction mixture was diluted with H₂O (100 mL) and the organic phase was separated. The aqueous phase was extracted with ethyl acetate (2×100 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by column chromatography [SiO₂, Petroleum ether/Ethyl acetate 1:0 to 10:1] to afford the title compound (38.8 g, 98% yield) as an off-white solid; ¹H NMR (400 MHz, CDCl₃) δ=7.68-7.57 (m, 4H), 7.49-7.36 (m, 6H), 4.58-4.36 (m, 2H), 3.74-3.63 (m, 3H), 3.59-3.35 (m, 2H), 2.31-2.15 (m, 1H), 1.95-1.80 (m, 1H), 1.50-1.39 (m, 9H), 1.06 (s, 9H).

(334) Step B. (4I)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyloxy)-2-(3-chloropropyl)pyrrolidine-1,2-dicarboxylate: To a solution of (2S,4R)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxy)pyrrolidine-1,2-dicarboxylate (36.0 g, 1.0 equiv.) in THF (200 mL) was added LDA (2.0 M in THF, 48.4 mL, 1.30 equiv.). The mixture was stirred at -70° C. for 1 hour. To the reaction mixture was added 1-bromo-3-chloro-propane (58.6 g, 5.0 equiv.) at -70° C. The mixture was stirred at 20° C. for 12 hours. The reaction mixture was diluted with H₂O (200 mL) and extracted with ethyl acetate (3×200 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (36.0 g, 86% yield) as a yellow oil; ¹H NMR (400 MHz, CDCl₃) δ=7.68-7.59 (m, 4H), 7.49-7.34 (m, 6H), 4.46-4.19 (m, 1H), 3.88-3.66 (m, 2H), 3.62-3.55 (m, 3H), 3.53-3.22 (m, 2H), 2.38-1.71 (m, 6H), 1.46-1.36 (m, 9H), 1.10-1.01 (m, 9H).

(335) Step C. (4R)-methyl 4-((tert-butyldiphenylsilyl)oxy)-2-(3-chloropropyl)pyrrolidine-2-carboxylate: To a solution of (4R)-1-tert-butyl 2-methyl 4-((tert-butyldiphenylsilyl)oxy)-2-(3-chloropropyl)pyrrolidine-1,2-dicarboxylate (36.0 g, 1.0 equiv.) in ACN (200 mL) was added HCl-dioxane (4.0 M, 200 mL) at 0° C. The mixture was stirred at 0° C. for 1 hour. The mixture was concentrated under reduced pressure to afford the title compound (34.0 g, crude, HCl) as a yellow solid.

(336) Step D. (2R,7aS)-methyl 2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizine-7a-carboxylate (peak A) and (2R,7aR)-methyl 2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizine-7a-carboxylate (peak B): To a solution of (4R)-methyl 4-((tert-butyldiphenylsilyl)oxy)-2-(3-chloropropyl)pyrrolidine-2-carboxylate (34.0 g, 1.0 equiv, HCl) in ACN (300 mL) was added NaHCO₃ (28.8 g, 5.0 equiv.) and KI (1.14 g, 0.1 equiv.). The mixture was stirred at 50° C. for 12 hours. The reaction mixture was diluted with H₂O (150 mL) and extracted with ethyl acetate (3×100 mL). The combined organic phase was dried over anhydrous sodium sulfate, concentrated and purified by prep-HPLC [column: Waters Xbridge C18 150×50 mm×10 μm; mobile phase: water (10 mM NH₄HCO₃)-ACN; B %: 73%-93%, 11.5 min] and another prep-HPLC [column: Phenomenex luna C18 250 mm×100 mm×10 μm; mobile phase: water (10 mM NH₄HCO₃)-ACN; B %: 55%-85% over 30 min] to afford the peak A (10.0 g, two steps 34% yield) and the peak B (6.0 g, two steps 21% yield) as yellow oil.

LCMS (ESI, M+1): m/z=424.1.

(337) Step E. ((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methanol: To a solution of (2R,7aS)-methyl 2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizine-7a-carboxylate (4.50 g, 1.0 equiv.) in THF (100 mL) was added LiAlH.sub.4 (1.21 g, 3.0 equiv.) at -40° C. The mixture was stirred at -40° C. for 1 hour. The reaction mixture was quenched by addition of H.sub.2O (4.0 mL) at 0° C. The mixture was dried over anhydrous sodium sulfate, filtered and concentrated to afford the title compound (4.20 g, 94% yield) as a yellow oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ=7.68-7.62 (m, 4H), 7.46-7.36 (m, 6H), 4.43-4.35 (m, 1H), 3.11 (s, 2H), 3.03-2.94 (m, 3H), 2.76-2.69 (m, 1H), 2.02-1.89 (m, 3H), 1.80-1.69 (m, 3H), 1.06 (s, 9H); LCMS (ESI, M+1): m/z=396.1.

Intermediate 27

(338) ##STR00073##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(339) ##STR00074##

(340) Step A. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (28.4 g, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (25.6 g, 1.25 equiv.) and Cs.sub.2CO.sub.3 (1.5 M in H.sub.2O, 129 mL, 3.0 equiv.) in methoxycyclopentane (300 mL) was added Ad.sub.2nBuP-Pd-G3 (7.07 g, 0.15 equiv.). The reaction was stirred at 100° C. for 3 hours under N.sub.2. The mixture was diluted with water (300 mL) and extracted with ethyl acetate (3×200 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (16.8 g, 40% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ=9.19 (d, J=10.4 Hz, 1H), 7.54 (dd, J=5.6, 8.8 Hz, 1H), 7.26-7.21 (m, 1H), 7.16-7.11 (m, 1H), 6.96-6.84 (m, 1H), 5.46-5.22 (m, 1H), 5.05-4.71 (m, 2H), 4.56-4.32 (m, 2H), 3.48-3.21 (m, 3H), 3.13-3.01 (m, 1H), 2.47-2.32 (m, 4H), 2.32-2.08 (m, 4H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=593.2.

Intermediate 28

(341) ##STR00075##

tert-butyl 2,4-dioxo-1,3,7-triazaspiro[4.6]undecane-7-carboxylate

(342) ##STR00076##

(343) Step A. tert-butyl 2,4-dioxo-1,3,7-triazaspiro[4.6]undecane-7-carboxylate: To a solution of tert-butyl 3-oxoazepane-1-carboxylate (2.00 g, 1.0 equiv.) and (NH.sub.4).sub.2CO.sub.3 (2.70 g, 3.0 equiv.) in EtOH (10 mL) and H.sub.2O (10 mL) was added KCN (1.12 g, 1.83 equiv.). The reaction was stirred at 85° C. for 16 hours. The mixture was cooled to 25° C., then diluted with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated to afford the title compound (2.20 g, 83% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.4) δ=11.39-9.68 (m, 1H), 7.97-7.52 (m, 1H), 3.45 (s, 2H), 3.29-3.12 (m, 2H), 1.82-1.65 (m, 3H), 1.64-1.47 (m, 3H), 1.45-1.31 (m, 9H); LCMS (ESI, M-55): m/z=228.0.

Intermediate 29

(344) ##STR00077##

5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-ylmethanol

(345) ##STR00078##

(346) Step A. tert-butyl 2-(hydroxymethyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5 (6H)-carboxylate: To a solution of 5-(tert-butoxycarbonyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (500 mg, 1.0 equiv.) in THF (10 mL) was added LiAlH.sub.4 (135

mg, 2.0 equiv.) at 0° C. The mixture was stirred at 0° C. for 1.5 hours and at 20° C. for 3 hours. The mixture was quenched with ice water (20 mL) and extracted with ethyl acetate (50 mL). The organic phase concentrated to give the title compound (260 mg, 33% yield) as a yellow oil; LCMS (ESI, M+1): m/z=268.2.

(347) Step B. 5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanol: To the solution of tert-butyl 2-(hydroxymethyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5 (6H)-carboxylate (260 mg, 1.0 equiv.) in ACN (4 mL) was added HCl.dioxane (4 M, 2.43 mL, 10 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The reaction mixture was concentrated to give a residue. The residue was dissolved in methanol (2 mL) and the pH was adjusted to 8 with saturated aqueous NaHCO₃. The mixture was filtered and concentrated to give a residue. The residue was dissolved with dichloromethane (5 mL), filtered and concentrated to afford the title compound (250 mg, crude) as a yellow oil; LCMS (ESI, M+1): m/z=167.9.

Intermediate 30

(348) ##STR00079##

3-(sulfamoylamino)piperidine

(349) ##STR00080##

(350) Step A. benzyl 3-(sulfamoylamino)piperidine-1-carboxylate: To a solution of benzyl 3-aminopiperidine-1-carboxylate (500 mg, 1.0 equiv.) in dioxane (10 mL) was added sulfamide (410 mg, 254 µL, 2.0 equiv.). After stirring at 80° C. for 12 hours, another portion of sulfamide (615 mg, 382 µL, 3.0 equiv.) was added. The mixture was stirred at 80° C. for 5 hours. The reaction mixture was concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (423 mg, 56% yield) as a colorless oil; LCMS (ESI, M+1): m/z=314.0.

(351) Step B. 3-(sulfamoylamino)piperidine: To a solution of benzyl 3-(sulfamoylamino)piperidine-1-carboxylate (420 mg, 1.0 equiv.) in methanol (10 mL) was added Pd/C (0.1 g, 10% purity) under N₂ atmosphere. The suspension was degassed and purged with H₂ 3 times. The mixture was stirred under H₂ (15 Psi) at 20° C. for 12 hours. The reaction mixture was filtered under N₂ and the filtrate was concentrated to afford the title compound (240 mg, 99% yield) as a colorless oil; ¹H NMR (400 MHz, DMSO-d₆) δ=7.65-7.01 (m, 5H), 6.70 (br d, J=7.2 Hz, 1H), 6.56 (s, 2H), 5.17-4.99 (m, 2H), 4.19-4.07 (m, 1H), 3.81 (br d, J=12.4 Hz, 1H), 3.11 (br s, 1H), 2.86-2.63 (m, 2H), 1.99-1.88 (m, 1H), 1.75-1.60 (m, 1H), 1.47-1.29 (m, 2H).

Intermediate 31

(352) ##STR00081##

pyrrolidin-1-yl(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanone

(353) ##STR00082##

(354) Step A. tert-butyl-2-(pyrrolidine-1-carbonyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (160 mg, 1.0 equiv.), pyrrolidine (121 mg, 3.0 equiv.) and triethylamine (74.8 mg, 103 µL, 1.3 equiv.) in DMF (1.5 mL) was added HATU (281 mg, 1.3 equiv.) at 0° C. The mixture was stirred at 25° C. for 0.5 hour. Upon reaction completion, the mixture was filtered and the filtrate was partitioned between ethyl acetate (10 mL) and water (10 mL). The organic phase was separated, washed with brine (2×5 mL), dried over sodium sulfate, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 1:1 to 0:1) to afford the title compound (110 mg, 58% yield) as a white solid; LCMS [ESI, M+1]: m/z=335.2.

(355) Step B. pyrrolidin-1-yl(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanone: To a solution of tert-butyl 2-(pyrrolidine-1-carbonyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (110 mg, 1.0 equiv.) in DCM (1.5 mL) was added trifluoroacetic acid (565 mg, 367 µL, 15.1 equiv.) at 0° C. The mixture was stirred at 25° C. for 0.5 hour. The reaction

mixture was concentrated under reduced pressure to afford the title compound (65.0 mg, 84% yield) as a yellow liquid; LCMS [ESI, M+1]: m/z=235.2.

(356) Intermediates 32, 33, 34 and 51 were synthesized according to the procedure described for Intermediate 31.

Intermediate 32

(357) ##STR00083##

N-ethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide Intermediate 33

(358) ##STR00084##

N-isopropyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide Intermediate 34

(359) ##STR00085##

N-ethyl-N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide Intermediate 51

(360) ##STR00086##

(4-methylpiperazin-1-yl)(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanone Intermediate 35

(361) ##STR00087##

(1-((4-methylpiperazin-1-yl)methyl)cyclopropyl)methanol

(362) ##STR00088##

(363) Step A. methyl 1-(chlorocarbonyl)cyclopropanecarboxylate: To a mixture of 1-(methoxycarbonyl)cyclopropanecarboxylic acid (2.0 g, 1.0 equiv.), DMF (101 mg, 0.1 equiv.) in DCM (15 mL) was added (COCl)₂ (2.64 g, 1.82 mL, 1.5 equiv.). The mixture was stirred at 0-20° C. for 1 hour. The mixture was concentrated and purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 5:1 to 1:1) to afford the title compound (2.2 g, 97% yield) as a yellow oil.

(364) Step B. methyl 1-(4-methylpiperazine-1-carbonyl)cyclopropanecarboxylate: To a mixture of methyl 1-chlorocarbonylcyclopropanecarboxylate (2.2 g, 1.0 equiv.), TEA (4.11 g, 3.0 equiv.) in DCM (15 mL) was added 1-methylpiperazine (1.63 g, 1.2 equiv.). The mixture was stirred at 0-20° C. for 1 hour. The mixture was concentrated and the residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 3:1 to 1:1) to afford the title compound (2.18 g, 71% yield) as a white solid; ¹H NMR (400 MHz, chloroform-d) δ=3.77-3.31 (m, 7H), 2.39-2.10 (m, 7H), 1.45-1.32 (m, 2H), 1.29-1.17 (m, 2H).

(365) Step C. (1-((4-methylpiperazin-1-yl)methyl)cyclopropyl)methanol: To a solution of methyl 1-(4-methylpiperazine-1-carbonyl)cyclopropanecarboxylate (1.5 g, 1.0 equiv.) in THF (40 mL) was added LiAlH₄ (503 mg, 2.0 equiv.) portion wise under N₂. The suspension was degassed and purged with N₂ for 3 times. The mixture was stirred at 0-15° C. for 2 hours. The mixture was quenched with water (500 μL), 15% NaOH aqueous (500 μL), water (1.5 mL), and filtered. The filter cake was washed with EtOAc (3×15 mL), and the filtrate was dried over anhydrous sodium sulfate, concentrated and purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 3:1 to 0:1) to afford the title compound (716 mg, 59% yield) as a colorless oil.

Intermediate 36

(366) ##STR00089##

4-(aminomethyl)imidazolidin-2-one

(367) ##STR00090##

(368) Step A. tert-butyl ((2-oxoimidazolidin-4-yl)methyl)carbamate: A mixture of tert-butyl N-[(2,5-dioxoimidazolidin-4-yl)methyl]carbamate (100 mg, 1.0 equiv.) and BH₃.sub.3.Math.Me.sub.2S (10.0 M, 2.0 equiv.) in THF (3 mL) at 0° C. was degassed and stirred at 60° C. for 1 hour under N₂ atmosphere. The reaction mixture was quenched by addition of MeOH (10 mL) at 0° C. Then it was extracted with ethyl acetate (3×20 mL). The combined organic layers were washed with brine (20 mL) and dried over Na₂SO₄, filtered and concentrated under reduced pressure to afford the title compound (60.0 mg, 64% yield) as a white solid; ¹H

NMR (400 MHz, DMSO-d.sub.6) δ =7.77 (br s, 1H), 6.97-6.78 (m, 1H), 6.37-5.98 (m, 1H), 4.02 (br dd, J=4.4, 6.4 Hz, 1H), 3.14-2.96 (m, 2H), 2.84-2.58 (m, 1H), 1.37 (s, 9H).

(369) Step B. 4-(aminomethyl)imidazolidin-2-one: To a solution of tert-butyl N-[(2-oxoimidazolidin-4-yl)methyl]carbamate (60.0 mg, 1.0 equiv.) in MeOH (2 mL) was added HCl (1 M, 5.0 equiv.). The mixture was stirred at 25° C. for 2 hours. The reaction mixture was concentrated under reduced pressure to afford the title compound (50.0 mg, crude) as a white solid. The crude product was used for the next step directly.

Intermediate 37

(370) ##STR00091##

4-(2-aminoethyl)imidazolidin-2-one

(371) ##STR00092##

(372) Step A. benzyl (2-(2,5-dioxoimidazolidin-4-yl)ethyl)carbamate: Ammonium carbonate (25.5 g, 11.0 equiv.) was added to benzyl N-(3-oxopropyl)carbamate (5.00 g, 1.00 equiv.) in methanol (40.0 mL) and water (36.0 mL) under a flow nitrogen gas. To the reaction mixture was added potassium cyanide (1.97 g, 1.25 equiv.) and the resulting was stirred at 20° C. for 12 hours. The suspension was filtered and the filter cake was dried under reduced to afford the title compound (3.10 g, 37.8% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =7.88 (s, 1H), 7.38-7.30 (m, 5H), 5.02 (s, 2H), 3.99 (dd, J=4.8, 8.0 Hz, 1H), 3.11 (d, J=6.0 Hz, 2H), 1.92-1.79 (m, 1H), 1.69-1.55 (m, 1H).

(373) Step B. benzyl (2-(2-oxoimidazolidin-4-yl) ethyl)carbamate: To a solution of benzyl N-[2-(2,5-dioxoimidazolidin-4-yl)ethyl]carbamate (1.00 g, 1.0 equiv.) in tetrahydrofuran (15.0 mL) was added borane dimethyl sulfide complex (10 M, 1.80 mL, 5.0 equiv.) at 0° C. The mixture was stirred at 50° C. for 1.5 hours. The mixture was quenched with methanol (10.0 mL). The mixture was concentrated in vacuum to produce a residue. The residue was purified by prep-HPLC (column: Welch Ultimate XB-CN 250×50×10 μ m; mobile phase: [Hexane-EtOH]; B %: 10%-35%, 12 min). The desired fraction was collected and lyophilized to give a residue. The residue was further re-purified by column chromatography on silica gel (ethyl acetate) to afford the title compound (150 mg, 13.9% yield) as a white solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =7.41-7.30 (m, 5H), 6.31 (s, 1H), 6.09 (s, 1H), 5.01 (s, 2H), 3.58 (q, J=6.8 Hz, 1H), 3.02 (s, 2H), 2.89 (s, 1H), 1.58-1.55 (m, 2H).

(374) Step C. 4-(2-aminoethyl)imidazolidin-2-one: To a solution of benzyl N-[2-(2-oxoimidazolidin-4-yl)ethyl]carbamate (80.0 mg, 1.00 equiv.) in tetrahydrofuran (5.00 mL) was added palladium/carbon (20 mg, 10% purity) under nitrogen atmosphere. The suspension was degassed and purged with hydrogen for 3 times. The mixture was stirred under hydrogen (15 psi) at 20° C. for 2 hours. The mixture was filtered and the filter cake was washed with methanol (10.0 mL). The filtrate was concentrated to afford the title compound (40 mg, crude) as a white solid.

Intermediate 38

(375) ##STR00093##

4-((methylamino)methyl)azetidin-2-one

(376) ##STR00094##

(377) Step A. 4-(benzyl(methyl)amino)methyl)azetidin-2-one: To a solution of 4-(iodomethyl)azetidin-2-one (100 mg, 1.0 equiv.) and N-methyl-1-phenyl-methanamine (86.2 mg, 1.5 equiv.) in acetonitrile (1.5 mL) was added potassium carbonate (196 mg, 3.0 equiv.). The mixture was stirred at 60° C. for 2 hours. After completion, the reaction mixture was cooled to 25° C. and filtered. Then the filtrate was partitioned between ethyl acetate (10 mL) and water (10 mL). The organic phase was separated, and then it was washed with saturated salt solution (2×5 mL), dried over sodium sulfate, filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate-1/1 to 0/1) to afford the title compound (42.0 mg, 43% yield) as a yellow liquid; LCMS [ESI, M+1]: m/z=205.0.

(378) Step B. 4-((methylamino)methyl)azetidin-2-one: A mixture of 4-[[benzyl(methyl)amino]methyl]azetidin-2-one (60.0 mg, 1.0 equiv.) and Pd/C (20.0 mg, 10% purity) in methyl alcohol (0.5 mL) was degassed and purged with hydrogen for 3 times, and then the mixture was stirred at 25° C. for 2 hours under hydrogen atmosphere (15 psi). The reaction mixture was concentrated under reduced pressure to afford the title compound (13.0 mg, 39% yield) as yellow liquid; The crude product was used for the next step directly.

(379) ##STR00095##

5-(aminomethyl)-1-methyl-1H-pyrazole-3-carboxamide

(380) ##STR00096##

(381) Step A. methyl 5-(((tert-butoxycarbonyl)amino)methyl)-1-methyl-1H-pyrazole-3-carboxylate: A mixture of methyl 5-bromo-1-methyl-1H-pyrazole-3-carboxylate (900 mg, 1 equiv.), Potassium [[(tert-Butoxycarbonyl)amino]methyl]trifluoroborate (1.46 g, 1.5 equiv.), Pd(OAc)₂ (73.8 mg, 0.08 equiv.) and XPhos (313 mg, 0.16 equiv.) and Cs₂CO₃ (4.02 g, 3 equiv.) in THF (50 mL) and water (5 mL) was degassed and stirred at 100° C. for 2 hours under N₂ atmosphere. The reaction mixture was diluted with water 20 mL and extracted with ethyl acetate (20 mL×3). The combined organic layers were washed with brine (20 mL×1), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue, which was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 5:1 to 3:1) to give the title compound (420 mg, 1.56 mmol, 37.96% yield) as a white solid; ¹H NMR (400 MHz, DMSO-d₆) δ=6.64 (s, 1H), 4.76 (s, 1H), 4.29 (d, J=8.0 Hz, 2H), 3.84 (d, J=8.0 Hz, 6H), 1.39 (s, 9H).

(382) Step B. tert-butyl ((3-carbamoyl-1-methyl-1H-pyrazol-5-yl)methyl)carbamate: A mixture of methyl 5-(((tert-butoxycarbonyl)amino)methyl)-1-methyl-1H-pyrazole-3-carboxylate (400 mg, 1 equiv.) in NH₃·H₂O (3.64 g, 4 mL, 25% NH₃, 17.48 equiv.) was stirred at 20° C. for 2 hours under N₂ atmosphere. The reaction mixture was diluted with water (20 mL) and extracted with ethyl acetate (20 mL×3). The combined organic layers were washed with brine (10 mL×1), dried over anhydrous sodium sulfate, filtered and concentrated to give the title compound (220 mg, 58% yield) as a white solid.

(383) Step C. 5-(aminomethyl)-1-methyl-1H-pyrazole-3-carboxamide: To a mixture of tert-butyl ((3-carbamoyl-1-methyl-1H-pyrazol-5-yl)methyl)carbamate (100 mg, 1 equiv.) in CH₂Cl₂ (0.5 mL) was added TFA (770 mg, 0.5 mL, 17.2 equiv.) at 0° C., the mixture was stirred at 25° C. for 12 hours. The reaction mixture was concentrated under reduced pressure to give the title compound (170 mg, crude, TFA) as a yellow oil. ¹H NMR (400 MHz, DMSO-d₆) δ=8.37 (s, 3H), 7.46 (s, 1H), 7.18 (m, 1H), 6.74 (s, 1H), 4.18 (d, J=5.2 Hz, 2H), 3.88 (s, 3H).

Intermediate 40

(384) ##STR00097##

5-ethynyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(385) ##STR00098##

(386) Step A. 5-ethynyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of 8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (1.50 g, 1 equiv.) in DMF (20 mL) was added CsF (2.53 g, 10 equiv.). The mixture was stirred at 20° C. for 2 hrs. The reaction mixture was concentrated under reduced pressure and purified by reversed phase HPLC (water (0.1% formic acid)-ACN) to afford the title compound (0.6 g, 57% yield) as a yellow solid; LCMS (ESI, M+1): m/z=589.3

Intermediate 41

(387) ##STR00099##

7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-

d]pyrimidine

(388) ##STR00100## ##STR00101##

(389) Step A. 5-chloro-6-fluoro-1,4-dihydro-1,4-epoxy)naphthalene: To a solution 1-bromo-3-chloro-2,4-difluorobenzene (250 g, 1 equiv.) and furan (150 g, 2 equiv.) in toluene (2.5 L) was added n-BuLi (2.5 M, 1.2 equiv.) dropwise over 0.5 hour at -15° C. The reaction mixture was stirred at 20° C. for 12 hours. After reaction completion, the mixture was quenched with water (2 L) and filtered. The filtrate was separated. The aqueous layer was extracted with ethyl acetate (2×2 L). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The residue was purified by reversed phase flash [C18, water (0.1% formic acid)-ACN, 0-80% MeCN] to afford the title compound (81 g, 37% yield) as a yellow oil; .sup.1H NMR (400 MHz, chloroform-d) δ 7.11-7.06 (m, 2H), 7.06-7.01 (m, 1H), 6.73 (dd, J=7.6, 9.6 Hz, 1H), 5.88 (s, 1H), 5.74 (s, 1H).

(390) Step B. 8-chloro-7-fluoronaphthalen-1-yl pivalate: A reaction mixture of 5-chloro-6-fluoro-1,4-dihydro-1,4-epoxy)naphthalene (162 g, 1 equiv.) in concentrated HCl (1.02 kg, 12.2 equiv.) and EtOH (1.2 L) was heated to 80° C. for 6 hours. The reaction mixture was concentrated in vacuum. The pH of the residue was adjusted to 7 with saturated aqueous NaHCO.sub.3 and extracted with ethyl acetate (2×2 L). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was triturated with petroleum ether (100 mL), filtered and the solid was dried under vacuum to afford the title compound (124 g, 76% yield) as a white solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.92 (s, 1H), 7.75 (dd, J=5.2, 8.8 Hz, 1H), 7.44-7.36 (m, 2H), 7.33-7.26 (m, 1H), 7.12-7.06 (m, 1H).

(391) Step C. 8-chloro-7-fluoronaphthalen-1-yl pivalate: A mixture of 8-chloro-7-fluoronaphthalen-1-ol (124 g, 1 equiv.) and DIEA (489 g, 6 equiv.), 4 Å molecular sieves (120 g) in dichloromethane (1.5 L) was stirred for 10 minutes at 20° C. Then PivCl (231 g, 1.3 equiv.) was added to the mixture dropwise at -40° C. The reaction mixture was stirred at -40° C. for 20 minutes. The reaction mixture was quenched with water (1 L) and the layers were separated. The aqueous layer was extracted with ethyl acetate (2×1 L). The combined organic layers were dried over Na.sub.2SO.sub.4 and concentrated under vacuum. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate 1:0 to 20:1) to afford the title compound (196 g, 92% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ'7.86 (d, J=8.0 Hz, 1H), 7.83-7.76 (m, 1H), 7.57 (d, J=8.0 Hz, 1H), 7.53-7.44 (m, 1H), 7.43-7.35 (m, 1H).

(392) Step D. 8-chloro-7-fluoro-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl pivalate: A mixture of 8-chloro-7-fluoronaphthalen-1-yl pivalate (8.00 g, 1.0 equiv.), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.24 g, 1.0 equiv.), (1,5-Cyclooctadiene) (methoxy)iridium(I) dimer (944 mg, 0.05 equiv.) and 4-tert-butyl-2-(4-tert-butyl-2-pyridyl)pyridine (918 mg, 0.12 equiv.) in n-hexane (220 mL) was degassed and stirred at 65° C. for 1 hour under N.sub.2 atmosphere. The mixture was filtered, concentrated and purified by column chromatography [SiO.sub.2, Petroleum ether/Ethyl acetate 10:1 to 3:1] to afford the title compound (18.8 g, crude) as a yellow solid.

(393) Step E. 8-chloro-7-fluoro-3-hydroxy)naphthalen-1-yl pivalate: To a solution of 8-chloro-7-fluoro-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl pivalate (50.0 g, 1.0 equiv.) and H.sub.2O.sub.2 (116 g, 98.3 mL, 30% purity, 8.3 equiv.) in THF (300 mL) was added AcOH (502 g, 478 mL, 68 equiv.) at 0° C. The solution was stirred at 20° C. for 2 hours. The reaction mixture was quenched by saturated sodium sulfite (500 mL), extracted with ethyl acetate (3×100 mL), washed with brine (200 mL), dried over Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography (C 18, water (0.1% NH.sub.3.Math.H.sub.2O)-ACN) to afford the title compound (5.4 g, 15% yield) as a gray solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=10.25 (s, 1H), 7.86 (dd, J=5.6, 9.2 Hz, 1H), 7.51 (t, J=9.2 Hz, 1H), 7.22 (d, J=2.4 Hz, 1H), 6.87 (d, J=2.1 Hz, 1H), 1.36 (s, 9H).

(394) Step F. 8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl pivalate: To a solution of 8-

chloro-7-fluoro-3-hydroxy)naphthalen-1-yl pivalate (5.4 g, 1.0 equiv.) in DCM (55 mL) were added DIPEA (7.06 g, 9.51 mL, 3 equiv.) and chloro(methoxy)methane (2.72 g, 2.57 mL, 1.8 equiv.) dropwise at 0° C. The mixture was stirred at 0° C. for 1.5 hours. The solution was diluted with water (50 mL), extracted with ethyl acetate (2×50 mL), dried over Na.sub.2SO.sub.4, concentrated and purified by column chromatography [SiO.sub.2, Petroleum ether/Ethyl acetate 20:1 to 5:1] to afford the title compound (6 g, 97% yield) as a yellow solid.

(395) Step G. 8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-ol: To a solution of 8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl pivalate (8.00 g, 1.0 equiv.) in MeOH (120 mL) was added KOH (5.38 g, 4.0 equiv.) at 0° C. The mixture was stirred at 20° C. for 1 hour. The solution was diluted with water (40 mL), extracted with ethyl acetate (3×40 mL), washed with brine (100 mL), dried over Na.sub.2SO.sub.4, concentrated and purified by column chromatography [SiO.sub.2, Petroleum ether/Ethyl acetate 10:1 to 5:1] to afford the title compound (5.6 g, 93% yield) as a yellow solid.

(396) Step H. 8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl trifluoromethanesulfonate: To a solution of 8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-ol (5.00 g, 1.0 equiv.) in DCM (60 mL) were added DIEA (2.52 g, 3.39 mL, 1.0 equiv.) and trifluoromethanesulfonic anhydride (8.24 g, 4.82 mL, 1.5 equiv.) dropwise at -40° C. The mixture was stirred at -40° C. for 1.5 hours. The solution was diluted with water (40 mL), extracted with ethyl acetate (3×40 mL), washed with brine (100 mL), dried over Na.sub.2SO.sub.4, concentrated and purified by column chromatography [SiO.sub.2, Petroleum ether/Ethyl acetate 50:1 to 10:1] to afford the title compound (6.9 g, 91% yield) as a yellow solid; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ =7.69 (dd, J=5.2, 9.2 Hz, 1H), 7.46 (d, J=2.0 Hz, 1H), 7.42-7.34 (m, 2H), 5.30 (s, 2H), 3.53 (s, 3H).

(397) Step I. (8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)trimethylstannane: To the mixture of [8-chloro-7-fluoro-3-(methoxymethoxy)-1-naphthyl]trifluoromethanesulfonate (2 g, 1 equiv.), trimethyl(trimethylstannyl)stannane (5.40 g, 3.2 equiv.), LiCl (654.37 mg, 3.0 equiv.) in toluene (80 mL) was added Pd(PPh.sub.3).sub.4 (595 mg, 0.1 equiv.) under N.sub.2. The mixture was stirred at 110° C. for 16 hours. The mixture was quenched with water (200 mL), extracted with ethyl acetate (100 mL×3), the combined organic phase was washed with brine (100 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by column chromatography (SiO.sub.2, Petroleum ether/Ethyl acetate 1:0 to 10:1) to afford the title compound (2 g, 96.35% yield) as a colorless oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =7.58-7.56 (m, 1H), 7.49-7.48 (m, 1H), 7.28-7.27 (d, J=2.4 Hz, 1H), 7.22-7.20 (m, 1H), 5.19 (s, 2H), 5.43 (s, 3H); 0.40-0.27 (m, 9H).

(398) Step J. 7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a to a solution of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (625 mg, 1 equiv.) and [8-chloro-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-trimethyl-stannane (1.72 g, 3 equiv.) in toluene (25 mL) was added Pd(dppf)Cl.sub.2 (104, 0.1 equiv.), BINAP (177 mg, 285 μ mol, 0.2 equiv.) and CuI (81.4 mg, 0.3 equiv.), the mixture was de-gassed and heated to 90° C. for 4 hours under N.sub.2. The resulting suspension was cooled to 20° C., filtered, the filtrate was concentrated under reduced pressure. The crude product was purified by reversed-phase HPLC (water (0.1% formic acid) to afford the title compound (776 mg, 1.21 mmol, 84.73% yield) as a yellow solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =9.25 (s, 1H), 7.79-7.75 (dd, J=5.6 Hz, 9.2 Hz, 1H), 7.57-7.56 (d, J=1.6 Hz, 1H), 7.38-7.27 (m, 2H), 5.34-5.29 (m, 3H), 5.09-5.05 (m, 2H), 4.38-4.12 (m, 2H); 3.53 (s, 3H); 3.29-3.28 (m, 2H); 3.19-3.18 (m, 1H); 3.05-3.01 (m, 1H); 2.23-2.22 (m, 31H); 2.00-1.95 (m, 3H); LCMS (ESI, M-55): m/z=642.8.

Intermediate 42

(399) ##STR00102##

7-(8-bromo-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (400) ##STR00103## ##STR00104##

(401) Step A. 5-(2-(3-bromo-4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione To a mixture of 2-(3-bromo-4-fluoro-phenyl)acetic acid (330 g, 210 mL, 1.0 equiv.) and 2,2-dimethyl-1,3-dioxane-4,6-dione (225 g, 1.1 equiv.) in MeCN (1.65 L) was added DMAP (14.7 g, 0.085 equiv.) in one portion at 15° C. under N.sub.2 atm. Then DIPEA (394 g, 530 mL, 2.15 equiv.) was carefully added to the reaction in four portions over a period of 1 hour while maintaining the temperature at 15~30° C. under to give a pale yellow suspension. Then pivaloyl chloride (188 g, 192 mL, 1.10 equiv.) was added slowly to the reaction mixture in five portions over a period of 1 hour maintaining 25° C.~40° C. After the addition was complete, a yellow suspension was obtained. The reaction mixture was heated at 45° C. for 3 hours under N.sub.2 atm. The mixture was cooled to 0° C. and slowly diluted with 4 N HCl (2.0 L) to adjust pH to 6-7 while maintaining the temperature between 0-15° C. The mixture was stirred at 0° C. for 1 hour and then filtered. The filter cake was concentrated in vacuum to afford the title compound (450 g, 88% yield) as a white solid; LCMS (ESI, M-57, M-55): m/z=300.8.

(402) Step B. 8-bromo-7-fluoro-1,3-dihydroxy-2-naphthoic acid: 5-[2-(3-bromo-4-fluorophenyl)acetyl]-2,2-dimethyl-1,3-dioxane-4,6-dione (600 g, 1.0 equiv.) was slowly added to CF.sub.3SO.sub.3H (1.30 L) in four portions maintaining the temperature at 25° C.~50° C. for 1 hour with ice-water cooled bath. Then the mixture was stirred at 20° C. for 2 hours and water (10 L) was added slowly to the reaction mixture. The mixture was filtered. The filter cake was collected and concentrated to afford the title compound (1200 g, crude) as a yellow solid.

(403) Step C. 8-bromo-7-fluoronaphthalene-1,3-diol: A solution of 8-bromo-7-fluoro-1,3-dihydroxy-naphthalene-2-carboxylic acid (1.3 kg, 1.0 equiv.) in water (700 mL) and ACN (700 mL) was stirred at 85° C. for 12 hours. The mixture was concentrated and the residue was diluted with water (1 L), extracted with ethyl acetate (2 L×2). The organic layer was dried over Na.sub.2SO.sub.4, concentrated in vacuum and the residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 3:1) and prep-HPLC (column: Phenomenex Luna C18 200×40 mm×10 um; mobile phase: [water (0.1% formic acid)-ACN]; B %: 27%-57%, 10 min). The desired fraction was collected and extracted with ethyl acetate (20 mL). The organic layer was dried over Na.sub.2SO.sub.4 and concentrated in vacuum to afford the title compound (55.0 g, 13% yield two steps) as a yellow solid.

(404) Step D. 8-bromo-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-ol: To a solution of 8-bromo-7-fluoro-naphthalene-1,3-diol (10.0 g, 1.0 equiv.) and DIEA (15.1 g, 20.3 mL, 3 equiv.) in DCM (200 mL) was added TIPSCl (6.75 g, 7.49 mL, 0.90 equiv.) portion wise at 0° C. The mixture was stirred at 0° C. for 0.5 hr. The mixture was concentrated to give a residue and the residue was purified by column chromatography (Silica gel, ethyl acetate/petroleum ether 0:1 to 1:50) to afford the title compound (14.7 g, 91% yield) as a yellow oil.

(405) Step E. 8-bromo-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl trifluoromethanesulfonate: To a solution of 8-bromo-7-fluoro-3-triisopropylsilyloxy-naphthalen-1-ol (14.7 g, 1.0 equiv.) and DIEA (13.8 g, 18.6 mL, 3.0 equiv.) in DCM (200 mL) was added Tf.sub.2O (12.0 g, 7.04 mL, 1.2 equiv.) at -40° C. The mixture was stirred at -40° C. for 0.5 hr. The mixture was quenched with water (200 mL) and the layers were separated. The organic layer was dried over Na.sub.2SO.sub.4 and concentrated. The residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 100:1 to 30:1) to afford the title compound (15 g, 77% yield) as a yellow oil; LCMS (ESI, M+1): m/z=547.0.

(406) Step F. ((5-bromo-6-fluoro-4-(trimethylstannyl)naphthalen-2-yl)oxy)triisopropylsilane: To a mixture of (8-bromo-7-fluoro-3-triisopropylsilyloxy-1-naphthyl) trifluoromethanesulfonate (5 g, 9.17 mmol, 1 equiv.), trimethyl(trimethylstannyl)stannane (9.41 g, 5.96 mL, 3.13 equiv.) and Pd(PPh.sub.3).sub.2Cl.sub.2 (643 mg, 0.1 equiv.) in toluene (50 mL) was added LiCl (1.17 g, 563

μL, 3.0 equiv.) under N.sub.2. The mixture was stirred at 100° C. for 12 hours under N.sub.2. The mixture was quenched with water (100 mL) and extracted with ethyl acetate (100 mL×3), the combined organic phase was washed with brine 100 mL, dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by column chromatography (Silica gel, petroleum ether) to afford the title compound (3 g, 58% yield) as a yellow oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.65 (dd, J=6.0, 9.2 Hz, 1H), 7.54 (d, J=2.4 Hz, 1H), 7.28-7.23 (m, 1H), 7.20 (d, J=2.4 Hz, 1H), 1.36-1.29 (m, 3H), 1.15 (d, J=7.2 Hz, 18H), 0.56-0.39 (m, 9H).

(407) Step G. 7-(8-bromo-7-fluoro-3-(((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (650 mg, 1.0 equiv.) and ((5-bromo-6-fluoro-4-(trimethylstannyl)naphthalen-2-yl)oxy)triisopropylsilane (1.24 g, 1.5 equiv.) in toluene (10 mL) was added BINAP (184 mg, 0.2 equiv.) and CuI (84.6 mg, 0.3 equiv.) at 25° C. The suspension was thoroughly degassed and Pd(dppf)Cl.sub.2 (108 mg, 0.1 equiv.) was added. The suspension was degassed and stirred at 100° C. and for 6 hours. The mixture was filtered and the residue was washed by ethyl acetate (50 mL×3). The filtrate was concentrated in vacuum and the residue was purified by prep-HPLC (column: Welch Xtimate C18 250*50 mm*10 μm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 27%-57%, 20 min) to afford the title compound (1.4 g, 36% yield) as a yellow solid; LCMS (ESI, M+1, M+3): m/z=799.2, 801.2.

Intermediate 43

(408) ##STR00105##

7-(8-chloro-7-fluoro-3-(((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(409) The title compounds was synthesized from 2-(3-chloro-4-fluoro-phenyl)acetic acid according to the procedure described for Intermediate 42. LCMS [ESI, M+1]: m/z=755.1.

Intermediate 44

(410) ##STR00106##

8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine

(411) ##STR00107##

(412) Step A. (8-bromonaphthalen-1-yl)methanol: To a solution of 8-bromonaphthalene-1-carboxylic acid (5 g, 1.0 equiv.) in 2-MeTHF (70 mL) was slowly added BH.sub.3.Math.Me.sub.2S (10 M, 3.0 equiv.) at 0° C. under N.sub.2. The reaction was stirred at 70° C. for 12 hours. The mixture was quenched with methanol (20 mL) at 0° C. and washed with 1 M HCl (30 mL). The mixture was diluted with water (50 mL) and extracted with ethyl acetate (2×50 mL). The combined organic layers were washed with brine (60 mL), dried over anhydrous sodium sulfate, concentrated to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 1:0 to 0:1 to methanol) to afford the title compound (5.4 g, 70% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 7.98 (dd, J=0.8, 8.0 Hz, 1H) 7.84-7.95 (m, 3H) 7.56 (t, J=7.6 Hz, 1H) 7.35 (t, J=7.6 Hz, 1H) 5.39-5.48 (m, 2H) 5.32-5.39 (m, 1H).

(413) Step B. 2-((8-bromonaphthalen-1-yl)methoxy)tetrahydro-2H-pyran: To a solution of 3,4-dihydropyran (1.22 g, 1.0 equiv.), 8-bromonaphthalen-1-yl)methanol (3.44 g, 1.0 equiv.) and 4-methylbenzenesulfonic acid (514 mg, 0.1 equiv.) in DCM (40 mL) was stirred at 25° C. for 5 hours. The reaction mixture was concentrated on vacuum. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 1:0 to 50:1) to afford the title compound (4.8 g, 43% yield) as a light yellow oil. .sup.1H NMR (400 MHz, chloroform-d) δ=7.76-7.94 (m, 4H) 7.48 (t, J=7.70 Hz, 1H) 7.27 (s, 1H) 5.58 (s, 2H) 4.88 (t, J=3.36 Hz, 1H) 3.93-4.02 (m, 1H) 3.56-3.64 (m, 1H) 1.73-1.98 (m, 3H) 1.64-1.69 (m, 1H) 1.61-1.56 (m, 2H)

(414) Step C. 4,4,5,5-tetramethyl-2-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-1,3,2-dioxaborolane: A mixture of 2-((8-bromonaphthalen-1-yl)methoxy)tetrahydro-2H-pyran

(300 mg, 1.0 equiv., 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (356 mg, 1.5 equiv.), KOAc (27.5 mg, 0.3 equiv.) and Pd(dppf)Cl.sub.2 (68.3 mg, 0.1 equiv.) in dioxane (1.5 mL) was degassed and stirred at 80° C. for 2 hrs under N.sub.2 atmosphere. The combined reaction mixture was diluted with ethyl acetate (10 mL) and water (20 mL), the aqueous layer was extracted with ethyl acetate (10 mL), the combined organic phase was washed with brine (15 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 100:1 to 15:1) to afford the title compound (290 mg, 85% yield) as a white solid; .sup.1H NMR (400 MHz, CDCl3) δ (ppm)=7.93-7.88 (m, 1H), 7.82-7.74 (m, 2H), 7.56 (d, J=7.2 Hz, 1H), 7.48-7.39 (m, 2H), 5.74 (d, J=13.2 Hz, 1H), 5.04-4.94 (m, 1H), 4.41-4.35 (m, 1H), 3.91-3.80 (m, 1H), 3.49-3.41 (m, 1H), 1.68-1.40 (m, 18H).

(415) Step D. 8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (350 mg, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-1,3,2-dioxaborolane (368 mg, 1.2 equiv.), CataCXium A Pd G3 (60.6 mg, 0.1 equiv.) and Cs.sub.2CO.sub.3 (1.5 M in water, 3.0 equiv.) in methoxycyclopentane (8 mL) was degassed and stirred at 100° C. for 2 hours under N.sub.2 atmosphere. The reaction mixture was diluted with ethyl acetate (50 mL) and water (60 mL), extracted with ethyl acetate (30 mL), the combined organic phase was washed with brine (40 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (234 g, 16% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ ppm=9.22 (s, 1H) 8.02 (d, J=7.95 Hz, 1H) 7.93 (d, J=7.95 Hz, 1H) 7.45-7.62 (m, 4H) 5.00-5.18 (m, 2H) 4.24-4.61 (m, 4H) 4.09-4.20 (m, 2H) 3.13-3.41 (m, 3H) 2.62-2.85 (m, 2H) 2.07-2.21 (m, 2H) 1.85-2.02 (m, 5H) 1.67-1.82 (m, 4H) 1.19-1.33 (m, 3H); LCMS (ESI, M+1): m/z=627.2.

Intermediate 45

(416) ##STR00108##

5,6-difluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(417) ##STR00109##

(418) Step A. 5,6-difluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol: To a solution of 2-[7,8-difluoro-3-(methoxymethoxy)-1-naphthyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (500 mg, 1.0 equiv.) in ACN (15 mL) was added HCl.dioxane (4 M, 7.50 mL, 21.0 equiv.) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The mixture was diluted with saturated aqueous NaHCO.sub.3 (15 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (340 mg, 74% yield) as a red solid; LCMS (ESI, M+1): m/z=307.4.

(419) Step B. 5,6-difluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (306 mg, 1.0 equiv.) and 5,6-difluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (320 mg, 1.50 equiv.) in water (1.4 mL) and methoxycyclopentane (4.0 mL) were added cataCXium A Pd G3 (50.7 mg, 0.1 equiv.) and Cs.sub.2CO.sub.3 (1.5 M, 1.39 mL, 3.0 equiv.) under N.sub.2. The mixture was stirred at 60° C. for 1 hour. The reaction mixture was diluted with water (5.0 mL), the aqueous phase was extracted with ethyl acetate (3×10 mL). The combined organic phase was washed with brine (15 mL), dried with anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography (C18, water (0.1% formic acid)-ACN) to afford the title compound (270 mg, 60%

yield) as a yellow solid; LCMS (ESI, M+1): m/z=583.2

Intermediate 46

(420) ##STR00110##

(R)-1-(7-chloro-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(421) ##STR00111##

(422) Step A. (R)-1-(7-chloro-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.20 g, 1.0 equiv.) and 4 Å molecular sieves (500 mg) in dioxane (15 mL) was added DIEA (2.58 g, 3.0 equiv.) and [(2S)-1-methylpyrrolidin-2-yl]methanol (1.53 g, 2.0 equiv.). The mixture was stirred at 95° C. for 20 hours. The reaction mixture was filtered and concentrated under reduced pressure to remove dioxane. The residue was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (1.80 g, 66% yield) as a yellow solid; LCMS (ESI, M+1): m/z=410.2.

Intermediate 47

(423) ##STR00112##

N,N-bis(4-methoxybenzyl)-4-methyl-6-(tributylstannyl)pyridin-2-amine

(424) ##STR00113##

(425) Step A. 6-bromo-N,N-bis(4-methoxybenzyl)-4-methylpyridin-2-amine: To a solution of 6-bromo-4-methylpyridin-2-amine (180 g, 1.0 equiv.) in DMAC (1.8 L) was added NaH (115 g, 60% purity, 3.0 equiv.) at 0° C. under nitrogen atmosphere. The mixture was stirred at 25° C. for 1 hour under nitrogen atmosphere. Then 1-(chloromethyl)-4-methoxybenzene (331 g, 2.2 equiv.) was added to the reaction mixture at 25° C. The mixture was stirred at 25° C. for 2 hours. After reaction completion, the mixture was quenched with ammonium chloride solution (3 L) at 0° C. and diluted with ethyl acetate (4 L). The layers were separated. The organic layer was washed with brine (3×2 L), dried with Na.sub.2SO.sub.4 then concentrated in vacuum. The residue was triturated with petroleum ether (1 L) at 25° C. for 30 mins to afford the title compound (320 g, 77% yield) as a white solid; ¹H NMR (400 MHz, CDCl₃-d) δ=7.16 (d, J=8.8 Hz, 4H), 6.85 (d, J=8.8 Hz, 4H), 6.60 (s, 1H), 6.17 (s, 1H), 4.64 (s, 4H), 3.80 (s, 6H), 2.13 (s, 3H); LCMS (ESI, M+1, M+3): m/z=427.1, 429.1.

(426) Step B. N,N-bis(4-methoxybenzyl)-4-methyl-6-(tributylstannyl)pyridin-2-amine: A mixture of 6-bromo-N,N-bis[(4-methoxyphenyl)methyl]-4-methylpyridin-2-amine (200 g, 1.0 equiv.), tributyl(tributylstannyl)stannane (651 g, 2.4 equiv.), Pd.sub.2(dba).sub.3 (42.9 g, 0.1 equiv.), PCy.sub.3 (26.2 g, 0.2 equiv.) and LiCl (99.2 g, 5.0 equiv.) in dioxane (1.8 L) was degassed and stirred at 110° C. for 5 hours under N.sub.2 atmosphere. After reaction completion, the mixture was concentrated in vacuum. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 1:0 to 10:1) followed by reversed phase flash chromatography (neutral condition) to afford the title compound (190 g, 62% yield) as a yellow oil. ¹H NMR (400 MHz, CDCl₃-d) δ=7.19 (d, J=8.8 Hz, 4H), 6.87-6.82 (m, 4H), 6.62-6.55 (m, 1H), 6.15 (s, 1H), 4.70 (s, 4H), 3.80 (s, 6H), 2.15 (s, 3H), 1.63-1.51 (m, 6H), 1.32 (qd, J=7.2, 14.7 Hz, 7H), 1.09-1.01 (m, 5H), 0.90-0.84 (m, 9H). LCMS (ESI, M+1): m/z=639.3.

Intermediate 48

(427) ##STR00114##

1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(428) ##STR00115##

(429) Step D. 1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(2,7-dichloro-8-

fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.22 g, 1.0 equiv.), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (1.60 g, 1.5 equiv.), 4 Å molecular sieves (1.00 g) and DIEA (2.60 g, 3.0 equiv.) in dioxane (30.0 mL) was stirred at 90° C. for 15 hours. The mixture was diluted with water (20 mL) and extracted with ethyl acetate (2×20 mL). The combined organic layers were washed with brine (30 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The crude product was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (1.90 g, 62% yield) as a yellow solid; LCMS (ESI, M+1): m/z=454.1.

Intermediate 49

(430) ##STR00116##

4,4,5,5-tetramethyl-2-(8-(methylthio)naphthalen-1-yl)-1,3,2-dioxaborolane

(431) ##STR00117##

(432) Step A. 4,4,5,5-tetramethyl-2-(8-(methylthio)naphthalen-1-yl)-1,3,2-dioxaborolane: To a solution of 1-bromo-8-methylsulfanyl-naphthalene (1.00 g, 1.0 equiv.) and 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (2.53 g, 5.0 equiv.) in ACN (20 mL) were added TEA (1.20 g, 3.0 equiv.) and Pd(dppf)Cl.sub.2 (433 mg, 0.15 equiv.). The mixture was stirred at 80° C. for 5 hours under N.sub.2 atmosphere. The mixture was filtered, the filtrate was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by column chromatography [Silica gel, Petroleum ether/Ethyl acetate 1:0 to 30:1] to afford the title compound (1.00 g, 84% yield, 99% purity) as a yellow oil; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ=7.85 (dd, J=1.2, 8.4 Hz, 1H), 7.81 (d, J=7.6 Hz, 2H), 7.64 (dd, J=1.2, 6.8 Hz, 1H), 7.52-7.48 (m, 1H), 7.45-7.41 (t, J=8.0 Hz, 1H), 2.42 (s, 3H), 1.46 (s, 12H)

Intermediate 50

(433) ##STR00118##

((6-((benzyloxy)methoxy)-7-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane

(434) ##STR00119##

(435) Step A. 1-bromo-2-chloro-4-fluoro-3,5-dimethoxybenzene: To a solution of 5-bromo-2-fluoro-1,3-dimethoxybenzene (50.0 g, 1.0 equiv.) and TMSCl (2.31 g, 0.1 equiv.) in MeCN (500 mL) was added NCS (34.1 g, 1.2 equiv.) at 10° C. The reaction mixture was stirred at 10° C. for 2 hours. The mixture was quenched with saturated brine (500 mL) at 0° C. and extracted with ethyl acetate (3×100 mL). The combined organic layers were dried over anhydrous sodium sulfate and filtered. The filtrate was concentrated under reduced pressure. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (85 g, 74% yield) as a yellow solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.05-6.95 (m, 1H), 3.98 (d, J=1.2 Hz, 3H), 3.89 (s, 3H).

(436) Step B. 6-fluoro-5,7-dimethoxy-9-methyl-1,4-dihydro-1,4-epiminonaphthalene: To a mixture of 1-bromo-2-chloro-4-fluoro-3,5-dimethoxybenzene (20.0 g, 1.0 equiv.) and 1-methylpyrrole (12.0 g, 2.0 equiv.) in THF (240 mL) was added n-BuLi (2.5 M, 32.65 mL, 1.1 equiv.) at -65° C. The reaction was stirred at -65° C. for 1 hour and then at 25° C. for 16 hours. The mixture was quenched with saturated aqueous NH.sub.4Cl solution (200 mL) and extracted with EtOAc (2×80 mL). The combined organic layers were concentrated under reduced pressure. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (7.8 g, 44% yield) as a yellow oil. LCMS (ESI, M+1): m/z=236.

(437) Step C. 2-fluoro-1,3-dimethoxynaphthalene: To a solution of 6-fluoro-5,7-dimethoxy-9-methyl-1,4-dihydro-1,4-epiminonaphthalene (5.00 g, 1.0 equiv.) in DCM (60.0 mL) was added m-CPBA (5.18 g, 85% purity, 1.2 equiv.) at 0-5° C. The mixture was stirred at 15° C. for 1 hr. The mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (100 mL). The organic layer was separated and concentrated under reduced pressure. The residue was purified by column

chromatography (SiO.sub.2, petroleum ether/ethyl acetate 1:0 to 20:1) to afford the title compound (2.4 g, 54% yield) as a light yellow oil.

(438) Step D. 2-fluoronaphthalene-1,3-diol: To a solution of 2-fluoro-1,3-dimethoxy)naphthalene (15.4 g, 1.0 equiv.) in DCM (250 mL) was added BBr.sub.3 (39.3 g, 2.1 equiv.) at -30° C. The mixture was stirred at -30° C. for 30 minutes. The mixture was quenched with MeOH (2.0 mL) and concentrated under reduced pressure. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 50:1 to 10:1) to afford the title compound (7.6 g, 57% yield) as a yellow solid. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 8.09 (d, J=8.4 Hz, 1H), 7.65 (d, J=8.0 Hz, 1H), 7.47-7.35 (m, 2H), 6.96 (d, J=8.4 Hz, 1H), 5.50 (br s, 1H), 5.21 (br s, 1H); LCMS (ESI, M+1): m/z=179.

(439) Step E. 2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol: To a solution of 2-fluoronaphthalene-1,3-diol (7.6 g, 1.0 equiv.) and 2-bromoethynyl(triisopropyl)silane (13.4 g, 1.2 equiv.) in 1,4-dioxane (150 mL) were added dichloro(p-cymene)ruthenium(II) dimer (2.61 g, 0.1 equiv.) and AcOK (8.37 g, 2.0 equiv.). The reaction was stirred at 110° C. for 3 hours under N.sub.2 atmosphere. The mixture was poured into H.sub.2O (200 mL) and extracted with EtOAc (3 $\times$ 60 mL). The combined organic layers were concentrated under reduced pressure. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (3.5 g, 22% yield) as a gray solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.65 (dd, J=0.8, 8.4 Hz, 1H), 7.52 (d, J=7.2 Hz, 1H), 7.27 (s, 2H), 6.94 (d, J=8.0 Hz, 1H), 1.24-1.17 (m, 21H); LCMS (ESI, M+1): m/z=359.

(440) Step F. 3-((benzyloxymethoxy)-2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-ol: To a solution of 2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalene-1,3-diol (2.8 g, 1.0 equiv.) and DIEA (1.51 g, 1.5 equiv.) in DCM (24.0 mL) was added ((chloromethoxy)methyl)benzene (1.22 g, 1.0 equiv.) at -40° C. The reaction was stirred at 0-10° C. for 5 hours. The mixture was concentrated under reduced pressure. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1 to 20:1) to afford the title compound (3.2 g, 77% yield) as a yellow solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 9.19 (s, 1H), 7.68 (dd, J=0.8, 8.4 Hz, 1H), 7.55 (d, J=7.2 Hz, 1H), 7.38-7.30 (m, 6H), 7.17 (d, J=7.6 Hz, 1H), 5.47 (s, 2H), 4.80 (s, 2H), 1.24-1.12 (m, 21H); LCMS (ESI, M+1): m/z=479.

(441) Step G. 3-((benzyloxy)methoxy)-2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate: To a solution of 3-((benzyloxy)methoxy)-2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-ol (550 mg, 1.0 equiv.) and DIEA (297 mg, 2.0 equiv.) in DCM (3.0 mL) was added Tf.sub.2O (486 mg, 1.5 equiv.) at -40° C. The reaction was stirred at -40° C. for 0.5 hour. The mixture was quenched with water (10 mL). The organic layer was separated and dried over anhydrous Na.sub.2SO.sub.4, and then filtered and concentrated under reduced pressure. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1 to 20:1) to afford the title compound (650 mg, 92% yield) as a yellow solid; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.71 (d, J=7.2 Hz, 1H), 7.62 (d, J=8.0 Hz, 1H), 7.54 (d, J=7.6 Hz, 1H), 7.37-7.33 (m, 1H), 7.30-7.23 (m, 5H), 5.44-5.38 (m, 2H), 4.76-4.68 (m, 2H), 1.19-1.13 (m, 3H), 1.12-1.03 (m, 18H).

(442) Step H. ((6-((benzyloxy)methoxy)-7-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane: To a solution of 3-((benzyloxy)methoxy)-2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl trifluoromethanesulfonate (50.0 mg, 1.0 equiv.) and TEA (24.8 mg, 3.0 equiv.) in 1,4-dioxane (1.0 mL) were added Pd(dppf)Cl.sub.2 (5.99 mg, 0.1 equiv.) and 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (31.4 mg, 3.0 equiv.). The reaction mixture was stirred under N.sub.2 at 100° C. for 2 hours. The mixture poured into water (10 mL) and extracted with EtOAc (3 $\times$ 5 mL). The combined organic layers were concentrated under reduced pressure. The residue was purified by column chromatography (SiO.sub.2, petroleum ether/ethyl acetate 100:1 to 50:1) to afford the title compound (32 mg, 66% yield) as a yellow oil; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ 7.66 (d, J=7.1 Hz, 1H), 7.63-7.55 (m, 2H), 7.51-7.47 (m,

1H), 7.29-7.27 (m, 5H), 5.40-5.37 (m, 2H), 4.73-4.69 (m, 2H), 1.42 (s, 12H), 1.11-1.06 (m, 21H).

Intermediate 52

(443) ##STR00120##

N-isopropyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2 (4H)-carboxamide

(444) ##STR00121##

(445) Step A. tert-butyl 2-(isopropylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[4,3-c]azepine-5-carboxylate: To a solution of tert-butyl 4,6,7,8-tetrahydro-2H-pyrazolo[4,3-c]azepine-5-carboxylate (4.0 g, 1.0 equiv.) in THF (20 mL) was added portion wise CDI (2.73 g, 1 equiv.) and isopropylamine (1.1 g, 1.1 equiv.) at 25° C. The reaction was stirred at 25° C. for 1 hour. The mixture was concentrated and purified by column chromatography (SiO₂, petroleum ether/ethyl acetate 3:1 to 0:1) to afford the title compound (4.5 g, 82% yield) as a white solid; LCMS (ESI, M+1): m/z=323.2.

(446) Step B. N-isopropyl-5,6,7,8-tetrahydro-4H-pyrazolo[4,3-c]azepine-2-carboxamide: A solution of tert-butyl 2-(isopropylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[4,3-c]azepine-5-carboxylate (6.0 g, 1 equiv.) in HCl-MeOH (30 mL) was stirred at 25° C. for 0.5 hour. The mixture was concentrated to afford the title compound (3.2 g, 77% yield) as a white solid; LCMS (ESI, M+1): m/z=223.2.

Intermediate 53

(447) ##STR00122##

3a-methyltetrahydropyrrolo[3,4-c]pyrrole-1,3(2H, 3aH)-dione

(448) ##STR00123##

(449) Step A. 5-benzyl-3a-methyltetrahydropyrrolo[3,4-c]pyrrole-1,3(2H, 3aH)-dione: To a mixture of 3-methylpyrrole-2,5-dione (608 mg, 1.0 equiv.) and N-(methoxymethyl)-1-phenyl-N-(trimethylsilylmethyl)methanamine (1.3 g, 1.0 equiv.) in CH₂Cl₂ (10 mL) was added a solution of TFA (62.4 mg, 40.5 uL, 0.1 equiv.) in CH₂Cl₂ (1.0 mL). The mixture was stirred at 25° C. for 1 hour. The reaction mixture was diluted with sat. aq. NaHCO₃ (40 mL) and extracted with CH₂Cl₂ (3×50 mL). The combined organic layers were dried, filtered and concentrated to give a residue. The residue was purified by flash silica gel chromatography (Ethyl acetate/Petroleum ether 2:1) to afford the title compound (0.8 g, 60% yield) as a colorless oil; ¹H NMR (400 MHz, DMSO-d₆) δ=11.19 (s, 1H), 7.33-7.27 (m, 2H), 7.26-7.20 (m, 3H), 3.54-3.49 (m, 2H), 3.12-3.00 (m, 2H), 2.82 (d, J=7.2 Hz, 1H), 2.40 (dd, J=8.0, 9.6 Hz, 1H), 1.96-1.91 (m, 1H), 1.24 (s, 3H); LCMS (ESI, M+1): m/z=245.2.

(450) Step B. tert-butyl N-[(5-carbamoyl-2-methyl-pyrazol-3-yl)methyl]carbamate: To a mixture of 5-benzyl-3a-methyltetrahydropyrrolo[3,4-c]pyrrole-1,3 (2H, 3aH)-dione (0.8 g, 1.0 equiv.) in THF (30 mL) was added Pd/C (300 mg, 10% purity), the mixture was degassed and purged with H₂ and stirred at 40° C. for 12 hours under H₂ atmosphere (15 psi). The reaction mixture was filtered and the filtrate was concentrated under the reduced pressure to afford the title compound (400 mg, 79% yield) as a colorless oil; LCMS (ESI, M+1): m/z=155.2.

Intermediate 54

(451) ##STR00124##

(2R,6R)-2-(hydroxymethyl)-6-methyl-1,4-oxazepan-6-ol

(452) ##STR00125##

(453) Step A. (R)-tert-butyl (2,3-dihydroxypropyl)carbamate: To a solution of (R)-3-aminopropane-1,2-diol (50.0 g, 1 equiv.) and Et₃N (84.3 g, 1.52 equiv.) in CH₃OH (1.5 L) was added Boc₂O (182 g, 1.52 equiv.) portion wise at 0° C. The reaction was stirred at 25° C. for 12 hours. The mixture was poured into H₂O (2.0 L) and extracted with ethyl acetate (1.0 L). The organic phase was washed with brine (500 mL), dried over Na₂SO₄, filtered and concentrated to dryness. The residue was triturated with petroleum ether (150 mL) at 25° C. for 1 hour and filtered. The filter cake was washed with petroleum ether (3×50 mL) and dried to afford the title compound (48.0 g, 45% yield) as a colorless oil; ¹H NMR (400 MHz, DMSO-

d.sub.6) δ =6.57 (t, J=4.8 Hz, 1H), 4.62 (d, J=4.8 Hz, 1H), 4.47 (t, J=6.0 Hz, 1H), 3.46-3.40 (m, 1H.sub.1), 3.31-3.25 (m, 2H), 3.06-2.99 (m, 1H), 2.87-2.80 (m, 2H), 1.37 (s, 9H).

(454) Step B. (R)-tert-butyl (3-(((tert-butyldiphenylsilyl)oxy)-2-hydroxypropyl)carbamate: To a solution of (R)-tert-butyl (2,3-dihydroxypropyl)carbamate (43.0 g, 1 equiv.) in CH.sub.2Cl.sub.2 (900 mL) were added imidazole (18.37 g, 1.2 equiv.) and TBDPSCl (67.9 g, 1.1 equiv.). The reaction mixture was stirred at 25° C. for 12 hours. The mixture was partitioned between CH.sub.2Cl.sub.2 (500 mL) and H.sub.2O (1.0 L). The organic phase was separated and washed with brine 1.0 L (2×500 mL). The organic layer was dried over Na.sub.2SO.sub.4 and concentrated. The residue was purified by flash silica gel chromatography (ethyl acetate in petroleum ether 0-20%) to afford the title compound (42.3 g, 37.9% yield) as a colorless oil; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =7.67-7.63 (m, 4H), 7.46-7.40 (m, 6H), 6.60 (t, J=5.2 Hz, 1H), 4.81 (s, 1H), 3.63 (s, 1H), 3.57-3.50 (m, 2H), 3.20-3.13 (m, 1H), 2.97-2.91 (m, 1H), 1.36 (s, 9H), 0.99 (s, 9H); LCMS (ESI, M+1): m/z=430.2.

(455) Step C. (R)-tert-butyl 2-(((tert-butyldiphenylsilyl)oxy)methyl)-6-methylene-1,4-oxazepane-4-carboxylate: To a solution of tert-butyl N-[(2R)-3-[tert-butyl(diphenyl)silyl]oxy-2-hydroxypropyl]carbamate (20 g, 1 equiv.) in THF (400 mL) was added NaH (4.10 g, 60% purity, 2.2 equiv.) at 0° C. under N.sub.2 atmosphere followed by 3-chloro-2-(chloromethyl)prop-1-ene (5.82 g, 1 equiv.). The reaction was stirred at 0° C. for 2 hours under N.sub.2 atmosphere. The mixture was quenched by addition of H.sub.2O (600 mL) at 0° C. and extracted with ethyl acetate (2×300 mL), the combined organic layers were washed with brine (20 mL) and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated. The residue was purified by reversed-phase HPLC (Column: I.D.100 mm×H350 mm Welch Ultimate XB_C18 20-40 μ m; mobile phase: water (0.1% formic acid)-ACN; B % 30-100% 50 min) to afford the title compound (1.8 g, 7.64% yield) as a colorless oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =7.68-7.66 (m, 4H), 7.42-7.36 (m, 6H), 5.03-4.90 (m, 2H), 4.66-4.22 (m, 3H), 4.04-3.98 (m, 1H), 3.78-3.49 (m, 4H), 2.85-2.77 (m, 1H), 1.47 (s, 9H), 1.07 (s, 9H); LCMS (ESI, M+1): m/z=482.2.

(456) Step D. (R)-tert-butyl 2-(((tert-butyldiphenylsilyl)oxy)methyl)-6-oxo-1,4-oxazepane-4-carboxylate: To a solution of tert-butyl (2R)-2-[[tert-butyl(diphenyl)silyl]oxymethyl]-6-methylene-1,4-oxazepane-4-carboxylate (1.8 g, 1 equiv.) in THF (9 mL) and H.sub.2O (9 mL) were added NaIO.sub.4 (1.84 g, 2.3 equiv.) and K.sub.2O.sub.sO.sub.4.Math.2H.sub.2O (68.8 mg, 0.05 equiv.). The reaction mixture was stirred at 25° C. for 1 hour. The mixture was quenched by addition of aqueous Na.sub.2SO.sub.3 (20 mL) at 0° C. and H.sub.2O (10 mL) and then extracted with ethyl acetate (20 mL×2). The combined organic layers were washed with brine (30 mL) and concentrated. The residue was purified by column chromatography (SiO.sub.2, Petroleum ether/Ethyl acetate 1:0 to 100:1) to afford the title compound (1.1 g, 57% yield) as a colorless oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =7.67 (d, J=6.8 Hz, 4H), 7.45-7.38 (m, 6H), 4.50-4.21 (m, 3H), 4.03-3.97 (m, 1H), 3.83-3.60 (m, 4H), 3.01-2.96 (m, 1H), 1.47 (d, J=10.0 Hz, 9H), 1.07 (m, 9H); LCMS (ESI, M+23): m/z=506.2.

(457) Step E. (2R,6S)tert-butyl 2-(((tert-butyldiphenylsilyl)oxy)methyl)-6-hydroxy-6-methyl-1,4-oxazepane-4-carboxylate and (2R,6R)-tert-butyl 2-(((tert-butyldiphenylsilyl)oxy)methyl)-6-hydroxy-6-methyl-1,4-oxazepane-4-carboxylate: To a solution of tert-butyl (2R)-2-[[tert-butyl(diphenyl)silyl]oxymethyl]-6-oxo-1,4-oxazepane-4-carboxylate (900 mg, 1 equiv.) in THF (13 mL) was added MeMgBr (3 M, 4.04 equiv.) under N.sub.2 atm. at 0° C. The reaction mixture was stirred at 25° C. for 2 hours under N.sub.2 atmosphere. The mixture was quenched by addition of sat.aq.NH.sub.4Cl (30 mL) at 0° C. and H.sub.2O (15 mL), and then was extracted with ethyl acetate (20 mL×2). The combined organic layers were washed with brine (20 mL) and concentrated. The residue was purified by flash silica gel chromatography (Ethyl acetate/Petroleum ether 0-15%) to afford two isomers: R,S (350 mg, 33% yield) as a colorless oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ =7.59-7.57 (m, 4H), 7.38-7.29 (m, 6H), 4.15-4.11 (m, 1H), 3.95-3.65 (m, 3H), 3.61-3.45 (m, 2H), 3.21 (d, J=12.4 Hz, 1H), 2.82 (d, J=15.2 Hz, 1H), 2.70-2.64 (m, 1H), 1.42 (s,

9H), 1.11 (s, 3H), 0.99 (s, 9H); LCMS (ESI, M-99): m/z=400.2. and R,R (190 mg, 18% yield) as a colorless oil; LCMS (ESI, M-99): m/z=400.2.

(458) Step F. (2R,6R)-2-(hydroxymethyl)-6-methyl-1,4-oxazepan-6-ol: To a solution of (2R,6R)-tert-butyl 2-(((tert-butyldiphenylsilyl)oxy)methyl)-6-hydroxy-6-methyl-1,4-oxazepane-4-carboxylate (520 mg, 1 equiv.) in CH₂Cl₂ (2 mL) was added HCl-dioxane (4 M, 10 mL). The mixture was stirred at 25° C. for 12 hours. The mixture was concentrated, the residue was diluted with H₂O (4 mL) and the pH of the mixture was adjusted to 7 with saturated NaHCO₃ aqueous. The mixture was extracted with ethyl acetate (2×5 mL). The combined organic phase was dried over anhydrous Na₂SO₄ and concentrated to afford the title compound (70 mg, crude) as a light-yellow oil.

Intermediate 55

(459) ##STR00126##

3,3-difluoro-7-azaspiro[4.5]decane

(460) ##STR00127##

(461) Step A. tert-butyl 3,3-difluoro-7-azaspiro[4.5]decane-7-carboxylate: To a solution of tert-butyl-3-oxo-7-azaspiro[4.5]decane-7-carboxylate (500 mg, 1.0 equiv.) in CH₂Cl₂ (8 mL) was added DAST (541 mg, 443 µL, 1.7 equiv.). The mixture was stirred at 25° C. for 15 hours. The reaction mixture was diluted with water (20 mL) and extracted with ethyl acetate (2×20 mL). The combined organic layers were dried over anhydrous Na₂SO₄, concentrated and purified by column chromatography (SiO₂, petroleum ether/ethyl acetate 50:1 to 30:1) to afford the title compound (150 mg, 28% yield) as a yellow oil. ¹H NMR (400 MHz, CHLOROFORM-d) δ=3.52-3.10 (m, 4H), 2.25-2.10 (m, 2H), 2.05-1.95 (m, 1H), 1.94-1.83 (m, 1H), 1.78-1.68 (m, 1H), 1.61-1.49 (m, 5H), 1.46 (s, 9H); ¹⁹F NMR (377 MHz, CHLOROFORM-d) δ=-87.62-90.03 (m, 1H).

(462) Step B. 3,3-difluoro-7-azaspiro[4.5]decane: To a solution of tert-butyl 3,3-difluoro-7-azaspiro[4.5]decane-7-carboxylate (100 mg, 1.0 equiv.) in dioxane (4 mL) was added HCl-dioxane (4 M, 4 mL, 44.1 equiv.). The mixture was stirred at 0° C. for 2 hours. The reaction mixture was concentrated under reduced pressure to remove solvent. The residue was diluted with MeOH (4 mL), the pH was adjusted 9 with NaHCO₃, the mixture was stirred for 0.3 hours. The reaction mixture was concentrated to afford the title compound (70.0 mg, 91.1% yield, HCl salt, crude) as a colorless oil.

Intermediate 56

(463) ##STR00128##

tert-butyl (3R,6S)-3-(((tert-butyldiphenylsilyl)oxy)-6-hydroxy-2,3,6,7-tetrahydro-1H-azepine-1-carboxylate

(464) ##STR00129##

(465) Step A. tert-butyl (3R,6S)-3-(((tert-butyldiphenylsilyl)oxy)-6-hydroxy-2,3,6,7-tetrahydro-1H-azepine-1-carboxylate: To a solution of tert-butyl 3,6-dihydroxy-2,3,6,7-tetrahydroazepine-1-carboxylate (3.00 g, 1 equiv.) in DMF (20 mL) was added imidazole (2.67 g, 3 equiv.) and TBDPSCl (4.32 g, 1.2 equiv.). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was concentrated under reduced pressure. The residue was diluted with water (20 mL) and extracted with ethyl acetate (20 mL×3). Combined organic phase was washed with brine (20.0 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by column chromatography (SiO₂, Petroleum ether/Ethyl acetate 10:1 to 5:1) to afford the title compound (1.8 g, 29.4% yield) as a white solid; ¹H NMR (400 MHz, CDCl₃) δ=7.67-7.55 (m, 4H), 7.40-7.27 (m, 6H), 5.64-5.45 (m, 2H), 4.33-4.25 (m, 1H), 4.22-4.07 (m, 1H), 3.82-3.46 (m, 2H), 3.40-2.91 (m, 2H), 2.54-2.14 (m, 1H), 1.38 (br s, 3H), 1.24 (s, 6H), 1.00 (s, 9H).

Intermediate 57

(466) ##STR00130##

((5-chloro-6-fluoro-4-(trimethylstannyl)naphthalen-2-yl)oxy)triisopropylsilane

(467) ##STR00131##

(468) Step A. 5-(2-(3-chloro-4-fluorophenylacetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione: To a mixture of 2-(3-chloro-4-fluoro-phenyl)acetic acid (330 g, 1 equiv.) and 2,2-dimethyl-1,3-dioxane-4,6-dione (277 g, 1.1 equiv.) in MeCN (1500 mL) was added DMAP (18.2 g, 0.09 equiv.) at 20° C. Then DIEA (486 g, 2.15 equiv.) was added into the mixture slowly over the course of 1 hour under 15-30° C. Following that 2,2-dimethylpropanoyl chloride (232.10 g, 1.1 equiv.) was added into the mixture over the course of 1 hour 1 hour while maintaining the temperature at 25-40° C. After the additions were complete the mixture was stirred at 45° C. for 3 hours. The mixture was cooled to 0° C., then the pH was adjusted to 3 with HCl (4N, 5 L) and mixture was stirred at 0° C. for 1 hour. The filter cake was triturated with MeCN (3 L) to afford the title compound (933 g, 84% yield) as a yellow solid that was used in next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ=15.36 (br s, 1H), 7.46 (dd, J=2.0, 6.8 Hz, 1H), 7.30-7.25 (m, 1H), 7.10 (t, J=8.8 Hz, 1H), 4.39-4.34 (m, 1H), 1.74 (s, 6H).

(469) Step B. 8-chloro-7-fluoro-1,3-dihydroxy-2-naphthoic acid: A mixture of 5-(2-(3-chloro-4-fluorophenyl)acetyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (650 g, 1.0 equiv.) in CF₃SO₃H (1300 mL) was stirred at 5-20° C. Then the mixture was stirred at 10° C. for 2 hours. After reaction completion, the mixture was poured into ice water (2 L) and filtered. The filter cake was washed with water (5 L) and dried to afford the title compound (2000 g, crude) as yellow solid and used in next step without further purification.

(470) Step C. 8-chloro-7-fluoronaphthalene-1,3-diol: A mixture of 8-chloro-7-fluoro-1,3-dihydroxy-2-naphthoic acid (1.2 kg, 1.0 equiv.) in MeCN (700 mL) and H₂O (700 mL) was stirred at 85° C. for 12 hours under N₂. The mixture was concentrated under vacuum to remove acetonitrile. The residue was extracted with ethyl acetate (2 L×2), dried over Na₂SO₄ and concentrated in vacuum. The residue was purified by column chromatography (SiO₂, petroleum ether/ethyl acetate 3:1) and prep-HPLC (column: Phenomenex Luna C18 200*40 mm*10 μm; mobile phase: water (0.1% formic acid)-ACN; B %: 27%-57%, 10 min). The desired fraction was collected and extracted with ethyl acetate (2 L), dried over Na₂SO₄ and concentrated under vacuum to afford the title compound (17 g, 16% yield two steps) as a yellow solid. ¹H NMR (400 MHz, DMSO-d₆) δ 10.17 (s, 1H), 9.71 (s, 1H), 7.58 (dd, J=5.6, 8.8 Hz, 1H), 7.32 (t, J=8.8 Hz, 1H), 6.67 (d, J=2.0 Hz, 1H), 6.63 (d, J=2.0 Hz, 1H).

(471) Step D. 8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-ol: To a solution of 8-chloro-7-fluoronaphthalene-1,3-diol (10 g, 1 equiv.) and DIEA (12.2 g, 2.0 equiv.) in DCM (150 mL) was added TIPSCl (8.16 g, 0.9 equiv.) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated in vacuum and the residue was purified by column chromatography (SiO₂, petroleum ether/ethyl acetate 10:1) to afford the title compound (15 g, 86% yield) as a yellow oil.

(472) Step E. 8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl trifluoromethanesulfonate: To a mixture of 8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-ol (15 g, 1.0 equiv.) and DIEA (15.8 g, 3.0 equiv.) in DCM (150 mL) was added Tf₂O (17.2 g, 1.5 equiv.) at 40° C. The mixture was stirred at -40° C. for 0.5 hour. The mixture was concentrated in vacuum and the residue was purified by column chromatography (SiO₂, petroleum ether/ethyl acetate 10:1) to afford the title compound (19 g, 90% yield) as a yellow oil.

(473) Example F. ((5-chloro-6-fluoro-4-(trimethylstannyl)naphthalen-2-yl)oxy)triisopropylsilane: To a mixture of 8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl trifluoromethanesulfonate (5 g, 1.0 equiv.), trimethyl(trimethylstannyl)stannane (12.7 g, 3.88 equiv.) and LiCl (1.27 g, 3.0 equiv.) in toluene (50 mL) was added Pd(PPh₃)₄ (1.15 g, 0.1 equiv.) under N₂. The mixture was stirred at 100° C. for 12 hours under N₂. The mixture was quenched with water (20 mL) and extracted with ethyl acetate (100 mL×3), the combined

organic phase was washed with brine (100 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue and the residue was purified by column chromatography (SiO₂, petroleum ether) and reversed phase flash [water (0.1%, FORMIC ACID)/acetonitrile] to afford the title compound (3 g, 55% yield) as yellow oil. ¹H NMR (400 MHz, CDCl₃) δ=7.60 (dd, J=5.6, 9.2 Hz, 1H), 7.52 (d, J=2.4 Hz, 1H), 7.31-7.26 (m, 1H), 7.20 (d, J=2.4 Hz, 1H), 1.35-1.30 (m, 3H), 1.15 (d, J=7.2 Hz, 18H), 0.51-0.35 (m, 9H)

Intermediate 58

(474) ##STR00132##

(R)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide Intermediate 59

(475) ##STR00133##

(S)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(476) ##STR00134##

(477) Step A. benzyl (R)-2-thia-1,3,7-triazaspiro[4.5]decane-7-carboxylate 2,2-dioxide and benzyl (S)-2-thia-1,3,7-triazaspiro[4.5]decane-7-carboxylate 2,2-dioxide: Racemic benzyl 2-thia-1,3,7-triazaspiro[4.5]decane-7-carboxylate 2,2-dioxide (46.5 g) was separated by chiral SFC (column: DAICEL CHIRALPAK IC (250 mm*50 mm, 10 μm); mobile phase: [0.1% NH₃.3H₂O-IPA]; B %: 55%-55%, 7 min) to afford the two title compound isomers: R-isomer (22 g, 41% yield, 99.9% ee) as a yellow solid and S-isomer (22 g, 42% yield, 99.9% ee) as a yellow solid. LCMS (ESI, M+1): m/z=326.1.

(478) Step B. (R)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of (R)-2-thia-1,3,7-triazaspiro[4.5]decane-7-carboxylate 2,2-dioxide (20.0 g, 1.00 equiv.) in methanol (160 mL) was added Pd/C (10%, 3.00 g) under nitrogen atmosphere. The suspension was degassed and stirred under hydrogen (15 Psi) at 25° C. for 2 hours. The reaction mixture was filtered and concentrated under reduced pressure to afford the title compound (11 g, 93% yield) as a yellow solid. ¹H NMR (400 MHz, MeOD-d₄) δ=3.37-3.32 (m, 1H), 3.13 (d, J=11.7 Hz, 1H), 2.93-2.84 (m, 1H), 2.82-2.72 (m, 2H), 2.72-2.62 (m, 1H), 1.82 (br dd, J=4.8, 6.7 Hz, 1H), 1.79-1.70 (m, 1H), 1.65 (td, J=4.2, 8.3 Hz, 1H), 1.60-1.49 (m, 1H)

(479) (S)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of benzyl benzyl (S)-2-thia-1,3,7-triazaspiro[4.5]decane-7-carboxylate 2,2-dioxide. (22.0 g, 1.00 equiv.) in methanol (160 mL) was added Pd/C (10%, 3.00 g) under nitrogen atmosphere. The suspension was degassed and stirred under hydrogen (15 Psi) at 25° C. for 2 hours. The reaction mixture was filtered and concentrated under reduced pressure to afford the title compound (11.8 g, 91% yield) as a yellow solid. ¹H NMR (400 MHz, MeOD-d₄) δ=3.37-3.32 (m, 1H), 3.15 (s, 1H), 3.19-3.07 (m, 1H), 2.92-2.83 (m, 1H), 2.82-2.72 (m, 2H), 2.72-2.62 (m, 1H), 1.82 (br dd, J=4.8, 6.7 Hz, 1H), 1.79-1.71 (m, 1H), 1.70-1.60 (m, 1H), 1.59-1.48 (m, 1H)

Intermediates 60

(480) ##STR00135##

3-chloro-4-cyclopropyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol

(481) ##STR00136##

(482) Step A. 3-chloro-4-cyclopropyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol: To a solution of 3-bromo-5-chloro-4-cyclopropyl-phenol (2.00 g, 1.0 equiv.) in dioxane (50 mL) was added KOAc (2.38 g, 3.0 equiv.), Pin₂B₂ (4.00 g, 2.0 equiv.) and Pd(dppf)Cl₂ (591 mg, 0.1 equiv.). The mixture was stirred at 100° C. for 4 hours under nitrogen atmosphere. The reaction mixture was poured into saturated aqueous NH₄Cl solution (5 mL) and extracted with EtOAc (3×2 mL). The combined organic layers were washed with brine (5 mL), dried over anhydrous sodium sulfate, filtered and concentrated. The residue was purified by flash silica gel chromatography [ethyl acetate in petroleum ether 0-35%] to afford the title compound (1.10 g, 46% yield) as a white solid; ¹H NMR (400 MHz, CHLOROFORM-d) δ=6.92 (d, J=2.6 Hz, 1H), 6.87 (d, J=2.6 Hz, 1H), 5.33 (br s, 1H), 1.99-1.90 (m, 1H), 1.39 (s, 12H), 1.00-0.94 (m, 2H), 0.53-0.47 (m, 2H).

Intermediate 61

(483) ##STR00137##

6-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(484) ##STR00138##

(485) Step A. 6-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 475 μ mol, 1.0 equiv.), 1,6-diazaspiro[3.5]nonan-2-one (60.0 mg, 428 μ mol, 0.9 equiv.), 4 Å molecular sieves (10.0 mg) in DMF (2 mL) was added DIEA (184 mg, 1.43 mmol, 248 μ L, 3.0 equiv.). The mixture was stirred at 40° C. for 4 hours. After reaction completion, the mixture was filtered to give a filtrate. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to give the title compound (60.0 mg, 130 μ mol, 27% yield) as a yellow solid; LCMS (ESI, M+1): m/z=461.3.

Intermediate 62

(486) ##STR00139##

4-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(487) ##STR00140##

(488) Step A. 4-(7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (0.45 g, 1 equiv.), 6-methyl-1,4-oxazepan-6-ol (210 mg, 1.5 equiv.), DIPEA (276 mg, 2 equiv.) and 4 Å molecular sieves (50 mg) in DMF (4 mL) was stirred at 40° C. for 14 hours under N.sub.2 atmosphere. The reaction mixture was filtered and the filtrate was concentrated to give a residue. The residue was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (205 mg, 40.3% yield) as a light-yellow solid; LCMS (ESI, M+1): m/z=452.0.

Example 1

(489) ##STR00141##

1-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(490) ##STR00142##

(491) Step A. 1-[2-chloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-3-methylpiperidin-3-ol: To a mixture of 2,4-dichloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine (1.48 g, 3.90 mmol) in dichloromethane (15 mL) was added DIEA (5.05 g, 39.0 mmol, 6.80 mL) and 3-methylpiperidin-3-ol (270 mg, 2.34 mmol) at -40° C. under N.sub.2. The mixture was stirred at -40° C. for 0.5 h. After completion, the mixture was quenched by water (10 mL). The aqueous phase was extracted with dichloromethane (2×8 mL), the combined organic layer was washed with brine (10 mL) and dried over with Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 1-[2-chloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-3-methylpiperidin-3-ol (128 mg, 35% yield) as yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.37 (s, 1H), 8.22 (d, J=8.0 Hz, 1H), 8.11 (d, J=8.0 Hz, 1H), 7.77-7.71 (m, 1H), 7.66 (t, J=7.6 Hz, 2H), 7.61-7.55 (m, 1H), 4.84 (d, J=16.4 Hz, 1H), 4.54-4.41 (m, 1H), 4.18 (br dd, J=6.0, 12.8 Hz, 1H), 3.68-3.54 (m, 1H), 2.08-2.00 (m, 1H), 1.75-1.66 (m, 3H), 1.20-1.17 (m, 3H); LCMS (ESI, M+1): m/z 457.1.

(492) Step B. 1-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 1-[2-chloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-3-methylpiperidin-3-ol (112 mg, 245 μ mol) and (tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (69.2 mg, 490 μ mol) in dioxane (2.0 mL)

was added DIEA (95.0 mg, 735 μ mol, 128 μ L). The mixture was stirred at 90° C. for 15 h. After completion, the mixture was diluted with ethyl acetate (8 mL) and water (8 mL). The aqueous phase was extracted with ethyl acetate (2 $\times$ 5 mL) and dichloromethane:methanol=10:1 (1 $\times$ 5 mL). The organic layers were concentrated in vacuum. The residue was purified by prep-HPLC (Water s Xbridge 150*25 mm*5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 22%-52%, 10 min) to afford the title compound (23.7 mg, 16% yield) as off-white solid. ¹H NMR (400 MHz, CDCl₃-d): δ 9.13 (d, J=2.4 Hz, 1H), 8.04-7.97 (m, 1H), 7.88 (d, J=7.6 Hz, 1H), 7.64-7.52 (m, 3H), 7.45-7.38 (m, 1H), 4.50-4.25 (m, 4H), 3.57-3.45 (m, 1H), 3.35 (br dd, J=6.8, 13.2 Hz, 1H), 3.26-2.98 (m, 3H), 2.75-2.61 (m, 2H), 2.18-2.03 (m, 3H), 1.96-1.84 (m, 5H), 1.78-1.71 (m, 4H), 1.36 (d, J=2.4 Hz, 3H); LCMS (ESI, M+1): m/z 562.1

Example 2

(493) ##STR00143##

7-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(494) ##STR00144##

(495) Step A. 7-[2-chloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 2,4-dichloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidine (0.15 g, 396 μ mol) in dichloromethane (5.0 mL) was added DIEA (512 mg, 3.96 mmol) at -40° C. After the mixture was stirred at -40° C. for 10 minutes, 2,9-diazaspiro[4.5]decan-3-one (73.3 mg, 475 μ mol) was added into the mixture. The mixture was stirred at -40° C. for 10 minutes. After completion, the mixture was diluted with water (10 mL) and extracted with dichloromethane (2 $\times$ 10 mL). The combined organic layer was washed with brine (10 mL), and then dried over Na₂SO₄. The mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 7-[2-chloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-2,7-diazaspiro[4.5]decan-3-one (120 mg, 45% yield) as a yellow Solid; LCMS [ESI, M+1]: m/z=496.0.

(496) Step B. 7-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: A mixture of 7-[2-chloro-7-(8-chloro-1-naphthyl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-2,7-diazaspiro[4.5]decan-3-one (100 mg, 201 μ mol), (tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (56.9 mg, 403 μ mol) and DIEA (130 mg, 1.01 mmol, 175 μ L) in dioxane (1.0 mL) was stirred at 80° C. for 9 hours. After completion, the mixture was diluted with water (5.0 mL) and extracted with ethyl acetate (2 $\times$ 10 mL). The combined organic layer was washed with brine (10 mL), and then dried over Na₂SO₄. The mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC (column: water s Xbridge 150*25 mm*5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 24%-54%, 10 min) to give the title compound (23.0 mg, 18% yield) as yellow solid; LCMS [ESI, M+1]: m/z=601. ¹H NMR (400 MHz, chloroform-d) δ 8.98 (d, J=2.0 Hz, 1H), 8.00 (dd, J=1.6, 7.8 Hz, 1H), 7.88 (td, J=1.2, 8.4 Hz, 1H), 7.63-7.52 (m, 3H), 7.42 (t, J=7.6 Hz, 1H), 6.00 (br d, J=8.8 Hz, 1H), 4.30-4.18 (m, 2H), 4.09 (q, J=12.4 Hz, 2H), 3.84-3.71 (m, 1H), 3.64 (dd, J=13.2, 18.0 Hz, 1H), 3.48-3.38 (m, 1H), 3.21 (dd, J=4.4, 10.0 Hz, 1H), 3.16-3.06 (m, 2H), 2.64 (td, J=6.8, 10.0 Hz, 2H), 2.38-2.23 (m, 2H), 2.15-2.04 (m, 2H), 1.94-1.86 (m, 7H), 1.74-1.60 (m, 3H).

Example 3

(497) ##STR00145##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,5-diazocan-2-one

(498) ##STR00146##

(499) Step A. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,5-diazocan-2-one: To a solution of 8-fluoro-7-(8-

fluorophthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 188 μ mol), 4 Å MS (50 mg) and 1,5-diazocan-2-one (48.3 mg, 377 μ mol) in DMF (2.00 mL) was added DIEA (73.1 mg, 565 μ mol). The mixture was stirred at 90° C. for 2 hours. After completion, the reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with brine (20 mL) and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 17%-47%, 9 min) affording 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,5-diazocan-2-one (18.6 mg, 17% yield) as white solid; ¹H NMR (400 MHz, CDCl.sub.3-d) δ 9.08 (s, 1H), 8.00 (dt, J=1.6, 8.0 Hz, 1H), 7.75 (d, J=7.6 Hz, 1H), 7.68-7.56 (m, 2H), 7.49-7.42 (m, 1H), 7.17-7.06 (m, 1H), 6.03-5.86 (m, 1H), 4.45-3.92 (m, 6H), 3.40-3.29 (m, 2H), 3.15-3.07 (m, 2H), 3.04-2.91 (m, 2H), 2.68-2.60 (m, 2H), 2.15-1.96 (m, 5H), 1.91-1.87 (m, 3H), 1.72-1.62 (m, 2H); LCMS (ESI, M+1): m/z=559.3.

Example 4

(500) ##STR00147##

(S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(501) ##STR00148##

(502) Step A. (S)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (103 mg, 286 μ mol, 1.3 equiv.) and (4)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 220 μ mol, 10 eq, synthesized according to example 5 step A replacing (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol with (S)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol) in THF (1.5 mL) was added cataCXium®-A-Pd-G3 (24.07 mg, 33.05 μ mol, 0.15 equiv.) in one portion at 25° C. under N.sub.2. Then K.sub.3PO.sub.4 (1.5 M, 440 μ L, 3.0 equiv.) was added under N.sub.2. The mixture was heated to 60° C. and stirred for 4 hours. After completion, the mixture was filtered and concentrated in vacuum. The crude product was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) to give title compound as yellow oil; ¹H NMR (400 MHz, CDCl.sub.3-d) δ =9.14 (d, J=8.0 Hz, 1H), 7.76 (dd, J=6.0, 8.8 Hz, 1H), 7.55-7.50 (m, 1H), 7.27-7.24 (m, 1H), 7.23-7.20 (m, 1H), 5.37-5.20 (m, 3H), 4.52-4.35 (m, 2H), 4.33-4.20 (m, 2H), 3.52 (s, 3H), 3.50-3.40 (m, 1H), 3.36-3.14 (m, 4H), 3.04-2.77 (m, 2H), 2.59-2.47 (m, 1H), 2.29-2.11 (m, 4H), 1.98-1.85 (m, 4H), 1.80-1.69 (m, 2H), 1.36 (s, 3H), 0.84 (q, J=7.6 Hz, 3H); LCMS (ESI, M+1-Boc): m/z=652.3.

(503) Step B. (S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (70 mg, 107 μ mol, 1 equiv.) in MeCN (1.4 mL) was added HCl.dioxane (4 M, 1.4 mL, 52 equiv.) at 5° C. The reaction mixture was stirred at 5° C. for 0.5 hour: Upon completion, the reaction mixture was diluted with MeCN (3 mL) and basified with saturated NaHCO.sub.3 solution to pH>7. The reaction mixture was extracted with ethyl acetate (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The residue was purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 40%-70%, 5 min) to give title compound as (32.5 mg, 50% yield) white solid; ¹H NMR δ =9.21-8.99 (m, 1H), 7.51 (dd, J=6.0, 9.2 Hz, 1H), 7.17 (t, J=9.2 Hz, 1H), 7.10 (dd, J=2.4, 15.6 Hz, 1H), 6.89-6.61 (m, 1H), 5.42-5.18 (m, 1H), 4.45-4.09 (m, 4H), 3.48-3.16 (m, 4H), 3.14-2.94 (m, 2H), 2.58-2.31 (m, 2H), 2.30-2.06 (m, 4H), 2.03-1.95 (m, 2H), 1.74-

1.70 (m, 1H), 1.67-1.40 (m, 3H), 1.23 (d, J=4.8 Hz, 3H), 0.81-0.75 (m, 3H); LCMS (ESI, M+1): m/z=608.2.

Example 5

(504) ##STR00149##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(505) ##STR00150##

(506) Step A. (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (20.0 g, 60.4 mmol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (10.6 g, 66.4 mmol), 4 Å molecular sieves (5.00 g) in dioxane (80 mL) was added DIEA (23.4 g, 181 mmol), and the mixture was stirred at 90° C. for 10 hours. Upon completion, the reaction mixture was filtered. The mixture was diluted with ethyl acetate (300 mL) and water (200 mL), and aqueous layer was then extracted with ethyl acetate (300 mL). The combined organic phase was washed with brine (200 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (19.5 g, 71% yield) as yellow solid; LCMS (ESI, M+1): m/z=454.2.

(507) Step B. (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (40.0 g, 88.1 mmol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (42.9 g, 119 mmol), K₂CO₃ (1.5 M in water, 117 mL) in THF (200 mL) was added Pd(PPh₃)₄ (6.42 g, 8.81 mmol) under N₂. The mixture was stirred at 65° C. for 8 hours. Upon completion, the reaction mixture was diluted with ethyl acetate (300 mL) and water (200 mL). The aqueous layer was extracted with ethyl acetate (300 mL). The combined organic phase was washed with brine (200 mL) and dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (37.4 g, 65% yield) as yellow solid; LCMS (ESI, M+1): m/z=652.3.

(508) Step C. (R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (38.0 g, 58.3 mmol) in ACN (190 mL) was added HCl.dioxane (4 M, 190 mL) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. Upon completion, the mixture was concentrated to give a residue. To the residue were added ethyl acetate (300 mL) and then sat. NaHCO₃ (to adjust the pH to 8). The aqueous layer was extracted with ethyl acetate (300 mL). The combined organic phase was washed with brine (300 mL) and dried over with Na₂SO₄. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, and the pH value was adjusted to 8 with NaHCO₃ (30 g). The mixture was concentrated in vacuum to remove acetonitrile, and then was extracted with dichloromethane (2×800 mL). The combined organic layer was washed with brine (600 mL) and dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated in vacuum. The

residue was dissolved in acetonitrile (100 mL) and water (200 mL), and lyophilized to afford (R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (25.3 g, 71% yield). Yellow Solid; ¹H NMR (400 MHz, CDCl₃) δ=9.20-8.94 (m, 1H), 7.52-7.44 (m, 1H), 7.18-7.11 (m, 1H), 7.07-7.01 (m, 1H), 6.39 (s, 1H), 5.49-5.15 (m, 1H), 4.46-3.94 (m, 5H), 3.49-2.90 (m, 7H), 2.00 (br s, 6H), 1.98-1.63 (m, 4H), 1.59-1.36 (m, 2H), 1.17 (d, J=5.6 Hz, 3H), 0.82-0.73 (m, 3H); ¹H NMR (400 MHz, METHANOL-d₄) δ=9.20 (d, J=0.8 Hz, 1H), 7.71-7.63 (m, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.39-5.22 (m, 1H), 4.57-4.22 (m, 4H), 3.68-3.54 (m, 1H), 3.50-3.39 (m, 1H), 3.29-3.13 (m, 3H), 3.05-2.96 (m, 1H), 2.53-2.11 (m, 6H), 2.04-1.94 (m, 2H), 1.94-1.73 (m, 4H), 1.28 (d, J=9.6 Hz, 3H), 0.84-0.77 (m, 3H); LCMS (ESI, M+1): m/z=608.3.

Example 6

(509) ##STR00151##

(3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol

(510) ##STR00152##

(511) Step A. 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl) methoxy)-7-chloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine: A solution of 2,7-dichloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (140 mg, 465 μmol) and ((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methanol (189 mg, 511 μmol) in toluene (3.00 mL) was added tBuONa (134 mg, 1.39 mmol). The mixture was stirred at 0° C. for 1 hr. The mixture was concentration in vacuum and was purified by prep-TLC (Silica gel, PE:EA=2:1) to give 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-chloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (200 mg, 68% yield) as a yellow oil.

(512) Step B. 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl) methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl) pyrido[4,3-d]pyrimidine: A mixture of 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-chloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (170 mg, 268 μmol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (193 mg, 536 μmol), cataCXium-A-Pd-G3cataCXium-A-Pd-G3 (19.5 mg, 26.8 μmol) and K₂CO₃ (1.5 M in water, 536 μL) in dioxane (5.00 mL) was degassed and purged with N₂ for 3 times, and then the mixture was stirred at 100° C. for 1.5 hour under N₂ atmosphere. The mixture was diluted with water (40.0 mL) and extracted with ethyl acetate (2×20.0 mL). The organic layers were dried over Na₂SO₄, concentrated in vacuum, and was purified by column (Silica gel, PE:EA=1:0 to 0:1) to give 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (200 mg, 54% yield, 60% purity) as a yellow oil.

(513) LCMS [ESI, M+1]: m/z=832.4.

(514) Step C. (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol: To a solution of 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (200 mg, 144 μmol, 60% purity) in DMF (6.0 mL) was added CsF (65.7 mg, 433 μmol, 15.9 μL). After stirred at 40° C. for 4 hours, the mixture was diluted with water (40 mL) and layers were separated. The aqueous phase was extracted with ethyl acetate (2×20 mL). Combined organic layers were dried over Na₂SO₄, filtered and concentrated in vacuum to give (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl) pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol (80 mg, crude) as a yellow oil. LCMS [ESI, M+1]: m/z=594.3.

(515) Step D. (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol: A solution of (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol (80.0 mg, 135 μ mol) in HCl-EtOAc (4 M, 1 mL) was stirring at 20° C. for 1 hour. The mixture was concentrated in vacuum. The pH of the mixture was adjusted to 8 with sat. NaHCO₃ (5 mL). The mixture was diluted with water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge BEH C18 100×25 mm×5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 30%-60%, 10 min) to give title compound (28.0 mg, 37% yield, 98% purity) as a white solid. ¹H NMR (400 MHz, DMSO) δ 9.93 (s, 1H), 9.05 (s, 1H), 7.78-7.75 (m, 1H), 7.39-7.30 (m, 2H), 7.03 (d, J=2.6 Hz, 1H), 4.78 (d, J=4.4 Hz, 1H), 4.43-4.34 (m, 1H), 4.28-4.12 (m, 2H), 3.94 (s, 4H), 3.20-3.17 (m, 1H), 2.81 (s, 1H), 2.34 (s, 4H), 2.19-2.06 (m, 2H), 1.90-1.72 (m, 8H), 0.73 (t, J=7.3 Hz, 3H); LCMS [ESI, M+1]: m/z=550.3.

Example 7

(516) ##STR00153##

(2R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-2-ol

(517) ##STR00154##

(518) Step A. 2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine: A solution of 2,7-dichloro-8-fluoro-4-(1-piperidyl)pyrido[4,3-d]pyrimidine (200 mg, 664 μ mol) and ((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (315 mg, 797 μ mol) in toluene (3.00 mL) was added drop-wise t-BuONa (191 mg, 1.99 mmol). The mixture was stirred at 0° C. for 1 hour under N₂. The reaction mixture was quenched by the addition of water (50.0 mL) at 0° C., and extracted with DCM (30.0 mL×3). The combined organic layers were washed with brine (40.0 mL), dried over Na₂SO₄, filtered and concentrated in vacuum. The residue was purified by prep-TLC (Silica gel, EtOAc/MeOH=50/1) to give 2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (300 mg, 68% yield) was obtained as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ =8.69 (s, 1H), 7.70-7.62 (m, 4H), 7.45-7.31 (m, 6H), 4.55-4.44 (m, 1H), 4.17-4.08 (m, 1H), 4.04-3.96 (m, 1H), 3.87-3.80 (m, 4H), 3.50 (s, 2H), 3.19-2.99 (m, 3H), 2.83-2.73 (m, 1H), 2.20-1.82 (m, 7H), 1.75-1.70 (m, 2H), 1.52-1.31 (m, 1H), 1.06 (s, 9H). LCMS [ESI, M+1]: m/z=660.4.

(519) Step B. 2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (240 mg, 363 μ mol), and 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (393 mg, 1.09 mmol) in dioxane (5.00 mL) were added K₃PO₄ (1.50 M, 727 μ L) and cataCXium-A-Pd-G3 (26.5 mg, 36.4 μ mol) under N₂. The mixture was stirred at 100° C. for 1 hour under N₂. The reaction mixture was quenched with water (60 mL) at 0° C. and extracted with EtOAc (30:0 mL×3). The combined organic layers were washed with brine (40.0 mL), dried over Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-TLC (Silica gel, EtOAc/MeOH=50/1) to give the title compound (300 mg, 77% yield) as a yellow solid; LCMS [ESI, M+1]: m/z=858.4.

(520) Step C. (2R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-2-ol: To a solution of 2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)

methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (250 mg, 262 μmol , 90% purity) in DMF (1.00 mL) was added CsF (398 mg, 2.62 mmol, 96.7 μL). The mixture was stirred at 40° C. for 4 hrs. The reaction mixture was quenched with water (40.0 mL) at 0° C. and extracted with EtOAc (30.0 mL $\times$ 3). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give the title compound (150 mg, crude) as a yellow solid. LCMS [ESI, M+1]m/z=620.4.

(521) Step D. (2R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-2-ol: To a solution of (2R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-2-ol (100 mg, 129 μmol , 80% purity) in EtOAc (1.00 mL) was added drop-wise HCl/EtOAc (4 M, 1.00 mL). The mixture was stirred at 20° C. for 12 hrs. The reaction mixture was quenched with sat. NaHCO.sub.3 (20.0 mL) at 0° C. and then extracted with EtOAc (15.0 mL $\times$ 3). The combined organic layers were washed with brine (20.0 mL) and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge BEH C18 100 $\times$ 25 mm $\times$ 5 μm ; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 25%-65%, 10 min) to give the title compound (19.85 mg, 26% yield, 99.0% purity) as a light-yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =9.93 (s, 1H), 9.04 (s, 1H), 7.81-7.73 (m, 1H), 7.38-7.30 (m, 2H), 7.04-7.00 (m, 1H), 4.80-4.72 (m, 1H), 4.36-4.24 (m, 1H), 4.07-3.86 (m, 6H), 3.10-3.01 (m, 1H), 2.97-2.88 (m, 1H), 2.82-2.71 (m, 1H), 2.45 (br s, 1H), 2.41-2.29 (m, 1H), 2.15-2.08 (m, 2H), 1.93-1.68 (m, 10H), 1.64-1.54 (m, 1H), 0.76%-0.68 (m, 3H). LCMS [ESI, M+1]: m/z=576.1

Example 8

(522) ##STR00155##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-4-ol

(523) ##STR00156##

(524) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-4-ol: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 189 μmol) and azepan-4-ol hydrochloride (42.9 mg, 283 μmol , HCl) in DMF (3.00 mL) was added DIEA (73.1 mg, 566 μmol , 98.5 μL , 3 equiv.). The mixture was stirred at 60° C. for 16 hours. Upon completion, the mixture was filtered. The filtrate was purified by prep-HPLC (column: water s Xbridge BEH C18 100 $\times$ 25 mm $\times$ 5 μm ; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 20%-60%, 10 min) and re-purified by prep-HPLC (column: Welch Xtimate C18 150 $\times$ 25 mm $\times$ 5 μm ; mobile phase: [water (0.04% HCl)-ACN]; B %: 10%-35%, 8 min) affording 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-4-ol (23.8 mg, 37% yield, HCl salt) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =11.04 (br s, 1H), 9.25 (s, 1H), 8.22 (d, J=8.0 Hz, 1H), 7.96 (d, J=8.0 Hz, 1H), 7.80-7.73 (m, 1H), 7.68 (d, J=6.4 Hz, 1H), 7.64-7.56 (m, 1H), 7.34 (dd, J=7.2, 13.2 Hz, 1H), 4.63 (s, 2H), 4.15-4.03 (m, 3H), 4.02-3.93 (m, 1H), 3.87-3.77 (m, 1H), 3.60-3.45 (m, 2H), 3.25-3.12 (m, 2H), 2.24-1.92 (m, 1H), 1.90-1.79 (m, 1H), 1.78-1.68 (m, 1H), 1.68-1.57 (m, 1H); .sup.1H NMR (400 MHz, MeOD-d.sub.4) δ =9.35 (s, 1H), 8.28-8.19 (m, 1H), 7.93 (d, J=8.0 Hz, 1H), 7.81-7.73 (m, 2H), 7.66-7.56 (m, 1H), 7.34-7.25 (m, 1H), 4.83 (s, 2H), 4.41-4.15 (m, 4H), 4.12-3.97 (m, 1H), 3.81-3.69 (m, 2H), 3.36-3.32 (m, 1H), 3.30-3.27 (m, 1H), 2.45-2.32 (m, 3H), 2.32-2.21 (m, 5H), 2.20-2.09 (m, 3H), 2.07-1.97 (m, 1H), 1.95-1.78 (m, 2H); LCMS [ESI, M+1]: m/z=546.2.

Example 9

(525) ##STR00157##

[(3R,5S)-5-[[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-4-[(3S)-3-hydroxy-3-methyl-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methyl-pyrrolidin-3-yl] N-methylcarbamate (526) ##STR00158## ##STR00159##

(527) Step A. 2,7-dichloro-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl) oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To the mixture of (S)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.00 g, 6.04 mmol), TsOH.Math.WATER (115 mg, 605 μ mol) in dichloromethane (30 mL) was added 3,4-dihydro-2H-pyran (1.02 g, 12.1 mmol) at 0° C. The mixture was stirred at 20° C. for 1.5 h. After completion, the mixture was diluted with dichloromethane (30 mL). The mixture was washed with sat. NaHCO.sub.3 (40 mL) and the aqueous solution was extracted with dichloromethane (30 mL). The combined organic phase was washed with brine (50 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by column chromatography (10% to 50% EA/PE) to give the title compound (1.95 g, 65% yield) as yellow oil; LCMS (ESI, M+1): m/z=415.4

(528) Step B. 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl) methoxy)-7-chloro-8-fluoro-4-(3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl) pyrido[4,3-d]pyrimidine: To the mixture of 2,7-dichloro-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine (1.48 g, 3.57 mmol) and ((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methanol (1.45 g, 3.92 mmol) in dioxane (12 mL) was added DIEA (1.38 g, 10.7 mmol). The mixture was stirred at 90° C. for 10 h. The mixture was concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (1.00 g, 36% yield) as yellow solid; .sup.1H NMR (400 MHz, CDCl.sub.3-d): δ 9.07-8.95 (m, 1H), 7.68-7.60 (m, 4H), 7.44-7.34 (m, 6H), 4.88-4.66 (m, 1H), 4.52-4.44 (m, 2H), 4.42-4.29 (m, 2H), 4.26-4.19 (m, 1H), 3.90-3.58 (m, 1H), 3.51-2.95 (m, 5H), 2.47 (d, J=1.2 Hz, 3H), 2.43-2.36 (m, 1H), 2.19-2.06 (m, 2H), 2.05-1.84 (m, 2H), 1.73-1.65 (m, 1H), 1.55-1.33 (m, 5H), 1.31-1.23 (m, 5H), 1.06 (s, 9H). LCMS (ESI, M+1): m/z=748.3.

(529) Step C. 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl) methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-4 (3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To the mixture of 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-chloro-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine (400 mg, 534 μ mol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (384 mg, 1.07 mmol), K.sub.3PO.sub.4 (1.50 M, 1.07 mL) in toluene (8 mL) was added [2-(2-aminophenyl)phenyl]palladium(1+); bis(1-adamantyl)-butyl-phosphane; methanesulfonate (38.9 mg, 53.4 μ mol). The mixture was degassed and stirred at 90° C. for 1.5 h. The reaction mixture was diluted with ethyl acetate (30 mL) and water (15 mL), and the aqueous layer was extracted with ethyl acetate (20 mL). The combined organic phase was washed with brine (10 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile, MeOH] to give title compound (400 mg, 78% yield) as yellow solid; LCMS (ESI, M+1): m/z=946.3.

(530) Step D. (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3 S)-3-methyl-3-((tetrahydro-2H-pyran-2-l)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol: To the mixture of 2-(((2S,4R)-4-((tert-butyldiphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine (950 mg, 1.00 mmol) in DMF (10 mL) was added CsF (2.28 g, 15.0 mmol). The mixture was stirred at 40° C. for 4 h. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic

acid)/acetonitrile] to give the title compound (600 mg, 83% yield) as yellow solid; LCMS (ESI, M+1): m/z=708.4.

(531) Step E. (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-yl (4-nitrophenyl) carbonate: To the mixture of (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol (100 mg, 141 μ mol), and (4-nitrophenyl) carbonochloridate (103 mg, 509 μ mol) in THF (4 mL) was added t-BuOK (1 M in THF, 424 μ l) at 0° C. The mixture was stirred at 15° C. for 0.5 h. The mixture was diluted with ethyl acetate (10 mL) and water (5 mL), and the aqueous layer was extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (110 mg, 84% yield) as yellow oil; LCMS (ESI, M+1): m/z 873.3.

(532) Step F. [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-4-[(3S)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methyl-pyrrolidin-3-yl] N-methylcarbamate: A mixture of (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3S)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-yl (4-nitrophenyl) carbonate (52.0 mg, 59.6 μ mol) and methanamine (2 M in THF, 745 μ l) in DMF (1 mL) was stirred at 25° C. for 0.5 h. After completion, the residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (29.0 mg, 51% yield) as yellow solid. LCMS (ESI, M+1): m/z=765.2.

(533) Step G. [(3R,5S).sub.5-[[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl]-8-fluoro-4-[(3S)-3-hydroxy-3-methyl-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methyl-pyrrolidin-3-yl] N-methylcarbamate: To a mixture of [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-4-[(3S)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methyl-pyrrolidin-3-yl] N-methylcarbamate (41.0 mg, 53.6 μ mol) in MeCN (0.7 mL) was added HCl-MeOH (4 M, 1.4 mL) at 0° C. The mixture was stirred at 0° C. for 1 h. After completion, the mixture was concentrated in vacuum. Then the pH value was adjusted to 8 with saturated NaHCO₃ solution. The mixture was triturated with methanol (2×10 mL) and filtered. The filtrate was concentrated in vacuum. The residue was purified by prep-HPLC (Phenomenex Gemini-NX C18 75×30 mm×3 μ m; mobile phase: [water (0.05% ammonia hydroxide v/v)-ACN]; B %: 23%-53%, 11.5 min) to afford the title compound (18.6 mg, 54% yield) as white solid. ¹H NMR (400 MHz, METHANOL-d₄): δ 9.21 (d, J=2.4 Hz, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.29 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.06 (t, J=2.0 Hz, 1H), 5.10-5.04 (m, 1H), 4.57-4.47 (m, 3H), 4.29 (br t, J=12.2 Hz, 1H), 3.68-3.57 (m, 1H), 3.53-3.42 (m, 2H), 3.09-3.00 (m, 1H), 2.69 (s, 3H), 2.52 (s, 3H), 2.50-2.41 (m, 2H), 2.22-2.08 (m, 4H), 1.92-1.75 (m, 3H), 1.28 (d, J=10.0 Hz, 3H), 0.81 (q, J=7.6 Hz, 3H). ¹⁹F NMR (400 MHz, METHANOL-d₄) δ =-121.268, -139.169. HPLC:>99% ee, Chiralcel OD-3 50×4.6 mm I.D., 3 μ m A: 60% MeOH+40% ACN (w/0.05% DEA), B: CO₂, 3 mL/min, 220 nm, t_R: 0.597 min; LCMS (ESI, M+1): m/z=637.0.

Example 10

(534) ##STR00160##

(S)-1-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(535) ##STR00161##

(536) Step A. (S)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-

yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (S)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (300 mg, 906 μ mol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (150 mg, 942 μ mol), DIPEA (311 mg, 2.41 mmol) and 4 Å molecular sieves (150 mg) in dioxane (1.8 mL) was stirred at 90° C. for 24 hours under N.sub.2 atmosphere. The reaction mixture was filtered, and the filter cake was washed with dichloromethane (20 mL). The filtrate was concentrated under reduced pressure and the residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=4/1] to give the title compound as light yellow foam (247 mg, 57% yield). LCMS (ESI, M+1): m/z=454.2.

(537) Step B. (S)-1-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (S)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (290 mg, 639 μ mol), K.sub.3PO.sub.4 (1.5 M in water, 1.3 mL) in THF (5.2 mL) was degassed and purged with N.sub.2 for 3 times. cataCXium-A-Pd-G3 (47 mg, 64.5 μ mol) and ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl) ethynyl)triisopropylsilane (460 mg, 898 μ mol) were added. The reaction mixture was stirred at 65° C. for 6 hours. The reaction mixture was diluted with water (1 mL) and brine (1 mL), and then extracted with ethyl acetate (2 mL×4). The combined organic layers were concentrated under reduced pressure and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile-9/11] to give title compound as light yellow foam (410 mg, 77% yield). LCMS (ESI, M+1): m/z=804.1.

(538) Step C. (S)-1-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (S)-1-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (400 mg, 497 μ mol) in DMF (3 mL) was added CsF (760 mg, 5.00 mmol). The mixture was stirred at 15° C. for 1 hour. The reaction mixture was filtered. The filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=3/2] to give the title compound as light yellow solid (310 mg, 95% yield). LCMS (ESI, M+1): m/z=648.2.

(539) Step D. (S)-1-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (S)-1-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 154 μ mol) in MeCN (1.5 mL) was dropwise added HCl/dioxane (4 M, 1.5 mL) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was concentrated under reduced pressure to give a residue. The residue was dissolved in ethyl acetate (5 mL) and treated with saturated NaHCO.sub.3 aqueous (5 mL). The mixture was extracted with ethyl acetate (5 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and concentrated to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75×30 mm×3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 9 min) to give the title compound as yellow solid (59.5 mg, 61% yield, 0.5 FORMIC ACID). .sup.1H NMR (400 MHz, methanol-d4) δ =9.16 (d, J=51.2, 1H); 7.88-7.84 (m, 1H), 7.36-7.29 (m, 2H), 7.23 (dd, J=2.4, 17.2 Hz, 1H), 5.43 (d, J=51.8, 1H), 4.65-4.38 (m, 4H), 3.66-3.35 (m, 6H), 3.27-3.20 (m, 1H), 2.45-1.75 (m, 10H), 1.27 (d, J=20.0, 3H). .sup.19F NMR (377 MHz, methanol-d4) δ =-111.68, -140.68, -173.93. LCMS (ESI, M+1): m/z=604.1.

Example 11

(540) ##STR00162##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-

d]pyrimidin-4-yl)piperidin-4-ol

(541) ##STR00163##

(542) Step A: 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-4-ol: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μ mol) in DMF (2.00 mL) were added piperidin-4-ol (19.1 mg, 189 μ mol) and DIEA (36.5 mg, 283 μ mol, 49.3 μ L). The reaction was degassed and purged with N.sub.2 for 3 times and stirred at 40° C. for 2 hours. Upon completion, the reaction was filtered and purified by prep-HPLC (column: water s Xbridge BEH C18 100 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 20%-60%, 10 min) affording the title compound (24.1 mg, 48% yield, 99.9% purity) as a white solid. .sup.1H NMR (400 MHz, MeOD) δ =9.04 (s, 1H), 8.11 (d, J=8.4 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.74-7.66 (m, 1H), 7.63-7.57 (m, 1H), 7.57-7.48 (m, 1H), 7.19 (dd, J=7.6, 12.8 Hz, 1H), 4.49-4.35 (m, 2H), 4.28 (s, 2H), 4.11-3.97 (m, 1H), 3.87-3.71 (m, 2H), 3.15-3.01 (m, 2H), 2.78-2.63 (m, 2H), 2.15-2.02 (m, 4H), 1.97-1.82 (m, 4H), 1.80-1.69 (m, 4H); LCMS (ESI, M+1): m/z 532.1.

Example 12

(543) ##STR00164##

(3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-yl methylcarbamate

(544) ##STR00165## ##STR00166##

(545) Step A. 2,7-dichloro-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl) oxy)piperidin-1-yl)pyrido[4,3-d]: A mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.00 g, 6.04 mmol), TsOH.Math.H.sub.2O (115 mg, 604 μ mol) and DHP (1.02 g, 12.1 mmol, 1.10 mL) in dichloromethane (20 mL) was stirred at 15° C. for 1 h. After completion, the mixture was concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (2.3 g, 77% yield). Yellow oil. LCMS (ESI, M+1): m/z 415.0

(546) Step B. tert-butyl-[(3R,5S)-5-[[7-chloro-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methyl-pyrrolidin-3-yl)oxy-diphenylsilane: A mixture of 2,7-dichloro-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidine (2 g, 4.82 mmol), [(2S,4R)-4-[tert-butyl (diphenyl)silyl]oxy-1-methyl-pyrrolidin-2-yl]methanol (3.56 g, 9.63 mmol) and DIEA (1.87 g, 14.5 mmol, 2.52 mL) in dioxane (20 mL) was stirred at 90° C. for 12 h. After completion, the mixture was diluted with water (30 mL) and extracted with ethyl acetate (50 mL). The organic layer was washed with brine (50 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography (water (0.1% formic acid)/acetonitrile] to give the title compound (1.7 g, 38% yield). Yellow oil. LCMS (ESI, M+1): m/z 748.2.

(547) Step C. 2-(((2S,4R)-4-((tert-butyl)diphenylsilyl)oxy)-1-methylpyrrolidin-2-yl) methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a mixture of tert-butyl-[(3R,5S)-5-[[7-chloro-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methyl-pyrrolidin-3-yl]oxy-diphenylsilane (1.20 g, 1.60 mmol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (869 mg, 2.41 mmol), K.sub.3PO.sub.4 (1.5 M, 3.21 mL) in toluene (10 mL) was added [cataCXium-A-Pd-G3 (117 mg, 160 μ mol) under N.sub.2. The mixture was stirred at 90° C. for 1.5 h. After completion, the reaction mixture was diluted with ethyl acetate (10 mL) and water (5 mL). The aqueous layer was extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography (water (0.1% formic

acid)/acetonitrile] to give 2-(((2S,4R)-4-((tert-butyl-diphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine (768 mg, 50% yield). Black oil; LCMS (ESI, M+1): m/z 946.5.

(548) Step D. (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol: To a mixture of 2-(((2S,4R)-4-((tert-butyl-diphenylsilyl)oxy)-1-methylpyrrolidin-2-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidine (650 mg, 687 μ mol) and DMF (7 mL) was added CsF (1.56 g, 10.3 mmol), and the mixture was stirred at 40° C. for 16 h. The mixture was filtered and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol (230 mg, 47% yield). Yellow Solid; LCMS [ESI, M+1]: m/z 708.5.

(549) Step E. [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy-naphthyl)]-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] (4-nitrophenyl) carbonate: To a mixture of (3R,5S)-5-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((3R)-3-methyl-3-((tetrahydro-2H-pyran-2-yl)oxy)piperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)-1-methylpyrrolidin-3-ol (40.0 mg, 56.5 μ mol) and (4-nitrophenyl) carbonochloridate (41.0 mg, 203 μ mol) in THF (1 mL) was added t-BuOK (1 M in THF, 170 μ L) at 0° C. The mixture was stirred at 25° C. for 1 h. After completion, the mixture was diluted with ethyl acetate (4 mL) and water (3 mL), and then separated. The aqueous phase was extracted with ethyl acetate (3 mL). The combined organic layer was washed with brine (4 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] (4-nitrophenyl) carbonate (49.0 mg, 73% yield). Yellow solid. LCMS (ESI, M+1): m/z 873.4.

(550) Step F. [(3R,5S)-5-[7-[8-ethyl-7-fluoro-3-(methoxymethoxy-1-naphthyl)]-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl) oxymethyl]-1-methylpyrrolidin-3-yl] N-methylcarbamate: A mixture of [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] (4-nitrophenyl) carbonate (49.0 mg, 56.1 μ mol) and methanamine (2 M in THF, 702 μ L) in DMF (1 mL) was stirred at 25° C. for 0.5 h. After completion, the residue was purified directly by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] N-methylcarbamate (33.0 mg, 59% yield). Yellow oil. LCMS (ESI, M+1): m/z 765.4.

(551) Step G. [(3R,5S)-5-[[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-4-[(3R)-3-hydroxy-3-methyl-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] N-methylcarbamate: To a mixture of [(3R,5S)-5-[[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-4-[(3R)-3-methyl-3-tetrahydropyran-2-yloxy-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] N-methylcarbamate (34.0 mg, 44.4 μ mol) and MeCN (0.5 mL) was added HCl-MeOH (4 M, 1 mL) at 0° C. and the mixture was stirred at a 0° C. for 0.5 h. After completion, the mixture was concentrated in vacuum. Then the pH value was adjusted to 9 with saturated NaHCO.sub.3 solution and the mixture was triturated with methanol (2×8 mL) The filtrate was concentrated in vacuum. The residue was purified by prep-HPLC

(column: Shim-pack C18 150*25*10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 20%-40%, 10 min) to afford [(3R,5S)-5-[[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-4-[(3R)-3-hydroxy-3-methyl-1-piperidyl]pyrido[4,3-d]pyrimidin-2-yl]oxymethyl]-1-methylpyrrolidin-3-yl] N-methylcarbamate (8.34 mg, 28% yield, 0.6 formic acid salt). ¹H NMR (400 MHz, METHANOL-d₄): δ 9.22 (br d, J=3.2 Hz, 1H), 7.68 (br dd, J=5.6, 8.4 Hz, 1H), 7.30 (br d, J=2.0 Hz, 1H), 7.25 (br t, J=9.2 Hz, 1H), 7.06 (br s, 1H), 5.13 (br s, 1H), 4.62-4.53 (m, 3H), 4.31 (br t, J=12.0 Hz, 1H), 3.67-3.56 (m, 2H), 3.49-3.43 (m, 1H), 2.67 (br d, J=15.2 Hz, 8H), 2.53-2.42 (m, 1H), 2.25-2.13 (m, 4H), 1.89-1.75 (m, 3H), 1.29 (br d, J=9.6 Hz, 3H), 0.81 (q, J=7.2 Hz, 3H). ¹⁹F NMR (400 MHz, METHANOL-d₄) δ =-121.123, -139.247. LCMS (ESI, M+1): m/z 637.3.

Example 13

(552) ##STR00167##

(R)-1-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(553) ##STR00168##

(554) Step A. (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (300 mg, 906 μ mol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (150 mg, 942 μ mol), DIPEA (334 mg, 2.58 mmol) and 4 Å molecular sieves (150 mg) in dioxane (1.8 mL) was stirred at 90° C. for 15 hours under N₂ atmosphere. The reaction mixture was filtered. The filter cake was washed with dichloromethane (20 mL). The filtrate was concentrated under reduced pressure, and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=1/9] to give the product as light yellow gum (260 mg, 61% yield). LCMS (ESI, M+1): m/z 454.1.

(555) Step B. (R)-1-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy).sub.8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (150 mg, 330 μ mol), and K₃PO₄ (1.5 M in water, 0.7 mL) in THE (3.5 mL) was degassed and purged with N₂ for 3 times. cataCXium-A-Pd-G3 (25 mg, 34.3 μ mol) was added, followed by ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (228 mg, 445 μ mol). The mixture was stirred at 65° C. for 3 hours. The reaction mixture was diluted with water (5 mL), and extracted with ethyl acetate (5 mL×4). The combined organic layers were concentrated under reduced pressure, and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=7/13] to give the product as yellow foam (219 mg, 79% yield). LCMS (ESI, M+1): m/z 804.4.

(556) Step C. (R)-1-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (375 mg, 466 μ mol) in DMF (3.5 mL) was added CsF (709 mg, 4.67 mmol). The mixture was stirred at 20° C. for 0.5 hour. The reaction mixture was filtered. The filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=11/9] to give the product as yellow solid (270 mg, 87% yield). LCMS (ESI, M+1): m/z 648.4.

(557) Step D. (R).sub.1-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (R)-1-(7-(8-ethynyl-7-fluoro-3-

(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 154 μ mol) in MeCN (3 mL) was added HCl/dioxane (4 M, 2 mL) dropwise at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was concentrated under reduced pressure to give a residue at room temperature (without heating). The residue was dissolved in ethyl acetate (5 mL) and saturated NaHCO₃ aqueous (5 mL). The mixture was extracted with ethyl acetate (5 mL $\times$ 3). The combined organic layers were dried over anhydrous Na₂SO₄, concentrated under reduced pressure, and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=3/2] to give the product as yellow solid (63.2 mg, 63% yield, 0.4 FORMIC ACID). ¹H NMR (400 MHz, methanol-d₄) δ =9.15 (d, J=51.6, 1H), 7.88-7.84 (m, 1H), 7.36-7.32 (m, 2H), 7.23 (dd, J=2.8, 18.4 Hz, 1H), 5.40 (d, J=53.2, 1H), 4.68-4.37 (m, 4H), 3.66-3.35 (m, 6H), 3.21-3.15 (m, 1H), 2.34-1.75 (m, 10H), 1.27 (d, J=20.0, 3H). ¹⁹F NMR (377 MHz, methanol-d₄) δ =-111.66, -140.59, -173.84. LCMS (ESI, M+1): m/z 604.3.

Example 14

(558) ##STR00169##

(R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(559) ##STR00170##

(560) Step A. (R)-1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (400 mg, 1.21 mmol), (hexahydro-1H-pyrrolizin-7a-yl)methanol (239 mg, 1.69 mmol), 4 Å molecular sieves (40.0 mg) in dioxane (8 mL) was added DIEA (468 mg, 3.62 mmol), the mixture was stirred at 90° C. for 3 hours. Upon completion, the reaction solution was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (244 mg, 44% yield). Yellow Solid; LCMS (ESI, M+1): m/z 436.3.

(561) Step B. (R)-1-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of (R)-1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 459 μ mol), ((2-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (291 mg, 642 μ mol), K₃PO₄ (1.5 M, 918 μ L) in THF (3 mL) was added cataCXium-A-Pd-G3 (33.4 mg, 45.9 μ mol) under N₂. The mixture was stirred at 60° C. for 1 hours. Upon completion, the mixture solution was diluted with ethyl acetate (15 mL) and water (5 mL), and the aqueous layer was extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (291 mg, 87% yield). Off-white Solid; LCMS (ESI, M+1): m/z 726.5.

(562) Step C. (R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of (R)-1-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 138 μ mol) in DMF (1.5 mL) was added CsF (209 mg, 1.38 mmol, 10 equiv.). The mixture was stirred at 20° C. for 1 hour. Upon completion, the mixture was filtered and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were lyophilized to afford (R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (51.72 mg, 60% yield, 0.77

FORMIC ACID). Yellow Solid; .sup.1H NMR (400 MHz, METHANOL-d4) δ =9.31-9.07 (m, 1H), 8.18-8.05 (m, 2H), 7.73-7.61 (m, 2H), 7.50-7.39 (m, 1H), 4.74-4.24 (m, 4H), 3.72-3.38 (m, 5H), 3.18-3.01 (m, 2H), 2.34-2.21 (m, 2H), 2.19-1.92 (m, 7H), 1.86-1.66 (m, 3H), 1.36-1.17 (m, 3H); LCMS (ESI, M+1): m/z 570.4.

Example 15

(563) ##STR00171##

(R)-1-(6-(8-ethyl-7-fluoronaphthalen-1-yl)-5-fluoro-3-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-1-yl)-3-methylpiperidin-3-ol

(564) ##STR00172##

(565) Step A. (R)-1-(6-(8-ethyl-7-fluoronaphthalen-1-yl)-5-fluoro-3-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-1-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50.0 mg, 87.8 μ mol) in MeOH (2 mL) was added Pd/C (20 mg, 10% h purity) under N.sub.2. The suspension was degassed in vacuum and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (15 psi) at 20° C. for 1 hour. Upon completion, the mixture was filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 50%-80/a, 10 min) to give (R)-1-(6-(8-ethyl-7-fluoronaphthalen-1-yl)-5-fluoro-3-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-1-yl)-3-methylpiperidin-3-ol (16.83 mg, 33% yield). White Solid; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ =9.14 (d, J=7.2 Hz, 1H), 7.97-7.92 (m, 1H), 7.83-7.75 (m, 1H), 7.52-7.42 (m, 2H), 7.33-7.27 (m, 1H), 4.50-4.37 (m, 2H), 4.32-4.20 (m, 2H), 3.52-3.41 (m, 1H), 3.38-3.27 (m, 1H), 3.17-3.08 (m, 2H), 3.04-2.88 (m, 1H), 2.70-2.52 (m, 3H), 2.32-2.21 (m, 1H), 2.13-2.02 (m, 3H), 1.93-1.83 (m, 5H), 1.79-1.68 (m, 4H), 1.35 (s, 3H), 0.90-0.83 (m, 3H); LCMS (ESI, M+1): m/z 574.4.

Example 16

(566) ##STR00173##

(R)-1-(7-(8-ethyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(567) ##STR00174##

(568) Step A. (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 604 μ mol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (145 mg, 911 μ mol), and 4 Å molecular sieves (20 mg) in dioxane (3 mL) was added DIEA (234 mg, 1.81 mmol). The mixture was stirred at 90° C. for 14 hours. Upon completion, the reaction solution was diluted with ethyl acetate (10 mL) and water (5 mL), and the aqueous layer was extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (168 mg, 56% yield). Yellow Solid; LCMS (ESI, M+1): m/z 454.3.

(569) Step B. (R)-1-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (180 mg, 397 μ mol), ((2-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (234 mg, 517 μ mol), and K.sub.3PO.sub.4 (1.5 M, 793 μ L) in THF (3 mL) was added cataCXium-A-Pd-G3cataCXium-A-Pd-G3 (28.9 mg, 39.7 μ mol) under N.sub.2. The mixture was stirred at 60° C. for 2 hours. Upon completion, the reaction mixture was diluted

with ethyl acetate (15 mL) and water (10 mL). The mixture was extracted with ethyl acetate (15 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (260 mg, 86% yield). Yellow Solid; LCMS (ESI, M+1): m/z 744.5.

(570) Step C. (R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of (R)-1-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (260 mg, 349 μ mol) in DMF (2 mL) was added CsF (531 mg, 3.49 mmol). The mixture was stirred at 20° C. for 0.5 hour. Upon completion, the mixture was filtered and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (190 mg, 89% yield). Yellow Solid; LCMS (ESI, M+1): m/z 588.4.

(571) Step D. (R)-1-(7-(8-ethyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50.0 mg, 85.1 μ mol) in MeOH (2.0 mL) was added Pd/C (20 mg, 10% purity) under N.sub.2. The suspension was degassed in vacuum and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (15 psi) at 20° C. for 1 hour. Upon completion, the mixture was filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 53%-83%, 10 min) to give (R)-1-(7-(8-ethyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (18.2 mg, 36% yield). White Solid; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ =9.15 (d, J=6.8 Hz, 1H), 8.01-7.89 (m, 1H), 7.85-7.75 (m, 1H), 7.55-7.40 (m, 2H), 7.30 (t, J=7.2 Hz, 1H), 5.40-5.16 (m, 1H), 4.50-4.37 (m, 2H), 4.35-4.29 (m, 1H), 4.26-4.18 (m, 1H), 3.54-3.41 (m, 1H), 3.38-3.09 (m, 4H), 3.06-2.85 (m, 2H), 2.64-2.47 (m, 1H), 2.31-2.05 (m, 5H), 2.01-1.86 (m, 4H), 1.80-1.67 (m, 2H), 1.35 (s, 3H), 0.94-0.79 (m, 3H); LCMS (ESI, M+1): m/z 592.4.

Example 17

(572) ##STR00175##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(573) ##STR00176##

(574) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 3-methylpiperidin-3-ol and DIEA (51.1 mg, 396 μ mol) and 4 Å molecular sieves (50 mg) in DMF (2 mL) was added 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (69.9 mg, 132 μ mol) in one portion at 20° C. under N.sub.2. The mixture was heated to 40° C. and stirred for 12 hours. Upon completion, the mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC (column: water s X-bridge 150*25 mm*5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 31%-61%, 10 min) to afford 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (24.34 mg, 33.5% yield); White solid. .sup.1H NMR (400 MHz, methanol-d4) δ 9.20 (d, J=6.0 Hz, 1H) 8.10 (d, J=8.0 Hz, 1H), 7.84 (d, J=8.0 Hz, 1H), 7.70 (t, J=7.2 Hz, 1H), 7.58 (t, J=8.0 Hz, 1H), 7.53 (td, J=5.2 Hz, 8.0

H_z, 1H), 7.19 (dd, J=5.2 Hz, 8.0 Hz, 1H), 4.54 (d, J=8.4 Hz, 1H), 4.33-4.25 (m, 3H), 3.63 (dd, J=2.8 Hz, 13.2 Hz, 1H), 3.50-3.40 (m, 1H), 3.14-3.06 (m, 2H), 2.76-2.68 (m, 2H), 2.23-2.03 (m, 3H), 1.97-1.73 (m, 9H), 1.29 (d, J=5.2 Hz, 3H); LCMS (ESI, M+1): m/z 546.2.

Example 18

(575) ##STR00177##

4-((1R,5S)-3-azabicyclo[3.2.1]octan-3-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(576) ##STR00178##

(577) Step A. 4-((1R,5S)-3-azabicyclo[3.2.1]octan-3-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μmol) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μmol, 49.3 μL) and (1R,5S)-3-azabicyclo[3.2.1]octane (31.4 mg, 283 μmol). The mixture was stirred at 40° C. for 1 hour. Upon completion, the residue was purified by prep-HPLC (column: 3_Phenomenex Luna C18 75×30 mm×3 μm; mobile phase: [water (0.2% formic acid salt)-MeOH]; B %: 25%-60%, 9 min.) affording 4-((1R,5S)-3-azabicyclo[3.2.1]octan-3-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine (22.7 mg, 42% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ 9.06 (s, 1H), 7.99 (br d, J=7.6 Hz, 1H), 7.74 (d, J=8.2 Hz, 1H), 7.66-7.54 (m, 2H), 7.48-7.40 (m, 1H), 7.15-7.08 (m, 1H), 4.70 (br d, J=12.0 Hz, 1H), 4.61 (br d, J=12.0 Hz, 1H), 4.46 (s, 2H), 3.59-3.43 (m, 4H), 2.82-2.72 (m, 2H), 2.43 (br s, 2H), 2.31-2.22 (m, 2H), 2.13-1.92 (m, 4H), 1.90-1.77 (m, 3H), 1.76-1.55 (m, 5H); LCMS [ESI, M+1]: m/z 542.1.

Example 19

(578) ##STR00179##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-((1-methylcyclobutyl)methyl)pyrido[4,3-d]pyrimidin-4-amine

(579) ##STR00180##

(580) Step A. 8-fluoro-7-(8-fluoronaphthalen-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-((1-methylcyclobutyl)methyl)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μmol) in DMF (1.0 mL) were added DIEA (60.9 mg, 471 μmol, 82.1 μL) and (1-methylcyclobutyl)methanamine (38.4 mg, 283 μmol, HCl). The mixture was stirred at 40° C. for 1 hour. Upon completion, the residue was purified by prep-HPLC (column: Phenomenex Luna C18 200×40 mm×0 μm; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 20%-50%, 8 min.) affording 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-((1-methylcyclobutyl)methyl)pyrido[4,3-d]pyrimidin-4-amine (10.4 mg, 19% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ 8.94 (s, 1H), 7.99 (d, J=8.2 Hz, 1H), 7.74 (d, J=8.0 Hz, 1H), 7.65-7.59 (m, 1H), 7.59-7.55 (m, 1H), 7.48-7.40 (m, 1H), 7.14-7.06 (m, 1H), 6.39 (br s, 1H), 4.42 (s, 2H), 3.81-3.66 (m, 2H), 3.49-3.38 (m, 2H), 2.80-2.70 (m, 2H), 2.28-2.14 (m, 2H), 2.07-1.89 (m, 8H), 1.87-1.71 (m, 4H), 1.32-1.20 (m, 3H); LCMS [ESI, M+1]: m/z 530.1.

Example 20

(581) ##STR00181##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(582) ##STR00182##

(583) Step A. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μmol), DIEA (60.9 mg, 471 μmol, 82.1

μL) and 4 Å molecular sieves (5.0 mg) in DMF (2.0 mL) was added 2,7-diazaspiro[4.5]decan-3-one (44.9 mg, 236 μmol, HCl). The mixture was stirred at 40° C. for 12 h. After completion, the mixture was filtered and concentrated. The residue was purified by prep-HPLC [Water s Xbridge 150×25 mm×5 μm; A: water (10 mM NH₄HCO₃), B: ACN; B %: 17%-50% over 9 min] and the desired fractions were lyophilized to afford 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (10.6 mg, 19% yield). Off-white solid; ¹H NMR (400 MHz, CDCl₃) δ=8.99 (s, 1H), 8.00 (br d, J=8.0 Hz, 1H), 7.74 (d, J=8.4 Hz, 1H), 7.66-7.54 (m, 2H), 7.49-7.40 (m, 1H), 7.17-7.06 (m, 1H), 6.03-5.87 (m, 1H), 4.31-4.16 (m, 2H), 4.15-4.02 (m, 2H), 3.80-3.68 (m, 1H), 3.61 (br dd, J=8.0, 12.8 Hz, 1H), 3.43 (dd, J=10.0, 12.8 Hz, 1H), 3.22 (d, J=10.0 Hz, 1H), 3.16-3.06 (m, 2H), 2.70-2.56 (m, 2H), 2.38-2.22 (m, 2H), 2.14-2.03 (m, 2H), 1.94-1.84 (m, 8H), 1.70-1.61 (m, 2H); LCMS (ESI, M+1): m/z 585.3.

Example 21

(584) ##STR00183##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine

(585) ##STR00184##

(586) Step A. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μmol) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μmol, 49.3 μL) and piperidine (12.0 mg, 141 μmol, 14.0 μL). The mixture was stirred at 40° C. for 1 hour. Upon completion, the residue was purified by prep-HPLC (column: water s Xbridge BEH C18 100×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 35%-70%, 10 mins.) affording 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(piperidin-1-yl)pyrido[4,3-d]pyrimidine (12.5 mg, 26% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ 9.01 (s, 1H), 8.02-7.96 (m, 1H), 7.74 (d, J=8.4 Hz, 1H), 7.66-7.56 (m, 2H), 7.48-7.41 m, 1H), 7.16-7.07 (m, 1H), 4.25-4.17 (m, 2H), 3.96 (br s, 4H), 3.16-3.07 (m, 2H), 2.68-2.60 (m, 2H), 2.16-2.06 (m, 2H), 1.91-1.84 (m, 4H), 1.82 (br s, 6H), 1.69-1.62 (m, 2H); LCMS [ESI, M+1]: m/z 516.1.

Example 22

(587) ##STR00185##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine

(588) ##STR00186##

(589) Step A. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)-4-(3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μmol) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μmol, 49.3 μL) and 3-methylpiperidine (28.0 mg, 283 μmol, 33.2 μL). The mixture was stirred at 40° C. for 1 hour. Upon completion, the residue was purified by prep-HPLC (column: water s Xbridge BEH C18 100×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 40%-80%, 10 mins.) affording 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine (7.87 mg, 16% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃) 9.00 (s, 1H), 7.99 (br d, J=7.6 Hz, 1H), 7.74 (d, J=8.2 Hz, 1H), 7.67-7.57 (m, 2H), 7.48-7.41 (m, 1H), 7.15-7.08 (m, 1H), 4.61 (br d, J=13.2 Hz, 1H), 4.49 (br d, J=12 Hz, 1H), 4.29-4.16 (m, 2H), 3.26-3.17 (m, 1H), 3.15-3.07 (m, 2H), 3.01-2.88 (m, 1H), 2.68-2.60 (m, 2H), 2.15-2.06 (m, 2H), 2.04-1.74 (m, 8H), 1.69-1.59 (m, 2H), 1.39-1.25 (m, 1H), 1.02 (d, J=6.6 Hz, 3H); LCMS [ESI, M+1]: m/z 530.1.

Example 23

(590) ##STR00187##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol

(591) ##STR00188##

(592) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μ mol) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μ mol, 49.3 μ L) and piperidin-3-ol (28.6 mg, 283 μ mol, 33.2 μ L). The mixture was stirred at 40° C. for 1 hour. Upon completion, the residue was purified by prep-HPLC (column: water s Xbridge BEH C18 100 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH₄sub.4HCO₃sub.3)-ACN]; B %: 20%-60%, 10 mins.) affording 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol (17.7 mg, 35% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ 9.10 (d, J=2.2 Hz, 1H), 7.99 (br d, J=8.0 Hz, 1H), 7.74 (d, J=8.2 Hz, 1H), 7.65-7.56 (m, 2H), 7.48-7.41 (m, 1H), 7.15-7.08 (m, 1H), 4.29-4.19 (m, 2H), 4.10 (br s, 1H), 4.06-3.88 (m, 4H), 3.24 (br s, 1H), 3.15-3.05 (m, 2H), 2.72-2.58 (m, 2H), 2.14-1.93 (m, 5H), 1.93-1.76 (m, 5H), 1.67-1.60 (m, 2H); LCMS [ESI, M+1]: m/z 532.1.

Example 24

(593) ##STR00189##

7-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(594) ##STR00190## ##STR00191##

(595) Step A. 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (2.0 g, 6.85 mmol, 86% purity) in dichloromethane (20 mL) were added DIEA (3.54 g, 27.4 mmol, 4.77 mL) and 2,7-diazaspiro[4.5]decan-3-one (1.44 g, 7.53 mmol, HCl) at -40° C. The mixture was stirred at -40° C. for 1 h. After completion, the mixture was added water (20 mL) and extracted with EtOAc (2 $\times$ 20 mL). The combined organic layer was dried over anhydrous Na₂SO₄ and filtered. The mixture was concentrated in vacuum. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (1.60 g, 63% yield). Yellow solid; LCMS (ESI, M+1): m/z 370.0.

(596) Step B. 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (300 mg, 810 μ mol), DIEA (314 mg, 2.43 mmol, 423 μ L) and 4 Å molecular sieves (10 mg) in dioxane (2.0 mL) was added ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (142 mg, 891 μ mol). The mixture was stirred at 95° C. for 16 h. After completion, water (5.0 mL) was added and the mixture was extracted with EtOAc (2 $\times$ 5.0 mL). The combined organic layer was dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuum. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording the title compound (120 mg, 19% yield). Yellow solid; LCMS (ESI, M+1): m/z 493.2.

(597) Step C. 7-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (100 mg, 203 μ mol), ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)

ethynyl)triisopropylsilane (208 mg, 406 μmol) and K.sub.3PO.sub.4 (1.5 M, 406 μL) in THF (3.0 mL) was cataCXium-A-Pd-G3 (14.8 mg, 20.3 μmol). The mixture was stirred at 60° C. for 2 h. After completion, water (5.0 mL) was added and the mixture was extracted with EtOAc (2 $\times$ 5.0 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4 and filtered. The mixture was concentrated in vacuum. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording 7-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (80 mg, 45% yield). Yellow solid; LCMS (ESI, M+1): m/z 843.5.

(598) Step D. 7-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (75.0 mg, 89.0 μmol) in DMF (1.5 mL) was added CsF (40.5 mg, 267 μmol , 9.84 μL). The mixture was stirred at 20° C. for 1 h. After completion, water (5.0 mL) was added and the mixture was extracted with EtOAc (2 $\times$ 5.0 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording 7-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (60 mg, 98% yield). Yellow solid; LCMS (ESI, M+1): m/z 687.3.

(599) Step E. 7-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (55.0 mg, 80.1 μmol) in MeCN (0.5 mL) was added HCl.dioxane (4 M, 1.0 mL). The mixture was stirred at 20° C. for 0.5 h. After completion, the mixture was concentrated at 20° C. The residue was purified by prep-HPLC [Water s Xbridge 150 $\times$ 25 mm $\times$ 5 μm ; A: water (10 mM NH.sub.4HCO.sub.3), B: ACN; B %: 33%-63% over 10 min] and the desired fractions were lyophilized to afford 7-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (6.16 mg, 12% yield). Yellow solid; ¹H NMR (400 MHz, DMSO-d.sub.6) δ =10.20 (s, 1H), 9.02 (d, J=5.6 Hz, 1H), 7.98 (dd, J=6.0, 9.2 Hz, 1H), 7.74-7.59 (m, 1H), 7.47 (t, J=9.2 Hz, 1H), 7.40 (d, J=2.4 Hz, 1H), 7.20 (t, J=2.4 Hz, 1H), 5.56-5.23 (m, 1H), 4.48-4.11 (m, 2H), 4.11-3.86 (m, 3H), 3.86-3.67 (m, 2H), 3.26-2.89 (m, 4H), 2.52 (br s, 2H), 2.34-2.09 (m, 4H), 2.07-1.71 (m, 8H); LCMS (ESI, M+1): m/z 643.3.

Example 25

(600) ##STR00192##

1-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(601) ##STR00193##

(602) Step A. 5,6-dimethyl-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indazole: To a solution of 4-bromo-5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazole (2.00 g, 6.47 mmol), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (3.29 g, 12.9 mmol) and KOAc (1.90 g, 19.4 mmol) in dioxane (40 mL) was added Pd(dppf)Cl.sub.2 (473 mg, 647 μmol). The mixture was stirred at 110° C. for 1 hour. After completion, water (30 mL) was added and the mixture was extracted with EtOAc (2 $\times$ 20 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and

concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, petroleum ether: ethyl acetate=100:1-15:1) affording 5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indazole (1.60 g, 67% yield). Yellow oil; LCMS (ESI, M+1): m/z=357.0.

(603) Step B. 1-(7-(5,6-dimethyl-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (290 mg, 665 μ mol) and 5,6-dimethyl-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indazole (355 mg, 998 μ mol) in THF (6 mL) was added K.sub.3PO.sub.4 (1.5 M, 1.33 mL) in one portion at 25° C. under N.sub.2. Then cataCXium-A-Pd-G3 (96.90 mg, 133 μ mol) was added. The mixture was heated to 60° C. and stirred for 2 h. The mixture was filtered and concentrated in vacuum. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (300 mg, 69% h yield). LCMS (ESI, M+1): m/z 630.5.

(604) Step C. 1-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (290 mg, 460 μ mol) and ACN (0.2 mL) was added HCl.dioxane (4 M, 1.98 μ L) in one portion at 25° C. under N.sub.2. The mixture was stirred at 25° C. for 10 minutes. The mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC to afford the title compound (76.4 mg, 30% yield). .sup.1H NMR 400 MHz, (DMSO-d.sub.6) δ 13.0 (br s, 1H), 9.30 (d, J=9.8 Hz, 1H), 8.23 (s, 1H), 7.33-7.23 (m, 2H), 4.20-4.15 (m, 1H), 4.17-4.02 (m, 3H), 3.31-3.27 (m, 2H), 3.05-2.95 (m, 2H), 2.35-2.27 (m, 2H), 2.44 (s, 3H), 2.13 (d, J=4.4 Hz, 3H), 2.04-1.27 (m, 12H), 1.18 (s, 3H). LCMS (ESI, M+1): m/z 546.1.

Example 26

(605) ##STR00194##

7-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(606) ##STR00195##

(607) Step A. 7-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (300 mg, 810 μ mol), 4 Å molecular sieves (10 mg) and DIEA (314 mg, 2.43 mmol, 423 μ L) in dioxane (2 mL) was added (hexahydro-1H-pyrrolizin-7a-yl)methanol (126 mg, 891 μ mol). The mixture was stirred at 95° C. for 16 h. After completion, water (5.0 mL) was added and the mixture was extracted with EtOAc (2×5.0 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) to afford 7-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (130 mg, 33% yield). Yellow solid; LCMS (ESI, M+1): m/z=475.3.

(608) Step B. 7-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (130 mg, 274 μ mol), 5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indazole (243.78 mg, 684.27 μ mol) and K.sub.3PO.sub.4 (1.5 M, 547 μ L) in THF (3 mL) was added cataCXium-A-Pd-G3 (39.9 mg, 54.7 μ mol). The mixture was stirred at 60° C. for 2 hour. After completion, to the reaction mixture was added water (5.0 mL), and the mixture extracted with EtOAc (2×5.0 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4,

filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording 7-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (120 mg, 65% yield). Yellow solid; LCMS (ESI, M+1): m/z=669.3.

(609) Step C. 7-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (30 mg, 44.9 μ mol) in MeCN (0.5 mL) was added HCl.dioxane (4 M, 1.0 mL). The mixture was stirred at 20° C. for 0.5 hour. After completion, the mixture was concentrated at 20° C. The residue was purified by prep-HPLC [Water s Xbridge 150×25 mm×5 μ m; A: water (10 mM NH₄sub.4HCO₃sub.3), B: ACN, B %: 16%-46% over 10 min) to afford the title compound (7.08 mg, 26% yield). White solid; ¹H NMR (400 MHz, CDCl₃) δ =10.87-10.17 (m, 1H), 8.98 (d, J=4.0 Hz, 1H), 7.56 (d, J=13.2 Hz, 1H), 7.31 (br d, J=4.8 Hz, 1H), 6.55-5.99 (m, 1H), 4.28-4.12 (m, 2H), 4.02 (br d, J=12.8 Hz, 2H), 3.66-3.52 (m, 1H), 3.42 (br d, J=12.0 Hz, 1H), 3.31 (br t, J=10.0 Hz, 1H), 3.14-3.00 (m, 3H), 2.64-2.52 (m, 2H), 2.46-2.36 (m, 3H), 2.19 (s, 2H), 2.13 (br d, J=5.2 Hz, 3H), 2.07-1.97 (m, 2H), 1.88-1.74 (m, 8H), 1.60-1.54 (m, 2H); LCMS (ESI, M+1): m/z=585.2.

Example 27

(610) ##STR00196##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(611) ##STR00197##

(612) Step A. (R)-1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (125 mg, 377 μ mol), (hexahydro-1H-pyrrolizin-7a-yl)methanol (56 mg, 396 μ mol), DIPEA (133 mg, 1.03 mmol) and 4 Å molecular sieves (56 mg) in dioxane (0.7 mL) was stirred at 90° C. for 21 hours under N₂ atmosphere. The reaction mixture was filtered and the filter cake was washed with dichloromethane (20 mL). The filtrate was concentrated under reduced pressure and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=7/3] to give the product as white solid (40 mg, 21% yield). LCMS (ESI, M+1): m/z 436.2.

(613) Step B. (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (R)-1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (40 mg, 91.8 μ mol), K₃PO₄ (1.5 M in water, 0.2 mL) in THF (1 mL) was degassed and purged with N₂ for 3 times. cataCXium-A-Pd-G3 (14 mg, 19.2 μ mol) was added and followed by 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (50 mg, 139 μ mol). The mixture was stirred at 65° C. for 4 hours. The reaction mixture was diluted with water (2 mL) and extracted with ethyl acetate (2 mL×4). The combined organic layers were concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=13/7] to give the product as brown gum (41 mg, 65% yield). LCMS (ESI, M+1): m/z 634.1.

(614) Step C. (R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (41 mg, 64.7 μ mol) in MeCN (2 mL) was added HCl/dioxane (4 M, 1.5 mL) dropwise at 0° C. The mixture was stirred

at 0° C. for 0.5 hour. The reaction mixture was concentrated under reduced pressure to give a residue at room temperature. The residue was dissolved in ethyl acetate (5 mL) and saturated NaHCO₃ aqueous (5 mL). The mixture was extracted with ethyl acetate (5 mL×4). The combined organic layers were dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 minutes) to give the product as yellow solid (9.02 mg, 22% yield, 0.7 FORMIC ACID). ¹H NMR (400 MHz, methanol-d₄) δ=9.26 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.06 (t, J=2.4 Hz, 1H), 4.62-3.55 (m, 3H), 4.34 (t, J=11.6 Hz, 1H), 3.67-3.61 (m, 1H), 3.55-3.40 (m, 3H), 3.14-3.11 (m, 2H), 2.25-2.15 (m, 1H), 2.13-2.03 (m, 10H), 2.02-1.73 (m, 3H), 1.30 (d, J=9.2 Hz, 3H), 0.81 (q, J=7.6 Hz, 3H). ¹⁹F NMR (377 MHz, methanol-d₄) δ=-121.11, -139.56. LCMS (ESI, M+1): m/z 590.4.

Example 28

(615) ##STR00198##

(3R)-1-[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol

(616) ##STR00199##

(617) Step A. (3R)-1-[7-chloro-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: A mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (150 mg, 453 µmol), [(2S)-1-methylpyrrolidin-2-yl]methanol (157 mg, 1.36 mmol, 161 µl), DIEA (176 mg, 1.36 mmol, 237 µl) and 4 Å molecular sieves (15.0 mg, 453 µmol) in dioxane (2 mL) was stirred at 90° C. for 15 hours. After completion, the mixture was diluted with ethyl acetate (5 mL) and water (5 mL), and then separated. The aqueous phase was extracted with ethyl acetate (5 mL). The combined organic layer was washed with brine (8 mL) and dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (3R)-1-[7-chloro-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (135 mg, 70% yield). Yellow oil. LCMS (ESI, M+1): m/z 410.2.

(618) Step B. (3R)-1-[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: To a mixture of (3R)-1-[7-chloro-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (135 mg, 329 µmol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (237 mg, 659 µmol) and K₃PO₄ (1.5 M, 659 µl) in THF (2.1 mL) was added [2-(2-aminophenyl)phenyl]palladium(1+); bis(1-adamantyl)-butyl-phosphane; methanesulfonate (24.0 mg, 32.9 µmol) under N₂. The mixture was degassed and then heated to 60° C. for 3 hours under N₂. After completion, the mixture was diluted with ethyl acetate (2 mL) and water (2 mL), and then separated. The aqueous phase was extracted with ethyl acetate (3 mL). The combined organic layer was washed with brine (4 mL) and dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (3R)-1-[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (100 mg, 50% yield). Yellow oil; LCMS (ESI, M+1): m/z 608.3.

(619) Step C. (3R)-1-[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-2-[[[(2S')-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: To a mixture of (3R)-1-[7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (40.0 mg, 65.8 µmol) in MeCN (0.5 mL) was added HCl.dioxane (4 M, 1 mL) at 0° C. and the mixture was

stirred at 0° C. for 0.5 hour. After completion, the mixture was concentrated in vacuum. Then the pH value was adjusted to 9 with saturated NaHCO₃ solution and the mixture was washed with methanol (2×8 mL). The mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC (Shim-pack C18 150×25×10 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 20%-40%, 10 min) to afford (3R)-1-[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-2-[[[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (14.4 mg, 36% yield, 0.6 FORMIC ACID). White solid. ¹H NMR (400 MHz, METHANOL-d₄): δ 9.24 (d, J=3.2 Hz, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (br s, 1H), 4.78-4.71 (m, 1H), 4.66-4.56 (m, 2H), 4.33 (br t, J=11.6 Hz, 1H), 3.66-3.59 (m, 1H), 3.54-3.41 (m, 3H), 3.00-2.93 (m, 1H), 2.89 (s, 3H), 2.51-2.41 (m, 1H), 2.35-2.27 (m, 1H), 2.23-2.13 (m, 2H), 2.09-1.96 (m, 3H), 1.89-1.76 (m, 3H), 1.29 (br d, J=8.8 Hz, 3H), 0.81 (q, J=7.6 Hz, 3H). ¹⁹F NMR (400 MHz, METHANOL-d₄) δ = -121.070, -139.482. HPLC: >99% ee, Chiralcel OJ-3 50×4.6 mm I.D., 3 µm; Mobile phase: Phase A for CO₂, and Phase B for MeOH (0.05% DEA); Gradient elution: MeOH (0.05% DEA) in CO₂ from 5% to 40%; Flow rate: 3 mL/min; Detector: PDA; Column Temp: 35° C.; Back Pressure: 100 Bar". LCMS (ESI, M+1): m/z 564.3.

Example 29

(620) ##STR00200##

4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one

(621) ##STR00201##

(622) Step A. 4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 µmol) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 µmol, 49.3 µL) and 4-(aminomethyl)pyrrolidin-2-one (42.6 mg, 283 µmol, HCl). The mixture was stirred at 40° C. for 1 hour. Upon completion, the residue was purified by prep-HPLC (column: Phenomenex Luna C18 200×40 mm×10 µm; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 20%-50%, 8 mins) to afford 4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one (18.5 mg, 33% yield, formic acid salt) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ 9.45-9.15 (m, 2H), 8.01-7.89 (m, 1H), 7.74-7.70 (m, 1H), 7.65-7.50 (m, 2H), 7.48-7.35 (m, 1H), 7.17-7.05 (m, 1H), 6.34 (br d, J=72.0 Hz, 1H), 4.55-4.34 (m, 2H), 3.79-3.64 (m, 2H), 3.60-3.52 (m, 1H), 3.48-3.33 (m, 2H), 3.20-3.03 (m, 1H), 2.96-2.71 (m, 3H), 2.47-2.35 (m, 1H), 2.35-2.17 (m, 2H), 2.16-2.03 (m, 3H), 2.02-1.89 (m, 2H), 1.89-1.75 (m, 2H); ¹H NMR (400 MHz, MeOD-d₄) δ 9.20 (s, 1H), 8.13 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.75-7.66 (m, 1H), 7.61-7.57 (m, 1H), 7.54 (dt, J=5.0, 8.0 Hz, 1H), 7.22-7.15 (m, 1H), 4.61-4.57 (m, 2H), 3.97-3.84 (m, 1H), 3.81-3.75 (m, 1H), 3.66-3.53 (m, 3H), 3.30-3.27 (m, 1H), 3.22-3.10 (m, 2H), 3.08-2.92 (m, 1H), 2.60-2.51 (m, 1H), 2.35-2.24 (m, 3H), 2.24-2.09 (m, 4H), 2.09-1.96 (m, 2H); LCMS (ESI, M+1): m/z 545.1.

Example 30

(623) ##STR00202##

4-(3-(1H-1,2,4-triazol-3-yl)piperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(624) ##STR00203##

(625) Step A. 4-(3-(1H-1,2,4-triazol-3-yl)piperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 µmol) in DMF (1.00 mL) was added DIEA (73.1 mg, 566 µmol) and 3-(1H-1,2,4-triazol-3-yl)piperidine (35.6 mg, 189 µmol, HCl). The

mixture was stirred at 40° C. for 8 hours. The mixture was filtered and was purified by prep-HPLC (column: 3_Phenomenex Luna C18 75×30 mm×3 µm; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 15%-45%, 8 min) to afford 4-(3-(1H-1,2,4-triazol-3-yl) piperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidine (23.4 mg, 42% yield) as a white solid. .sup.1H NMR (400 MHz, MeOD) δ 9.13 (s, 1H), 8.55 (s, 0.6H), 8.30 (s, 1H), 8.13 (d, J=7.9 Hz, 1H), 7.86 (d, J=8.1 Hz, 1H), 7.71 (t, J=7.7 Hz, 1H), 7.62 (d, J=7.1 Hz, 1H), 7.56-7.51 (m, 1H), 7.22-7.17 (m, 1H), 4.65-4.49 (m, 3H), 3.91-3.67 (m, 2H), 3.49-3.45 (m, 2H), 3.36 (d, J=5.5 Hz, 2H), 3.14-3.05 (m, 2H), 2.34 (d, J=9.2 Hz, 1H), 2.26-2.21 (m, 2H), 2.16-2.04 (m, 6H), 2.02-1.91 (m, 3H); LCMS (ESI, M+1): m/z 583.2.

Example 31

(626) ##STR00204##

4-(2-azabicyclo[4.1.0]heptan-2-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(627) ##STR00205##

(628) Step A. 4-(2-azabicyclo[4.1.0]heptan-2-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 µmol) in DMF (1.00 mL) were added DIEA (36.5 mg, 283 µmol) and 2-azabicyclo[4.1.0]heptane (25.2 mg, 189 µmol, HCl). The mixture was stirred at 50° C. for 16 hours. The mixture was filtered and was purified by prep-HPLC (column: water s Xbridge BEH C18 100×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 25%-65%, 12 min) to afford 4-(2-azabicyclo[4.1.0]heptan-2-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine (12.1 mg, 24% yield) as a white solid. .sup.1H NMR (400 MHz, MeOD) δ 9.89 (s, 1H), 8.11 (d, J=8.2 Hz, 1H), 7.85 (d, J=8.1 Hz, 1H), 7.73-7.68 (m, 1H), 7.63-7.58 (m, 1H), 7.57-7.49 (m, 1H), 7.24-7.16 (m, 1H), 4.35-4.22 (m, 2H), 3.43-3.40 (m, 2H), 3.15-3.06 (m, 2H), 3.02-2.96 (t, J=12.5 Hz, 1H), 2.76-2.69 (m, 2H), 2.12-2.05 (m, 4H), 1.97-1.83 (m, 5H), 1.80-1.69 (m, 3H), 1.59-1.54 (d, J=2.8 Hz, 1H), 1.29-1.24 (td, J=5.8, 9.6 Hz, 1H), 0.88-0.80 (m, 1H); LCMS (ESI, M+1): m/z 528.2.

Example 32

(629) ##STR00206##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(octahydro-1H-cyclopenta[b]pyridin-1-yl)pyrido[4,3-d]pyrimidine

(630) ##STR00207##

(631) Step A. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)-4-(octahydro-1H-cyclopenta[b]pyridin-1-yl)pyrido[4,3-d]pyrimidine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 µmol) in DMF (1 mL) were added DIEA (36.5 mg, 283 µmol) and octahydro-1H-cyclopenta[b]pyridine (30.5 mg, 243 µmol). The mixture was stirred at 50° C. for 16 hours. The mixture was filtered and was purified by prep-HPLC (column: water s Xbridge BEH C18 100×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 40%-80%, 12 min) to afford 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(octahydro-1H-cyclopenta[b]pyridin-1-yl)pyrido[4,3-d]pyrimidine (2.07 mg, 4% yield) as a white solid. .sup.1H NMR (400 MHz, MeOD) δ 9.02 (s, 1H), 8.11 (d, J=8.3 Hz, 1H), 7.85 (d, J=8.2 Hz, 1H), 7.72-7.68 (m, 1H), 7.63-7.56 (m, 1H), 7.55-7.51 (m, 1H), 7.22-7.17 (m, 1H), 5.04-4.99 (m, 1H), 4.65-4.62 (m, 1H), 4.35-4.26 (m, 2H), 3.58-3.43 (m, 1H), 3.17-3.07 (m, 2H), 2.81-2.69 (m, 2H), 2.22-2.05 (m, 5H), 2.00-1.84 (m, 7H), 1.83-1.74 (m, 5H), 1.67-1.52 (m, 2H); LCMS (ESI, M+1): m/z 556.2.

Example 33

(632) ##STR00208##

(3R,5S)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-methylpiperidin-3-ol

(633) ##STR00209##

(634) Step A. (3R,5S)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-methylpiperidin-3-ol: To a solution of (3R,5S)-5-methylpiperidin-3-ol (15 mg, 98.9 μ mol, HCl) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (52.5 mg, 98.9 μ mol) in DMF (0.1 mL) was added DIEA (63.9 mg, 495 μ mol). The mixture was stirred at 40° C. for 1 hour. The reaction mixture was quenched by water (10 mL) at 20° C., and extracted with EtOAc (4 mL $\times$ 3). The combined organic layers were washed with brine (5 mL), dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge BEH C18 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (0.05% NH₃.Math.H₂O+10 mM NH₄HCO₃)-ACN]; B %: 30%-60%, 8 min) to give the title compound (8.92 mg, 16% yield) as a light yellow solid. ¹H NMR (400 MHz, DMSO-d₆): δ 9.07 (d, J=2.2 Hz, 1H), 8.18 (br d, J=8.2 Hz, 1H), 7.93 (d, J=8.2 Hz, 1H), 7.78-7.70 (m, 1H), 7.67-7.54 (m, 2H), 7.31 (dd, J=7.6, 13.2 Hz, 1H), 5.20 (d, J=4.2 Hz, 1H), 4.60 (br d, J=10.3 Hz, 1H), 4.42 (br d, J=10.9 Hz, 1H), 4.11-4.00 (m, 2H), 3.75 (td, J=4.8, 9.8 Hz, 1H), 2.96-2.88 (m, 3H), 2.88-2.80 (m, 1H), 2.58-2.51 (m, 3H), 2.07 (brd, J=12.1 Hz, 1H), 1.93-1.84 (m, 3H), 1.84-1.69 (m, 4H), 1.62-1.51 (m, 2H), 1.16 (q, J=11.7 Hz, 1H), 0.98 (d, J=6.6 Hz, 3H); LCMS (ESI, M+1): m/z 546.2.

Example 34

(635) ##STR00210##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(imidazo[1,2-a]pyrazin-3-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(636) ##STR00211##

(637) Step A: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(imidazo[1,2-a]pyrazin-3-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of imidazo[1,2-a]pyrazin-3-amine (3.0 equiv.) in DMF (1 mL) were added 4 Å molecular sieves (20 mg), t-BuONa (2.00 equiv.) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (0.09 mmol, 1.00 equiv.) at 0° C. The resulting mixture was stirred at 0° C. for 0.5 hour. After completion, the reaction mixture was diluted with water (10.0 mL) and extracted with ethyl acetate (3 $\times$ 10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 27%-57%, 10 min) to afford the title compound. LCMS (ESI, M+1): m/z 565.3.

Example 35

(638) ##STR00212##

4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-4-methylpyrrolidin-2-one

(639) ##STR00213##

(640) Step A. 4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-4-methylpyrrolidin-2-one: To a mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 mg, 37.7 μ mol) and 4-(aminomethyl)-4-methylpyrrolidin-2-one (96.6 mg, 754 μ mol) in DMF (2.00 mL) was added DIEA (24.4 mg, 189 μ mol) at 25° C. under N₂. The mixture was stirred at 40° C. for 2 hours. The reaction mixture was quenched with water (10.0 mL) and extracted with ethyl acetate (3 $\times$ 10.0 mL). The combined organic layers were washed with brine (20.0 mL), dried over sodium sulfate. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by

prep-HPLC (column: Phenomenex Luna C18 75×30 mm×3 μm; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 20%-60%, 8 min) to give 4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-4-methylpyrrolidin-2-one (7.29 mg, 34% yield) as white solid. ¹H NMR (400 MHz, DMSO-d₆): δ 9.39 (d, J=2.42 Hz, 1H), 8.90 (s, 1H), 8.18 (d, J=7.72 Hz, 1H), 7.93 (d, J=7.94 Hz, 1H), 7.77-7.69 (m, 1H), 7.65-7.48 (m, 3H), 7.30 (dd, J=12.58, 8.16 Hz, 1H), 4.12 (s, 2H), 3.69 (d, J=5.07 Hz, 2H), 3.00-2.96 (m, 2H), 2.61 (d, J=6.84 Hz, 2H), 2.39 (d, J=18.08 Hz, 2H), 2.00-1.52 (m, 8H), 1.64-1.60 (m, 2H), 1.19 (s, 3H); LCMS (ESI, M+1): m/z 559.1

Example 36

(641) ##STR00214##

7-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(642) ##STR00215##

(643) Step A. 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (2.0 g, 6.85 mmol) in dichloromethane (20 mL) were added DIEA (3.54 g, 27.4 mmol) and 2,7-diazaspiro[4.5]decan-3-one (1.44 g, 7.53 mmol, HCl) at -40° C. The mixture was stirred at -40° C. for 1 hour. The mixture was diluted with water (20 mL), and extracted with EtOAc (2×20 mL). The combined organic layer was dried over Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) to afford 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (1.6 g, 63% yield) as yellow solid. LCMS (ESI, M+1): m/z 370.0

(644) Step B. 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (300 mg, 810 μmol), DIEA (314 mg, 2.43 mmol) and 4 Å molecular sieves (10 mg) in dioxane (2 mL) was added ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (142 mg, 891 μmol). The mixture was stirred at 95° C. for 16 hours. The mixture was diluted with water (5 mL), and extracted with EtOAc (2×5 mL). The combined organic layer was dried over Na₂SO₄, filtered and concentrated. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) to afford 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (120 mg, 19% yield) as yellow solid. LCMS (ESI, M+1): m/z 493.2

(645) Step C. 7-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (300 mg, 608 μmol), ((2-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (413 mg, 913 μmol) and K₃PO₄ (1.5 M, 1.22 mL) in THF (6 mL) was added cataCXium-A-Pd-G3 (66.5 mg, 91.3 μmol). The mixture was stirred at 60° C. for 2 hours. The mixture was diluted with water (10 mL), extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (100 mL), and dried over Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, 0.1% formic acid in water, 0-40% ACN) to afford 7-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (350 mg, 70% yield) as yellow solid. LCMS (ESI, M+1): m/z 783.4

(646) Step D. 7-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-

fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (350 mg, 447 μ mol) in DMF (4 mL) was added CsF (339 mg, 2.23 mmol). The mixture was stirred at 25° C. for 1 hour. The mixture was diluted with water (10 mL), extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were washed with brine (100 mL), and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) to give a crude product. The crude product was purified by prep-HPLC (column: Phenomenex luna C18 150 $\times$ 25 mm $\times$ 10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 min) to afford 7-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (205 mg, 0.5 FA, 70% yield) as yellow solid. .sup.1H NMR (400 MHz, methanol-d₄): δ 9.06 (dd, J=1.2, 2.8 Hz, 1H), 8.50 (s, 1H), 8.18-8.08 (m, 2H), 7.71-7.61 (m, 2H), 7.45 (dt, J=0.8, 9.2 Hz, 1H), 5.56-5.31 (m, 1H), 4.53-4.14 (m, 4H), 4.13-3.79 (m, 2H), 3.73-3.43 (m, 4H), 3.29-3.16 (m, 2H), 2.56-2.23 (m, 5H), 2.20-2.01 (m, 3H), 1.99-1.80 (m, 4H); LCMS (ESI, M+1): m/z 627.3

Example 37

(647) ##STR00216##

7-(7-(8-ethyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(648) ##STR00217##

(649) Step A. 7-(7-(8-ethyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (50.0 mg, 77.0 μ mol, 0.5 FORMIC ACID) in MeOH (1 mL) was added Pd/C (5.0 mg, 10% h purity). The suspension was degassed and purged with H.sub.2 3 times. The mixture was stirred at 25° C. for 1 hour under H.sub.2 (15 psi). The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 33%-63%, 10 min) to afford the title compound (17.0 mg, 34% yield) as off-white solid. .sup.1H NMR (400 MHz, CHLOROFORM-d) δ 8.98 (d, J=4.0 Hz, 1H), 7.95 (dd, J=1.2, 8.0 Hz, 1H), 7.80 (dd, J=6.0, 9.2 Hz, 1H), 7.53-7.40 (m, 2H), 7.30 (t, J=9.2 Hz, 1H), 5.84 (br d, J=14.8 Hz, 1H), 5.41-5.19 (m, 1H), 4.34-4.17 (m, 2H), 4.11-3.96 (m, 2H), 3.90-3.63 (m, 2H), 3.40 (dd, J=4.8, 9.6 Hz, 1H), 3.32-3.15 (m, 4H), 3.04-2.92 (m, 1H), 2.65-2.49 (m, 1H), 2.39-2.27 (m, 3H), 2.26-2.14 (m, 3H), 2.00-1.93 (m, 2H), 1.93-1.84 (m, 5H), 0.86 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z 631.3

Example 38

(650) ##STR00218##

7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(651) ##STR00219## ##STR00220##

(652) Step A. 7-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl).sub.2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (100 mg, 202.86 μ mol) and ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (156 mg, 304 μ mol) in THF (2 mL) were added K.sub.3PO.sub.4 (1.5 M, 406 μ L) and cataCXium-A-Pd-G3 (22.2 mg, 30.4 μ mol). The mixture was stirred at 60° C. for 2 hours. The reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (3 $\times$ 20 mL). The combined organic layers were washed with brine (100 mL), and dried over

Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, 0.1% formic acid in water, 0-40% ACN) to afford the title compound (120 mg, 67% yield) as yellow solid. LCMS (ESI, M+1): m/z 843.4

(653) Step B. 7-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (110 mg, 130 μ mol) and DMF (2 mL) was added CsF (198 mg, 1.30 mmol) in one portion at 25° C. under N.sub.2. The mixture was stirred at 25° C. for 0.5 hour. The residue was filtered. The filtrate was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) to afford 7-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (75.0 mg, 80% yield) as yellow solid. LCMS (ESI, M+1): m/z 687.4

(654) Step C. 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (75.0 mg, 109 μ mol) and MeOH (2 mL) was added Pd/C (100 mg, 10% purity) in one portion at 25° C. under N.sub.2. The suspension was degassed and purged with H.sub.2 3 times. The mixture was stirred at 25° C. for 1 hour under H.sub.2 (15 psi). The reaction mixture was filtered and concentrated under reduced pressure to afford 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (65.0 mg, 86% yield) as yellow solid. LCMS (ESI, M+1): m/z 691.3

(655) Step D. 7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a mixture of 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (65.0 mg, 94.1 μ mol) and ACN (1 mL) was added HCl-dioxane (4 M, 862 μ L) in one portion at 25° C. under N.sub.2. The mixture was stirred at 25° C. for 10 minutes. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 7 min) to afford 7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (15.3 mg, 24% yield, 0.1 FORMIC ACID) as white solid. .sup.1H NMR (400 MHz, methanol-d4): δ 9.10 (s, 1H), 7.70 (dd, J=5.6 Hz, J=8.8 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.27 (t, J=9.2 Hz, 1H), 7.07 (d, J=2.4 Hz, 1H), 5.37 (d, J=5.4 Hz, 1H), 4.60 (s, 1H), 4.45-3.90 (m, 6H), 3.51-3.35 (m, 3H), 3.30-3.26 (m, 1H), 3.16-3.06 (m, 1H), 2.55-2.26 (m, 5H), 2.25-2.15 (m, 2H), 2.10-1.85 (m, 7H), 0.87-0.75 (m, 3H); LCMS (ESI, M+1): m/z 647.4

Example 39

(656) ##STR00221##

1-(7-(2-amino-5,6-dimethylbenzo[d]thiazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(657) ##STR00222##

(658) Step A. 1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (0.200 g, 604 μ mol), ((2R,7aS)-2-

fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (144 mg, 906 μ mol), DIEA (234 mg, 1.81 mmol, 315 μ L) and 4 Å molecular sieves (20 mg) in dioxane (1.5 mL) was stirred at 90° C. for 15 h. After completion, the mixture was filtered. Then the mixture was diluted with ethyl acetate (3 mL) and water (3 mL), and then separated. The aqueous phase was extracted with ethyl acetate (5 mL). Then the aqueous phase was extracted with dichloromethane (4 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (184 mg, 62% yield). Yellow solid. LCMS (ESI, M+1): m/z 454.2.

(659) Step B. tert-butyl (4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate: To a mixture of 1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50.0 mg, 110 μ mol), [2-(tert-butoxycarbonylamino)-5,6-dimethyl-1,3-benzothiazol-4-yl]boronic acid (49.7 mg, 154 μ mol) and K.sub.3PO.sub.4 (1.5 M in water, 220 μ L) in THF (1 mL) was added [2-(2-aminophenyl)phenyl]-methylsulfonyloxy-palladium; dicyclohexyl-[3,6-dimethoxy-2-(2,4,6-triisopropylphenyl)phenyl]phosphane (9.99 mg, 11.0 μ mol) under N.sub.2. The mixture was de-gassed and then heated to 60° C. for 20 h under N.sub.2. After completion, the mixture was diluted with ethyl acetate (2 mL) and water (2 mL), and then separated. The aqueous phase was extracted with ethyl acetate (2 mL). The combined organic layer was washed with brine (3 mL) and dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (40.0 mg, 52% yield). Yellow solid. LCMS (ESI, M+1): m/z 696.2.

(660) Step C. 1-(7-(2-amino-5,6-dimethylbenzo[d]thiazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of tert-butyl (4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate (37.0 mg, 53.2 μ mol) and dichloromethane (0.7 mL) was added TFA (2.16 g, 18.9 mmol, 1.4 mL) at 0° C. and the mixture was stirred at 0-28° C. for 1.5 h. After completion, the mixture was concentrated in vacuum and pH value was adjusted to 8 with cold saturated NaHCO.sub.3 solution. Then the mixture was diluted with ethyl acetate (5 mL) and separated. The aqueous phase was extracted with ethyl acetate (2×4 mL). The combined organic layer was washed with brine (6 mL) and dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC (Phenomenexluna C18 150*25 mm*10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 min) to afford the title compound (16.4 mg, 50% yield, 0.3 FORMIC ACID). White solid. .sup.1H NMR (400 MHz, METHANOL-d4): 89.30-9.18 (m, 1H), 7.54 (s, 1H), 5.51-5.30 (m, 1H), 4.55 (br d, J=13.2 Hz, 1H), 4.50-4.38 (m, 2H), 4.34-4.23 (m, 1H), 3.68-3.61 (m, 1H), 3.60-3.36 (m, 4H), 3.24-3.14 (m, 1H), 2.52-2.30 (m, 5H), 2.28-2.07 (m, 7H), 2.05-1.95 (m, 1H), 1.90-1.72 (m, 3H), 1.29 (s, 3H). LCMS (ESI, M+1): m/z 596.2.

Example 40

(661) ##STR00223##

1-(7-(2-amino-5,6-dimethylbenzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(662) ##STR00224##

(663) Step A. tert-butyl (4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-5,6-dimethylbenzo[d]thiazol-2-yl) carbamate: To the mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (150 mg, 344 μ mol), (2-((tert-

butoxycarbonyl)amino)-5,6-dimethylbenzo[d]thiazol-4-yl)boronic acid (130 mg, 403 μ mol), and K.sub.3PO.sub.4 (1.5 M, 688 μ L) in THF (2 mL) was added BrettPhos Pd G3 (31.2 mg, 34.4 μ mol) under N.sub.2. The mixture was stirred at 60° C. for 4 hours. Upon completion, the mixture was partitioned between ethyl acetate (10 mL) and water (3 mL), and the aqueous layer was extracted with ethyl acetate (5 mL). The combined organic phase was washed with brine (5 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give tert-butyl (4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate (132 mg, 52% yield). White Solid; LCMS (ESI, M+1): m/z 678.3.

(664) Step B. 1-(7-(2-amino-5,6-dimethylbenzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of tert-butyl (4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-5,6-dimethylbenzo[d]thiazol-2-yl)carbamate (60.0 mg, 88.5 μ mol) in dichloromethane (0.5 mL) was added TFA (770 mg, 6.75 mmol), and the mixture was stirred at 20° C. for 0.5 hour. Upon completion, to the mixture was added sat. Na.sub.2CO.sub.3 to adjust the pH to 7 and the mixture was extracted with dichloromethane (5 mL). The mixture was dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150 $\times$ 25 mm $\times$ 10 μ m; mobile phase: [water (0.225% formic acid formic acid)-ACN]; B %: 15%-45%, 10 min) to afford the title compound (32.46 mg, 61% yield, 0.4 formic acid salt). Off-white Solid; .sup.1H NMR (400 MHz, METHANOL-d4) δ =9.33-9.18 (m, 1H), 7.55 (s, 1H), 4.58 (s, 3H), 4.32 (br d, J=13.2 Hz, 1H), 3.69-3.52 (m, 3H), 3.50-3.39 (m, 1H), 3.23-3.10 (m, 2H), 2.39 (s, 3H), 2.33-2.23 (m, 2H), 2.21-1.96 (m, 10H), 1.92-1.71 (m, 3H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z 578.3.

Example 41

(665) ##STR00225##

4-(azocan-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(666) ##STR00226##

(667) Step A 4-(azocan-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: To a mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.2 μ mol) and azocane (21.3 mg, 188 μ mol) in DMF (1 mL) were added DIEA (48.7 mg, 377 μ mol) and 4 Å molecular sieves (30 mg) in one portion at 25° C. under N.sub.2. The mixture was stirred at 40° C. for 12 h. After completion, the mixture was filtered. The filtrate was purification by prep-HPLC (column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 20%-50%, 7 min) to afford 4-(azocan-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidine (9.4 mg, 16.8 μ mol, 18% yield, 99% purity, 0.24 FORMIC ACID). White solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.13 (s, 1H), 8.17 (d, J=8.0 Hz, 1H), 7.93 (d, J=8.4 Hz, 1H), 7.74 (t, J=7.6 Hz, 1H), 7.65-7.55 (m, 2H), 7.30 (d d, J=7.6 Hz, 12.8 Hz, 1H), 4.11 (s, 2H), 4.06-3.99 (m, 4H), 2.93-3.02 (m, 2H), 2.64-2.56 (m, 2H), 2.00-1.70 (m, 10H), 1.68-1.49 (m, 8H); LCMS (ESI, M+1): m/z 544.3.

Example 42

(668) ##STR00227##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane

(669) ##STR00228##

(670) Step A. 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane: To a mixture of 8-fluoro-7-(8-

fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 mg, 34.7 μ mol, formic acid salt), 1,4-oxazepane (5.73 mg, 41.6 μ mol, HCl), and 4 Å molecular sieves (5 mg) in DMF (0.5 mL) was added DIEA (17.9 mg, 139 μ mol), and the mixture was stirred at 40° C. for 14 hours. Upon completion, the mixture was filtered and the filtrate was purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 16%-46%, 10 min) to afford 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (2.05 mg, 10% yield, 0.51 formic acid salt). Off-white Gum; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ =9.12 (s, 1H), 8.03-7.97 (m, 1H), 7.74 (d, J=7.6 Hz, 1H), 7.67-7.58 (m, 2H), 7.48-7.42 (m, 1H), 7.16-7.08 (m, 1H), 4.51 (s, 2H), 4.26-4.18 (m, 4H), 4.08-4.01 (m, 2H), 3.91-3.85 (m, 2H), 3.61-3.50 (m, 2H), 2.85-2.76 (m, 2H), 2.31-2.23 (m, 4H), 2.04-1.96 (m, 4H), 1.90-1.83 (m, 2H); LCMS (ESI, M+1): m/z 532.4.

Example 43

(671) ##STR00229##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-thiazepane

(672) ##STR00230##

(673) Step A. 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-thiazepane: To a mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50 mg, 94.2 μ mol) and 1,4-thiazepane (33.1 mg, 283 μ mol) in DMF (1 mL) were added DIEA (48.7 mg, 377 μ mol, 65.0 μ L) and 4 Å molecular sieves (30 mg) in one portion at 25° C. under N.sub.2. The mixture was stirred at 40° C. for 12 h. After completion, the residue was filtered. The filtrate was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75×30 mm×3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 20%-50%, 7 min) to afford 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-thiazepane (5.63 mg, 9.88 μ mol, 10% yield, 99% purity, 0.37 FORMIC ACID); Yellow solid; 1H NMR (400 MHz, DMSO-d.sub.6) δ 9.16 (s, 1H) 8.18 (d, J=7.6 Hz, 1H), 7.93 (d, J=7.6 Hz, 1H), 7.77-7.71 (m, 1H), 7.65-7.55 (m, 2H), 7.31 (d, J=13.2 Hz, 7.2 Hz, 1H), 4.30-4.23 (m, 2H), 4.19-4.13 (m, 2H), 4.12-4.06 (m, 2H), 3.14-3.10 (m, 2H), 3.00-2.90 (m, 2H), 2.76-2.70 (m, 2H), 2.60-2.54 (m, 2H), 2.22-2.15 (m, 2H), 1.95-1.70 (m, 6H), 1.65-1.55 (m, 2H); LCMS (ESI, M+1): m/z 548.3.

Example 44

(674) ##STR00231##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-3-ol

(675) ##STR00232##

(676) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-3-ol: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 mg, 37.7 μ mol) and DIEA (14.6 mg, 113 μ mol, 19.7 μ L) in DMF (1.0 mL) was added azepan-3-ol (8.68 mg, 75.4 μ mol). The mixture was stirred at 40° C. for 24 h. After completion, the mixture was concentrated at 20° C. The residue was purified by prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 μ m; A: water (0.225% formic acid)-formic acid, B: ACN, B %: 18%-38% over 10 min) to afford 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-3-ol (6.83 mg, 11.9 μ mol, 32% yield, 95.5% purity). Yellow solid; .sup.1H NMR (400 MHz, METHANOL-d) δ =9.30 (d, J=4.4 Hz, 1H), 8.53 (br s, 1H), 8.13 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.71 (t, J=8.0 Hz, 1H), 7.64-7.58 (m, 1H), 7.57-7.50 (m, 1H), 7.24-7.14 (m, 1H), 4.44 (br d, J=13.2 Hz, 1H), 4.22-4.10 (m, 3H), 3.87-3.77 (m, 1H), 3.62-3.52 (m, 2H), 3.21-3.10 (m, 2H), 2.35-2.23 (m, 2H),

2.22-2.07 (m, 6H), 2.06-1.99 (m, 5H), 1.98-1.86 (m, 1H), 1.74-1.60 (m, 1H), 1.54-1.39 (m, 1H); LCMS (ESI, M+1): m/z 546.3.

Example 45

(677) ##STR00233##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(678) ##STR00234##

(679) Step A. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1 H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 mg, 37.7 μ mol) and 1,3,7-triazaspiro[4.5]decan-2-one (8.78 mg, 56.6 μ mol) in DMF (0.2 mL) was added DIEA (48.7 mg, 377 μ mol). The mixture was stirred at 40° C. for 1 hour. The reaction mixture was quenched with water (10 mL) at 20° C. and then extracted with EtOAc (5 mL $\times$ 3). The combined organic layers were washed with brine (3 mL $\times$ 2), and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: 3_Phenomenex Luna C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 20%-40%, 10 min) affording 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one (2.65 mg, 12% yield). White solid; .sup.1H NMR (400 MHz, MeOD) δ =9.16 (d, J=1.5 Hz, 1H), 8.54 (s, 1H), 8.13 (d, J=8.2 Hz, 1H), 7.87 (d, J=7.9 Hz, 1H), 7.74-7.69 (m, 1H), 7.65-7.59 (m, 1H), 7.54 (d, J=5.1, 7.9 Hz, 1H), 7.20 (d, J=7.7, 13.2 Hz, 1H), 4.65-4.60 (m, 1H), 4.57-4.50 (m, 1H), 4.44-4.30 (m, 1H), 4.30-4.18 (m, 1H), 4.01 (d, J=6.8, 13.9 Hz, 1H), 3.96-3.84 (m, 1H), 3.66-3.55 (m, 2H), 3.44 (d, J=5.7, 9.3 Hz, 1H), 3.32 (s, 1H), 3.24-3.14 (m, 2H), 2.33-2.24 (m, 2H), 2.15 (d, J=6.7, 13.9 Hz, 4H), 2.09-1.95 (m, 5H), 1.88 (s, 1H). LCMS (ESI, M+1): m/z 586.1.

Example 46

(680) ##STR00235##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-one

(681) ##STR00236##

(682) Step A. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-one: To a solution of 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-ol (30.0 mg, 51.2 μ mol) in dichloromethane (2.0 mL) was added Dess-Martin Reagent (43.5 mg, 102 μ mol). The mixture was stirred at 20° C. for 10 hours. After completion, the reaction mixture was quenched with saturated 10 mL Na.sub.2SO.sub.3 at 20° C., and then extracted with EtOAc (5 mL $\times$ 3). The combined organic layer was washed with saturated NaHCO.sub.3 (3 mL $\times$ 2) and brine (5 mL). The mixture was dried over anhydrous Na.sub.2SO.sub.4 and filtered. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (Phenomenex Luna C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 15%-45%, 8 min) affording 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-one (6.51 mg, 22% yield). White solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =9.10 (s, 1H), 8.18 (d, J=8.0 Hz, 1H), 7.93 (d, J=8.0 Hz, 1H), 7.78-7.71 (m, 1H), 7.63 (d, J=7.2 Hz, 1H), 7.59 (d, J=5.2, 8.0 Hz, 1H), 7.31 (d, J=7.6, 13.2 Hz, 1H), 4.09-4.02 (m, 2H), 3.96 (s, 2H), 3.89-3.81 (m, 2H), 2.99-2.90 (m, 2H), 2.56 (d, J=7.0, 9.6 Hz, 2H), 2.34-2.20 (m, 3H), 2.17-2.10 (m, 1H), 1.94-1.83 (m, 4H), 1.83-1.69 (m, 8H), 1.59-1.54 (m, 2H). LCMS (ESI, M+1): m/z 584.1.

Example 47

(683) ##STR00237##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-ol

(684) ##STR00238##

(685) Step A. 7-azaspiro[4.5]decan-2-ol hydrochloride: To a solution of tert-butyl 2-hydroxy-7-azaspiro[4.5]decane-7-carboxylate (70 mg, 274.13 μ mol) in MeOH (1 mL) was added HCl-MeOH (4 M, 1 mL). The mixture was stirred at 20° C. for 2 hours. The reaction mixture was concentrated under reduced pressure to afford 7-azaspiro[4.5]decan-2-ol hydrochloride (55 mg, HCl salt).

Colorless oil; LCMS (ESI, M+1): m/z 156.2.

(686) Step B. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-ol: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (10 mg, 18.9 μ mol) and 7-azaspiro[4.5]decan-2-ol hydrochloride (8.8 mg, 56.6 μ mol) in DMF (0.2 mL) was added DIEA (24.4 mg, 188 μ mol, 32.8 μ L). The mixture was stirred at 40° C. for 1 hr. The reaction mixture was quenched with 10 mL water at 20° C., and then extracted with EtOAc (5 mL $\times$ 3). The combined organic layers were washed with brine (3 mL $\times$ 2), and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge BEH C18 100 $\times$ 30 mm $\times$ 10 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %:

30%-55%, 10 min) to afford 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-ol (2.43 mg, 21.0% yield).

White solid; HPLC: 95.3%. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =9.11 (s, 1H), 8.15 (d, J=8.4 Hz, 1H), 7.91 (d, J=8.2 Hz, 1H), 7.75-7.68 (m, 1H), 7.60 (td, J=3.3, 6.7 Hz, 1H), 7.58-7.53 (m, 1H), 7.28 (d, J=7.6, 13.1 Hz, 1H), 4.52 (s, 1H), 4.11 (s, 1H), 4.04 (s, 2H), 4.02-3.96 (m, 1H), 3.93-3.84 (m, 1H), 3.83-3.72 (m, 2H), 2.94-2.86 (m, 2H), 2.54-2.50 (m, 2H), 1.96 (s, 1H), 1.90-1.82 (m, 2H), 1.81-1.69 (m, 8H), 1.60-1.50 (m, 6H), 1.47-1.41 (m, 1H), 1.39-1.30 (m, 1H). LCMS (ESI, M+1): m/z 586.3.

Example 48

(687) ##STR00239##

8-fluoro-4-(2-fluoro-7-azaspiro[4.5]decan-7-yl)-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(688) ##STR00240##

(689) Step A. 8-fluoro-4-(2-fluoro-7-azaspiro[4.5]decan-7-yl)-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-ol (30.0 mg, 51.2 μ mol) in DCM (2 mL) was added bis(2-methoxyethyl)aminosulfur trifluoride (113 mg, 512 μ mol, 112 μ L) at 0° C. The mixture was stirred at 20° C. for 10 hrs. The reaction mixture was quenched with 10 mL water at 20° C., and then extracted with EtOAc (5 mL $\times$ 3). The combined organic layers were washed with saturated NaHCO.sub.3 (3 mL $\times$ 2) and brine (3 mL), dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Luna C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (0.2% FORMIC ACID)-ACN]; B %: 30%-60%, 8 min) affording 8-fluoro-4-(2-fluoro-7-azaspiro[4.5]decan-7-yl)-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine (5.11 mg, 16.9% yield). White solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =9.07 (s, 1H), 8.34 (s, 1H), 8.18 (d, J=8.4 Hz, 1H), 7.93 (d, J=7.9 Hz, 1H), 7.79-7.70 (m, 1H), 7.67-7.52 (m, 2H), 7.31 (d, J=7.5, 13.3 Hz, 1H), 5.28-5.07 (m, 1H), 4.10-3.96 (m, 3H), 3.89-3.76 (m, 2H), 3.63 (d, J=13.0 Hz, 1H), 2.98-2.87 (m, 2H), 2.58-2.52 (m, 2H), 1.98-1.64 (m, 15H), 1.62-1.50 (m, 3H). .sup.1H NMR (400 MHz, MeOD-d.sub.4) δ =9.04 (s, 1H), 8.11 (d, J=8.4 Hz, 1H), 7.85 (d, J=7.9 Hz, 1H), 7.74-7.66 (m, 1H), 7.61 (d, J=1.0, 7.2 Hz, 1H), 7.53 (d, J=5.0, 8.0 Hz, 1H), 7.19 (d, J=7.7, 13.0 Hz, 1H), 5.27-5.05 (m, 1H), 4.28 (s, 2H), 4.22-4.07 (m, 1H), 4.02-3.87 (m, 2H), 3.78 (d, J=2.8, 12.9 Hz,

1H), 3.18-3.08 (m, 2H), 2.76 (d, J=6.6, 10.5 Hz, 2H), 2.13-1.82 (m, 14H), 1.81-1.70 (m, 4H); LCMS (ESI, M+1): m/z 588.2.

Example 49

(690) ##STR00241##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)morpholine

(691) ##STR00242##

(692) Step A. 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)morpholine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (30.0 mg, 52.0 μ mol, formic acid salt), morpholine (9.07 mg, 104 μ mol, 9.16 μ L), DIEA (26.9 mg, 208 μ mol, 36.3 μ L) and 4 Å molecular sieves (5 mg) in DMF (0.5 mL) was stirred at 40° C. for 1.5 h. After completion, the mixture was filtered and purified by prep-HPLC (Shim-pack C18 150*25*10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 min) to afford the title compound (7.45 mg, 27% yield, 0.4 FORMIC ACID). Off-white Solid. ¹H NMR (400 MHz, CDCl₃-d₃): δ 9.04 (s, 1H), 7.99 (td, J=1.6, 8.0 Hz, 1H), 7.74 (d, J=8.0 Hz, 1H), 7.66-7.57 (m, 2H), 7.45 (dt, J=4.8, 8.0 Hz, 1H), 7.15-7.08 (m, 1H), 4.56 (s, 2H), 4.12-4.01 (m, 4H), 3.94-3.89 (m, 4H), 3.65-3.59 (m, 2H), 2.84 (td, J=6.8, 10.8 Hz, 2H), 2.34-2.23 (m, 2H), 2.14-2.07 (m, 2H), 2.05-1.98 (m, 2H), 1.92-1.83 (m, 2H). ¹⁹F NMR (400 MHz, CDCl₃-d₃) δ =-112.256. -138.683. LCMS (ESI, M+1): m/z 518.2.

Example 50

(693) ##STR00243##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)thiomorpholine

(694) ##STR00244##

(695) Step A. 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)thiomorpholine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (30 mg, 56.6 μ mol), thiomorpholine (18 mg, 174 μ mol), DIPEA (30 μ L), and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was stirred at 40° C. for 6 hours under N₂ atmosphere. The reaction mixture was filtered and purified by prep-HPLC (column: Shim-pack C18 150×25×10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 19%-49%, 10 minutes) to give the product as white solid (14.5 mg, 45% yield, 0.3 FORMIC ACID). ¹H NMR (400 MHz, DMSO-d₆) δ =9.05 (s, 1H), 8.18 (br d, J=8.0 Hz, 1H), 7.93 (br d, J=8.0 Hz, 1H), 7.74 (br t, J=7.6 Hz, 1H), 7.64-7.58 (m, 2H), 7.31 (br dd, J=7.6, 12.8 Hz, 1H), 4.19 (br s, 4H), 4.10 (s, 2H), 2.98-2.91 (m, 6H), 2.60-2.57 (m, 2H), 1.92-1.78 (m, 6H), 1.61-1.58 (m, 2H). LCMS (ESI, M+1): m/z 534.2.

Example 51

(696) ##STR00245##

(R)-1-(7-(8-ethynyl-7-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(697) Synthesized according to Example 16, step A to C. Yellow Solid; ¹H NMR (400 MHz, methanol-d₄) δ 9.24-9.03 (m, 1H), 8.19-8.06 (m, 2H), 7.74-7.58 (m, 2H), 7.51-7.36 (m, 1H), 5.47-5.21 (m, 1H), 4.67-4.57 (m, 2H), 4.41-4.23 (m, 3H), 3.68-3.62 (m, 1H), 3.52-3.34 (m, 2H), 3.29-3.18 (m, 2H), 3.10-2.98 (m, 1H), 2.44-2.11 (m, 4H), 2.08-1.97 (m, 2H), 1.95-1.65 (m, 4H), 1.27 (d, J=19.6 Hz, 3H); LCMS (ESI, M+1): m/z 588.3.

Example 52

(698) ##STR00246##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(((1-hydroxycyclohexyl)methyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(699) ##STR00247##

(700) Step A 1-(((2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol: To a mixture of 1-(aminomethyl)cyclohexanol (256 mg, 1.98 mmol) and 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (500 mg, 1.98 mmol) in DMA (6 mL) was added DIEA (768 mg, 5.94 mmol) in one portion at 0° C. under N.sub.2. The mixture was stirred at 0° C. for 1 h. After completion, the mixture was filtered and the filtrate was purification by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) affording 1-(((2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol (530 mg, 77% yield). Yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.37 (br s, 1H), 9.31 (s, 1H), 4.53 (br s, 1H), 3.58 (s, 2H), 1.63-1.35 (m, 9H), 1.30-1.15 (m, 1H); LCMS (ESI, M+1): m/z 345.1.

(701) Step B 1-(((7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol: To a mixture of ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (346 mg, 2.17 mmol) and 1-(((2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol (500 mg, 1.45 mmol) in dioxane (5 mL) was added DIEA (561 mg, 4.35 mmol) in one portion at 25° C. under N.sub.2. The mixture was heated to 95° C. and stirred for 12 h. After completion, the mixture was filtered and concentrated in vacuum. The residue was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) affording 1-(((7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol (630 mg, 91% yield). Off-white solid; LCMS (ESI, M+1): m/z 468.2.

(702) Step C 1-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol: To a mixture of 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (200 mg, 427 μmol) and 1-(((7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol (200 mg, 555 μmol) in THE (4 mL) was added K.sub.3PO.sub.4 (1.5 M, 855 μL) in one portion at 25° C. under N.sub.2. Then cataCXium-A-Pd-G3 (46.7 mg, 64.1 μmol) was added. The mixture was heated to 60° C. and stirred for 2 h. After completion, the mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)-ACN) affording 1-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol (110 mg, 36% yield). Yellow solid; LCMS (ESI, M+1): m/z 666.2.

(703) Step D 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(((1-hydroxycyclohexyl)methyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: A mixture of 1-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)cyclohexanol (90 mg, 135 μmol) and HCl-dioxane (4 M, 4.50 mL) in ACN (0.5 mL) was stirred at 20° C. for 10 min. After completion, the residue was concentrated in vacuum. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75×30 mm×3 μm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 18%-48%, 5 min) affording 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(((1-hydroxycyclohexyl)methyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (52.3 mg, 60% yield, 0.35 FORMIC ACID). White solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.93 (br s, 1H), 9.40 (s, 1H), 8.75 (s, 1H), 7.76 (dd, J=6.0 Hz, 9.2 Hz, 1H), 7.38-7.30 (m, 2H), 8.99 (s, 1H), 5.28 (d, J=54.4 Hz, 1H), 4.56 (br s, 1H), 4.15 (d, J=10.4 Hz, 1H), 4.08 (d, J=10.4 Hz, 1H), 3.71-3.64 (m, 1H), 3.58-3.53 (m, 1H), 3.12-3.05 (m, 2H), 3.04-2.99 (m, 1H), 2.88-2.78 (m, 1H), 2.40-2.35 (m, 1H), 2.20-1.95 (m, 4H), 1.89-1.72 (m, 3H), 1.65-1.35 (m, 10H), 1.28-1.18 (m, 1H), 0.72 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=622.3.

Example 53

(704) ##STR00248##

4-(3,3-dimethylpiperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(705) ##STR00249##

(706) Step A. 4-(3,3-dimethylpiperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 188 μ mol) and DIEA (73.1 mg, 565 μ mol, 98.5 μ L) in DMF (1.5 mL) was added 3,3-dimethylpiperidine (42.7 mg, 377 μ mol). The mixture was stirred at 40° C. for 12 h. After completion, the mixture was concentrated at 20° C. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150*25 mm*10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 27%-57%, 10 min) affording 4-(3,3-dimethylpiperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine (32.8 mg, 29% yield). White solid; ¹H NMR (400 MHz, METHANOL-d₄) δ =9.09 (s, 1H), 8.12 (br d, J=8.0 Hz, 1H), 7.85 (br d, J=8.4 Hz, 1H), 7.74-7.66 (m, 1H), 7.65-7.59 (m, 1H), 7.57-7.49 (m, 1H), 7.19 (dd, J=7.6, 12.8 Hz, 1H), 4.47 (s, 2H), 4.08-3.94 (m, 2H), 3.83 (s, 2H), 3.44-3.36 (m, 2H), 3.06-2.95 (m, 2H), 2.26-2.15 (m, 2H), 2.13-1.98 (m, 4H), 1.98-1.86 (m, 4H), 1.69-1.59 (m, 2H), 1.03 (br d, J=4.0 Hz, 6H); LCMS (ESI, M+1): m/z=544.3.

Example 54

(707) ##STR00250##

1-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(708) ##STR00251##

(709) Step A. 1-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(7-(8-ethynyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (65 mg, 118 μ mol) in MeOH (4 ML) was added Pd/C (10 mg, 10% purity) under N.sub.2 atmosphere. The mixture was degassed and purged with H.sub.2 for 3 times and stirred at 25° C. for 18 hours under H.sub.2 atmosphere (15 psi). The reaction mixture was filtered. The filtrate was concentrated under reduced pressure, purified by prep-HPLC (column: Phenomenex luna C18 150*25 mm*10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 22%-52%, 10 minutes) to give the product as yellow solid (13.1 mg, 18% yield, 0.9 FORMIC ACID). ¹H NMR (400 MHz, DMSO+D.sub.2O) δ =9.20 (d, J=7.2 Hz, 1H), 8.05 (d, J=8.4 Hz, 1H), 7.88 (d, J=8.0 Hz, 1H), 7.57 (t, J=7.6 Hz, 1H), 7.49 (t, J=7.6 Hz, 1H), 7.39 (dd, J=6.8, 12.8 Hz, 2H), 4.46-4.39 (m, 3H), 4.09 (br dd, J=13.6, 19.2 Hz, 1H), 3.57 (dd, J=13.2, 42.4 Hz, 1H), 3.40-3.35 (m, 3H), 3.06-3.02 (m, 2H), 2.32-2.22 (m, 2H), 2.05-1.90 (m, 12H), 1.15 (d, J=11.6 Hz, 3H), 0.83-0.78 (m, 3H). LCMS (ESI, M+1): m/z=556.3.

Example 55

(710) ##STR00252##

1-(7-(8-ethynyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(711) ##STR00253##

(712) Step A. 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (115 mg, 264 μ mol) and K.sub.3PO.sub.4 (1.5 M in water, 550 μ L) in THF (3 mL) was degassed and purged with N.sub.2 for 3 times. Triisopropyl-[2-[8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-naphthyl]ethynyl]silane (130 mg, 299 μ mol) was added, followed by cataCXium-A-Pd-G3 (20 mg, 27.5 μ mol). The mixture was stirred at 60° C. for 3 hours under N.sub.2 atmosphere. The combined reaction mixture was diluted with water (15 mL), extracted with

ethyl acetate (5 mL×4). The combined organic phase was dried with anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile-1/1] to give the product as a yellow gum (131 mg, 65%). LCMS (ESI, M+1): m/z=708.4.

(713) Step B. 1-(7-(8-ethynyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (59 mg, 83.3 μmol), and CsF (127 mg, 833 μmol) in DMF (1 mL) was stirred at 25° C. for 1 hour. The reaction mixture was filtered. The filtrate was purified by prep-HPLC (column: water s Xbridge 150×25 mm×Sum; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 29%-59%, 10 min) to give the product as a yellow solid (19.6 mg, 42%). ¹H NMR (400 MHz, DMSO-d₆) δ=9.14 (d, J=60.4 Hz, 1H), 8.14 (t, J=8.0 Hz, 2H), 7.73-7.56 (m, 4H), 4.78-4.72 (m, 1H), 4.46-4.26 (m, 1H), 4.11-4.03 (m, 3H), 3.68 (d, J=4.4 Hz, 1H), 3.59 (t, J=13.6 Hz, 1H), 2.95-2.91 (m, 2H), 2.56-2.54 (m, 3H), 1.90-1.70 (m, 10H), 1.58-1.55 (m, 2H), 1.16 (d, J=15.2 Hz, 3H). LCMS (ESI, M+1): m/z=552.3.

Example 56

(714) ##STR00254##

8-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-1-naphthonitrile

(715) ##STR00255##

(716) Step A. 8-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-1-naphthonitrile: To a solution of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50.0 mg, 115 μmol) and 8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-naphthonitrile (48.0 mg, 172 μmol) in toluene (2.0 mL) was added K₃PO₄ (1.5 M, 229 μl) and cataCXium-A-Pd-G3 (12.53 mg, 17.20 μmol). The mixture was stirred at 90° C. for 2 hours. After completion, the reaction mixture was diluted with water (5 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150*25 mm*10 μm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 18%-48%, 10 min) to give 8-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)-1-naphthonitrile (22.0 mg, 33% yield, 0.6 FORMIC ACID); Yellow solid. ¹H NMR (400 MHz, DMSO-d₆) δ 9.26 (s, 1H), 8.46 (d, J=8.4 Hz, 1H), 8.29 (dd, J=2.0, 7.2 Hz, 1H), 8.24 (s, 1H), 8.14 (dd, J=1.2, 7.2 Hz, 1H), 7.87-7.78 (m, 2H), 7.77-7.70 (m, 1H), 4.33-4.25 (m, 1H), 4.13 (s, 2H), 4.03 (br d, J=13.2 Hz, 2H), 3.47-3.35 (m, 2H), 3.05-2.97 (m, 2H), 2.66-2.57 (m, 2H), 2.08-1.99 m, 1H), 1.98-1.90 (m, 2H), 1.88-1.76 (m, 4H), 1.75-1.58 (m, 5H), 1.17 (br s, 3H). LCMS (ESI, M+1): m/z=553.3.

Example 57

(717) ##STR00256##

6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(718) ##STR00257## ##STR00258##

(719) Step A. 7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol: To a mixture of 7-chloro-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol, 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (14.0 g, 39.0 mmol) and K₃PO₄ (1.5 M, 46.4 mL) in EtOH (140 mL) was added cataCXium-A-Pd-G3 (1.39 g, 1.90 mmol) under N₂. The mixture was de-gassed and then heated to 78° C. for 9.5 h under N₂. After completion, the mixture was concentrated in vacuum. Then the mixture was diluted with ethyl acetate (500 mL) then filtered. 100 mL water was added, and the organic

layer was separated. Then the aqueous phase was extracted with ethyl acetate (80 mL). The combined organic layer was washed with brine (120 mL), dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (2.34 g, 24% yield). Yellow solid. LCMS [ESI, M+1]: m/z=414.1.

(720) Step B. 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine. A mixture of POCl₃ (278 mg, 1.81 mmol, 169 μ L) in toluene (3 mL) were added DIEA (141 mg, 1.09 mmol, 190 μ L) and 7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine-2,4-diol (150 mg, 363 μ mol) and the mixture was stirred at 110° C. for 25 mins. After completion, the mixture was concentrated in vacuum and the pH value was adjusted to 8 with cold saturated NaHCO₃ solution. Then the mixture was extracted with ethyl acetate (8 mL) twice. The combined organic layer was washed with brine (10 mL) and dried over Na₂SO₄. The mixture was filtered and concentrated in vacuum to give 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine (164 mg, crude) which was used to next step without further purification. Brown oil.

(721) Step C. 8-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-1,8-diazaspiro[3.5]nonan-2-one. To a mixture of 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine (147 mg, 326 μ mol) and DIEA (127 mg, 979 μ mol, 171 μ L) in dichloromethane (2 mL) was added 1,8-diazaspiro[3.5]nonan-2-one (45.8 mg, 326 μ mol) at -40° C. and the mixture was stirred at -40° C. for 20 mins. After completion, the mixture was quenched by water (10 mL) and filtered. Then the filtrate was extracted with dichloromethane twice. The combined organic layer was washed with brine (6 mL) and dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 8-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-1,8-diazaspiro[3.5]nonan-2-one (54.0 mg, 27% yield over two steps). Yellow solid. LCMS [ESI, M+1]: m/z=554.2.

(722) Step D. 6-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one. A mixture of 8-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-1,8-diazaspiro[3.5]nonan-2-one (50.0 mg, 90.3 μ mol), ((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (43.1 mg, 271 μ mol), DIEA (35.0 mg, 271 μ mol, 47.2 μ L) and 4 Å molecular sieves (10 mg) in dioxane (1.5 mL) was stirred at 90° C. for 22 h. After completion, the mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (36.0 mg, 55% yield). Yellow solid. LCMS [ESI, M+1]: m/z=677.4.

(723) Step E. 6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one. To a mixture of 6-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (37.0 mg, 54.7 μ mol) and dichloromethane (0.5 mL) was added TFA (770 mg, 6.75 mmol, 0.5 mL) at 0° C. and the mixture was stirred at 0° C. for 1 h. After completion, the mixture was concentrated in vacuum. The pH value was adjusted to 9 with saturated NaHCO₃ solution and the mixture was extracted with ethyl acetate twice. The combined organic layer was dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by prep-HPLC (Water s Xbridge 150*25 mm*5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 35%-65%, 10 min) to afford the title compound (14.3 mg,

41% yield). Yellow solid. ¹H NMR (400 MHz, METHANOL-d₄): 6.9.08 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.05 (d, J=2.0 Hz, 1H), 5.39-5.22 (m, 1H), 4.42-4.24 (m, 4H), 4.02-3.92 (m, 1H), 3.85-3.68 (m, 1H), 3.28-3.11 (m, 3H), 3.05-2.97 (m, 1H), 2.92-2.83 (m, 1H), 2.78-2.70 (m, 1H), 2.51-2.09 (m, 6H), 2.04-1.90 (m, 6H), 0.79 (td, J=2.4, 7.2 Hz, 3H). ¹⁹F NMR (400 MHz, METHANOL-d₄) δ=−121.066, −138.847, −173.641; LCMS [ESI, M+1]: m/z=633.3.

Example 58

(724) ##STR00259##

6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol

(725) ##STR00260##

(726) Step A. 6-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol: To a solution of 2,4-dichloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidine (163 mg, 362 μmol) and DIPEA (297 mg, 2.30 mmol) in DCM (2 mL) was added 6-azabicyclo[3.2.1]octan-3-ol (33 mg, 259 μmol) in DCM (1 mL) at −40° C. slowly. The mixture was stirred at this temperature for 10 minutes. The reaction mixture was diluted with water (5 mL), and extracted with ethyl acetate (10 mL×4). The combined organic phase was dried with anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to afford a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=2/3] to give the product as a red oil (88 mg, 39%). LCMS [ESI, M+1]: m/z=541.2.

(727) Step B. 6-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol: A mixture of 6-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol (66 mg, 122 μmol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (78 mg, 490 μmol), DIPEA (52 mg, 402 μmol), and 4 Å molecular sieves (40 mg, 40.7 μmol) in dioxane (1 mL) was degassed and purged with N₂ for 3 times, and then the mixture was stirred at 95° C. for 48 hours under N₂ atmosphere. The reaction mixture was filtered. The residue was washed with DCM (1 mL×5). The combined DCM phase was concentrated under reduced pressure to afford a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=12/13] to give the product as a yellow oil (48 mg, 59%). LCMS [ESI, M+1]: m/z=664.3.

(728) Step C. 6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol: To a solution of 6-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol (58 mg, 87.4 μmol) in DCM (2 mL) was added dropwise TFA (2.5 mL) at 0° C. The mixture was stirred between 0 and 15° C. for 1.5 hours. The reaction mixture was diluted with DCM (4 mL), and quenched with saturated NaHCO₃ aqueous (8 mL) at 0° C. The residue was washed with DCM (4 mL×4) and the combined DCM phase was concentrated under reduced pressure to afford a residue. The filtrate was purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 42%-72%, 10 min) to give the product as a yellow solid. (10 mg, 18%). ¹H NMR (400 MHz, METHANOL-d₄) δ=9.27-9.24 (m, 1H), 7.68-7.66 (m, 1H), 7.29-7.24 (m, 2H), 7.04 (br dd, J=2.0, 12.4 Hz, 1H), 5.31 (d, J=54.4 Hz, 1H), 4.98 (br s, 1H), 4.34-4.22 (m, 4H), 4.13 (br s, 1H), 3.22 (br d, J=14.0 Hz, 2H), 3.02-3.00 (m, 1H), 2.77-2.64 (m, 2H), 2.47-1.81 (m, 14H), 0.78 (t, J=7.2 Hz, 3H). F NMR (376 MHz, METHANOL-d₄) δ=−121, −138, −173. LCMS [ESI, M+1]: m/z=620.3.

Example 59

(729) ##STR00261##

(3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol

(730) ##STR00262##

(731) Step A. (3R,5R)-1-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol: To a mixture of 2,4-dichloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidine (200 mg, 444 μ mol), and DIEA (287 mg, 2.22 mmol) in dichloromethane (3 mL) was added (3R,5R)-piperidine-3,5-diol (47.8 mg, 311 μ mol, HCl) at -40° C. The mixture was stirred at -40° C. for 0.5 hour. Upon completion, the reaction mixture was partitioned between dichloromethane (6 mL) and water (5 mL), and the aqueous layer was extracted with dichloromethane (5 mL) one more time. The combined organic phase was washed with brine (5 mL) and dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (3R,5R)-1-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl) piperidine-3,5-diol (85.0 mg, 34% yield). Yellow Oil; LCMS [ESI, M+1]: m/z=531.2.

(732) Step B. (3R,5R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl) piperidine-3,5-diol. To the mixture of (3R,5R)-1-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol (75.0 mg, 141 μ mol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (45.0 mg, 282 μ mol), and 4 Å molecular sieves (30.0 mg) in dioxane (1.5 mL) was added DIEA (54.8 mg, 424 μ mol), and the mixture was stirred at 90° C. for 15 hours. Upon completion, the reaction mixture was partitioned between ethyl acetate (15 mL) and water (10 mL), and the aqueous layer was extracted with ethyl acetate (15 mL). The combined organic phase was washed with brine (20 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give (3R,5R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol (45.0 mg, 45% yield). Yellow Solid; LCMS [ESI, M+1]: m/z=654.1.

(733) Step C. (3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl) piperidine-3,5-diol. To the solution of (3R,5R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol (42.0 mg, 64.3 μ mol) in dichloromethane (0.8 mL) was added TFA (1.47 g, 12.9 mmol), and the mixture was stirred at 0° C. for 1 hour. Upon completion, the mixture was concentrated, and the pH was adjusted to 8 with sat. NaHCO.sub.3. The mixture was extracted with ethyl acetate (8 mL $\times$ 2), and the combined organic phase was dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 29%-59%, 10 min) to afford (3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol (15.71 mg, 40% yield). White Solid; .sup.1H NMR (400 MHz, METHANOL-d4) δ =9.26 (d, J=7.2 Hz, 1H), 7.72-7.65 (m, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.07-7.02 (m, 1H), 5.46-5.31 (m, 1H), 4.51-4.34 (m, 2H), 4.30-4.21 (m, 4H), 3.86-3.76 (m, 2H), 3.55-3.47 (m, 1H), 3.56-3.35 (m, 2H), 3.20-3.10 (m, 1H), 2.52-2.05 (m, 1H), 2.04-1.93 (m, 3H), 0.84-0.75 (m, 3H); LCMS [ESI, M+1]: m/z 610.2.

Example 60

(734) ##STR00263##

4-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one (735) ##STR00264##

(736) Step A. 4-(((2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one: To the mixture of 2,4-dichloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidine (170 mg, 378 μ mol), and DIEA (244 mg, 1.89 mmol) in dichloromethane (3 mL) was added 4-(aminomethyl)pyrrolidin-2-one (34.5 mg, 302 μ mol) at -40° C., and the mixture was stirred at -40° C. for 0.5 hour. Upon completion, the reaction mixture was quenched with water and extracted with dichloromethane twice. The combined organic phase was washed with brine (5 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (120 mg, 59% yield). Off-White Solid; LCMS [ESI, M+1]: m/z 528.2.

(737) Step B. 4-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one: To the mixture of 4-(((2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one (110 mg, 208 μ mol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (66.3 mg, 417 μ mol), and 4 Å molecular sieves (5.0 mg) in dioxane (1.5 mL) was added DIEA (80.8 mg, 625 μ mol), and the mixture was stirred at 90° C. for 15 hours. Upon completion, the reaction mixture was quenched with water and extracted with ethyl acetate twice. The combined organic phase was washed with brine (20 mL) and dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give the title compound (83.0 mg, 59% yield). Yellow Solid; LCMS [ESI, M+1]: m/z 651.2.

(738) Step C. 4-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one: To the solution of 4-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one (78.0 mg, 120 μ mol) in dichloromethane (0.8 mL) was added TFA (1.54 g, 13.5 mmol), and the mixture was stirred at 0° C. for 1 hour. Upon completion, the mixture was concentrated, and the pH was adjusted with sat. NaHCO_3 to 8. The mixture was extracted with ethyl acetate (8 mL $\times$ 2). The combined organic phase dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH_4HCO_3)-ACN]; B %: 29%-59%, 10 min) to afford 4-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one (25.04 mg, 34% yield). White Solid; ^1H NMR (400 MHz, METHANOL-d_4) δ =9.18 (s, 1H), 7.73-7.66 (m, 1H), 7.33 (d, J=2.4 Hz, 1H), 7.27 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.8 Hz, 1H), 5.46-5.28 (m, 1H), 4.46-4.31 (m, 2H), 3.89-3.76 (m, 2H), 3.67-3.58 (m, 1H), 3.51-3.34 (m, 3H), 3.30 (br s, 1H), 3.17-3.00 (m, 2H), 2.62-2.15 (m, 7H), 2.11-1.94 (m, 3H), 0.81 (t, J=7.2 Hz, 3H); LCMS [ESI, M+1]: m/z 607.2.

Example 61

(739) ##STR00265##

4-(4-(2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5 (6H)-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(740) ##STR00266##

(741) Step A. 5-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To the mixture of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (280 mg, 1.1 mmol) and DIEA (717 mg, 5.5 mmol, 966 μ L) in dichloromethane (1 mL) was added 5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (118 mg, 776 μ mol) at -40° C. and the mixture was stirred at -40° C. for 0.5 hour. After completion, the mixture was quenched by water (3 mL) and filtered. Then the mixture was extracted with dichloromethane twice. The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum to give 5-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (474 mg, crude). Brown oil; LCMS [ESI, M+1]: m/z 368.0.

(742) Step B. 5-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: A mixture of 5-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (454 mg, 1.2 mmol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (589 mg, 3.7 mmol), DIEA (478 mg, 3.7 mmol, 644 μ L) and 4 Å molecular sieves (400 mg) in dioxane (1 mL) was stirred at 90° C. for 16 hours. After completion, the mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 5-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (80 mg, 9% yield over two steps). Yellow solid; LCMS [ESI, M+1]: m/z 491.2.

(743) Step C. 5-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To the mixture of 5-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (70 mg, 143 μ mol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (67 mg, 185 μ mol), K.sub.3PO.sub.4 (1.5 M, 285 μ L) in THF (1 mL) was added cataCXium-A-Pd-G3 (10 mg, 14 μ mol) under N.sub.2, and the mixture was stirred at 60° C. for 4 hours. After completion, the mixture was quenched by water (2 mL). Then the mixture was extracted with ethyl acetate twice. The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 5-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (53 mg, 43% yield). Yellow solid; LCMS [ESI, M+1]: m/z 689.4.

(744) Step D. 4-(4-(2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol. 5-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (14 mg, 20 μ mol) was added to HCl-MeOH (4.0 mmol, 1 mL) at 0° C. and the mixture was stirred at 0° C. for 20 min. After completion, the mixture was concentrated in vacuum. The pH value was adjusted to 9.0 with saturated NaHCO.sub.3 solution and the mixture was extracted with ethyl acetate (15 mL $\times$ 2). The aqueous phase was extracted with ethyl acetate (2 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 20%-50%, 10 min) to afford 4-(4-(2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-

6-fluoronaphthalen-2-ol (4.58 mg, 34% yield). White solid; ¹H NMR (400 MHz, METHANOL-d₄): 8=9.19 (s, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.8 Hz, 1H), 5.78 (s, 1H), 5.39-5.22 (m, 1H), 5.18-5.07 (m, 2H), 4.40-4.21 (m, 6H), 3.24-3.11 (m, 2H), 3.01 (m, 1H), 2.43-2.30 (m, 4H), 2.27-2.07 (m, 4H), 2.05-1.86 (m, 3H), 0.83-0.75 (m, 3H); LCMS [ESI, M+1]: m/z 645.3.

Example 62

(745) ##STR00267##

4-(4-(((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(746) ##STR00268##

(747) Step A. N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-amine: To a solution of 2,4-dichloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidine (163 mg, 362 μmol) and DIPEA (148 mg, 1.15 mmol) in DCM (2 mL) were added (1H-pyrrolo[2,3-b]pyridin-3-yl)methanamine (44 mg, 200 μmol, 2HCl) and DIPEA (74.2 mg, 574 μmol) in DMF (1.5 mL) dropwise at -40° C. The mixture was stirred at -40° C. for 0.1 h. The reaction mixture was diluted with water (50 mL) and brine (10 mL), and was extracted with ethyl acetate (20 mL×4). The combined organic layers were dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=2/3] to afford the product as light yellow solid (100 mg, 57% yield). LCMS (ESI, M+1): m/z 561.1.

(748) Step B. N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: A mixture of N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-amine (150 mg, 267 μmol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (128 mg, 804 μmol), DIPEA (104 mg, 804 μmol) and 4 Å molecular sieves (60 mg) in dioxane (2.5 mL) was degassed and purged with N₂ for 3 times. The mixture was stirred at 95° C. for 48 hours under N₂ atmosphere. The reaction mixture was filtered through a pad of Celite. The filter cake was washed with DCM (10 mL). The combined organic phase was concentrated under reduced pressure, and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=1/1] to afford the product as white solid (100 mg, 54% yield). LCMS (ESI, M+1): m/z 684.3.

(749) Step C. 4-(4-(((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To a solution of N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine (100 mg, 146 μmol) in DCM (2 mL) was added TFA (2.5 mL) dropwise at 0° C. The mixture was stirred between 0 and 15° C. for 1.5 hours. The reaction mixture was concentrated under reduced pressure to give a residue at room temperature. The residue was dissolved in DCM (20 mL) and water (5 mL). The pH of the mixture was adjusted to 9 with NaHCO₃ solid below 5° C. The mixture was extracted with DCM (10 mL×4). The combined organic layers were dried over anhydrous Na₂SO₄ and filtered. The mixture was concentrated under reduced pressure. The residue was purified by prep-HPLC (column: water s Xbridge 150×25 mm×Sum; A: [water (10 mM NH₄HCO₃)]; B: CAN, B %: 40%-70% over 10 minutes) to afford the product as white solid (53.4 mg, 55% yield).

HPLC: >99% ee, Chiralcel OJ-3 50×4.6 mm I.D., 3 μm column A: CO₂, B: MeOH (w/0.05% DEA), 5% to 40%, 3 mL/min, 220 nm, t_R: 2.009 min; ¹H NMR (400 MHz, DMSO-d₆): δ 11.59 (s, 1H), 9.91 (br s, 1H), 9.39-9.29 (m, 2H), 8.23-8.14 (m, 2H), 7.55-7.54 (m, 1H), 7.77-7.73 (m, 1H), 7.39-7.24 (m, 2H), 7.11-6.98 (m, 2H), 5.29 (d, J=53.6 Hz, 1H), 4.94-4.90 (m,

2H), 4.20-4.09 (m, 2H), 3.10-3.08 (m, 2H), 3.08-3.07 (m, 1H), 3.07-3.01 (m, 1H), 2.10-1.78 (m, 8H), 0.73-0.64 (m, 3H); .sup.19F NMR (376 MHz, DMSO-d.sub.6): δ -119, -139, -172; LCMS (ESI, M+1): m/z 640.3.

Example 63

(750) ##STR00269##

4-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(751) ##STR00270##

(752) Step A. 4-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-6-methyl-1,4-oxazepan-6-ol: To a mixture of 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine (189 mg, 420 μ mol) and DIEA (217 mg, 1.68 mmol, 292 μ L) in dichloromethane (2 mL) was added 6-methyl-1,4-oxazepan-6-ol (49.5 mg, 378 μ mol) at -40° C. and the mixture was stirred at -40° C. for 20 mins. After completion, the mixture was quenched by water (8 mL) and filtered. Then the mixture was extracted with dichloromethane (8 mL $\times$ 2). The combined organic layer was dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 4-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-6-methyl-1,4-oxazepan-6-ol (103 mg, 44% yield). Yellow solid. LCMS (ESI, M+1): m/z 545.2.

(753) Step B. 4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of 4-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-6-methyl-1,4-oxazepan-6-ol (103 mg, 189 μ mol), ((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (90.3 mg, 567 μ mol), DIEA (73.3 mg, 567 μ mol, 98.8 μ L) and 4 Å molecular sieves (10 mg) in dioxane (1 mL) was stirred at 90° C. for 21 h. After completion, the mixture was filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to the title compound (92.0 mg, 73% yield). Yellow solid. LCMS (ESI, M+1): m/z 668.3.

(754) Step C. 4-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To a mixture of 4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (84.0 mg, 125.8 μ mol) in dichloromethane (0.5 mL) was added TFA (770 mg, 6.75 mmol, 0.5 mL) at 0° C. and the mixture was stirred at 25° C. for 0.5 h. After completion, the mixture was concentrated in vacuum. The pH value was adjusted to 9 with saturated NaHCO.sub.3 solution and the mixture was extracted with ethyl acetate twice. The combined organic layer was dried over Na.sub.2SO.sub.4 and filtered. The residue was concentrated in vacuum. The residue was purified by prep-HPLC (Shim-pack C18 150*25*10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 19%-39%, 10 min) to afford the title compound (26.9 mg, 33% yield, 0.2 formic acid salt). White solid. .sup.1H NMR (400 MHz, METHANOL-d4): δ 9.61-9.56 (m, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.33-7.30 (m, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (dd, J=2.4, 13.6 Hz, 1H), 5.54-5.37 (m, 1H), 4.63-4.44 (m, 5H), 4.25-4.14 (m, 1H), 4.07-3.85 (m, 3H), 3.74-3.66 (m, 2H), 3.65-3.56 (m, 2H), 3.29-3.23 (m, 1H), 2.60-2.37 (m, 3H), 2.33-2.03 (m, 5H), 1.28 (d, J=2.4 Hz, 3H), 0.84-0.76 (m, 3H). LCMS (ESI, M+1): m/z 624.3.

Example 64

(755) ##STR00271##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(756) ##STR00272##

(757) Step A. 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)

methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To the mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30 mg, 56 μ mol), 6-methyl-1,4-oxazepan-6-ol (15 mg, 113 μ mol) and 4 Å molecular sieves (5.0 mg) in DMF (0.5 mL) was added DIEA (22 mg, 169 mmol). The mixture was stirred at 40° C. for 12 hours. After completion, the reaction mixture was filtered and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 min) and (column: Phenomenex Gemini-NX C18 75×30 mm×3 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 25%-55%, 8 min) to afford 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (3.93 mg, 12% yield). Yellow solid; ¹H NMR (400 MHz, CHLOROFORM-d): δ 9.29 (d, J=2.4 Hz, 1H), 8.04-8.00 (m, 1H), 7.76 (d, J=8.0 Hz, 1H), 7.67-7.44 (m, 3H), 7.14 (br dd, J=7.6, 12.8 Hz, 1H), 5.64-5.21 (m, 1H), 4.73-4.61 (m, 1H), 4.56-4.43 (m, 1H), 4.39-4.22 (m, 2H), 4.05-3.80 (m, 3H), 3.76-3.52 (m, 3H), 3.26-3.08 (m, 2H), 2.79-2.57 (m, 2H), 2.14-2.01 (m, 2H), 1.93-1.84 (m, 4H), 1.81-1.74 (m, 2H), 1.42-1.34 (m, 3H); LCMS (ESI, M+1): m/z 562.2.

Example 65

(758) ##STR00273##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylazepan-3-ol

(759) ##STR00274##

(760) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylazepan-3-ol: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.3 μ mol, 1.0 equiv.), DIEA (60.9 mg, 471 μ mol, 82.1 μ L, 5.0 equiv.) and 4 Å molecular sieves (5.00 mg) in DMF (1.0 mL) was added 3-methylazepan-3-ol (24.4 mg, 189 μ mol, 2.0 equiv.). The mixture was stirred at 40° C. for 12 h. Upon completion, the reaction solution was filtered and purified by prep-HPLC [Phenomenex luna C18 150×25 mm×10 μ m; A: water (0.225% formic acid), B: ACN, B %: 19%-49% over 10 min] to afford 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylazepan-3-ol (12.6 mg, 15% yield). Off-white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ 9.45 (br d, J=8.4 Hz, 1H), 8.54 (s, 1H), 8.11 (br d, J=8.4 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.73-7.65 (m, 1H), 7.65-7.57 (m, 1H), 7.56-7.48 (m, 1H), 7.23-7.14 (m, 1H), 4.57-4.51 (m, 1H), 4.50-4.38 (m, 2H), 4.37-4.24 (m, 1H), 3.98-3.81 (m, 2H), 3.50-3.38 (m, 2H), 3.11-2.97 (m, 2H), 2.29-2.13 (m, 3H), 2.13-1.98 (m, 5H), 1.98-1.88 (m, 3H), 1.87-1.67 (m, 3H), 1.42-1.31 (m, 3H); LCMS (ESI, M+1): m/z 560.2.

Example 66

(761) ##STR00275##

1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(quinolin-8-yl)pyrido[4,3-d]pyrimidin-1-yl)-3-methylpiperidin-3-ol

(762) ##STR00276##

(763) Step A. 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(quinolin-8-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 229 μ mol), quinolin-8-ylboronic acid (79.4 mg, 459 μ mol, 79.4 μ L), Na₂CO₃ (72.9 mg, 688 μ mol) and tetrabutylammonium bromide (74.0 mg, 229 μ mol) in 1,2-dimethoxyethane (3.0 mL), EtOH (3.0 mL) and water (1.0 mL) was added Pd(PPh₃)₄ (53.0 mg, 45.9 μ mol, 0.2 equiv.). The mixture was stirred at 80° C. for 12 h. Upon completion, the mixture reaction was diluted water (5.0 mL) and was extracted with EtOAc (3×5.0 mL). The combined organic layer was dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated in vacuum. The reaction was purified by prep-HPLC [Phenomenex Gemini-NX C18

75×30 mm×3 µm; A: water (0.225% formic acid), B: ACN, B %: 8%-38% over 7 min] to afford 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(quinolin-8-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (7.25 mg, 5.8% yield). White solid; ¹H NMR (400 MHz, METHANOL-d₄) δ 9.29 (s, 1H), 8.85-8.79 (m, 1H), 8.54 (s, 1H), 8.50-8.45 (m, 1H), 8.15 (d, J=8.0 Hz, 1H), 7.98-7.71 (m, 1H), 7.83-7.76 (m, 1H), 7.63-7.57 (m, 1H), 4.59 (br d, J=12.8 Hz, 3H), 4.34 (br d, J=13.2 Hz, 1H), 3.72-3.62 (m, 1H), 3.52-3.39 (m, 3H), 3.10-2.98 (m, 2H), 2.27-2.19 (m, 2H), 2.16-1.91 (m, 7H), 1.89-1.75 (m, 3H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z 529.3.

Example 67

(764) ##STR00277##

1-(7-(2-aminobenzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(765) ##STR00278##

(766) Step A. tert-butyl (4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)benzo[d]thiazol-2-yl)carbamate: To a mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (150 mg, 344 µmol), (2-((tert-butoxycarbonyl)amino)benzo[d]thiazol-4-yl)boronic acid (202 mg, 688 µmol) and K₂CO₃ (1.5 M, 688 µL) in THF (2.0 mL) was added BrettPhos Pd G3 (31.2 mg, 34.4 µmol). The mixture was stirred at 60° C. for 16 h. After completion, water (10 mL) was added and the mixture was extracted with EtOAc (2×10 mL). The combined organic layer was dried over Na₂SO₄. The mixture was filtered and concentrated. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording tert-butyl (4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)benzo[d]thiazol-2-yl)carbamate (190 mg, 79% yield). Yellow solid; LCMS (ESI, M+1): m/z 650.3.

(767) Step B. 1-(7-(2-aminobenzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of tert-butyl (4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)benzo[d]thiazol-2-yl)carbamate (90.0 mg, 139 µmol) in dichloromethane (1.0 mL) was added TFA (1.54 g, 13.5 mmol, 1.0 mL). The mixture was stirred at 20° C. for 0.5 h. After completion, the pH of the mixture was adjusted to -7 with saturated NaHCO₃ aqueous solution. The mixture was extracted with EtOAc (2×10 mL). The combined organic layer was dried over Na₂SO₄. The mixture was filtered and concentrated. The residue was purified by prep-HPLC [Phenomenex luna C18 150×25 mm×10 µm; A: water (0.225% formic acid), B: ACN, B %: 9%-39% over 10 min], followed by prep-HPLC [Water s Xbridge 150×5 mm×5 µm; A: water (10 mM NH₄HCO₃), B: ACN, B %: 20%-50% over 10 min] to afford 1-(7-(2-aminobenzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (9.54 mg, 12% yield). Yellow solid; ¹H NMR (400 MHz, METHANOL-d₄) δ 9.26 (s, 1H), 7.77 (dd, J=1.2, 8.0 Hz, 1H), 7.45 (d, J=6.8 Hz, 1H), 7.24 (t, J=8.0 Hz, 1H), 4.66-4.56 (m, 3H), 4.33 (br d, J=13.2 Hz, 1H), 3.76-3.60 (m, 3H), 3.49-3.39 (m, 1H), 3.30-3.24 (m, 2H), 2.39-2.28 (m, 2H), 2.27-2.07 (m, 7H), 1.92-1.73 (m, 3H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z 550.1.

Example 68

(768) ##STR00279##

1-(7-(benzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(769) ##STR00280##

(770) Step A. 1-(7-(benzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 229 µmol), benzo[d]thiazol-4-ylboronic acid (53.4 mg, 298 µmol), K₂CO₃ (1.5

M, 0.5 mL) and [2-(2-aminophenyl)phenyl]palladium(1+); bis (1-adamantyl)-butyl-phosphane; methanesulfonate (25.1 mg, 34.4 μ mol) in THF (2.00 mL) was degassed and purged with N.sub.2 for 3 times, and then the mixture was stirred at 60° C. for 10 hours under N.sub.2 atmosphere. Upon completion, the reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (3 $\times$ 10 mL). The combined organic layers were washed with brine (50 mL) and dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150 $\times$ 25 mm $\times$ 10 μ m; mobile phase: [water (0.225% formic acid formic acid)-ACN]; B %: 11%-41%, 10 min) to give 1-(7-(benzo[d]thiazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (52.32 mg, 42% yield). Yellow solid; .sup.1H NMR (400 MHz, METHANOL-d4) δ 9.36-9.25 (m, 2H), 8.53 (s, 1H), 8.27 (dd, J=0.8 Hz, 8.0 Hz, 1H), 7.79 (dd, J=0.8 Hz, 7.2 Hz, 1H), 7.69 (t, J=8.0 Hz, 1H), 4.67-4.59 (m, 3H), 4.40-4.30 (m, 1H), 3.72-3.62 (m, 3H), 3.48-3.39 (m, 1H), 3.30-3.20 (m, 2H), 2.36-2.28 (m, 2H), 2.27-2.05 (m, 7H), 1.90-1.74 (m, 3H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z 535.3.

Example 69

(771) ##STR00281##

1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(772) ##STR00282##

(773) Step A. 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50.0 mg, 115 μ mol), naphthalen-1-ylboronic acid (39.5 mg, 229 μ mol) and K.sub.3PO.sub.4 (1.50 M in water, 229 μ L) in THF (1.0 mL) was added cataCXium-A-Pd-G3 (8.35 mg, 11.5 μ mol). The mixture was stirred at 60° C. for 12 h. Upon completion, the reaction mixture was diluted water (2.0 mL) and was extracted with EtOAc (3 $\times$ 2 mL). The combined organic layer was dried over Na.sub.2SO.sub.4. The mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC [Phenomenex luna C18 150 $\times$ 25 mm $\times$ 10 μ m; A: water (0.225% formic acid formic acid), B: ACN, B %: 17/6-47% over 10 min] to afford 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (14.9 mg, 24% yield). Yellow solid; .sup.1H NMR (400 MHz, METHANOL-d4) δ 9.34 (s, 1H), 8.52 (s, 1H), 8.09-8.04 (m, 1H), 8.01 (d, J=8.4 Hz, 1H), 7.70 (br d, J=8.8 Hz, 1H), 7.67-7.62 (m, 2H), 7.59-7.53 (m, 1H), 7.52-7.46 (m, 1H), 4.61-4.55 (m, 4H), 4.40-4.34 (m, 1H), 3.72-3.62 (m, 3H), 3.50-3.39 (m, 1H), 3.28-3.22 (m, 1H), 2.38-2.28 (m, 2H), 2.27-2.04 (m, 7H), 1.92-1.74 (m, 3H), 1.30 (s, 3H); LCMS (ESI, M+1): m/z 528.3.

Example 70

(774) ##STR00283##

N-((2-aminopyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(775) ##STR00284##

(776) Step A. N-((2-aminopyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (30 mg, 56.55 μ mol) and 3-(aminomethyl)pyridin-2-amine (34.8 mg, 283 μ mol) in DMF (3 mL) were added DIPEA (21.9 mg, 170 μ mol) and 4 Å molecular sieves (50 mg). The mixture was stirred at 40° C. for 12 hours. The mixture was filtered and purified by prep-HPLC (column: Unisil 3-100 C18 Ultra 150 $\times$ 50 mm $\times$ 3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-35%, 10 min) to give N-((2-aminopyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine (11.1 mg, 35% yield, 2 FORMIC ACID) as white solid.

.sup.1H NMR (400 MHz, DMSO-d₆): δ 9.57-9.26 (m, 2H), 8.30 (br s, 2H), 8.19 (br d, J=8.4 Hz, 1H), 7.98-7.85 (m, 2H), 7.80-7.69 (m, 1H), 7.65-7.52 (m, 2H), 7.47-7.36 (m, 1H), 7.35-7.22 (m, 1H), 6.56 (dd, J=5.0, 7.2 Hz, 1H), 6.06-5.88 (m, 2H), 4.65-4.55 (m, 2H), 4.16-4.02 (m, 3H), 3.02-2.87 (m, 3H), 1.97-1.66 (m, 7H), 1.64-1.49 (m, 2H); LCMS (ESI, M+1): m/z 554.2.

Example 71

(777) ##STR00285##

1-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(778) ##STR00286## ##STR00287##

(779) Step A. 1-bromo-3-chloro-2-cyclopropyl-benzene: A mixture of cyclopropylboronic acid (1.62 g, 18.91 mmol), 1-bromo-3-chloro-2-iodo-benzene (2 g, 6.30 mmol), K₃PO₄ (4.82 g, 22.7 mmol), and Pd(dppf)Cl₂ (461 mg, 630 μmol) in 1,4-dioxane (12 mL) and water (3 mL) was degassed and purged with N₂ for 3 times. The mixture was stirred at 100° C. for 7 hours under N₂ atmosphere. Upon completion, the reaction mixture was extracted with ethyl acetate (10 mL×2). The combined organic layers were washed with brine 15 mL and dried over Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=I/O) to give 1-bromo-3-chloro-2-cyclopropyl-benzene as a colorless liquid (1.6 g, 88% yield); .sup.1H NMR (400 MHz, CDCl₃) δ=7.38 (d, J=8.0 Hz, 1H), 7.22 (d, J=8.0 Hz, 1H), 6.91 (t, J=8.0 Hz, 1H), 2.31 (tt, J=5.4, 8.5 Hz, 1H), 1.69 (tt, J=5.8, 8.4 Hz, 1H), 1.29-1.16 (m, 1H), 1.15-1.06 (m, 3H), 0.84-0.74 (m, 2H), 0.74-0.66 (m, 3H).

(780) Step B. 2-(3-bromo-5-chloro-4-cyclopropyl-phenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: A mixture of 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (995 mg, 7.77 mmol, 1.13 mL), 1-bromo-3-chloro-2-cyclopropyl-benzene (600 mg, 2.59 mmol), [Ir(OMe)(COD)]₂ (172 mg, 259 μmol) and 4-tert-butyl-2-(4-tert-butyl-2-pyridyl)pyridine (83.5 mg, 311 μmol) in hexane (10 mL) was degassed and purged with N₂ for 3 times. The reaction mixture was stirred at 60° C. for 3 hours under N₂ atmosphere. The reaction mixture was concentrated under reduced pressure to remove solvent. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=50/1) to give 2-(3-bromo-5-chloro-4-cyclopropyl-phenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (782 mg, 84% yield) as a colorless oil; .sup.1H NMR (400 MHz, CDCl₃) δ=7.79 (s, 1H), 7.63 (s, 1H), 1.71 (tt, J=5.8, 8.5 Hz, 1H), 1.26 (s, 12H), 1.15-1.09 (m, 2H), 0.72-0.68 (m, 2H).

(781) Step C. 3-bromo-5-chloro-4-cyclopropyl-phenol: To a solution of 2-(3-bromo-5-chloro-4-cyclopropyl-phenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (1.12 g, 3.13 mmol) in water (4 mL) and THF (8 mL) were added AcOH (12.1 g, 200 mmol, 11.5 mL) and water 2 (2.05 g, 18.1 mmol, 1.74 mL, 30% purity). The mixture was stirred at 0° C. for 1 hour. The reaction mixture was quenched with Na₂S₂O₃ (10% aq, 40 mL) at 0° C. The mixture was extracted with ethyl acetate 60 mL (30 mL×2). The combined organic layers were washed with brine 100 mL and dried over Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=10/1) to give 3-bromo-5-chloro-4-cyclopropyl-phenol as a colorless oil (700 mg, 90% yield); .sup.1H NMR (400 MHz, CDCl₃) δ=6.93 (d, J=2.6 Hz, 1H), 6.77 (d, J=2.6 Hz, 1H), 5.73-5.49 (m, 1H), 1.64-1.55 (m, 1H), 1.08-1.00 (m, 2H), 0.67-0.60 (m, 2H).

(782) Step D. 1-bromo-3-chloro-2-cyclopropyl-5-(methoxymethoxy)benzene: To a solution of chloro(methoxy)methane (1.62 g, 20.12 mmol, 1.53 mL), 3-bromo-5-chloro-4-cyclopropyl-phenol (1.2 g, 4.85 mmol) and DIEA (1.88 g, 14.5 mmol, 2.53 mL) in DCM (15 mL) was added chloro(methoxy)methane (1.62 g, 20.1 mmol, 1.53 mL) at 0° C. The reaction mixture was stirred at 25° C. for 2 hours. The reaction mixture was quenched with water (10 mL) at 0° C. The mixture was extracted with EtOAc (10 mL×3). The combined organic layers were washed with brine (30 mL) and dried over Na₂SO₄. The mixture was filtered and concentrated under reduced

pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=30/1) to give 1-bromo-3-chloro-2-cyclopropyl-5-(methoxymethoxy)benzene as a colorless oil (923 mg, 65% yield); ¹H NMR (400 MHz, CDCl₃) δ=7.20 (d, J=2.6 Hz, 1H), 7.04 (d, J=2.5 Hz, 1H), 5.12 (s, 2H), 3.47 (s, 3H), 1.70 (tt, J=5.7, 8.4 Hz, 1H), 1.17-1.11 (m, 2H), 0.76-0.69 (m, 2H).

(783) Step E. 2-[3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: To a mixture of 1-bromo-3-chloro-2-cyclopropyl-5-(methoxymethoxy)benzene (300 mg, 1.03 mmol), KOAc (303 mg, 3.09 mmol) and 4,4,5,5-tetramethyl-2-(4,5,5-trimethyl-4-methyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (521 mg, 2.06 mmol) in dioxane (4.5 mL) was added Pd(dppf)Cl₂·DCE (75.37 mg, 103 μmol). The mixture was stirred at 100° C. for 1 hour. Upon completion, to the reaction mixture was added water (5 mL) and the mixture was extracted with ethyl acetate 15 mL (3×5 mL). The organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated. The residue was purified by prep-TLC (Silica gel, Petroleum ether/Ethyl acetate=15:1) to give 2-[3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane as a colorless oil (135 mg, 39% yield); ¹H NMR (400 MHz, CDCl₃) δ=7.11-7.08 (m, 2H), 5.14 (s, 2H), 3.46 (s, 3H), 1.98 (tt, J=5.6, 8.5 Hz, 1H), 1.38 (s, 12H), 1.01-0.95 (m, 2H), 0.55-0.50 (m, 2H).

(784) Step F. 1-(7-(3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 2-[3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (77.7 mg, 229 μmol), 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50 mg, 115 μmol) and K₂CO₃ (1.5 M in water, 229.40 μL) in THE (0.5 mL) was added Pd(PPh₃)₄ (8.35 mg, 11.5 μmol). The mixture was degassed and purged with N₂ for 3 times. The mixture was stirred at 60° C. for 2.5 hours. The reaction mixture was diluted with water (5 mL) and the mixture was extracted with ethyl acetate (3×5 mL). The organic layer was dried over Na₂SO₄, filtered and concentrated. The residue was purified by prep-TLC (Silica gel, DCM:MeOH=10:1) to give the title compound as a brown oil (43 mg, 61% yield); ¹H NMR (400 MHz, CDCl₃) δ=9.15-9.13 (m, 1H), 7.20 (d, J=2.5 Hz, 1H), 7.08-7.06 (m, 1H), 5.19-5.16 (m, 2H), 4.46-4.39 (m, 2H), 4.39-4.34 (m, 3H), 3.55-3.50 (m, 1H), 3.49 (s, 2H), 3.47 (s, 3H), 3.40 (d, J=13.5 Hz, 1H), 3.30 (br d, J=6.6 Hz, 2H), 2.72 (td, J=6.8, 10.4 Hz, 2H), 2.21-2.11 (m, 3H), 1.97-1.91 (m, 6H), 1.79-1.71 (m, 4H), 1.70-1.64 (m, 1H), 0.67 (br s, 2H), 0.19-0.06 (m, 2H).

(785) Step G. 1-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(7-(3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (40 mg, 65.3 μmol) in MeCN (0.8 mL) was added HCV/EtOAc (4 M, 1.8 mL). The mixture was stirred at 20° C. for 0.5 hour. After completion, the mixture was concentrated to give a residue. The residue was purified by prep-HPLC (FA condition, column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 16%-46%, 11 min) to give the title compound (19.68 mg, 48% yield, FA salt); ¹H NMR (400 MHz, METHANOL-d₄) δ=9.26 (s, 1H), 8.53 (br s, 1H), 6.98 (d, J=2.5 Hz, 1H), 6.80 (d, J=2.6 Hz, 1H), 4.61 (s, 2H), 4.58 (br s, 1H), 4.33 (br d, J=13.5 Hz, 1H), 3.68-3.59 (m, 3H), 3.43 (br t, J=11.0 Hz, 1H), 3.27-3.20 (m, 2H), 2.35-2.26 (m, 2H), 2.25-2.02 (m, 8H), 1.91-1.75 (m, 4H), 1.32-1.28 (m, 3H), 0.71-0.56 (m, 2H), 0.15-0.00 (m, 2H). LCMS (ESI, M+1): m/z 568.4.

Example 72

(786) ##STR00288##

8-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,8-triazaspiro[5.5]undecan-2-one

(787) ##STR00289##

(788) Step A. 2-(1-benzylpiperidin-3-ylidene)acetonitrile: A mixture of 2-diethoxyphosphorylacetonitrile (9.27 g, 52.3 mmol, 8.42 mL), K₂CO₃ (8.89 g, 64.3 mmol) in THF (25 mL) was stirred for 10 min at 25° C. and 20 min at 70° C. under N₂ atmosphere. The reaction mixture was cooled to 25° C. A mixture of 1-benzylpiperidin-3-one (10 g, 52.84 mmol) and K₂CO₃ (7.30 g, 52.84 mmol) in THF (25 mL) was added into the mixture and the reaction was stirred at 70° C. for further 18 hours. The mixture was diluted with water (350 mL) and extracted with ethyl acetate (200 mL×2). The combined organic layers were dried over Na₂SO₄ and concentrated to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=10/1 to 4/1) to give 2-(1-benzyl-3-piperidylidene)acetonitrile (7.8 g, crude) as a yellow oil. ¹H NMR (400 MHz, CDCl₃) δ=7.27 (br s, 5H), 5.17-5.01 (m, 1H), 4.09 (dq, J=1.5, 7.1 Hz, 1H), 3.65-3.47 (m, 2H), 2.96 (s, 2H), 2.64-2.39 (m, 4H), 2.07-1.95 (m, 1H), 1.84-1.58 (m, 2H). LCMS (ESI, M+1): m/z 213.2.

(789) Step B. 2-(3-amino-1-benzyl-3-piperidyl)acetonitrile: To a mixture of 2-(1-benzyl-3-piperidylidene)acetonitrile (4 g, 18.8 mmol) in MeOH (6 mL) was added ammonium hydroxide (39.6 g, 339 mmol, 43.5 mL, 30% purity). The mixture was heated to 80° C. and stirred for 18 hours in a sealed tube. The reaction mixture was concentrated under reduced pressure to give 2-(3-amino-1-benzyl-3-piperidyl)acetonitrile (3.5 g, crude) as a yellow oil. ¹H NMR (400 MHz, CDCl₃) δ=7.31-7.09 (m, 5H), 3.50-3.33 (m, 2H), 2.58-2.11 (m, 8H), 1.76-1.29 (m, 4H). LCMS (ESI, M+1): m/z 230.1.

(790) Step C. tert-butyl N-[1-benzyl-3-[2-(tert-butoxycarbonylamino)ethyl]-3-piperidyl]carbamate: To a mixture of 2-(3-amino-1-benzyl-3-piperidyl)acetonitrile (4.0 g, 17.4 mmol), tert-butoxycarbonyl tert-butyl carbonate (15.2 g, 69.8 mmol, 16.0 mL) and NiCl₂·6H₂O (2.07 g, 8.72 mmol) in MeOH (40 mL) was added NaBH₄ (3.35 g, 88.5 mmol) at 0° C. in portions. The reaction mixture was stirred at 27° C. for 3 hours under N₂ atmosphere. The mixture was quenched with saturated NH₄Cl solution at 25° C. and diluted with brine (80 mL). The mixture was extracted with ethyl acetate (80 mL×3). The combined organic layers were dried over Na₂SO₄, then filtered and concentrated to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=0 to 20/1) to give tert-butyl N-[1-benzyl-3-[2-(tert-butoxycarbonylamino)ethyl]-3-piperidyl]carbamate (2.2 g, 25% yield) as a colorless liquid. ¹H NMR (400 MHz, CDCl₃) δ=7.34-7.12 (m, 5H), 5.12-4.99 (m, 1H), 4.78-4.66 (m, 1H), 4.63-4.42 (m, 3H), 3.45-3.31 (m, 2H), 3.18-3.01 (m, 1H), 3.01-2.87 (m, 1H), 2.71-2.60 (m, 2H), 2.26-2.02 (m, 2H), 1.94-1.73 (m, 2H), 1.68-1.47 (m, 2H), 1.40-1.36 (m, 23H), 1.36-1.33 (m, 8H). LCMS (ESI, M+1): m/z 434.5.

(791) Step D. 3-(2-aminoethyl)-1-benzyl-piperidin-3-amine: A mixture of tert-butyl N-[1-benzyl-3-[2-(tert-butoxycarbonylamino)ethyl]-3-piperidyl]carbamate (500 mg, 1.15 mmol) and HCl/EtOAc (4 M, 7.21 mL) was stirred at 25° C. for 18 hours. The reaction mixture was concentrated under reduced pressure to give 3-(2-aminoethyl)-1-benzyl-piperidin-3-amine (250 mg, crude, HCl) as a white solid. ¹H NMR (400 MHz, METHANOL-d₄) δ=7.79-7.63 (m, 2H), 7.56-7.29 (m, 3H), 4.57-4.35 (m, 2H), 3.69-3.42 (m, 3H), 3.25-2.99 (m, 4H), 3.24-2.95 (m, 3H), 2.22-2.03 (m, 4H), 2.00-1.94 (m, 3H). LCMS (ESI, M+1): m/z 234.0.

(792) Step E. 8-benzyl-1,3,8-triazaspiro[5.5]undecan-2-one: To a mixture of 3-(2-aminoethyl)-1-benzyl-piperidin-3-amine (250 mg, 926 μmol, HCl) in toluene (25 mL) was added TEA (281 mg, 2.78 mmol, 387 μL) and di(imidazol-1-yl)methanone (300 mg, 1.85 mmol). The reaction mixture was stirred at 80° C. for 16 hours. The reaction mixture was concentrated under reduced pressure to remove toluene. The residue was diluted with brine 5 mL and extracted with EtOAc (5 mL×3). The combined organic layers were dried over Na₂SO₄ and filtered. The mixture was concentrated to give a residue. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 2%-32%, 10 min) to give 8-benzyl-1,3,8-triazaspiro[5.5]undecan-2-one (70 mg, 269.91 μmol, 29% yield) as a white

solid. ¹H NMR (400 MHz, METHANOL-d₄) δ=7.62-7.46 (m, 5H), 7.44-7.27 (m, 1H), 4.48-4.28 (m, 2H), 3.59-3.38 (m, 2H), 3.28-3.12 (m, 2H), 3.09-2.72 (m, 3H), 2.27-2.19 (m, 1H), 2.31-2.16 (m, 1H), 2.14-1.57 (m, 9H). LCMS (ESI, M+1): m/z 260.2.

(793) Step F. 1,3,8-triazaspiro[5.5]undecan-2-one: To a mixture of 8-benzyl-1,3,8-triazaspiro[5.5]undecan-2-one (70 mg, 270 μmol) in MeOH (1 mL) was added Pd/C (34.5 mg, 32.4 μmol, 10% purity). The reaction mixture was stirred at 25° C. for 16 hours under H.sub.2 (15 psi). The reaction mixture was filtered and concentrated under reduced pressure to give 1,3,8-triazaspiro[5.5]undecan-2-one (45 mg, crude) as a colorless oil. ¹H NMR (400 MHz, METHANOL-d₄) δ=3.29-3.22 (m, 3H), 3.21-3.03 (m, 4H), 2.92-2.84 (m, 1H), 2.92-2.78 (m, 1H), 2.06-1.83 (m, 5H), 1.82-1.67 (m, 2H). LCMS (ESI, M+1): m/z 170.1.

(794) Step G. 8-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,8-triazaspiro[5.5]undecan-2-one: To a mixture of 1,3,8-triazaspiro[5.5]undecan-2-one (45 mg, 266 μmol) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (155 mg, 292 μmol) in DMF (3 mL) was added DIEA (103 mg, 798 μmol, 139 μL) and 4 Å molecular sieves (100 mg) in one portion at 25° C. The mixture was heated to 40° C. and stirred for 16 hours under N₂ atmosphere. The mixture was filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 12%-42%, 11 min) to give the title compound (9.06 mg, 5.63% yield) as an off-white solid. ¹H NMR (400 MHz, DMSO-d₆) δ=9.13 (d, J=1.9 Hz, 1H), 8.28 (s, 1H), 8.21-8.14 (m, 1H), 7.92 (d, J=7.6 Hz, 1H), 7.74 (dd, J=7.2, 8.2 Hz, 1H), 7.64-7.52 (m, 2H), 7.30 (dd, J=7.4, 13.6 Hz, 1H), 4.32-4.19 (m, 2H), 4.05-3.80 (m, 3H), 3.79-3.66 (m, 1H), 3.24-3.05 (m, 4H), 2.86-2.75 (m, 2H), 2.04-1.69 (m, 14H), 1.65-1.52 (m, 1H). LCMS (ESI, M+1): m/z 600.2.

Example 73

(795) ##STR00290##

7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one

(796) ##STR00291## ##STR00292##

(797) Step A. 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (400 mg, 1.58 mmol), DIEA (614 mg, 4.75 mmol) and 4 Å molecular sieves (10 mg) in DCM (8.0 mL) was added 1,7-diazaspiro[3.5]nonan-2-one (244 mg, 1.74 mmol) at -40° C. The mixture was stirred at -40° C. for 1 hour. The mixture was quenched with water (10 mL) and separated. The aqueous layer was extracted with EtOAc (2×10 mL). The combined organic layer was dried over anhydrous Na₂SO₄. The mixture was filtered and concentrated under reduced pressure to give 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (350 mg, crude) as yellow solid. (M+1). LCMS (ESI, M+1): m/z 356.1.

(798) Step B. 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one: To a solution of 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (300 mg, 842 μmol), 4 Å molecular sieves (50 mg) and DIEA (326 mg, 2.53 mmol) in dioxane (6.0 mL) was added ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (161 mg, 1.01 mmol). The mixture was stirred at 90° C. for 12 hours. The mixture was quenched with water (10 mL) and extracted with EtOAc (2×10 mL). The combined organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)-ACN) affording 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (150 mg, 311 μmol, 37% yield over two steps, 99.3% purity) as yellow solid. LCMS (ESI, M+1): m/z 479.2.

(799) Step C. 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one: To a mixture of 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (90.0 mg, 188 μ mol), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (135 mg, 376 μ mol) and Cs.sub.2CO.sub.3 (1.5 M, 376 μ L) in methoxycyclopentane (2.0 mL) was added cataCXium-A-Pd-G3 (27.4 mg, 37.6 μ mol). The mixture was stirred at 100° C. for 3 hours. The mixture was quenched with water (5 mL) and extracted with EtOAc (2 $\times$ 5 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) affording 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (100 mg, 78% yield) as yellow solid. LCMS (ESI, M+1): m/z 677.3.

(800) Step D. 7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one: To a solution of 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (90 mg, 133 μ mol) in DCM (1.5 mL) was added TFA (2.31 g, 20.3 mmol, 1.50 mL). The mixture was stirred at 20° C. for 0.5 hour. The mixture was concentrated. The residue was purified by prep-HPLC [Water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; A: water (10 mM NH.sub.4HCO.sub.3), B: ACN, B %: 30%-60/over 10 min) to afford 7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (16.7 mg, 19% yield) as white solid. ¹H NMR (400 MHz, METHANOL-d.sub.4): δ 9.06 (s, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 5.39-5.23 (m, 1H), 4.79-4.65 (m, 1H), 4.37-4.23 (m, 2H), 4.22-4.13 (m, 1H), 4.09-3.98 (m, 1H), 4.22-3.97 (m, 1H), 3.29-3.16 (m, 1H), 3.29-3.15 (m, 2H), 3.05-2.98 (m, 1H), 2.87 (s, 2H), 2.57-2.42 (m, 1H), 2.41-2.28 (m, 1H), 2.26-2.20 (m, 1H), 2.19-2.04 (m, 5H), 2.03-1.83 (m, 4H), 0.80 (br t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z 633.3.

Example 74

(801) ##STR00293##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(802) ##STR00294## ##STR00295##

(803) Step A. tert-butyl 6-(((tert-butyldiphenylsilyl)oxy)methyl)-1,4-oxazepane-4-carboxylate: To a solution of tert-butyl 6-(hydroxymethyl)-1,4-oxazepane-4-carboxylate (750 mg, 3.24 mmol) in THF (20 mL) were added imidazole (662 mg, 9.73 mmol) and TBDPSCl (1.34 g, 4.86 mmol, 1.25 mL) at 0° C. The mixture was stirred at 25° C. for 2 hours. The reaction mixture was diluted with water (100 mL) and extracted with ethyl acetate (3 $\times$ 50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give tert-butyl 6-(((tert-butyldiphenylsilyl)oxy)methyl)-1,4-oxazepane-4-carboxylate (1.10 g, 71% yield) as yellow oil. ¹H NMR (400 MHz, CDCl.sub.3) δ 7.74-7.60 (m, 4H), 7.50-7.32 (m, 6H), 3.85-3.56 (m, 8H), 3.41-3.21 (m, 1H), 3.19-3.01 (m, 1H), 2.41-2.20 (m, 1H), 1.52-1.39 (m, 9H), 1.06 (s, 9H). LCMS (ESI, M-55): m/z 414.2.

(804) Step B. 6-(((tert-butyldiphenylsilyl)oxy)methyl)-1,4-oxazepane: To a solution of tert-butyl 6-(((tert-butyldiphenylsilyl)oxy)methyl)-1,4-oxazepane-4-carboxylate (900 mg, 1.92 mmol) in DCM (5 mL) was added TFA (13.9 g, 121 mmol, 9.00 mL) at 0° C. The mixture was stirred at 0° C. for 1

hour. The reaction mixture was concentrated under reduced pressure to remove solvent. The residue was diluted with water 10 mL and pH was adjusted to 8 with NaHCO₃. The mixture was extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over Na₂SO₄ and filtered. The mixture was concentrated to give 6-(((tert-butyl diphenylsilyl)oxy)methyl)-1,4-oxazepane (700 mg, crude) as yellow solid. LCMS (ESI, M+1): m/z 370.3.

(805) Step C. 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (500 mg, 1.98 mmol) in DCM (10 mL) were added DIEA (768 mg, 5.94 mmol, 1.03 mL) and 6-(((tert-butyl diphenylsilyl)oxy)methyl)-1,4-oxazepane (439 mg, 1.19 mmol) at -40° C. The mixture was stirred at -40° C. for 0.5 hour. The reaction mixture was diluted with water (20 mL) and extracted with DCM (3×20 mL). The combined organic layers were dried over Na₂SO₄, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (700 mg, 60% yield); Yellow solid. LCMS (ESI, M+1): m/z 585.2.

(806) Step D. 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(7-chloro-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane: To a solution of 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (680 mg, 1.16 mmol) in dioxane (8 mL) were added DIEA (450 mg, 3.48 mmol), 4 Å molecular sieves (300 mg) and ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (370 mg, 2.32 mmol). The mixture was stirred at 95° C. for 12 hours. After completion, the reaction mixture was filtered and concentrated under reduced pressure to give a residue. The crude product was triturated with DMF (10 mL) at 20° C. for 1 hour to give 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(7-chloro-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (550 mg, 63% yield) as yellow solid. LCMS (ESI, M+1): m/z 708.4.

(807) Step E. 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane: To a solution of 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(7-chloro-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (500 mg, 706 µmol) and 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (381 mg, 1.06 mmol) in methoxycyclopentane (8 mL) were added K₃PO₄ (1.5 M, 1.41 mL) and cataCXium-A-Pd-G3 (77.1 mg, 106 µmol). The mixture was stirred at 90° C. for 2 hours. The reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over Na₂SO₄, filtered and concentrated to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to give 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (500 mg, 67% yield) as yellow solid. LCMS (ESI, M+1): m/z 906.5.

(808) Step F. 4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol: To a solution of 6-(((tert-butyl diphenylsilyl)oxy)methyl)-4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (300 mg, 285 µmol, 86% purity) in DMF (5 mL) was added CsF (432 mg, 2.85 mmol). The mixture was stirred at 25° C. for 2 hours. The reaction mixture was filtered and purified by reversed phase flash chromatography (C18, 0.1% formic acid in water, 0-60% ACN) to give 4-(7-(8-ethyl-7-fluoro-3-

(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol (180 mg, 91% yield); Yellow solid. LCMS (ESI, M+1): m/z 668.3.

(809) Step G. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of (4-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol (100 mg, 150 μ mol) in ACN (1 mL) was added HCl-dioxane (4.0 M, 2 mL). The mixture was stirred at 25° C. for 0.5 hour. After The reaction mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH₄sub.4HCO₃sub.3)-ACN]; B %: 32%-62%, 10 min) to give 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (58.0 mg, 62% yield); White solid. ¹H NMR (400 MHz, METHANOL-d₄) δ 9.19 (d, J=4.4 Hz, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (dd, J=2.4, 10.0 Hz, 1H), 5.41-5.19 (m, 1H), 4.59 (br s, 1H), 4.46-4.22 (m, 3H), 4.16-3.91 (m, 4H), 3.89-3.74 (m, 1H), 3.70-3.55 (m, 3H), 3.36-3.12 (m, 3H), 3.05-2.96 (m, 1H), 2.56-2.42 (m, 2H), 2.40-2.07 (m, 4H), 2.04-1.85 (m, 3H), 0.86-0.73 (m, 3H). LCMS (ESI, M+1): m/z 624.4.

Example 75

(810) ##STR00296##

4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(811) ##STR00297##

(812) Step A. 4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 94.0 μ mol) in DMF (0.5 mL) was added 4 Å molecular sieves (50.0 mg, 94.0 μ mol), DIEA (48.7 mg, 377 μ mol), 1-(azetidin-3-yl)imidazole (15.0 mg, 94.0 μ mol, HCl salt) and stirred at 40° C. for 16 hours under N₂ atmosphere. The mixture solution was filtered and purified by reverse-phase preparatory HPLC (15% to 25% water/ACN with 0.225% formic acid over 7 minutes) to give 4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine (10.81 mg, 20% yield). ¹H NMR (400 MHz CDCl₃): δ 8.94 (s, 1H), 8.04-7.99 (m, 1H), 7.79-7.72 (m, 2H), 7.68-7.62 (m, 1H), 7.61-7.57 (m, 1H), 7.47 (m, 1H), 7.29 (br s, 1H), 7.24 (s, 1H), 7.14 (dd, J=7.6, 12.8 Hz, 1H), 5.31-5.27 (m, 1H), 5.19 (m, 2H), 4.82 (m, 2H), 4.54 (s, 2H), 3.61-3.42 (m, 2H), 2.88-2.73 (m, 2H), 2.26 (m, 2H), 2.08-2.01 (m, 4H), 1.87-1.82 (m, 2H); LCMS (ESI, M+1): m/z 554.3

Example 76

(813) ##STR00298##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol

(814) ##STR00299##

(815) Step A. 6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 189 μ mol), DIEA (122 mg, 943 μ mol) and 4 Å molecular sieves (10.0 mg) in DMF (5.0 mL) was added 6-azabicyclo[3.2.1]octan-3-ol (48.0 mg, 377 μ mol). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was diluted water (5.0 mL) and was extracted with EtOAc (3 $\times$ 5.0 mL). The combined organic layer was dried over anhydrous

Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The reaction was purified by prep-HPLC [Phenomenex Synergi C18 150×25 mm×10 µm; A: water (0.225% formic acid formic acid), B: ACN, B %: 12%-42%] to afford 6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol (3.89 mg, 4% yield) as yellow solid. .sup.1H NMR (400 MHz, METHANOL-d.sub.4): δ 9.43-9.24 (m, 1H), 8.61-8.44 (m, 1H), 8.17-8.03 (m, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.75-7.65 (m, 1H), 7.63-7.48 (m, 2H), 7.23-7.14 (m, 1H), 5.07-4.97 (m, 1H), 4.64-4.59 (m, 2H), 4.55 (br s, 1H), 4.41-4.34 (m, 1H), 4.3-4.22 (m, 1H), 4.15 (br d, J=2.0 Hz, 1H), 3.65-3.53 (m, 2H), 3.26-3.10 (m, 2H), 2.81 (br s, 1H), 2.75-2.57 (m, 1H), 2.38-1.97 (m, 1H), 1.90-1.81 (m, 1H); LCMS (ESI, M+1): m/z 558.2

Example 77

(816) ##STR00300##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one

(817) ##STR00301##

(818) Step A. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 189 µmol), DIEA (122 mg, 943 µmol) and 4 Å molecular sieves (10 mg) in DMF (5 mL) was added 1,7-diazaspiro[3.5]nonan-2-one (50.0 mg, 357 µmol). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was diluted water (5.0 mL) and extracted with EtOAc (3×5.0 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4. The mixture was filtered and concentrated under reduced pressure to give a residue. The reaction was purified by prep-HPLC [Phenomenex Synergi C18 150×25 mm×10 µm; A: water (0.225% formic acid formic acid), B: ACN, B %: 11%-41%] to afford 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.5]nonan-2-one (16.0 mg, 14% yield) as yellow solid. .sup.1H NMR (400 MHz, METHANOL-d.sub.4): δ 9.10 (s, 1H), 8.52 (br s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.74-7.68 (m, 1H), 7.62 (dd, J=0.8, 7.2 Hz, 1H), 7.57-7.50 (m, 1H), 7.24-7.15 (m, 1H), 4.63 (s, 2H), 4.29-4.14 (m, 2H), 4.12-4.00 (m, 2H), 3.72-3.57 (m, 2H), 3.27-3.17 (m, 2H), 2.87 (s, 2H), 2.36-2.26 (m, 2H), 2.24-2.12 (m, 4H), 2.11-2.01 (m, 6H); LCMS (ESI, M+1): m/z 571.3

Example 78

(819) ##STR00302##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine

(820) ##STR00303##

(821) Step A. 5-(8-fluoro-7-(8-fluoronaphthalen-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To the mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (42 mg, 79 µmol), 5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (18 mg, 119 µmol) and 4 Å molecular sieves (40 mg, 79 µmol) in DMF (0.6 mL) was added DIEA (31 mg, 237 µmol). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 10%-40%, 10 min) to afford 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (5.57 mg, 11% yield, 0.422 FA) as white solid. .sup.1H NMR (400 MHz, METHANOL-d.sub.4): δ 9.23 (s, 1H), 8.55 (s, 1H), 8.13 (br d, J=8.8 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.71 (dd, J=7.2, 8.0 Hz, 1H), 7.61 (dd, J=1.2, 7.2 Hz, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.24-7.13 (m, 1H), 5.79 (s, 1H), 5.17-5.06 (m, 2H), 4.49 (s, 2H), 4.38 (br t, J=5.6 Hz, 2H), 4.28-4.19 (m, 2H), 3.48-

3.39 (m, 2H), 3.10-2.96 (m, 2H), 2.43-2.32 (m, 2H), 2.25-2.16 (m, 2H), 2.15-2.02 (m, 4H), 2.00-1.89 (m, 2H); LCMS (ESI, M+1): m/z 583.2

Example 79

(822) ##STR00304##

N-((2-aminopyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-methyl-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(823) ##STR00305##

(824) Step A: N-((2-aminopyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-methyl-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 189 μ mol, 1.0 equiv.), DIEA (122 mg, 943 μ mol, 164 μ L, 5.0 equiv.) and 4 Å molecular sieves (50 mg) in DMF (1.0 mL) was added 3-((methylamino)methyl)pyridin-2-amine (51.7 mg, 377 μ mol, 2.0 equiv.). The mixture was stirred at 40° C. for 12 h. Upon completion, the reaction solution was filtered and purified by prep-HPLC [Phenomenex Gemini-NX C18 75×30 mm×3 μ m; A: water (0.225% formic acid formic acid), B: ACN, B %: 8%-38% over 7 min] to afford the title compound (6.38 mg, 6% yield, 99% purity). White solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.22 (s, 1H), 8.48 (br d, J=0.4 Hz, 1H), 8.12 (br d, J=8.2 Hz, 1H), 7.99-7.91 (m, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.75-7.66 (m, 1H), 7.63-7.58 (m, 1H), 7.57-7.50 (m, 2H), 7.24-7.13 (m, 1H), 6.79-6.67 (m, 1H), 5.00 (s, 2H), 4.59 (s, 2H), 3.71-3.61 (m, 5H), 3.30-3.22 (m, 2H), 2.31-2.17 (m, 4H), 2.16-2.01 (m, 4H); LCMS [ESI, M+1]: 568.4.

Example 80

(825) ##STR00306##

N-((1H-benzo[d]imidazol-5-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(826) ##STR00307##

(827) Step A. N-((1H-benzo[d]imidazol-5-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 189 μ mol), DIEA (122 mg, 943 μ mol) and 4 Å molecular sieves (10 mg) in DMF (1.0 mL) was added (1H-benzo[d]imidazol-5-yl)methanamine (55.5 mg, 377 μ mol). The mixture was stirred at 40° C. for 12 hours. The reaction solution was filtered and purified by prep-HPLC [Phenomenex Gemini-NX C18 75×30 mm×3 μ m; A: water (0.225% formic acid formic acid), B: ACN, B %: 8%-38% over 7 min] to afford N-((1H-benzo[d]imidazol-5-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine (41.0 mg, 38% yield) as white solid. .sup.1H NMR (400 MHz, METHANOL-d.sub.4): δ 9.27 (s, 1H), 8.55 (br s, 1H), 8.19 (s, 1H), 8.11 (br d, J=8.4 Hz, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.75-7.67 (m, 2H), 7.64-7.57 (m, 2H), 7.56-7.49 (m, 1H), 7.43-7.38 (m, 1H), 7.23-7.14 (m, 1H), 5.03 (s, 2H), 4.55 (s, 2H), 3.63-3.52 (m, 2H), 3.24-3.13 (m, 2H), 2.26-2.18 (m, 2H), 2.17-2.11 (m, 2H), 2.10-2.02 (m, 2H), 2.01-1.93 (m, 2H); LCMS (ESI, M+1): m/z 578.2

Example 81

(828) ##STR00308##

3-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol

(829) ##STR00309##

(830) Step A. 3-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: To the mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50 mg, 94 μ mol), 3-azabicyclo[3.2.1]octan-6-ol (26 mg, 207 μ mol) and 4

A molecular sieves (40 mg) in DMF (1 mL) was added DIEA (49 mg, 377 μ mol). The mixture was stirred at 40° C. for 16 hours. The reaction mixture was filtered and purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 27%-57%, 10 min) to afford 3-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (12.5 mg, 23% yield) as white solid. ¹H NMR (400 MHz, DMSO-d₆): δ 9.32 (d, J=12.4 Hz, 1H), 8.17 (br d, J=8.0 Hz, 1H), 7.93 (d, J=8.0 Hz, 1H), 7.73 (dt, J=3.2, 7.6 Hz, 1H), 7.67-7.52 (m, 2H), 7.35-7.24 (m, 1H), 4.84-4.66 (m, 2H), 4.59 (m, 1H), 4.16 (br d, J=2.4 Hz, 1H), 4.08-3.96 (m, 2H), 3.72 (br t, J=12.4 Hz, 1H), 2.95-2.88 (m, 2H), 2.56-2.53 (m, 2H), 2.33 (br s, 1H), 2.17-2.04 (m, 2H), 1.88 (m, 2H), 1.83-1.70 (m, 5H), 1.69-1.61 (m, 1H), 1.61-1.49 (m, 2H), 1.26 (br d, J=13.2 Hz, 1H); LCMS (ESI, M+1): m/z 558.3

Example 82

(831) ##STR00310##

3-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol

(832) ##STR00311## ##STR00312##

(833) Step A. 3-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (150 mg, 594 μ mol), DIPEA (231 mg, 1.79 mmol) in DCM (1.5 mL) was added 3-azabicyclo[3.2.1]octan-6-ol (54.0 mg, 330 μ mol, HCl). DIPEA (231 mg, 1.79 mmol) in DMF (1.0 mL) was dropwise added at -40° C. The mixture was stirred at -40° C. for 1.5 h. The reaction mixture was diluted with water (20 mL) and extracted DCM (10 mL $\times$ 3). The combined organic phase was dried with anhydrous Na₂SO₄ and filtered. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate=5/1 to 1.5/1). TLC (PE/EA=1/1, R_f=0.41) to afford 3-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (90 mg, 31% yield) as a yellow solid; LCMS (ESI, M+1): m/z 343.0.

(834) Step B. 3-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: A mixture of 3-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (90 mg, 262 μ mol), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (50 mg, 314 μ mol), 4 Å molecular sieves (20 mg), DIPEA (102 mg, 792 μ mol) in dioxane (1.0 mL) was degassed and purged with N₂ for 3 times. The mixture was stirred at 95° C. for 16 h under N₂ atmosphere. After completion, the reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=3/1] to afford 3-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (96 mg, 71% yield) as a yellow gum; LCMS (ESI, M+1): m/z 466.1.

(835) Step C. 3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: A mixture of 3-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (96 mg, 206 μ mol), and K₃PO₄ (1.5 M in water, 412 μ L) in THF (2.0 mL) was degassed and purged with N₂ for 3 times. 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (112 mg, 311 μ mol) was added. Then CataCXium A Pd G3 (15 mg, 20.6 μ mol) was added into the reaction mixture. The mixture was stirred at 65° C. for 4 h under N₂ atmosphere. After completion, the reaction mixture was diluted with water (3.0 mL) and extracted with ethyl acetate (1.0 mL $\times$ 4). The combined organic phase was dried over anhydrous Na₂SO₄ and filtered. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile=13/7] to afford 3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-

yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (104 mg, 75% yield) as a yellow oil; LCMS (ESI, M+1): m/z 664.3.

(836) Step D. 3-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: To a solution of 3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (99 mg, 149 μ mol) in MeCN (4.0 mL) was added HCl/dioxane (4 M, 2.0 mL) at 0° C. The mixture was stirred between 0° C. and 10° C. for 0.5 h. After completion, the reaction mixture was quenched with saturated NaHCO₃ aqueous (10 mL). The mixture was extracted with ethyl acetate (10 mL×4). The combined organic phase was dried over anhydrous Na₂SO₄ and filtered. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC [Water s Xbridge 150×25 mm×5 gm; A: water (10 mM NH₄HCO₃), B: ACN, B %: 41%-71% over 10 min] to afford 3-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (33.5 mg, 36% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ 9.25-9.01 (m, 1H), 7.70-7.62 (m 1H), 7.30 (d, J=2.8 Hz, 1H), 7.27-7.20 (m, 1H), 7.09-7.01 (m, 1H), 5.41-5.19 (m, 1H), 5.04-4.97 (m, 1H), 4.77-4.73 (m, 1H), 4.38-4.16 (m, 3H), 3.97-3.68 (m, 1H), 3.54-3.39 (m, 1H), 3.29-3.07 (m, 3H), 3.04-2.81 (m, 1H), 2.58-2.28 (m, 3H), 2.28-2.11 (m, 5H), 2.03-1.85 (m, 4H), 1.85-1.76 (m, 1H), 1.40 (br d, J=13.6 Hz, 1H), 0.83-0.75 (m, 3H). F NMR (376 MHz, METHANOL-d₄) δ = -121, -139, -173. LCMS (ESI, M+1): m/z 620.3.

Example 83

(837) ##STR00313##

(4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol

(838) ##STR00314##

(839) Step A. 6-(((tert-butyl)diphenylsilyl)oxy)methyl)-4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy))pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 189 μ mol), 6-(((tert-butyl)diphenylsilyl)oxy)methyl)-1,4-oxazepane (153 mg, 377 μ mol, HCl) and 4 Å molecular sieves (10.0 mg) in DMF (1 mL) was added DIEA (73.1 mg, 566 μ mol). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was diluted water (5 mL) and was extracted with EtOAc (3×5.0 mL). The combined organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)/ACN) to afford 6-(((tert-butyl)diphenylsilyl)oxy)methyl)-4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (46.0 mg, 29.8% yield) as colorless oil. LCMS (ESI, M+1): m/z 800.4.

(840) Step B. (4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol: 6-(((tert-butyl)diphenylsilyl)oxy)methyl)-4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane (45.0 mg, 56.3 μ mol) in DMF (1 mL) was added CsF (85.4 mg, 562 μ mol). The mixture was stirred at 25° C. for 12 hours. The reaction solution was filtered and purified by prep-HPLC [Phenomenex Gemini-NX C18 75×30 mm×3 μ m; A: water (0.225% formic acid), B: ACN; B %: 12%-42% over 7 min] to afford (4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol (9.78 mg, 30.7% yield) as yellow solid. ¹H NMR (400 MHz, METHANOL-d₄): δ 9.24 (s, 1H), 8.54 (s, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H),

7.75-7.67 (m, 1H), 7.66-7.57 (m, 1H), 7.57-7.50 (m, 1H), 7.24-7.15 (m, 1H), 4.66 (br d, J=14.0 Hz, 1H), 4.55-4.55 (m, 1H), 4.55-4.39 (m, 2H), 4.19-3.92 (m, 4H), 3.84 (dd, J=10.0, 14.0 Hz, 1H), 3.68-3.55 (m, 3H), 3.52-3.40 (m, 2H), 3.12-2.99 (m, 2H), 2.59-2.44 (m, 1H), 2.28-2.17 (m, 2H), 2.16-2.02 (m, 4H), 2.00-1.92 (m, 2H); LCMS (ESI, M+1): m/z 562.3.

Example 84

(841) ##STR00315##

N-(3-(1H-pyrazol-3-yl) propyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d] pyrimidin-4-amine

(842) ##STR00316##

(843) Step A. N-(3-(1H-pyrazol-3-yl)propyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50 mg, 94.2 μ mol), 3-(1H-pyrazol-3-yl)propan-1-amine (18.7 mg, 94.2 μ mol, 2 HCl salt) and 4 Å molecular sieves (50 mg) in DMF (0.5 mL) was added DIEA (60.9 mg, 471 μ mol). The mixture was stirred at 40° C. for 16 hours under N.sub.2. The mixture was filtered through a pad of Celite. The filtrate was poured into 10 mL water and extracted with ethyl acetate (10 mL $\times$ 2). The combined organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated in vacuum. The residue was purified by reverse-phase preparatory HPLC (20% to 40% water/ACN with 0.225% formic acid over 10 minutes) to give N-(3-(1H-pyrazol-3-yl)propyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-amine (2.08 mg, 3.67% yield). LCMS (ESI, M+1): m/z 556.3. ¹H NMR (400 MHz, METHANOL-d.sub.4) δ : 9.17 (s, 1H), 8.63-8.47 (m, 1H), 8.13 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.75-7.67 (m, 1H), 7.59 (d, J=6.4 Hz, 1H), 7.57-7.47 (m, 2H), 7.23-7.15 (m, 1H), 6.21 (d, J=2.0 Hz, 1H), 4.58 (br s, 2H), 3.76 (m, 2H), 3.67-3.57 (m, 2H), 3.27-3.16 (m, 2H), 2.84 (t, J=7.6 Hz, 2H), 2.66 (s, 1H), 2.35-2.03 (m, 10H).

Example 85

(844) ##STR00317##

(3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-yl)methanol

(845) ##STR00318##

(846) Step A. (3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-yl)methanol: To a solution of (3-amino-1-bicyclo[1.1.1]pentanyl)methanol (9.60 mg, 84.8 μ mol) and 4 Å molecular sieves (20 mg) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μ mol, 49.2 μ L) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30 mg, 56.5 μ mol). The mixture was stirred at 40° C. for 12 hours. Upon completion, The reaction mixture was filtered through Celite and the filtrate was concentrated. The residue was purified by prep-HPLC (column: water s Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)-ACN]; B %: 32%-62%, 10 min) to afford (3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-yl)methanol (9.82 mg, 32% yield) as a white solid. LCMS (ESI, M+1): m/z 544.2.

Example 86

(847) ##STR00319##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3-carboxamide

(848) ##STR00320##

(849) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3-carboxamide: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)

pyrido[4,3-d]pyrimidine (30 mg, 56.6 μ mol), DIEA (21.9 mg, 169 μ mol, 29.6 μ L) and 4 Å molecular sieves (1.0 mg) in DMF (1.0 mL) was added (3R)-piperidine-3-carboxamide (14.5 mg, 113 μ mol). The mixture was stirred at 40° C. for 2 hours. After completion, the mixture was concentrated at 20° C. The residue was purified by prep-HPLC (column: Shim-pack C18 Ultra 150×25 mm×10 μ m; A: water (0.225% formic acid formic acid), B: ACN, B %: 17%-37% over 10 min) to afford 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3-carboxamide (8.41 mg, 26% yield, 98.2% purity). Yellow solid; LCMS (EST, M+1): m/z 559.3.

Example 87

(850) ##STR00321##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-4-carbonitrile

(851) ##STR00322##

(852) Step A. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-4-carbonitrile: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30.0 mg, 56.6 μ mol), DIEA (21.9 mg, 169 μ mol, 29.5 μ L) and 4 Å molecular sieves (1.0 mg) in DMF (1.0 mL) was added (4R)-azepane-4-carbonitrile (14.1 mg, 113 μ mol HCl). The mixture was stirred at 40° C. for 12 hours. After completion, the mixture was concentrated at 20° C. The residue was purified by prep-HPLC [column: water s Xbridge C18 150×25 mm×5 μ m; A: water (10 mM NH₄HCO₃), B: ACN, B %: 30%-60% over 9 min] to afford 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-4-carbonitrile (3.64 mg, 12% yield, 100% purity). Yellow solid; LCMS (ESI, M+1): m/z 555.2.

Example 88

(853) ##STR00323##

5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-methylpyridin-2 (1H)-one

(854) ##STR00324##

(855) Step A. 5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1-methylpyridin-2 (1H)-one: To a mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30.0 mg, 56.5 μ mol) and 5-(aminomethyl)-1-methylpyridin-2-one (19.7 mg, 113 μ mol, HCl) in DMF (1.0 mL) was added DIEA (29.2 mg, 226 μ mol) and 4 Å molecular sieves (30 mg) in one portion at 25° C. under N₂. The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75*30 mm*3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 7 min) to afford 5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1-methylpyridin-2 (1H)-one (12.3 mg, 22% yield); Off-white solid. LCMS (ESI, M+1): m/z 569.3.

Example 89

(856) ##STR00325##

(3S,5R)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol

(857) ##STR00326##

(858) Step A. (3S,5R)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol: To a solution of (3S,5R)-piperidine-3,5-diol (9.94 mg, 84.8 μ mol) and 4 Å molecular sieves (20 mg) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μ mol, 49.2 μ L) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30 mg, 56.5

μmol). The mixture was stirred at 40° C. for 12 hours. Upon completion, the reaction mixture was filtered through Celite and the filtrate was concentrated. The residue was purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 25%-55%, 0 min) to afford (3S,5R)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol (13.03 mg, 42% yield, 100% purity) as a white solid. LCMS (ESI, M+1): m/z 548.2.

Example 90

(859) ##STR00327##

3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-ol

(860) ##STR00328##

(861) Step A. 3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-ol: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (30.0 mg, 56.5 μmol) in DMF (1.0 mL) were added DIEA (29.2 mg, 226 μmol, 39.4 μL), 4 Å molecular sieves (10 mg), and 3-aminobicyclo[1.1.1]pentan-1-ol (11.2 mg, 113 μmol). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and purified by prep-HPLC (column: Phenomenex luna C18 150*25 mm*10 μm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 min) to afford 3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-ol (6.53 mg, 12.3 μmol, 100% purity); White solid. LCMS (ESI, M+J): m/z 530.3.

Example 91

(862) ##STR00329##

trans-3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)cyclobutane-1-carboxylic acid

(863) ##STR00330##

(864) Step A. trans-3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)cyclobutanecarboxylic acid: To a mixture of trans-3-aminocyclobutanecarboxylic acid (9 mg, 59.4 μmol, HCl), DIPEA (49.0 mg, 379 μmol) and 4 Å molecular sieves (10 mg) in DMSO (0.8 mL) was added 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (20 mg, 37.7 μmol). The mixture was stirred at 40° C. for 39 hours under N₂ atmosphere. The reaction mixture was filtered and purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 14%-44%, 10 minutes) to give the product as white solid (7.35 mg, 35% yield). LCMS (ESI, M+1): m/z 546.3.

Example 92

(865) ##STR00331##

N-((7H-pyrrolo[2,3-d]pyrimidin-2-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(866) ##STR00332##

(867) Step A. N-((7H-pyrrolo[2,3-d]pyrimidin-2-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 7H-pyrrolo[2,3-d]pyrimidin-2-ylmethanamine (12.6 mg, 84.8 μmol) and 4 Å molecular sieves (20 mg) in DMF (1.0 mL) were added DIEA (36.5 mg, 283 μmol, 49.25 μL) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine. The mixture was stirred at 40° C. for 12 hours. Upon completion, the reaction mixture was filtered and purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 30%-60%, 10 min) to afford N-((7H-pyrrolo[2,3-d]pyrimidin-2-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-

yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine (13.53 mg, 41.3% yield, 100% purity) as an off-white solid. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.31 (s, 1H), 8.95 (s, 1H), 8.12 (br d, J=8.1 Hz, 1H), 7.86 (d, J=8.2 Hz, 1H), 7.71 (t, J=7.7 Hz, 1H), 7.61 (d, J=6.6 Hz, 1H), 7.54 (dt, J=5.0, 7.9 Hz, 1H), 7.44 (d, J=3.5 Hz, 1H), 7.20 (dd, J=7.8, 13.0 Hz, 1H), 6.64 (d, J=3.5 Hz, 1H), 5.19-5.08 (m, 2H), 4.24 (s, 2H), 3.29-3.21 (m, 2H), 2.97-2.84 (m, 2H), 2.06-1.92 (m, 4H), 1.90-1.72 (m, 4H). LCMS (ESI, M+1): m/z 579.3.

Example 93

(868) ##STR00333##

N-((1H-pyrazol-4-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(869) ##STR00334##

(870) Step A. N-((1H-pyrazol-4-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30.0 mg, 56.5 μmol) and 1H-pyrazol-4-ylmethanamine (19.2 mg, 113 μmol, 2 HCl) in DMF (1 mL) was added DIEA (29.2 mg, 226 μmol) and 4 Å molecular sieves (30 mg) in one portion at 25° C. under N₂. The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and the filtrate was purified. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75*30 mm*3 μm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 7 min) to afford N-((1H-pyrazol-4-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine (2.14 mg, 6.6% yield); Yellow solid. LCMS (ESI, M+1): m/z 528.3.

Example 94

(871) ##STR00335##

N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(872) ##STR00336##

(873) Step A. N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-v)methoxy)pyrido[4,3-d]pyrimidin-4-amine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 86.6 μmol, formic acid salt), 1H-pyrrolo[2,3-b]pyridin-3-ylmethanamine (28.6 mg, 130 μmol, 2 HCl), DIEA (78.3 mg, 606 μmol, 105 μL) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was stirred at 40° C. for 16 hours. After completion, the mixture was filtered and purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 33%-63%, 8 min) to afford the title compound, 18% yield). White Solid. ¹H NMR (400 MHz, CDCl₃-d₃): δ 9.33-9.22 (m, 1H), 8.83 (s, 1H), 8.33 (br dd, J=4.0, Hz, 1H), 8.00 (br dd, J=8.0, 18.8 Hz, 2H), 7.72 (d, J=7.6 Hz, 1H), 7.62-7.57 (m, 1H), 7.56-7.52 (m, 1H), 7.46-7.40 (m, 2H), 7.13-7.06 (m, 2H), 6.54-6.37 (m, 1H), 5.09-5.01 (m, 2H), 4.33 (s, 2H), 3.19-3.09 (m, 2H), 2.71-2.61 (m, 2H), 2.18-2.09 (m, 2H), 1.93-1.86 (m, 4H), 1.73-1.69 (m, 2H). LCMS (ESI, M+1): m/z 578.3.

Example 95

(874) ##STR00337##

3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)thietane1,1-dioxide

(875) ##STR00338##

(876) Step A. 3-4 (8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl amino)thietane1,1-dioxide: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30.0 mg, 56.5 μmol) in DMF (1 mL) was added DIEA (29.2 mg, 226 μmol, 39.4 μL), 4 Å molecular sieves (10 mg), 1,1-dioxothietan-3-amine (13.7 mg, 113 μmol). The

mixture was stirred at 60° C. for 12 hr. The reaction mixture was filtered and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 11%-47%, 11 min) to afford 3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl) amino)thietane1,1-dioxide (3.25 mg, 5.89 µmol, 10% yield, 100% purity); White solid. LCMS (ESI, M+1): m/z 552.2.

Example 96

(877) ##STR00339##

5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyridin-2 (1H)-one

(878) ##STR00340##

(879) Step A. 5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyridin-2 (1H)-one: To a mixture of 5-(aminomethyl)-1H-pyridin-2-one (10.53 mg, 84.83 µmol) and 4 Å molecular sieves (20 mg) in DMF (1 mL) were added DIEA (36.54 mg, 282.75 µmol, 49.25 µL) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30 mg, 56.55 µmol). The mixture was stirred at 40° C. for 12 hours.

Upon completion, the reaction mixture was filtered. The residue was purified by prep-HPLC (column: water s Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 23%-53%, 10 min) to afford 5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyridin-2 (1H)-one (8.11 mg, 26% yield, 100% purity) as a white solid. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.15 (s, 1H), 8.11 (br d, J=7.8 Hz, 1H), 7.85 (d, J=8.1 Hz, 1H), 7.76 (dd, J=2.5, 9.4 Hz, 1H), 7.69 (t, J=7.7 Hz, 1H), 7.64-7.49 (m, 3H), 7.18 (dd, J=7.6, 13.0 Hz, 1H), 6.57 (d, J=9.4 Hz, 1H), 4.69 (s, 2H), 4.28 (s, 2H), 3.14-3.05 (m, 2H), 2.72 (td, J=6.5, 10.4 Hz, 2H), 2.06 (td, J=6.1, 12.1 Hz, 2H), 1.97-1.82 (m, 4H), 1.80-1.71 (m, 2H). LCMS (ESI, M+1): m/z 555.3.

Example 97

(880) ##STR00341##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(881) ##STR00342##

(882) Step A. 6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: A solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30 mg, 56.55 µmol) in DMF (1 mL) was added DIEA (29.3 mg, 226 µmol, 39.4 µL), 4 Å molecular sieves (10 mg) and 1,6-diazaspiro[3.5]nonan-2-one (15.8 mg, 113 µmol). The mixture was stirred at 60° C. for 2 hours. After completion, the mixture was filtered and concentrated in vacuum. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150×40 mm×15 µL; mobile phase: [water (0.225% formic acid)-ACN]; B %: 15%-45%, 10 min) to give 6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (9.11 mg, 28% yield); White solid. ¹H NMR (400 MHz, METHANOL-d₄) δ 9.14 (s, 1H), 8.53 (br s, 1H), 8.13 (br d, J=8.4 Hz, 1H), 7.87 (d, J=8.4 Hz, 1H), 7.71 (t, J=7.6 Hz, 1H), 7.62 (d, J=7.2 Hz, 1H), 7.58-7.49 (m, 1H), 7.25-7.15 (m, 1H), 4.60-7.57 (m, 2H), 4.48-4.27 (m, 2H), 4.03 (t, J=12.8 Hz, 1H), 3.95-3.80 (m, 1H), 3.66-3.52 (m, 2H), 3.25-3.10 (m, 2H), 2.97-2.87 (m, 1H), 2.83-2.74 (m, 1H), 2.35-2.23 (m, 2H), 2.23-2.09 (m, 5H), 2.09-1.96 (m, 5H). LCMS (ESI, M+1): m/z 571.2.

Example 98

(883) ##STR00343##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,6-diazaspiro[3.5]nonan-1-one

(884) ##STR00344##

(885) Step A. (S)-6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,6-diazaspiro[3.5]nonan-1-one: To a solution 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (30.0 mg, 56.5 μ mol) in DMF (1 mL) was added DIEA (29.2 mg, 226 μ mol, 39.4 μ L), 4 Å molecular sieves (10 mg) and (4R)-2,8-diazaspiro[3.5]nonan-3-one (15.8 mg, 113 μ mol, 2.0 equiv.). The mixture was stirred at 40° C. for 12 hr. The reaction mixture was filtered and the filtrate was concentrated. The residue was purified by prep-HPLC (column: waters Xbridge 150×25 mm×5 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 32%-62%, 10 min) to afford the title compound (5.18 mg, 8.91 μ mol, 16% yield, 98.1% purity); White solid. LCMS (ESI, M+1): m/z 571.3.

Example 99

(886) ##STR00345##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1-methyl-1H-indazol-6-yl)pyrido[4,3-d]pyrimidin-4-amine

(887) ##STR00346##

(888) Step A. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)-N-(1-methyl-1H-indazol-6-yl)pyrido[4,3-d]pyrimidin-4-amine: To a mixture of 1-methylindazol-6-amine (11.1 mg, 75.4 μ mol) in THF (1 mL) was added NaH (4.52 mg, 113 μ mol, 60% purity) in one portion at 0° C. under N₂ for 15 minutes. Then 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 mg, 37.7 μ mol) was added at 0° C. The mixture was stirred at 0° C. for 15 minutes. The residue was quenched with saturated NH₄Cl aqueous solution (0.1 mL) dropwise, and then filtered. The filtrate was concentrated in vacuum. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75×30 mm×3 μ m; mobile phase: [water (0.225% formic acid)-ACN]; B %: 18%-48%, 7 min) to afford 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1-methyl-1H-indazol-6-yl)pyrido[4,3-d]pyrimidin-4-amine (5.29 mg, 22% yield); Off-white solid. LCMS (ESI, M+1): m/z 578.4.

Example 100

(889) ##STR00347##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1H-indazol-5-yl)pyrido[4,3-d]pyrimidin-4-amine

(890) ##STR00348##

(891) Step A. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)-N-(1H-indazol-5-yl)pyrido[4,3-d]pyrimidin-4-amine: To the mixture of 1H-indazol-5-amine (6.93 mg, 52.0 μ mol), 4 Å MS (5 mg) in THF (0.5 mL) was added LiHMDS (1 M, 105 μ L) at 0° C. After 0.5 hour, to the mixture was added 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (20.0 mg, 34.7 μ mol, formic acid salt). The mixture was stirred at 20° C. for 14 hours. The reaction mixture was diluted with sat. NH₄Cl (3 mL) and extracted with ethyl acetate twice. The combined organic phase was washed with brine (5 mL), dried over anhydrous sodium sulfate. The mixture was filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Gemini—NX C18 75×30 mm×3 μ m; mobile phase: [water (10 mM NH₄HCO₃)-ACN]; B %: 40%-70%, 8 min) to afford 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1H-indazol-5-yl)pyrido[4,3-d]pyrimidin-4-amine (3.98 mg, 20% yield). Yellow Solid; LCMS (ESI, M+1): m/z 564.4.

(892) Synthesis of Example 101 to 131: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine or 7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (0.04 mmol, 1 equiv.), amine (HNR₂, 2 equiv.) and DIEA (3 eq or 5/7 eq for amine hydro/dihydro chlorides) in DMSO (1 mL) was heated

with stirring at 40° C. for 16 hours. The resulting solution was cooled to room temperature and subjected to HPLC purification (deionized water/HPLC-grade acetonitrile or methanol, ammonia) to give the product.

(893) ##STR00349##

(894) TABLE-US-00001 TABLE 1 Mass Spectrum Data of Example 103 to 131 Example No. Obs. M + 1 Example No. Obs. M + 1 101 570 117 562.2 102 561.2 118 557.4 103 589 119 558.2 104 584.1 120 566.2 105 575.1 121 580.2 106 574.2 122 580.2 107 562.2 123 514.2 108 599.2 124 567.2 109 598.1 125 580.2 110 547 126 569.2 111 548.4 127 566 112 546.4 128 581.2 113 546.4 129 546.2 114 542.2 130 560.2 115 543.2 131 582.2 116 560.2

Example 101

(895) ##STR00350##

4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 102

(896) ##STR00351##

4-(((7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one

Example 103

(897) ##STR00352##

3-(3-(((7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)propyl)-1H-1,2,4-triazol-5-ol

Example 104

(898) ##STR00353##

4-(2-(1H-pyrazol-4-yl)pyrrolidin-1-yl)-7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 105

(899) ##STR00354##

5-(((7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-1,2,4-triazol-3-yl)methanol

Example 106

(900) ##STR00355##

6-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol

Example 107

(901) ##STR00356##

3-(((7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)azetidine-1-carboxamide

Example 108

(902) ##STR00357##

4-(3-(1H-1,2,3-triazol-4-yl)piperidin-1-yl)-7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 109

(903) ##STR00358##

4-(3-(1H-pyrazol-1-yl)piperidin-1-yl)-7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 110

(904) ##STR00359##

5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)oxazolidin-2-one

Example 111

(905) ##STR00360##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-ol

Example 112

(906) ##STR00361##

(3R,5R)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-methylpiperidin-3-ol

Example 113

(907) ##STR00362##

(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanol

Example 114

(908) ##STR00363##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-4-(5-azaspiro[2.5]octan-5-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 115

(909) ##STR00364##

N-(2-(1H-1,2,3-triazol-4-yl)ethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 116

(910) ##STR00365##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylazepan-4-ol

Example 117

(911) ##STR00366##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-(hydroxymethyl)piperidin-4-ol

Example 118

(912) ##STR00367##

N-(3-(1H-1,2,3-triazol-1-yl)propyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 119

(913) ##STR00368##

N-(3-(1H-tetrazol-1-yl)propyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 120

(914) ##STR00369##

2-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)isoindolin-4-ol

Example 121

(915) ##STR00370##

N-([1,2,4]triazolo[4,3-a]pyrimidin-3-ylmethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 122

(916) ##STR00371##

N-([1,2,4]triazolo[4,3-b]pyridazin-3-ylmethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 123

(917) ##STR00372##

4-(3,6-dihydropyridin-1 (2H)-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-

pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 124

(918) ##STR00373##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(3-(pyridin-3-yl)propyl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 125

(919) ##STR00374##

N-(benzo[c][1,2,5]oxadiazol-5-ylmethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 126

(920) ##STR00375##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydropyrazolo[1,5-a]pyrazin-2-amine

Example 127

(921) ##STR00376##

3-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide

Example 128

(922) ##STR00377##

(R)-3-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)isothiazolidine 1,1-dioxide

Example 129

(923) ##STR00378##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-4-ol

Example 130

(924) ##STR00379##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-3-yl)methanol

Example 131

(925) ##STR00380##

4-(3-(1H-pyrazol-3-yl)piperidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 132

(926) ##STR00381##

1-(7-(3-chloro-5-hydroxy-2-((1S,2R)-2-methylcyclopropyl)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(927) ##STR00382## ##STR00383##

(928) Step A. (3-bromo-5-chloro-phenoxy)-tert-butyl-dimethyl-silane: To the mixture of 3-bromo-5-chlorophenol (50 g, 1.0 equiv.) and imidazole (36.1 g, 2.2 equiv.) in DMF (250 mL) was added TBSCl (40 g, 1.1 equiv.) at 0° C. The reaction was stirred and allowed to warm up to 25° C. over the course of 15 hours. The mixture was diluted with water (3 L) and extracted with ethyl acetate (400 mL×2). The combined organic layers were washed with saturated aqueous NH₄Cl solution (150 mL), dried over anhydrous Na₂SO₄, concentrated, and purified by silica gel column chromatography (petroleum ether) to afford the title compound (89.5 g, crude) as a colorless oil; ¹H NMR (400 MHz, DMSO-d₆) δ=7.31 (d, J=1.6 Hz, 1H), 7.03 (t, J=1.2 Hz, 1H), 6.94 (d, J=2.0 Hz, 1H), 0.94 (s, 9H), 0.21 (s, 6H).

(929) Step B. 2-bromo-6-chloro-4-hydroxy-benzaldehyde: To the solution of (3-bromo-5-chloro-phenoxy)-tert-butyl-dimethyl-silane (84.5 g, crude) in THE (850 mL) was added LDA (2 M in hexane, 184 mL) dropwise at -65° C. The reaction was stirred at -65° C. for 2 hours. DMF (40.5

mL) was added dropwise at -65°C . and the reaction was stirred at -65°C . for 0.1 hour. The mixture was quenched with 1.5N aqueous HCl solution between -65°C . and 0°C . until the pH was 2-3 and then extracted with ethyl acetate (300 mL $\times$ 3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and concentrated. The residue was dispersed in petroleum ether/ethyl acetate 20:1 (210 mL), stirred for 0.5 hour and filtered. The filter cake was washed with petroleum ether (50 mL) and dried under reduced pressure to afford the title compound (42.7 g, 86% yield over two steps) as a light yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =11.38 (br s, 1H), 10.13 (s, 1H), 7.11 (d, J=2.4 Hz, 1H), 6.95 (d, J=2.0 Hz, 1H); LCMS (ESI, M+1): m/z=236.9.

(930) Step C. 2-bromo-6-chloro-4-(methoxymethoxy)benzaldehyde: To the solution of 2-bromo-6-chloro-4-hydroxy-benzaldehyde (49 g, 1 equiv.) and DIEA (54.2 g, 2.0 equiv.) in DCM (500 mL) was added MOMCl (21.8 g, 1.3 equiv.) dropwise at 0°C . under N.sub.2 atmosphere. The reaction was stirred between 0°C . and 15°C . for 1 hour. The mixture was quenched with saturated aqueous NaHCO.sub.3 solution (100 mL) and water (100 mL) maintaining the temperature below 10°C . The organic layer was washed with saturated aqueous NH.sub.4Cl solution (200 mL $\times$ 5) and water (100 mL), dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by silica gel column chromatography (petroleum ether/ethyl acetate 5:1 to 9:2) to afford the title compound (56 g, 94% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-4) δ =10.18 (s, 1H), 7.42 (d, J=2.4 Hz, 1H), 7.29 (d, J=2.4 Hz, 1H), 5.36 (s, 2H), 3.40 (s, 3H).

(931) Step D. 1-bromo-3-chloro-5-(methoxymethoxy)-2-(prop-1-en-1-yl)benzene: To the mixture of ethyl(triphenyl)phosphonium bromide (59.8 g, 1.5 equiv.) in THF (350 mL) was added t-BuOK (1 M in THF, 140 mL, 1.3 equiv.) dropwise at 0°C . The mixture was stirred between 0 and 10°C . for 1 hour. A solution of 2-bromo-6-chloro-4-(methoxymethoxy)benzaldehyde (30 g, 1.0 equiv.) in THE (300 mL) was added dropwise maintaining the temperature below 10°C . The mixture was stirred between 0 and 10°C . for 0.5 hour. The mixture was quenched with water (200 mL) maintaining the temperature below 10°C . The mixture was concentrated and extracted with ethyl acetate (100 mL $\times$ 3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by silica gel column chromatography (petroleum ether) to afford the title compound (30 g, 95% yield) as a light yellow liquid.

(932) Step E. 1-bromo-3-chloro-5-(methoxymethoxy)-2-[(Z)-prop-1-enyl]benzene: 1-bromo-3-chloro-5-(methoxymethoxy)-2-(prop-1-en-1-yl)benzene (30 g, 1 equiv.) was purified by prep-HPLC [column: Phenomenex luna C18 (250 $\times$ 70 mm, 10 μ m); mobile phase: [water (0.225% Formic acid)-ACN] B %: 80%-100%, 13 min) to afford the title compound (11.5 g, 38% yield) as a light yellow liquid; .sup.1H NMR (400 MHz, chloroform-d) δ =7.24 (d, J=2.4 Hz, 1H), 7.10 (d, J=2.4 Hz, 1H), 6.20-6.18 (m, 1H), 5.94 (qd, J=6.8, 11.2 Hz, 1H), 5.15 (s, 2H), 3.49 (s, 3H), 1.58-1.56 (m, 3H).

(933) Step F. cis-1-bromo-3-chloro-5-(methoxymethoxy)-2-(2-methylcyclopropyl)benzene: To a solution of ZnEt.sub.2 (1 M in n-hexane, 175 mL, 6 equiv.) in DCM (180 mL) was added TFA (20.0 g, 6 equiv.) in DCM (20 mL) dropwise at -40°C . The reaction was stirred at -40°C . for 30 minutes. A solution of CH.sub.2I.sub.2 (46.9 g, 6 equiv.) in DCM (10 mL) was added dropwise at -40°C . and the reaction was stirred at -40°C . for 30 minutes. 1-bromo-3-chloro-5-(methoxymethoxy)-2-[(Z)-prop-1-enyl]benzene (8.5 g, 1.0 equiv.) in DCM (10 mL) was added dropwise at -40°C . and the reaction was stirred and allowed to warm up to 25°C . for 16 hours. The mixture was quenched with saturated aqueous NaHCO.sub.3 solution maintaining the temperature below 10°C . until pH 7-8. The mixture was filtered through a pad of celite. The filter cake was washed with DCM (100 mL $\times$ 2). The layers of the filtrate were separated, and the aqueous layer was extracted with DCM (100 mL $\times$ 3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified reversed phase flash chromatography (mobile phase: [water (0.1% Formic acid)/ACN 1:9]) to afford the title compound (1.5 g, 16% yield) as a yellow oil; .sup.1H NMR (400 MHz, chloroform-d) δ =7.22 (d, J=2.4 Hz, 1H), 7.06 (d,

J=2.4 Hz, 1H), 5.13 (s, 2H), 3.48 (s, 3H), 1.82 (dt, J=6.8, 8.4 Hz, 1H), 1.58-1.56 (m, 2H), 1.54-1.37 (m, 1H), 0.85 (d, J=6.0 Hz, 3H).

(934) Step G. cis-2-[3-chloro-5-(methoxymethoxy)-2-(2-methylcyclopropyl)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: To a mixture of cis-1-bromo-3-chloro-5-(methoxymethoxy)-2-(2-methylcyclopropyl)benzene (1.4 g, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (2.33 g, 2.0 equiv.) and KOAc (1.57 g, 3.5 equiv.) in dioxane (22 mL) was added Pd(dppf)Cl.sub.2 (336 mg, 0.1 equiv.). The reaction was stirred 80° C. for 5 hours under N.sub.2 atmosphere. The reaction mixture was diluted with water (100 mL) and extracted with ethyl acetate (20 mL×4). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by silica gel column chromatography [petroleum ether/ethyl acetate 30:1 to 10:1]. The crude product was purified again by reversed phase flash chromatography (mobile phase: [water (0.1% Formic acid)]) to afford the title compound (0.4 g, crude) as a light yellow oil; .sup.1H NMR (400 MHz, chloroform-d) δ=7.13-7.10 (m, 2H), 5.16-5.15 (m, 2H), 3.48-3.46 (m, 3H), 2.11-2.05 (m, 1H), 1.53-1.51 (m, 1H), 1.39-1.36 (m, 12H), 1.28-1.25 (m, 2H), 0.72 (d, J=6.0 Hz, 3H); LCMS (ESI, M+1): m/z=353.2.

(935) Step H. cis-1-[7-[3-chloro-5-(methoxymethoxy)-2-(2-methylcyclopropyl)phenyl]-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: A mixture of 1-[7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol, Intermediate 13 (65 mg, 1.0 equiv.), cis-2-[3-chloro-5-(methoxymethoxy)-2-(2-methylcyclopropyl)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (95 mg, crude), cataCXium A Pd G3 (17 mg, 0.15 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 300 μL, 3 equiv.) in methoxycyclopentane (1.5 mL) was stirred at 90° C. for 1 hours under N.sub.2 atmosphere. The mixture was diluted with water (1 mL) and extracted with ethyl acetate (2 mL×4). The combined organic layers were concentrated and purified by reversed phase flash chromatography (Mobile phase: [water (0.1% formic acid)/acetonitrile 1:1] to afford the title compound (69 mg, 70% yield) as a brown solid; LCMS (ESI, M+1): m/z=626.4.

(936) Step I. 1-(7-(3-chloro-5-hydroxy-2-((1S,2R)-2-methylcyclopropyl)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of cis-1-[7-[3-chloro-5-(methoxymethoxy)-2-(2-methylcyclopropyl)phenyl]-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (69 mg, 1.0 equiv.) in MeCN (2 mL) was added HCl in dioxane (4 M, 1.5 mL) dropwise at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The mixture was concentrated at room temperature. The residue was dissolved in ethyl acetate (20 mL) and water (5 mL). The pH of the mixture was adjusted to 8 with NaHCO.sub.3 solid maintaining the temperature below 5° C. The mixture was extracted with ethyl acetate (10 mL×4). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by prep-HPLC (column: water s Xbridge 150×25 mm×Sum; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/CAN], B %: 29%-59%, 9 min] to afford the title compound (24.9 mg, 37% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d4) δ=9.21-9.17 (m, 1H), 7.00-6.94 (m, 1H), 6.81-6.76 (m, 1H), 4.51 (d, J=12 Hz, 1H), 4.28-4.25 (m, 3H), 3.66-3.59 (m, 1H), 3.46-3.44 (m, 1H), 3.11-3.09 (m, 2H), 2.73 (td, J=6.4, 10.4 Hz, 2H), 2.09-2.06 (m, 3H), 1.94-1.87 (m, 6H), 1.79-1.74 (m, 4H), 1.28-1.27 (m, 3H), 1.21-1.04 (m, 1H), 0.82 (br s, 1H), 0.65 (br d, J=5.6 Hz, 3H), 0.52-0.15 (m, 1H); LCMS (ESI, M+1): m/z=582.4.

yl)-8-Example 133

(937) ##STR00384##

7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(938) ##STR00385##

(939) Step A. benzyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate: To the mixture of benzyl 3-oxopiperidine-1-carboxylate (7.00 g, 1.0 equiv.), (NH.sub.4).sub.2CO.sub.3 (8.65 g, 3.0

equiv.) in EtOH (35 mL) and water (35 mL) was added KCN (2.93 g, 1.5 equiv.), the reaction was stirred at 85° C. for 16 hours. The reaction was cooled to 25° C. and concentrated. The residue was diluted with water (300 mL), extracted with ethyl acetate (300 mL×3). The combined organic phase was washed with brine (100 mL), dried over anhydrous sodium sulfate, concentrated to give the title compound (6.70 g, 63% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ 7.42-7.19 (m, 5H), 5.19-4.97 (m, 2H), 3.84 (br s, 2H), 3.54-3.34 (m, 1H), 3.26-3.06 (m, 1H), 2.10-1.94 (m, 1H), 1.91-1.50 (m, 3H); LCMS (ESI, M+Na): m/z=326.1.

(940) Step B. Benzyl 2-oxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate: To the solution of benzyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate (1.00 g, 1.0 equiv.) in THF (20 mL) was added BH.sub.3-THF (1 M in THF, 19.8 mL, 6.0 equiv.) at 0° C. The reaction was then stirred at 80° C. for 2 hours. The mixture was quenched by addition of MeOH (20 mL) slowly at 0° C. and stirred at 20° C. for 0.5 hour and then concentrated. The residue was purified reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to provide the title compound (800 mg, 82% yield) as a yellow solid; LCMS (ESI, M+1): m/z=290.1.

(941) Step C. 1,3,7-triazaspiro[4.5]decan-2-one: To a solution of benzyl 2-oxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate (480 mg, 1.0 equiv.) in t-BuOH (8 mL) was added Pd(OH).sub.2C (70 mg, 10% purity) under N.sub.2. The suspension was degassed and purged with H.sub.2 several times. The reaction was stirred under H.sub.2 (15 psi) at 20° C. for 2 hours. The mixture was filtered and concentrated to give the title compound (255 mg, crude) as a white solid. (942) Step D. 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To a mixture of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (500 mg, 1.0 equiv.), DIEA (768 mg, 3.0 equiv.) in dichloromethane (8 mL) was added 1,3,7-triazaspiro[4.5]decan-2-one (250 mg, 0.8 equiv.) at -40° C. The reaction was stirred at -40° C. for 0.5 hour. The mixture was diluted with dichloromethane (10 mL) and water (10 mL), extracted with dichloromethane (5 mL). The combined organic phase was washed with brine (8 mL), dried over anhydrous sodium sulfate, concentrated to give the title compound (650 mg, crude) as a yellow solid.

(943) Step E. 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To the mixture of 7-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one (650 mg, crude), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (502 mg), 4 Å molecular sieves (300 mg) in dioxane (7 mL) was added DIEA (679 mg). The reaction was stirred at 90° C. for 12 hours. The reaction mixture was diluted with ethyl acetate (15 mL) and water (10 mL). The aqueous layer was extracted with ethyl acetate (15 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to give the title compound (405 mg, 38% yield over three steps) as a yellow solid; ¹H NMR (400 MHz, chloroform-d) δ 8.74-8.40 (m, 1H), 5.47-5.21 (m, 1H), 4.61-4.12 (m, 4H), 3.97-2.74 (m, 8H), 2.51-2.05 (m, 4H), 2.01-1.56 (m, 6H); LCMS (ESI, M+1): m/z=494.2.

(944) Step F. 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To a mixture of 7-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one (210 mg, 1.0 equiv.), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (184 mg, 1.2 equiv.), K.sub.3PO.sub.4 (1.5 M in water, 3.0 equiv.) in THF (3 mL) was added CataCXium A Pd G3 (31.0 mg, 0.1 equiv.) under N.sub.2. The reaction was stirred at 60° C. for 28 hours. The reaction mixture was diluted with ethyl acetate (15 mL) and water (10 mL). The aqueous layer was extracted with ethyl acetate (15 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) and prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 μm; mobile phase: [water

(0.05% ammonia hydroxide v/v)/ACN], B %: 28%-58%, 10 min] to give the title compound (9.00 mg, 3% yield) as a white solid; LCMS (ESI, M+1): m/z=692.4.

(945) Step G. 7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To the solution of 7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one (8.00 mg, 1.0 equiv.) in MeOH (1 mL) was added HCl.dioxane (4 M, 2 mL) at 0° C., the mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated, pH was adjusted to 7 by NaHCO₃ and MeOH (1 mL) was added. The residue was purified by prep-HPLC [column: Shim-pack C18 150×25×10 µm; mobile phase: [water (0.225% formic acid formic acid)/ACN], B %: 15%-45%, 10 min] to afford the title compound (6.07 mg, 78% yield, 0.50 formic acid) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ 9.13 (s, 1H), 7.75-7.61 (m, 1H), 7.36-7.19 (m, 2H), 7.05 (t, J=3.2 Hz, 1H), 5.56-5.34 (m, 1H), 4.59-4.45 (m, 2H), 4.43-4.14 (m, 2H), 4.07-3.86 (m, 2H), 3.74-3.49 (m, 3H), 3.46-3.34 (m, 2H), 3.27-3.21 (m, 1H), 2.63-2.35 (m, 3H), 2.32-2.11 (m, 4H), 2.10-1.79 (m, 5H), 0.90-0.73 (m, 3H); LCMS (ESI, M+1): m/z=648.3.

Example 134

(946) ##STR00386##

4-(4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(947) ##STR00387##

(948) Step A. 4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine: To the solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (60.0 mg, 1.0 equiv.) and DIEA (60.9 mg, 5.0 equiv.) in DMF (2.0 mL) was added 1-(azetidin-3-yl)imidazole (18.1 mg, 1.20 equiv., HCl salt). The mixture was stirred at 40° C. for 12 hours. The mixture treated with water (5 mL) and extracted with ethyl acetate (3×5 mL). The organic layer was washed with brine (2×10 mL), dried over anhydrous sodium sulfate, filtered and concentrated to afford the title compound (60 mg, crude) as a yellow liquid; LCMS (ESI, M+1): m/z=660.3.

(949) Step B. 4-(4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To the solution of 4-(3-(1H-imidazol-1-yl)azetidin-1-yl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 1.0 equiv.) in DCM (0.5 mL) was added TFA (7.70 g, 891 equiv.). The reaction was stirred at 20° C. for 0.5 hour. The mixture was concentrated. The residue was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: [water (10 mM NH₄HCO₃)/ACN], B %: 34/a-64%, 8 min] to afford the title compound (9.23 mg, 19% yield over two steps) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ 8.96 (s, 1H), 7.97 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.58 (s, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.11 (s, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.55-5.45 (m, 1H), 5.41-4.90 (m, 5H), 4.38-4.23 (m, 2H), 3.29-3.13 (m, 3H), 3.05-2.96 (m, 1H), 2.56-2.43 (m, 1H), 2.39-2.19 (m, 2H), 2.18-2.08 (m, 2H), 2.05-1.83 (m, 3H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=616.4.

Example 135

(950) ##STR00388##

4-(4-((3-(1H-pyrazol-3-yl)propyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(951) ##STR00389##

(952) Step A. N-(3-(1H-pyrazol-3-yl)propyl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50 mg, 1.0 equiv.) and 3-(1H-pyrazol-3-yl)propan-1-amine (14.7 mg, 1.5 equiv.) in DMF (2 mL) was added DIEA (50.8 mg, 5 equiv.). The mixture was stirred at 40° C. for 16 hours. The mixture was poured into water (10 mL) and extracted with ethyl acetate (10 mL×2). The combined organic phase was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuum to afford the title compound (40 mg, 77% yield) as a yellow solid; LCMS (ESI, M+1): m/z=662.4.

(953) Step B. 4-(4-((3-(1H-pyrazol-3-yl)propyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To a solution of N-(3-(1H-pyrazol-3-yl)propyl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-amine (40 mg, 1.0 equiv.) in ACN (2 mL) was added HCl.dioxane (4 M, 1.36 mL, 90 equiv.). The mixture was stirred at 0° C. for 1 hour. The mixture was concentrated in vacuum. The residue was purified by Prep-HPLC [column: water s Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN], B %: 35/0-65%, 8 min) to afford the title compound (13.7 mg, 34% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ 9.12 (s, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.50 (br s, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 6.20 (d, J=1.6 Hz, 1H), 5.43-5.24 (m, 1H), 4.38-4.24 (m, 2H), 3.75 (t, J=7.2 Hz, 2H), 3.25 (br s, 1H), 3.11-3.00 (m, 1H), 2.83 (t, J=7.2 Hz, 2H), 2.54-1.86 (m, 12H), 0.79 (t, J=7.2 Hz, 3H). LCMS (ESI, M+1): m/z=618.3.

Example 136

(954) ##STR00390##

4-(4-(((2-aminopyridin-3-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(955) ##STR00391##

(956) Step A. N-((2-aminopyridin-3-yl)methyl)-2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-amine: To a mixture of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (300 mg, 1.0 equiv.), DIEA (461 mg, 3.0 equiv) and 4 Å molecular sieves (20 mg) in DCM (5 mL) was added 3-(aminomethyl)pyridin-2-amine (205 mg, 1.4 equiv.) at -40° C. The mixture was stirred at -40° C. for 1 hour. The mixture was poured into water (10 mL) and the aqueous layer was extracted with ethyl acetate (2×10 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated to afford the title compound (150 mg, 37% yield) as a yellow solid; LCMS (ESI, M+1): m/z=339.1.

(957) Step B. N.sub.4 (2-aminopyridin-3-yl)methyl)-7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a mixture of N-((2-aminopyridin-3-yl)methyl)-2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-amine (150 mg, 1.0 equiv.), 4 Å molecular sieves (15 mg) and DIEA (172 mg, 3.0 equiv.) in dioxane (3 mL) was added ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (84.5 mg, 1.2 equiv.). The reaction was stirred at 90° C. for 12 hours. The mixture was poured into water (10 mL) and extracted with ethyl acetate (2×10 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified by reversed phase flash column chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (50 mg, 24% yield) as a yellow solid; LCMS (ESI, M+1): m/z=462.2.

(958) Step C. N-((2-aminopyridin-3-yl)methyl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a mixture of N-((2-aminopyridin-3-yl)methyl)-7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-

d]pyrimidin-4-amine (50 mg, 1.0 equiv.), 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (78.0 mg, 2.0 equiv.) and Cs.sub.2CO.sub.3 (1.5 M in water, 216 μ L, 3.0 equiv.) in methoxycyclopentane (2.0 mL) was added cataCXium-A-Pd-G3 (15.8 mg, 0.2 equiv.). The reaction was stirred at 100° C. for 3 hours. The mixture was poured into water (5 mL) and extracted with ethyl acetate (2 $\times$ 5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified reversed phase flash column chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (40 mg, 56% yield) as a yellow solid; LCMS (ESI, M+1): m/z=660.3.

(959) Step D. 4-(4-(((2-aminopyridin-3-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To a mixture of N-((2-aminopyridin-3-yl)methyl)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine (25.0 mg, 1.0 equiv.) in DCM (1 mL) was added TFA (1.54 g, 356 equiv.). The reaction was stirred at 20° C. for 1 hour. The mixture was concentrated. The residue was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (0.225% formic acid)/ACN], B %: 8%-38%, 7 min] to afford the title compound (7.44 mg, 32% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.22 (s, 1H), 8.55-8.38 (m, 1H), 7.92 (br d, J=5.2 Hz, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.62 (d, J=7.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (s, 1H), 6.71 (dd, J=5.2, 7.2 Hz, 1H), 5.57-5.32 (m, 1H), 4.76 (br s, 2H), 4.56-4.45 (m, 2H), 3.74-3.58 (m, 3H), 3.29-3.23 (m, 1H), 2.60-2.38 (m, 3H), 2.36-2.24 (m, 1H), 2.23-2.11 (m, 3H), 2.10-1.96 (m, 1H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=616.4.

Example 137

(960) ##STR00392##

4-(4-(3-(1H-1,2,4-triazol-5-yl)piperidin-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5,6-difluoronaphthalen-2-ol

(961) ##STR00393## ##STR00394##

(962) Step A. benzyl 3-(1H-1,2,4-triazol-5-yl)piperidine-1-carboxylate: A mixture of benzyl 3-carbamoylpiperidine-1-carboxylate (5.0 g, 1.0 equiv.) in DMF-DMA (45.4 g, 20.0 equiv.) was heated to 110° C. for 2 hours. After removal of the volatiles, the residue was re-dissolved in AcOH (30 mL) and treated with N.sub.2H.sub.4-WATER (1.17 g, 1.20 equiv.). The reaction was heated to 90° C. for 1 hour. The mixture was concentrated under reduced pressure to remove AcOH. The residue was diluted with water (50 mL) and extracted with ethyl acetate (3 $\times$ 50 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-100%]) to afford the title compound (5.30 g, 97% yield) as a white solid; LCMS (ESI, M+1): m/z=287.2.

(963) Step B. benzyl 3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl) piperidine-1-carboxylate: To a solution of benzyl 3-(1H-1,2,4-triazol-5-yl)piperidine-1-carboxylate (4.40 g, 1.0 equiv.) in THF (40 mL) was added NaH (1.23 g, 60% purity, 2.0 equiv.). The mixture was stirred at 0° C. for 0.5 hour before SEM-Cl (3.84 g, 1.50 equiv.) was added at 0° C. The mixture was stirred at 0° C. for 1 hour. The reaction mixture was quenched by addition of water (50 mL) at 0° C., and extracted with ethyl acetate (3 $\times$ 50 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-100%]) to afford the title compound (3.30 g, 49% yield) as a yellow liquid; LCMS (ESI, M+1): m/z=417.3.

(964) Step C. 3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidine: To a solution of benzyl 3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidine-1-carboxylate (3.00 g, 1.0 equiv.) in MeOH (20 mL) was added Pd/C (300 mg, 10% purity). The suspension was degassed in vacuum and purged with H.sub.2 several times. The reaction was stirred under H.sub.2 (15 psi) at 25° C. for 1 hour. The mixture was filtered and the filtrate was

concentrated to afford the title compound (1.80 g, crude) as a colourless liquid.

(965) Step D. 2,7-dichloro-8-fluoro 1 (3 (1 ((2 (trimethylsilyl)ethoxy)methyl) 1H 1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidine (1.51 g, 0.90 equiv.) in DCM (15 mL) was added DIEA (2.30 g, 3.0 equiv.) and 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (1.50 g, 1.0 equiv.) at -60°C . The mixture was stirred at -60°C . for 0.5 hour. The reaction mixture was diluted with water (20 mL) and extracted with DCM (3 $\times$ 20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-100%]) to afford the title compound (2.30 g, 81% yield over two steps) as a yellow solid; LCMS (ESI, M+1): m/z=498.2.

(966) Step E. 7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxyl-4-(3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2,7-dichloro-8-fluoro-4-(3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine (2.00 g, 1.0 equiv.) in dioxane (8 mL). was added DIEA (1.56 g, 3.0 equiv.), 4 Å molecular sieves (500 mg) and (hexahydro-1H-pyrrolizin-7a-yl)methanol (1.13 g, 2.0 equiv.). The reaction was stirred at 95°C . for 12 hours. The mixture was filtered. The filtrate was concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-100%]) to afford the title compound (1.60 g, 66% yield) as a yellow solid; LCMS (ESI, M+1): m/z=603.3.

(967) Step F. 7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine (500 mg, 1.0 equiv.) and 2-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (580 mg, 2.0 equiv.) in THF (10 mL) was added K.sub.3PO.sub.4 (1.5 M, 3.0 equiv.) and CataCXium A Pd G3 (90.5 mg, 0.15 equiv.). The mixture was stirred at 60°C . for 2 hours. The reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (3 $\times$ 10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-100%]) to afford the title compound (400 mg, 58% yield) as a yellow solid; LCMS (ESI, M+1): m/z=791.4.

(968) Step G. 4-(4-(3-(1H-1,2,4-triazol-5-yl)piperidin-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5,6-difluoronaphthalen-2-ol: To a solution of 7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(1-((2-(trimethylsilyl)ethoxy)methyl)-1H-1,2,4-triazol-5-yl) piperidin-1-yl)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) in DCM (1.0 mL) was added TEA (1.45 g, 56.8 equiv.). The reaction was stirred at 25°C . for 2 hours. The reaction mixture was concentrated under reduced pressure to remove solvent. The residue was diluted with water (10 mL) and pH was adjusted to 8 with aqueous Na.sub.2CO.sub.3. The mixture was extracted with DCM (3 $\times$ 10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μm ; mobile phase: [water (0.225% formic acid)/CAN]; B %: 20%-30%, 7 min] to afford the title compound (46.8 mg, 27% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d₄) δ 9.12 (s, 1H), 8.33 (s, 1H), 7.77-7.66 (m, 1H), 7.61-7.50 (m, 1H), 7.43-7.35 (m, 1H), 7.28-7.20 (m, 1H), 4.77-4.63 (m, 1H), 4.51-4.40 (m, 1H), 4.39-4.26 (m, 2H), 3.57-3.51 (m, 1H), 3.30-3.17 (m, 3H), 2.94-2.81 (m, 2H), 2.27-2.17 (m, 1H), 2.08-1.72 (m, 12H). LCMS (ESI, M+1): m/z=617.3.

Example 138

(969) ##STR00395##

5,6-difluoro-4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(970) ##STR00396## ##STR00397##

(971) Step A. benzyl 3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidine-t-carboxylate: A mixture of benzyl 3-carbamoylpiperidine-1-carboxylate (1.50 g, 1.0 equiv.) in 1,1-dimethoxy-N, N-dimethylethanamine (15.2 g, 20.0 equiv.) was heated at 110° C. for 2 hours. After removal of the volatiles, the residue was re-dissolved in AcOH (3 mL). hydrazine hydrate (438 mg, 1.50 equiv.) was added, and the reaction was heated to 90° C. for 1 hour. The mixture was concentrated under reduced pressure to remove AcOH. The residue was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with brine (50 mL), dried over anhydrous sodium sulfate, concentrated, and purified by silica gel column chromatography (petroleum ether/ethyl acetate 10:1 to 1:1) to afford the title compound (1.50 g, 87% yield) as a yellow liquid; ¹H NMR (400 MHz, CDCl₃) δ 7.39-7.28 (m, 5H), 5.21-5.07 (m, 2H), 4.40-4.18 (m, 1H), 4.11-3.88 (m, 1H), 3.34-2.82 (m, 3H), 2.39 (s, 3H), 2.18-2.08 (m, 1H), 1.93-1.49 (m, 3H). LCMS (ESI, M+1): m/z=301.2.

(972) Step B. 3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidine: To a solution of benzyl 3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidine-1-carboxylate (1.50 g, 1.0 equiv.) in MeOH (20 mL) was added Pd/C (200 mg, 10% purity) under N₂. The suspension was degassed in vacuum and purged with H₂ several times. The mixture was stirred under H₂ (15 psi) at 25° C. for 2 hours. The reaction mixture was filtered and the filtrate was concentrated to afford the title compound (800 mg, crude) as a by yellow liquid.

(973) Step C. 2,7-dichloro-8-fluoro-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidine (592 mg, 1.50 equiv.) in THF (20 mL) was added DIEA (921 mg, 3.0 equiv.) and 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (600 mg, 1.0 equiv.) at -60° C. The mixture was stirred at -60° C. for 0.5 hour. The reaction mixture was diluted with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-100%]) to afford the title compound (600 mg, 66% yield) as yellow liquid; ¹H NMR (400 MHz, CDCl₃) δ 8.92 (s, 1H), 4.65 (br dd, J=3.2, 13.2 Hz, 1H), 4.50-4.39 (m, 1H), 3.94 (dd, J=9.2, 13.2 Hz, 1H), 3.74-3.62 (m, 1H), 3.32-3.20 (m, 1H), 2.48 (s, 3H), 2.37-2.26 (m, 1H), 2.19-2.10 (m, 1H), 2.04-1.98 (m, 1H), 1.92-1.79 (m, 1H); LCMS (ESI, M+1): m/z=382.1.

(974) Step D. 7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxyl-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2,7-dichloro-8-fluoro-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine (600 mg, 1.0 equiv.) in dioxane (6 mL) was added DIEA (609 mg, 3.0 equiv.) and (hexahydro-1H-pyrrolizin-7a-yl)methanol (665 mg, 3.0 equiv.). The reaction was stirred at 95° C. for 3 hours. The reaction mixture was diluted with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-20%]) to afford the title compound (400 mg, 47% yield) as a yellow solid; LCMS (ESI, M+1): m/z=487.3.

(975) Step E. 7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) and 2-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (288 mg, 2.0 equiv.) in methoxycyclopentane (6.0 mL) was added K₃PO₄ (1.5 M, 3.0 equiv.) and CataCXium A Pd G3 (44.9 mg, 0.15 equiv.). The reaction was stirred at 90° C. for 2 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN, 0-20%]) to afford the title compound (80 mg, 29% yield) as a yellow solid; LCMS (ESI, M+1): m/z=675.4.

(976) Step F. 5,6-difluoro-4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of 5,6-difluoro-4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(3-methyl-1H-1,2,4-triazol-5-yl)piperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (80.0 mg, 1.0 equiv.) in ACN (1 mL) was added HCl.dioxane (4.0 M, 72.0 equiv.). The reaction was stirred at 25° C. for 0.5 hour. The mixture was concentrated and purified by reversed phase HPLC [column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: water (0.225% formic acid)/ACN; B %: 12/a-42%, 10 min] to afford the title compound (29.5 mg, 39% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d4) δ=9.14 (s, 1H), 8.54 (s, 1H), 7.66-7.59 (m, 1H), 7.45-7.37 (m, 1H), 7.36-7.31 (m, 1H), 7.29-7.22 (m, 1H), 4.62-4.51 (m, 3H), 3.79-3.66 (m, 2H), 3.65-3.56 (m, 2H), 3.28-3.15 (m, 3H), 2.41 (s, 3H), 2.34-2.25 (m, 3H), 2.24-2.00 (m, 9H), 1.97-1.86 (m, 1H). LCMS (ESI, M+1): m/z=631.3.

Example 139

(977) ##STR00398##

4-(4-(((1H-benzo[d]imidazol-5-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(978) The title compound was synthesized according to the procedure described for example 134. .sup.1H NMR (400 MHz, methanol-d4) δ=9.23 (s, 1H), 8.52 (br d, J=1.2 Hz, 1H), 8.18 (s, 1H), 7.72 (s, 1H), 7.70-7.60 (m, 2H), 7.41 (dd, J=1.2, 8.4 Hz, 1H), 7.32-7.22 (m, 2H), 7.04 (d, J=2.8 Hz, 1H), 5.40-5.22 (m, 1H), 5.03 (br d, J=1.2 Hz, 2H), 4.42-4.28 (m, 2H), 3.51-3.34 (m, 3H), 3.18-2.99 (m, 1H), 2.50-2.38 (m, 1H), 2.30-2.09 (m, 4H), 2.07-1.99 (m, 2H), 1.96-1.81 (m, 1H), 0.79 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=640.3.

Example 140

(979) ##STR00399##

3-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol

(980) ##STR00400##

(981) Step A. 3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: A mixture of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (130 mg, 1.0 equiv., 1.7 FORMIC ACID), 3-azabicyclo[3.2.1]octan-6-ol (60 mg, 2.0 equiv., HCl), DIPEA (200 µL, 6.3 equiv.), 4 Å molecular sieves (30 mg) in DMF (1.0 mL) was degassed and stirred at 40° C. for 15 hours. The mixture was filtered. The filtrate was purified reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile 11:9]) to afford the title compound (115 mg, 89% yield) as a yellow oil; LCMS (ESI, M+1): m/z=664.3.

(982) Step B. 3-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: To a solution of 3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (142 mg, 1.0 equiv.) in MeCN (3.0 mL) was added HCl.dioxane (4 M, 1.5 mL). The reaction was stirred at 0° C. for 0.5 hour. The mixture was quenched with saturated aqueous NaHCO₃ (15 mL) and extracted with ethyl acetate (15 mL×4). The combined organic phase was dried over anhydrous Na₂SO₄, concentrated and purified by prep-HPLC [column: Water s Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH₄HCO₃)/ACN]; B %: 41%-71%, 9 min] to afford the title compound (49.1 mg, 36% yield) as a white solid.

(983) Step C 3-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxyl)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo

[3.2.1]octan-6-ol: 3-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol (72 mg, 1 equiv.) was purified by SFC (column: DAICEL CHIRALPAK AD (250 mm×30 mm, 10 μm); mobile phase: [IPA-ACN]; B %: 35%-35%, 30 minutes) to give the crude product as the second eluting peak. The crude product was further re-purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄sub.4HCO₃sub.3)/ACN]; B %: 43%-73%, 10 min] to afford the title compound (13.6 mg, 18% yield) as a white solid; SFC: >99% ee, Chiralpak AD-3 50×4.6 mm I.D., 3 μm column A: IPA+ACN (0.05% DEA), B: CO₂, 3 mL/min, t_{sub.R}: 0.900 min; ¹H NMR (400 MHz, methanol-d₄) δ=9.24 (d, J=19.6 Hz, 1H), 7.67 (dd, J=6.0, 2.8 Hz, 1H), 7.31-7.30 (m, 1H), 7.26-7.21 (m, 1H), 7.09-7.03 (m, 1H), 5.39 (d, J=53.2 Hz, 1H), 5.00-4.78 (m, 1H), 4.45-4.42 (m, 1H), 4.35-4.31 (m, 3H), 3.80 (br dd, J=12.8, 18.8 Hz, 1H), 3.57-3.50 (m, 1H), 3.49-3.34 (m, 3H), 3.22-3.09 (m, 1H), 2.51-2.36 (m, 3H), 2.34-2.18 (m, 4H), 2.16-1.88 (m, 5H), 1.86-1.75 (m, 1H), 1.45-1.26 (m, 1H), 0.83-0.76 (m, 3H); LCMS (ESI, M+1): m/z=620.3.

Example 141

(984) ##STR00401##

3-((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)propenamide

(985) The title compound was synthesized according to the procedure described for example 134. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.13 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.6 Hz, 1H), 7.03 (d, J=2.4 Hz, 1H), 5.42-5.22 (m, 1H), 4.41-4.24 (m, 2H), 4.02-3.90 (m, 2H), 3.27-3.13 (m, 3H), 3.05-2.98 (m, 1H), 2.71 (t, J=6.4 Hz, 2H), 2.50 (br s, 1H), 2.39-2.19 (m, 2H), 2.18-2.10 (m, 2H), 2.04-1.90 (m, 3H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=581.3.

Example 142

(986) ##STR00402##

4-(4-(((7H-pyrrolo[2,3-d]pyrimidin-2-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(987) The title compound was synthesized according to the procedure described for example 61. ¹H NMR (400 MHz, methanol-d₄): δ=9.31 (s, 1H), 8.95 (s, 1H), 8.51 (br s, 1H), 7.69 (dd, J=5.6, 8.8 Hz, 1H), 7.49-7.43 (m, 1H), 7.35-7.30 (m, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (s, 1H), 6.65 (d, J=3.6 Hz, 1H), 5.35-5.19 (m, 1H), 5.13 (s, 2H), 4.16-4.08 (m, 1H), 4.24-4.06 (m, 1H), 3.49-3.34 (m, 2H), 3.14-3.00 (m, 1H), 2.57-2.44 (m, 1H), 2.07-1.91 (m, 4H), 2.35-1.89 (m, 3H), 1.86-1.73 (m, 1H), 0.82 (br t, J=6.8 Hz, 3H); LCMS (ESI, M+1): m/z=641.3.

Example 143

(988) ##STR00403##

(989) 5-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyridin-2 (1H)-one

(990) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₄) δ=9.23 (s, 1H), 7.77 (dd, J=2.4, 9.2 Hz, 1H), 7.69 (dd, J=6.0, 9.2 Hz, 1H), 7.61 (d, J=2.0 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 6.58 (d, J=9.2 Hz, 1H), 5.70-5.49 (m, 1H), 4.78-4.65 (m, 4H), 4.12-3.84 (m, 3H), 3.53-3.43 (m, 1H), 2.80-2.53 (m, 2H), 2.51-2.30 (m, 4H), 2.24-2.07 (m, 2H), 0.78 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=617.3.

Example 144

(991) ##STR00404##

1-(7-(5-cyclopropyl-6-methyl-1H-indazol-4-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(992) ##STR00405## ##STR00406##

(993) Step A. 6-methyl-5-nitro-1-tetrahydropyran-2-yl)-indazole: To a solution of 6-methyl-5-nitro-1H-indazole (8.60 g, 1 equiv.) in DCM (86 mL) was added THP (12.5 g, 3 equiv.), TsOH (834 mg, 0.1 equiv.) and resulting mixture was stirred at 20° C. for 2 hours under N.sub.2 atmosphere. The mixture was diluted with water (100 mL) and extracted with ethyl acetate (2×100 mL). The combined organic layers were washed with brine (1×100 mL), dried over anhydrous sodium sulfate, concentrated and purified by flash silica gel chromatography (ethyl acetate/petroleum 0 to 20%) to afford the title compound (12 g, 95% yield) as a yellow solid; ¹HNMR (400 MHz, CDCl₃) δ 8.49 (s, 1H), 8.12 (s, 1H), 7.49 (s, 1H), 5.73 (dd, J=2.7, 9.2 Hz, 1H), 4.08-3.95 (m, 1H), 3.81-3.72 (m, 1H), 2.74 (s, 3H), 2.59-2.46 (m, 1H), 2.21-2.06 (m, 2H), 1.85-1.69 (m, 3H).

(994) Step B. 6-methyl-1-tetrahydropyran-2-yl)-indazol-5-amine: To a mixture of 6-methyl-5-nitro-1-tetrahydropyran-2-yl)-indazole (6.00 g, 1 equiv.) and NH₄Cl (4.91 g, 91.9 mmol, 4 equiv.) in EtOH (50 mL) and water (10 mL) was added iron powder (6.41 g, 5 equiv.). The reaction was stirred at 80° C. for 3 hours. The mixture was filtered to remove solids. The filtrate was concentrated in vacuum. The residue was diluted with water (30 mL) and filtered. Then the filter cake was triturated with petroleum ether/ethyl acetate 50:1 (20 mL) at 25° C. for 30 min to afford the title compound (3.7 g, 70% yield) as a light red solid; ¹HNMR (400 MHz, CDCl₃) δ 7.81 (s, 1H), 7.32 (s, 1H), 6.94 (s, 1H), 5.64 (dd, J=2.8, 9.2 Hz, 1H), 4.10-3.97 (m, 1H), 3.81-3.70 (m, 1H), 3.69-3.45 (m, 2H), 2.64-2.47 (m, 1H), 2.35 (s, 3H), 2.22-2.01 (m, 2H), 1.83-1.72 (m, 2H), 1.70-1.61 (m, 1H).

(995) Step C. 4-bromo-6-methyl-1-tetrahydropyran-2-yl)-indazol-5-amine: To a solution of 6-methyl-1-tetrahydropyran-2-yl)-indazol-5-amine (4.50 g, 1 equiv.) in THF (130 mL) was added NBS (3.81 g, 1.1 equiv.). The reaction was stirred at 25° C. for 2 hours under N.sub.2 atmosphere. The mixture was diluted with water (150 mL) and extracted with ethyl acetate (2×150 mL). The combined organic layers were washed with brine (1 x 150 mL), dried over anhydrous sodium sulfate, concentrated and purified by flash silica gel chromatography (ethyl acetate/petroleum ether 0 to 50%) to afford the title compound (3.5 g, 57% yield) as a yellow solid; ¹HNMR (400 MHz, CDCl₃) δ 7.83 (s, 1H), 7.29 (s, 1H), 5.63 (dd, J=2.8, 9.2 Hz, 1H), 4.07-3.90 (m, 3H), 3.82-3.67 (m, 1H), 2.60-2.47 (m, 1H), 2.38 (d, J=0.8 Hz, 3H), 2.20-2.02 (m, 2H), 1.83-1.63 (m, 3H).

(996) Step D. 4-bromo-5-iodo-6-methyl-1-tetrahydropyran-2-yl)-indazole: To a solution of 4-bromo-6-methyl-1-tetrahydropyran-2-yl)-indazol-5-amine (1.80 g, 1 equiv.) in THF (20 mL) was added CuI (1.33 g, 1.2 equiv.), CH₂I₂ (77.7 g, 50 equiv.) and isopentyl nitrite (2.04 g, 3 equiv.). The reaction was stirred at 70° C. for 2 hours. The mixture was diluted with water (100 mL), extracted with ethyl acetate (2×200 mL). The combined organic layer was washed with brine (100 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with solid NaHCO₃ (5 g), concentrated in vacuum to remove acetonitrile, and extracted with dichloromethane (2×100 mL). The combined organic layer was washed with brine (100 mL), dried over anhydrous sodium sulfate, concentrated and purified by column chromatography (Silica gel, ethyl acetate/petroleum ether 3/1) to afford the title compound (1.4 g, 53% yield) as a yellow solid; LCMS (ESI, M+1): m/z=421.0.

(997) Step E. 4-bromo-5-cyclopropyl-6-methyl-1-tetrahydropyran-2-yl)-indazole: A mixture of 4-bromo-5-iodo-6-methyl-1-tetrahydropyran-2-yl)-indazole (500 mg, 1 equiv.), cyclopropylboronic acid (112 mg, 1.1 equiv.), K₂CO₃ (492 mg, 3 equiv.) and Pd(dppf)Cl₂ (86.9 mg, 0.1 equiv.) in dioxane (10 mL) and water (3 mL) was stirred at 90° C. for 24 hours under N.sub.2. The mixture was diluted with ethyl acetate (40 mL), washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) twice to afford the title compound (350 mg, 80% yield) as a yellow oil; LCMS (ESI, M+1): m/z=337.0.

(998) Step F. 5-cyclopropyl-6-methyl-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-

dioxaborolan-2-yl)indazole: A mixture of 4-bromo-5-cyclopropyl-6-methyl-1-tetrahydropyran-2-yl)-indazole (140 mg, 1 equiv.), 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (107 mg, 2 equiv.), PCy.sub.3 Pd G2 (24.7 mg, 0.1 equiv.), TEA (169 mg, 4 equiv.) in dioxane (1 mL) was stirred at 70° C. for 1 hour under N.sub.2. The mixture was concentrated in vacuum. The residue was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (35 mg, 19% yield) as a yellow oil; LCMS (ESI, M+1): m/z=383.3.

(999) Step G: 1-[7-(5-cyclopropyl-6-methyl-1-tetrahydropyran-2-yl)-indazol-4-yl]-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: A mixture of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (40 mg, 1 equiv.), 5-cyclopropyl-6-methyl-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl) indazole (28.1 mg, 0.8 equiv.), CataCXium A Pd G3 (6.68 mg, 0.1 equiv.), K.sub.3PO.sub.4 (1.5 M in water, 3 equiv.) in THF (1 mL) was stirred at 60° C. for 8 hours. The mixture was concentrated and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (16 mg, 23% yield) as a yellow oil.

(1000) Step H. 1-(7-(5-cyclopropyl-6-methyl-1H-indazol-4-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-[7-(5-cyclopropyl-6-methyl-1-tetrahydropyran-2-yl)-indazol-4-yl]-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (14 mg, 1 equiv.) in MeCN (0.5 mL) was added HCl-dioxane (4 M, 131 equiv.). The reaction was stirred at 20° C. for 0.5 hour. The mixture was concentrated in vacuum, neutralized with saturated NaHCO.sub.3 solution (0.5 mL), purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) and then prep-HPLC [column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)/ACN] B %: 9%-39%, 10 min] to afford the title compound (2.89 mg, 23% yield) as a white solid; .sup.1H NMR (400 MHz, CDCl.sub.3-d) δ 9.36 (br d, J=4.4 Hz, 1H), 8.55 (br s, 1H), 7.74 (br s, 1H), 7.52 (s, 1H), 4.69-4.54 (m, 3H), 4.38 (br dd, J=4.4, 12.8 Hz, 1H), 3.72-3.61 (m, 1H), 3.58-3.45 (m, 3H), 3.13 (br dd, J=4.0, 4.8 Hz, 2H), 2.69 (s, 3H), 2.27 (br dd, J=6.0, 12.0 Hz, 2H), 2.22-1.97 (m, 8H), 1.94-1.77 (m, 3H), 1.33 (br d, J=7.6 Hz, 3H), 0.92-0.53 (m, 2H), 0.25--0.02 (m, 2H); LCMS (ESI, M+1): m/z=572.4.

Example 145

(1001) ##STR00407##

4-(4-(3-(1H-1,2,4-triazol-5-yl)piperidin-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1002) The title compound was synthesized according to the procedure described for example 134. .sup.1H NMR (400 MHz, CDCl.sub.3) δ=13.99-13.61 (m, 1H), 10.16-9.75 (m, 1H), 9.16-9.02 (m, 1H), 7.83-7.69 (m, 1H), 7.41-7.26 (m, 2H), 7.09-6.96 (m, 1H), 5.38-5.16 (m, 1H), 4.82-4.61 (m, 1H), 4.53-4.34 (m, 1H), 4.23-4.00 (m, 2H), 3.78-3.53 (m, 2H), 3.13-2.99 (m, 3H), 2.88-2.79 (m, 1H), 2.45-2.29 (m, 2H), 2.27-1.69 (m, 12H), 0.81-0.66 (m, 311). LCMS (ESI, M+1): m/z=645.3.

Example 146

(1003) ##STR00408##

6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,6-diazaspiro[3.5]nonan-1-one

(1004) The title compound was synthesized according to the procedure described for example 134. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ 9.07 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (t, J=2.0 Hz, 1H), 5.40-5.23 (m, 1H), 4.41-4.32 (m, 2H), 4.31-4.18 (m, 2H), 4.16-3.96 (m, 2H), 3.38-3.33 (m, 1H), 3.29-3.25 (m, 2H), 3.24-3.18 (m, 2H), 3.08-3.00 (m, 1H), 2.52-2.41 (m, 1H), 2.40-2.27 (m, 1H), 2.25-1.99 (m, 8H), 1.95-1.86 (m, 2H), 0.80 (br t, J=7.2 Hz, 3H); .sup.19F NMR (376 MHz, methanol-d4) δ -121.158, -138.976, -173.778; LCMS (ESI, M+1): m/z=633.4.

Example 147

(1005) ##STR00409##

1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3-carboxamide

(1006) The title compound was synthesized according to the procedure described for example 134. ¹H NMR (400 MHz, DMSO-d₆) δ=9.93 (br s, 1H), 9.09 (s, 1H), 7.76 (dd, J=6.0, 9.2 Hz, 1H), 7.45 (br d, J=4.4 Hz, 1H), 7.39-7.29 (m, 2H), 7.03 (d, J=2.4 Hz, 1H), 6.97 (br s, 1H), 5.40-5.15 (m, 1H), 4.55-4.42 (m, 1H), 4.41-4.31 (m, 1H), 4.19-4.10 (m, 1H), 4.09-4.01 (m, 1H), 3.58-3.42 (m, 2H), 3.17-3.04 (m, 1H), 3.16-3.03 (m, 1H), 3.01 (s, 1H), 2.88-2.76 (m, 1H), 2.66-2.54 (m, 2H), 2.22-2.09 (m, 2H), 2.08-1.95 (m, 3H), 1.93-1.82 (m, 2H), 1.81-1.66 (m, 4H), 0.72 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=621.3.

Example 148

(1007) ##STR00410##

6-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1008) ##STR00411##

(1009) Step A: 8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (410 mg, 1.0 equiv., formic acid salt), ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl) ethynyl)triisopropylsilane (563 mg, 1.3 equiv.) and Cs₂CO₃ (1.5 M in water, 1.69 mL, 3.0 equiv.) in methoxycyclopentane (5.1 mL) was added CataCXium A Pd G3 (61.6 mg, 0.1 equiv.) under N₂. The reaction was de-gassed and then heated to 100° C. for 1 hour under N₂. The mixture was poured into saturated NH₄Cl solution (10 mL). Then the mixture was diluted with ethyl acetate (10 mL) and layers were separated. The aqueous phase was extracted with ethyl acetate (10 mL). The combined organic layer was washed with brine (10 mL), dried over Na₂SO₄, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (547 mg, 76% yield, formic acid salt) as a brown oil; ¹H NMR (400 MHz, CDCl₃-d) δ=9.31-9.27 (m, 1H), 7.81 (dd, J=5.6, 9.2 Hz, 1H), 7.55 (d, J=2.4 Hz, 1H), 7.37-7.28 (m, 2H), 5.69-5.43 (m, 1H), 5.36-5.23 (m, 3H), 5.01-4.79 (m, 2H), 4.30-4.17 (m, 1H), 3.70-3.56 (m, 1H), 3.54-3.50 (m, 3H), 3.32 (br s, 1H), 2.92-2.72 (m, 2H), 2.68-2.55 (m, 2H), 2.42-2.29 (m, 4H), 0.95-0.80 (m, 18H), 0.70-0.44 (m, 3H); LCMS (ESI, M+1): m/z=789.3.

(1010) Step B: 6-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: A mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (0.1 g, 1.0 equiv., formic acid salt), 1,8-diazaspiro[3.5]nonan-2-one (58.8 mg, 3.5 equiv.), DIEA (46.4 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (1 mL) was stirred at 40-60° C. for 105 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (27 mg, 26% yield) as a yellow solid; LCMS (ESI, M+1): m/z=829.5.

(1011) Step C: 6-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: A mixture of 6-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl) ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (27.0 mg, 1.0 equiv.) and CsF (74.2 mg, 15 equiv.) in DMF (0.5 mL) was stirred at 25° C. for 1 hour. The mixture was

purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (15.0 mg, 68% yield) as a yellow solid; LCMS (ESI, M+1): m/z=673.3.

(1012) Step D: 6-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: A mixture of 6-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (15.0 mg, 1.0 equiv.) and TFA (462 mg, 182 equiv.) in dichloromethane (0.3 mL) was stirred at 25° C. for 0.5 hour. The mixture was concentrated in vacuum and the pH value was adjusted to around 8 with ice-cold saturated Na.sub.2CO.sub.3 solution. The mixture was diluted with ethyl acetate/methanol 7:1 (8 mL) and the layers were separated. The aqueous phase was extracted with ethyl acetate (6 mL). The combined organic layers were washed with brine (8 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuum. The residue was purified by prep-HPLC [column: water s Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN] B %: 29%-59%, 9 min] to afford the title compound (4.50 mg, 31% yield) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d4) δ=9.03 (d, J=6.4 Hz, 1H), 7.87 (dd, J=5.6, 9.2 Hz, 1H), 7.37-7.30 (m, 2H), 7.22 (d, J=2.4 Hz, 1H), 5.41-5.25 (m, 1H), 4.46-4.23 (m, 4H), 4.05-3.96 (m, 1H), 3.88-3.64 (m, 1H), 3.41-3.32 (m, 3H), 3.26-3.02 (m, 2H), 2.92-2.73 (m, 2H), 2.41-2.23 (m, 2H), 2.19-2.09 (m, 2H), 2.04-1.92 (m, 6H); .sup.19F NMR (376 MHz, methanol-d4) δ=-111.662, -139.984, -173.693; LCMS (ESI, M+1): m/z=629.3.

Example 149

(1013) ##STR00412##

4-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1014) The title compound was synthesized according to the procedure described for example 24. .sup.1H NMR (400 MHz, methanol-d4) δ=9.49-9.36 (m, 1H), 7.88-7.83 (m, 1H), 7.36-7.29 (m, 2H), 7.22 (dd, J=2.4, 10.0 Hz, 1H), 5.39-5.22 (m, 1H), 4.64-4.45 (m, 2H), 4.35-4.16 (m, 3H), 4.05-3.84 (m, 3H), 3.74-3.61 (m, 2H), 3.47-3.41 (m, 1H), 3.29-3.11 (m, 3H), 3.06-2.98 (m, 1H), 2.40-2.18 (m, 2H), 2.17-2.09 (m, 1H), 2.05-1.95 (m, 2H), 1.94-1.85 (m, 1H), 1.27 (d, J=13.6 Hz, 3H); .sup.19F NMR (376 MHz, methanol-d4) δ=-111.763, -139.876, -173.658; LCMS (ESI, M+1): m/z=620.3.

Example 150

(1015) ##STR00413##

4-(4-(((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethynyl-6-fluoronaphthalen-2-ol

(1016) The title compound was synthesized according to the procedure described for example 24. .sup.1H NMR (400 MHz, CD3OD) δ 9.12 (s, 1H), 8.22 (s, 1H), 8.21-8.19 (m, 1H), 7.87-8.82 (m, 1H), 7.54 (s, 1H), 7.35-7.28 (m, 2H), 7.21-7.09 (m, 2H), 5.39-5.20 (d, J=56.4 Hz, 1H), 5.12-5.00 (m, 2H), 4.38-4.26 (m, 2H), 3.30-3.11 (m, 4H), 3.09-2.92 (m, 1H), 2.39-2.25 (m, 1H), 2.24-2.18 (m, 1H), 2.17-2.08 (m, 1H), 2.04-1.93 (m, 2H), 1.93-1.82 (m, 1H); LCMS (ESI, M+1): m/z=636.4.

Example 151

(1017) ##STR00414##

1-(8-fluoro-7-(3-hydroxy)naphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1018) ##STR00415##

(1019) Step A. 1-[8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-7-[4-(methoxymethoxy)-2-naphthyl]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: To a solution of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50 mg, 1.0 equiv.), 2-[3-(methoxymethoxy)-1-naphthyl]-4,4,5,5-

tetramethyl-1,3,2-dioxaborolane (72.1 mg, 2.0 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 3.0 equiv.) in THF (0.5 mL) was added CataCXium A Pd G3 (8.35 mg, 0.1 equiv.). The mixture was degassed and stirred at 60° C. for 2 hours. Upon reaction completion, the mixture was diluted with water (5 mL) and extracted with ethyl acetate (5 mL×3). The combined organic layers were washed with brine 20 mL, dried over Na.sub.2SO.sub.4, filtered, concentrated and purified by silica gel column chromatography (ethyl acetate and then NH.sub.4OH/MeOH 1:1000) to afford the title compound as a brown liquid (45 mg, 67% yield); LCMS (ESI, M+1): m/z=588.3.

(1020) Step B. 1-(8-fluoro-7-(3-hydroxynaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-[8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-7-[4-(methoxymethoxy)-2-naphthyl]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (45 mg, 76.57 µmol, 1 equiv.) in MeCN (0.5 mL) was added HCl/dioxane (4 M, 0.5 mL, 26.12 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The mixture was concentrated under reduced pressure and purified by prep-HPLC (column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 7%-37%, 11.5 min) to afford the title compound as a yellow oil (26.86 mg, 59% yield, formic acid salt); .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.33 (s, 1H), 8.55 (s, 1H), 7.79 (d, J=8.4 Hz, 1H), 7.56 (br d, J=8.0 Hz, 1H), 7.46 (dt, J=1.0, 7.6 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.30-7.24 (m, 2H), 4.66 (s, 3H), 4.38 (br d, J=13.5 Hz, 1H), 3.72-3.62 (m, 3H), 3.53-3.40 (m, 1H), 3.28-3.21 (m, 2H), 2.33 (ddd, J=3.2, 6.7, 12.4 Hz, 2H), 2.25-2.07 (m, 7H), 1.89-1.76 (m, 3H), 1.34-1.28 (m, 3H), 1.31 (s, 3H); .sup.19F NMR (377 MHz, METHANOL-d.sub.4) δ=-140.42; LCMS (ESI, M+1): m/z=544.5.

Example 152

(1021) ##STR00416##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(isoxazol-3-ylamino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1022) The title compound was synthesized according to the procedure described for example 134. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.50 (s, 1H), 8.70 (d, J=1.6 Hz, 1H), 8.47 (br s, 1H), 7.69 (dd, J=6.0, 9.2 Hz, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.30-7.22 (m, 2H), 7.07 (d, J=2.4 Hz, 1H), 5.56-5.31 (m, 1H), 4.60-4.45 (m, 2H), 3.67-3.47 (m, 3H), 3.28-3.17 (m, 1H), 2.60-2.37 (m, 3H), 2.36-2.24 (m, 1H), 2.23-2.11 (m, 3H), 2.09-1.96 (m, 1H), 0.80 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=577.2.

Example 153

(1023) ##STR00417##

(3R)-1-(7-(6-chloro-5-(2-methylcyclopropyl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1024) ##STR00418## ##STR00419##

(1025) Step A. 4,4,5,5-tetramethyl-2-[(Z)-prop-1-enyl]-1,3,2-dioxaborolane: A mixture of [(Z)-prop-1-enyl]boronic acid (3.00 g, 1 equiv.), pinacol (4.13 g, 1 equiv.) and MgSO.sub.4 (12.6 g, 3 equiv.) in DCM (30 mL) was stirred at 20° C. for 12 hours. After reaction completion, the mixture was filtered and concentrated in vacuum to afford the title compound (5 g, 85% yield) as a yellow liquid. .sup.1H NMR (400 MHz, CDCl.sub.3-d) δ 6.59-6.39 (m, 1H), 5.39-5.27 (m, 1H), 1.97 (dd, J=1.6, 6.8 Hz, 3H), 1.32-1.25 (m, 12H).

(1026) Step B. rac-4,4,5,5-tetramethyl-2-[2-methylcyclopropyl]-1,3,2-dioxaborolane: To a solution of ZnEt.sub.2 (2 M in hexane, 3 equiv.) in DCM (10 mL) was added TFA (4.07 g, 3 equiv.) at -40° C. After stirred at -40° C. for 10 mins, CH.sub.2I.sub.2 (9.56 g, 3 equiv.) was added into the mixture. The reaction was stirred at -40° C. for 10 mins, 4,4,5,5-tetramethyl-2-[(Z)-prop-1-enyl]-1,3,2-dioxaborolane (2.00 g, 1 equiv.) was added into the mixture. The reaction was stirred at 25° C. for 16 hours. After reaction completion, the mixture was diluted with DCM (100 mL) and filtered. The filtrate was washed with water (100 mL) and brine (100 mL), dried over anhydrous

sodium sulfate, filtered and concentrated in vacuum to afford the title compound (2.00 g, 92% yield) as a yellow oil and used into next step without further purification. ¹H NMR (400 MHz, CDCl₃-d) δ 1.23 (d, J=7.6 Hz, 12H), 1.15-1.11 (m, 3H), 1.10-1.03 (m, 1H), 0.80-0.71 (m, 1H), 0.40-0.30 (m, 1H), -0.10 (dt, J=6.8, 9.2 Hz, 1H).

(1027) Step C. 6-chloro-5-nitro-1H-indazole: To a solution of 5-chloro-2-methyl-4-nitro-aniline (10.0 g, 1 equiv.) in AcOH (300 mL) was added dropwise the solution of NaNO₂ (4.07 g, 1.1 equiv.) in water (40 mL) at 0° C. The reaction was stirred at 25° C. for 2 hours. After completion, the mixture was poured into 1000 mL water and filtered. The filter cake was washed by 1000 mL water, then dried in vacuum to afford the title compound (8 g, 76% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ 8.44 (s, 1H), 8.25 (s, 1H), 7.71 (s, 1H).

(1028) Step D. 6-chloro-5-nitro-1-(tetrahydropyran-2-yl)-indazole: To a solution of 6-chloro-5-nitro-1H-indazole (6.6 g, 1 equiv.) in DCM (60 mL) was added TsOH·H₂O (635 mg, 0.1 equiv.), then 3,4-dihydro-2H-pyran (8.43 g, 3 equiv.) slowly at 25° C. The mixture was stirred at 25° C. for 1 hour before being concentrated in vacuum and purified by flash silica gel chromatography (ethyl acetate in petroleum ether 0-30%) to afford the title compound (7.20 g, 76% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ 8.38 (s, 1H), 8.15 (s, 1H), 7.81 (s, 1H), 5.73 (dd, J=2.4, 8.9 Hz, 1H), 4.06-3.97 (m, 1H), 3.82-3.71 (m, 1H), 2.55-2.41 (m, 1H), 2.22-2.08 (m, 2H), 1.90-1.64 (m, 4H).

(1029) Step E. 6-chloro-1-(tetrahydropyran-2-yl)-indazol-5-amine: To a solution of 6-chloro-5-nitro-1-(tetrahydropyran-2-yl)-indazole (7.2 g, 1 equiv.) in EtOH (80 mL) and water (16 mL) was added NH₄Cl (5.74 g, 4.2 equiv.) and iron powder (5.99 g, 4.2 equiv.). The reaction was stirred at 80° C. for 6 hours. After completion, the mixture was filtered, the filtrate was concentrated and purified by flash silica gel chromatography (ethyl acetate in petroleum ether 0-30%) to afford the title compound (6 g, 93% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ 7.80 (s, 1H), 7.59 (s, 1H), 7.00 (s, 1H), 5.59 (dd, J=2.4, 9.3 Hz, 1H), 4.06-3.92 (m, 3H), 3.80-3.69 (m, 1H), 2.58-2.44 (m, 1H), 2.18-2.05 (m, 2H), 1.81-1.57 (m, 4H).

(1030) Step F. 4-bromo-6-chloro-1-(tetrahydropyran-2-yl)-indazol-5-amine: To a solution of 6-chloro-1-(tetrahydropyran-2-yl)-indazol-5-amine (6.10 g, 1 equiv.) in THF (120 mL) was added NBS (5.18 g, 1.2 equiv.). The reaction was stirred at 25° C. for 2 hours. After completion, the mixture was filtered and the filtrate was concentrated in vacuum and purified by flash silica gel chromatography (ethyl acetate in petroleum ether 0-30%) to afford the title compound (3 g, 34% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ 7.83 (d, J=0.8 Hz, 1H), 7.59 (s, 1H), 5.60 (dd, J=2.8, 9.2 Hz, 1H), 4.37 (br s, 2H), 4.04-3.96 (m, 1H), 3.80-3.66 (m, 1H), 2.52-2.41 (m, 1H), 2.22-2.09 (m, 2H), 1.84-1.66 (m, 4H).

(1031) Step G. 4-bromo-6-chloro-5-iodo-1-(tetrahydropyran-2-yl)-indazole: A reaction mixture of 4-bromo-6-chloro-1-(tetrahydropyran-2-yl)-indazol-5-amine (2.40 g, 1 equiv.), CuI (1.80 g, 1.3 equiv.), isopentyl nitrite (2.55 g, 3 equiv.) and CH₂I₂ (97.2 g, 50 equiv.) in THE (20 mL) was stirred at 70° C. for 2 hours. After completion, the mixture was diluted with water (100 mL), extracted with ethyl acetate (2×200 mL). The combined organic layers were washed with brine (100 mL), dried over anhydrous sodium sulfate, concentrated in vacuum and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with solid NaHCO₃ (5 g), concentrated in vacuum to remove acetonitrile, and extracted with ethyl acetate (2×50 mL). The combined organic layer was washed with brine (50 mL), dried over anhydrous Na₂SO₄, concentrated and purified by silica gel column chromatography (ethyl acetate in petroleum ether 30%) to afford the title compound (1.05 g, 32% yield) as a yellow solid. LCMS (ESI, M-84): m/z=358.8.

(1032) Step H. 4-bromo-6-chloro-5-[2-methylcyclopropyl]-1-(tetrahydropyran-2-yl)-indazole: To a solution of 4-bromo-6-chloro-5-iodo-1-(tetrahydropyran-2-yl)-indazole (500 mg, 1 equiv.), 4,4,5,5-tetramethyl-2-[2-methylcyclopropyl]-1,3,2-dioxaborolane (309 mg, 1.5 equiv.) in dioxane (10 mL) and water (3 mL) was added K₂CO₃ (469 mg, 3 equiv.) and Pd(dppf)Cl₂ (82.9 mg,

0.1 equiv.). The reaction was stirred at 90° C. for 12 hours. After completion, the mixture was diluted with ethyl acetate (40 mL), washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (90 mg, 21% yield) as a yellow oil. LCMS (ESI, M-84): m/z=286.9.

(1033) Step I. 6-chloro-5-[2-methylcyclopropyl]-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indazole: A reaction mixture of 4-bromo-6-chloro-5-[2-methylcyclopropyl]-1-tetrahydropyran-2-yl)-indazole (80 mg, 1 equiv.), 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (83.1 mg, 3 equiv.), TEA (131 mg, 6 equiv.), Pd(OAc)₂.sub.2 (4.86 mg, 0.1 equiv.) in dioxane (2 mL) was stirred at 70° C. for 5 hours. After completion, the mixture was concentrated in vacuum and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (70 mg, 78% yield) as a yellow liquid. LCMS (ESI, M+1): m/z=417.1.

(1034) Step J. (3R)-1-(7-(6-chloro-5-(2-methylcyclopropyl)-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A reaction mixture of (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (40 mg, 1 equiv.), 6-chloro-5-[2-methylcyclopropyl]-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indazole (36.7 mg, 1 equiv.), CataCXium A Pd G3 (6.42 mg, 0.1 equiv.) and K₂CO₃.sub.4 (1.5 M in water, 3 equiv.) in THF (1 mL) was stirred at 60° C. for 2 hours. After completion, the mixture was concentrated and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (7 mg, 11% yield) as a yellow solid. LCMS (ESI, M+1): m/z=708.3.

(1035) Step K. (3R)-1-(7-(6-chloro-5-(2-methylcyclopropyl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A reaction mixture of (3R)-1-(7-(6-chloro-5-(2-methylcyclopropyl)-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (7 mg, 1 equiv.) in HCl/dioxane (4 M, 142 equiv.) and MeCN (0.1 mL) was stirred at 15° C. for 10 mins. After reaction completion, the mixture was concentrated in vacuum and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) and then re-purified by prep-HPLC: [column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/CAN]; B %: 18%-48%, 10 min] to afford the title compound (1.75 mg, 27% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃-d) 8.936-9.25 (m, 1H), 8.57-8.46 (m, 1H), 8.17-7.68 (m, 2H), 5.46-5.28 (m, 1H), 4.91 (br d, J=1.2 Hz, 1H), 4.64-4.52 (m, 1H), 4.46-4.23 (m, 3H), 3.70-3.63 (m, 1H), 3.63-3.57 (m, 1H), 3.55-3.34 (m, 4H), 3.17-3.09 (m, 1H), 2.49-2.27 (m, 2H), 2.26-2.03 (m, 5H), 2.02-1.75 (m, 5H), 1.36-1.22 (m, 4H), 1.18 (t, J=7.2 Hz, 2H), 0.71-0.64 (m, 3H). LCMS (ESI, M+1): m/z=624.3.

Example 154

(1036) ##STR00420##

(3R)-1-(7-(1H-benzo[f]indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1037) ##STR00421##

(1038) Step A. 4-bromo-1H-benzo[f]indazole: To a mixture of 1-bromo-3-fluoro-2-naphthaldehyde (500 mg, 1.00 equiv.) in dimethyl sulfoxide (5.00 mL) was added hydrazine hydrate (0.60 g, 582 μL, 98.0% purity, 5.9 equiv.) and diisopropylethylamine (2.04 g, 2.75 mL, 8.00 equiv.) in one portion under nitrogen. The reaction was stirred at 130° C. for 8 hours. The reaction mixture was concentrated under reduced pressure and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 44%-74%, 10 min)

to give the title compound (150 mg, 30% yield) as a white solid. ¹H NMR (400 MHz, CD₃OD) δ=8.32-8.25 (m, 2H), 8.03 (s, 1H), 8.00-7.93 (m, 1H), 7.51-7.44 (m, 2H); LCMS (ESI, M+1): m/z=247.1.

(1039) Step B. 4-bromo-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazole: To a solution of 4-bromo-1H-benzo[f]indazole (150 mg, 1.0 equiv.) and 3,4-dihydro-2H-pyran (153 mg, 167 μL, 3.0 equiv.) in dichloromethane (2.00 mL) was added 4-methylbenzenesulfonic acid hydrate (5.23 mg, 0.05 equiv.) in one portion at 25° C. under nitrogen. The reaction was stirred at 25° C. for 12 hours. The reaction mixture was concentrated under reduced pressure and purified by prep-TLC (silica gel, petroleum ether/ethyl acetate 4:1) to give the title compound (180 mg, 543 μmol, 89.5% yield, 100% purity) as a yellow solid. LCMS (ESI, M+1): m/z=330.9.

(1040) Step C. 1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-benzo[f]indazole: To a solution of 4-bromo-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazole (180 mg, 1.0 equiv.) and 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (166 mg, 1.2 equiv.) in dioxane (3.00 mL) was added potassium acetate (160 mg, 3.00 equiv.), Pd(dppf)Cl₂.sub.2 (39.8 mg, 0.1 equiv.), then the mixture was degassed and stirred at 90° C. for 1 hour under nitrogen. After completion, the reaction mixture was concentrated under reduced pressure and purified by prep-TLC (silica gel, petroleum ether/ethyl acetate 4:1) to give the title compound (150 mg, 54% yield) as a yellow solid. LCMS (ESI, M+1): m/z=379.3.

(1041) Step D. (3R)-1-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxyl-7-(1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-benzo[f]indazole (150 mg, 1.0 equiv.) and (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (180 mg, 1.0 equiv.) in toluene (3.00 mL) was added CataCXium A Pd G3 (43.3 mg, 0.15 equiv.), potassium phosphate (1.5 M, 793 μL, 3.0 equiv.), the reaction mixture was degassed and purged with nitrogen for 3 times, then it was stirred at 90° C. for 2 hours under nitrogen. The reaction mixture was concentrated under reduced pressure and purified by prep-TLC (silica gel, petroleum ether/ethyl acetate 4:1) to give the title compound (200 mg, 36% yield) as a yellow solid. LCMS (ESI, M+1): m/z=670.5.

(1042) Step E. (3R)-1-(7-(1H-benzo[f]indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (3R)-1-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)-7-(1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 47.5% purity, 1.0 equiv.) in dichloromethane (4.00 mL) was added trifluoroacetic acid (6.16 g, 4.00 mL, equiv.) in one portion at 0° C. under nitrogen. The reaction was stirred at 20° C. for 10 min. The reaction mixture was concentrated under reduced pressure and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 8%-38%, 10 min) to give the title compound (27.9 mg, 32% yield, formic acid salt) as a yellow solid. ¹H NMR (400 MHz, CD₃OD) δ=9.41 (d, J=9.2 Hz, 1H), 8.45 (s, 1H), 8.21 (s, 1H), 8.11-8.00 (m, 2H), 7.80 (d, J=8.8 Hz, 1H), 7.54-7.46 (m, 1H), 7.41-7.32 (m, 1H), 5.67-5.33 (m, 1H), 4.72-4.62 (m, 1H), 4.62-4.48 (m, 2H), 4.39 (d, J=13.6 Hz, 1H), 3.84-3.54 (m, 4H), 3.52-3.39 (m, 1H), 3.29-3.25 (m, 1H), 2.68-2.38 (m, 2H), 2.37-2.28 (m, 1H), 2.28-2.13 (m, 3H), 2.12-1.98 (m, 1H), 1.94-1.72 (m, 3H), 1.31 (s, 3H); ¹⁹F NMR (377 MHz, CD₃OD) δ=-139.13, δ=-173.98; LCMS (ESI, M+1): m/z=586.3.

Example 155

(1043) ##STR00422##

2-(4-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperazin-1-yl) nicotinonitrile

(1044) The title compound was synthesized according to the procedure described for example 134. ¹H NMR (400 MHz, METHANOL-d₄) 5-9.17 (s, 1H), 8.40 (dd, J=1.6, 4.8 Hz, 1H), 7.98

(dd, J=1.6, 7.6 Hz, 1H), 7.66 (dd, J=5.6, 9.2 Hz, 1H), 7.33-7.18 (m, 2H), 7.06 (d, J=2.4 Hz, 1H), 6.91 (dd, J=4.8, 7.6 Hz, 1H), 5.41-5.18 (m, 1H), 4.40-4.20 (m, 6H), 4.08-3.95 (m, 4H), 3.29-3.11 (m, 3H), 3.05-2.91 (m, 1H), 2.55-2.43 (m, 1H), 2.40-2.28 (m, 1H), 2.27-2.11 (m, 4H), 2.05-1.96 (m, 2H), 0.80 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=681.4.

Example 156

(1045) ##STR00423##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((imidazo[1,2-a]pyridin-6-ylmethyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1046) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₃): δ=9.20 (s, 1H), 8.60 (s, 1H), 8.55 (s, 1H), 7.85 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.60-7.53 (m, 2H), 7.45 (dd, J=1.6, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.34-5.16 (m, 1H), 4.92 (br d, J=4.4 Hz, 3H), 4.59 (br s, 2H), 4.32-4.19 (m, 2H), 3.26-3.12 (m, 3H), 2.98 (dt, J=6.0, 9.6 Hz, 1H), 2.54-2.42 (m, 1H), 2.34-1.89 (m, 7H), 1.87-1.73 (m, 1H), 0.79 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=640.3.

Example 157

(1047) ##STR00424##

4-(4-((2-(1H-pyrrolo[2,3-b]pyridin-3-yl)ethyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1048) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, CD₃OD): δ 9.07 (s, 1H), 8.20-8.16 (dd, J=1.6, 4.8 Hz, 1H), 8.13-8.07 (dd, J=1.6, 7.6 Hz, 1H), 7.71-7.66 (dd, J=5.6, 8.8 Hz, 1H), 7.33-7.29 (m, 2H), 7.26 (t, J=9.2 Hz, 1H), 7.10-7.01 (m, 2H), 5.43-5.20 (m, 1H), 4.31-4.18 (m, 2H), 4.09-3.94 (m, 2H), 3.29-3.12 (m, 5H), 3.07-2.97 (m, 1H), 2.54-2.41 (m, 1H), 2.38-2.25 (m, 1H), 2.25-2.19 (m, 1H), 2.18-2.08 (m, 2H), 2.05-1.95 (m, 2H), 1.95-1.84 (m, 1H), 0.84-0.76 (dt, J=2.0, 7.6 Hz, 3H); LCMS (ESI, M+1): m/z=654.3.

Example 158

(1049) ##STR00425##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1050) ##STR00426##

(1051) Step A. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (150 mg, 1.0 equiv.), 1,3,7-triazaspiro[4.5]decane-2,4-dione (76.5 mg, 1.6 equiv.) in DMF (1 mL) was added DIEA (183 mg, 5.0 equiv.). The reaction was stirred at 40° C. for 10 hours. The mixture was filtered to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75×30 mm×3 μm; mobile phase: [water (10 mM NH₄HCO₃)/ACN]; B %: 20%-50%, 8 min) and lyophilized to afford the title compound (75.13 mg, 42% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₃) δ=9.10 (s, 1H), 8.11 (br d, J=8.8 Hz, 1H), 7.84 (d, J=8.0 Hz, 1H), 7.73-7.65 (m, 1H), 7.63-7.58 (m, 1H), 7.56-7.48 (m, 1H), 7.23-7.14 (m, 1H), 4.62 (br d, J=13.2 Hz, 1H), 4.53-4.41 (m, 1H), 4.35-4.20 (m, 2H), 3.85-3.62 (m, 2H), 3.18-3.05 (m, 2H), 2.78-2.66 (m, 2H), 2.31-2.17 (m, 1H), 2.13-1.99 (m, 4H), 1.98-1.83 (m, 5H), 1.80-1.67 (m, 2H); LCMS (ESI, M+1): m/z=600.3.

Example 159

(1052) ##STR00427##

4-(4-(((1H-pyrazol-3-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1053) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.18 (s, 1H), 7.70-7.60 (m, 2H), 7.30 (d, J=2.8

Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.8 Hz, 1H), 6.39 (br s, 1H), 5.37-5.21 (m, 1H), 4.94 (s, 2H), 4.29 (q, J=10.4 Hz, 2H), 3.25-3.12 (m, 3H), 3.04-2.97 (m, 1H), 2.53-2.42 (m, 1H), 2.27-2.20 (m, 1H), 2.16-2.08 (m, 2H), 2.03-1.83 (m, 4H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=590.3.

Example 160

(1054) ##STR00428##

4-(4-(((1H-pyrazol-4-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1055) The title compound was synthesized according to the procedure described for example 134. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.13 (s, 1H), 7.73 (br s, 2H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.29 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.46-5.16 (m, 1H), 4.80 (s, 2H), 4.44-4.22 (m, 2H), 3.29-3.13 (m, 3H), 3.05-2.97 (m, 1H), 2.53-2.40 (m, 1H), 2.39-2.19 (m, 2H), 2.18-2.07 (m, 2H), 2.05-1.94 (m, 2H), 1.94-1.82 (m, 1H), 0.78 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=590.3.

Example 161

(1056) ##STR00429##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(((4-methyl-1H-pyrimidin-3-yl)methyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1057) ##STR00430##

(1058) Step A. ethyl 4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazole-3-carboxylate: To a solution of ethyl 4-methyl-1H-pyrazole-3-carboxylate (1.50 g, 1.0 equiv.) in THF (15 mL) was added NaH (428 mg, 60% purity, 1.1 equiv.) at 0° C. The mixture was stirred at 0° C. for 30 minutes. Then SEMCl (1.92 g, 1.2 equiv.) was added to the mixture. The reaction was stirred at 20° C. for 1 hour. Upon reaction completion, the mixture was quenched by addition of water (20 mL) at 0° C., then extracted with ethyl acetate (40 mL×3). The combined organic layers were washed with brine (80 mL), dried over anhydrous sodium sulfate, concentrated and purified by silica gel column chromatography (petroleum ether/ethyl acetate 20:1 to 5:1) to afford the title compound (2.15 g, 65% yield) as a yellow liquid; ¹H NMR (400 MHz, CDCl₃): δ 7.44-7.39 (d, J=0.8 Hz, 1H), 5.42 (s, 2H), 4.46-4.32 (dd, J=7.2, 14.0 Hz, 2H), 3.58-3.48 (m, 2H), 2.31-2.27 (d, J=0.8 Hz, 3H), 1.42-1.35 (t, J=7.2 Hz, 3H), 0.93-0.85 (m, 2H), -0.02-0.08 (m, 9H); LCMS (ESI, M+1): m/z=285.2.

(1059) Step B. 4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazole-3-carboxylic acid: To a solution of ethyl 4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazole-3-carboxylate (2.0 g, 1.0 equiv.) in EtOH (18.0 mL) was added LiOH hydrate (885 mg, 3.0 equiv.) and water (6.0 mL). The reaction was stirred at 20° C. for 2 hours. Upon completion, the reaction mixture was concentrated under reduced pressure to remove EtOH. The residue was diluted with HCl (30 mL, 1 M) and extracted with ethyl acetate (20 mL×3). The combined organic layers were washed with brine (50 mL), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure to afford the title compound (1.70 g, 93% yield) as a yellow solid; ¹H NMR (400 MHz, CDCl₃): δ 7.44 (s, 1H), 5.46 (s, 2H), 3.62-3.51 (m, 2H), 2.38-2.27 (m, 3H), 0.97-0.83 (m, 2H), 0.04-0.09 (m, 9H).

(1060) Step C. 4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazole-3-carboxamide: To a solution of 4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazole-3-carboxylic acid (1.50 g, 1.0 equiv.) and NH₄Cl (1.5 g, 5.0 equiv.) in DMF (20.0 mL) was added HATU (4.50 g, 2.0 equiv.) and DIEA (6.12 g, 8.0 equiv.). The reaction was stirred at 40° C. for 2 hours. Upon completion, the mixture was diluted with water (50 mL) and extracted with ethyl acetate (20 mL×3). The combined organic layers were washed with brine (40 mL), dried over anhydrous sodium sulfate, filtered, concentrated and purified by reversed phase flash chromatography [C18, mobile phase: [water (0.1% formic acid)/acetonitrile]] to afford the title compound (1.40 g, 81% yield) as a yellow liquid; ¹H NMR (400 MHz, DMSO-D₆): δ 7.71 (d, J=0.8 Hz, 1H),

7.35-7.06 (m, 2H), 5.35 (s, 2H), 3.56-3.47 (m, 2H), 2.22-2.14 (d, J=0.4 Hz, 3H), 0.88-0.75 (m, 2H), -0.02-0.11 (m, 9H).

(1061) Step D. (4-methyl-1-((2-(trimethylsilylethoxy)methyl)-1H-pyrazol-3-yl) methanamine: A mixture of 4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazole-3-carboxamide (1.30 g, 1.0 equiv.) and THF (15 mL) was degassed and purged with N.sub.2 for 3 times, and then the mixture was stirred at 0° C. for 5 minutes under N.sub.2 atmosphere. Following that LiAlH.sub.4 (386 mg, 2.0 equiv.) was added into the mixture in portions during a period of 10 minutes. The mixture was stirred at 40° C. for 2 hours. Upon completion, the mixture was quenched by addition of saturated sodium sulfate (386 µL) at 0° C., filtered and the filtrate was diluted with water (10 mL) and extracted with ethyl acetate (10 mL×3). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (1.10 g, 87% yield) as a yellow solid; .sup.1H NMR (400 MHz, CD3OD): δ 7.47 (s, 1H), 5.31 (s, 2H), 3.75 (s, 2H), 3.56-3.50 (m, 2H), 2.07 (s, 3H), 0.89-0.84 (m, 2H), -0.01-0.07 (m, 9H).

(1062) Step E. 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-N((4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazol-3-yl)methyl)pyrido[4,3-d]pyrimidin-4-amine: To a solution of (4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazol-3-yl) methanamine (150 mg, 1.0 equiv.) and 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (395 mg, 1.0 equiv.) in DMF (3 mL) was added DIEA (401 mg, 541 µL, 5.0 equiv.) and 4 Å molecular sieves (50 mg). The reaction was stirred at 40° C. for 2 hours. Upon completion, the mixture was diluted with water (10 mL) and extracted with ethyl acetate (10 mL×3). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate. The mixture was filtered and concentrated and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (100 mg, 20.2% yield) as a yellow oil; LCMS (ESI, M+1): m/z=778.5.

(1063) Step F. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(((4-methyl-1H-pyrazol-3-yl)methyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy) naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-((4-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1H-pyrazol-3-yl)methyl)pyrido[4,3-d]pyrimidin-4-amine (90.0 mg, 1.0 equiv.) in DCM (1 mL) was added TFA (769 mg, 499 µL, 58 equiv.). The mixture was stirred at 20° C. for 0.5 hour. Upon reaction completion, the mixture was concentrated under reduced pressure to remove DCM, the residue was diluted with water (10 mL) and pH was adjusted to 9 with solid Na.sub.2CO.sub.3. Then the mixture was extracted with ethyl acetate (10 mL×3) and the combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by prep-HPLC [column: waters Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN]; B %: 42%-72%, 10 minutes] to afford the title compound (16 mg, 52.6% yield, 98.2% purity) as a yellow solid; .sup.1H NMR (400 MHz, CD3OD): δ 9.22-9.15 (d, J=0.8 Hz, 1H), 7.71-7.64 (dd, J=5.6, 9.2 Hz, 1H), 7.41 (s, 1H), 7.31-7.26 (d, J=2.4 Hz, 1H), 7.26-7.20 (t, J=9.2 Hz, 1H), 7.06-7.03 (d, J=2.0 Hz, 1H), 5.38-5.17 (m, 1H), 4.89 (s, 2H), 4.34-4.23 (m, 2H), 3.29-3.11 (m, 3H), 3.04-2.94 (m, 1H), 2.53-2.39 (m, 1H), 2.37-2.06 (m, 7H), 2.02-1.83 (m, 3H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=604.4.

Example 162

(1064) ##STR00431##

3-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-pyrazole-4-carboxamide

(1065) ##STR00432##

(1066) ##STR00433##

(1067) Step A. di-tert-butyl 3-methyl-1H-pyrazole-1,4-dicarboxylate: To a solution of 3-methyl-1H-pyrazole-4-carboxylic acid (4.50 g, 1.0 equiv.) in 2-methylpropan-2-ol (34.9 g, 13.2 equiv.) was added DMAP (872 mg, 0.2 equiv.) and Boc.sub.2O (31.1 g, 4.0 equiv.) under nitrogen. The reaction was stirred at 40° C. for 12 hours. The reaction mixture was filtered, concentrated and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN], 0-70% ACN) to afford the title compound (6.0 g, 59% yield) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =8.44 (s, 1H), 2.44 (s, 3H), 1.65 (s, 9H), 1.57 (s, 9H). LCMS (ESI, M-55): m/z=227.2.

(1068) Step B. di-tert-butyl 3-methyl-1H-pyrazole-1,4-dicarboxylate: To a solution of di-tert-butyl 3-methylpyrazole-1,4-dicarboxylate (3.0 g, 1.0 equiv.) in CCl.sub.4 (20 mL) was added NBS (2.27 g, 1.20 equiv.) and AIBN (523 mg, 0.3 equiv.) under nitrogen. The mixture was stirred at 80° C. for 12 hours. The reaction mixture was filtered, concentrated and purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 1:0 to 50:1) to afford the title compound (3.0 g, 65% purity, 51% yield) as a colorless liquid; LCMS (ESI, M-99): m/z=261.0.

(1069) Step C. di-tert-butyl 3-((1,3-dioxoisindolin-2-yl)methyl)-1H-pyrazole-1,4-dicarboxylate: To a solution of di-tert-butyl 3-(bromomethyl)pyrazole-1,4-dicarboxylate (2.80 g, 65% purity, 1.0 equiv.) in DMF (20 mL) was added (1,3-dioxoisindolin-2-yl)potassium (1.87 g, 2.0 equiv.). The reaction was stirred at 100° C. for 2 hours. The reaction mixture was filtered, concentrated and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN], 0-40% ACN) to afford the title compound (1.10 g, 55% yield) as a red solid; LCMS (ESI, M-155): m/z=272.1.

(1070) Step D. 3-((1,3-dioxoisindolin-2-yl)methyl)-1H-pyrazole-4-carboxylic acid: To a solution of di-tert-butyl 3-[(1,3-dioxoisindolin-2-yl)methyl]pyrazole-1,4-dicarboxylate (1.10 g, 1.0 equiv.) in DCM (10 mL) was added TFA (15.4 g, 52.5 equiv.) under nitrogen. The mixture was stirred at 20° C. for 2 hours. The reaction mixture was filtered and concentrated to afford the title compound (0.9 g, crude, TFA salt) as a yellow solid; LCMS (ESI, M+1): m/z=272.1.

(1071) Step E. 3-((1,3-dioxoisindolin-2-yl)methyl)-1H-pyrazole-4-carboxamide: To a solution of 3-[(1,3-dioxoisindolin-2-yl)methyl]-1H-pyrazole-4-carboxylic acid (100 mg, 1.0 equiv.) in DMF (2 mL) was added DIEA (476 mg, 10.0 equiv.), HATU (420 mg, 3.0 equiv.) and NH.sub.4Cl (98.6 mg, 5.0 equiv.) under nitrogen. The reaction was stirred at 40° C. for 12 hours. The reaction mixture was filtered, the filtrate was concentrated and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN], 0-40% ACN) to afford the title compound (30 mg, 30% yield) as a red solid; LCMS (ESI, M+1): m/z=271.1.

(1072) Step F. 3-(aminomethyl)-1H-pyrazole-4-carboxamide: To a solution of 3-[(1,3-dioxoisindolin-2-yl)methyl]-1H-pyrazole-4-carboxamide (20.0 mg, 1.0 equiv.) in EtOH (0.5 mL) was added hydrazine hydrate (37.8 mg, 10.0 equiv.). The reaction was stirred at 40° C. for 1 hour. The reaction mixture was filtered and the filtrate was concentrated to afford the title compound (5.0 mg, crude) as a yellow solid; LCMS (ESI, M+23): m/z=163.1.

(1073) Step G. 3-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-pyrazole-4-carboxamide: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (60.0 mg, 1.0 equiv., 0.5 formic acid salt) and DIEA (35.3 mg, 3.0 equiv.) in DMF (1 mL) was added 4 Å MS (20.0 mg) and 3-(aminomethyl)-1H-pyrazole-4-carboxamide (12.7 mg, 1.0 equiv.). The reaction was stirred at 40° C. for 12 hours. The reaction mixture was filtered, concentrated and purified by prep-TLC [Silica gel, DCM/MeOH 10:1] to afford the title compound (50.0 mg, 80% yield) as a light yellow solid; LCMS (ESI, M+1): m/z=677.1.

(1074) Step H. 3-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-

fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl) amino)methyl)-1H-pyrazole-4-carboxamide: To a solution of 3-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl) methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-pyrazole-4-carboxamide (40.0 mg, 1.0 equiv.) in MeOH (0.5 mL) was added HCl in MeOH (4 M, 0.5 mL). The reaction was stirred at 20° C. for 0.5 hour. The reaction mixture filtered, concentrated and purified by reversed phase flash chromatography [column: Phenomenex Gemini-NX C18 75*30 mm*3 um; mobile phase: [water (0.225% formic acid)/ACN]; B %: 12%-42%, 7 min] to afford the title compound (8.14 mg, 20% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d4) δ 9.20 (d, J=1.6 Hz, 1H), 8.51 (br s, 1H), 8.16 (br s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.05 (t, J=2.8 Hz, 1H), 5.50-5.33 (m, 1H), 5.19 (s, 2H), 4.54-4.35 (m, 2H), 3.69-3.43 (m, 3H), 3.21 (br d, J=5.4 Hz, 1H), 2.57-2.32 (m, 3H), 2.30-2.10 (m, 4H), 2.05-1.94 (m, 1H), 0.78 (dt, J=2.0, 7.2 Hz, 3H). LCMS (ESI, M+1): m/z=633.3.

Example 163

(1075) ##STR00434##

4-(4-(((1H-pyrazol-4-yl)methyl)(methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1076) The title compound was synthesized according to the procedure described for example 134.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.21 (s, 1H), 7.77 (br d, J=2.0 Hz, 2H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.4 Hz, 1H), 7.05 (d, J=2.8 Hz, 1H), 5.39-5.21 (m, 1H), 5.06 (s, 2H), 4.39-4.23 (m, 2H), 3.59 (s, 3H), 3.27-3.12 (m, 3H), 3.06-2.93 (m, 1H), 2.53-2.42 (m, 1H), 2.38-2.20 (m, 2H), 2.19-2.09 (m, 2H), 2.03-1.85 (m, 3H), 0.80 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=604.3.

Example 164

(1077) ##STR00435##

4-(4-(((1H-imidazol-4-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1078) The title compound was synthesized according to the procedure described for example 134.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.18 (s, 1H), 7.70-7.64 (m, 2H), 7.30 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.15 (s, 1H), 7.04 (d, J=2.8 Hz, 1H), 5.38-5.21 (m, 1H), 4.77-4.45 (m, 2H), 4.36-4.26 (m, 2H), 3.28-3.13 (m, 3H), 3.06-2.96 (m, 1H), 2.51-2.41 (m, 1H), 2.38-2.19 (m, 2H), 2.18-2.10 (m, 2H), 2.03-1.86 (m, 3H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=590.2.

Example 165

(1079) ##STR00436##

4-(4-(((1H-imidazol-2-yl)methyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1080) The title compound was synthesized according to the procedure described for example 135.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.20 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.09-6.97 (m, 3H), 5.38-5.19 (m, 1H), 4.94 (d, J=4.4 Hz, 2H), 4.28-4.15 (m, 2H), 3.29-3.12 (m, 3H), 3.06-2.95 (m, 1H), 2.54-2.41 (m, 1H), 2.35-2.06 (m, 4H), 2.03-1.83 (m, 3H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=590.3.

Example 166

(1081) ##STR00437##

3-((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)bicyclo[1.1.1]pentan-1-ol

(1082) The title compound was synthesized according to the procedure described for example 135.

.sup.1H NMR (400 MHz, methanol-d4) δ 9.24 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.0 Hz, 1H), 5.67-5.51 (m, 1H), 4.71 (qd, J=4.4, 12.0 Hz, 2H), 4.10-3.85 (m, 3H), 3.54-3.44 (m, 1H), 2.80-2.56 (m, 2H), 2.45 (s, 6H), 2.43-2.31 (m, 4H),

2.28-2.07 (m, 2H), 0.77 (br t, J=6.4 Hz, 3H); LCMS (ESI, M+1): m/z=592.3.

Example 167

(1083) ##STR00438##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((3-(hydroxymethyl)bicyclo[1.1.1]pentan-1-yl)amino)pyrido[4,3-d]pyrimidin-7-yl) naphthalen-2-ol
(1084) The title compound was synthesized according to the procedure described for example 135.
.sup.1H NMR (400 MHz, METHANOL-d₄): δ=9.24 (s, 1H), 7.69 (dd, J=6.0, 9.2 Hz, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.66-5.51 (m, 1H), 4.76-4.62 (m, 2H), 4.11-3.86 (m, 3H), 3.73 (s, 2H), 3.54-3.43 (m, 1H), 2.79-2.59 (m, 2H), 2.49-2.30 (m, 5H), 2.28 (s, 5H), 2.21-2.07 (m, 2H), 0.78 (br t, J=6.8 Hz, 3H); LCMS (ESI, M+1): m/z=606.3.

Example 168

(1085) ##STR00439##

3-((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)thietane 1,1-dioxide
(1086) The title compound was synthesized according to the procedure described for example 134.
.sup.1H NMR (400 MHz, CD₃OD) δ 9.28 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.43-5.18 (m, 1H), 4.98-4.91 (m, 1H), 4.79-4.67 (m, 2H), 4.53-4.40 (m, 2H), 4.38-4.22 (m, 2H), 3.29-3.19 (m, 3H), 3.03 (dt, J=5.6, 9.2 Hz, 1H), 2.50-2.39 (m, 1H), 2.38-2.20 (m, 2H), 2.19-2.09 (m, 2H), 2.06-1.87 (m, 3H), 0.78 (t, J=7.2 Hz, 3H). LCMS (ESI, M+1): m/z=614.3.

Example 169

(1087) ##STR00440##

4-(4-((2-(1H-1,2,3-triazol-4-yl)ethyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol
(1088) The title compound was synthesized according to the procedure described for example 135.
.sup.1H NMR (400 MHz, METHANOL-d₄): δ=9.10 (s, 1H), 7.69 (s, 1H), 7.68-7.63 (m, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.40-5.22 (m, 1H), 4.34-4.23 (m, 2H), 4.05-3.95 (m, 2H), 3.28-3.10 (m, 5H), 3.07-2.95 (m, 1H), 2.49-2.12 (m, 5H), 2.03-1.88 (m, 3H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=605.2.

Example 170

(1089) ##STR00441##

1-(1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide
(1090) The title compound was synthesized according to the procedure described for example 135.
.sup.1H NMR (400 MHz, methanol-d₄) δ 9.08 (s, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.05 (dd, J=2.4, 6.0 Hz, 1H), 5.40-5.23 (m, 1H), 5.03-4.93 (m, 1H), 4.60-4.50 (m, 1H), 4.41-4.26 (m, 2H), 3.68-3.55 (m, 1H), 3.43-3.34 (m, 1H), 3.30-3.07 (m, 5H), 3.06-2.97 (m, 1H), 2.56-2.43 (m, 2H), 2.41-2.08 (m, 5H), 2.04-1.89 (m, 4H), 1.86-1.74 (m, 1H), 1.68-1.55 (m, 1H), 0.80 (q, J=6.8 Hz, 3H); .sup.19F NMR (376 MHz, methanol-d₄) δ=-121.192, -138.964, -173.392; LCMS (ESI, M+1): m/z=671.3.

Example 171

(1091) ##STR00442##

4-(4-((1H-indazol-6-yl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1092) ##STR00443##

(1093) Step A. 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1H-indazol-6-yl)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) and 1H-indazol-6-amine (41.8 mg, 1.0 equiv.) in DMF (2 mL)

was added 4 Å MS (50mg) and t-BuONa (60.4 mg, 2.0 equiv.). The mixture was stirred at 0° C. for 0.5 hour. Upon reaction completion, the mixture was quenched with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography (C18, mobile phase: [water (0.1% formic acid)/ACN]) to afford the title compound (80 mg, 29% yield) as a yellow solid; LCMS (ESI, M+1): m/z=670.3.

(1094) Step B. 4-(4-((1H-indazol-6-yl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1H-indazol-6-yl)pyrido[4,3-d]pyrimidin-4-amine (90 mg, 1.0 equiv.) in DCM (1 mL) was added TFA (1.15 g, 750 µL, 90 equiv.). The mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was concentrated under reduced pressure to remove TFA. The residue was dissolved in water (5 mL), neutralized with solid Na.sub.2CO.sub.3, then extracted with ethyl acetate (3×5 mL). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by prep-HPLC [column: waters Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN]; B %: 39%-69%, 9 min] to afford the title compound (12 mg, 17% yield) as a white solid. .sup.1H NMR (400 MHz, CD.sub.3OD): δ 9.65 (s, 1H), 8.42 (s, 1H), 8.13 (s, 1H), 7.91-7.87 (d, J=8.4 Hz, 1H), 7.82-7.76 (dd, J=6.0, 10.2 Hz, 1H), 7.64-7.57 (dd, J=0.8, 8.4 Hz, 1H), 7.43-7.31 (m, 2H), 7.16-7.13 (d, J=2.4 Hz, 1H), 5.49-5.20 (m, 1H), 4.41-4.22 (m, 3H), 3.27-3.15 (m, 3H), 3.06-2.93 (m, 1H), 2.55-2.45 (m, 1H), 2.35-2.11 (m, 4H), 2.03-1.82 (m, 3H), 0.86 (t, J=7.2 Hz, 3H). LCMS (ESI, M+1): m/z=626.3.

Example 172

(1095) ##STR00444##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(pyridin-4-yl)azetidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1096) The title compound was synthesized according to the procedure described for example 135. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.94 (s, 1H), 9.01 (s, 1H), 8.58 (br d, J=4.8 Hz, 2H), 7.77 (dd, J=6.4, 8.6 Hz, 1H), 7.55 (br d, J=4.8 Hz, 2H), 7.39-7.30 (m, 2H), 7.00 (s, 1H), 5.40-5.15 (m, 2H), 5.03-4.72 (m, 2H), 4.47-4.27 (m, 1H), 4.24-4.11 (m, 2H), 4.09-4.00 (m, 1H), 3.08 (br d, J=9.2 Hz, 2H), 3.01 (br s, 1H), 2.87-2.76 (m, 1H), 2.41-2.31 (m, 1H), 2.15-2.07 (m, 2H), 2.06-1.95 (m, 2H), 1.87-1.73 (m, 3H), 0.73 (br t, J=7.2 Hz, 3H); LCMS [ESI, M+1]: 627.4

Example 173

(1097) ##STR00445##

1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-4-carbonitrile

(1098) The title compound was synthesized according to the procedure described for example 134. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.15 (s, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (dd, J=2.8, 5.2 Hz, 1H), 5.40-5.22 (m, 1H), 4.40-3.95 (m, 6H), 3.29-3.12 (m, 4H), 3.10-2.95 (m, 1H), 2.55-2.44 (m, 1H), 2.41-2.20 (m, 5H), 2.19-2.06 (m, 3H), 2.05-1.86 (m, 5H), 0.87-0.71 (m, 3H); LCMS (ESI, M+1): m/z=617.3.

Example 174

(1099) ##STR00446##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((1-methyl-1H-indazol-6-yl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1100) The title compound was synthesized according to the procedure described for example 171. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.49 (s, 1H), 8.31 (s, 1H), 8.02 (d, J=0.8 Hz, 1H), 7.82 (d, J=8.4 Hz, 1H), 7.69 (dd, J=6.0, 9.2 Hz, 1H), 7.51 (dd, J=1.6, 8.8 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.09 (d, J=2.8 Hz, 1H), 5.37-5.18 (m, 1H), 4.41-4.24 (m, 2H), 4.11 (s, 3H), 3.26-3.14 (m, 3H), 2.99 (dt, J=5.6, 9.6 Hz, 1H), 2.58-2.44 (m, 1H), 2.37-2.20 (m,

2H), 2.18-2.03 (m, 21H), 2.02-1.82 (m, 3H), 0.82 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=640.2.

Example 175

(1101) ##STR00447##

4-(4-((benzo[c][1,2,5]oxadiazol-5-ylmethyl)amino)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1102) The title compound was synthesized according to the procedure described for example 135.

.sup.1H NMR (400 MHz, methanol-d.sub.4) δ 9.23 (s, 1H), 7.97-7.88 (m, 2H), 7.71-7.61 (m, 2H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 5.34-5.13 (m, 1H), 5.04-4.94 (m, 2H), 4.27-4.14 (m, 2H), 3.25-3.08 (m, 3H), 3.01-2.90 (m, 1H), 2.55-2.40 (m, 1H), 2.30-2.11 (m, 3H), 2.10-1.89 (m, 3H), 1.84-1.70 (m, 1H), 0.80 (br t, J=7.2 Hz, 3H); LCMS [ESI, M+1]: 642.3.

Example 176

(1103) ##STR00448##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((3-(pyridin-4-yl)propyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1104) The title compound was synthesized according to the procedure described for example 134.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.09 (s, 1H), 8.41 (d, J=6.0 Hz, 2H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.37 (d, J=6.0 Hz, 2H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.8 Hz, 1H), 5.39-5.22 (m, 1H), 4.32-4.21 (m, 2H), 3.80-3.72 (m, 2H), 3.27-3.13 (m, 3H), 3.05-2.97 (m, 1H), 2.86 (t, J=7.6 Hz, 2H), 2.53-2.42 (m, 1H), 2.38-2.20 (m, 2H), 2.19-2.09 (m, 4H), 2.03-1.85 (m, 3H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=629.3.

Example 177

(1105) ##STR00449##

3-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide

(1106) The title compound was synthesized according to the procedure described for example 135.

.sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.27 (s, 1H), 9.23-9.11 (m, 1H), 8.18 (s, 1H), 7.76 (dd, J=6.0, 8.8 Hz, 1H), 7.41-7.28 (m, 2H), 6.99 (d, J=2.4 Hz, 1H), 5.38-5.19 (m, 1H), 4.35 (br t, J=11.2 Hz, 2H), 4.19-4.03 (m, 4H), 3.94-3.76 (m, 2H), 3.15-3.08 (m, 2H), 3.05-2.95 (m, 2H), 2.89-2.79 (m, 1H), 2.28-1.96 (m, 5H), 1.90-1.70 (m, 3H), 0.81-0.64 (m, 3H); LCMS (ESI, M+1): m/z=628.

Example 178

(1107) ##STR00450##

(3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-fluoropiperidin-3-ol

(1108) The title compound was synthesized according to the procedure described for example 135.

.sup.1H NMR (400 MHz, CD3OD): δ 9.14-9.12 (d, J=6.8 Hz, 1H), 7.71-7.63 (dd, J=6.0, 9.2 Hz, 1H), 7.32-7.29 (d, J=2.8 Hz, 1H), 7.28-7.21 (t, J=9.6 Hz, 1H), 7.08-7.01 (m, 1H), 5.54 (s, 1H), 5.40 (s, 1H), 5.37 (s, 1H), 5.24 (s, 1H), 4.98 (s, 1H), 4.46-4.15 (m, 5H), 3.82-3.76 (dd, J=2.0, 11.6 Hz, 1H), 3.29-3.10 (m, 3H), 3.08-2.96 (m, 1H), 2.56-2.40 (m, 3H), 2.40-2.10 (m, 4H), 2.05-1.85 (m, 3H), 0.86-0.74 (td, J=7.4, 19.2 Hz, 3H); LCMS (ESI, M+1): m/z=612.3.

Example 179

(1109) ##STR00451##

(3S,4S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-4-fluoropiperidin-3-ol

(1110) The title compound was synthesized according to the procedure described for example 135

except for HCl-MeOH (4 M, 49.2 equiv.) was used in the deprotection step. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.24 (dd, J=1.2, 6.4 Hz, 1H), 8.49 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.35-7.19 (m, 1H), 7.06 (t, J=2.8 Hz, 1H), 5.55-5.28 (m, 1H), 4.79-4.60 (m, 1H), 4.55-4.38 (m, 2H), 4.32-4.12 (m, 2H), 4.11-3.88 (m, 3H), 3.69-3.41 (m, 3H), 3.21 (dt, J=5.6, 9.6 Hz, 1H), 2.56-

2.23 (m, 5H), 2.22-2.08 (m, 3H), 2.07-1.92 (m, 2H), 0.80 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=612.2.

Example 180

(1111) ##STR00452##

(3R,5S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-methylpiperidin-3-ol

(1112) The title compound was synthesized according to the procedure described for example 135 except for HCl/EtOAc (4 M, 1.0 mL, 27 equiv.) was used in the deprotection step. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.05 (d, J=1.6 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.38-7.15 (m, 2H), 7.06 (s, 1H), 5.44-5.15 (m, 1H), 4.81-4.72 (m, 1H), 4.55 (br d, J=12.8 Hz, 1H), 4.38-4.18 (m, 2H), 3.88 (br dd, J=4.4, 10.4 Hz, 1H), 3.29-3.12 (m, 3H), 3.09-2.90 (m, 3H), 2.54-2.10 (m, 6H), 2.06-1.86 (m, 4H), 1.35-1.21 (m, 1H), 1.06 (d, J=6.5 Hz, 3H), 0.80 (br t, J=7.3 Hz, 3H); LCMS (ESI, M+1): m/z=608.4.

Example 181

(1113) ##STR00453##

(2R,3R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol

(1114) ##STR00454##

(1115) Step A. (2R,3R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol: To a mixture of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (450 mg, 1.0 equiv.), DIEA (2.30 g, 10 equiv.) and 4 Å molecular sieves (45 mg) in DMAc (4 mL) was added (2R,3R)-2-methylpiperidin-3-ol (135 mg, 0.5 equiv., HCl) at 15° C. The reaction was stirred at 15° C. for 1 hour. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with NaHCO₃ solid, concentrated in vacuum to remove acetonitrile, and extracted with ethyl acetate (2×20 mL). The combined organic layer was washed with brine (15 mL) dried over anhydrous sodium sulfate, concentrated in vacuum to afford the title compound (265 mg, 44% yield) as a red solid; ¹H NMR (400 MHz, DMSO-d₆) δ=8.90 (s, 1H), 5.12 (d, J=4.0 Hz, 1H), 4.92 (br s, 1H), 4.25 (br d, J=11.6 Hz, 1H), 3.77-3.67 (m, 1H), 3.46 (br s, 1H), 1.82-1.58 (m, 4H), 1.31 (br d, J=6.8 Hz, 3H); LCMS (ESI, M+1): m/z=331.0.

(1116) Step B. (2R,3R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol: A mixture of (2R,3R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol (257 mg, 1.0 equiv.), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (371 mg, 3.0 equiv.), DIEA (501 mg, 5.0 equiv.) and 4 Å molecular sieves (30 mg) in dioxane (3 mL) was stirred at 90° C. for 14 hours. The mixture was filtered and the filtrate was concentrated, purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with NaHCO₃ solid, extracted with ethyl acetate (2×40 mL). The combined organic layer was washed with brine (40 mL), dried over anhydrous Na₂SO₄, and concentrated in vacuum to afford the title compound (296 mg, 74% yield) as a yellow solid; LCMS (ESI, M+1): m/z=454.1.

(1117) Step C. (2R,3R)-1-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol: To a mixture of (2R,3R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol (100 mg, 1.0 equiv.), ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (156 mg, 1.5 equiv.) and Cs₂CO₃ (144 mg, 2.0 equiv.) in dioxane (1.5 mL) and water (0.5 mL) was added Pd(PPh₃)₄ (25.5 mg, 0.1

equiv.) under N.sub.2. The mixture was de-gassed and stirred at 100° C. for 8 hours under N.sub.2. The mixture was diluted with water (3 mL) and extracted with ethyl acetate (3×3 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, concentrated in vacuum. The residue was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with NaHCO.sub.3 solid, concentrated in vacuum to remove acetonitrile, and extracted with ethyl acetate (15 mL). The combined organic layer was washed with brine (15 mL), dried over anhydrous Na.sub.2SO.sub.4, concentrated in vacuum to afford the title compound (53.0 mg, 31% yield) as a yellow solid; LCMS (ESI, M+1): m/z=764.3.

(1118) Step D. (2R,3R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol: A mixture of (2R,3R)-1-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methylpiperidin-3-ol (50.0 mg, 1.0 equiv.) and CsF (99.4 mg, 10 equiv.) in DMF (0.6 mL) was stirred at 15° C. for 1 hour. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with solid NaHCO.sub.3, concentrated in vacuum to remove acetonitrile and extracted with ethyl acetate (2×10 mL). The combined organic layer was washed with brine (15 mL), dried over anhydrous Na.sub.2SO.sub.4 and concentrated in vacuum. The residue was purified by prep-HPLC [Waters Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN]; B %: 39%-69%, 9 min] twice to afford the title compound (22.0 mg, 55% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=8.97 (d, J=2.8 Hz, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.05 (dd, J=2.4, 6.8 Hz, 1H), 5.30 (d, J=54.4 Hz, 1H), 5.11-5.02 (m, 1H), 4.40 (br d, J=13.6 Hz, 1H), 4.34-4.21 (m, 2H), 3.96-3.88 (m, 1H), 3.58-3.47 (m, 1H), 3.29-3.13 (m, 3H), 3.06-2.96 (m, 1H), 2.55-2.43 (m, 1H), 2.38-2.10 (m, 4H), 2.03-1.95 (m, 2H), 1.94-1.79 (m, 5H), 1.45 (t, J=6.0 Hz, 3H), 0.85-0.75 (m, 3H); .sup.19F NMR (376 MHz, methanol-d.sub.4) δ=-121.191, -138.888, -173.675; LCMS (ESI, M+1): m/z=608.3.

Example 182

(1119) ##STR00455##

(3R,6R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol

(1120) ##STR00456##

(1121) Step A. (3R,6R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) and (3R,6R)-6-methylpiperidin-3-ol (84.1 mg, 0.7 equiv., HCl) in DMAc (4 mL) was added DIEA (307 mg, 414 µL, 3 equiv.) and 4 Å molecular sieves (300 mg). The reaction was stirred at 20° C. for 2 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (10 mL×3). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, and concentrated in vacuum. The residue was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (205 mg, 78% yield) as a yellow solid; .sup.1H NMR (400 MHz, DMSO-D.sub.6): δ 9.32 (s, 1H), 5.13-4.96 (m, 2H), 4.28-4.21 (d, J=13.6 Hz, 1H), 4.00 (s, 1H), 3.76-3.68 (d, J=14.0 Hz, 1H), 2.24-2.14 (m, 1H), 2.03-1.92 (m, 1H), 1.64-1.55 (d, J=12.8 Hz, 1H), 1.47-1.40 (m, 1H), 1.37-1.30 (d, J=6.8 Hz, 3H); LCMS (ESI, M+1): m/z=331.1.

(1122) Step B. (3R,6R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol: To a solution of (3R,6R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol (205 mg, 1.0 equiv.) and ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (296 mg, 3.0 equiv.) in DMF (5 mL)

was added DIEA (400 mg, 539 μ L, 5.0 equiv.) and 4 Å molecular sieves (400 mg). The reaction was stirred at 90° C. for 12 hours. The mixture was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (210 mg, 71.0% yield) as a yellow solid; LCMS (ESI, M+1): m/z=454.1.

(1123) Step C. (3R,6R)-14 (74(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol: A mixture of (3R,6R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol (190 mg, 1.0 equiv.), ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (395 mg, 2.0 equiv.), K.sub.3PO.sub.4 (1.5 M, 3.0 equiv.) and CataCXium A Pd G3 (30.5 mg, 0.1 equiv.) in THF (3 mL) was degassed and stirred at 60° C. for 2 hours under N.sub.2 atmosphere. The mixture was diluted with water (20 mL) and extracted with ethyl acetate (20 mL $\times$ 3). The combined organic layers were washed with brine (30 mL), dried over anhydrous sodium sulfate, concentrated under reduced pressure. The residue was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (210 mg, 63% yield) as a black solid; LCMS (ESI, M+1): m/z=764.4.

(1124) Step D. (3R,6R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol: To a solution of (3R,6R)-1-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol (100 mg, 1.0 equiv.) in DMF (2 mL) was added CsF (199 mg, 10 equiv.). The reaction was stirred at 40° C. for 1 hour. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (10 mL $\times$ 3). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μ m; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN]; B %: 34-64%, 8 minutes] to afford the title compound (72 mg, 90% yield) as a white solid; .sup.1H NMR (400 MHz, CD3OD): δ 9.41-9.32 (d, J=4.4 Hz, 1H), 7.69-7.64 (dd, J=5.6, 8.8 Hz, 1H), 7.32-7.29 (d, J=2.0 Hz, 1H), 7.27-7.21 (t, J=9.6 Hz, 1H), 7.07-7.03 (dd, J=2.4, 6.8 Hz, 1H), 5.42-5.21 (d, J=54.4 Hz, 1H), 5.20-5.12 (m, 1H), 4.86 (s, 9H), 4.52-4.44 (d, J=14.4 Hz, 1H), 4.34-4.23 (m, J=10.5 Hz, 2H), 4.13 (s, 1H), 3.84-3.76 (m, 1H), 3.29-3.16 (m, 3H), 3.05-2.97 (m, 1H), 2.57-2.08 (m, 7H), 2.04-1.73 (m, 4H), 1.59-1.52 (m, 1H), 1.48-1.44 (m, 3H), 0.84-0.77 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=608.3.

Example 183

(1125) ##STR00457##

1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-(trifluoromethyl)piperidin-3-ol

(1126) The title compound was synthesized according to the procedure described for example 135. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.21 (dd, J=3.2, 14.8 Hz, 1H), 7.67 (dd, J=5.8, 8.9 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.07 (t, J=2.4 Hz, 1H), 5.30 (d, J=54.4 Hz, 1H), 4.72-4.64 (m, 2H), 4.35-4.25 (m, 2H), 3.72 (t, J=14.4 Hz, 1H), 3.37 (t, J=12.8 Hz, 1H), 3.28-3.18 (m, 3H), 3.01 (dt, J=5.2, 9.2 Hz, 1H), 2.24-2.20 (m, 6H), 2.07-1.90 (m, 6H), 0.80 (dt, J=3.2, 7.2 Hz, 3H); F NMR (400 MHz, methanol-d.sub.4) δ =-84.2, -121, -139, -174; LCMS (ESI, M+1): m/z=662.4.

Example 184

(1127) ##STR00458##

3-(difluoromethyl)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol

(1128) The title compound was synthesized according to the procedure described for example 135. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.25 (dd, J=2.4, 6.8 Hz, 1H), 8.50 (br s, 1H),

7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.32-7.22 (m, 2H), 7.06 (d, J=2.4 Hz, 1H), 5.96-5.60 (m, 1H), 5.52-5.31 (m, 1H), 4.69 (br d, J=13.2 Hz, 1H), 4.57-4.40 (m, 3H), 3.78-3.64 (m, 1H), 3.64-3.45 (m, 3H), 3.45-3.37 (m, 1H), 3.25-3.15 (m, 1H), 2.54-2.31 (m, 3H), 2.29-2.09 (m, 5H), 2.06-1.85 (m, 4H), 0.85-0.74 (m, 3H); LCMS (ESI, M+1): m/z=644.3.

Example 185

(1129) ##STR00459##

(2S,3R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-methylpiperidin-3-ol

(1130) The title compound was synthesized according to the procedure described for example 182. ¹H NMR (400 MHz, methanol-d₄): δ ppm 0.74-0.86 (m, 3H) 1.55 (m, 3H) 1.61-1.72 (m, 1H) 1.77-2.04 (m, 4H) 2.04-2.39 (m, 6H) 2.41-2.56 (m, 1H) 2.96-3.05 (m, 1H) 3.12-3.29 (m, 3H) 3.14-3.27 (m, 3H) 3.47-3.63 (m, 1H) 3.82-3.96 (m, 1H) 4.18-4.39 (m, 2H) 4.49-4.59 (m, 1H) 4.88-5.11 (m, 1H) 5.21-5.39 (m, 1H) 7.05 (dd, J=16.8, 2.8 Hz, 1H) 7.21-7.28 (m, 1H) 7.29-7.32 (m, 1H) 7.67 (dd, J=8.8, 6.4 Hz, 1H) 9.03 (t, J=2.4 Hz, 1H); LCMS (ESI, M+1): m/z=608.2.

Example 186

(1131) ##STR00460##

(3R,6S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methylpiperidin-3-ol

(1132) The title compound was synthesized according to the procedure described for example 182 except for BrettPhos-Pd-G3 (0.1 equiv.) was used instead of cataCXium A Pd G3 in step C. ¹H NMR (400 MHz, CHLOROFORM-d): δ 8.99-8.72 (m, 1H), 8.54 (s, 1H), 7.59-7.43 (m, 1H), 7.25-6.78 (m, 3H), 5.50-5.26 (m, 1H), 4.84-4.64 (m, 2H), 4.61-4.40 (m, 2H), 3.91-3.60 (m, 3H), 3.49-3.34 (m, 1H), 3.21-2.96 (m, 2H), 2.64-2.33 (m, 4H), 2.22-2.03 (m, 4H), 1.92 (br s, 4H), 1.51 (br t, J=6.8 Hz, 3H), 0.85-0.73 (m, 3H); LCMS (ESI, M+1): m/z=608.2.

Example 187

(1133) ##STR00461##

(3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-methylpiperidin-3-ol

(1134) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₄) δ 9.22 (dd, J=2.8, 8.4 Hz, 1H), 7.67 (dd, J=6.0, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.08-7.04 (m, 1H), 5.39-5.19 (m, 1H), 4.69-4.50 (m, 2H), 4.37-4.21 (m, 2H), 4.15 (br s, 1H), 3.64 (br d, J=14.0 Hz, 1H), 3.29-3.16 (m, 3H), 3.06-2.92 (m, 2H), 2.51-2.33 (m, 2H), 2.29-2.10 (m, 4H), 2.04-1.86 (m, 4H), 1.63-1.51 (m, 1H), 1.03 (br d, J=6.4 Hz, 3H), 0.85-0.76 (m, 3H); ¹⁹F NMR (400 MHz, methanol-d₄) δ -121.176, -139.194, -173.708; LCMS (ESI, M+1): m/z=608.3.

Example 188

(1135) ##STR00462##

(3R,4S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-4-methylpiperidin-3-ol

(1136) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₄) δ 9.06 (d, J=2.0 Hz, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.34-7.19 (m, 2H), 7.05 (t, J=2.4 Hz, 1H), 5.42-5.19 (m, 1H), 4.75-4.63 (m, 1H), 4.57 (br d, J=13.6 Hz, 1H), 4.36-4.19 (m, 2H), 3.54-3.35 (m, 2H), 3.28-3.12 (m, 4H), 3.01 (dt, J=5.6, 9.2 Hz, 1H), 2.54-2.40 (m, 1H), 2.39-2.09 (m, 4H), 2.05-1.84 (m, 4H), 1.78-1.65 (m, 1H), 1.55-1.40 (m, 1H), 1.14 (d, J=6.4 Hz, 3H), 0.87-0.72 (m, 3H); LCMS (ESI, M+1): m/z=608.4.

Example 189

(1137) ##STR00463##

(3R,4R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-4-methylpiperidin-3-ol

(1138) ##STR00464##

(1139) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₄) δ=9.30 (dd, J=2.8, 9.2 Hz, 1H), 8.50 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.37-7.18 (m, 2H), 7.13-6.99 (m, 1H), 5.52-5.33 (m, 1H), 4.81-4.69 (m, 2H), 4.53-4.38 (m, 2H), 3.91 (br s, 1H), 3.67-3.45 (m, 4H), 3.35 (br d, J=3.6 Hz, 1H), 3.21 (dt, J=5.6, 10.0 Hz, 1H), 2.56-2.32 (m, 3H), 2.30-2.10 (m, 4H), 2.07-1.83 (m, 3H), 1.66 (br d, J=12.2 Hz, 1H), 1.07 (d, J=6.4 Hz, 3H), 0.85-0.75 (m, 3H); LCMS (ESI, M+1): m/z=608.4.

Example 190

(1140) ##STR00465##

1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,5-difluoropiperidin-3-ol

(1141) ##STR00466##

(1142) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₄): δ=9.26 (dd, J=1.2, 10.8 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.07 (t, J=2.4 Hz, 1H), 5.40-5.21 (m, 1H), 4.46-4.32 (m, 3H), 4.31-4.13 (m, 3H), 3.78-3.59 (m, 1H), 3.28-3.13 (m, 3H), 3.01 (dt, J=5.6, 9.6 Hz, 1H), 2.63-2.38 (m, 2H), 2.33-2.10 (m, 5H), 2.05-1.89 (m, 3H), 0.80 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=630.2.

Example 191

(1143) ##STR00467##

1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-4,4-difluoropiperidin-3-ol

(1144) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d) 0.79 (t, J=7.2 Hz, 3H) 1.87-2.04 (m, 3H) 2.11-2.38 (m, 5H) 2.41-2.62 (m, 2H) 3.01 (m, 1H) 3.16-3.29 (m, 3H) 3.61-3.70 (m, 1H) 3.96-4.07 (m, 2H) 4.25-4.37 (m, 2H) 4.38-4.60 (m, 2H) 5.21-5.39 (m, 1H) 7.06 (dd, J=4.4, 2.4 Hz, 1H) 7.25 (t, J=9.2 Hz, 1H) 7.30 (d, J=2.8 Hz, 1H) 7.68 (dd, J=8.8, 5.6 Hz, 1H) 9.28 (dd, J=9.6, 2.0 Hz, 1H); ¹⁹F NMR (400 MHz, DMSO-d₆) δ=-110.901, -121.183, -138.993, 173.712. LCMS (ESI, M+1): m/z=630.3.

Example 192

(1145) ##STR00468##

(3R,5S)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-fluoropiperidin-3-ol

(1146) ##STR00469##

(1147) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₄) 6-9.36 (d, J=5.8 Hz, 1H), 8.48 (br s, 1H), 7.68 (dd, J=5.8, 9.2 Hz, 1H), 7.35-7.19 (m, 1H), 7.06 (t, J=2.8 Hz, 1H), 5.55-5.32 (m, 1H), 5.08-4.91 (m, 1H), 4.58-4.43 (m, 2H), 4.43-4.28 (m, 1H), 4.25-3.95 (m, 4H), 3.72-3.46 (m, 3H), 3.24 (dt, J=5.6, 10.0 Hz, 1H), 2.60-2.24 (m, 5H), 2.23-1.95 (m, 5H), 0.80 (dt, J=2.0, 7.2 Hz, 3H); LCMS (ESI, M+1): m/z=612.3.

Example 193

(1148) ##STR00470##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol

(1149) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₄) δ=9.18-9.08 (m, 1H), 7.73-7.61 (m, 1H), 7.32-7.28 (m,

1H), 7.27-7.19 (m, 1H), 7.08-7.03 (m, 1H), 5.39-5.20 (m, 1H), 4.36-4.29 (m, 1H), 4.28-4.19 (m, 2H), 4.08-3.91 (m, 3H), 3.87-3.74 (m, 1H), 3.29-3.12 (m, 3H), 3.06-2.95 (m, 1H), 2.56-2.41 (m, 1H), 2.40-1.84 (m, 10H), 1.82-1.67 (m, 2H), 0.86-0.74 (m, 3H). LCMS (ESI, M+1): m/z=594.2.

Example 194

(1150) ##STR00471##

(S)-6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1151) ##STR00472##

(1152) 6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: The title compound was synthesized according to the procedure described for example 134. LCMS (ESI, M+1): m/z=633.3.

(1153) Step A. (S)-6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: Stereoisomeric mixture 6-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (70 mg, 1.0 equiv.) was separated by SFC [column: Daicel Chiralpak AS (250 mm×30 mm, 10 μm); mobile phase: [CO.sub.2/0.1% NH.sub.3.Meth.H.sub.2O in MeOH] 45%-45%; 25 min]. The desired fractions of product (2^{sup}.nd eluting peak) were collected and concentrated in vacuum to give an impure product. The impure product was purified by prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 15%-45%, 10 min]. The desired fractions were collected and concentrated in vacuum to remove ACN. The aqueous layer was lyophilized to afford title compound (8.09 mg, 11% yield, 0.1 formic acid salt) as a white solid; LCMS (ESI, M+1): m/z=633.4; SFC: 97.2% ee Chiralpak AS-3 50×4.6 mm I.D., 3 μm column A: 60% MeOH+40% ACN (w/0.05% DEA), B: CO.sub.2, 3 mL/min, 220 nm, t_{sub}.R: 2.070 min; ¹H NMR (400 MHz, MeOD): δ 9.09 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.08-7.04 (m, 1H), 5.44-5.23 (m, 1H), 4.47-4.23 (m, 4H), 3.99 (t, J=11.6 Hz, 1H), 3.90-3.68 (m, 1H), 3.42-3.35 (m, 1H), 3.28-3.20 (m, 1H), 3.16-3.00 (m, 1H), 2.95-2.84 (m, 1H), 2.80-2.71 (m, 1H), 2.56-2.24 (m, 3H), 2.24-1.86 (m, 10H), 0.79 (t, J=7.6 Hz, 3H).

Example 195

(1154) ##STR00473##

(S)-4-(8-fluoro-2-((1-methylpyrrolidin-2-yl)methoxy)-4-morpholinopyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1155) ##STR00474##

(1156) Step A. 4-(2,7-dichloro-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl)morpholine: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (600 mg, 2.38 mmol, 1.0 equiv.) in DCM (10.0 mL) were added DIEA (921 mg, 7.13 mmol, 1.24 mL, 3.0 equiv.) and 4 Å molecular sieves (200 mg). The mixture was stirred at 25° C. for 10 minutes. Then a solution of morpholine (103 mg, 1.19 mmol, 104 μL, 0.5 equiv.) in DCM (2.0 mL) was added dropwise at -40° C. The mixture was stirred at -40° C. for 0.5 h. After completion, the reaction mixture was filtered and the filtrate was concentrated. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate 30:1 to 1:1) to afford the title compound (310 mg, 84% yield) as a yellow solid; ¹H NMR (400 MHz, CDCl₃): δ 8.87 (s, 1H), 4.13-4.07 (m, 4H), 3.92-3.86 (m, 4H); LCMS [ESI, M+1]: 303.1.

(1157) Step B. 4-[7-chloro-8-fluoro-2-(((2S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)morpholine: To a solution of 4-(2,7-dichloro-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl)morpholine (530 mg, 1.75 mmol, 1.0 equiv.) in dioxane (10.0 mL) were added DIEA (678 mg, 5.25 mmol, 914 μL, 3.0 equiv.) and [(2S)-1-methylpyrrolidin-2-yl]methanol (604 mg, 5.25 mmol, 623 μL, 3.0 equiv.). The mixture was stirred at 60° C. for 1.5 hrs. After completion, water (10.0

mL) was added and the mixture was extracted with ethyl acetate (3×10.0 mL). The combined the organic layers were dried over Na.sub.2SO.sub.4 and concentrated. The residue was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) to afford the title compound (480 mg, 67% yield) as a yellow solid; .sup.1H NMR (400 MHz, CDCl.sub.3): δ 8.73 (s, 1H), 4.52 (dd, J=4.8, 10.8 Hz, 1H), 4.33 (dd, J=6.8, 10.8 Hz, 1H), 4.01-3.95 (m, 4H), 3.88-3.81 (m, 4H), 3.13-3.06 (m, 1H), 2.74-2.64 (m, 1H), 2.48 (s, 3H), 2.31-2.25 (m, 1H), 2.09-2.03 (m, 1H), 1.88-1.71 (m, 3H); LCMS [ESI, M+1]: 382.1.

(1158) Step C. 4-[8-fluoro-2-[(2S)-1-methylpyrrolidin-2-yl]methoxy]-4-morpholino-pyrido[4,3-d]pyrimidin-7-yl]naphthalen-2-ol: To a solution of 4-[7-chloro-8-fluoro-2-[(2S)-1-methylpyrrolidin-2-yl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]morpholine (150 mg, 393 μmol, 1.0 equiv.) and 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (127 mg, 471 μmol, 1.2 equiv.) in THF (3.0 mL) was added CataCXium A Pd G3 (28.6 mg, 39.3 μmol, 0.1 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 786 μL, 3.0 equiv.). The mixture was stirred at 60° C. for 2 hrs. After reaction completion, water (5.0 mL) was added and the mixture was extracted with ethyl acetate (3×5.0 mL). Combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and concentrated. The residue was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected, neutralized with solid NaHCO.sub.3, and concentrated in vacuum to remove ACN. The aqueous phase was extracted with ethyl acetate (2×10.0 mL). The combined organic phase was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuum. Then the residue was purified by prep-HPLC (column: Phenomenex Gemini-NX C18 75×30 mm×3 μm, mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN]; B %: 22%-52%, 8 min) to afford the title compound (51.1 mg, 26% yield) as a yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.21-10.00 (m, 1H), 9.22 (s, 1H), 7.80 (d, J=8.4 Hz, 1H), 7.54 (br d, J=8.4 Hz, 1H), 7.47-7.40 (m, 1H), 7.29 (d, J=2.0 Hz, 1H), 7.26-7.21 (m, 2H), 4.41 (dd, J=4.4, 10.8 Hz, 1H), 4.23 (dd, J=6.4, 10.8 Hz, 1H), 4.05-3.97 (m, 4H), 3.84-3.75 (m, 4H), 2.99-2.90 (m, 1H), 2.64-2.54 (m, 1H), 2.35 (s, 3H), 2.17 (q, J=8.8 Hz, 1H), 2.02-1.88 (m, 1H), 1.74-1.57 (m, 3H); HRMS (ESI+) calcd for C.sub.27H.sub.29FN.sub.5O.sub.3* (M+H⁺): 490.2249, found 490.2235.

Example 196

(1159) ##STR00475##

7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1160) The title compound was synthesized according to the procedure described for example 134. .sup.1H NMR (400 MHz, CD3OD) δ (ppm)=9.18-9.04 (m, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.8 Hz, 1H), 5.40-5.15 (m, 1H), 4.69-4.57 (m, 1H), 4.55-4.39 (m, 1H), 4.35-4.17 (m, 2H), 3.88-3.63 (m, 2H), 3.29-3.12 (m, 3H), 3.06-2.94 (m, 1H), 2.56-2.30 (m, 2H), 2.29-2.07 (m, 5H), 2.05-1.87 (m, 5H), 0.79 (q, J=7.8 Hz, 3H). LCMS (ESI, M+1): 662.4.

Example 197

(1161) ##STR00476##

5-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H, 3aH)-dione

(1162) The title compound was synthesized according to the procedure described for example 135. .sup.1H NMR (400 MHz, MeOD) δ 9.29 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.8 Hz, 1H), 5.57-5.37 (m, 1H), 4.68-4.60 (m, 2H), 4.59-4.48 (m, 2H), 4.40-4.30 (m, 2H), 3.80-3.55 (m, 5H), 2.67-1.97 (m, 9H), 0.78 (br t, J=7.2 Hz, 3H). LCMS (ESI, M+1): m/z=633.3.

Example 198

(1163) ##STR00477##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((3-(pyridin-3-yl)propyl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1164) The title compound was synthesized according to the procedure described for example 134. ¹H NMR (400 MHz, CD₃OD-d.sub.4) δ 9.10 (s, 1H), 8.46 (d, J=2.0 Hz, 1H), 8.36 (dd, J=1.6, 5.2 Hz, 1H), 7.79 (br d, J=8.0 Hz, 1H), 7.66 (dd, J=6.0, 9.2 Hz, 1H), 7.37 (dd, J=4.8, 8.0 Hz, 1H), 7.28 (d, J=2.8 Hz, 1H), 7.25 (s, 1H), 7.03 (d, J=2.8 Hz, 1H), 5.45-5.18 (m, 1H), 4.35-4.19 (m, 2H), 3.76 (br t, J=7.2 Hz, 2H), 3.28-3.12 (m, 3H), 3.01 (dt, J=5.6, 9.4 Hz, 1H), 2.85 (t, J=7.6 Hz, 2H), 2.52-2.38 (m, 1H), 2.38-2.19 (m, 2H), 2.19-2.07 (m, 4H), 2.03-1.82 (m, 3H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=629.3.

Example 199

(1165) ##STR00478##

5-(((7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1-methylpyridin-2 (1H)-one

(1166) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, CD₃OD-d.sub.4) δ 9.15 (s, 1H), 7.81 (s, 1H), 7.74-7.61 (m, 2H), 7.34-7.17 (m, 2H), 7.03 (d, J=2.4 Hz, 1H), 6.57 (d, J=9.2 Hz, 1H), 5.41-5.17 (m, 1H), 4.67 (s, 2H), 4.37-4.19 (m, 2H), 3.59 (s, 3H), 3.28-3.15 (m, 3H), 3.01 (dt, J=6.0, 9.2 Hz, 1H), 2.53-2.39 (m, 1H), 2.37-2.18 (m, 2H), 2.18-2.07 (m, 2H), 2.04-1.94 (m, 2H), 1.93-1.81 (m, 1H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=631.3.

Example 200

(1167) ##STR00479##

(S)-4-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1168) ##STR00480##

(1169) Step A. 4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of 6-methyl-1,4-oxazepan-6-ol (2.66 g, 1.0 equiv.), 4 Å molecular sieves (850 mg) and DIEA (10.5 g, 4.0 equiv.) in DMF (90 mL) was stirred at 15° C. for 0.5 hour. Then 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (8.90 g, 1.0 equiv.) was added into the above mixture and the mixture was stirred at 40° C. for 4 hours. The mixture was filtered. The filtrate was diluted with ethyl acetate (120 mL) and water (200 mL), extracted with ethyl acetate (2×50 mL) and dichloromethane/methanol=10/1 (2×50 mL). The combined organic layers were dried over Na₂SO₄, concentrated and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]) twice. The desired fractions were collected, neutralized with solid NaHCO₃, concentrated in vacuum to remove acetonitrile, and extracted with ethyl acetate (3×500 mL). The combined organic phase was dried over anhydrous Na₂SO₄ and concentrated in vacuum to afford the title compound (6.63 g, 68% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d.sub.4) δ=9.34 (d, J=2.0 Hz, 1H), 5.38-5.21 (m, 1H), 4.52 (td, J=5.2, 14.4 Hz, 1H), 4.43 (dd, J=3.6, 14.8 Hz, 1H), 4.33-4.19 (m, 2H), 4.17-4.09 (m, 1H), 3.98 (ddd, J=3.6, 6.0, 12.4 Hz, 1H), 3.91 (d, J=15.2 Hz, 1H), 3.86-3.78 (m, 1H), 3.71-3.60 (m, 2H), 3.28-3.16 (m, 3H), 3.05-2.97 (m, 1H), 2.36-2.23 (m, 1H), 2.23-2.17 (m, 1H), 2.16-2.08 (m, 1H), 2.03-1.94 (m, 2H), 1.93-1.83 (m, 1H), 1.25 (s, 3H); LCMS (ESI, M+1): m/z=470.1.

(1170) Step B. (S)-4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: 4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (7.38 g) was separated by SFC [column: DAICEL CHIRALPAK IG (250 mm×50 mm, 10 μm); mobile phase: [0.1% NH₃·H₂O in EtOH/CO₂]; 45%-45%, 8.7; 300 min]. The first peak was collected and concentrated in vacuum to afford the title compound (3.24 g, 43% yield) as a white solid; ¹H NMR (400 MHz, methanol-d.sub.4)

δ =9.33 (s, 1H), 5.39-5.21 (m, 1H), 4.52 (td, J=5.2, 14.4 Hz, 1H), 4.43 (d, J=14.8 Hz, 1H), 4.31-4.21 (m, 2H), 4.17-4.09 (m, 1H), 3.98 (ddd, J=4.0, 6.4, 12.8 Hz, 1H), 3.90 (d, J=14.8 Hz, 1H), 3.82 (ddd, J=4.0, 7.2, 14.4 Hz, 1H), 3.65 (q, J=12.8 Hz, 2H), 3.29-3.15 (m, 3H), 3.05-2.95 (m, 1H), 2.36-2.23 (m, 1H), 2.23-2.17 (m, 1H), 2.16-2.07 (m, 1H), 2.03-1.93 (m, 2H), 1.93-1.82 (m, 1H), 1.25 (s, 3H); SFC: >99% ee, Chiralpak IG-3 50×4.6 mm I.D., 3 μ m column; A: 60% EtOH+40% ACN (w/0.05% DEA), B: CO.sub.2, 3 mL/min, 220 nm, t.sub.R: 1.072 min; LCMS (ESI, M+1): m/z=470.1.

(1171) Step C. (S)-4-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To a mixture of (S)-4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (1.47 g, 1.0 equiv.), ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (1.92 g, 1.3 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 6.26 mL, 3.0 equiv.) in methoxycyclopentane (19 mL) was added CataCXium A Pd G3 (228 mg, 0.1 equiv.) under N.sub.2. The reaction was de-gassed and stirred at 90° C. for 3 hours under N.sub.2. The mixture was diluted with water (20 mL) and extracted with ethyl acetate (20 mL). The organic layer was concentrated in vacuum and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected and neutralized with NaHCO.sub.3 solid, concentrated in vacuum to remove acetonitrile, and extracted with ethyl acetate (2×80 mL). The combined organic layer was washed with brine (60 mL). The organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated in vacuum to afford the title compound (2.08 g, 77% yield) as a white solid; LCMS (ESI, M+1): m/z=780.5.

(1172) Step D. (S)-4-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of (S)-4-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (1.97 g, 1.0 equiv.) and CsF (5.75 g, 15 equiv.) in DMF (10 mL) was stirred at 15° C. for 1 hour. After completion, the mixture was filtered and the filtrate was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected and lyophilized to afford the title compound (1.28 g, 75% yield, formic acid salt) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) 5-9.58 (s, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.32-7.29 (m, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (dd, J=2.4, 18.4 Hz, 1H), 5.46 (br d, J=52.8 Hz, 1H), 4.62-4.50 (m, 4H), 4.24-4.14 (m, 1H), 4.06-3.82 (m, 3H), 3.80-3.54 (m, 5H), 3.30-3.23 (m, 1H), 2.63-2.27 (m, 4H), 2.26-2.01 (m, 4H), 1.27 (s, 3H), 0.80 (q, J=8.0 Hz, 3H); .sup.19F NMR (376 MHz, methanol-d.sub.4) δ =-121.041, -139.083, -173.810; SFC: >99% ee, Chiralpak IG-3 50×4.6 mm I.D., 3 μ m column, A: 60% IPA+40% ACN (w/0.05% DEA), B: CO.sub.2, 3 mL/min, 220 nm, t.sub.R: 2.337 min; LCMS (ESI, M+1): m/z=624.3.

Example 201

(1173) ##STR00481##

(S)-4-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1174) ##STR00482##

(1175) Step A. (S)-4-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyloxy)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To a mixture of (S)-4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (1.47 g, 1.0 equiv.), ((6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)ethynyl)naphthalen-2-yl)oxy)triisopropylsilane (2.54 g, 1.3 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 6.26 mL, 3.0 equiv.) in THF (19 mL) was added

CataCXium A Pd G3 (228 mg, 0.1 equiv.) under N.sub.2. The reaction was de-gassed and stirred at 60° C. for 3 hours under N.sub.2. The mixture was diluted with water (20 mL), extracted with ethyl acetate (20 mL), concentrated in vacuum, and purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected and concentrated in vacuum to afford the title compound (2.12 g, 70% yield) as a yellow solid; LCMS (ESI, M+1): m/z=932.6.

(1176) Step B. (S)-4-(7-(8-ethynyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of (S)-4-(8-fluoro-7-(7-fluoro-8-((triisopropylsilyl) ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (1.87 g, 1.0 equiv.) and CsF (4.57 g, 15 equiv.) in DMF (10 mL) was stirred at 15° C. for 7 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography (mobile phase: [water (0.1% formic acid)/acetonitrile]). The desired fractions were collected and lyophilized to afford the title compound (1.19 g, 89% yield, formic acid salt) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.56-9.43 (m, 1H), 7.86 (ddd, J=1.6, 5.6, 8.8 Hz, 1H), 7.37-7.29 (m, 2H), 7.23 (dd, J=2.4, 9.6 Hz, 1H), 5.56-5.39 (m, 1H), 4.69-4.51 (m, 4H), 4.29-4.17 (m, 1H), 4.05-3.85 (m, 3H), 3.81-3.59 (m, 5H), 3.45 (d, J=11.6 Hz, 1H), 3.30-3.25 (m, 1H), 2.64-2.40 (m, 2H), 2.36-2.28 (m, 1H), 2.26-2.16 (m, 2H), 2.13-2.01 (m, 1H), 1.28 (d, J=14.0 Hz, 3H); .sup.19F NMR (376 MHz, methanol-d.sub.4) δ=-111.749, -140.066, -173.900; SFC: >99% ee, Chiralpak IE-3 50×4.6 mm I.D., 3 μm; A: heptane (0.05% DEA); B: EtOH (0.05% DEA); Gradient: 25% B in A; 1 mL/min, 254 nm, t.sub.R: 4.443 min; LCMS (ESI, M+1): m/z=620.1.

Example 202

(1177) ##STR00483##

(3R)-1-[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-2-[[1-(pyrrolidin-1-ylmethyl)cyclopropyl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol

(1178) ##STR00484##

(1179) Step A. (3R)-[7-chloro-8-fluoro-2-[[1-(pyrrolidin-1-ylmethyl)cyclopropyl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: A mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (150 mg, 1 equiv.), [1-(pyrrolidin-1-ylmethyl)cyclopropyl]methanol (141 mg, 2.0 equiv.), DIPEA (178 mg, 3 equiv.) and 4 Å molecular sieves (50 mg) in dioxane (1.5 mL) was stirred at 95° C. for 12 hours under N.sub.2 atmosphere. The reaction mixture was filtered. The filter cake was washed with DCM (20 mL). The filtrate was concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 7:3] to afford the title compound (130 mg, 62% yield) as a light yellow gum; LCMS (ESI, M+1): m/z=450.2.

(1180) Step B. (3R)-1-[7-(8-ethyl-7-fluoro-3-triisopropylsilyloxy-1-naphthyl)-8-fluoro-2-[[1-(pyrrolidin-1-ylmethyl)cyclopropyl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: A mixture of (3R)-1-[7-chloro-8-fluoro-2-[[1-(pyrrolidin-1-ylmethyl)cyclopropyl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (130 mg, 1 equiv.), ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (165 mg, 1.2 equiv.), CataCXium A (22 mg, 0.1 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 0.6 mL, 3.1 equiv.) in methoxycyclopentane (2.4 mL) was degassed and purged with N.sub.2 for 3 times. The mixture was stirred at 90° C. for 2.5 hours. The reaction mixture was diluted with water (1 mL), extracted with ethyl acetate (3 mL×4), dried over anhydrous Na.sub.2SO.sub.4 and concentrated to afford the title compound (300 mg, crude) as a red gum; LCMS (ESI, M+1): m/z=760.5.

(1181) Step C. (3R)-1-[7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-2-[[1-(pyrrolidin-1-yl)methyl)cyclopropyl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol: A mixture of (3R)-1-[7-(8-ethyl-7-fluoro-3-triisopropylsilyloxy-1-naphthyl)-8-fluoro-2-[[1-(pyrrolidin-1-

ylmethyl)cyclopropyl]methoxy]pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (300 mg, crude) and CsF (300 mg) in DMF (2.5 mL) was stirred at 25° C. for 13 hours. The reaction mixture was filtered. The filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 13:7] and prep-HPLC [column: water s Xbridge 150×25 mm×Sum; mobile phase: [water (10 mM NH₄HCO₃)/CAN]; B %: 33%-66%, 9 minutes] to give a crude product. The crude product was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 um; mobile phase: [water (0.225% formic acid)/ACN]; B %: 20%-50%, 7 minutes] to afford the title compound (60.7 mg, 33% yield over two steps, 0.58 formic acid salt) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.23 (d, J=3.2 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.06 (s, 1H), 4.54 (br d, J=12.4 Hz, 1H), 4.48-4.42 (m, 2H), 4.29 (t, J=11.2 Hz, 1H), 3.61 (q, J=13.6 Hz, 1H), 3.45-3.43 (m, 1H), 3.17-3.08 (m, 6H), 2.45-2.42 (m, 1H), 2.19-2.17 (m, 2H), 1.98 (br s, 4H), 1.85-1.76 (m, 3H), 1.29 (d, J=9.2 Hz, 3H), 0.85-0.75 (m, 7H); LCMS (ESI, M+1): m/z=604.3.

Example 203

(1182) ##STR00485##

1-(1-(7-(7,8-difluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1183) ##STR00486##

(1184) Step A. 7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (300 mg, 1.0 equiv.), 2-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (359 mg, 1.5 equiv.) and K₃PO₄ (1.5 M in water, 1.37 mL, 3.0 equiv.) in THF (5 mL) was added CataCXium A Pd G3 (49.8 mg, 0.1 equiv.) under N₂. The reaction was de-gassed and stirred at 60° C. for 1 hour under N₂. The mixture was diluted with water (10 mL), extracted with ethyl acetate (2×8 mL), washed with brine (10 mL), dried over Na₂SO₄, concentrated, and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with NaHCO₃ solid, and extracted with ethyl acetate (2×25 mL). The combined organic phase was washed with brine (30 mL), dried over anhydrous Na₂SO₄ and concentrated in vacuum to afford the title compound (323 mg, 73% yield, 0.5 formic acid salt) as a yellow solid; ¹H NMR (400 MHz, chloroform-d) δ=9.29 (s, 1H), 7.59 (br dd, J=4.4, 9.2 Hz, 1H), 7.56-7.52 (m, 1H), 7.41-7.38 (m, 1H), 7.37-7.31 (m, 1H), 5.69-5.43 (m, 1H), 5.36-5.28 (m, 2H), 5.20-5.06 (m, 2H), 5.04-4.95 (m, 1H), 4.87-4.69 (m, 1H), 4.23-4.15 (m, 1H), 4.12-3.96 (m, 1H), 3.64-3.49 (m, 4H), 3.35-3.26 (m, 1H), 2.91-2.69 (m, 1H), 2.63-2.36 (m, 3H), 2.35-2.28 (m, 2H); LCMS (ESI, M+1): m/z=627.1.

(1185) Step B. 1-(7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: A mixture of piperidin-3-ylmethanesulfonamide (65.9 mg, 2.0 equiv.), DIEA (95.5 mg, 4.0 equiv.) and 4 Å molecular sieves (20 mg) in DMF (1.5 mL) was stirred at 15° C. for 0.5 hour. Then 7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (120 mg, 0.5 formic acid salt) was added into the above mixture and the resulting was stirred at 40° C. for 14 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with NaHCO₃ solid, concentrated in vacuum to remove acetonitrile, and extracted with ethyl acetate (2×15 mL). The combined organic phase was washed with brine (15 mL), dried over anhydrous Na₂SO₄ and concentrated in vacuum to afford the title compound (120 mg, 90% yield) as a yellow solid; LCMS (ESI, M+1, M/2+1): m/z=705.3, 353.2.

(1186) Step C. 1-(1-(7-(7,8-difluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a mixture of (1-(7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide (60.0 mg, 1.0 equiv.) in MeCN (0.5 mL) was added HCl.dioxane (4 M, 1.0 mL, 47 equiv.) at 0° C. and the mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated in vacuum, neutralized with saturated NaHCO₃ solution, extracted with ethyl acetate (2×8 mL), concentrated in vacuum and purified by prep-HPLC [column: PhenomenexSynergi C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 8%-38%, 10 min]. The desired fraction was collected and lyophilized to afford the title compound (30.4 mg, 51% yield, 0.6 formic acid salt) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.13 (s, 1H), 7.62 (br dd, J=4.4, 8.8 Hz, 1H), 7.44-7.36 (m, 1H), 7.34 (s, 1H), 7.25 (dd, J=2.0, 15.2 Hz, 1H), 5.48 (br d, J=52.8 Hz, 1H), 5.18-5.05 (m, 1H), 4.73-4.47 (m, 3H), 3.85-3.55 (m, 4H), 3.36 (br s, 1H), 3.29-3.10 (m, 3H), 2.68-2.36 (m, 4H), 2.25-2.01 (m, 4H), 1.96-1.87 (m, 1H), 1.83-1.54 (m, 2H); ¹⁹F NMR (376 MHz, methanol-d₄) δ=-141.320, -145.127, -146.874, -173.734; LCMS (ESI, M+1, M/2+1): m/z=661.3, 331.2.

Example 204

(1187) ##STR00487##

N-((2-aminopyridin-3-yl)methyl)-7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1188) ##STR00488##

(1189) Step A. N-((2-aminopyridin-3-yl methyl)-7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine: To a solution of 7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (60.0 mg, 1.0 equiv.), 3-(aminomethyl)pyridin-2-amine (27.3 mg, 2.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was added DIEA (71.7 mg, 5.0 equiv.). The reaction was stirred at 40° C. for 72 hours. The mixture was filtered and the filter cake was washed with DMF (1 mL). The filtrate was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 μm; mobile phase: [water (0.225% formic acid), ACN]; B %: 18%-28%, 7 min] to afford title compound (24.2 mg, 39% yield) as an off-white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.24 (s, 1H), 8.61-8.29 (m, 1H), 8.06 (d, J=7.2 Hz, 1H), 7.94-7.90 (m, 1H), 7.87 (d, J=7.6 Hz, 1H), 7.64 (dd, J=1.2, 7.2 Hz, 1H), 7.57 (t, J=7.6 Hz, 1H), 7.52-7.46 (m, 1H), 7.45-7.38 (m, 2H), 6.72 (dd, J=5.2, 7.2 Hz, 1H), 4.77 (s, 2H), 4.65 (s, 2H), 3.72-3.62 (m, 21H), 3.30-3.22 (m, 2H), 2.45-2.37 (m, 1H), 2.36-2.26 (m, 3H), 2.26-2.12 (m, 4H), 2.11-2.04 (m, 2H), 0.91 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=564.2.

Example 205

(1190) ##STR00489##

N-((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1191) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.15 (s, 1H), 8.23-8.17 (m, 2H), 8.06-8.00 (m, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.57-7.52 (m, 2H), 7.50-7.45 (m, 1H), 7.43-7.36 (m, 2H), 7.15-7.11 (m, 1H), 5.13-5.02 (m, 2H), 4.39 (s, 2H), 3.15 (br dd, J=5.6, 9.6 Hz, 2H), 2.84-2.73 (m, 2H), 2.44-2.23 (m, 2H), 2.10 (ddd, J=2.4, 6.4, 12.4 Hz, 2H), 2.00-1.87 (m, 4H), 1.84-1.76 (m, 2H), 0.90 (t, J=7.5 Hz, 3H); LCMS (ESI, M+1): m/z=588.4.

Example 206

(1192) ##STR00490##

(S)-7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2S,7aR)-2-hydroxyhexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(1193) ##STR00491## ##STR00492##

(1194) Step A. 7-(2-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 2,9-diazaspiro[4.5]decan-3-one (3.36 g, 1.0 equiv.) in DMF (50 mL) was added DIEA (8.45 g, 11.38 mL, 3.0 equiv.) and 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (5.5 g, 1.0 equiv.). The reaction was stirred at -40° C. for 3 hours. The mixture was diluted with water 50 mL and extracted with ethyl acetate 150 mL (50 mL×3). The combined organic layers were washed with brine 300 mL, dried over Na.sub.2SO.sub.4, concentrated, and purified by reversed phase flash chromatography (Silica gel, DCM/MeOH 10:1) to afford the title compound (7.00 g, 87% yield) as a yellow solid.

(1195) Step B. 7-(2-(((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of 7-(2,7-dichloro-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (3.00 g, 1.0 equiv.) and DIPEA (5.24 g, 7.06 mL, 5.0 equiv.) in dioxane (30 mL) was added ((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (3.21 g, 1.0 equiv.). The reaction was stirred at 100° C. for 6 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (5 mL×3). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, concentrated, and purified by prep-HPLC (mobile phase: [water (0.1% formic acid)/acetonitrile]; 5-70%) to afford the title compound (1.63 g, 15% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ=8.76 (d, J=4.8 Hz, 1H), 7.66-7.64 (m, 1H), 7.66-7.58 (m, 3H), 7.51-7.36 (m, 6H), 4.74-4.52 (m, 2H), 4.35 (dd, J=18.8, 10.9 Hz, 1H), 4.27-3.66 (m, 6H), 3.61-3.27 (m, 3H), 3.19-2.96 (m, 2H), 2.53-2.09 (m, 8H), 1.91-1.65 (m, 4H), 1.09 (s, 9H); LCMS (ESI, M+1): m/z=729.3.

(1196) Step C. (S)-7-(2-(((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: 7-(2-(((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (3.80 g) was purified by chiral prep-HPLC (column: DAICEL CHIRALCEL OD (250 mm×30 mm, 10 μm); mobile phase: [0.1% NH₃.3H₂O/MeOH]; B %: 50%-50%, 4.5; 180 min) to give the title compound (1.7 g, 32% yield, 98.9% ee) as a light-yellow solid. ¹H NMR (400 MHz, CDCl₃) δ=8.62 (s, 1H), 7.59-7.54 (m, 4H), 7.34-7.24 (m, 6H), 4.49-4.40 (m, 1H), 4.13-3.83 (m, 6H), 3.65-3.59 (m, 1H), 3.30 (d, J=7.0 Hz, 3H), 3.04 (s, 3H), 2.18-1.98 (m, 8H), 1.80-1.71 (m, 4H), 0.98 (s, 9H); LCMS (ESI, M+1): m/z=729.3.

(1197) Step D. (S)-7-(2-(((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: A mixture of 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (356 mg, 2.0 equiv.), (S)-7-(2-(((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (500 mg, 1.0 equiv.), CataCXium A Pd G3 (35.95 mg, 0.1 equiv.) and K₃PO₄ (1.5 M, 987 μL, 3.0 equiv.) in THF (1 mL) was degassed and stirred at 60° C. for 2 hours under nitrogen atmosphere. The mixture was diluted with water (5 mL) and extracted with ethyl acetate (5 mL×2). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, concentrated and purified by column chromatography (silica gel, DCM/MeOH 1:1) to afford (349 mg, 76% yield) as an orange oil. LCMS (ESI, M+1): m/z=927.8.

(1198) Step E. (S)-7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2S,7aR)-2-hydroxyhexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of (S)-7-(2-(((2S,7aR)-2-((tert-butyl diphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (100 mg, 1.0 equiv.) in DMF (1 mL) was added CsF (246 mg, 15 equiv.), the reaction was stirred at 40° C. for 16 hours. The mixture was diluted with ethyl acetate

(5 mL), washed with brine (15 mL), dried over Na₂SO₄, concentrated, and purified by column chromatography (Silica gel, Ethyl acetate/MeOH (0.1% NH₄OH) 0-100%) to afford the title compound (51 mg, 69% yield) as a yellow solid. ¹H NMR (400 MHz, CDCl₃) δ=9.03-8.90 (m, 1H), 7.67 (dd, J=8.8, 6.0, Hz, 1H), 7.51 (d, J=2.8 Hz, 1H), 7.25-7.17 (m, 2H), 5.31-5.26 (m, 2H), 4.68 (d, J=5.2 Hz, 1H), 4.49-4.34 (m, 1H), 4.22 (d, J=7.2 Hz, 1H), 4.09 (s, 1H), 3.89 (br s, 1H), 3.76 (t, J=14.4 Hz, 2H), 3.50 (s, 3H), 3.50-3.40 (m, 2H), 3.22 (d, J=9.6 Hz, 1H), 3.05-2.91 (m, 1H), 2.84-2.69 (m, 1H), 2.55-2.42 (m, 2H), 2.41-2.14 (m, 5H), 1.94-1.82 (m, 7H), 0.84 (dt, J=7.4, 2.8 Hz, 4H); LCMS (ESI, M+1): m/z=689.6.

(1199) Step F. (S)-7-(7-(8-ethyl-7-fluoro-3-hydroxy)naphthalen-1-yl)-8-fluoro-2-(((2S,7aR)-2-hydroxyhexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one: To a solution of (S)-7-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2S,7aR)-2-hydroxyhexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one (37.00 mg, 1.0 equiv.) in MeCN (0.4 mL) was added HCl/dioxane (4 M, 376 μL, 28 equiv.). The mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 14%-44%, 11.5 min) to afford the title compound (16.7 mg, 47% yield, 96.7% ee, formic acid salt) as a yellow solid. ¹H NMR (400 MHz, CD₃OD) δ=9.12 (s, 1H), 8.58-8.48 (m, 1H), 7.73-7.64 (m, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.05 (t, J=3.2 Hz, 1H), 4.66 (s, 1H), 4.56 (dd, J=11.8, 2.4 Hz, 1H), 4.51-4.37 (m, 2H), 4.36-4.20 (m, 1H), 4.12-3.95 (m, 1H), 3.83-3.59 (m, 3H), 3.57-3.43 (m, 1H), 3.29-3.19 (m, 3H), 2.51-2.09 (m, 10H), 1.94-1.83 (m, 4H), 0.85-0.75 (m, 3H); ¹⁹F NMR (377 MHz, CD₃OD) δ=-121.09, -139.15; LCMS (ESI, M+1): m/z=645.5.

Example 207

(1200) ##STR00493##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-4-(3-(pyridin-3-yl)propoxy)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

(1201) Step A: 8-fluoro-7-(8-fluoronaphthalen-1-yl)-4-(3-(pyridin-3-yl)propoxy)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine: To a mixture of 3-(pyridin-3-yl)propan-1-ol (1.5 equiv.) and 4 Å molecular sieves (20 mg) in THF (2 mL) was added LiHMDS (1 M, 141 μL, 1.5 equiv.) at 0° C. After addition, the mixture was stirred at 0° C. for 20 minutes, and then 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50 mg, 1 equiv.) was added at 0° C. The resulting mixture was stirred at 20° C. for 40 minutes. The reaction mixture was quenched with saturated aqueous NH₄Cl (3 mL) and extracted with ethyl acetate (3×5 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Shim-pack C18 150×25×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 31%-51%, 10 min) and lyophilized to afford the title compound. LCMS (ESI, M+1): m/z 568.4.

(1202) ##STR00494##

(1203) Example 208 to 223 were synthesized according to the procedure described for example 34.

(1204) TABLE-US-00002 TABLE 2 Mass Spectrum Data of Example 208 to 223

Example No.	Obs. M + 1	Example No.	Obs. M + 1
208	578.3	216	618.3
209	619.3	217	586.3
210	591.2	218	564.3
211	568.3	219	543.2
212	592.3	220	558.3
213	565.3	221	609.2
214	579.2	222	584.3
215	542.3	223	592.2

(1205) ##STR00495##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(imidazo[1,2-a]pyridin-6-ylmethyl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 209

(1206) ##STR00496##

2-(4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)

methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperazin-1-yl)nicotinonitrile

Example 210

(1207) ##STR00497##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(1-(pyridin-4-yl)-1H-pyrazol-4-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 211

(1208) ##STR00498##

N1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N.SUB.2.-(pyridin-3-yl)ethane-1,2-diamine

Example 212

(1209) ##STR00499##

N-(2-(1H-pyrrolo[2,3-b]pyridin-3-yl)ethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 213

(1210) ##STR00500##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-4-(3-(pyridin-4-yl)azetidin-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidine

Example 214

(1211) ##STR00501##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(pyrazolo[1,5-a]pyrimidin-3-ylmethyl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1212) ##STR00502##

N-(2-(1H-pyrazol-5-yl)ethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 216

(1213) ##STR00503##

2-(4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperazin-1-yl)benzonitrile

Example 217

(1214) ##STR00504##

8-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,8-triazaspiro[4.5]decan-2-one

Example 218

(1215) ##STR00505##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(1H-indazol-6-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 219

(1216) ##STR00506##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-(2-(isoxazol-4-yl)ethyl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 220

(1217) ##STR00507##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-oxa-6-azaspiro[3.5]nonane

Example 221

(1218) ##STR00508##

1-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

Example 222

(1219) ##STR00509##

N-(((6,7-dihydro-5H-pyrazolo[5,1-b][1,3]oxazin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 223

(1220) ##STR00510##

N-(((1H-pyrrolo[2,3-b]pyridin-3-yl)methyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-N-methyl-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

Example 224

(1221) ##STR00511##

8-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-oxa-1,8-diazaspiro[4.5]decan-2-one

(1222) The title compound was synthesized according to the procedure described for example 85. LCMS (ESI, M+1): m/z 587.3.

Example 225

(1223) ##STR00512##

3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)propanamide

(1224) The title compound was synthesized according to the procedure described for example 84. ¹H NMR (400 MHz, MeOD-d₄) δ 9.17 (s, 1H), 8.12 (br d, J=8.2 Hz, 1H), 7.86 (d, J=8.2 Hz, 1H), 7.70 (t, J=7.6 Hz, 1H), 7.59 (d, J=6.6 Hz, 1H), 7.57-7.50 (m, 1H), 7.19 (dd, J=7.6, 13.0 Hz, 1H), 4.66-4.54 (m, 2H), 4.07-3.89 (m, 2H), 3.60-3.48 (m, 2H), 3.19-3.07 (m, 2H), 2.71 (t, J=6.8 Hz, 2H), 2.33-2.23 (m, 2H), 2.21-2.06 (m, 4H), 2.05-1.95 (m, 2H); LCMS (ESI, M+1): m/z=519.1.

Example 226

(1225) ##STR00513##

3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)(methyl)amino)propanamide

(1226) The title compound was synthesized according to the procedure described for example 84. ¹H NMR (400 MHz, DMSO-d₆) δ 9.26 (s, 1H), 8.20 (s, 1H), 8.18 (d, J=8.6 Hz, 1H), 7.93 (d, J=8.2 Hz, 1H), 7.78-7.68 (m, 1H), 7.65-7.54 (m, 2H), 7.52-7.42 (m, 1H), 7.33-7.28 (m, 1H), 7.01-6.88 (m, 1H), 4.13 (s, 2H), 4.06-4.05 (d, J=6.2 Hz, 2H), 3.53 (s, 3H), 3.02-2.97 (m, 2H), 2.63-2.61 (m, 3H), 1.93-1.89 (dd, J=5.5, 11.5 Hz, 3H), 1.87-1.74 (m, 5H), 1.68-1.57 (m, 2H); LCMS (ESI, M+1): m/z=533.2.

Example 227

(1227) ##STR00514##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-oxa-1,7-diazaspiro[4.5]decan-2-one

(1228) ##STR00515##

(1229) Step A. 7-benzyl 1,3-di-tert-butyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-1,3,7-tricarboxylate: A mixture of benzyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate (3 g, 1 equiv.), (Boc).sub.20 (8.63 g, 4 equiv.), TEA (1.0 g, 1 equiv.) and DMAP (31 mg, 0.03 equiv.) in DME (100 mL) was stirred at 25° C. for 18 hours. The reaction mixture was concentrated to give a yellow solid. The solid was dispersed in isopropyl ether (20 mL) and the mixture was stirred for 10 minutes. The mixture was filtered and the solid was dried under reduced pressure to afford the title compound (4 g, crude) as a light-yellow solid; LCMS [ESI, M+H]: m/z=504.3.

(1230) Step B. 1-((benzyloxycarbonyl)-3-((tert-butoxycarbonyl)amino)piperidine-3-carboxylic acid: To a solution of 7-benzyl 1,3-di-tert-butyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-1,3,7-tricarboxylate (2 g, 1 equiv.) in THF (20 mL) was added LiOH.Math.H.sub.2O (1 M in water, 31.8 mL, 8 equiv.) at 25° C. The mixture was stirred at 25° C. for 18 hours. The pH of the reaction mixture was adjusted to 10 with 2N HCl while maintaining the reaction mixture temperature below 5° C. (Boc).sub.20 (4 mL) was added and the mixture was stirred at 25° C. for 18 hours. The pH of

the reaction mixture was adjusted to 3-4 with 2N HCl while maintaining the reaction mixture temperature below 5° C. The mixture was extracted with ethyl acetate (4×30 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 1:1] to afford the title compound (1.9 g, crude) as a brown gum; LCMS [ESI, M-55]: m/z=323.3.

(1231) Step C. benzyl 3-((tert-butoxycarbonyl)amino)-3-(hydroxymethyl)piperidine-1-carboxylate: To a solution of 1-((benzyloxy)carbonyl)-3-((tert-butoxycarbonyl)amino)piperidine-3-carboxylic acid (1.9 g, crude) in THF (30 mL) was added LiAlH.sub.4 (250 mg, 1.3 equiv.) at 0° C. under N.sub.2 atmosphere. The mixture was stirred at 20° C. for 15 hours. The reaction mixture was quenched with Na.sub.2SO.sub.4.10H.sub.2O (2 g) and saturated aqueous NaHCO.sub.3 (0.1 mL) at 0° C. Na.sub.2SO.sub.4 (3 g) was added and the mixture was stirred for 5 minutes. The mixture was filtered through a pad of Celite and the filter cake was washed with ethyl acetate/methanol 10:1 (20 mL) and methanol (100 mL). The filtrate was concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 1:4] to afford the title compound (0.69 g, crude) as a white gum; LCMS [ESI, M-55]: m/z=309.2.

(1232) Step D. benzyl 2-oxo-3-oxa-1,7-diazaspiro[4.5]decane-7-carboxylate: To a solution of benzyl 3-((tert-butoxycarbonyl)amino)-3-(hydroxymethyl)piperidine-1-carboxylate (0.69 g, crude) in DMF (10 mL) was added NaH (60% in mineral oil, 114 mg, 1.5 equiv.) in portions at 0° C. The mixture was stirred at 20° C. for 16 hours. The mixture was quenched with water (150 mL) at 0° C. and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 1:1] to afford the title compound (270 mg, crude) as a white gum; LCMS [ESI, M-55]: m/z=291.2.

(1233) Step E. 3-oxa-1,7-diazaspiro[4.5]decan-2-one: To a solution of benzyl 2-oxo-3-oxa-1,7-diazaspiro[4.5]decane-7-carboxylate (270 mg, 1 equiv.) in CF.sub.3COOH (9 mL) was added Pd(OH).sub.2 (40 mg, 10% purity) under N.sub.2 atmosphere. The mixture was degassed and purged with H.sub.2 for 3 times. The mixture was stirred at 20° C. for 2 hours under H.sub.2 atmosphere (15 psi). The reaction mixture was filtered through a pad of Celite. The filter cake was washed with THF/MeOH 1:1 (30 mL). The combined organic phase was concentrated under reduced pressure. The residue was washed with THF (5 mL×5) and the combined THF phase was concentrated to afford the title compound (145 mg, 36% yield over five steps) as a colorless oil.

(1234) Step F. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-oxa-1,7-diazaspiro[4.5]decan-2-one: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy) pyrido[4,3-d]pyrimidine (200 mg, 1 equiv.), 3-oxa-1,7-diazaspiro[4.5]decan-2-one (145 mg, 2.5 equiv.), DIPEA (171 mg, 3.5 equiv.) and 4 Å molecular sieves (40 mg) in DMF (2 mL) was stirred at 40° C. for 18 hours under N.sub.2 atmosphere. The reaction mixture was filtered. The filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 7:3] and then prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN]; B %: 13%-43%, 10 minutes] to afford the title compound (82.6 mg, 36% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d6) δ=9.09 (d, J=1.6 Hz, 1H), 8.41 (br s, 1H), 8.18 (br d, J=8.4 Hz, 1H), 7.94 (d, J=8.4 Hz, 1H), 7.74-7.62 (m, 1H), 7.59-7.58 (m, 2H), 7.31 (dd, J=7.2, 12.8 Hz, 1H), 4.29-4.05 (m, 6H), 3.69-3.66 (m, 2H), 2.94-2.89 (m, 2H), 2.59-2.51 (m, 2H), 1.89-1.77 (m, 10H), 1.61-1.58 (m, 2H); LCMS [ESI, M+H]: m/z=587.4.

Example 228

(1235) ##STR00516##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1236) The title compound was synthesized according to the procedure described for example 84.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.29 (s, 1H), 8.12 (br d, J=7.6 Hz, 1H), 7.85 (br d, J=8.0 Hz, 1H), 7.70 (br t, J=7.6 Hz, 1H), 7.61 (br d, J=7.2 Hz, 1H), 7.57-7.48 (m, 1H), 7.18 (br dd, J=7.6, 12.8 Hz, 1H), 4.64 (br d, J=13.2 Hz, 2H), 4.56 (s, 2H), 4.35 (br d, J=3.6 Hz, 2H), 3.78-3.76 (m, 2H), 3.60-3.47 (m, 2H), 3.20-3.07 (m, 2H), 2.26 (td, J=6.4, 12.4 Hz, 2H), 2.11 (ddd, J=6.4, 12.4, 18.8 Hz, 4H), 2.01 (br dd, J=6.4, 12.8 Hz, 2H) LCMS [ESI, M+1]: m/z=571.2

Example 229

(1237) ##STR00517##

N-(1-(1H-pyrrolo[2,3-b]pyridin-3-yl)ethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1238) The title compound was synthesized according to the procedure described for example 84.

.sup.1H NMR (400 MHz, methanol-d₄) δ 9.29 (d, J=3.2 Hz, 1H), 8.23-8.09 (m, 3H), 7.86 (d, J=8.2 Hz, 1H), 7.70 (td, J=4.4, 7.6 Hz, 1H), 7.62-7.49 (m, 2H), 7.23-7.07 (m, 2H), 6.10-6.00 (m, 11H), 4.64-4.51 (m, 2H), 3.64-3.52 (m, 2H), 3.24-3.14 (m, 2H), 2.29-1.95 (m, 8H), 1.86 (dd, J=3.2, 6.8 Hz, 3H); .sup.19F NMR (376 MHz, methanol-d₄) δ =-115.427, -141.83; LCMS [ESI, M+1, M/2+1]: m/z=592.3, 296.7.

Example 230

(1239) ##STR00518##

4-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-pyrazol-3(2H)-one

(1240) ##STR00519##

(1241) Step A. Ethyl 3-hydroxy-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazole-4-carboxylate: A mixture of ethyl 3-oxo-2,3-dihydro-1H-pyrazole-4-carboxylate (2 g, 1 equiv.) in DCM (20 mL), DHP (3.31 g, 3 equiv.), TosOH (220 mg, 0.1 equiv.) was stirred at 25° C. for 2 hours under N.sub.2 atmosphere. The mixture was diluted with water (50 mL) and extracted with ethyl acetate (2×50 mL). The combined organic layers were washed with brine (50 mL), dried over sodium sulfate, concentrated and purified by silica gel column chromatography [petroleum ether/ethyl acetate] to afford the title compound (2.5 g, 80% yield) as a yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =10.42 (s, 1H), 8.10 (s, 1H), 5.20 (dd, J=2.4, 9.6 Hz, 1H), 4.15 (q, J=7.2 Hz, 2H), 3.92 (m, 1H), 3.65-3.50 (m, 1H), 1.99 (br s, 1H), 1.93-1.79 (m, 2H), 1.60 (m, 1H), 1.55-1.45 (m, 2H), 1.23 (t, J=7.2 Hz, 3H); LCMS (ESI, M+Na): m/z=263.0.

(1242) Step B. Ethyl 3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazole-4-carboxylate: To a mixture of ethyl 3-hydroxy-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazole-4-carboxylate (800 mg, 1 equiv.), K.sub.2CO.sub.3 (920 mg, 2 equiv.) in DMF (12 mL) was added PMBCl (626 mg, 1.2 equiv.). The mixture was stirred at 60° C. for 16 hours under N.sub.2 atmosphere. The mixture was diluted with water (20 mL) and extracted with ethyl acetate (2×20 mL). The combined organic layers were washed with brine (20 mL), dried over sodium sulfate, concentrated and purified by silica gel column chromatography [petroleum ether/ethyl acetate] to afford the title compound (1.0 g, 75.6% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =8.25 (s, 1H), 7.40 (d, J=8.8 Hz, 2H), 6.95 (d, J=8.8 Hz, 2H), 5.28 (dd, J=2.4, 10.0 Hz, 1H), 5.16 (s, 2H), 4.15 (q, J=7.2 Hz, 2H), 3.97-3.89 (m, 1H), 3.76 (s, 3H), 3.64-3.56 (m, 1H), 2.13-2.02 (m, 1H), 1.96-1.83 (m, 2H), 1.70-1.57 (m, 1H), 1.56-1.48 (m, 2H), 1.25-1.19 (m, 3H); LCMS (ESI, M+1): m/z=361.1.

(1243) Step C. (3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methanol: To a solution of ethyl 3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazole-4-carboxylate (1 g, 1 equiv.) in THF (20 mL) was added LiAlH.sub.4 (316 mg, 3 equiv.) at 0° C. and the mixture was stirred at 25° C. for 30 minutes. The mixture was quenched with saturated aqueous NH.sub.4Cl (10 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4 and concentrated to afford the title compound (0.7 g, 69.4% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =7.60 (s, 1H), 7.38 (d, J=8.8 Hz, 2H), 6.93 (d, J=8.8 Hz, 2H), 5.18-5.12 (m, 1H), 5.08 (s,

2H), 4.68 (s, 1H), 4.19 (d, J=5.2 Hz, 2H), 3.88 (br s, 1H), 3.75 (s, 3H), 3.61-3.53 (m, 1H), 2.09-2.00 (m, 1H), 1.91 (br d, J=1.6 Hz, 1H), 1.86-1.77 (m, 1H), 1.70-1.56 (m, 1H), 1.54-1.47 (m, 2H); LCMS (ESI, M+1): m/z=319.2.

(1244) Step D. 2-((3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methyl)isoindoline-1,3-dione: A solution of DIAD (381 mg, 1.5 equiv.), PPh.sub.3 (494 mg, 1.5 equiv.) in THF (8 mL) was stirred at 0° C. for 30 minutes. A solution of (3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methanol (400 mg, 1 equiv.) and isoindoline-1,3-dione (277 mg, 1.5 equiv.) in THF (8 mL) was added at 0° C. and the mixture was stirred at 25° C. for 1 hour. The filtrate was diluted with water (15 mL) and extracted with ethyl acetate (2×15 mL). The combined organic layers were washed with brine (15 mL), dried over sodium sulfate, concentrated and purified by silica gel column chromatography [petroleum ether/ethyl acetate] to afford the title compound (150 mg, 22.1% yield) as a colorless oil; LCMS (ESI, M+Na): m/z=447.9.

(1245) Step E. (3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methanamine: A solution of 2-((3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methyl)isoindoline-1,3-dione (50 mg, 1 equiv.) and N.sub.2H.sub.4.Math.H.sub.2O (237 mg, 98% purity, 41 equiv.) in EtOH (1 mL) was stirred at 40° C. for 16 hours. The mixture was concentrated and triturated with ethyl acetate (10 mL) for 10 minutes to afford the title compound (30 mg, 82.1% yield) as a colorless oil; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=8.92 (br s, 2H), 7.55 (s, 1H), 7.37 (d, J=8.4 Hz, 2H), 6.93 (d, J=8.4 Hz, 2H), 5.15-5.11 (m, 1H), 5.07 (s, 2H), 3.87 (br d, J=1.6 Hz, 1H), 3.75 (s, 3H), 3.61-3.52 (m, 1H), 3.40 (s, 2H), 2.07-1.92 (m, 2H), 1.86-1.79 (m, 1H), 1.69-1.58 (m, 1H), 1.54-1.45 (m, 2H); LCMS (ESI, M+1): m/z=318.2.

(1246) Step F. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-((3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methyl)pyrido[4,3-d]pyrimidin-4-amine: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (40 mg, 1 equiv.), DIEA (39.0 mg, 4 equiv.), (3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methanamine (23.9 mg, 1 equiv.) and 4 Å molecular sieves (50 mg) in DMF (0.5 mL) was stirred at 40° C. for 16 hours under N.sub.2 atmosphere. The mixture was filtered and purified by reverse phase flash chromatography to afford the title compound (25 mg, 35.1% yield) as a yellow oil; LCMS [ESI, M+1]: m/z=748.3.

(1247) Step G. 4-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-pyrazol-3(2H)-one: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-((3-((4-methoxybenzyl)oxy)-1-(tetrahydro-2H-pyran-2-yl)-1H-pyrazol-4-yl)methyl)pyrido[4,3-d]pyrimidin-4-amine (20 mg, 1 equiv.) in DCM (1 mL) was added TFA (770 mg, 252 equiv.) at 0° C. and the mixture was stirred at 20° C. for 16 hours under N.sub.2 atmosphere. The mixture was concentrated and purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 14%-44%, 10 minutes] to afford the title compound (10.5 mg, 63.6% yield) as a white solid. .sup.1HNMR (400 MHz, METHANOL-d.sub.4) δ=9.19 (s, 1H), 8.11 (br d, J=8.4 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.69 (t, J=7.6 Hz, 1H), 7.59 (dd, J=1.2, 7.2 Hz, 1H), 7.56-7.47 (m, 2H), 7.18 (dd, J=7.6, 13.2 Hz, 1H), 4.71 (s, 2H), 4.56 (s, 2H), 3.75 (m, 2H), 3.31-3.26 (m, 2H), 2.40-2.31 (m, 2H), 2.29-2.16 (m, 4H), 2.11 (m, 2H); LCMS (ESI, M+1): m/z=544.2.

Example 231

(1248) ##STR00520##

5-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1H-pyrazol-3(2H)-one

(1249) The title compound was synthesized according to the procedure described for example 85. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=11.78-11.35 (m, 1H), 9.79-9.16 (m, 3H), 8.17 (br d,

J=8.0 Hz, 1H), 7.93 (d, J=8.0 Hz, 1H), 7.76-7.70 (m, 1H), 7.63-7.55 (m, 2H), 7.30 (dd, J=7.6, 13.2 Hz, 1H), 5.65-5.23 (m, 1H), 4.70 (br s, 2H), 4.06 (s, 2H), 2.95-2.89 (m, 2H), 2.56-2.52 (m, 2H), 1.92-1.84 (m, 2H), 1.82-1.71 (m, 4H), 1.60-1.52 (m, 2H); ^{sup}.19F NMR (400 MHz, DMSO-d.sub.6) δ-113.230, -141.701; LCMS (ESI, M+1): m/z=544.3.

Example 232

(1250) ##STR00521##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-azaspiro[3.5]nonan-6-yl)pyrido[4,3-d]pyrimidine

(1251) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1HNMR (400 MHz, METHANOL-d.sub.4) δ=9.10 (s, 1H), 8.12 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.74-7.67 (m, 1H), 7.62 (d, J=7.2 Hz, 1H), 7.57-7.50 (m, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 4.56 (s, 2H), 4.13-3.93 (m, 4H), 3.51 (br s, 2H), 3.18-3.06 (m, 2H), 2.30-2.21 (m, 2H), 2.20-2.05 (m, 4H), 2.04-1.93 (m, 4H), 1.90-1.76 (m, 8H) LCMS [ESI, M+1]: m/z=556.4

Example 233

(1252) ##STR00522##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-1-ol

(1253) The title compound was synthesized according to the procedure described for example 85. ^{sup}.1H NMR (400 MHz, methanol-d₄) δ 9.13-9.02 (m, 1H), 8.11 (br d, J=8.4 Hz, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.73-7.67 (m, 1H), 7.61 (br d, J=7.2 Hz, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.19 (br dd, J=7.6, 13.2 Hz, 1H), 4.49-4.37 (m, 0.5H), 4.35-4.27 (m, 2H), 4.21-3.80 (m, 4.5H), 3.16-3.06 (m, 2H), 2.79-2.66 (m, 2H), 2.31-2.21 (m, 1H), 2.13-1.82 (m, 10H), 1.81-1.59 (m, 4H), 1.53-1.31 (m, 1H); LCMS [ESI, M+1]: 572.3.

Example 234

(1254) ##STR00523##

8-fluoro-4-(2-fluoro-6-azaspiro[3.5]nonan-6-yl)-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(1255) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.05 (d, J=8.4 Hz, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.71 (t, J=7.6 Hz, 1H), 7.68 (d, J=8.0 Hz, 1H), 7.54-7.49 (m, 11H), 7.25-7.21 (m, 1H), 5.26-5.00 (m, 1H), 4.34 (d, J=16.0 Hz, 2H), 4.11-3.95 (m, 4H), 3.40-3.12 (m, 2H), 2.85-2.70 (m, 2H), 2.50-2.35 (m, 1H), 2.27-2.22 (m, 1H), 2.18-2.05 (m, 3H), 2.04-1.75 (m, 11H); LCMS (ESI, M+1): m/z=574.4.

Example 235

(1256) ##STR00524##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1257) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1H NMR (400 MHz, METHANOL-d.sub.4) δ 9.09 (d, J=6.8 Hz, 1H), 8.54 (s, 1H), 8.12 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.74-7.67 (m, 1H), 7.62 (d, J=6.8 Hz, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 12.8 Hz, 1H), 4.54 (d, J=8.0 Hz, 2H), 4.40-4.20 (m, 1H), 4.17-3.92 (m, 4H), 3.60-3.42 (m, 2H), 3.21-2.98 (m, 2H), 2.38-2.19 (m, 4H), 2.10 (tdd, J=6.8, 13.2, 19.6 Hz, 4H), 2.03-1.94 (m, 2H), 1.88-1.77 (m, 5H), 1.77-1.66 (m, 1H); LCMS [ESI, M+1]: m/z=572.2.

Example 236

(1258) ##STR00525##

3-(((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidine-2,5-dione

(1259) The title compound was synthesized according to the procedure described for example 85. ^{sup}.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.19 (s, 1H), 8.11 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.76 (t, J=7.6 Hz, 1H), 7.60 (d, J=7.2 Hz, 1H), 7.54-7.52 (m, 1H), 7.21-7.18 (m,

1H), 4.61-4.59 (m, 2H), 4.24-4.12 (m, 1H), 4.05-3.90 (m, 1H), 3.60-3.40 (m, 3H), 3.20-3.08 (m, 2H), 3.00-2.88 (m, 1H), 2.80-2.62 (m, 1H), 2.40-2.27 (m, 2H), 2.21-2.07 (m, 4H), 2.04-1.90 (m, 2H); LCMS (ESI, M+1): m/z=559.2.

Example 237

(1260) ##STR00526##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-methylpiperidine-3-sulfonamide

(1261) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1H NMR (400 MHz, methanol-d₄) δ=9.13 (s, 1H), 8.12 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.70-7.68 (m, 1H), 7.63-7.60 (m, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.21-7.18 (m, 1H), 4.95 (br d, J=12.4 Hz, 1H), 4.56 (s, 3H), 3.69-3.68 (m, 3H), 3.47-3.45 (m, 2H), 3.09-3.06 (m, 2H), 2.79 (d, J=2.4 Hz, 3H), 2.24-2.22 (m, 3H), 2.12-1.97 (m, 8H), 1.86-1.75 (m, 1H). LCMS [ESI, M+1]: m/z=609.3.

Example 238

(1262) ##STR00527##

N-(2-(3-amino-1H-1,2,4-triazol-5-yl)ethyl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1263) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.16 (s, 1H), 8.53 (br s, 1H), 8.12 (br d, J=8.8 Hz, 1H), 7.86 (d, J=7.6 Hz, 1H), 7.70 (dd, J=7.2, 8.4 Hz, 1H), 7.61-7.49 (m, 2H), 7.19 (ddd, J=1.2, 7.6, 13.2 Hz, 1H), 4.63 (d, J=1.2 Hz, 2H), 4.01 (dt, J=2.8, 6.8 Hz, 2H), 3.71-3.60 (m, 2H), 3.27-3.19 (m, 2H), 3.00 (t, J=6.8 Hz, 2H), 2.33 (br dd, J=6.4, 12.4 Hz, 2H), 2.25-2.13 (m, 4H), 2.09 (br dd, J=6.4, 12.4 Hz, 2H) LCMS [ESI, M+1]: m/z=558.4

Example 239

(1264) ##STR00528##

2-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)isothiazolidine 1,1-dioxide

(1265) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.13 (d, J=3.2 Hz, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.70 (dt, J=8.0, 3.6 Hz, 1H), 7.66-7.57 (m, 1H), 7.53 (dt, J=8.0, 4.8 Hz, 1H), 7.19 (ddd, J=12.4 Hz, 7.6, 4.8, 1H), 4.93 (d, J=10.8 Hz, 1H), 4.69-4.56 (m, 2H), 4.47 (dd, J=11.6, 6.6 Hz, 1H), 3.88-3.77 (m, 1H), 3.69-3.58 (m, 1H), 3.53-3.37 (m, 5H), 3.28-3.17 (m, 2H), 3.06-2.97 (m, 2H), 2.39 (quin, J=7.0 Hz, 2H), 2.31-2.22 (m, 2H), 2.13-2.03 (m, 5H), 2.00 (d, J=11.6 Hz, 2H), 1.97-1.87 (m, 2H), 1.84-1.70 (m, 1H); ^{sup}.19F NMR (377 MHz, CD₃OD) δ=-115.32, -141.27; LCMS: [ESI, M+1].^{sup}+: 635.5.

Example 240

(1266) ##STR00529##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-(1,2,3,5,6,7-hexahydropyrrolizin-8-ylmethoxy)-4-[3-(sulfamoylamino)piperidin-1-yl]pyrido[4,3-d]pyrimidine

(1267) The title compound was synthesized according to the procedure described for example 84. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.18 (s, 1H), 8.56 (s, 1H), 8.14 (br d, J=8.4 Hz, 1H), 7.88 (d, J=8.4 Hz, 1H), 7.78-7.69 (m, 1H), 7.63 (dd, J=7.2, 10.2 Hz, 1H), 7.55 (dt, J=5.2, 8.0 Hz, 1H), 7.21 (ddd, J=2.8, 7.6, 12.8 Hz, 1H), 4.78 (br d, J=16.4 Hz, 1H), 4.68-4.54 (m, 2H), 4.53-4.40 (m, 1H), 3.80-3.63 (m, 2H), 3.61-3.45 (m, 3H), 3.19-3.04 (m, 2H), 2.35-2.23 (m, 2H), 2.22-2.05 (m, 5H), 2.05-1.93 (m, 3H), 1.84-1.70 (m, 2H); LCMS (ESI, M+1): m/z=610.4.

Example 241

(1268) ##STR00530##

1-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)urea

(1269) The title compound was synthesized according to the procedure described for example 84.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.18 (d, J=1.6 Hz, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.73-7.67 (m, 1H), 7.65-7.59 (m, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 12.8 Hz, 1H), 4.72-4.64 (m, 1H), 4.63-4.56 (m, 1H), 4.55-4.43 (m, 1H), 4.42-4.29 (m, 1H), 3.96-3.74 (m, 2H), 3.73-3.54 (m, 3H), 3.25-3.15 (m, 2H), 2.37-2.26 (m, 2H), 2.24-2.11 (m, 4H), 2.11-1.93 (m, 4H), 1.89-1.66 (m, 2H); LCMS [ESI, M+1]: m/z=574.4.

Example 242

(1270) ##STR00531##

N-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)formamide

(1271) ##STR00532##

(1272) Step A. Benzyl 3-formamidopiperidine-1-carboxylate: A mixture of benzyl 3-oxopiperidine-1-carboxylate (500 mg, 1.0 equiv.), formamide (2.49 g, 25.7 equiv.), ammonium formate (3.30 g, 24.4 equiv.) and formic acid (2.44 g, 2 mL, 98% purity, 24.2 equiv.) was stirred at 160° C. for 7 hours. The reaction mixture was diluted with water (20.0 mL), and then was extracted with ethyl acetate (30 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The crude product was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (198 mg, 34% yield) as a yellow oil; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =8.12 (br d, J=6.4 Hz, 1H), 7.96 (s, 1H), 7.47-7.25 (m, 5H), 5.16-4.99 (m, 2H), 3.85-3.45 (m, 3H), 3.16-2.74 (m, 2H), 1.87-1.60 (m, 2H), 1.52-1.30 (m, 2H); LCMS (ESI, M+1): m/z=263.0.

(1273) Step B. N-(piperidin-3-yl)formamide: To a solution of benzyl 3-formamidopiperidine-1-carboxylate (140 mg, 1.0 equiv.) in MeOH (10 mL) was added NH.sub.3.MeOH (1 mL, 20% purity) and Pd/C (10.0 mg, 10% purity) under N.sub.2. The suspension was degassed in vacuum and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (15 psi) at 15° C. for 1 hour. The mixture was filtered and the filtrate was concentrated in vacuum to afford the title compound (77.0 mg, crude) as a colorless solid.

(1274) Step C. N-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)formamide: A mixture of N-(piperidin-3-yl)formamide (72.5 mg, 2.0 equiv.), DIEA (110 mg, 3.0 equiv.) and 4 Å molecular sieves (100 mg) in DMF (5 mL) was stirred at 15° C. for 0.5 hour. Then to the mixture was added 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (150 mg, 1.0 equiv.) at 15° C. The mixture was stirred at 40° C. for 14 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (15 mL). The organic layer was washed with water (2×10 mL), dried over sodium sulfate, filtered and concentrated in vacuum. The residue was purified by prep-HPLC (column: Phenomenex Luna C18 150×25 mm×10 μ m; mobile phase: water (0.225% formic acid, v/v)/ACN]B %: 16%-46%, 10 min) and lyophilized to afford the title compound (66.2 mg, 38% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =9.16 (d, J=3.2 Hz, 1H), 8.35 (br d, J=7.6 Hz, 1H), 8.26 (s, 1H), 8.18 (br d, J=8.4 Hz, 1H), 8.02 (s, 1H), 7.93 (d, J=8.0 Hz, 1H), 7.74 (dt, J=2.4, 8.0 Hz, 1H), 7.68-7.55 (m, 2H), 7.36-7.25 (m, 1H), 4.32-4.13 (m, 4H), 4.09-3.99 (m, 1H), 3.68-3.44 (m, 2H), 3.08-2.97 (m, 2H), 2.72-2.59 (m, 2H), 2.02-1.89 (m, 4H), 1.88-1.73 (m, 5H), 1.71-1.59 (m, 3H); LCMS (ESI, M+1): m/z=559.3.

Example 243

(1275) ##STR00533##

1-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-4-yl)urea

(1276) The title compound was synthesized according to the procedure described for example 84. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.09 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.75-7.67 (m, 1H), 7.61 (d, J=6.8 Hz, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 4.70-4.53 (m, 4H), 3.99-3.85 (m, 1H), 3.68-3.51 (m, 4H), 3.18 (td, J=6.0, 11.6

Hz, 2H), 2.37-2.22 (m, 2H), 2.20-1.96 (m, 8H), 1.68 (q, J=11.2 Hz, 2H); LCMS [ESI, M+1]: m/z=574.4.

Example 244

(1277) ##STR00534##

N-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-4-yl)formamide

(1278) The title compound was synthesized according to the procedure described for example 84.

¹H NMR (400 MHz, DMSO-d₆) δ=9.09 (s, 1H), 8.27-8.16 (m, 3H), 8.02 (s, 1H), 7.94 (d, J=8.4 Hz, 1H), 7.78-7.70 (m, 1H), 7.66-7.55 (m, 2H), 7.31 (dd, J=7.6, 13.2 Hz, 1H), 4.53-4.39 (m, 2H), 4.20 (s, 2H), 4.13-4.00 (m, 1H), 3.57 (br t, J=11.2 Hz, 2H), 3.18-3.02 (m, 2H), 2.77-2.64 (m, 2H), 2.04-1.93 (m, 4H), 1.92-1.77 (m, 4H), 1.72-1.55 (m, 4H); LCMS (ESI, M+1): m/z=559.3.

Example 245

(1279) ##STR00535##

4-(7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(1280) The title compound was synthesized according to the procedure described for example 84.

¹H NMR (400 MHz, CD₃OD-d₆) δ=9.19 (s, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.73-7.67 (m, 1H), 7.61 (d, J=7.2 Hz, 1H), 7.53 (dt, J=8.0, 5.2 Hz, 1H), 7.42 (d, J=2.0 Hz, 1H), 7.19 (dd, J=13.2, 8.0 Hz, 1H), 6.48 (d, J=1.6 Hz, 1H), 5.34-5.18 (m, 2H), 4.55-4.49 (m, 2H), 4.42 (t, J=5.2 Hz, 2H), 4.32 (s, 2H), 3.21-3.13 (m, 2H), 2.85-2.74 (m, 2H), 2.45-2.37 (m, 2H), 2.08 (dd, J=12.4, 6.8 Hz, 2H), 2.01-1.87 (m, 4H), 1.84-1.75 (m, 2H); ¹⁹F NMR (377 MHz, CD₃OD) δ=-115.18, -140.71; LCMS (ESI, M+1): m/z=568.3.

Example 246

(1281) ##STR00536##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6,7,8,9-tetrahydro-5H-pyrido[3,2-c]azepine

(1282) The title compound was synthesized according to the procedure described for example 84.

¹H NMR (400 MHz, CD₃OD) δ=9.14 (s, 1H), 8.36 (dd, J=4.8, 1.2 Hz, 1H), 8.11 (br d, J=8.4 Hz, 1H), 7.99 (d, J=6.8 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.73-7.64 (m, 1H), 7.64-7.57 (m, 1H), 7.53 (dt, J=7.6, 4.8 Hz, 1H), 7.32 (dd, J=7.6, 4.8 Hz, 1H), 7.19 (dd, J=12.8, 7.6 Hz, 1H), 5.34-5.14 (m, 2H), 4.49-4.37 (m, 2H), 4.33-4.22 (m, 2H), 3.29-3.20 (m, 4H), 2.93-2.84 (m, 2H), 2.41-2.31 (m, 2H), 2.15-1.81 (m, 8H); ¹⁹F NMR (377 MHz, CD₃OD) δ=-115.23, -140.94; LCMS (ESI, M+1): m/z=579.2.

Example 247

(1283) ##STR00537##

8-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,8-triazaspiro[5.5]undecane 2,2-dioxide

(1284) ##STR00538##

(1285) Step A. Tert-butyl 3-(cyanomethylene)piperidine-1-carboxylate: To a solution of 2-diethoxyphosphorylacetonitrile (10.7 g, 1.20 equiv.) in 2-MeTHF (45 mL) was added K₂CO₃ (8.32 g, 1.20 equiv.). The mixture was stirred at 20° C. for 1 hours, then tert-butyl 3-oxopiperidine-1-carboxylate (10.0 g, 50.2 mmol, 1.0 equiv.) was added to the mixture. The mixture was stirred at 70° C. for 13 hours. After completion, the reaction mixture was diluted with water (40 mL) and extracted with ethyl acetate (20 mL×3). The combined organic layers were washed with brine (20.00 mL×2), dried over Na₂SO₄, concentrated and purified by silica gel column chromatography [Petroleum ether/Ethyl acetate 5:1 to 4:1] to afford the title compound (9.0 g, 81% yield) as a yellow oil; ¹H NMR (400 MHz, CDCl₃) δ=5.24-5.15 (m, 1H), 4.25-3.97 (m, 2H), 3.51-3.47 (m, 2H), 2.64-2.38 (m, 2H), 1.76-1.71 (m, 2H), 1.48-1.44 (m, 9H). (1286) Step B. tert-butyl 3-amino-3-(cyanomethyl)piperidine-1-carboxylate: The mixture of tert-butyl 3-(cyanomethylene)piperidine-1-carboxylate (4.0 g, 1.0 equiv.) and NH₄OH (20.0 g,

21.98 mL, 30% purity, 9.51 equiv.) in MeOH (6.00 mL) was stirred at 90° C. for 4 hours in a sealed tube. After completion, the mixture was concentrated to remove MeOH and then the residue was extracted with ethyl acetate (120 mL×3). The combined organic phase was washed with brine (100 mL×3), dried with anhydrous Na₂SO₄, concentrated and purified by silica gel column chromatography [petroleum ether/ethyl acetate 1:0 to 1:10 then dichloromethane/methyl alcohol 10:1] to afford the title compound (550 mg, 13% yield) as a yellow oil. ¹H NMR (400 MHz, CD₃OD) δ=3.56-3.52 (m, 2H), 3.32-3.00 (m, 2H), 2.65-2.54 (m, 2H), 1.80-1.71 (m, 2H), 1.66-1.53 (m, 2H), 1.49 (s, 9H).

(1287) Step C. tert-butyl 3-amino-3-(2-aminoethyl)piperidine-1-carboxylate: A mixture of tert-butyl 3-amino-3-(cyanomethyl)piperidine-1-carboxylate (200 mg, 1.0 equiv.), NH₄OH (97.6 mg, 107 μL, 30% purity, 1.0 equiv.) and Raney Nickel (7.16 mg, 0.10 equiv.) in MeOH (5 mL) was stirred at 35° C. for 12 hours under H₂ (50 psi). After completion, the mixture was filtered and concentrated to afford the title compound (160 mg, 79% yield) as a yellow gum; ¹H NMR (400 MHz, CD₃OD) δ=3.32-3.21 (m, 2H), 3.20-2.95 (m, 2H), 2.69-2.66 (m, 2H), 1.39-1.36 (m, 15H).

(1288) Step D. tert-butyl 2-thia-1,3,8-triazaspiro[5.5]undecane-8-carboxylate 2,2-dioxide: A mixture of tert-butyl 3-amino-3-(2-aminoethyl)piperidine-1-carboxylate (160 mg, 1.00 equiv.) and sulfamide (94.8 mg, 1.50 equiv.) in pyridine (0.30 mL) was stirred at 110° C. for 12 hours under N₂ atmosphere. After completion, the mixture was concentrated and purified by prep-HPLC [column: Phenomenex luna CIS 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 22%-52%, 10 min] to afford the title compound (80.0 mg, 40% yield) as a yellow gum; ¹H NMR (400 MHz, CD₃OD) δ=4.22-4.12 (m, 1H), 3.62-3.56 (m, 2H), 3.48-3.42 (m, 1H), 3.16-3.12 (m, 2H), 1.95-1.41 (m, 15H).

(1289) Step E. 2-thia-1,3,8-triazaspiro[5.5]undecane 2,2-dioxide: A mixture of tert-butyl 2-thia-1,3,8-triazaspiro[5.5]undecane-8-carboxylate 2,2-dioxide (20.0 mg, 1.00 equiv.) and TFA (231 mg, 30.9 equiv.) in DCM (0.5 mL) was stirred at 20° C. for 1 hour. After reaction completion, the mixture was concentrated to remove DCM to afford the title compound (20.0 mg, crude, TFA salt) as a yellow oil.

(1290) Step E. 8-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,8-triazaspiro[5.5]undecane 2,2-dioxide: A mixture of 2-thia-1,3,8-triazaspiro[5.5]undecane 2,2-dioxide (20.0 mg, 1.00 equiv., TFA), 8-fluoro-7-(8-fluoro-1-naphthyl)-2-(1,2,3,5,6,7-hexahydropyrrolizin-8-ylmethoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (33.2 mg, 1.00 equiv.), DIPEA (32.4 mg, 4.00 equiv.) and 4 Å molecular sieves (10.0 mg, 1.00 equiv.) in DMF (1.00 mL) was stirred at 40° C. for 12 hours under N₂. After reaction completion, the mixture was concentrated and purified by prep-HPLC [column: Phenomenex Luna C18 150×25 mm×10 μm; mobile phase: [water (0.225% formic acid)/ACN]; B %: 13%-43%, 10 min] to afford the title compound (2.50 mg, 5.7% yield) as an off-white solid; ¹H NMR (400 MHz, CD₃OD) δ=9.15 (d, J=2.8 Hz, 1H), 8.14 (d, J=8.8 Hz, 1H), 7.88 (d, J=8.0 Hz, 1H), 7.73-7.69 (m, 1H), 7.64 (t, J=7.2 Hz, 1H), 7.57-7.55 (m, 1H), 7.22-7.20 (m, 1H), 4.66-4.53 (m, 3H), 4.42-4.20 (m, 2H), 4.01-3.81 (m, 1H), 3.75-3.45 (m, 4H), 3.23-3.13 (m, 2H), 2.37-2.02 (m, 10H), 1.90-1.72 (m, 3H), 1.61-1.50 (m, 1H); LCMS (ESI, M+1): m/z=636.0.

Example 248

(1291) ##STR00539##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl sulfamate

(1292) ##STR00540##

(1293) Step A. benzyl 3-(sulfamoyloxy)piperidine-1-carboxylate: To a solution of benzyl 3-hydroxypiperidine-1-carboxylate (500 mg, 1.0 equiv.) and sulfamoyl chloride (442 mg, 1.8 equiv.) in DCM (10 mL) was added TEA (1.08 g, 1.48 mL, 5.0 equiv.), the mixture was stirred at 15° C. for 8 hours. After completion, the reaction mixture was diluted with water (10 mL) and extracted

with DCM (20 mL). The organic layer was dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Waters Xbridge C18 150×50 mm×10 µm; mobile phase: [water (10 mM NH.sub.4HCO.sub.3)/ACN], B %: 24%-54%, 11 min) to afford the title compound (230 mg, 33% yield) as a light yellow oil; LCMS [ESI, M+1]: m/z=315.2.

(1294) Step B. piperidin-3-yl sulfamate: To a mixture of benzyl 3-(sulfamoyloxy)piperidine-1-carboxylate (100 mg, 1.0 equiv.) in MeOH (2 mL) was added Pd/C (100 mg, 10% purity), the reaction was stirred at 15° C. for 2 hours under H.sub.2 atmosphere (15 psi). After completion, the reaction mixture was filtered and concentrated under reduced pressure to afford the title compound (59 mg, crude) as a light yellow solid.

(1295) Step C. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl sulfamate: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 1.0 equiv.), piperidin-3-yl sulfamate (25.5 mg, 1.5 equiv.) in DMF (1 mL) was added DIEA (36.5 mg, 49.2 µL, 3.0 equiv.), the reaction was stirred at 40° C. for 4 hours. After completion, the reaction mixture was filtered to give a residue and the residue was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) and then prep-HPLC (column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: water (0.225% formic acid)/ACN], B %: 13%-43%, 10 min) and lyophilized to afford the title compound (3.85 mg, 6% yield over two steps) as a white solid; .sup.1H NMR (400 MHz, methanol-d4): δ=9.15 (s, 1H), 8.53 (s, 1H), 8.13 (br d, J=8.8 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.74-7.68 (m, 1H), 7.64-7.59 (m, 1H), 7.57-7.50 (m, 1H), 7.24-7.15 (m, 1H), 4.62 (br s, 2H), 4.39-3.99 (m, 3H), 3.52-3.37 (m, 4H), 3.16-3.00 (m, 2H), 2.31-2.17 (m, 3H), 2.10 (br dd, J=6.0, 12.4 Hz, 8H), 1.87-1.74 (m, 1H); LCMS [ESI, M+1]: m/z=611.3.

Example 249

(1296) ##STR00541##

1-(4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)morpholin-2-yl)methanesulfonamide

(1297) ##STR00542##

(1298) Step A. benzyl 2-(hydroxymethyl)morpholine-4-carboxylate: A mixture of morpholin-2-ylmethanol (1 g, 1 equiv., HCl salt) and NaHCO.sub.3 (1.37 g, 2.5 equiv.) in water (10 mL) and THF (10 mL) was stirred at 20° C. for 0.1 hour. Benzyl carbonochloridate (1.20 g, 1.1 equiv.) was added dropwise and the mixture was stirred at 20° C. for 16 hours. The mixture was diluted with water (30 mL) and extracted with ethyl acetate (10 mL×4). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 3:1] to afford the title compound (1.18 g, 71% yield) as a yellow oil; .sup.1H NMR (400 MHz, methanol-d4) δ=7.37-7.31 (m, 5H), 5.14 (s, 2H), 4.02 (br d, J=13.2 Hz, 1H), 3.91 (br d, J=12.8 Hz, 2H), 3.56-3.43 (m, 4H), 3.01 (br s, 1H), 2.78 (br s, 1H); LCMS [ESI, M+Na]: m/z=274.2.

(1299) Step B. benzyl 2-(((methylsulfonyl)oxy)methyl)morpholine-4-carboxylate: To a solution of benzyl 2-(hydroxymethyl)morpholine-4-carboxylate (1.18 g, 1 equiv) and TEA (1.02 g, 2.1 equiv.) in DCM (15 mL) was added MsCl (1.03 g, 1.9 equiv.) dropwise at 0° C. The mixture was stirred at 10° C. for 1 hour. The mixture was quenched with saturated aqueous NaHCO.sub.3 (20 mL) at 0° C. The DCM phase was separated. The aqueous phase was extracted with DCM (20 mL). The combined organic layers were washed with saturated aqueous NH.sub.4Cl (20 mL), brine (20 mL), dried over Na.sub.2SO.sub.4 and concentrated to afford the title compound (1.65 g, crude) as a light yellow oil.

(1300) Step C. benzyl 2-((acetylthio)methyl)morpholine-4-carboxylate: A mixture of benzyl 2-(((methylsulfonyl)oxy)methyl)morpholine-4-carboxylate (1.65 g, crude) and acetylsulfanylpotassium (660 mg, 1.2 equiv.) in DMF (20 mL) was stirred at 95° C. for 1.5 hours.

The mixture was diluted with ethyl acetate (30 mL), water (300 mL) and brine (30 mL). The mixture was extracted with ethyl acetate (4×20 mL). The organic layer was washed with brine (3×20 mL), dried over Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 1:9] to afford the title compound (1.15 g, 79% yield over two steps) as a light-yellow liquid; LCMS (EST, M+Na): m/z=332.1.

(1301) Step D. benzyl 2-((chlorosulfonyl)methyl)morpholine-4-carboxylate: To a solution of benzyl 2-((acetylthio)methyl)morpholine-4-carboxylate (1.15 g, 1 equiv.) in AcOH (18 mL) and water (2 mL) was added NCS (1.62 g, 3.3 equiv.) portion wise. The mixture was poured into water (100 mL). The mixture was extracted with ethyl acetate (4×20 mL). The combined organic layers were washed with water (50 mL), saturated aqueous NaHCO.sub.3 (3×30 mL), brine (20 mL), dried over anhydrous Na.sub.2SO.sub.4 and concentrated to afford the title compound (1.4 g, crude) as a light-yellow oil.

(1302) Step E. benzyl 2-(sulfonylmethyl)morpholine-4-carboxylate: Benzyl 2-((chlorosulfonyl)methyl)morpholine-4-carboxylate (1.4 g, crude) in THF (15 mL) was added into NH.sub.3—H.sub.2O (20 mL, 25% purity) dropwise maintaining the temperature below 5° C. The mixture was stirred between 5 and 15° C. for 0.5 hour. The mixture was diluted with water (150 mL) and extracted with DCM (4×15 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 8:17] to afford the title compound (1 g, 85% yield over two steps) as a white solid; .sup.1H NMR (400 MHz, chloroform-d) δ=7.40-7.32 (m, 5H), 5.15 (br s, 2H), 4.96 (br s, 2H), 4.14-3.94 (m, 4H), 3.63 (br t, J=10.8 Hz, 1H), 3.31-3.25 (m, 2H), 3.05 (br s, 1H), 2.83 (br s, 1H); LCMS [ESI, M+Na]: m/z=315.1.

(1303) Step F. morpholin-2-ylmethanesulfonamide: To a solution of benzyl 2-(sulfonylmethyl)morpholine-4-carboxylate (300 mg, 1 equiv.) in MeOH (10 mL) was added Pd/C (60 mg, 10% purity) under N.sub.2 atmosphere. The mixture was degassed and purged with H.sub.2. The mixture was stirred at 10° C. for 2.5 hours under H.sub.2 atmosphere (15 psi) before being filtered through a pad of Celite. The filter cake was washed with THF/MeOH 1:1 (20 mL). The combined filtrate was concentrated to afford the title compound (180 mg, 100% yield, crude) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=6.79 (br s, 2H), 3.80-3.71 (m, 1H), 3.70 (td, J=2.4, 11.2 Hz, 1H), 3.45 (dt, J=3.2, 10.8 Hz, 1H), 3.34 (br s, 1H), 3.10-3.06 (m, 2H), 2.90 (dd, J=2.0, 12.0 Hz, 1H), 2.64-2.62 (m, 2H), 2.41 (br dd, J=10.0, 12.0 Hz, 1H).

(1304) Step G. 1-(4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)morpholin-2-yl)methanesulfonamide: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (80 mg, 1 equiv.), morpholin-2-ylmethanesulfonamide (55 mg, crude), DIPEA (58.6 mg, 3.0 equiv.) and 4 Å molecular sieves (20 mg) in DMF (1 mL) was stirred at 40° C. for 14 hours under N.sub.2 atmosphere. The mixture was filtered. The filtrate was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 12%-42%, 7 minutes] to afford the title compound (29.7 mg, 30% yield, 0.5 formic acid salt) as a white solid; .sup.1H NMR (400 MHz, methanol-d4) δ=9.16 (s, 1H), 8.12 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.71-7.64 (m, 1H), 7.61 (dd, J=7.2, 14.8 Hz, 1H), 7.53 (dt, J=5.6, 7.6 Hz, 1H), 7.19 (br dd, J=7.6, 13.2 Hz, 1H), 5.02 (br t, J=14.0 Hz, 1H), 4.63 (br dd, J=6.4, 12.0 Hz, 1H), 4.55-4.46 (m, 2H), 4.31-4.18 (m, 1H), 4.05 (br d, J=10.8 Hz, 1H), 3.82-3.73 (m, 2H), 3.54-3.43 (m, 5H), 3.17-3.03 (m, 2H), 2.29 (td, J=6.4, 12.8 Hz, 2H), 2.14-2.10 (m, 4H), 2.04-1.99 (m, 2H); LCMS [ESI, M+1]: m/z=611.4.

Example 250

(1305) ##STR00543##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidine

(1306) ##STR00544##

(1307) Step A. 2,7-dichloro-8-fluoro-4-(3-(methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (215 mg, 1.0 equiv.), 3-methylsulfonylazepane (200 mg, 1.1 equiv., HCl) in THF (1 mL) was added DIEA (550 mg, 5.0 equiv.) at -40°C . The mixture was stirred at -40°C for 2 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc ($3\times 20\text{ mL}$). The combined organic layers were dried over anhydrous sodium sulfate, concentrated to afford the title compound (500 mg, crude) as a yellow solid; LCMS (ESI, M+1): $m/z=393.3$.

(1308) Step B. 7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2,7-dichloro-8-fluoro-4-(3-methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidine (470 mg, 1.0 equiv.), 1,2,3,5,6,7-hexahydropyrrolizin-8-ylmethanol (337 mg, 2.0 equiv.) and 4 Å molecular sieves (200 mg) in dioxane (2.5 mL) was added DIEA (463 mg, 3.0 equiv.). The reaction was stirred at 95°C for 16 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc ($3\times 20\text{ mL}$). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (180 mg, 29.9% yield) as a yellow solid; LCMS (ESI, M+1): $m/z=498.1$.

(1309) Step C. 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-(1,2,3,5,6,7-hexahydropyrrolizin-8-ylmethoxy)-4-(3-methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), Cs.sub.2CO.sub.3 (1.5 M, 402 μL , 3.0 equiv.) and cataCXium® A Pd G3 (14.6 mg, 0.1 equiv.) in methoxycyclopentane (1.5 mL) was added 2-(8-fluoro-1-naphthyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (71.0 mg, 1.3 equiv.). The reaction was stirred at 90°C for 3 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc ($3\times 20\text{ mL}$). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to give a crude product. The crude product was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75 $\times$ 30 mm $\times$ 3 μm ; mobile phase: [water (0.225% formic acid)/ACN]; B %: 18%-48%, 7 min] and lyophilized to afford the title compound (21.4 mg, 17% yield) as a light yellow solid; ^1H NMR (400 MHz, methanol- d_4) $\delta=9.22$ (s, 1H), 8.11 (d, $J=8.4\text{ Hz}$, 1H), 7.85 (d, $J=8.0\text{ Hz}$, 1H), 7.75-7.65 (m, 1H), 7.65-7.49 (m, 2H), 7.25-7.10 (m, 1H), 4.93 (br d, $J=13.6\text{ Hz}$, 1H), 4.57-4.44 (m, 2H), 4.33-4.21 (m, 1H), 4.20-4.05 (m, 1H), 3.91-3.71 (m, 2H), 3.36-3.31 (m, 1H), 3.30-3.26 (m, 1H), 3.15 (d, $J=2.0\text{ Hz}$, 3H), 2.99-2.85 (m, 2H), 2.40-2.28 (m, 1H), 2.27-2.14 (m, 3H), 2.14-2.02 (m, 3H), 2.02-1.92 (m, 3H), 1.91-1.78 (m, 3H), 1.52-1.36 (m, 1H); LCMS (ESI, M+1): $m/z=608.0$.

Example 251

(1310) ##STR00545##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azocan-1-yl)pyrido[4,3-d]pyrimidine

(1311) The title compound was synthesized according to the procedure described for example 250. ^1H NMR (400 MHz, methanol- d_4) $\delta=9.25$ (d, $J=0.8\text{ Hz}$, 1H), 8.12 (br d, $J=8.4\text{ Hz}$, 1H), 7.86 (d, $J=8.0\text{ Hz}$, 1H), 7.75-7.67 (m, 1H), 7.61 (dd, $J=4.4, 6.8\text{ Hz}$, 1H), 7.53 (dt, $J=5.2, 8.0\text{ Hz}$, 1H), 7.25-7.12 (m, 1H), 5.12 (br d, $J=12.4\text{ Hz}$, 1H), 4.64-4.51 (m, 3H), 3.98-3.79 (m, 3H), 3.47-3.36 (m, 2H), 3.12 (s, 3H), 3.07-2.97 (m, 2H), 2.52-2.38 (m, 1H), 2.27-2.18 (m, 2H), 2.16-2.08 (m, 3H), 2.07-2.00 (m, 4H), 2.00-1.88 (m, 3H), 1.88-1.69 (m, 2H), 1.59-1.45 (m, 1H); LCMS (ESI, M+1): $m/z=622.2$.

Example 252

(1312) ##STR00546##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(1313) ##STR00547##

(1314) Step A. benzyl 3-amino-3-cyano-piperidine-1-carboxylate: A mixture of benzyl 3-

oxopiperidine-1-carboxylate (10.0 g, 1.0 equiv.) and NH₄Cl (9.17 g, 4.0 equiv.) in isopropyl alcohol (60 mL) and NH₄OH (120 mL) was added KCN (10.1 g, 3.61 equiv.) in one portion at 25° C. under N₂. The mixture was stirred at 25° C. for 12 hours. The reaction mixture was poured into water (150 mL), and the aqueous phase was extracted with ethyl acetate (3×50 mL). The combined organic phase was washed with brine (100 mL), and dried with anhydrous Na₂SO₄, filtered and concentrated in vacuum. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 20:1 to 0:1) to afford the title compound (10.0 g, 89% yield) as a yellow oil; LCMS (ESI, 2M+1): m/z=519.2.

(1315) Step B. benzyl 3-amino-3-(aminomethyl)piperidine-1-carboxylate: A mixture of benzyl 3-amino-3-cyano-piperidine-1-carboxylate (200 mg, 1.0 equiv.), NH₃·MeOH (1.00 mL, 20% purity, 1.0 equiv.) in MeOH (5 mL) and Raney Ni (30.0 mg) was stirred at 25° C. for 5 hours under H₂ atmosphere (15 psi). The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)/ACN], B %: 9%-39%, 10 min] and lyophilized to afford the title compound (100 mg, 45% yield) as a colorless oil; ¹H NMR (400 MHz, chloroform-d₄) δ=7.38-7.30 (m, 5H), 5.23-5.03 (m, 2H), 3.45 (br s, 3H), 3.19 (br d, J=13.2 Hz, 1H), 2.78-2.63 (m, 1H), 2.49 (d, J=13.2 Hz, 1H), 1.66-1.40 (m, 4H).

(1316) Step C. benzyl 2-thia-1,3,7-triazaspiro[4.5]decane-7-carboxylate 2,2-dioxide: To a refluxing solution of sulfamide (109 mg, 10.0 equiv.) in Pyridine (2 mL) was added benzyl 3-amino-3-(aminomethyl)piperidine-1-carboxylate (30.0 mg, 1.0 equiv.). The resulting mixture was stirred at 120° C. for further 12 hours under nitrogen. The reaction mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 μm; mobile phase: water (0.225% formic acid)/ACN], B %: 22%-52%, 7 min] and lyophilized to afford the title compound (15.0 mg, 40% yield) as a yellow solid; ¹H NMR (400 MHz, chloroform-d₄) δ=7.35 (s, 5H), 5.62 (br d, J=0.8 Hz, 1H), 5.13 (br s, 2H), 5.03 (br s, 1H), 3.65 (br d, J=12.0 Hz, 1H), 3.58-3.31 (m, 4H), 3.22-3.11 (m, 11H), 1.95-1.84 (m, 1H), 1.81-1.67 (m, 2H), 1.54 (br dd, J=2.8, 7.2 Hz, 1H).

(1317) Step D. 2λ,6-thia-1,3,9-triazaspiro[4.5]decane 2,2-dioxide: To a mixture of benzyl 2,2-dioxo-2λ,6-thia-1,3,9-triazaspiro[4.5]decane-9-carboxylate (15.0 mg, 1.0 equiv.) in MeOH (2 mL) was added Pd/C (3.00 mg, 10% purity) and then the mixture was stirred at 25° C. for 1 hour under H₂ atmosphere (15 psi). The mixture was filtered and concentrated under reduced pressure to afford the title compound (17.0 mg, crude) as a yellow solid.

(1318) Step E. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (15.0 mg, 1.0 equiv.) in DMF (1.0 mL) was added DIEA (11.0 mg, 3.0 equiv.) and 2λ,6-thia-1,3,9-triazaspiro[4.5]decane 2,2-dioxide (16.2 mg, 3.0 equiv.). The mixture was stirred at 45° C. for 12 hours. The reaction mixture was filtered and concentrated under reduced pressure. The residue was purified by prep-HPLC [column: Phenomenex C18 75×30 mm×3 μm; mobile phase: [water (0.1% formic acid)/ACN] B %: 15%-45%, 7 min] and lyophilized to afford the title compound (4.74 mg, 26% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.14 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.76-7.66 (m, 1H), 7.61 (t, J=5.2 Hz, 1H), 7.54 (dt, J=4.8, 7.6 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 4.68-4.58 (m, 2H), 4.55-4.33 (m, 2H), 3.90-3.72 (m, 2H), 3.56-3.39 (m, 3H), 3.28-3.19 (m, 1H), 3.16-3.04 (m, 2H), 2.26 (td, J=6.4, 12.4 Hz, 2H), 2.17-1.89 (m, 10H); LCMS (ESI, M+1): m/z=622.3.

Example 253

(1319) ##STR00548##

4-(3-(1H-1,2,4-triazol-1-yl)azetidin-1-yl)-8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(1320) The title compound was synthesized according to the procedure described for example 84.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =8.96 (s, 1H), 8.28 (s, 1H), 8.10 (d, J=8.4 Hz, 1H), 7.84 (t, J=4.0 Hz, 2H), 7.72-7.66 (m, 1H), 7.60 (dd, J=0.8, 7.2 Hz, 1H), 7.52 (dt, J=5.2, 8.0 Hz, 1H), 7.18 (dd, J=7.6, 13.2 Hz, 1H), 5.90-5.78 (m, 1H), 5.55-4.94 (m, 4H), 4.43 (s, 2H), 3.37-3.32 (m, 2H), 2.97-2.86 (m, 2H), 2.20-2.11 (m, 2H), 2.09-1.90 (m, 4H), 1.90-1.81 (m, 2H);
.sup.19F NMR (377 MHz, METHANOL-d.sub.4) δ =-115.14, -140.68; LCMS (ESI, M+1): m/z=555.3.

Example 254

(1321) ##STR00549##

8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-methyl-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)pyrido[4,3-d]pyrimidine

(1322) The title compound was synthesized according to the procedure described for example 84.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.25 (s, 1H), 8.13 (br d, J=8.0 Hz, 1H), 7.87 (d, J=8.0 Hz, 1H), 7.71 (t, J=8.0 Hz, 1H), 7.62 (dd, J=0.8, 7.2 Hz, 1H), 7.58-7.50 (m, 1H), 7.20 (dd, J=7.2, 13.2 Hz, 1H), 6.26 (s, 1H), 5.29-5.14 (m, 2H), 4.71-4.61 (m, 2H), 4.49-4.37 (m, 4H), 3.76-3.65 (m, 2H), 3.37-3.32 (m, 1H), 3.30-3.26 (m, 1H), 2.46-2.37 (m, 2H), 2.36-2.29 (m, 2H), 2.28-2.07 (m, 9H); LCMS (ESI, M+1): m/z=582.3.

Example 255

(1323) ##STR00550##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-7-ol

(1324) The title compound was synthesized according to the procedure described for example 85.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.42 (d, J=16.0 Hz, 1H), 8.11 (d, J=8.0 Hz, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.70 (t, J=7.2 Hz, 1H), 7.64-7.58 (m, 1H), 7.57-7.49 (m, 1H), 7.43 (d, J=1.6 Hz, 1H), 7.22-7.15 (m, 1H), 6.60-6.50 (m, 1H), 5.60-5.45 (m, 1H), 5.14-5.00 (m, 1H), 4.67-4.55 (m, 2H), 4.54-4.44 (m, 1H), 4.43-4.27 (m, 4H), 3.20-3.09 (m, 21H), 2.82-2.73 (m, 2H), 2.15-2.04 (m, 2H), 2.02-1.85 (m, 4H), 1.84-1.74 (m, 2H); LCMS (ESI, M+1): m/z=584.3.

Example 256

(1325) ##STR00551##

1-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoro-7-(8-fluoronaphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1326) ##STR00552##

(1327) Step A. 1-(7-chloro-2-((1-((dimethylamino)methyl)cyclopropyl)methoxyl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 1 equiv.), (1-((dimethylamino)methyl)cyclopropyl)methanol (78 mg, 1 equiv.), DIPEA (195 mg, 2.5 equiv.) and 4 Å molecular sieves (40 mg) in dioxane (2.5 mL) was stirred at 90° C. for 14 hours under N.sub.2 atmosphere. The reaction mixture was filtered. The filter cake was washed with DCM (20 mL). The combined organic phase was concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile 3:1] to afford the title compound (65 mg, 24% yield) as a light yellow solid; LCMS (ESI, M+1): m/z=424.3.

(1328) Step B. 1-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoro-7-(8-fluoronaphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 1-(7-chloro-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (65 mg, 1 equiv.), 2-(8-fluoronaphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (50 mg, 184 μ mol, 1.2 equiv.), cataCXium® A Pd G3 (12 mg, 0.1 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 307 μ L, 3 equiv.) in methoxycyclopentane (1.5 mL) was degassed and stirred at 90° C. for 3 hours. The mixture was diluted with water (1 mL) and extracted with ethyl acetate (4×2 mL). The organic layer was dried over anhydrous Na.sub.2SO.sub.4 and concentrated to give a residue. The residue was purified by reversed phase flash chromatography

[water (0.1% formic acid)/acetonitrile 13:7] and then prep-HPLC [column: waters Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH₄HCO₃)/ACN]; B %: 43%-73%, 9 minutes] to afford the title compound (47.5 mg, 57% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.20 (d, J=5.2 Hz, 1H), 8.11 (d, J=8.4 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.70 (t, J=7.6 Hz, 1H), 7.62-7.52 (m, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=8.0, 12.8 Hz, 1H), 4.51 (br dd, J=4.4, 9.6 Hz, 1H), 4.39-4.36 (m, 2H), 4.26 (br d, J=13.2 Hz, 1H), 3.63 (dd, J=2.0, 13.2 Hz, 1H), 3.49-3.46 (m, 1H), 2.46-2.44 (m, 2H), 2.30 (s, 6H), 2.18-2.14 (m, 1H), 1.85-1.75 (m, 3H), 1.29 (d, J=5.6 Hz, 3H), 0.75-0.72 (m, 2H), 0.54-0.52 (m, 2H); ¹⁹F NMR (400 MHz, methanol-d₄) δ=-115, -141; LCMS (ESI, M+1): 534.4.

Example 257

(1329) ##STR00553##

1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1330) ##STR00554##

(1331) Step A. 1-(7-chloro-8-fluoro-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 1.0 equiv.), (1-(pyrrolidin-1-ylmethyl)cyclopropyl)methanol (140 mg, 1.5 equiv.), 4 Å molecular sieves (10.0 mg) in dioxane (1.5 mL) was added DIEA (234 mg, 315 µL, 3.0 equiv.). The reaction was stirred at 95° C. for 12 hours. After completion, the reaction mixture was diluted with water (1 mL) and extracted with Ethyl acetate (2 mL). The combined organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed-phase flash chromatography (water (0.1% formic acid)/ACN) to afford the title compound (93.0 mg, 33% yield) as a yellow oil; LCMS [ESI, M+1]: m/z=450.1.

(1332) Step B. 1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(7-chloro-8-fluoro-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50.0 mg, 1.0 equiv.), 2-(8-fluoronaphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (36.3 mg, 1.2 equiv.), K₃PO₄ (1.5 M, 222 µL, 3.0 equiv.) in Methoxycyclopentane (1 mL) was added CataCXium A Pd G3 (8.09 mg, 11.1 µmol, 0.1 equiv.) under N₂. The reaction was stirred at 90° C. for 2 hours. After completion, the reaction mixture was diluted with water (1 mL) and extracted with Ethyl acetate (2 mL). The combined organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure to give a residue. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)/ACN) and prep-HPLC (column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)/ACN] B %: 12%-42%, 10 min) and lyophilized to afford the title compound (19.9 mg, 32% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d₄): δ 9.23 (d, J=8.0 Hz, 1H), 8.54 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.73-7.68 (m, 1H), 7.61 (dt, J=1.2, 8.0 Hz, 1H), 7.56-7.51 (i, 1H), 7.20 (ddd, J=1.6, 7.6, 13.2 Hz, 1H), 4.58-4.46 (m, 2H), 4.43-4.37 (m, 1H), 4.30 (br d, J=13.2 Hz, 1H), 3.62 (dd, J=2.0, 13.2 Hz, 1H), 3.43 (br t, J=11.6 Hz, 1H), 3.30-3.05 (m, 6H), 2.22-2.10 (m, 1H), 2.00 (br s, 4H), 1.88-1.74 (m, 3H), 1.29 (d, J=4.4 Hz, 3H), 0.90-0.75 (m, 4H); LCMS [ESI, M+1]: m/z=560.2.

Example 258

(1333) ##STR00555##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1334) ##STR00556##

(1335) Step A. tert-butyl-2-carbamoyl-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-(6H)-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (1.00 g, 1.0 equiv.), TEA (2.16 g, 6.0 equiv.) and HATU (2.70 g,

2.0 equiv.) in THF (10 mL) was added NH₄Cl (570 mg, 3.0 equiv.). The mixture was stirred at 25° C. for 12 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (800 g, 80% yield) as a yellow solid; LCMS (ESI, M+1): m/z=281.2.

(1336) Step B. 5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of tert-butyl 2-carbamoyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (800 mg, 1.0 equiv.) in MeCN (8 mL) was added HCl.dioxane (4 M, 16 mL, 22 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The mixture was concentrated to afford the title compound (800 mg, crude, HCl) as a yellow solid; LCMS (ESI, M+1): m/z=180.1.

(1337) Step C. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), 5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (408 mg, 10 equiv., HCl) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was added DIEA (487 mg, 20 equiv.). The reaction was stirred at 40° C. for 24 hours. The residue was filtered and purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: water (0.225% formic acid)/ACN] B %: 15%-45%, 7 minutes] to afford the title compound (15.2 mg, 13% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.27-9.16 (m, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.74-7.67 (m, 1H), 7.61 (dd, J=1.2, 7.2 Hz, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.24-7.14 (m, 1H), 6.88 (s, 1H), 5.35-5.16 (m, 2H), 4.58-4.45 (m, 6H), 3.56-3.43 (m, 2H), 3.17-3.04 (m, 2H), 2.57-2.35 (m, 2H), 2.28-2.19 (m, 2H), 2.18-2.03 (m, 4H), 2.02-1.92 (m, 2H); LCMS (ESI, M+1): m/z=611.2.

Example 259

(1338) ##STR00557##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1339) ##STR00558##

(1340) Step A. tert-butyl-2-(methylcarbamoyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (1.00 g, 1.0 equiv.), TEA (2.16 g, 6.0 equiv.) and HATU (2.70 g, 2.0 equiv.) in THF (10 mL) was added methylamine (720 mg, 3.0 equiv., HCl). The reaction was stirred at 25° C. for 3 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (900 g, 83% yield) as a yellow solid; LCMS (ESI, M+1): m/z=295.0.

(1341) Step B. N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of tert-butyl-2-(methylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (900 mg, 1.0 equiv.) in MeCN (9 mL) was added HCl.dioxane (4 M, 18 mL, 23 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The mixture was concentrated to afford the title compound (900 mg, crude) as a yellow solid; LCMS (ESI, M+): m/z=194.2.

(1342) Step C. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (435 mg, 10 equiv., HCl) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was added DIEA (487 mg, 20 equiv.). The reaction was stirred at 40° C. for 12 hours. The residue was filtered and

purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: water (0.225% formic acid)/ACN] B %: 15%-45%, 7 minutes] to afford the title compound (60.1 mg, 50% yield) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.22 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.70 (t, J=7.6 Hz, 1H), 7.61 (d, J=7.2 Hz, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 6.84 (s, 1H), 5.37-5.13 (m, 2H), 4.56 (s, 2H), 4.54-4.42 (m, 4H), 3.65-3.54 (m, 2H), 3.20 (td, J=6.0, 11.6 Hz, 2H), 2.88 (s, 3H), 2.45 (br s, 2H), 2.33-2.23 (m, 2H), 2.23-2.09 (m, 4H), 2.09-2.00 (m, 2H); LCMS (ESI, M+1): m/z=625.1.

Example 260

(1343) ##STR00559##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1344) ##STR00560##

(1345) Step A. tert-butyl-2-(dimethylcarbamoyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (1.00 g, 1.0 equiv.), TEA (1.08 g, 3.0 equiv.) and HATU (2.70 g, 7.11 mmol, 2.0 equiv.) in THF (10 mL) was added dimethylamine (2 M in THF, 2.0 equiv.). The mixture was stirred at 25° C. for 12 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (1.00 g, 89% yield) as a yellow solid; LCMS (ESI, M+1): m/z=309.0.

(1346) Step B. N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of tert-butyl-2-(dimethylcarbamoyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate (1.00 g, 1.0 equiv.) in MeCN (10 mL) was added HCl.dioxane (4 M, 20 mL, 25 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The mixture was concentrated to afford the title compound (1.00 g, crude, HCl) as a yellow liquid; LCMS (ESI, M+): m/z=208.3.

(1347) Step C. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (461 mg, 10 equiv., HCl) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was added DIEA (487 mg, 20 equiv.). The reaction was stirred at 40° C. for 24 hours. The mixture was filtered and purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; A: water (0.225% formic acid)/ACN] B %: 15%-45%, 7 minutes] to afford the title compound (7.28 mg, 6.02% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-&) δ=9.19 (s, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.74-7.66 (m, 1H), 7.61 (dd, J=1.2, 7.2 Hz, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.23-7.13 (m, 1H), 6.79 (s, 1H), 5.38-5.15 (m, 2H), 4.58-4.52 (m, 2H), 4.46 (br t, J=4.8 Hz, 2H), 4.36 (s, 2H), 3.33 (s, 3H), 3.29-3.20 (m, 2H), 3.08 (s, 3H), 2.94-2.80 (m, 2H), 2.50-2.38 (m, 2H), 2.18-2.08 (m, 2H), 2.06-1.91 (m, 4H), 1.90-1.79 (m, 2H); LCMS (ESI, M+1): m/z=639.1.

Example 261

(1348) ##STR00561##

(5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanol

(1349) The title compound was synthesized according to the procedure described for example 84. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.24 (s, 1H), 8.13 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.3 Hz, 1H), 7.75-7.66 (m, 1H), 7.61 (d, J=7.2 Hz, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 6.46 (s, 1H), 5.3-5.17 (m, 2H), 4.56 (d, J=17.6 Hz, 4H), 4.44 (br d, J=6.0 Hz,

4H), 3.64-3.56 (m, 2H), 3.20 (td, J=6.0, 11.6 Hz, 2H), 2.42 (br s, 2H), 2.33-2.25 (m, 2H), 2.25-2.18 (m, 1H), 2.18-2.09 (m, 3H), 2.09-2.01 (m, 2H); LCMS (ESI, M+1): m/z=598.3.

Example 262

(1350) ##STR00562##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-6-azaspiro[3.5]nonane

(1351) The title compound was synthesized according to the procedure described for example 84. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.14 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.72-7.69 (m, 1H), 7.62 (dd, J=0.8, 7.2 Hz, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.2, 13.2 Hz, 1H), 4.57 (s, 2H), 4.43-4.36 (m, 2H), 4.02 (br dd, J=4.4, 6.0 Hz, 2H), 3.46-3.43 (m, 2H), 3.16 (d, J=9.2 Hz, 2H), 3.06-3.03 (m, 2H), 2.90 (t, J=9.2 Hz, 2H), 2.22-2.17 (m, 2H), 2.15-2.05 (m, 4H), 1.97-1.93 (m, 4H), 1.83-1.77 (m, 2H); LCMS (ESI, M+1): m/z=574.3.

Example 263

(1352) ##STR00563##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine

(1353) ##STR00564##

(1354) Step A. tert-butyl 2-(benzyloxycarbonylamino)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate: A mixture of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (500 mg, 1.0 equiv.), 4 Å MS (300 mg) and TEA (539 mg, 3.0 equiv.) in toluene (5 mL) and BnOH (1.15 g, 6.0 equiv.) was stirred at 110° C. for 0.5 hour under nitrogen. And then the mixture was cooled to 15° C. and DPPA (734 mg, 1.5 equiv.) was added at 15° C. The mixture was stirred at 110° C. for 5 hours. The reaction mixture was diluted with water (15 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were washed with brine (20 mL), and dried over sodium sulfate, filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (400 mg, 42% yield) as a gray solid; LCMS (ESI, M+1): m/z=387.2.

(1355) Step B. tert-butyl 2-(benzyloxycarbonyl(methyl)amino)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepin-5-carboxylate: A mixture of tert-butyl 2-(benzyloxycarbonylamino)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (100 mg, 1.0 equiv.) in DMF (3 mL) was added K₂CO₃ (179 mg, 5.0 equiv.) and MeI (220 mg, 6.0 equiv.). The mixture was stirred at 50° C. for 3 hours. The reaction mixture was quenched with saturated NH₄Cl solution (20 mL). The mixture was extracted with ethyl acetate (2×10 mL). The combined organic extracts were washed with brine (10 mL), dried over sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 20:1 to 1:1) to afford the title compound (85.0 mg, 82% yield) as a yellow oil; LCMS (ESI, M+1): m/z=401.3.

(1356) Step C. benzyl N-methyl-N-(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)carbamate: To a solution of tert-butyl 2-[benzyloxycarbonyl(methyl)amino]-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (85.0 mg, 1.0 equiv.) in MeCN (1 mL) was added HCl.dioxane (4 M, 0.5 mL, 9.42 equiv.). The mixture was stirred at 25° C. for 2 hours. The reaction mixture was concentrated under reduced pressure to give a residue. The pH of the mixture was adjusted to 8 with saturated NaHCO₃ solution and the resulting was extracted with ethyl acetate (15 mL×3). The combined organic layers were washed with brine (5 mL), dried over sodium sulfate, filtered and concentrated under reduced pressure to afford the title compound (60.0 mg, 94% yield) as a yellow solid.

(1357) Step D. N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To a mixture of benzyl N-methyl-N-(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)carbamate

(60.0 mg, 1.0 equiv.) in MeOH (3 mL) was added Pd/C (15.0 mg, 10% purity). The mixture was stirred at 25° C. for 2 hours under H.sub.2 atmosphere (15 psi) before being filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-TLC (Silica gel, DCM/MeOH 5:1) to afford the title compound (32.0 mg, 88% yield) as a yellow solid; ¹H NMR (400 MHz, chloroform-d) δ=5.41 (s, 1H), 4.24-4.11 (m, 2H), 3.78 (s, 2H), 3.21-3.09 (m, 2H), 2.80 (s, 3H), 1.82-1.74 (m, 2H); LCMS [ESI, M+1]: m/z=167.3.

(1358) Step E. 5-[8-fluoro-7-(8-fluoro-1-naphthyl)-2-(1,2,3,5,6,7-hexahydropyrrolizin-8-yl)methoxy]pyrido[4,3-d]pyrimidin-4-yl]-N-methyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepin-2-amine: To a solution of N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (31.3 mg, 10 equiv.), 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (10.0 mg, 1.0 equiv.), DIEA (7.31 mg, 3.0 equiv.) in DMF (0.05 mL) was degassed and stirred at 50° C. for 12 hours under N.sub.2 atmosphere. The reaction mixture was filtered and concentrated under reduced pressure. The residue was purified by prep-HPLC [column: Phenomenex C 18 75×30 mm×3 μm; mobile phase: water (0.1% formic acid)/ACN], B %: 12%-42%, 7 min] to afford the title compound (3.84 mg, 33% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄): δ=9.23 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.71 (t, J=7.2 Hz, 1H), 7.63-7.59 (m, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.19 (dd, J=8.0, 13.2 Hz, 1H), 5.77 (s, 1H), 5.18-5.06 (m, 2H), 4.45 (s, 2H), 4.37 (br s, 2H), 4.30-4.21 (m, 2H), 3.41-3.34 (m, 2H), 3.03-2.93 (m, 2H), 2.77 (s, 3H), 2.39 (br s, 2H), 2.17 (br dd, J=6.0, 12.0 Hz, 2H), 2.04 (dt, J=6.4, 13.2 Hz, 4H), 1.91 (br dd, J=6.8, 12.8 Hz, 2H); LCMS [ESI, M+1]: m/z=597.4.

Example 264

(1359) ##STR00565##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine

(1360) ##STR00566##

(1361) Step A. tert-butyl 2-(dimethylamino)-7,8-dihydro-4H-pyrazolo[1,5-a]diazepine-5(6H)-carboxylate: To a solution of tert-butyl-2-(((benzyloxy)carbonyl)amino)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate (589 mg, 1.0 equiv.) and HCHO (371 mg, 37% purity, 3.0 equiv.) in MeOH (20.0 mL) was added Pd/C (500 mg, 10% purity) under N.sub.2. The mixture was degassed and purged with H.sub.2 for 3 times, and then stirred at 25° C. for 1 hour under H.sub.2 (15 psi). The mixture was filtered, the filtrate was concentrated in vacuum. The crude product was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (202 mg, 43% yield) as a yellow oil. LCMS (ESI, M+1): m/z=281.2.

(1362) Step B. N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To a solution of tert-butyl 2-(dimethylamino)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate (170 mg, 1.0 equiv.) in MeCN (3.00 mL) was added HCl-dioxane (4 M, 6.00 mL, 39.6 equiv.). The mixture was stirred at 25° C. for 0.5 hour. The mixture was concentrated in vacuum to afford the title compound (131 mg, crude, HCl) as a white solid.

(1363) Step C. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To a mixture of N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (129 mg, 4.5 equiv., HCl) and DIEA (170 mg, 10.0 equiv.) was added 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (70.0 mg, 1.0 equiv.) and DMF (0.5 mL). The mixture was stirred at 50° C. for 14 hours. The mixture was diluted with DMF (1.5 mL) and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] and further purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μm; mobile phase: water (0.1% formic acid)/ACN] B %: 15%-45%, 10 mins) to afford the title compound (10.2 mg, 2.7% yield

over two steps) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.23 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.74-7.67 (m, 1H), 7.61 (dd, J=0.8, 6.8 Hz, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.19 (ddd, J=0.8, 7.6, 13.2 Hz, 1H), 5.90 (s, 1H), 5.21-5.06 (m, 2H), 4.43-4.33 (m, 4H), 4.32-4.24 (m, 2H), 3.29-3.23 (m, 2H), 2.96-2.86 (m, 2H), 2.81 (s, 6H), 2.45-2.32 (m, 2H), 2.20-2.08 (m, 2H), 2.07-1.92 (m, 4H), 1.91-1.80 (m, 2H); LCMS (ESI, M+1): m/z=611.4.

Example 265

(1364) ##STR00567##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decane-1,3-dione

(1365) The title compound was synthesized according to the procedure described for example 84. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.12 (d, J=2.4 Hz, 1H), 8.13 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.74-7.68 (m, 1H), 7.62 (ddd, J=0.8, 7.2, 10.8 Hz, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (ddd, J=1.6, 7.6, 13.2 Hz, 1H), 4.66 (br dd, J=6.0, 13.2 Hz, 1H), 4.52 (s, 2H), 4.42 (br dd, J=5.2, 12.8 Hz, 1H), 3.95 (dt, J=3.2, 10.4 Hz, 1H), 3.82 (d, J=13.2 Hz, 1H), 3.59-3.47 (m, 2H), 3.18-3.08 (m, 2H), 2.97 (dd, J=12.0, 18.0 Hz, 1H), 2.68 (d, J=18.0 Hz, 1H), 2.30-2.19 (m, 3H), 2.17-2.04 (m, 5H), 2.04-1.83 (m, 4H); LCMS [ESI, M+1]: m/z=599.4.

Example 266

(1366) ##STR00568##

6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.6]decan-2-one

(1367) ##STR00569##

(1368) Step A. 3-methyleneazepane: To a solution of tert-butyl 3-methyleneazepane-1-carboxylate (1.50 g, 1.0 equiv.) in MeCN (5 mL) was added HCl.dioxane (4 M, 10.0 mL, 5.63 equiv.). The mixture was stirred at 20° C. for 1 hour. After completion, the residue was concentrated to afford the title compound (1.50 g, crude, HCl) as a white solid.

(1369) Step B. benzyl 3-methyleneazepane-1-carboxylate: To a solution of 3-methyleneazepane (1.50 g, 1.0 equiv., HCl) and DIEA (6.57 g, 5.0 equiv.) in dichloromethane (20.0 mL) was added CbzCl (2.60 g, 1.50 equiv.). The mixture was stirred at 20° C. for 3 hours. The residue was concentrated to afford the title compound (2.70 g, crude) as a yellow solid.

(1370) Step C. benzyl 2-oxo-1,6-diazaspiro[3,6]decane-6-carboxylate: To a solution of benzyl 3-methyleneazepane-1-carboxylate (2.70 g, 1.0 equiv.) in toluene (60.0 mL) was added a solution of sulfurisocyanatidic chloride (1.71 g, 1.10 equiv.) in toluene (20.0 mL) dropwise at 0° C. After addition the mixture was stirred at 20° C. for 20 hours. After reaction completion, the mixture was slowly added to a vigorously stirred solution of Na₂SO₃ (3.0 g) and K₂CO₃ (20 g) in water (60.0 mL). The mixture was stirred at 20° C. for 4 hours. Then the mixture was extracted with ethyl acetate (3×15 mL). The organic layer was dried over Na₂SO₄, filtered and concentrated. The residue was purified by column chromatography (Al₂O₃, petroleum ether/ethyl acetate 10:1 then ethyl acetate/methanol 10:1) to afford the title compound (1.20 g, three steps 36% yield) as a yellow oil; ¹H NMR (400 MHz, CHLOROFORM-d) δ=7.46-7.29 (m, 5H), 5.26-5.02 (m, 2H), 3.87-3.52 (m, 3H), 3.36-3.15 (m, 1H), 2.98-2.59 (m, 2H), 1.96-1.70 (m, 4H), 1.65-1.43 (m, 2H); LCMS (ESI, M-41, 2M+1): m/z=247.1, 577.4.

(1371) Step D. 1,6-diazaspiro[3.6]decan-2-one: To a solution of benzyl 2-oxo-1,6-diazaspiro[3.6]decane-6-carboxylate (200 mg, 1.0 equiv.) in MeOH (5.0 mL) was added Pd/C (50 mg, 10% purity) under N₂. The suspension was degassed in vacuum and purged with H₂ several times. The mixture was stirred under H₂ (15 Psi) at 20° C. for 12 hours. The mixture was filtered and concentrated to afford the title compound (120 mg, crude) as a yellow oil.

(1372) Step E. 6-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3,6]decan-2-one: To a solution of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (65.0 mg, 1.0 equiv.) and DIEA (110 mg, 149 μL, 7.0

equiv.) in DMF (2 mL) was added 1,6-diazaspiro[3.6]decan-2-one (94.5 mg, 5.0 equiv.). The mixture was stirred at 40° C. for 12 hours. After reaction completion, the mixture was filtered and purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water (0.1% formic acid)/ACN], B %: 20%.-50%, 7 min) to afford the title compound (32.1 mg, 43% yield over two steps) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.22 (d, J=4.0 Hz, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.70 (dd, J=7.2, 8.0 Hz, 1H), 7.63-7.58 (m, 1H), 7.56-7.50 (m, 1H), 7.19 (dd, J=7.6, 12.8 Hz, 1H), 4.85-4.78 (m, 2H), 4.46-4.35 (m, 2H), 4.32-4.22 (m, 11H), 4.21-4.05 (m, 2H), 3.38-3.33 (m, 1H), 3.10 (dd, J=3.6, 14.8 Hz, 1H), 2.98-2.89 (m, 2H), 2.82 (dd, J=3.6, 14.8 Hz, 1H), 2.25-2.09 (m, 4H), 2.08-1.94 (m, 6H), 1.94-1.84 (m, 3H), 1.83-1.72 (m, 1H); LCMS (ESI, M+1): m/z=585.4.

Example 267

(1373) ##STR00570##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.6]undecane-2,4-dione

(1374) ##STR00571##

(1375) Step A. tert-butyl 2,4-dioxo-1,3,7-triazaspiro[4.6]undecane-7-carboxylate: To a solution of tert-butyl 3-oxoazepane-1-carboxylate (2.00 g, 1.0 equiv.), (NH.sub.4).sub.2CO.sub.3 (2.70 g, 3.0 equiv.) in EtOH (10 mL) and water (10 mL) was added KCN (1.12 g, 1.8 equiv.). The reaction was stirred at 85° C. for 16 hours. The mixture was diluted with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate and concentrated to afford the title compound (2.20 g, 83% yield) as a white solid; .sup.1H NMR (400 MHz, dimethylsulfoxide-d.sub.6) δ=11.39-9.68 (m, 1H), 7.97-7.52 (m, 1H), 3.45 (s, 2H), 3.29-3.12 (m, 2H), 1.82-1.65 (m, 3H), 1.64-1.47 (m, 3H), 1.45-1.31 (m, 9H); LCMS (ESI, M+1): m/z=284.0.

(1376) Step B. 1,3,7-triazaspiro[4.6]undecane-2,4-dione: To a solution of tert-butyl 2,4-dioxo-1,3,7-triazaspiro[4.6]undecane-7-carboxylate (500 mg, 1.0 equiv.) in MeCN (5 mL) was added HCl.dioxane (4 M, 2.5 mL, 5.6 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. The mixture was concentrated to afford the title compound (500 mg, crude, HCl salt) as a yellow liquid; LCMS (ESI, M+): m/z=183.1.

(1377) Step C. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.6]undecane-2,4-dione: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), 1,3,7-triazaspiro[4.6]undecane-2,4-dione (345 mg, 10 equiv.), DIEA (487 mg, 20 equiv.) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was stirred at 40° C. for 24 hours. The residue was filtered and purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: [water (0.225% formic acid)/ACN] B %: 15%-45%, 7 minutes] to afford the title compound (32.7 mg, 52.9 µmol, 28% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.30 (d, J=5.6 Hz, 1H), 8.13 (d, J=8.4 Hz, 11H), 7.86 (d, J=8.0 Hz, 1H), 7.75-7.68 (m, 11H), 7.65-7.59 (m, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 11H), 7.20 (dd, J=7.2, 13.2 Hz, 11H), 4.98 (dd, J=11.6, 14.2 Hz, 11H), 4.60-4.47 (m, 2H), 4.43-4.28 (m, 11H), 4.20-4.00 (m, 2H), 3.61 (qd, J=6.0, 11.6 Hz, 2H), 3.24-3.10 (m, 2H), 2.32-2.19 (m, 4H), 2.19-2.08 (m, 6H), 2.07-1.98 (m, 2H), 1.95-1.70 (m, 2H); LCMS (ESI, M+1): m/z=614.1.

Example 268

(1378) ##STR00572##

(R)-(1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanol

(1379) The title compound was synthesized according to the procedure described for example 84. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.10 (s, 1H), 8.11 (br d, J=8.0 Hz, 11H), 7.85 (d, J=8.0 Hz, 1H), 7.70 (t, J=7.6 Hz, 1H), 7.61 (d, J=7.2 Hz, 11H), 7.52 (dt, J=5.2, 8.0 Hz, 11H), 7.18 (dd, J=7.6, 13.2 Hz, 1H), 4.76-4.58 (m, 2H), 4.47 (s, 2H), 3.60 (dd, J=4.8, 11.2 Hz, 1H), 3.53-3.44 (m, 2H), 3.42-3.32 (m, 3H), 3.03-2.94 (m, 2H), 2.24-2.14 (m, 2H), 2.12-1.98 (m, 5H), 1.98-

1.88 (m, 4H), 1.84-1.73 (m, 1H), 1.53-1.41 (m, 1H); LCMS [ESI, M+1]: m/z=546.3.

Example 269

(1380) ##STR00573##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-2-yl)methanol

(1381) The title compound was synthesized according to the procedure described for example 84.

¹H NMR (400 MHz, methanol-d₄) δ=9.22 (s, 1H), 8.12 (br d, J=8.0 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.71 (dt, J=2.8, 7.2 Hz, 1H), 7.65-7.57 (m, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.24-7.14 (m, 1H), 4.70-4.64 (m, 11H), 4.63-4.55 (m, 2H), 4.39-4.29 (m, 1H), 4.28-4.20 (m, 1H), 4.16 (td, J=4.0, 12.8 Hz, 1H), 4.00-3.91 (m, 1H), 3.84-3.73 (m, 1H), 3.73-3.64 (m, 2H), 3.64-3.53 (m, 3H), 3.23-3.13 (m, 2H), 2.36-2.22 (m, 3H), 2.20-2.08 (m, 5H), 2.06-1.99 (m, 2H); LCMS (ESI, M+1): m/z=562.3.

Example 270

(1382) ##STR00574##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane-6-carboxamide

(1383) The title compound was synthesized according to the procedure described for example 84.

¹H NMR (400 MHz, methanol-d₄) δ=9.22 (s, 1H), 8.13 (br d, J=8.4 Hz, 1H), 7.87 (d, J=8.0 Hz, 1H), 7.71 (dt, J=2.8, 7.6 Hz, 1H), 7.66-7.58 (m, 1H), 7.54 (dt, J=5.2, 7.6 Hz, 1H), 7.19 (td, J=6.8, 13.2 Hz, 1H), 4.73-4.66 (m, 1H), 4.60 (br s, 2H), 4.44-4.33 (m, 1H), 4.26-4.05 (m, 5H), 3.93-3.79 (m, 1H), 3.72-3.59 (m, 2H), 3.28-3.18 (m, 3H), 2.37-2.25 (m, 2H), 2.25-2.12 (m, 4H), 2.12-2.02 (m, 2H); LCMS (ESI, M+1): m/z=575.2.

Example 271

(1384) ##STR00575##

4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-(methylsulfonyl)-1,4-oxazepane

(1385) ##STR00576##

(1386) Step A. benzyl 6-hydroxy-1,4-oxazepane-4-carboxylate: To a solution of 1,4-oxazepan-6-ol (539 mg, 1.0 equiv.) and DIEA (1.78 g, 3.0 equiv.) in DCM (10 mL) at 0° C. was added CbzCl (1.18 g, 1.5 equiv.). The mixture was stirred at 0° C. for 1 hour and then warmed to 25° C. for 12 hours. The reaction mixture was diluted with ethyl acetate (10 mL) and water (20 mL), and the aqueous layer was extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×8 mL). The combined organic layers were dried over anhydrous sodium sulfate and concentrated in vacuum to afford the title compound (660 mg, 56.40% yield) as a yellow oil;

¹H NMR (400 MHz, DMSO) δ=7.43-7.27 (m, 5H), 5.09 (s, 2H), 5.03-4.96 (m, 1H), 3.89-3.43 (m, 7H), 3.32-3.19 (m, 1H), 3.09 (dt, J=8.0, 13.9 Hz, 1H); LCMS (ESI, M+1): m/z=252.2.

(1387) Step B. benzyl 6-((methylsulfonyl)oxy)-1,4-oxazepane-4-carboxylate: To a solution mixture of benzyl 6-hydroxy-1,4-oxazepane-4-carboxylate (400 mg, 1.0 equiv.) and TEA (483 mg, 3.0 equiv.) in DCM (8 mL) at 0° C. was added MsCl (645 mg, 3.5 equiv.). The mixture was stirred at 0° C. for 1 hour. The mixture was quenched with water (20 mL) and extracted with dichloromethane (2×8 mL). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (524 mg, crude) as a yellow solid which was used to next step without further purification.

(1388) Step C. benzyl 6-(methylthio)-1,4-oxazepane-4-carboxylate: To a solution of benzyl 6-((methylsulfonyl)oxy)-1,4-oxazepane-4-carboxylate (524 mg, 1.0 equiv.) in DMF (10 mL) at 0° C. was added MeSNa (335 mg, 3.0 equiv.). The reaction mixture was heated to 60° C. for 2 hours. The

mixture was diluted with ethyl acetate (8 mL) and water (15 mL), and the aqueous layer was extracted with ethyl acetate (8 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃ and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×5 mL). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (175 mg, 39.0% yield) as a yellow oil; ¹H NMR (400 MHz, DMSO) δ=7.42-7.28 (m, 5H), 5.14-5.04 (m, 2H), 3.98-3.82 (m, 2H), 3.75-3.62 (m, 3H), 3.53 (m, 1H), 3.43-3.32 (m, 1H), 3.30-3.22 (m, 1H), 2.99-2.83 (m, 1H), 2.10 (s, 1H), 2.08-1.97 (m, 2H) LCMS (ESI, M+1): m/z=282.1.

(1389) Step D. benzyl 6-(methylsulfonyl)-1,4-oxazepane-4-carboxylate: To a solution of benzyl 6-methylsulfonyl-1,4-oxazepane-4-carboxylate (170 mg, 1.0 equiv.) in DCM (3 mL) was added m-CPBA (368 mg, 3.0 equiv.) at 0° C. The reaction mixture was stirred at 25° C. for 2 hours. The mixture was quenched with saturated Na₂SO₃ solution (10 mL). The organic phase was separated, and then washed with NaHCO₃ (10 mL) and brine (10 mL). The organic phase was dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×5 mL). The combined organic layers were dried over anhydrous sodium sulfate and concentrated in vacuum to afford the title compound (144 mg, 76.0% yield) as a colorless oil; ¹H NMR (400 MHz, DMSO) δ=7.45-7.29 (m, 5H), 5.17-5.08 (m, 2H), 4.20-4.09 (m, 2H), 3.85-3.75 (m, 2H), 3.67-3.48 (m, 3H), 3.42-3.34 (m, 1H), 3.30-3.21 (m, 1H), 2.98 (br d, J=16.0 Hz, 3H); LCMS (ESI, M+1): m/z=314.0.

(1390) Step E. 6-(methylsulfonyl)-1,4-oxazepane: To a mixture of benzyl 6-(methylsulfonyl)-1,4-oxazepane-4-carboxylate (130 mg, 1.0 equiv.) in MeOH (3 mL) was added Pd/C (20 mg, 10% purity) under N₂. The suspension was degassed in vacuum and purged with H₂ for three times. The mixture was stirred at 25° C. under H₂ (15 Psi) for 3 hours. The mixture was filtered and the filtrate was concentrated in vacuum to afford the title compound (74 mg, crude) as a colorless oil; ¹H NMR (400 MHz, METHANOL-d₄) δ=4.33-4.25 (m, 1H), 4.13 (dd, J=5.6, 13.6 Hz, 1H), 3.87-3.80 (m, 1H), 3.66-3.58 (m, 1H), 3.48-3.38 (m, 2H), 3.28-3.19 (m, 1H), 2.99-2.95 (m, 3H), 2.95-2.86 (m, 2H).

(1391) Step F. 4-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-(methylsulfonyl)-1,4-oxazepane: To a solution of 6-(methylsulfonyl)-1,4-oxazepane (54.1 mg, 2.9 equiv.) in DMF (0.3 mL) were added DIEA (40.4 mg, 3.0 equiv.), 4 Å molecular sieves (10 mg) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (60 mg, 1.0 equiv., formic acid salt). The reaction mixture was stirred at 40° C. for 12 hours. The mixture was filtered. The filtrate was purified by prep-HPLC [column: Phenomenex Luna C18 150×25 mm×10 μm; mobile phase: [water (0.1% formic acid)/acetonitrile] B %: 12/-42%, 10 min] to afford the title compound (25.4 mg, 39.3% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₆) δ=9.22 (s, 1H), 8.54 (br s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.71 (dt, J=3.8, 7.6 Hz, 1H), 7.62 (dd, J=7.2, 19.8 Hz, 1H), 7.57-7.49 (m, 1H), 7.23-7.14 (m, 1H), 5.18-5.05 (m, 1H), 4.72-4.65 (m, 2H), 4.53-4.41 (m, 1H), 4.32 (td, J=4.8, 12.4 Hz, 1H), 4.24-4.10 (m, 5H), 4.05 (br d, J=3.2 Hz, 1H), 3.63-3.52 (m, 2H), 3.23-3.14 (m, 2H), 3.11 (s, 3H), 2.37-2.23 (m, 2H), 2.21-2.08 (m, 4H), 2.07-1.98 (m, 2H); LCMS (ESI, M+1): m/z=610.3.

Example 272

(1392) ##STR00577##

10-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-

d]pyrimidin-4-yl)-7-oxa-1,3,10-triazaspiro[4.6]undecane-2,4-dione

(1393) ##STR00578##

(1394) Step A. tert-butyl 2,4-dioxo-10-oxa-1,3,7-triazaspiro[4.6]undecane-7-carboxylate: To a solution of tert-butyl 6-oxo-1,4-oxazepane-4-carboxylate (2.00 g, 1.0 equiv.) and (NH.sub.4).sub.2CO.sub.3 (2.68 g, 3.0 equiv.) in EtOH (10 mL) and water (10 mL) was added KCN (907 mg, 1.5 equiv.). The reaction was stirred at 85° C. for 16 hours. The mixture was diluted with water (50 mL) and extracted with ethyl acetate (3×50 mL). The combined organic layers were dried over anhydrous sodium sulfate and concentrated to afford the title compound (2.40 g, 90% yield) as a white solid; .sup.1H NMR (400 MHz, dimethylsulfoxide-d.sub.6) δ=11.55-9.81 (m, 1H), 8.23-8.09 (m, 1H), 3.82-3.62 (m, 5H), 3.61-3.47 (m, 2H), 3.46-3.41 (m, 1H), 1.44-1.34 (m, 9H); LCMS (ESI, M-100): m/z=186.2.

(1395) Step B. 7-oxa-1,3,10-triazaspiro[4.6]undecane-2,4-dione: To a solution of tert-butyl 2,4-dioxo-10-oxa-1,3,7-triazaspiro[4.6]undecane-7-carboxylate (500 mg, 1.0 equiv.) in MeCN (2.5 mL) was added HCl.dioxane (4 M, 5 mL, 11.4 equiv.) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated to afford the title compound (450 mg, crude, HCl) as a white liquid; LCMS (ESI, M+1): m/z=186.2.

(1396) Step C. 10-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-oxa-1,3,10-triazaspiro[4.6]undecane-2,4-dione: A mixture of 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), 10-oxa-1,3,7-triazaspiro[4.6]undecane-2,4-dione (349 mg, 10 equiv.), DIEA (487 mg, 20 equiv.) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was stirred at 40° C. for 96 hours. The residue was filtered and purified by prep-HPLC [column: Phenomenex C18 75×30 mm×3 μm; mobile phase: [water (0.1% formic acid)/acetonitrile]; B %: 12%-42%, 7 minutes] to afford the title compound (68.0 mg, 58% yield) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.28 (d, J=2.9 Hz, 1H), 8.13 (d, J=8.3 Hz, 1H), 7.86 (d, J=8.1 Hz, 1H), 7.75-7.67 (m, 1H), 7.66-7.58 (m, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 5.04 (dd, J=14.4, 17.6 Hz, 1H), 4.59-4.54 (m, 1H), 4.53-4.47 (m, 1H), 4.46-4.37 (m, 1H), 4.36-4.31 (m, 1H), 4.31-4.24 (m, 2H), 4.24-4.16 (m, 1H), 4.11-3.99 (m, 1H), 3.89 (d, J=12.6 Hz, 1H), 3.66-3.52 (m, 2H), 3.24-3.10 (m, 2H), 2.32-2.21 (m, 2H), 2.15 (quind, J=7.2, 13.6 Hz, 4H), 2.09-1.97 (m, 2H); LCMS (ESI, M+1): m/z=616.4.

Example 273

(1397) ##STR00579##

(S)-1-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3-carboxamide

(1398) The title compound was synthesized according to the procedure described for example 84. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.10 (s, 1H), 8.53 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.74-7.67 (m, 1H), 7.61 (d, J=7.2 Hz, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 4.73-4.62 (m, 1H), 4.60-4.56 (m, 2H), 4.56-4.46 (m, 1H), 3.74-3.50 (m, 4H), 3.21-3.10 (m, 2H), 2.82-2.72 (m, 1H), 2.33-2.22 (m, 2H), 2.21-2.07 (m, 5H), 2.07-1.92 (m, 4H), 1.92-1.77 (m, 1H); HPLC:>99% ee, Chiralcel IC-3 50×4.6 mm I.D., 3 μm Mobile phase:A=Heptane; B=40% EtOH (0.05% DEA), Flow rate:1 mL/min, 254 nm, t.sub.R=6.421 min; LCMS [ESI, M-55]: m/z=559.3.

Example 274

(1399) ##STR00580##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2(4H)-carboxamide

(1400) ##STR00581##

(1401) Step A. tert-butyl 2-(dimethylcarbamoyl-4,6,7,8-tetrahydropyrazolo[4,3-c]azepine-5(2H)-carboxylate: A solution of tert-butyl 4,6,7,8-tetrahydropyrazolo[4,3-c]azepine-5(2H)-carboxylate

(890 mg, 1.0 equiv.) in THF (18 mL) was added NaH (300 mg, 2.0 equiv.) at 0° C. and the mixture was stirred at 0° C. for 0.5 hour. Then NN-dimethylcarbamoyl chloride (605 mg, 1.5 equiv.) was added into the mixture and the resulting was stirred at 25° C. for 2 hours. The mixture was quenched with water (50 ml) at 0° C. and extracted with ethyl acetate (20 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by column chromatography (silica gel, petroleum ether/ethyl acetate 10:1 to 3:1) to afford the title compound (860 mg, 74.4% yield) as a yellow solid; .sup.1H NMR (400 MHz, CHLOROFORM-d) δ=7.88 (br s, 1H), 4.42-4.24 (m, 2H), 3.75-3.60 (m, 2H), 3.33-3.13 (m, 6H), 2.94-2.84 (m, 2H), 1.91-1.77 (m, 2H), 1.50-1.33 (m, 9H).

(1402) Step B. N,N-dimethyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2(4H)-carboxamide: To a solution of tert-butyl 2-(dimethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[4,3-c]azepine-5(2H)-carboxylate (860 mg, 1.0 equiv.) in ACN (5 mL) at 0° C. was added HCl.dioxane (4 M, 14.3 equiv.). The reaction mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated in vacuum to afford the title compound (682 mg, crude, HCl) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=8.15 (s, 1H), 4.29 (s, 2H), 3.57-3.52 (m, 2H), 3.19 (br s, 6H), 3.05-2.95 (m, 2H), 2.09-2.02 (m, 2H).

(1403) Step C. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2(4H)-carboxamide: To a solution of N,N-dimethyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2(4H)-carboxamide (46.1 mg, 2.4 equiv.) in DMF (0.5 mL) were added DIEA (60.9 mg, 5.0 equiv.), 4 Å molecular sieves (5 mg) and 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50 mg, 1.0 equiv.). The reaction mixture was stirred at 40° C. for 12 hours. The mixture was filtered and the filtrate was purified by prep-HPLC [column: Phenomenex Synergi Polar-RP 100×25 mm×4 μm; mobile phase: [water (0.01% TFA)/acetonitrile]; B %: 35%-55% over 7 min]. The desired fractions were collected and concentrated in vacuum to remove acetonitrile. The mixture was lyophilized to give a residue. The residue was re-purified by prep-HPLC (column: Phenomenex C18 75×30 mm×3 μm; mobile phase: [water (0.1% formic acid)/acetonitrile]; B %: 18%-48%, 7 min]. The desired fractions were collected and lyophilized to afford the title compound (10.2 mg, 16.8% yield) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.15 (s, 1H), 8.54 (s, 1H), 8.23 (s, 1H), 8.11 (br d, J=8.4 Hz, 1H), 7.85 (d, J=8.4 Hz, 1H), 7.74-7.66 (m, 1H), 7.60 (dd, J=0.8, 7.2 Hz, 1H), 7.52 (dt, J=5.2, 8.0 Hz, 1H), 7.18 (dd, J=7.2, 13.2 Hz, 1H), 5.20-5.03 (m, 2H), 4.46-4.39 (m, 2H), 4.36 (s, 2H), 3.29-3.23 (m, 2H), 3.18 (br s, 5H), 3.05-2.98 (m, 2H), 2.87 (td, J=6.4, 10.8 Hz, 2H), 2.31-2.23 (m, 2H), 2.19-2.08 (m, 2H), 1.98 (tq, J=6.4, 12.6 Hz, 4H), 1.89-1.79 (m, 2H); LCMS (ESI, M+1): m/z=639.4.

Example 275

(1404) ##STR00582##

N-(5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)acetamide

(1405) ##STR00583##

(1406) Step A. tert-butyl 2-(((benzyloxy)carbonyl)amino)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: A mixture of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (3 g, 1.0 equiv.), 4 Å molecular sieves (1.00 g), TEA (3.24 g, 3.0 equiv.) and BnOH (6.92 g, 6.0 equiv.) in toluene (50 mL) was stirred at 110° C. for 0.5 hour under nitrogen. Then the mixture was cooled to 15° C. and DPPA (4.40 g, 1.5 equiv.) was added at 15° C. The mixture was stirred at 110° C. for 5 hours. The reaction mixture was diluted with ethyl acetate (70 mL) and water (200 mL), and extracted with ethyl acetate (50 mL). The combined organic phase was washed with brine (50 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by column

chromatography (Silica gel, petroleum ether/ethyl acetate 10:1 to 1:1) to afford the title compound (2.1 g, 50.0% yield) as a yellow solid; LCMS [ESI, M-55]: $m/z=387.3$.

(1407) Step B. tert-butyl 2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: tert-butyl 2-(benzyloxycarbonylamino)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (470 mg, 1.0 equiv.) was dissolved in MeOH (10 mL) and charged with N.sub.2. To the solution was added Pd/C (50 mg, 10% purity) under nitrogen. The mixture was stirred at 25° C. for 12 hours under H.sub.2 (15 psi). The mixture was filtered and the filtrate was concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO.sub.3 and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×15 mL). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (270 mg, 84.9% yield) as a yellow oil; ¹H NMR (400 MHz, CHLOROFORM-d) $\delta=5.66-5.45$ (m, 1H), 4.41-4.26 (m, 2H), 4.21-4.16 (m, 2H), 3.76-3.40 (m, 4H), 1.90-1.83 (m, 2H), 1.42 (s, 9H); LCMS [ESI, M+1]: 253.1.

(1408) Step C. tert-butyl 2-acetamido-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: To a solution of tert-butyl 2-amino-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (270 mg, 1.0 equiv.) in DCM (2 mL) were added TEA (325 mg, 3.0 equiv.) and acetylchloride (126 mg, 1.5 equiv.) at 0° C. The reaction mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was diluted with ethyl acetate (15 mL) and water (30 mL), and extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (15 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO.sub.3 and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (273 mg, 85.8% yield) as a white solid; LCMS [ESI, M+1]: 295.2.

(1409) Step D. N-(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)acetamide: To a solution of tert-butyl 2-acetamido-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (310 mg, 1.0 equiv.) in dioxane (1.5 mL) at 0° C. was added HCl.dioxane (4 M, 11 equiv.). The reaction mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated in vacuum to afford the title compound (200 mg, crude) as a yellow solid; ¹H NMR (400 MHz, methanol-d.sub.4) $\delta=6.45$ (s, 1H), 4.45 (s, 2H), 4.34-4.26 (m, 2H), 3.70 (br d, $J=4.4$ Hz, 2H), 2.09 (s, 3H), 1.83 (br s, 2H).

(1410) Step E. N-(5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)acetamide: A mixture of N-(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)acetamide (110 mg, 2.0 equiv.), DIEA (183 mg, 5.0 equiv.) and 4 Å molecular sieves (20 mg) in DMF (1 mL) were stirred at 25° C. for 0.5 hour. Then 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (150 mg, 1.0 equiv.) was added into the reaction mixture and the resulting was stirred at 40° C. for 12 hours. The mixture was filtered and the filtrate was purified by prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 μ m; mobile phase: [water (0.1% formic acid)/acetonitrile], B %: 5%-35%, 10 min]. The desired fraction was collected and lyophilized to afford the title compound (26.8 mg, 15.1% yield) as a white solid; ¹H NMR (400 MHz, methanol-d.sub.4) $\delta=9.22$ (s, 1H), 8.54 (s, 1H), 8.12 (br d, $J=8.4$ Hz, 1H), 7.86 (d, $J=8.0$ Hz, 1H), 7.73-7.67 (m, 1H), 7.61 (dd, $J=1.2, 7.2$ Hz, 1H), 7.53 (dt, $J=5.2, 8.0$ Hz, 1H), 7.19 (ddd, $J=0.8, 7.6, 13.2$ Hz, 1H), 6.71 (s, 1H), 5.27-5.12 (m, 2H), 4.52 (s, 2H), 4.46 (br t, $J=5.2$ Hz, 2H), 4.37-4.29 (m, 2H), 3.56-3.45 (m, 2H), 3.15-3.05 (m, 2H), 2.48-2.38 (m, 2H), 2.30-2.20 (m, 2H), 2.19-2.03 (m, 7H), 2.03-1.92 (m, 2H); LCMS (ESI, M+1): $m/z=625.4$.

Example 276

(1411) ##STR00584##

(5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)

(morpholino)methanone

(1412) ##STR00585##

(1413) Step A. tert-butyl 2-(morpholine-4-carbonyl)-7,8-dihydro-4H-pyrazolo[1,5-a]

[1,4]diazepine-5(6H)-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (300 mg, 1.0 equiv.), morpholine (278 mg, 3.0 equiv.) and DIEA (1.65 g, 12 equiv.) in ethyl acetate (3 mL) was added T3P (2.04 g, 6.0 equiv.) at 0° C. The mixture was stirred at 0-25° C. for 2 hours. The mixture was concentrated under reduced pressure to give a residue and the residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (320 mg, 85% yield) as a white oil; LCMS (ESI, M+1): m/z=351.3.

(1414) Step B. morpholino(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanone: To a solution of tert-butyl 2-(morpholine-4-carbonyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (310 mg, 1.0 equiv.) in MeCN (3 mL) was added HCl-dioxane (4 M, 6.8 equiv.) at 0° C. The mixture was stirred at 0-25° C. for 0.5 hour. The mixture was concentrated under reduced pressure to afford the title compound (215 mg, crude) as a white solid;

(1415) Step C. (5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl) (morpholino)methanone: To a solution of morpholino(5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanone (212 mg, 3.0 equiv.) in DMF (3 mL) was added DIEA (219 mg, 6.0 equiv.) and 4 Å molecular sieves (40 mg). 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (150 mg, 1.0 equiv.). The mixture was stirred at 40° C. for 16 hours. DIEA (365 mg, 10 equiv.) was added, the mixture was stirred at 40° C. for 20 hours. The reaction mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] and re-purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 µm, mobile phase: [water (0.1% formic acid)/acetonitrile], B %: 22%-52%, 7 min] and lyophilized to afford the title compound (8.66 mg, 4.3% yield) as a yellow solid; .sup.1H NMR (400 MHz, METHANOL-&) δ=9.20 (s, 1H), 8.12 (d, J=8.4 Hz, 1H), 7.85 (d, J=8.0 Hz, 1H), 7.72-7.67 (m, 1H), 7.61 (dd, J=1.2, 7.2 Hz, 1H), 7.53 (dt, J=4.8, 8.0 Hz, 1H), 7.19 (ddd, J=0.8, 7.6, 13.2 Hz, 1H), 6.80 (s, 1H), 5.33-5.21 (m, 2H), 4.56-4.51 (m, 2H), 4.50-4.45 (m, 2H), 4.44 (s, 2H), 4.04 (br s, 2H), 3.75-3.65 (m, 6H), 3.42-3.34 (m, 2H), 3.03-2.95 (m, 2H), 2.48-2.39 (m, 2H), 2.23-2.15 (m, 2H), 2.10-1.97 (m, 4H), 1.96-1.88 (m, 2H), 1.37-1.31 (m, 2H); LCMS [ESI, M+1]:681.4.

Example 277

(1416) ##STR00586##

azetidin-1-yl(5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)methanone

(1417) The title compound was synthesized according to the procedure described for example 276. .sup.1H NMR (400 MHz, methanol-d.sub.4): δ=9.21 (s, 1H), 8.53 (s, 1H), 8.12 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.70 (t, J=7.6 Hz, 1H), 7.61 (d, J=7.2 Hz, 1H), 7.53 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=7.6, 13.2 Hz, 1H), 6.84 (s, 1H), 5.32-5.19 (m, 2H), 4.60 (br t, J=7.6 Hz, 2H), 4.55-4.44 (m, 6H), 4.16 (br t, J=7.6 Hz, 2H), 3.60-3.50 (m, 2H), 3.20-3.11 (m, 2H), 2.47-2.33 (m, 4H), 2.30-2.22 (m, 2H), 2.20-2.08 (m, 4H), 2.06-1.97 (m, 2H); LCMS [ESI, M+1]: m/z=651.5.

Example 278

(1418) ##STR00587##

5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-

d]pyrimidin-4-yl)-3-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (1419) ##STR00588##

(1420) Step A. tert-butyl 2-((tert-butoxycarbonyl)amino)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: A mixture of 5-(tert-butoxycarbonyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (2.00 g, 1.0 equiv.), 4 Å molecular sieves (2.00 g) and TEA (2.16 g, 3.0 equiv.) in toluene (20 mL) and 2-methylpropan-2-ol (15.5 g, 29.4 equiv.) was stirred at 110° C. for 0.5 hour under nitrogen. The mixture was cooled to 25° C. and DPPA (2.93 g, 1.5 equiv.) was added at 25° C. The mixture was stirred at 110° C. for 5 hours before being quenched by addition of water (30 mL) and extracted with ethyl acetate (80 mL). The organic layer was dried over anhydrous sodium sulfate, concentrated, and purified by reversed-phase HPLC [water (0.1% formic acid)/ACN] to afford the title compound (1.50 g, 60% yield) as a yellow solid; LCMS (ESI, M+1): m/z=353.1.

(1421) Step B. tert-butyl 2-((tert-butoxycarbonyl)amino)-3-iodo-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: To a solution of tert-butyl 2-((tert-butoxycarbonyl)amino)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate (1.50 g, 1.0 equiv.) in MeCN (50 mL) was added NIS (1.92 g, 2.0 equiv.) at 25° C., The mixture was stirred at 25° C. for 1 hour. The mixture was concentrated in vacuum. The crude product was purified by reversed phase flash chromatography [water (0.1% formic acid)/ACN] to afford the title compound (1.90 g, 93% yield) as a yellow solid; LCMS (ESI, M+1): 479.2.

(1422) Step C. tert-butyl 2-((tert-butoxycarbonyl)amino)-3-methyl-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: A mixture of tert-butyl 2-((tert-butoxycarbonyl)amino)-3-iodo-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate (1.90 g, 1.0 equiv.), 2,4,6-trimethyl-1,3,5,2,4,6-trioxatriborinane (3.5 M, 2.27 mL, 2.0 equiv.), CataCXium A Pd G3 (578 mg, 0.2 equiv.) and K.sub.3PO.sub.4 (1.5 M, 7.94 mL, 3.0 equiv.) in THF (80 mL) was degassed and stirred at 60° C. for 24 hours under N.sub.2 atmosphere. The mixture was quenched by addition of water (100 mL) and extracted with ethyl acetate (2×60 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, and concentrated. The crude product was purified by reversed phase flash chromatography [water (0.1% formic acid)/ACN] to afford the title compound (900 mg, 46% yield) as a yellow solid; LCMS (ESI, M+1): 367.2.

(1423) Step D. 3-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To a solution of tert-butyl 2-((tert-butoxycarbonyl)amino)-3-methyl-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate (300 mg, 1.0 equiv.) in MeCN (1 mL) was added HCl-dioxane (4 M, 2 mL, 9.77 equiv.) at 25° C. The mixture was stirred at 25° C. for 0.5 hour. The mixture was concentrated in vacuum. The residue was diluted with MeOH (5 mL), treated with solid NaHCO.sub.3 (1 g), filtered and the filtrate was concentrated in vacuum to afford the title compound (210 mg, crude, 2HCl) as a yellow solid; LCMS (ESI, M+1): m/z=167.2.

(1424) Step E. 5-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine: To a mixture of 3-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-amine (75.2 mg, 4.0 equiv.) and DIEA (73.1 mg, 5.0 equiv.) was added 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (60.0 mg, 1.0 equiv.) and DMF (0.5 mL). The mixture was stirred at 40° C. for 14 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/ACN], and further re-purified by prep-HPLC (column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: [water (0.1% formic acid)/ACN], B %: 11%-41%, 10 min) to afford the title compound (2.05 mg, 2.9% yield) as an off-white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.15 (s, 1H), 8.53 (br s, 1H), 8.13 (br d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.71 (t, J=7.2 Hz, 1H), 7.61 (d, J=6.4 Hz, 1H), 7.54 (dt, J=5.2, 8.0 Hz, 1H), 7.19 (dd, J=8.0, 13.2 Hz, 1H), 5.17-5.12 (m, 2H), 4.59 (br s, 2H), 4.40-4.35 (m, 2H), 4.23-4.19 (m, 2H), 3.65-3.55 (m, 2H), 3.25-3.18 (m, 2H), 2.41-2.04 (m, 10H), 2.04 (s, 3H);

LCMS (ESI, M+1): m/z=597.4.

Example 279

(1425) ##STR00589##

3-((8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)tetrahydrothiophene 1,1-dioxide

(1426) The title compound was synthesized according to the procedure described for example 85.

¹H NMR (400 MHz, methanol-d₄) δ ppm 9.26-9.24 (m, 1H), 9.25 (s, 1H), 8.12 (br d, J=8.2 Hz, 1H), 7.85 (d, J=8.2 Hz, 1H), 7.73-7.67 (m, 11H), 7.59 (d, J=6.8 Hz, 11H), 7.53 (m, 7.9 Hz, 1H), 7.19 (dd, J=7.2, 13.2 Hz, 1H), 5.23-5.14 (m, 1H), 4.32 (s, 2H), 3.81-3.72 (m, 1H), 3.49-3.40 (m, 1H), 3.30-3.22 (m, 2H), 3.16-3.07 (m, 2H), 2.81-2.70 (m, 3H), 2.56-2.44 (m, 1H), 2.15-2.03 (m, 2H), 1.96-1.92 (m, 4H), 1.82-1.72 (m, 2H). HPLC:>99%. SFC: Chiralcel OD-3 50×4.6 mm I.D., 3 μm column A: 40% (MeOH:ACN=4:1) (w/0.05% DEA), B: CO₂, 3 mL/min, 220 nm. LCMS [ESI, M+1]: m/z=566.2.

Example 280

(1427) ##STR00590##

(S)-7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.4]nonan-3-one

(1428) The title compound was synthesized according to the procedure described for example 85.

LCMS (ESI, M+1): m/z=571.3.

Example 281

(1429) ##STR00591##

1-(1-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1430) ##STR00592##

(1431) Step A. 7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.), (8-chloronaphthalen-1-yl)trimethylstannane (464 mg, 1.43 mmol, 3.0 equiv.), CuI (27.2 mg, 0.3 equiv.) and BINAP (59.2 mg, 0.2 equiv.) in toluene (8 mL) was added Pd(dppf)Cl₂ (34.8 mg, 0.1 equiv.) under N₂. The reaction was de-gassed and stirred at 90° C. for 2 hours under N₂. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (156 mg, 25% yield) as a yellow solid; LCMS (ESI, M+1): m/z=547.1.

(1432) Step B. 1-(1-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a solution of 7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (40.0 mg, 1.0 equiv.), piperidin-3-ylmethanesulfonamide (26.1 mg, 2.0 equiv.) and DIEA (28.4 mg, 3.0 equiv.) in DMF (0.5 mL) was added 4 Å molecular sieves (10 mg). The reaction was stirred at 40° C. for 2 hours. The mixture was filtered and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)/ACN] B %: 42%-72%, 8 min] and lyophilized to afford the title compound (13.8 mg, 29% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.08 (s, 1H), 8.13 (d, J=8.0 Hz, 1H), 8.01 (d, J=7.6 Hz, 1H), 7.69 (dt, J=2.0, 7.6 Hz, 1H), 7.63-7.58 (m, 2H), 7.50 (t, J=7.6 Hz, 1H), 4.99 (br d, J=11.2 Hz, 1H), 4.56 (br d, J=13.6 Hz, 1H), 4.47-4.36 (m, 2H), 3.70-3.58 (m, 1H), 3.39-3.33 (m, 1H), 3.29-3.21 (m, 2H), 3.21-3.09 (m, 2H), 2.90-2.81 (m, 2H), 2.58-2.46 (m, 1H), 2.22-2.08 (m, 3H), 2.04-1.90 (m, 5H), 1.87-1.74 (m, 3H), 1.68-1.56 (m, 1H); ¹⁹F NMR (400 MHz, methanol-d₄) δ=-139.695; LCMS (ESI, M+1): m/z=625.2.

Example 282

(1433) ##STR00593##

7-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-

d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1434) The title compound was synthesized according to the procedure described for example 281. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.09 (d, J=3.6 Hz, 1H), 8.14 (d, J=7.9 Hz, 1H), 8.01 (d, J=8.0 Hz, 1H), 7.68 (t, J=5.6 Hz, 1H), 7.63-7.57 (m, 2H), 7.50 (t, J=5.2 Hz, 13.6 Hz, 1H), 4.63 (dd, J=5.2, 13.2 Hz, 1H), 4.48-4.45 (m, 1H), 4.37-4.30 (m, 2H), 3.85-3.68 (m, 2H), 3.25-3.14 (m, 2H), 2.89-2.74 (m, 2H), 2.31-2.19 (m, 1H), 2.18-2.00 (m, 4H), 1.99-1.87 (m, 5H), 1.85-1.76 (m, 2H); LCMS (ESI, M+1): m/z=616.2.

Example 283

(1435) ##STR00594##

6-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1436) The title compound was synthesized according to the procedure described for example 281. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.07 (s, 1H), 8.14 (dd, J=1.2, 8.0 Hz, 1H), 8.04-7.98 (m, 1H), 7.72-7.65 (m, 1H), 7.63-7.57 (m, 2H), 7.54-7.48 (m, 1H), 4.38 (d, J=13.6 Hz, 1H), 4.34-4.22 (m, 3H), 3.99 (d, J=13.2 Hz, 1H), 3.88-3.70 (m, 1H), 3.16-3.04 (m, 2H), 2.92-2.86 (m, 11H), 2.80-2.65 (m, 3H), 2.18-2.01 (m, 4H), 2.01-1.84 (m, 6H), 1.80-1.69 (m, 2H); LCMS (ESI, M+1): m/z=587.2.

Example 284

(1437) ##STR00595##

7-(7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1438) The title compound was synthesized according to the procedure described for example 281. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.14 (s, 1H), 8.16-8.14 (d, J=8.0 Hz, 1H), 8.2 (d, J=8.0 Hz, 1H), 7.70 (t, J=7.2 Hz, 1H), 7.66-7.57 (m, 2H), 7.54-7.50 (t, J=8.0 Hz, 1H), 4.81 (s, 1H), 4.65-4.50 (m, 3H), 4.38-4.21 (m, 1H), 4.04-3.81 (m, 2H), 3.61-3.55 (m, 2H), 3.45-3.41 (m, 1H), 3.22-3.08 (m, 2H), 2.32-2.23 (m, 2H), 2.18-2.14 (m, 4H), 2.08-1.94 (m, 5H), 1.94-1.80 (m, 1H); LCMS (ESI, M+1): m/z=602.3.

Example 285

(1439) ##STR00596##

1-(1-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1440) The title compound was synthesized according to the procedure described for example 204. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.08 (s, 1H), 8.04 (d, J=8.0 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.59-7.52 (m, 1H), 7.51-7.38 (m, 3H), 4.99-4.88 (m, 1H), 4.60-4.50 (m, 1H), 4.37-4.27 (m, 2H), 3.66-3.51 (m, 1H), 3.48-3.33 (m, 1H), 3.22-3.08 (m, 4H), 2.76-2.67 (m, 2H), 2.55-2.29 (m, 3H), 2.10 (dd, J=6.4, 5.2 Hz, 3H), 1.98-1.86 (m, 5H), 1.82-1.70 (m, 3H), 1.67-1.58 (m, 1H), 0.93 (dd, J=7.6, 2.0 Hz, 3H); ^{sup}.19F NMR (377 MHz, METHANOL-d_{sub}.4) δ=-134.15, -147.26. LCMS (ESI, M+1): m/z=619.2.

Example 286

(1441) ##STR00597##

7-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(1442) The title compound was synthesized according to the procedure described for example 204. ^{sup}.1H NMR (400 MHz, METHANOL-d_{sub}.4) δ=9.10 (s, 1H), 8.05 (dd, J=1.2, 8.0 Hz, 1H), 7.87 (d, J=7.6 Hz, 1H), 7.59-7.54 (m, 1H), 7.52-7.47 (m, 1H), 7.46-7.39 (m, 2H), 4.54-4.38 (m, 2H), 4.54-3.98 (m, 2H), 3.95-3.79 (m, 1H), 3.46-3.35 (m, 2H), 3.34 (br s, 1H), 3.30-3.24 (m, 2H), 3.01 (br d, J=9.2 Hz, 2H), 2.49-2.17 (m, 6H), 2.07 (td, J=6.3, 12.4 Hz, 4H), 1.91 (br s, 6H), 0.93 (q, J=7.6 Hz, 3H). LCMS [ESI, M+1]: m/z=595.4.

Example 287

(1443) ##STR00598##

N-((1H-pyrazol-4-yl)methyl)-7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1444) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, dimethylsulfoxide-*d*.₆) δ=9.36 (br t, J=5.2 Hz, 1H), 8.07 (d, J=7.6 Hz, 11H), 7.91 (d, J=8.0 Hz, 11H), 7.67 (s, 2H), 7.57 (t, J=7.6 Hz, 11H), 7.51 (t, J=7.6 Hz, 1H), 7.39 (t, J=7.6 Hz, 2H), 4.66 (br d, J=5.2 Hz, 2H), 4.16 (s, 2H), 3.05-2.97 (m, 2H), 2.69-2.58 (m, 2H), 2.41-2.17 (m, 3H), 2.00-1.90 (m, 2H), 1.88-1.75 (m, 4H), 1.68-1.58 (m, 2H), 0.83 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=538.3.

Example 288

(1445) ##STR00599##

6-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1446) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, dimethylsulfoxide-*d*.₆) δ=8.58 (br d, J=7.6 Hz, 1H), 8.08 (d, J=7.6 Hz, 1H), 7.92 (d, J=8.0 Hz, 1H), 7.63-7.55 (m, 11H), 7.52 (t, J=7.6 Hz, 1H), 7.45 (br d, J=7.2 Hz, 1H), 7.39 (d, J=7.2 Hz, 1H), 4.28-4.12 (m, 4H), 3.94-3.79 (m, 2H), 3.09-3.00 (m, 2H), 2.88-2.76 (m, 1H), 2.71-2.60 (m, 3H), 2.45-2.20 (m, 3H), 2.02-1.92 (m, 3H), 1.91-1.74 (m, 7H), 1.70-1.60 (m, 2H), 0.85 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=581.1.

Example 289

(1447) ##STR00600##

(4-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-yl)methanol

(1448) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, METHANOL-*d*.₄): δ=9.26-9.21 (m, 1H), 8.53 (s, 1H), 8.05 (d, J=8.0 Hz, 1H), 7.87 (d, J=8.0 Hz, 1H), 7.61-7.53 (m, 1H), 7.52-7.36 (m, 3H), 4.71-4.53 (m, 3H), 4.47-4.37 (m, 1H), 4.18-3.93 (m, 4H), 3.89-3.77 (m, 1H), 3.71-3.49 (m, 5H), 3.19-3.08 (m, 2H), 2.56-2.22 (m, 5H), 2.19-1.96 (m, 6H), 0.98 (br s, 3H); LCMS [ESI, M+1]: m/z=572.2.

Example 290

(1449) ##STR00601##

3-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol

(1450) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, methanol-*d*.₄) δ 9.23-9.06 (m, 1H), 8.04 (d, J=8.4 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.59-7.53 (m, 1H), 7.51-7.37 (m, 3H), 5.01-4.92 (m, 1H), 4.84-4.76 (m, 1H), 4.36-4.26 (m, 3H), 3.78 (br dd, J=7.6, 11.6 Hz, 1H), 3.53-3.45 (m, 1H), 3.20-3.08 (m, 2H), 2.81-2.72 (m, 2H), 2.47-2.17 (m, 5H), 2.15-2.06 (m, 2H), 2.02-1.85 (m, 5H), 1.85-1.74 (m, 3H), 1.45-1.27 (m, 1H), 0.95-0.89 (m, 3H); LCMS [ESI, M+1]: m/z=568.4.

Example 291

(1451) ##STR00602##

4-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1452) The title compound was synthesized according to the procedure described for example 204 except for water (10 mM NH₄HCO₃)/ACN was used as a mobile phase for prep-HPLC. ¹H NMR (400 MHz, methanol-*d*.₄) δ=9.53 (d, J=5.2 Hz, 1H), 8.07-8.03 (m, 1H), 7.88-7.84 (m, 1H), 7.59-7.53 (m, 1H), 7.51-7.38 (m, 3H), 4.61-4.49 (m, 2H), 4.35-4.26 (m, 2H), 4.23-4.14 (m, 1H), 4.07-3.86 (m, 3H), 3.74-3.63 (m, 2H), 3.16-3.07 (m, 2H), 2.79-2.70 (m, 2H), 2.49-2.30 (m, 2H), 2.14-2.03 (m, 2H), 2.00-1.83 (m, 4H), 1.82-1.73 (m, 2H), 1.28 (d, J=3.2 Hz, 3H), 0.97-0.89 (m, 3H); LCMS [ESI, M+1]: m/z=572.2

Example 292

(1453) ##STR00603##

5-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1454) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, methanol-d₃) δ=9.32-9.23 (m, 1H), 8.05 (dd, J=1.2, 8.4 Hz, 1H), 7.86 (d, J=7.6 Hz, 1H), 7.57 (t, J=7.6 Hz, 1H), 7.51-7.46 (m, 1H), 7.44 (dd, J=1.2, 7.2 Hz, 1H), 7.40 (d, J=7.2 Hz, 1H), 4.65 (br d, J=12.4 Hz, 2H), 4.48-4.42 (m, 2H), 4.40-4.29 (m, 2H), 3.80-3.58 (m, 2H), 3.36-3.32 (m, 1H), 3.29 (br s, 1H), 3.01-2.85 (m, 2H), 2.48-2.26 (m, 2H), 2.17 (dtd, J=2.4, 6.4, 12.4 Hz, 2H), 2.11-2.04 (m, 1H), 2.04-2.00 (m, 2H), 1.99-1.93 (m, 1H), 1.93-1.83 (m, 2H), 0.96-0.87 (m, 3H); LCMS (ESI, M+1): m/z=581.1.

Example 293

(1455) ##STR00604##

7-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1456) The title compound was synthesized according to the procedure described for example 291. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.11-9.10 (m, 1H), 8.05-8.03 (d, J=7.2 Hz, 1H), 7.87-7.85 (d, J=8.0 Hz, 1H), 7.58-7.54 (m, 1H), 7.50-7.38 (m, 3H), 4.65-4.60 (m, 1H), 4.54-4.39 (m, 1H), 4.37-4.27 (m, 2H), 3.89-3.80 (m, 1H), 3.77-3.68 (m, 1H), 3.24-3.08 (m, 2H), 2.79-2.73 (m, 2H), 2.50-2.29 (m, 2H), 2.28-2.19 (m, 1H), 2.16-2.00 (m, 4H), 1.99-1.88 (m, 5H), 1.83-1.72 (m, 2H), 0.94-0.88 (m, 3H); LCMS [EST, M+1]: m/z=610.3.

Example 294

(1457) ##STR00605##

7-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1458) The title compound was synthesized according to the procedure described for example 204. ¹H NMR (400 MHz, methanol-d₃) δ=9.14 (d, J=1.6 Hz, 1H), 8.05 (dd, J=0.8, 8.0 Hz, 1H), 7.87 (d, J=8.0 Hz, 1H), 7.59-7.55 (m, 1H), 7.51-7.47 (m, 1H), 7.46-7.38 (m, 2H), 4.68-4.36 (m, 3H), 4.41-4.33 (m, 1H), 4.30-4.17 (m, 1H), 4.07-3.80 (m, 2H), 3.66-3.51 (m, 2H), 3.42 (dd, J=4.2, 9.6 Hz, 1H), 3.22-3.10 (m, 2H), 2.47-2.32 (m, 2H), 2.32-2.25 (m, 2H), 2.19-2.11 (m, 4H), 2.07-1.94 (m, 5H), 1.90-1.86 (m, 1H), 0.95-0.90 (m, 3H); LCMS [ESI, M+1]: m/z=596.3.

Example 295

(1459) ##STR00606##

4-(3-(1H-1,2,4-triazol-3-yl)piperidin-1-yl)-7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine

(1460) The title compound was synthesized according to the procedure described for example 291. ¹H NMR (400 MHz, CD₃OD, 298 K) δ (ppm)=9.09 (d, J=2.4 Hz, 1H), 8.25 (d, J=4.4 Hz, 1H), 8.07-8.00 (m, 1H), 7.86 (d, J=7.6 Hz, 1H), 7.56 (t, J=7.6 Hz, 1H), 7.52-7.37 (m, 3H), 4.55 (br t, J=12.0 Hz, 1H), 4.35-4.24 (m, 2H), 3.87-3.65 (m, 2H), 3.41-3.33 (m, 1H), 3.14-3.06 (m, 2H), 2.73 (m, 2H), 2.47-2.26 (m, 3H), 2.14-2.01 (m, 4H), 2.01-1.70 (m, 8H), 0.92 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=593.3.

Example 296

(1461) ##STR00607##

(3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidine-3,5-diol

(1462) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₃) δ=9.24 (dd, J=2.8, 7.2 Hz, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.05 (dd, J=2.4, 8.8 Hz, 1H), 5.39-5.22 (m, 1H), 4.37-4.31 (m, 1H), 4.30-4.19 (m, 5H), 3.84-3.75 (m, 2H), 3.21 (br d, J=19.6 Hz, 3H), 3.01 (m, 1H), 2.54-2.42 (m, 1H), 2.39-2.12 (m, 4H), 2.04-1.88 (m, 5H), 0.84-0.76 (m, 3H). ¹⁹F NMR (376 MHz, METHANOL-d₃) δ=-121.237, -139.064, -173.716. HPLC:>99%. LCMS (ESI, M+1): m/z=610.2.

Example 297

(1463) ##STR00608##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((1-(pyridin-4-yl)-1H-pyrazol-4-yl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1464) ##STR00609##

(1465) Step A. 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1-(pyridin-4-yl)-1H-pyrazol-4-yl)pyrido[4,3-d]pyrimidin-4-amine: To the mixture of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (143 mg, 1.0 equiv.), 1-(pyridin-4-yl)-1H-pyrazol-4-amine (110 mg, 2.5 equiv., HCl), 4 Å molecular sieves (50.0 mg) in DMAC (1.5 mL) was added CsF (340 mg, 10.0 equiv.), and the mixture was stirred at 40° C. for 14 hours. After reaction completion, the mixture was filtered and diluted with water (10 mL), extracted with ethyl acetate (2×20 mL), dried over anhydrous sodium sulfate, filtered and concentrated. The residue was purified by reversed-phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (106 mg, 62% yield over two steps) as a yellow solid; LCMS [ESI, M+1]: m/z=697.2.

(1466) Step B. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-((1-(pyridin-4-yl)-1H-pyrazol-4-yl)amino)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-N-(1-(pyridin-4-yl)-1H-pyrazol-4-yl)pyrido[4,3-d]pyrimidin-4-amine (50.0 mg, 1.0 equiv.) in ACN (1 mL) was added HCl.dioxane (4 M, 2 mL, 111.5 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. After reaction completion, the mixture was concentrated under reduced pressure to give a residue. pH was adjusted with sat. aq. NaHCO₃ to 9 and the mixture was extracted with ethyl acetate (3×2 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered and concentrated. The residue was purified by prep-HPLC (column: Waters Xbridge 150×25 mm×5 µm; mobile phase: [water (10 mM NH₄HCO₃)/ACN,] B %: 45%-75%, 10 min) and lyophilized to afford the title compound (14.9 mg, 31% yield) as an off-white solid; ¹H NMR (400 MHz, methanol-d₄): δ=9.39 (s, 1H), 9.06 (s, 1H), 8.67-8.61 (m, 2H), 8.23 (s, 1H), 7.93-7.88 (m, 2H), 7.69 (dd, J=5.6, 9.2 Hz, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.08 (d, J=2.8 Hz, 1H), 5.43-5.24 (m, 1H), 4.50-4.36 (m, 2H), 3.29-3.20 (m, 3H), 3.12-2.97 (m, 1H), 2.54-2.23 (m, 3H), 2.22-1.93 (m, 5H), 0.81 (t, J=7.6 Hz, 3H); LCMS [ESI, M+1]: m/z=653.3.

Example 298

(1467) ##STR00610##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(pyridin-3-yl)propoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1468) ##STR00611##

(1469) Step A. 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(pyridin-3-yl)propoxy)pyrido[4,3-d]pyrimidine: To a mixture of 3-(pyridin-3-yl)propan-1-ol (15.1 mg, 110 µmol, 14.2 µL, 1.5 equiv.) and 4 Å molecular sieves (5 mg) in THF (1 mL) was added LiHMDS (1.0 M in THF, 220 µL, 3.0 equiv.) at -40° C. and the mixture was stirred at -40° C. for 0.5 hour. Then 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 73.3 µmol, 1.0 equiv., formic acid salt) in THF (0.5 mL) was added into the above mixture and the mixture was stirred at 25° C. for 1 hour. The mixture was quenched by water (3 mL) at 0° C. The mixture was diluted with ethyl acetate (4 mL), the layers were separated and the aqueous phase was extracted with ethyl acetate (2×5 mL). The combined organic layers were dried over Na₂SO₄, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (35.0 mg, 64% yield) as a yellow

solid; LCMS (ESI, M+1): m/z=674.3.

(1470) Step B. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(pyridin-3-yl)propoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a mixture of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(pyridin-3-yl)propoxy)pyrido[4,3-d]pyrimidine (60.0 mg, 89.1 μ mol, 1.0 equiv.) in dichloromethane (0.60 mL) was added TFA (924 mg, 91 equiv.) at 0° C. and the mixture was stirred at 0-25° C. for 1.5 hours. The mixture was concentrated in vacuum and the pH value was adjusted to 8 with saturated NaHCO₃ solution. The mixture was diluted with ethyl acetate (4 mL) then separated. The aqueous layer was extracted with ethyl acetate (2×4 mL) and the combined organic layers were washed with brine (4 mL) and dried over Na₂SO₄. The mixture was filtered and concentrated in vacuum. The mixture was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μ m; mobile phase: water (10 mM NH₄HCO₃)/ACN], B %: 43%-73%, 9 min] to afford the title compound (13.3 mg, 23% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ =9.05 (s, 1H), 8.37 (dd, J=1.6, 4.8 Hz, 1H), 7.82 (br d, J=7.6 Hz, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.38 (dd, J=4.8, 7.6 Hz, 1H), 7.31 (d, J=3.0 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.05 (s, 1H), 5.38-5.22 (m, 1H), 4.75 (t, J=6.0 Hz, 2H), 4.35 (q, J=10.4 Hz, 2H), 3.29-3.17 (m, 3H), 3.05-2.95 (m, 3H), 2.49-2.39 (m, 1H), 2.38-2.26 (m, 3H), 2.24-2.19 (m, 1H), 2.17-2.09 (m, 2H), 2.04-1.96 (m, 2H), 1.94-1.86 (m, 1H), 0.77 (t, J=7.2 Hz, 3H); ¹⁹F NMR (376 MHz, methanol-d₄) δ =a -121.108, -138.808, -173.817; SFC condition: Column: Chiralcel 01-3 50×4.6 mm I.D., 3 μ m; Mobile phase: Phase A for CO₂, and Phase B for MeOH (0.05% DEA); Gradient elution: MeOH (0.05% DEA) in CO₂ from 5% to 40%; Flow rate: 3 mL/min; Detector: PDA; Column Temp: 35° C.; Back Pressure: 100 Bar; LCMS (ESI, M+1): m/z=630.3.

Example 299

(1471) ##STR00612##

1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol

(1472) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₄) δ =9.21-9.13 (m, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.09-7.02 (m, 1H), 5.51-5.32 (m, 1H), 4.52-4.37 (m, 2H), 4.27-4.19 (m, 1H), 4.13-3.94 (m, 3H), 3.93-3.83 (m, 1H), 3.64-3.41 (m, 3H), 3.24-3.14 (m, 1H), 2.56-2.30 (m, 3H), 2.28-2.05 (m, 6H), 2.03-1.95 (m, 1H), 1.84-1.69 (m, 2H), 0.84-0.75 (m, 3H); LCMS (ESI, M+1): m/z=594.3.

Example 300

(1473) ##STR00613##

(R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1474) ##STR00614##

(1475) Step A. (R)-benzyl 2-oxo-1,3,7-triazaspiro[4.5]decan-7-carboxylate: SFC separation of benzyl 2-oxo-1,3,7-triazaspiro[4.5]decan-7-carboxylate (650 mg) [column: DAICEL CHIRALPAK AY-H (250 mm×30 mm, 10 μ m); mobile phase: water (0.1% NH₄OH)-EtOH]; B %: 50%-50%, 4.8 min; 110 min]. The crude product was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (200 mg, 54% yield) as a white solid; ¹H NMR (400 MHz, DMSO-d₆) δ =7.43-7.27 (m, 5H), 6.65 (s, 1H), 6.21 (s, 1H), 5.07 (br s, 2H), 3.59-3.43 (m, 2H), 3.18-2.91 (m, 4H), 1.71-1.52 (m, 3H), 1.47-1.30 (m, 1H).

(1476) Step B. (S)-1,3,7-triazaspiro[4.5]decan-2-one: To a solution of benzyl (R)-benzyl 2-oxo-1,3,7-triazaspiro[4.5]decan-7-carboxylate (200 mg, 1.0 equiv.) in MeOH (30 mL) was added Pd/C (50.0 mg, 10% purity) under N₂. The mixture was stirred at 10° C. for 2 hours under H₂ (15 psi). The reaction mixture was filtered and concentrated to afford the title compound (120 mg,

crude) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =6.36 (s, 1H), 6.08 (br s, 1H), 3.18-3.13 (m, 4H), 2.96 (d, J=8.8 Hz, 1H), 2.20-2.10 (m, 1H), 1.64-1.41 (m, 4H).

(1477) Step C. 7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-5-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (1.0 g, 1.0 equiv.), CataCXium A Pd G3 (166 mg, 0.1 equiv.) and ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (1.29 g, 1.20 equiv.) in THF (17 mL) was added K.sub.3PO.sub.4 (1.5 M, 3.0 equiv.) under N.sub.2. The mixture was stirred at 60° C. for 1 hour. The reaction mixture was diluted with saturated NH.sub.4Cl (20 mL), the aqueous phase was extracted with ethyl acetate (3×10 mL). The combined organic phase was washed with brine (15 mL), dried with anhydrous Na.sub.2SO.sub.4, concentrated and purified by column chromatography [Silica gel, Petroleum ether/Ethyl acetate 1:0 to 3:1] to afford the title compound (crude) as a yellow solid; LCMS (ESI, M+1): m/z=749.4.

(1478) Step D. (R)-7-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To a solution of 7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (380 mg, 1.0 equiv.), 4 Å molecular sieves (200 mg) and (S)-1,3,7-triazaspiro[4.5]decan-2-one (78.7 mg, 1.0 equiv.) in DMF (4.0 mL) was added DIEA (197 mg, 3.0 equiv.). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered. The crude product was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (180 mg, 44% yield) as a brown solid; LCMS (ESI, M+1): m/z=804.3.

(1479) Step E. (R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one: To a solution of (R)-7-(7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one (180 mg, 1.0 equiv.) in DMF (1.0 mL) was added CsF (340 mg, 10 equiv.). The mixture was stirred at 10° C. for 0.5 hour. The reaction mixture was filtered and purified by prep-HPLC [Waters Xbridge 150×25 mm×5 μ m; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN; B %: 34%-64%, 9 min] to afford the title compound (125 mg, 86% yield) as a white solid; .sup.1H NMR (400 MHz, CD3OD) δ =9.11 (d, J=4.0 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.05 (d, J=2.8 Hz, 1H), 5.40-5.21 (m, 1H), 4.33-4.24 (m, 2H), 4.18-3.84 (m, 4H), 3.47-3.38 (m, 1H), 3.28-3.14 (m, 3H), 3.05-2.96 (m, 1H), 2.55-2.08 (m, 5H), 2.06-1.81 (m, 7H), 0.87-0.72 (m, 3H); LCMS (ESI, M+1): m/z=648.2.

Example 301

(1480) ##STR00615##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((3R,7aS)-3-(hydroxymethyl)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1481) ##STR00616##

(1482) Step A. (R)-1-(2-(((3R,7aS)-3-((tert-butyl)di(phenyl)silyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of ((3R,7aR)-3-(((tert-butyl)di(phenyl)silyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (408 mg, 1.1 equiv.) and 4 Å molecular sieves (30 mg) in toluene (3 mL) was added t-BuONa (261 mg, 3.0 equiv.) at 0° C. The reaction was stirred at 0° C. for 10 mins. Then (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (300 mg, 906 μ mol, 1.0 equiv.) was added. The reaction was stirred at 0° C. for 1 hour. The mixture was quenched with

water (4 mL) at 0° C. and extracted with ethyl acetate (3×8 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃, concentrated in vacuum to remove acetonitrile, and extracted with dichloromethane (2×20 mL). The combined organic layer was washed with brine (20 mL), dried over anhydrous Na₂SO₄, and concentrated in vacuum to afford the title compound (580 mg, 89% yield) as a brown solid; ¹H NMR (400 MHz, methanol-d₄) δ 8.97 (s, 1H), 7.74-7.67 (m, 4H), 7.45-7.35 (m, 6H), 4.44 (br d, J=13.2 Hz, 1H), 4.23-4.14 (m, 3H), 3.70-3.57 (m, 2H), 3.52 (d, J=13.2 Hz, 1H), 3.30-3.25 (m, 1H), 3.04-2.88 (m, 2H), 2.83-2.73 (m, 1H), 2.19-2.12 (m, 1H), 2.10-2.02 (m, 2H), 1.93-1.83 (m, 3H), 1.81-1.73 (m, 3H), 1.72-1.60 (m, 3H), 1.22 (s, 3H), 1.04 (s, 9H); LCMS [ESI, M+1]: m/z=704.2.

(1483) Step B. (R)-1-(2-(((3R,7aS)-3-(((tert-butyl)diphenylsilyl)oxy methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(2-(((3R,7aR)-3-(((tert-butyl)diphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (530 mg, 1.0 equiv.), ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (462 mg, 1.3 equiv.) and K₃PO₄ (1.5 M in water, 1.50 mL, 3.0 equiv.) in methoxycyclopentane (4.6 mL) was added CataCXium A Pd G3 (54.8 mg, 0.1 equiv.) under N₂. The reaction was stirred at 90° C. for 2.5 hours. The layers of the mixture were separated and the aqueous phase was extracted with ethyl acetate (2×5 mL). The combined organic layers were concentrated and purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃, concentrated in vacuum to remove acetonitrile, and extracted with dichloromethane (2×15 mL). The combined organic layers were washed with brine (15 mL), dried over anhydrous Na₂SO₄, and concentrated in vacuum to afford the title compound (193 mg, 23% yield) as a brown solid; LCMS [ESI, M+1, M/2+1]: m/z=1014.7, 508.1.

(1484) Step C. (R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((3R,7aS)-3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidine-4-yl)-3-methylpiperidin-3-ol: To a mixture of (R)-1-(2-(((3R,7aR)-3-(((tert-butyl)diphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (163 mg, 1.0 equiv.) in DMF (0.8 mL) was added CsF (366 mg, 15 equiv.). The reaction was stirred at 20° C. for 15 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃, concentrated in vacuum to remove acetonitrile, and extracted with dichloromethane (2×15 mL). The combined organic layer was washed with brine (15 mL), dried over anhydrous Na₂SO₄, and concentrated in vacuum. The residue was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: [water (10 mM NH₄HCO₃)-AC,] B %: 35%-65%, 8 min] and lyophilized to afford the title compound (46.7 mg, 47% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d₄) δ 9.20 (d, J=1.2 Hz, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 11H), 7.06 (t, J=2.0 Hz, 1H), 4.54 (br d, J=12.4 Hz, 1H), 4.34-4.24 (m, 3H), 3.68-3.59 (m, 1H), 3.59-3.52 (m, 2H), 3.50-3.42 (m, 0.1H), 3.13-3.01 (m, 1H), 2.98-2.84 (m, 2H), 2.54-2.41 (m, 1H), 2.26-2.11 (m, 3H), 2.06-1.99 (m, 1H), 1.98-1.92 (m, 2H), 1.92-1.84 (m, 2H), 1.83-1.66 (m, 5H), 1.29 (d, J=10.0 Hz, 3H), 0.82 (q, J=7.2 Hz, 3H); ¹⁹F NMR (376 MHz, methanol-d₄) δ -121.198, -139.196; SFC: >99% ee, (S,S)Whelk-O1 100×4.6 mm I.D., 3.5 μm column, A: CO₂, B: 40% [IPA+ACN (w/0.05% DEA)], 3 mL/min, 220 nm, t_R: 1.705 min; LCMS [ESI, M+1, M/2+1]: m/z=620.2, 310.7.

Example 302

(1485) ##STR00617##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((3S,7aR)-3-(hydroxymethyl)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1486) The title compound was synthesized according to the procedure described for example 301. ^{sup}.1H NMR (400 MHz, methanol-d.sub.4) δ ppm=9.21 (s, 1H), 7.68 (dd, J=9.2, 6.0 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.28-7.21 (m, 1H), 7.08-7.04 (m, 1H), 4.55 (br d, J=13.2 Hz, 1H), 4.40-4.20 (m, 3H), 3.68-3.59 (m, 1H), 3.59-3.50 (m, 2H), 3.50-3.40 (m, 1H), 3.16-3.06 (m, 1H), 3.02-2.87 (m, 2H), 2.48 (qd, J=7.2, 14.4 Hz, 1H), 2.28-2.10 (m, 3H), 2.09-2.02 (m, 1H), 2.01-1.94 (m, 2H), 1.93-1.88 (m, 1H), 1.88-1.67 (m, 6H), 1.34-1.24 (m, 3H), 0.82 (q, J=7. Hz, 3H); HPLC:>99% ee, Chiralcel IC-3 50×4.6 mm I.D., 3 μm column A: 50% (MeOH:ACN=4:1) (w/0.05% DEA), B: CO.sub.2, 3 mL/min, 220 nm, t.sub.R: 0.519 min; LCMS (ESI, M+1): m/z=620.2.

Example 303

(1487) ##STR00618##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((3R,7aR)-3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1488) The title compound was synthesized according to the procedure described for example 301. ^{sup}.1H NMR (400 MHz, methanol-d.sub.4) δ=9.21 (s, 1H), 7.67 (dd, J=6.0, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.06 (t, J=2.4 Hz, 1H), 4.56 (br d, J=12.8 Hz, 1H), 4.42-4.36 (m, 1H), 4.35-4.26 (m, 2H), 3.85 (dd, J=6.8, 11.2 Hz, 1H), 3.73 (dd, J=6.0, 11.2 Hz, 1H), 3.68-3.55 (m, 1H), 3.51-3.36 (m, 2H), 3.06-2.96 (m, 1H), 2.93-2.81 (m, 1H), 2.55-2.40 (m, 1H), 2.29-2.11 (m, 3H), 2.09-1.97 (m, 1H), 1.95-1.83 (m, 5H), 1.82-1.65 (m, 4H), 1.29 (d, J=10.4 Hz, 3H), 0.87-0.75 (m, 3H); LCMS (ESI, M+1): m/z=620.4.

Example 304

(1489) ##STR00619##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((3S,7aS)-3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1490) The title compound was synthesized according to the procedure described for example 301. ^{sup}.1H NMR (400 MHz, methanol-d.sub.4) δ=9.22 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.37-7.18 (m, 2H), 7.06 (s, 1H), 4.61-4.51 (m, 1H), 4.49-4.39 (m, 1H), 4.39-4.24 (m, 2H), 3.88-3.79 (m, 1H), 3.79-3.70 (m, 1H), 3.69-3.56 (m, 1H), 3.52-3.42 (m, 2H), 3.15-3.03 (m, 1H), 3.02-2.86 (m, 11H), 2.57-2.38 (m, 1H), 2.29-2.12 (m, 3H), 2.11-2.02 (m, 1H), 1.99-1.84 (m, 5H), 1.84-1.69 (m, 4H), 1.35-1.25 (m, 3H), 0.90-0.72 (m, J=7.2, 3H); LCMS (ESI, M+1): m/z=620.2.

Example 305

(1491) ##STR00620##

(R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2S,7aR)-2-hydroxytetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(1492) The title compound was synthesized according to the procedure described for example 206 except for second eluted peak ((R)-7-(2-(((2S,7aR)-2-((tert-butyl)diphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one) was collected during chiral prep HPLC in step C and carried through steps D, E and F. ^{sup}.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.15 (s, H), 7.69 (dd, J=9.2, 6.0 Hz, 1H), 7.32 (d, J=2.4 Hz, 11H), 7.26 (t, J=9.2 Hz, 1H), 7.05 (dd, J=2.8, 4.4 Hz, 1H), 4.74-4.68 (m, 1H), 4.65-4.52 (m, 3H), 4.45-4.29 (m, 1H), 4.10-3.94 (m, 1H), 3.88-3.79 (m, 2H), 3.75-3.60 (m, 2H), 3.41 (d, J=12 Hz, 1H), 3.29 (s, 2H), 2.58-2.19 (m, 10H), 1.97-1.82 (m, 4H), 0.86-0.76 (m, 3H); ^{sup}.19F NMR (377 MHz, MeOD) δ=121.08, 139.33; LCMS (ESI, M+1): m/z=645.1

Example 306

(1493) ##STR00621##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(hydroxymethyl)piperidin-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol
(1494) The title compound was synthesized according to the procedure described for example 134.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4): δ =9.08 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.28-7.22 (m, 1H), 7.06 (s, 1H), 5.41-5.20 (m, 1H), 4.76-4.64 (m, 1H), 4.63-4.50 (m, 1H), 4.37-4.20 (m, 2H), 3.63-3.56 (m, 1H), 3.54-3.44 (m, 2H), 3.38-3.33 (m, 1H), 3.29-3.12 (m, 3H), 3.01 (dt, J=5.6, 9.6 Hz, 1H), 2.51-2.09 (m, 5H), 2.05-1.78 (m, 7H), 1.55-1.41 (m, 1H), 0.81 (t, J=7.6 Hz, 3H); LCMS [ESI, M+1]: m/z=608.3.

Example 307

(1495) ##STR00622##

(R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione
(1496) ##STR00623##

(1497) Step A. (R-benzyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate: Enantiomers of benzyl 2,4-dioxo-1,3,7-triazaspiro[4.5]decane-7-carboxylate were separated by chiral SFC [DAICEL CHIRALPAK IG (250 mm*30 mm, 10 μ m); A: 0.1% NH.sub.3*H.sub.2O B: MeOH, B %: 45%-45% over 4.3 min; 150 min] to afford the title compound (200 mg, 50% yield) as a yellow liquid; LCMS (ESI, M+1): m/z=304.3.

(1498) Step B. (R)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of benzyl 2-oxo-1,3,9-triazaspiro[4.5]decane-9-carboxylate (200 mg, 1.0 equiv.) in MeOH (10 mL) was added Pd/C (200 mg, 10% purity) under N.sub.2. The suspension was degassed in vacuum and purged with H.sub.2 several times. The reaction was stirred under H.sub.2 (15 psi) at 20° C. for 2 hours. The reaction mixture was filtered and the filtrate was concentrated to afford the title compound (150 mg, 71% yield) as a colorless liquid; LCMS (ESI, M+1): m/z=170.2.

(1499) Step C. (R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (80.0 mg, 1.0 equiv.), (R)-1,3,7-triazaspiro[4.5]decane-2,4-dione (45.7 mg, 2.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (1 mL) was added DIEA (52.3 mg, 3.0 equiv.). The mixture was stirred at 40° C. for 12 hours and then concentrated. The residue was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75*30 mm*3 μ m; mobile phase: water (0.225% formic acid)/ACN] B %: 15%-45%, 7 min] and lyophilized to afford the title compound (49.0 mg, 54.6% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.14 (d, J=6.4 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.56-5.31 (m, 1H), 4.64 (br t, J=12.8 Hz, 1H), 4.53-4.36 (m, 3H), 3.90-3.71 (m, 2H), 3.70-3.58 (m, 1H), 3.58-3.45 (m, 2H), 3.28-3.18 (m, 1H), 2.57-2.46 (m, 1H), 2.45-2.33 (m, 2H), 2.33-2.21 (m, 2H), 2.07 (br s, 4H), 2.06-1.90 (m, 3H), 0.90-0.70 (m, 3H); LCMS (ESI, M+1): m/z=662.3.

Example 308

(1500) ##STR00624##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azepan-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1501) The title compound was synthesized according to the procedure described for example 135.
.sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.24 (dd, J=2.0, 3.2 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.33-7.21 (m, 2H), 7.04 (dd, J=2.4, 14.4 Hz, 1H), 5.55-5.28 (m, 1H), 5.05-4.92 (m, 1H), 4.64-4.46 (m, 2H), 4.36-4.23 (m, 1H), 4.21-4.04 (m, 1H), 3.97-3.73 (m, 2H), 3.65-3.39 (m, 3H), 3.25-3.15 (m, 1H), 3.14-3.08 (m, 3H), 2.61-2.41 (m, 2H), 2.40-2.23 (m, 4H), 2.21-1.99 (m, 6H), 1.93-1.80 (m, 1H), 1.56-1.36 (m, 1H), 0.94-0.66 (m, 3H); LCMS (ESI, M+1): m/z=670.3.

Example 309

(1502) ##STR00625##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azocan-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1503) ##STR00626##

(1504) Step A. 2,7-dichloro-8-fluoro-4-(3-(methylsulfonyl)azocan-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (130 mg, 1 equiv.), 3-methylsulfonylazocane (129.00 mg, 1.1 equiv., HCl) in THE (1.5 mL) was added DIEA (333 mg, 5.0 equiv.). The mixture was stirred at -40° C. for 2 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated to afford the title compound (220 mg, crude) as a yellow solid; LCMS (ESI, M+1): m/z=407.0.

(1505) Step B. 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(34methylsulfonyl)azocan-1-yl)pyrido[4,3-d]pyrimidine: To a solution of 2,7-dichloro-8-fluoro-4-(3-methylsulfonylazocan-1-yl)pyrido[4,3-d]pyrimidine (210 mg, 1.0 equiv.), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (164 mg, 2.0 equiv.) and 4 Å MOLECULAR SIEVES (100 mg) in dioxane (1.5 mL) was added DIEA (200 mg, 3.0 equiv.). The reaction was stirred at 95° C. for 12 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (90.0 mg, 32.5% yield) as a yellow solid; LCMS (ESI, M+1): m/z=530.1.

(1506) Step C. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azocan-1-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a solution of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3-(methylsulfonyl)azocan-1-yl)pyrido[4,3-d]pyrimidine (80.0 mg, 1.0 equiv.), Cs.sub.2CO.sub.3 (1.5 M, 302 µL, 3.0 equiv.) and CataCXium A Pd G3 (11.0 mg, 0.1 equiv.) in methoxycyclopentane (1 mL) was added 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (62.0 mg, 1.3 equiv.). The reaction was stirred at 95° C. for 3 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated, and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to give a crude product. The crude product was purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: water (0.225% formic acid)/ACN] B %: 18%-48%, 7 min] and lyophilized to afford the title compound (28.4 mg, 26.8% yield,) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.23 (d, J=4.3 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.34-7.18 (m, 2H), 7.05 (d, J=2.0 Hz, 1H), 5.47-5.25 (m, 1H), 5.19-5.07 (m, 1H), 4.59 (dt, J=4.8, 9.6 Hz, 1H), 4.48-4.32 (m, 2H), 3.96-3.74 (m, 3H), 3.48-3.33 (m, 2H), 3.26 (br s, 1H), 3.15-3.03 (m, 4H), 2.53-2.32 (m, 3H), 2.30-2.21 (m, 1H), 2.20-2.13 (m, 2H), 2.11-1.91 (m, 7H), 1.89-1.68 (m, 2H), 1.57-1.41 (m, 1H), 0.86-0.70 (m, 3H); LCMS (ESI, M+1): m/z=684.3.

Example 310

(1507) ##STR00627##

3-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)azetidine-1-carboxamide

(1508) The title compound was synthesized according to the procedure described for example 135. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.29 (s, 1H), 8.43 (s, 1H), 7.82-7.63 (m, 1H), 7.44-7.19 (m, 2H), 7.15-6.98 (m, 1H), 5.63-5.38 (m, 1H), 5.02-4.97 (m, 1H), 4.64-4.48 (m, 2H), 4.41 (br t, J=8.0 Hz, 2H), 4.30-4.13 (m, 2H), 3.93-3.60 (m, 3H), 2.74-2.00 (m, 9H), 0.91-0.70 (m, 3H); LCMS (ESI, M+1): m/z=608.2.

Example 311

(1509) ##STR00628##

(2R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-2-ol (1510) ##STR00629##

(1511) Step A. (R)-1-(2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (146 mg, 1.0 equiv.) in dioxane (2 mL) were added ((2S,7aR)-2-((tert-butyldiphenylsilyl)oxy)tetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (209 mg, 1.2 equiv.), DIEA (171 mg, 3.0 equiv.) and 4 Å molecular sieves (15 mg). The reaction was stirred at 90° C. for 16 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃ and concentrated in vacuum to remove acetonitrile. Then the mixture was extracted with ethyl acetate (30 mL). The organic phase was washed with brine (30 mL), dried over anhydrous sodium sulfate, filtered and concentrated to afford the title compound (171 mg, 48% yield) as a yellow oil; ¹H NMR (400 MHz, methanol-d₄) δ=8.98 (s, 1H), 7.65-7.65 (m, 1H), 7.65-7.64 (m, 1H), 7.70-7.61 (m, 2H), 7.47-7.30 (m, 6H), 4.64-4.51 (m, 3H), 4.21-4.13 (m, 2H), 3.54-3.47 (m, 1H), 3.27-3.18 (m, 2H), 3.16-3.03 (m, 3H), 2.16-2.04 (m, 4H), 1.92-1.59 (m, 6H), 1.24 (s, 3H), 1.08-1.03 (m, 9H).

(1512) Step B. (R)-1(2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpyridin-3-ol: A mixture of (R)-1-(2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (205 mg, 1.0 equiv.), ((5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-yl)oxy)triisopropylsilane (168 mg, 1.2 equiv.), K₃PO₄ (1.5 M in water, 3.0 equiv.) and CataCXium A Pd G3 (21.6 mg, 0.1 equiv.) in methoxycyclopentane (2 mL) was de-gassed and stirred at 90° C. for 2 hours under N₂. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, concentrated in vacuum to remove acetonitrile. Then the mixture was diluted with ethyl acetate (20 mL) and water (30 mL), extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, filtered and concentrated to afford the title compound (160 mg, 54% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.21 (d, J=3.4 Hz, 1H), 7.78-7.72 (m, 1H), 7.69-7.62 (m, 4H), 7.45-7.26 (m, 8H), 7.09 (dd, J=2.4, 5.6 Hz, 1H), 4.61-4.54 (m, 1H), 4.44 (br d, J=13.2 Hz, 1H), 4.28-4.13 (m, 3H), 3.64-3.49 (m, 1H), 3.21-3.12 (m, 3H), 2.87-2.79 (m, 1H), 2.55-2.42 (m, 1H), 2.13 (br d, J=7.6 Hz, 4H), 1.99-1.66 (m, 6H), 1.39-1.30 (m, 3H), 1.29-1.21 (m, 5H), 1.14 (dd, J=2.0, 7.2 Hz, 18H), 1.08-1.04 (m, 9H), 0.81 (t, J=7.2 Hz, 3H); LCMS [ESI, M/2+1]: m/z=501.0.

(1513) Step C. (2R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-2-ol: A solution of (R)-1-(2-(((2R,7aS)-2-((tert-butyldiphenylsilyl)oxy)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (120 mg, 1.0 equiv.) and CsF (273 mg, 15 equiv.) in DMF (0.6 mL) was stirred at 18° C. for 7 hrs. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and lyophilized to afford the title compound (39 mg, 53% yield) as a white solid; ¹H NMR (400 MHz, CD₃OD, 298K) δ=9.24 (s, 1H), 8.54 (br s, 1H), 7.68 (br dd, J=5.6, 8.4 Hz, 1H), 7.31 (br d, J=2.4 Hz, 1H), 7.25 (br t, J=9.2 Hz, 11H), 7.05 (br s, 1H), 4.67-4.56 (m, 2H), 4.54-4.41 (m, 2H), 4.38-4.27 (m, 1H), 3.68-3.41 (m, 5H), 3.17-3.03 (m, 1H), 2.51-2.41 (m, 2H), 2.32-2.16 (m, 5H), 2.11-1.98 (m, 2H), 1.93-1.74 (m, 3H), 1.30 (br d, J=9.2 Hz, 3H), 0.90-0.75 (m, 3H); SFC analysis: Column: Chiralpak AS-3 50×4.6 mm I.D., 3 μm; Mobile phase:

Phase A for CO.sub.2, and Phase B for MeOH (0.05% DEA); Gradient elution: B in A from 5% to 40%; Flow rate: 3 mL/min; Detector: DAD; Column Temp: 35° C.; Back Pressure: 100 Bar; LCMS [EST, M+1]: m/z=606.3.

Example 312

(1514) ##STR00630##

N-(1-(dimethylamino)propan-7-yl)-7-(R-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carboxamide

(1515) ##STR00631##

(1516) Step A. (R)-7-chloro-8-fluoro-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carbonitrile: To a mixture of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (800 mg, 1.0 equiv.), NaCN (280 mg, 2.4 equiv.) and DABCO (27.1 mg, 0.1 equiv.) in DMSO (10 mL) was stirred at 60° C. for 12 hours. After reaction completion, the mixture was diluted with ethyl acetate (20 mL) and extracted with water (3×10 mL). The organic layer was washed with brine (20 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (170 mg, 19% yield) as a black brown oil. LCMS [ESI, M+1]: m/z=322.2.

(1517) Step B. (R)-methyl 7-chloro-8-fluoro-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carboxylate: To a solution of (R)-7-chloro-8-fluoro-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carbonitrile (170 mg, 1.0 equiv.) in MeOH (0.5 mL) was added HCl-MeOH (4 M, 5 mL, 37.8 equiv.). The mixture was stirred at 50° C. for 1 hour. After completion, the reaction mixture was concentrated under reduced pressure to give a residue, diluted sat. aq. NaHCO.sub.3 to adjust the pH to 9, extracted with ethyl acetate (3×5 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated to afford the title compound (143 mg, crude) as a black brown oil; LCMS [ESI, M+1]: m/z=355.2.

(1518) Step C. 7-chloro-N-(1-(dimethylamino)propan-2-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carboxamide: To a solution of (R)-methyl 7-chloro-8-fluoro-4-(3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carboxylate (135 mg, 1.0 equiv.) in MeOH (0.5 mL) was added N1,N1-dimethylpropane-1,2-diamine (777 mg, 20.0 equiv.). The reaction was stirred at 25° C. for 1 hour. After completion, the reaction mixture was filtered. The filtrate was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (33 mg, 20% yield over two steps) as a yellow oil; LCMS [EST, M+1]: m/z=425.2.

(1519) Step D. N-(1-(dimethylamino)propan-2-yl)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carboxamide: To a mixture of 7-chloro-N-(1-(dimethylamino)propan-2-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidine-2-carboxamide (15.0 mg, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (14.5 mg, 1.3 equiv.), K.sub.3PO.sub.4 (1.5 M, 70.6 µL, 3.0 equiv.) in methoxycyclopentane (1 mL) was added CataCXium A Pd G3 (2.57 mg, 0.1 equiv.). The reaction was stirred at 90° C. for 2 hours. After reaction completion, the mixture was quenched by water (2 mL) and extracted with ethyl acetate (5 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: water (0.1% formic acid)/ACN, B %: 9%-39%, 10 min) and lyophilized to afford the title compound (6.75 mg, 33% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4): δ=9.41 (dd, J=3.2, 4.8 Hz, 1H), 8.53 (s, 1H), 7.70 (dd, J=5.6, 9.2 Hz, 1H), 7.33 (d, J=2.4 Hz, 1H), 7.27 (t, J=9.2 Hz, 1H), 7.11-7.08 (m, 1H), 4.78 (br d, J=13.6 Hz, 1H), 4.44 (br s, 2H), 3.63-3.45 (m, 2H), 3.04-2.92 (m, 1H.sub.4), 2.76-2.68 (m, 1H), 2.56 (s, 6H), 2.49-2.37 (m, 1H), 2.28-2.11 (m, 2H), 1.93-1.75 (m, 3H), 1.36-1.29 (m, 6H), 0.84-0.76 (m, 3H); LCMS [ESI, M+1]: m/z=579.4.

(1520) ##STR00632##

((3S,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate

(1521) ##STR00633##

(1522) Step A. (R)-1-(2-(((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of ((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (408 mg, 1.1 equiv.uiv), 4 Å molecular sieves (20 mg) and t-BuONa (261 mg, 3.0 equiv.) in toluene (3 mL) was stirred at 0° C. for 10 mins. Then (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (300 mg, 1.0 equiv.) was added and the mixture was stirred at 0° C. for 1 hour. The reaction mixture was quenched by water (40 mL) at 0° C. and extracted with ethyl acetate (3×20 mL). Then the combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃ and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (3×30 mL). The combined organic layers were dried over anhydrous sodium sulfate and concentrated in vacuum to afford the title compound (484 mg, 66.4% yield) as a black solid; LCMS (ESI, M+1): m/z=704.2.

(1523) Step B. (R)-1-(2-(((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(2-(((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-chloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (495 mg, 1.0 equiv.) in methoxycyclopentane (8 mL) was added 2-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (329 mg, 1.3 equiv.), K₃PO₄ (1.5 M in water, 3.0 equiv.) and CataCXium A Pd G3 (51.2 mg, 0.1 equiv.) at 18° C. under N₂. The mixture was degassed and then heated to 90° C. for 3 hours. The reaction mixture was diluted with ethyl acetate (30 mL) and water (50 mL), the aqueous phase was extracted with ethyl acetate (20 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃ and concentrated in vacuum to remove ACN. Then the mixture was extracted with ethyl acetate (20 mL×2). The combined organic phase was dried over anhydrous sodium sulfate and concentrated to afford the title compound (327 mg, 48.8% yield,) as a yellow solid; LCMS (ESI, M+1): m/z=902.3.

(1524) Step C. (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((3S,7aR).sub.3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(2-(((3S,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 1.0 equiv.) in DMF (2 mL) was added CsF (505 mg, 15 equiv.). The reaction mixture was stirred at 40° C. for 12 hrs. The reaction mixture was diluted with ethyl acetate (15 mL) and water (30 mL), the aqueous phase was extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃ and concentrated in vacuum to remove acetonitrile. Then the mixture was diluted

with ethyl acetate (20 mL×2), dried over anhydrous sodium sulfate and concentrated to afford the title compound (107 mg, 72.6% yield) as a yellow solid.

(1525) Step D. ((3S,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yloxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate: To a solution of (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((3S,7aR)-3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (107 mg, 1.0 equiv.) in THF (1 mL) was added NaH (12.9 mg, 2.0 equiv.) at 0° C. After stirring at 0° C. for 0.5 hour, N,N-dimethylcarbamoyl chloride (34.7 mg, 2.0 equiv.) in THF (1 mL) was added to the above mixture and the reaction was stirred at 0° C. for 2 hours. The mixture was quenched with water (10 mL) and extracted with ethyl acetate (8 mL×2). The combined organic layers were washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by prep-HPLC [column: Water s Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH₄HCO₃)/ACN; B %: 56%-86%, 8 min]. The desired fraction was collected and lyophilized to afford the title compound (64 mg, 53.5% yield) as white solid; LCMS (ESI, M+1): m/z=735.5.

(1526) Step E. ((3S,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate: To a solution of ((3S,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate (59 mg, 1.0 equiv.) in ACN (0.2 mL) was added HCl.dioxane (4 M, 20 equiv.) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. Then the mixture was diluted with ethyl acetate (20 mL×2), dried over anhydrous sodium sulfate and concentrated in vacuum. Then the residue was purified by SFC (REGIS(S,S)WHELK-O1(250 mm×25 mm, 10 µm); A: [0.1% NH₄HCO₃]/EtOH; B %: 50%-50%, 4.0 min; 30 min). The desired fraction was collected and concentrated in vacuum to afford the title compound (19.9 mg, 16% yield) as a yellow solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.18 (s, 1H), 8.55 (s, 1H), 7.62 (dd, J=6.0, 9.0 Hz, 1H), 7.23 (d, J=2.4 Hz, 1H), 7.22 (s, 1H), 7.04 (t, J=2.6 Hz, 1H), 4.59-4.44 (m, 1H), 4.32-4.20 (m, 3H), 4.11-4.02 (m, 1H), 4.00-3.92 (m, 1H), 3.70-3.56 (m, 1H), 3.51-3.43 (m, 1H), 3.10-2.99 (m, 2H), 2.98-2.86 (m, 6H), 2.85-2.77 (m, 1H), 2.55-2.39 (m, 1H), 2.23-2.13 (m, 3H), 2.07-1.99 (m, 1H), 1.98-1.84 (m, 4H), 1.83-1.72 (m, 4H), 1.71-1.62 (m, 1H), 1.29 (d, J=10.8 Hz, 3H), 0.81 (q, J=7.2 Hz, 3H); ¹⁹F NMR (400 MHz, DMSO, 273 K) δ=-122.415, -139.163; HPLC: >99% ee, Chiralcel ID-3 100×4.6 mm I.D., 3 µm column A: 20% (IPA/ACN=4/1) (w/0.05% DEA), B: Heptane (0.05% DEA), 254 nm, t_R: 0.817 min; LCMS [ESI, M+1]: m/z=691.5.

Example 314

(1527) ##STR00634##

((3S,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate

(1528) The title compound was synthesized according to the procedure described for example 313. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.24-9.19 (m, 1H), 7.71-7.65 (m, 1H), 7.32-7.29 (m, 1H), 7.28-7.22 (m, 1H), 7.07 (t, J=2.4 Hz, 1H), 4.58-4.51 (m, 1H), 4.42-4.38 (m, 1H), 4.35-4.25 (m, 4H), 3.69-3.59 (m, 1H), 3.52-3.48 (m, 3H), 2.98-2.91 (m, 6H), 2.90-2.82 (m, 1H), 2.55-2.42 (m, 1H), 2.28-2.12 (m, 3H), 2.07-1.98 (m, 1H), 1.94-1.84 (m, 5H), 1.83-1.75 (m, 3H), 1.74-1.65 (m, 1H), 1.29 (d, J=10.0 Hz, 3H), 0.86-0.77 (m, 3H); LCMS (ESI, M+1): m/z=691.5.

Example 315

(1529) ##STR00635##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1530) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₄) δ=9.19 (s, 1H), 7.69 (dd, J=6.0, 9.2 Hz, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.8 Hz, 1H), 5.64-5.42 (m, 1H), 4.67-4.55 (m, 2H), 4.36-4.20 (m, 4H), 4.05 (t, J=4.8 Hz, 2H), 3.98-3.71 (m, 5H), 3.44-3.34 (m, 1H), 2.72-2.50 (m, 2H), 2.50-2.33 (m, 2H), 2.32-2.24 (m, 2H), 2.23-2.05 (m, 4H), 0.80 (dt, J=1.6, 7.6 Hz, 3H); LCMS (ESI, M+1): m/z=594.3.

Example 316

(1531) ##STR00636##

(R)-1-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1532) ##STR00637##

(1533) Step A. (R)-1-(7-chloro-2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (350 mg, 1.0 equiv.) and (1-((dimethylamino)methyl)cyclopropyl)methanol (205 mg, 1.5 equiv.) in DMF (4 mL) was added DIEA (410 mg, 3.0 equiv.) and 4 Å molecular sieves (110 mg). The mixture was stirred at 40° C. for 12 hours. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (360 mg, 80% yield) as a yellow solid; LCMS [ESI, M+1]: m/z=424.2.

(1534) Step B. (R)-1-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (R)-1-(7-chloro-2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50 mg, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (44.7 mg, 1.2 equiv.), K₃PO₄ (1.5 M, 236 IL, 3.0 equiv.) and CataCXium A Pd G3 (8.59 mg, 0.1 equiv.) in methoxycyclopentane (1 mL) was degassed and stirred at 90° C. for 2 hours under N₂ atmosphere. The mixture was diluted with water (5 mL) and extracted with ethyl acetate (4×5 mL). The combined organic layers were dried over anhydrous Na₂SO₄, concentrated and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH₄HCO₃)/ACN; B %: 44%-74%, 10 min] to afford the title compound (24.6 mg, 36% yield) as a yellow solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.21 (s, 1H), 7.69-7.66 (m, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.27-7.22 (m, 1H), 7.05 (d, J=2.4 Hz, 1H), 4.54-4.50 (m, 1H), 4.44-4.34 (m, 2H), 4.27 (t, J=12.8 Hz, 1H), 3.70-3.55 (m, 1H), 3.52-3.40 (m, 1H), 2.62-2.41 (m, 3H), 2.36 (s, 6H), 2.25-2.10 (m, 2H), 1.93-1.71 (m, 3H), 1.29 (d, J=10.4 Hz, 3H), 0.84-0.78 (m, 3H), 0.77-0.72 (m, 2H), 0.63-0.50 (m, 2H); LCMS [ESI, M+1]: m/z=578.4.

Example 317

(1535) ##STR00638##

4-(4-(azepan-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1536) ##STR00639##

(1537) 4-(4-(azepan-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (100 mg, 1.0 equiv.) and azepane (25.1 mg, 1.5 equiv) in DMF (1 mL) were added DIEA (65.4 mg, 3 equiv.) and 4 Å molecular sieves (30 mg). The mixture was stirred at 40° C. for 12 hours. The mixture was filtered and the filtrate was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase:

[water (10 mM NH₄CO₃/ACN)] B %: 60%-90%, 10 min) to afford the title compound (49.1 mg, 47% yield) as a yellow solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.13 (s, H), 7.69-7.65 (m, 1H), 7.30 (d, J=2.6 Hz, 1H), 7.24 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.40-5.20 (m, 1H), 4.35-4.21 (m, 2H), 4.17-4.03 (m, 4H), 3.30-3.20 (m, 2H), 3.20-3.13 (m, 1H), 3.05-2.95 (m, 1H), 2.52-2.46 (m, 1H), 2.38-2.10 (m, 4H), 2.08-1.95 (m, 6H), 1.93-1.82 (m, 1H), 1.73-1.71 (s, 4H), 0.82-0.78 (m, 3H); LCMS [ESI, M+1]: m/z=592.3.

Example 318

(1538) ##STR00640##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decane-1,3-dione

(1539) The title compound was synthesized according to the procedure described for example 135.

¹H NMR (400 MHz, METHANOL-d₄) δ=9.10 (d, J=1.6 Hz, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (dd, J=2.4, 4.8 Hz, 1H), 5.51-5.30 (m, 1H), 4.67-4.59 (m, 1H), 4.52-4.32 (m, 3H), 4.01-3.89 (m, 1H), 3.89-3.77 (m, 1H), 3.64-3.39 (m, 3H), 3.23-3.12 (m, 1H), 2.95 (dd, J=14.4, 18.0 Hz, 1H), 2.67 (d, J=18.0 Hz, 1H), 2.53-2.29 (m, 3H), 2.27-1.79 (m, 9H), 0.87-0.72 (m, 3H); LCMS [ESI, M+1]: m/z=661.4.

Example 319

(1540) ##STR00641##

(S)-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1541) The title compound was synthesized according to the procedure described for example 309

except for K₃PO₄ was used in step C instead of Cs₂CO₃. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.07 (s, 1H), 7.69-7.65 (m, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=12.4 Hz, 1H), 7.06-7.04 (m, 1H), 4.45-4.39 (m, 2H), 4.38-4.21 (m, 2H), 4.02-3.96 (m, 1H), 3.85-3.70 (m, 1H), 2.92-2.86 (m, 1H), 2.76-2.73 (m, 1H), 2.69-2.55 (m, 6H), 2.53-2.38 (m, 1H), 2.23-2.14 (m, 1H), 2.13-2.08 (m, 1H), 2.06-1.99 (m, 1H), 1.98-1.91 (m, 2H), 1.83-1.81 (m, 4H), 0.82-0.77 (m, 3H), 0.74-0.68 (m, 2H), 0.59-0.52 (m, 2H); LCMS [ESI, M+1]: m/z=629.3.

Example 320

(1542) ##STR00642##

(1R,5R,6R)-3-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol

(1543) The title compound was synthesized according to the procedure described for example 309.

¹H NMR (400 MHz, METHANOL-d₄) δ=9.29-9.19 (m, 1H), 8.54 (s, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.28-7.21 (m, 1H), 7.06 (dd, J=2.4, 15.2 Hz, 1H), 4.98 (br d, J=13.2 Hz, 1H), 4.82-4.68 (m, 1H), 4.51-4.37 (m, 2H), 4.36-4.27 (m, 1H), 3.85-3.71 (m, 1H), 3.55-3.38 (m, 1H), 3.23-2.95 (m, 6H), 2.56-2.35 (m, 2H), 2.31-2.11 (m, 3H), 2.04-1.88 (m, 5H), 1.82 (br s, 1H), 1.37-1.36 (m, 1H), 0.89-0.68 (m, 7H); LCMS [ESI, M+1]: 616.4.

Example 321

(1544) ##STR00643##

(S)-4-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(pyrrolidin-1-ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1545) The title compound was synthesized according to the procedure described for example 309.

¹H NMR (400 MHz, METHANOL-d₄) δ=9.59 (d, J=5.6 Hz, 1H), 7.69 (dd, J=5.6, 9.2 Hz, 1H), 7.32 (d, J=2.0 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (dd, J=2.4, 14.2 Hz, 1H), 4.63-4.52 (m, 2H), 4.51-4.42 (m, 2H), 4.25-4.15 (m, 1H), 4.07-3.84 (m, 3H), 3.74-3.64 (m, 2H), 3.60-3.42 (m, 3H), 3.35 (s, 3H), 2.55-2.40 (m, 1H), 2.29-2.15 (m, 1H), 2.14-2.04 (m, 4H), 1.28 (d, J=2.0 Hz, 3H), 0.98-0.85 (m, 4H), 0.84-0.76 (m, 3H); LCMS (ESI, M+1): m/z=620.3.

Example 322

(1546) ##STR00644##

(R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(pyrrolidin-1-

ylmethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1547) ##STR00645##

(1548) Step A. 7-chloro-8-fluoro-2-((1-(pyrrolidin-1-yl)methyl)cyclopropyl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a suspension of 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (3.00 g, 1.0 equiv.) and 4 Å molecular sieves (3.0 g) in THF (600 mL) was added a solution of [1-(pyrrolidin-1-ylmethyl)cyclopropyl]methanol (1.47 g, 1.0 equiv.) and DIEA (3.68 g, 3.0 equiv.) in THF (120 mL) at 0° C. The mixture was stirred at 0° C. for 1 hour, then 20° C. for 2 hours. The mixture was quenched by addition of brine (200 mL) and extracted with ethyl acetate (3×100 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (260 mg, 12% yield) as a yellow solid; LCMS (ESI, M+1): m/z=435.0.

(1549) Step B. 5-ethyl-6-fluoro-4-(8-fluoro-2-((1-(pyrrolidin-1-yl)methyl)cyclopropyl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: To a mixture of 7-chloro-8-fluoro-2-[[1-(pyrrolidin-1-ylmethyl)cyclopropyl]methoxy]-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (218 mg, 1.5 equiv.) and Cs.sub.2CO.sub.3 (1.5 M, 920 µL, 3.0 equiv.) in methoxycyclopentane (3 mL) was added CataCXium A Pd G3 (33.5 mg, 0.10 equiv.). The reaction mixture was stirred at 90° C. for 3 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (160 mg, 57% yield) as a yellow solid; LCMS (ESI, M+1): m/z=589.2.

(1550) Step C. (R)-7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(pyrrolidin-1-yl)methyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of 5-ethyl-6-fluoro-4-[8-fluoro-2-[[1-(pyrrolidin-1-ylmethyl)cyclopropyl]methoxy]-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (100 mg, 1.0 equiv.), (5R)-1,3,9-triazaspiro[4.5]decane-2,4-dione (57.5 mg, 2.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (1 mL) was added DIEA (110 mg, 5.0 equiv.). The reaction was stirred at 40° C. for 12 hours. The residue was filtered and washed with DMF (1 mL), and purified by prep-HPLC [column: Phenomenex C18 75×30 mm×3 µm; mobile phase: water (0.1% formic acid)/ACN; B %: 15%-45%, 7 min] and lyophilized to afford the title compound (47.9 mg, 43% yield) as a yellow solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.14 (d, J=10.0 Hz, 1H), 7.69 (dd, J=6.0, 8.4 Hz, 1H), 7.32 (t, J=2.0 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (dd, J=2.8, 7.2 Hz, 1H), 4.60 (br t, J=14.2 Hz, 3H), 4.50-4.29 (m, 3H), 3.91-3.80 (m, 1H), 3.80-3.56 (m, 1H), 3.30-3.25 (m, 2H), 3.20 (s, 2H), 2.57-2.29 (m, 1H), 2.29-2.10 (m, 2H), 2.10-1.98 (m, 5H), 1.97 (br d, J=3.6 Hz, 2H), 0.95-0.84 (m, 2H), 0.83-0.69 (m, 5H); LCMS (ESI, M+1): m/z=658.3.

Example 323

(1551) ##STR00646##

trans-(2R,4r)-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1552) ##STR00647##

(1553) Step A. 6-(2,7-dichloro-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol: To a solution of 2,4,7-trichloro-8-fluoro-pyrido[4,3-d]pyrimidine (2.0 g, 1.0 equiv.) in DCM (50 mL) was added DIEA (5.12 g, 6.90 mL, 5.0 equiv.) and 6-azaspiro[3.5]nonan-2-ol (1.13 g, 0.8 equiv., HCl). The reaction was stirred at -40° C. for 0.5 hour. The mixture was diluted with water (20 mL) and the layers were separated. The aqueous phase was extracted with ethyl acetate (2×10 mL). Combined the organic layers were dried over anhydrous Na.sub.2SO.sub.4 and concentrated.

The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate 50:1 to 1:1). The product was further purified by chiral SFC separation (DAICEL CHIRALPAK AY-H (250 mm×30 mm, 10 μm); mobile phase: [0.1% NH₃.Math.H₂O EtOH]; B %: 60%-60%, 8.5; 80 min) to afford (2s,4r)-6-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol (trans product) as the first eluting peak (650 mg) as a yellow solid; .sup.1H NMR (400 MHz, dimethylsulfoxide-d₆) δ 8.98 (s, 1H), 5.07-4.99 (d, J=6.2 Hz, 1H), 4.16-4.05 (m, 1H), 3.87-3.94 (m, 4H), 2.20-2.10 (m, 2H), 1.74-1.62 (m, 4H), 1.58-1.50 (m, 2H); LCMS [ESI, M+1]: 357.0. The product was purified by SFC separation (DAICEL CHIRALPAK AY-H (250 mm×30 mm, 10 μm); mobile phase: [0.1% NH₃.Math.H₂O EtOH]; B %: 60%-60%, 8.5; 80 min) to afford the (2r,4s)-6-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol, (cis product) as the second eluting peak (650 mg) as a yellow solid; .sup.1H NMR (400 MHz, dimethylsulfoxide-d₆) δ 9.03 (s, 1H), 4.97 (d, J=6.4 Hz, 1H), 4.12-4.02 (m, 1H), 3.94 (s, 2H), 3.90-3.85 (m, 2H), 2.08-1.99 (m, 2H), 1.74-1.67 (br s, 4H), 1.65-1.59 (m, 2H); LCMS [ESI, M+1]: m/z=357.1.

(1554) Step B. (2R,4r)-6-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol: To a solution of (2s,4r)-6-(2,7-dichloro-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol (100 mg, 1.0 equiv.) in dioxane (2.0 mL) was added DIEA (108 mg, 146 μL, 3.0 equiv.) and ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (66.8 mg, 1.5 equiv.). The reaction was stirred at 90° C. for 16 hours. The mixture was concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, concentrated in vacuum to remove ACN. The aqueous phase was extracted with ethyl acetate (2×6.0 mL). The combined organic phase was dried with anhydrous Na₂SO₄, filtered and concentrated in vacuum to afford the title compound (100 mg, 71% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ 8.73 (s, 1H), 5.44-5.21 (m, 1H), 4.43 (quin, J=7.2 Hz, 1H), 4.38-4.27 (m, 2H), 3.86-3.80 (m, 4H), 3.48-3.21 (m, 3H), 3.08-2.97 (m, 1H), 2.38-2.27 (m, 4H), 2.26-2.19 (m, 2H), 2.03-1.94 (m, 3H), 1.81-1.72 (m, 5H); LCMS [ESI, M+1]: m/z=480.3.

(1555) Step C. trans-(2R,4r)-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol: A mixture of (2R,4r)-6-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol (60 mg, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (47.4 mg, 1.2 equiv.), K₃PO₄ (1.5 M, 250 μL, 3.0 equiv.) and cataCXium-A-Pd-G3 (9.10 mg, 0.1 equiv.) in THF (1.5 mL) was degassed and purged with N₂ for 3 times, and then the reaction was stirred at 60° C. for 2 hours under N₂ atmosphere. The mixture was diluted with water (2.0 mL) and extracted with ethyl acetate (3×2.0 mL). The combined organic layers were dried over anhydrous Na₂SO₄ and concentrated. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and concentrated in vacuum to remove ACN. The aqueous layers was lyophilized. The residue was purified by prep-HPLC [Unisil 3-100 C18 Ultra 150×50 mm×3 μm; mobile phase: water (0.1% formic acid)/ACN, B %: 14%-44% over 15 min]. The desired fractions were collected and lyophilized to afford the title compound (9.89 mg, 11% yield, 0.62FORMIC ACID) as a white solid; .sup.1H NMR (400 MHz, methanol-d₄) δ 9.07 (s, 1H), 7.72-7.72 (m, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.29-7.22 (m, 1H), 7.07 (d, J=2.4 Hz, 1H), 5.49-5.30 (m, 1H), 4.49-4.43 (m, 1H), 4.42-4.37 (m, 1H), 4.32-4.24 (m, 1H), 4.18-4.08 (m, 2H), 4.03-3.86 (m, 2H), 3.59-3.38 (m, 3H), 3.21-3.10 (m, 1H), 2.53-2.46 (m, 1H), 2.44-2.29 (m, 3H), 2.28-2.19 (m, 2H), 2.18-2.07 (m, 3H), 2.05-1.95 (m, 1H), 1.86-1.78 (m, 4H), 1.74-1.65 (m, 2H), 0.84-0.77 (m, 3H); LCMS [ESI, M+1]: m/z=634.4.

Example 324

(1556) ##STR00648##

cis-(2S,4s)-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1557) The title compound was synthesized according to the procedure described for example 323 except using second eluting peak from chiral SFC separation (step A) in further steps B and C.

¹H NMR (400 MHz, methanol-*d*.sub.4) δ=9.06 (s, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.07 (d, J=2.4 Hz, 1H), 5.40-5.22 (m, 1H), 4.36-4.20 (m, 3H), 4.12-3.91 (m, 4H), 3.30-3.11 (m, 3H), 3.06-2.96 (m, 1H), 2.55-2.42 (m, 1H), 2.38-2.11 (m, 6H), 2.04-1.88 (m, 3H), 1.86-1.72 (m, 6H), 0.81 (br t, J=6.4 Hz, 3H); ¹⁹F NMR (376 MHz, methanol-*d*.sub.4) δ=-121.085, -138.973, -173.678; SFC: >99% ee, Chiralpak IF-3 50×4.6 mm I.D., 3 μm column, A: CO₂, B: 30% MeOH (w/0.05% DEA), 3 mL/min, 220 nm, ta: 3.449 min; LCMS (ESI, M+1): m/z=634.3.

Example 325

(1558) ##STR00649##

4-(4-(azocan-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1559) The title compound was synthesized according to the procedure described for example 135.

¹H NMR (400 MHz, DMSO-*d*.sub.6) δ=9.96-9.90 (m, 1H), 9.16-9.08 (m, 11H), 7.76 (dd, J=6.0, 9.1 Hz, 1H), 7.39-7.29 (m, 2H), 7.03 (d, J=2.4 Hz, 1H), 5.42-5.18 (m, 1H), 4.23-3.95 (m, 6H), 3.18-3.01 (m, 3H), 2.90-2.80 (m, 1H), 2.43-2.34 (m, 1H), 2.21-2.05 (m, 3H), 2.04-1.85 (m, 6H), 1.84-1.75 (m, 2H), 1.62 (br d, J=5.2 Hz, 4H), 1.57-1.47 (m, 2H), 0.77-0.67 (m, 3H); LCMS (ESI, M+1): m/z=606.4.

Example 326

(1560) ##STR00650##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1561) The title compound was synthesized according to the procedure described for example 135.

¹H NMR (400 MHz, methanol-*d*.sub.4) δ=9.21 (s, 1H), 7.68 (dd, J=5.6, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.05 (d, J=2.8 Hz, 1H), 6.76 (s, 1H), 5.50-5.19 (m, 3H), 4.66-4.52 (m, 4H), 4.52-4.33 (m, 4H), 3.52-3.43 (m, 2H), 3.34 (s, 3H), 3.23-3.13 (m, 1H), 3.08 (s, 3H), 2.52-2.42 (m, 3H), 2.36 (br dd, J=9.2, 14.0 Hz, 1H), 2.31-2.21 (m, 1H), 2.20-2.07 (m, 3H), 2.06-1.93 (m, 1H), 0.90-0.67 (m, 3H); LCMS (ESI, M+1): m/z=701.3.

Example 327

(1562) ##STR00651##

(3R,5R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-(hydroxymethyl)-3-methylpiperidin-3-ol

(1563) ##STR00652## ##STR00653##

(1564) Step A. benzyl 3-hydroxy-5-(hydroxymethyl)piperidine-1-carboxylate: To a solution of 1-benzyl 3-methyl 5-hydroxypiperidine-1,3-dicarboxylate (5.00 g, 1.0 equiv.) in THF (100 mL) was added LiAlH₄.sub.4 (970 mg, 1.5 equiv.) under N₂ at -40° C. The reaction was stirred at -40° C. for 0.5 hour. The reaction mixture was quenched with water (970.5 mg), 15% NaOH aqueous (970.5 mg), and water (3×970.5 mg) at 0° C. sequentially, then filtered. The filter cake was washed with ethyl acetate (2×25 mL), the filtrate was concentrated to give title compound (4.41 g, crude) as a colorless oil; LCMS (ESI, M+1): m/z=266.2.

(1565) Step B. benzyl 3-(((tert-butyl)diphenylsilyl)oxy)methyl)-5-hydroxypiperidine-1-carboxylate: To a solution of benzyl 3-hydroxy-5-(hydroxymethyl)piperidine-1-carboxylate (4.41 g, 1.0 equiv.) in DCM (45 mL) was added imidazole (2.26 g, 2.0 equiv.) and TBDPSCl (5.48 g, 1.2 equiv.) at 0° C. The reaction mixture was stirred at 25° C. for 1 hour. The mixture was quenched by addition of

water (50 mL), and extracted with DCM (3×50 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to give title compound (4.50 g, 52% yield) as a yellow oil; LCMS (ESI, M+1): m/z=504.4.

(1566) Step C. benzyl 3-(((tert-butyldiphenylsilyl)oxy)methyl)-5-oxopiperidine-1-carboxylate: To a solution of benzyl 3-(((tert-butyldiphenylsilyl)oxy)methyl)-5-hydroxypiperidine-1-carboxylate (4.50 g, 1.0 equiv.) in DCM (50 mL) was added Dess-Martin reagent (7.58 g, 2.0 equiv.). The reaction mixture was stirred at 25° C. for 12 hours. The reaction mixture was quenched by addition of saturated sodium sulfite solution (300 mL) and filtered, then extracted with DCM (150 mL). The organic layer was dried over anhydrous sodium sulfate, concentrated and purified by column chromatography (Silica gel, Petroleum ether/Ethyl acetate 20:1 to 3:1) to give the title compound (2.90 g, 64% yield) as a yellow oil; LCMS (ESI, M+1): m/z=502.1.

(1567) Step D. (3RS,5RS)-benzyl 5-(((tert-butyldiphenylsilyl)oxy)methyl)-3-hydroxy-3-methylpiperidine-1-carboxylate: To a solution of benzyl 3-(((tert-butyldiphenylsilyl)oxy)methyl)-5-oxopiperidine-1-carboxylate (2.90 g, 1.0 equiv.) in THE (20 mL) was added MeMgBr (3 M, 1.2 equiv.) at 0° C. The reaction mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was quenched by addition of water (20 mL) at 0° C., extracted with ethyl acetate (3×20 mL). The combined organic layer was dried over anhydrous sodium sulfate, concentrated and purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) and then column chromatography (Silica gel, Petroleum ether/Ethyl acetate 20:1 to 1:1) to produce a yellow oil, which was further separated by SFC (column: DAICEL CHIRALPAK IC (250 mm×30 mm, 5 µm); mobile phase: [0.1% NH.sub.3H.sub.2O IPA]; B %: 35%-35%, 4.2; 60 min) and (column: REGIS(S,S)WHELK-O1 (250 mm×25 mm, 10 µm); mobile phase: [0.1% NH.sub.3H.sub.2O IPA]; B %: 30%-30%, 7.4; 50 min) to give the title compound (30.0 mg, 1% yield) as a yellow oil; LCMS (ESI, M+23): m/z=540.4.

(1568) Step E. (3S,5S)-5-(((tert-butyldiphenylsilyl)oxy)methyl)-3-methylpiperidin-3-ol: To a suspension of Pd/C (80 mg, 10% purity) in MeOH (1 mL) was added (3RS,5RS)-5-(((tert-butyldiphenylsilyl)oxy)methyl)-3-methylpiperidin-3-ol (30 mg, 57.95 µmol, 1.0 equiv.) under N₂. The suspension was degassed in vacuum and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (15 psi) at 25° C. for 2 hours. The reaction mixture was filtered, the filtrate was concentrated under reduced pressure to give the title compound (15 mg, crude) as a white solid.

(1569) Step F. (3S,5S)-5-(((tert-butyldiphenylsilyl)oxy)methyl)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (20.0 mg, 1.0 equiv.), (3RS,5RS)-5-(((tert-butyldiphenylsilyl)oxy)methyl)-3-methylpiperidin-3-ol (12.9 mg, 1 equiv.), 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was added DIEA (4.36 mg, 1.0 equiv.). The mixture was stirred at 40° C. for 2 hours. The reaction mixture was filtered to give a solution which was purified by reversed-phase flash chromatography (C18, water (0.1% formic acid)/ACN) to give the title compound (20.0 mg, 53% yield) as a yellow solid. LCMS (ESI, M+1): m/z=876.2.

(1570) Step G. (3RS,5RS)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5-(hydroxymethyl)-3-methylpiperidin-3-ol: To a solution of (3RS,5RS)-5-(((tert-butyldiphenylsilyl)oxy)methyl)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (20 mg, 1.0 equiv.) in DMF (0.5 mL) was added CsF (34.7 mg, 10 equiv.). The reaction was stirred at 25° C. for 2 hours. The reaction mixture was filtered to give a solution which was purified by prep-HPLC(column: Phenomenex Luna C18 150×25 mm×10 µm; mobile

phase: water (0.1% formic acid)/ACN, B %: 10%-40%, 10 min) and lyophilized to afford the title compound (6.56 mg, 43% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₃): δ=9.27 (s, 1H), 8.51 (s, 1H), 7.68 (dd, J=5.6, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (dd, J=2.4, 8.0 Hz, 1H), 5.50-5.33 (m, 1H), 4.68-4.56 (m, 2H), 4.53-4.39 (m, 2H), 3.65-3.44 (m, 5H), 3.43-3.34 (m, 1H), 3.22-2.87 (m, 2H), 2.54-2.33 (m, 4H), 2.29-2.09 (m, 4H), 2.05-1.80 (m, 2H), 1.46 (dt, J=5.2, 13.2 Hz, 1H), 1.31 (d, J=6.4 Hz, 3H), 0.86-0.76 (m, 3H); LCMS (ESI, M+1): m/z=638.4.

Example 328

(1571) ##STR00654##

trans-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methyl-6-azaspiro[3.5]nonan-2-ol

(1572) ##STR00655##

(1573) Step A. 6-azaspiro[3.5]nonan-2-one: To a solution of tert-butyl 2-oxo-6-azaspiro[3.5]nonane-6-carboxylate (1.00 g, 1.0 equiv.) in MeCN (5 mL) was added HCl.dioxane (4 M, 9.57 equiv.). The mixture was stirred at 25° C. for 0.5 hour. The mixture was concentrated in vacuum to afford the title compound (734 mg, crude, HCl salt) as a yellow oil.

(1574) Step B. benzyl 2-oxo-6-azaspiro[3.5]nonane-6-carboxylate: To a mixture of 6-azaspiro[3.5]nonan-2-one (3.67 g, 1.0 equiv., HCl) in ethyl acetate (50 mL) was added TEA (10.6 g, 5.0 equiv.) and CbzCl (5.35 g, 1.5 equiv.) at 25° C. The mixture was stirred at 25° C. for 12 hours. The mixture was diluted with water (50 mL) and extracted with ethyl acetate (2×60 mL). The combined organic layers were dried over sodium sulfate, filtered and concentrated in vacuum. The crude product was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (4.70 g, 81% yield over two steps) as a yellow oil; ¹H NMR (400 MHz, chloroform-d) δ=7.40-7.29 (m, 5H), 5.14 (s, 2H), 3.60-3.45 (m, 4H), 2.97-2.75 (m, 2H), 2.74-2.65 (m, 2H), 1.82-1.74 (m, 2H), 1.61 (br s, 2H); LCMS (ESI, M+1): m/z=274.1.

(1575) Step C. trans-benzyl 2-hydroxy-2-methyl-6-azaspiro[3.5]nonane-6-carboxylate: To a solution of benzyl 2-oxo-6-azaspiro[3.5]nonane-6-carboxylate (500 mg, 1.0 equiv.) in acetonitrile (10 mL) was added MeMgBr (3 M, 1.22 mL, 2.0 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hours. The mixture was quenched by addition of water (10 mL) and extracted with ethyl acetate (2×10 mL). The combined organic layers were dried over Na₂SO₄, concentrated and purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN), and stereoisomers were further separated by SFC (column: daicel chiralpak IG 250 mm×30 mm×10 μm; mobile phase: 0.1% NH₃—H₂O in IPA—CO₂, 40%, 2.6 min over 40 min) to afford the title compound (100 mg, 19% yield) as a colorless oil; LCMS (ESI, M+1): m/z=290.2.

(1576) Step D. trans-2-methyl-6-azaspiro[3.5]nonan-2-ol: To a solution of trans-benzyl 2-hydroxy-2-methyl-6-azaspiro[3.5]nonane-6-carboxylate (100 mg, 1.0 equiv.) in MeOH (5 mL) was added Pd(OH)₂/C (20.0 mg, 20% purity) under N₂. The mixture was degassed and purged with H₂ for 3 times, and then stirred at 25° C. for 0.5 hours under H₂ (15 psi). The mixture was filtered and the filtrate was concentrated to afford the title compound (60.0 mg, crude) as a colorless oil.

(1577) Step E. 5-ethyl-6-fluoro-4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (420 mg, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (410 mg, 1.3 equiv.), CataCXium A Pd G3 (145 mg, 0.2 equiv.) and Cs₂CO₃ (1.5 M, 2.00 mL, 3.0 equiv.) in methoxycyclopentane (15 mL) were degassed and stirred at 100° C. for 3 hours under N₂ atmosphere. The mixture was diluted with water (20 mL) and extracted with ethyl acetate (30 mL). The organic layer was dried over sodium sulfate, filtered and concentrated in vacuum. The crude product was purified by reversed phase flash chromatography (water (0.1% formic acid)-ACN), to afford the title compound (210 mg, 32% yield) as a yellow solid; LCMS

(ESI, M+1): m/z=575.3.

(1578) Step F. trans-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methyl-6-azaspiro[3.5]nonan-2-ol: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (100 mg, 1.0 equiv.) and trans-2-methyl-6-azaspiro[3.5]nonan-2-ol (54.0 mg, 2.0 equiv.) in DMF (2 mL) was added 4 Å molecular sieves (20 mg) and DIEA (67.5 mg, 3.0 equiv.). The mixture was stirred at 40° C. for 14 hours. The mixture was filtered and the filtrate was purified by prep-HPLC (column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH₄HCO₃)-ACN, B %: 48%-78%, 10 min) to afford the title compound (35.7 mg, 31% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.05 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 4.34-4.26 (m, 2H), 4.21-4.11 (m, 2H), 3.99 (d, J=12.8 Hz, 1H), 3.90-3.78 (m, 1H), 3.17-3.06 (m, 2H), 2.57-2.43 (m, 1H), 2.20-2.02 (m, 4H), 1.98-1.86 (m, 9H), 1.86-1.73 (m, 4H), 1.36 (s, 3H), 0.81 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=630.5.

Example 329

(1579) ##STR00656##

cis-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-methyl-6-azaspiro[3.5]nonan-2-ol

(1580) The title compound was synthesized according to the procedure described for example 328 except for using the other eluting peak from SFC separation in step C in further steps D and F.

¹H NMR (400 MHz, methanol-d₄) δ=9.09 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 4.31 (s, 2H), 4.22-4.12 (m, 2H), 4.07-3.91 (m, 2H), 3.17-3.04 (m, 2H), 2.81-2.69 (m, 2H), 2.55-2.40 (m, 1H), 2.27-2.03 (m, 3H), 2.02-1.86 (m, 8H), 1.86-1.71 (m, 6H), 1.33 (s, 3H), 0.81 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=630.5.

Example 330

(1581) ##STR00657##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(1582) ##STR00658##

(1583) Step A. 7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a mixture of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (50.0 mg, 1.0 equiv.), 2λ,6-thia-1,3,9-triazaspiro[4.5]decane 2,2-dioxide (32.3 mg, 2.0 equiv.) and 4 Å molecular sieves (10 mg, 1.0 equiv.) in DMF (0.5 mL) was added DIEA (32.7 mg, 3.0 equiv.). The reaction was stirred at 40° C. for 12 hours. The residue was filtered and washed with DMF (1 mL). The filtrate was purified by prep-HPLC [column: Phenomenex C18 75×30 mm×3 µm; mobile phase: water (0.1% formic acid)/ACN] B %: 15%-45%, 7 min] and lyophilized to afford the title compound (26.4 mg, 45% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.14 (s, 1H), 7.68 (ddd, J=3.2, 5.6, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (dt, J=1.6, 9.6 Hz, 1H), 7.04 (t, J=3.2 Hz, 1H), 5.62-5.38 (m, 1H), 4.79-4.61 (m, 3H), 4.59-4.42 (m, 2H), 3.88-3.74 (m, 2H), 3.74-3.60 (m, 3H), 3.41 (td, J=2.4, 12.0 Hz, 1H), 3.22 (dd, J=1.6, 12.4 Hz, 1H), 2.71-2.49 (m, 2H), 2.48-2.30 (m, 2H), 2.28-2.15 (m, 3H), 2.15-2.00 (m, 3H), 2.00-1.81 (m, 2H), 0.80 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=684.4.

Example 331

(1584) ##STR00659##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(piperidin-1-

(methyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol
(1585) The title compound was synthesized according to the procedure described for example 309.
.sup.1H NMR (400 MHz, chloroform-d) δ =9.22 (s, 1H), 8.54 (s, 1H), 7.65 (s, 1H), 7.31 (s, 1H), 7.29-7.22 (m, 1H), 7.03 (br d, J=1.2 Hz, 1H), 4.55 (br d, J=13.2 Hz, 1H), 4.48-4.41 (m, 2H), 4.35-4.25 (m, 1H), 3.69-3.57 (m, 1H), 3.50-3.41 (m, 2H), 3.13-2.79 (m, 5H), 2.55-2.38 (m, 1H), 2.26-2.12 (m, 2H), 1.86-1.75 (m, 6H), 1.66-1.53 (m, 2H), 1.29 (d, J=10.4 Hz, 3H), 0.89-0.78 (m, 5H), 0.75-0.66 (m, 2H); LCMS [ESI, M+1] m/z=618.4.

Example 332

(1586) ##STR00660##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-((1-(morpholinomethyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol
(1587) The title compound was synthesized according to the procedure described for example 309.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.25 (s, 1H), 8.59-8.47 (m, 1H), 7.74-7.66 (m, 1H), 7.36-7.22 (m, 2H), 7.08 (br d, J=19.1 Hz, 1H), 4.41 (s, 3H), 4.33-4.23 (m, 1H), 3.72-3.61 (m, 5H), 3.53-3.46 (m, 1H), 2.62-2.52 (m, 4H), 2.51-2.43 (m, 3H), 2.29-2.12 (m, 2H), 1.96-1.72 (m, 3H), 1.33-1.29 (m, 3H), 0.89-0.79 (m, 3H), 0.78-0.71 (m, 2H), 0.47 (s, 2H); LCMS [ESI, M+1] m/z=620.1.

Example 333

(1588) ##STR00661##

5-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1589) The title compound was synthesized according to the procedure described for example 319.
.sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.16 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.29 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 6.72 (s, 1H), 5.37-5.17 (m, 2H), 4.58-4.54 (m, 2H), 4.48-4.32 ((m, 4H), 3.33 (s, 3H), 3.13-3.02 (m, 3H), 2.50-2.37 (m, 5H), 2.31 (s, 6H), 2.21-2.11 (m, 1H), 0.78 (t, J=7.2 Hz, 3H), 0.75-0.70 (m, 2H), 0.56-0.51 (m, 2H); LCMS (ESI, M+1): m/z=671.1.

Example 334

(1590) ##STR00662##

4-(7-(8-ethylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepane

(1591) The title compound was synthesized according to the procedure described for example 204.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.19-9.15 (m, 1H), 8.09-8.04 (m, 1H), 7.91-7.85 (m, 1H), 7.62-7.55 (m, 1H), 7.53-7.39 (m, 3H), 4.40-4.34 (m, 2H), 4.33-4.23 (m, 4H), 4.11-4.04 (m, 2H), 3.92-3.84 (m, 2H), 3.27-3.17 (m, 2H), 2.89-2.80 (m, 2H), 2.51-2.31 (m, 2H), 2.27-2.19 (m, 2H), 2.18-2.09 (m, 2H), 2.05-1.90 (m, 4H), 1.88-1.79 (m, 2H), 1.00-0.91 (m, 3H); LCMS (ESI, M+1): m/z=542.2.

(1592) ##STR00663##

7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)hexahydro-2,7-naphthyridine-1,3(2H,4H)-dione

(1593) ##STR00664##

(1594) Step A. 4H-2,7-naphthyridine-1,3-dione: A mixture of methyl 2-(3-cyano-4-pyridyl)acetate (500 mg, 1.0 equiv.), (1E)-acetaldehyde oxime (838 mg, 5.0 equiv.) and RhCl(PPh.sub.3).sub.3 (263 mg, 0.1 equiv.) in toluene (15 mL) was degassed and stirred at 110° C. for 2 hours under N.sub.2 atmosphere. The mixture was concentrated under reduced pressure and then diluted with water (20 mL). The mixture was refluxed at 100° C. for 1 hour. After cooling to 20° C., the mixture was filtered. The filter cake was triturated with EtOAc (10 mL) and MeOH (10 mL) to afford the title compound (300 mg, 52.2% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =12.41-11.59 (m, 1H), 10.52 (br s, 1H), 8.19 (s, 1H), 7.36 (br d, J=7.2 Hz, 1H), 6.66 (br d, J=7.2

H_z, 1H), 5.14 (s, 1H) LCMS (ESI, M+1): m/z=163.0.

(1595) Step B. tert-butyl 6,8-dioxooctahydro-2,7-naphthyridine-2(1H)-carboxylate: A mixture of 4H-2,7-naphthyridine-1,3-dione (400 mg, 1.0 equiv.), Boc.sub.2O (1.62 g, 3.0 equiv.) and Pd/C (50 mg, 10% purity) in MeOH (10 mL) was degassed and purged with H.sub.2 for 3 times before being stirred at 60° C. for 6 hours under H.sub.2 atmosphere (1 atm). The mixture was filtered and concentrated under reduced pressure to afford the title compound (400 mg, crude) as a green oil; LCMS (ESI, M+1): m/z=269.0.

(1596) Step C. hexahydro-2,7-naphthyridine-1,3(2H,4H)-dione: A mixture of tert-butyl 6,8-dioxo-1,3,4,4a,5,8a-hexahydro-2,7-naphthyridine-2-carboxylate (400 mg, 1.0 equiv.) in DCM (3 mL) and TFA (1 mL) was stirred at 25° C. for 2 hours under N.sub.2 atmosphere. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC [column: Waters Atlantis T3 150×30 mm×5 μm; mobile phase: water (0.1% formic acid)/ACN; B %: 1%-20%, 10 min] to afford the title compound (100 mg, 25% yield) as a green oil; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=4.06-3.93 (m, 1H), 3.51-3.33 (m, 1H), 3.23-3.16 (m, 1H), 3.12-2.90 (m, 2H), 2.80-2.64 (m, 1H), 2.62-2.35 (m, 2H), 2.11-1.83 (m, 1H), 1.70-1.48 (m, 1H); LCMS (ESI, M+1): m/z=169.2.

(1597) Step D. 7-(8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)hexahydro-2,7-naphthyridine-1,3(2H,4H)-dione: A mixture of 4a,5,6,7,8,8a-hexahydro-4H-2,7-naphthyridine-1,3-dione (63.0 mg, 2.0 equiv.), 8-fluoro-7-(8-fluoronaphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), DIEA (243 mg, 10.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (0.01 mL) was stirred at 40° C. for 48 hours under N.sub.2atmosphere. The mixture was filtered. The residue was purified by prep-HPLC (neutral condition) to afford the title compound (6.2 mg, 5.3% yield) as an off-white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.13 (s, 1H), 8.54 (s, 1H), 8.13 (br d, J=8.4 Hz, 1H), 7.87 (d, J=8.4 Hz, 1H), 7.76-7.67 (m, 1H), 7.62 (d, J=7.2 Hz, 1H), 7.54 (dt, J=5.2, 7.8 Hz, 1H), 7.20 (dd, J=8.2, 13.2 Hz, 1H), 5.24 (br d, J=14.0 Hz, 1H), 4.84-4.79 (m, 1H), 4.59 (s, 2H), 3.61-3.42 (m, 3H), 3.40-3.33 (m, 1H), 3.22-3.10 (m, 2H), 2.84-2.66 (m, 2H), 2.55-2.44 (m, 1H), 2.41-2.23 (m, 3H), 2.22-2.11 (m, 3H), 2.10-1.98 (m, 4H), 1.76-1.53 (m, 1H); LCMS (ESI, M+1): m/z=599.4.

Example 336

(1598) ##STR00665##

(R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((1-((4-methylpiperazin-1-yl)methyl)cyclopropyl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1599) The title compound was synthesized according to the procedure described for example 309. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.20 (d, J=4.0 Hz, 1H), 7.68 (dd, J=8.8, 6.0 Hz, 1H), 7.32-7.30 (m, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.09-7.01 (m, 1H), 4.56-4.38 (m, 3H), 4.29-4.26 (m, 1H), 3.68-3.60 (m, 1H), 3.51-3.40 (m, 1H), 3.10-2.38 (m, 14H), 2.27-2.08 (m, 2H), 1.91-1.71 (m, 3H), 1.31-1.26 (m, 3H), 0.85-0.81 (m, 3H), 0.79-0.76 (m, 2H), 0.55-0.53 (m, 2H); LCMS (ESI, M+1): m/z=633.5;

Example 337

(1600) ##STR00666##

10-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-oxa-1,3,10-triazaspiro[4.6]undecane-2,4-dione

(1601) The title compound was synthesized according to the procedure described for example 330. .sup.1H NMR (400 MHz, METHANOL-d) δ=9.35 (d, J=1.6 Hz, 1H), 7.76 (dd, J=6.0, 8.8 Hz, 1H), 7.39 (d, J=2.8 Hz, 1H), 7.34 (t, J=9.6 Hz, 1H), 7.13 (dd, J=2.4, 7.2 Hz, 1H), 5.63-5.42 (m, 1H), 5.20-5.00 (m, 1H), 4.62-4.25 (m, 7H), 4.18-4.09 (m, 1H), 4.02-3.96 (m, 1H), 3.81-3.50 (m, 3H), 3.36-3.25 (m, 1H), 2.69-2.39 (m, 3H), 2.38-2.06 (m, 5H), 0.96-0.82 (m, 3H); LCMS (ESI, M+1): m/z=678.1.

Example 338

(1602) ##STR00667##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.6]undecane-2,4-dione

(1603) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, methanol-d₃) δ=9.28 (t, J=3.2 Hz, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.32-7.24 (m, 2H), 7.05 (d, J=2.4 Hz, 1H), 5.54-5.33 (m, 1H), 5.00-4.95 (m, 1H), 4.52-4.40 (m, 3H), 4.21-4.05 (m, 2H), 3.63-3.61 (m, 3H), 3.26 (br d, J=5.2 Hz, 1H), 2.50-2.45 (m, 3H), 2.20-2.11 (m, 9H), 1.88-1.84 (m, 2H), 0.84-0.77 (m, 3H); LCMS (ESI, M+1): m/z=676.3.

Example 339

(1604) ##STR00668##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1605) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.19 (s, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.33-7.20 (m, 2H), 7.05 (d, J=2.4 Hz, 1H), 6.82 (s, 1H), 5.52-5.12 (m, 3H), 4.60-4.30 (m, 6H), 3.65-3.38 (m, 3H), 3.18 (dt, J=5.6, 9.2 Hz, 1H), 2.88 (s, 3H), 2.57-2.21 (m, 6H), 2.20-1.92 (m, 4H), 0.78 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=687.1.

Example 340

(1606) ##STR00669##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-(methylamino)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1607) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, methanol-d₃) δ=9.21 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.8 Hz, 1H), 5.75 (s, 1H), 5.32 (d, J=54.4, 1H), 5.17-5.03 (m, 2H), 4.39-4.25 (m, 6H), 3.36 (br d, J=2.8 Hz, 1H), 3.26-3.25 (m, 2H), 3.06 (dt, J=6.0, 9.6 Hz, 1H), 2.76 (s, 3H), 2.49-2.46 (m, 1H), 2.38-2.03 (m, 6H), 2.02-1.87 (m, 3H), 0.79 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=659.4.

Example 341

(1608) ##STR00670##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-(hydroxymethyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1609) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.24-9.16 (m, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.34-7.19 (m, 2H), 7.06 (d, J=2.4 Hz, 1H), 6.44 (s, 1H), 5.51-5.33 (m, 1H), 5.32-5.06 (m, 2H), 4.73-4.24 (m, 8H), 3.72-3.40 (m, 3H), 3.26-3.15 (m, 1H), 2.58-2.21 (m, 6H), 2.19-1.96 (m, 4H), 0.78 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=660.1.

Example 342

(1610) ##STR00671##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(1,4-thiazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1611) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, Methanol-d₃) δ ppm 0.81 (td, J=7.2, 1.6 Hz, 3H) 1.86-2.00 (m, 1H) 2.01-2.11 (m, 2H) 2.13-2.25 (m, 2H) 2.27-2.56 (m, 5H) 2.79 (t, J=6.0 Hz, 2H) 3.10 (dt, J=9.2, 4.8 Hz, 1H) 3.16 (br t, J=5.6 Hz, 2H) 3.35-3.51 (m, 2H) 4.23-4.43 (m, 6H) 4.61 (br s, 1H) 5.27-5.46 (m, 1H) 7.07 (d, J=2.4 Hz, 1H) 7.21-7.29 (m, 1H) 7.31 (d, J=2.4 Hz, 1H) 7.68 (dd, J=9.2, 6.0 Hz,

1H) 8.53 (s, 1H) 9.16 (s, 1H); LCMS (ESI, M+1): m/z=610.3

Example 343

(1612) ##STR00672##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1613) The title compound was synthesized according to the procedure described for example 330.

.sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.20 (d, J=2.8 Hz, 1H), 8.55 (s, 1H), 7.70 (dd, J=6.0, 9.2 Hz, 1H), 7.33 (d, J=2.8 Hz, 1H), 7.27 (t, J=9.6 Hz, 1H), 7.08 (dd, J=2.4, 8.8 Hz, 1H), 5.48-5.29 (m, 1H), 4.74-4.56 (m, 4H), 4.48-4.29 (m, 3H), 4.28-4.12 (m, 2H), 4.04-3.89 (m, 1H), 3.74-3.58 (m, 3H), 3.19-3.07 (m, 1H), 3.13-2.64 (m, 1H), 2.60-1.93 (m, 10H), 2.55-1.90 (m, 1H), 0.87-0.77 (m, 3H); LCMS (ESI, M+1): m/z=624.4

Example 344

(1614) ##STR00673##

5-ethyl-6-fluoro-4-(8-fluoro-4-(2-fluoro-6-azaspiro[3.5]nonan-6-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1615) The title compound was synthesized according to the procedure described for example 330.

.sup.1H NMR (400 MHz, chloroform-d) δ =9.06-8.97 (m, 1H), 7.58-7.51 (m, 1H), 7.23-6.87 (m, 3H), 5.96-5.73 (m, 1H), 5.45-5.09 (m, 3H), 4.66-4.48 (m, 2H), 4.43-4.19 (m, 2H), 3.44-2.93 (m, 6H), 2.57-2.35 (m, 3H), 2.25-2.16 (m, 2H), 2.03-1.85 (m, 4H), 1.82-1.62 (m, 5H), 0.89-0.74 (m, 3H); LCMS (ESI, M+1): m/z=636.3.

Example 345

(1616) ##STR00674##

4-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,4-oxazepan-6-ol

(1617) The title compound was synthesized according to the procedure described for example 330.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.33 (d, J=11.6 Hz, 1H), 8.55 (s, 1H), 7.67 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.37 (br s, 1H), 4.46-4.39 (m, 1H), 4.37-4.21 (m, 5H), 4.17-4.00 (m, 3H), 4.00-3.89 (m, 1H), 3.79-3.67 (m, 1H), 3.24 (br s, 1H), 3.18 (s, 11H), 3.01 (dt, J=6.0, 9.6 Hz, 11H), 2.56-2.45 (m, 1H), 2.36-2.11 (m, 4H), 2.08-1.81 (m, 4H), 0.87-0.73 (m, 3H); LCMS (ESI, M+1): m/z=610.4.

Example 346

(1618) ##STR00675##

1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-4-ol

(1619) The title compound was synthesized according to the procedure described for example 330.

.sup.1H NMR (400 MHz, MeOH-d.sub.4) δ =9.17 (s, 1H), 8.49 (br s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.05 (d, J=2.8 Hz, 1H), 5.51-5.33 (m, 1H), 4.52-4.46 (m, 1H), 4.44-4.37 (m, 1H), 4.25-4.17 (m, 11H), 4.25-4.15 (m, 1H), 4.13 (br t, J=5.6 Hz, 1H), 4.06-3.95 (m, 2H), 3.67-3.45 (m, 3H), 3.25-3.17 (m, 1H), 2.55-2.35 (m, 3H), 2.34-2.18 (m, 4H), 2.17-2.05 (m, 3H), 2.02-1.84 (m, 3H), 1.83-1.70 (m, 1H), 0.80 (t, J=7.2 Hz, 3H). LCMS (ESI, M+1): m/z=608.1

Example 347

(1620) ##STR00676##

1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepan-3-ol

(1621) ##STR00677##

(1622) The title compound was synthesized according to the procedure described for example 330.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.35-9.18 (m, 1H), 8.56-8.41 (m, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.36-7.19 (m, 2H), 7.05 (dd, J=2.8, 4.4 Hz, 1H), 5.53-5.31 (m, 1H), 4.59 (s, 1H), 4.54-4.37 (m, 3H), 4.19 (td, J=4.0, 8.4 Hz, 1H), 4.14 (t, J=6.4 Hz, 1H), 3.88-3.76 (m, 1H),

3.46 (s, 3H), 3.26-3.18 (m, 1H), 2.54-1.92 (m, 12H), 1.68 (d, J=10.8 Hz, 1H), 1.55-1.40 (m, 1H), 0.88-0.72 (m, 3H); LCMS [ELSD, M+1]: m/z=608.1

Example 348

(1623) ##STR00678##

2-(1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)isothiazolidine 1,1-dioxide

(1624) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.09 (s, 1H), 7.76 (br dd, J=6.0, 8.8 Hz, 1H), 7.44-7.25 (m, 2H), 7.02 (br s, 11H), 5.47-5.11 (m, 1H), 4.60 (br t, J=13.6 Hz, 1H), 4.50-4.35 (m, 1H), 4.21-4.03 (m, 2H), 3.71-3.59 (m, 2H), 3.22 (br d, J=3.6 Hz, 3H), 3.14-2.94 (m, 4H), 2.86-2.65 (m, 2H), 2.37-2.30 (m, 1H), 2.29-2.19 (m, 2H), 2.18-2.08 (m, 2H), 2.03 (br d, J=16.4 Hz, 3H), 1.94-1.71 (m, 6H), 0.72 (q, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=697.4.

Example 349

(1625) ##STR00679##

3-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)tetrahydrothiophene 1,1-dioxide

(1626) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.36 (d, J=2.0 Hz, 1H), 9.10 (br d, J=6.8 Hz, 1H), 8.35 (s, 1H), 7.77 (dd, J=6.0, 9.2 Hz, 1H), 7.41-7.28 (m, 2H), 6.99 (d, J=2.4 Hz, 1H), 5.41-5.17 (m, 1H), 5.14-5.02 (m, 1H), 4.23-4.03 (m, 2H), 3.76-3.59 (m, 2H), 3.06-2.98 (m, 2H), 2.86-2.78 (m, 1H), 2.72-2.65 (m, 2H), 2.35-2.29 (m, 3H), 2.13-1.97 (m, 4H), 1.95-1.68 (m, 4H), 0.71 (t, J=7.6 Hz, 3H), LCMS (ESI, M+1): m/z=628.2.

Example 350

(1627) ##STR00680##

4-(4-(3-azabicyclo[3.2.1]octan-3-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(1628) The title compound was synthesized according to the procedure described for example 330. LCMS (ESI, M+1): m/z=604.0.

Example 351

(1629) ##STR00681##

6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-1-ol

(1630) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.14 (d, J=6.0 Hz, 1H), 8.61-8.44 (m, 1H), 7.75-7.62 (m, 1H), 7.31 (d, J=2.4 Hz, 11H), 7.25 (t, J=9.6 Hz, 1H), 7.07 (t, J=3.2 Hz, 1H), 5.49-5.27 (m, 1H), 4.52-4.27 (m, 3H), 4.25-4.11 (m, 1H), 4.10-3.81 (m, 3H), 3.57-3.37 (m, 3H), 3.20-3.08 (m, 1H), 2.55-2.16 (m, 6H), 2.14-2.05 (m, 2H), 2.00-1.84 (m, 4H), 1.83-1.72 (m, 2H), 1.72-1.55 (m, 1H), 1.38-1.30 (m, 1H), 0.86-0.74 (m, 3H). LCMS (ESI, M+1): m/z=634.5.

Example 352

(1631) ##STR00682##

7-(8-ethyl-7-fluoro-3-hydroxy-1-naphthyl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-[3-(sulfamoylamino)-1-piperidyl]pyrido[4,3-d]pyrimidine

(1632) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.12 (s, 1H), 7.69-7.60 (m, 1H), 7.28 (d, J=2.4 Hz, 1H), 7.23 (t, J=9.2 Hz, 1H), 7.05 (t, J=3.2 Hz, 1H), 5.41-5.20 (m, 1H), 4.79-4.67 (m, 1H), 4.46-4.23 (m, 3H), 3.75-3.60 (m, 2H), 3.58-3.45 (m, 1H), 3.28-3.11 (m, 3H), 3.04-2.95 (m, 1H), 2.57-2.43 (m, 1H), 2.41-2.12 (m, 5H), 2.05-1.87 (m, 4H), 1.85-1.67 (m, 2H), 0.87-0.73 (m, 3H); LCMS [ESI, M+1]: m/z=672.2.

Example 353

(1633) ##STR00683##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2(4H)-carboxamide

(1634) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.97 (s, 1H), 9.16 (s, 1H), 8.20 (s, 1H), 7.76 (dd, J=6.0, 9.2 Hz, 1H), 7.38-7.31 (m, 2H), 7.01 (d, J=2.4 Hz, 1H), 5.46-5.26 (m, 1H), 5.12-5.00 (m, 2H), 4.32 (br s, 5H), 3.10 (br s, 6H), 2.99-2.82 (m, 3H), 2.43-1.74 (m, 1H), 1.23 (s, 1H), 0.70 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=701.4.

(1635) ##STR00684##

(5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)(morpholino)methanone

(1636) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, CD₃OD) δ=9.25 (s, 1H), 7.75-7.65 (m, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 6.79 (s, 1H), 5.68-5.51 (m, 1H), 5.38-5.30 (m, 1H), 5.28-5.20 (m, 1H), 4.83-4.75 (m, 2H), 4.73-4.59 (m, 2H), 4.53 (br d, J=5.6 Hz, 2H), 4.12-4.03 (m, 2H), 3.95-3.87 (m, 2H), 3.77-3.64 (m, 6H), 3.52-3.42 (m, 1H), 2.83-2.05 (m, 1H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=743.3.

Example 355

(1637) ##STR00685##

(5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)(pyrrolidin-1-yl)methanone

(1638) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, CD₃OD) δ=9.24 (s, 1H), 7.73-7.65 (m, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 6.85 (s, 1H), 5.65-5.43 (m, 1H), 5.39-5.19 (m, 2H), 4.59-4.42 (m, 6H), 3.98-3.87 (m, 2H), 3.81-3.71 (m, 2H), 3.60 (t, J=6.8 Hz, 2H), 2.71-2.06 (m, 1H), 2.03-1.81 (m, 5H), 0.78 (t, J=7.2 Hz, 3H); LCMS [ESI, M+1]: m/z=727.4.

Example 356

(1639) ##STR00686##

N-ethyl-5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1640) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.94 (s, 1H), 9.20 (s, 1H), 8.09 (br t, J=6.0 Hz, 1H), 7.83-7.69 (m, 1H), 7.39-7.30 (m, 2H), 7.00 (d, J=2.4 Hz, 1H), 6.65 (s, 1H), 5.53-5.00 (m, 3H), 4.56-4.47 (m, 2H), 4.42-4.25 (m, 2H), 4.22-3.96 (m, 2H), 3.27-3.15 (m, 3H), 3.13-2.74 (m, 3H), 2.33 (br s, 2H), 2.21-1.97 (m, 5H), 1.88-1.73 (m, 3H), 1.05 (t, J=7.2 Hz, 3H), 0.70 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=701.1.

Example 357

(1641) ##STR00687##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-isopropyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1642) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=11.09-10.83 (m, 1H), 9.27 (s, 1H), 7.81-7.73 (m, 2H), 7.40-7.32 (m, 2H), 7.02 (d, J=2.4 Hz, 1H), 6.77 (s, 1H), 5.68-5.50 (m, 1H), 5.36-5.19 (m, 2H), 4.64-4.38 (m, 6H), 4.09-4.00 (m, 1H), 3.91-3.84 (m, 1H), 3.17-3.08 (m, 1H), 2.23-2.00 (m, 5H), 1.29-1.24 (m, 7H), 1.12 (d, J=6.4 Hz, 6H), 0.70 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=715.5.

Example 358

(1643) ##STR00688##

N-ethyl-5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-methyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1644) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.26 (s, 1H), 8.09 (s, 1H), 7.69 (dd, J=6.0, 9.0 Hz, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 6.76 (br d, J=4.0 Hz, 1H), 5.70-5.48 (m, 1H), 5.37-5.19 (m, 2H), 4.76-4.43 (m, 6H), 4.03-3.73 (m, 3H), 4.05-3.70 (m, 1H), 3.62-3.43 (m, 2H), 3.06 (s, 11H), 2.84-2.02 (m, 11H), 1.22 (td, J=7.2, 11.8 Hz, 3H), 0.78 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=715.5

Example 359

(1645) ##STR00689##

7-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1646) ##STR00690##

(1647) Step A. 7-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of 2,4-dichloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) in DCM (2 mL) was added DIEA (172 mg, 3.0 equiv.) and 1,3,7-triazaspiro[4.5]decane-2,4-dione (75.2 mg, 1.0 equiv.) at -40° C. The reaction mixture was stirred at -40° C. for 1 hour. The mixture was poured dropwise into ice-cold saturated NH₄Cl solution (8 mL). Then the mixture was extracted with ethyl acetate (2×5 mL). The combined organic layers were washed with brine (5 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (10 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (170 mg, 64.7% yield) as a white solid; LCMS (ESI, M+1): m/z=583.2.

(1648) Step B. 7-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of 7-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione (170 mg, 1.0 equiv.) in THF (2 mL) were added (1-((dimethylamino)methyl)cyclopropyl)methanol (113 mg, 3.0 equiv.), DIEA (113 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg). The reaction mixture was stirred at 40° C. for 12 hours. The mixture was filtered. The filter cake was triturated with dichloromethane/methanol 10:1 (10 mL) at 25° C. for 15 mins to afford the title compound (140 mg, 68.5% yield) as a white solid; LCMS (ESI, M+1): m/z=676.4.

(1649) Step C. 7-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of 7-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione (130 mg, 1.0 equiv.) in MeCN (1 mL) was added HCl.dioxane (4 M, 6.0 equiv.) at 0° C. The reaction mixture was stirred at 0° C. for 1 hour. The mixture was concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and lyophilized to afford the title compound (92.7 mg, 69.1% yield, formic acid salt) as a white solid; ¹H NMR (400 MHz, methanol-d₃) δ=9.14 (d, J=4.0 Hz, 1H), 8.50 (br s, 1H), 7.69 (dd,

J=6.0, 9.2 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 4.70-4.53 (m, 2H), 4.45-4.38 (m, 2H), 3.95-3.74 (m, 2H), 3.23-3.09 (m, 2H), 2.87 (s, 6H), 2.55-2.35 (m, 1H), 2.32-2.03 (m, 3H), 2.03-1.89 (m, 2H), 1.00-0.90 (m, 2H), 0.87-0.71 (m, 5H); LCMS (ESI, M+1): m/z=632.4.

Example 360

(1650) ##STR00691##

(1R,5R,6R)-3-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol

(1651) The title compound was synthesized according to the procedure described for example 359 except for TsOH (10 equiv.) in DCM was used in step C. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.28-9.18 (m, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.34-7.28 (m, 1H), 7.27-7.20 (m, 1H), 7.06 (dd, J=2.8, 13.2 Hz, 1H), 4.63-4.51 (m, 2H), 4.37 (s, 2H), 4.35-4.28 (m, 1H), 3.82-3.7 (m, 1H), 3.52-3.43 (m, 1H), 2.55-2.44 (m, 3H), 2.423-2.38 (m, 1H), 2.37-2.29 (m, 6H), 2.28-2.10 (m, 3H), 1.96-1.88 (m, 1H), 1.85-1.77 (m, 11H), 1.46-1.37 (m, 1H), 0.85-0.76 (m, 3H), 0.76-0.70 (m, 2H), 0.57-0.49 (m, 2H); LCMS (ESI, M+1): m/z=590.2.

Example 361

(1652) ##STR00692##

(S)-4-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1653) The title compound was synthesized according to the procedure described for example 360. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.53 (d, J=4.4 Hz, 1H), 7.67 (dd, J=6.0, 8.98 Hz, 1H), 7.33-7.19 (m, 2H), 7.10-7.00 (m, 1H), 4.57-4.44 (m, 2H), 4.39 (dd, J=5.6, 10.4 Hz, 2H), 4.24-4.13 (m, 1H), 4.08-3.94 (m, 2H), 3.93-3.84 (m, 1H), 3.74-3.61 (m, 2H), 2.5-2.40 (m, 3H), 2.31 (s, 6H), 2.24-2.13 (m, 1H), 1.27 (d, J=2.8 Hz, 3H), 0.84-0.76 (m, 3H), 0.76-0.69 (m, 2H), 0.57-0.51 (m, 2H); LCMS (ESI, M+1): m/z=594.1.

Example 362

(1654) ##STR00693##

((3R,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate

(1655) ##STR00694##

(1656) Step A. (R)-1-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine (5.00 g, 1.0 equiv.) in DCM (50 mL) were added 4 Å molecular sieves (30 mg, 1.0 equiv.), DIEA (7.18 g, 5.0 equiv.) and (3R)-3-methylpiperidin-3-ol (1.85 g, 1.1 equiv., HCl) at -40° C. The reaction mixture was stirred at -40° C. for 15 mins. The reaction mixture was diluted with DCM (30 mL) and water (40 mL), and extracted with DCM (20 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (4.9 g, 81.4% yield) as an off-yellow solid; LCMS (ESI, M+1): m/z=529.2.

(1657) Step B. (R)-1-(2-(((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of ((3R,7aR)-3-(((tert-butyldiphenylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methanol (465 mg, 1.2 equiv.) in toluene (5 mL) were added 4 Å molecular sieves (20 mg) and t-BuONa (273 mg, 3.0 equiv.) at 0°

C. and the resulting mixture was stirred for 10 minutes. Then (R)-1-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (500 mg, 1.0 equiv.) was added and the mixture was stirred at 0° C. for 1 hour. The reaction mixture was diluted with ethyl acetate (15 mL) and water (20 mL), and extracted with ethyl acetate (10 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to give a residue. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (10 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum to afford the title compound (600 mg, 67.5% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d₄) 9.22 (d, J=2.2 Hz, 1H), 7.80 (dd, J=5.6, 9.2 Hz, 1H), 7.69 (qd, J=1.8, 6.0 Hz, 4H), 7.62 (d, J=2.8 Hz, 1H), 7.47-7.38 (m, 6H), 7.31 (t, J=9.2 Hz, 1H), 7.22 (t, J=3.2 Hz, 1H), 5.33 (s, 2H), 4.65-4.48 (m, 2H), 4.42-4.23 (m, 3H), 3.97 (dd, J=5.6, 10.8 Hz, 1H), 3.90-3.78 (m, 1H), 3.69-3.55 (m, 1H), 3.53-3.39 (m, 5H), 2.96-2.82 (m, 2H), 2.58-2.42 (m, 1H), 2.29-2.06 (m, 3H), 1.96-1.73 (m, 8H), 1.71-1.60 (m, 1H), 1.31-1.26 (m, 3H), 1.08-1.03 (m, 9H), 0.82 (dt, J=4.8, 7.2 Hz, 3H); LCMS (ESI, M+1): m/z=902.6.

(1658) Step C. (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((3R,7aR)-3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(2-(((3R,7aR)-3-((tert-butyl)dimethylsilyl)oxy)methyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (300 mg, 1.0 equiv.) in DMF (3 mL) was added CsF (758 mg, 15 equiv.). The reaction mixture was stirred at 40° C. for 12 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (5 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, and concentrated in vacuum to afford the title compound (190 mg, 85.5% yield) as a yellow solid; LCMS (ESI, M+1): m/z=664.4.

(1659) Step D. ((3R,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate: To a solution of (R)-1-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((3R,7aR)-3-(hydroxymethyl)hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (190 mg, 1.0 equiv.) in THF (2 mL) was added NaH (22.9 mg, 60% purity, 2.0 equiv.) at 0° C. After stirring at 0° C. for 0.5 hour, N,N-dimethylcarbamoyl chloride (46.17 mg, 1.5 equiv.) in THF (1 mL) was added and the resulting was stirred at 0° C. for 2 hours. The mixture was quenched with water (10 mL) and extracted with ethyl acetate (2×10 mL). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, and concentrated in vacuum to remove acetonitrile. The aqueous layer was extracted with ethyl acetate (2×8 mL). The combined organic layers were dried over anhydrous sodium sulfate, and concentrated in vacuum to give a residue. Then the residue was purified by prep-TLC (silica gel, DCM/MeOH 10:1) to afford the title compound (45 mg, 21.4% yield) as a yellow solid; LCMS (ESI, M+1): m/z=735.6.

(1660) Step E. ((3R,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate: To a solution of ((3R,7aR)-7a-(((7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate (32 mg, 1.0

equiv.) in MeCN (0.3 mL) was added HCl.dioxane (4 M, 55 equiv.) at 0° C. The reaction mixture was stirred at 0° C. for 0.5 hour. The mixture was concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and lyophilized to afford the title compound (15.6 mg, 45.4% yield, formic acid salt) as a white solid; .sup.1H NMR (400 MHz, methanol-d4) δ =9.26-9.21 (m, 1H), 8.55 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.34-7.20 (m, 2H), 7.06 (s, 1H), 4.82-4.70 (m, 2H), 4.52-4.22 (m, 3H), 4.14-3.97 (m, 1H), 3.88-3.74 (m, 1H), 3.69-3.54 (m, 1H), 3.52-3.35 (m, 1H), 3.34-2.30 (m, 2H), 2.95-2.91 (m, 5H), 2.51-2.40 (m, 1H), 2.35-2.09 (m, 4H), 2.07-1.89 (d, J=14. Hz, 5H), 1.86-1.67 (m, 4H), 1.29 (d, J=9.2 Hz, 3H), 0.86-0.74 (m, 3H); LCMS (ESI, M+1): m/z=691.4.

Example 363

(1661) ##STR00695##

(4aR,7aS)-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)hexahydro-1H-pyrrolo[3,4-d]pyridazine-1,4(4aH)-dione

(1662) ##STR00696##

(1663) Step A. (4aR,7aS)-6-benzylhexahydro-1H-pyrrolo[3,4-d]pyridazine-1,4(4aH)-dione: To a solution of 1,2-dihydropyridazine-3,6-dione (10.0 g, 1.0 equiv.) in dichloromethane (90.0 mL) was added N-benzyl-1-methoxy-N-((trimethylsilyl)methyl)methanamine (42.4 g, 2.0 equiv.) at 0° C. The resulting mixture was stirred at 0° C. for 5 minutes, then TFA (1.02 g, 0.1 equiv.) in dichloromethane (10 mL) was added dropwise at 0° C. The resulting mixture was stirred at 25° C. for 2 hours. The mixture was quenched by addition of water (50 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (3×50 mL), dried over anhydrous sodium sulfate, concentrated and triturated with dichloromethane/methyl alcohol 10:1 (3×100 mL) at 25° C. for 30 minutes and filtered to afford the title compound (10 g, crude) as a yellow solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 7.36-7.26 (m, 5H), 4.59 (d, J=13.2 Hz, 1H), 4.31 (d, J=12.8 Hz, 1H), 3.21-3.10 (m, 2H), 3.01-2.91 (m, 2H), 2.75 (dt, J=8.0, 9.6 Hz, 2H).

(1664) Step B. (4aR,7aS)-hexahydro-1H-pyrrolo[3,4-d]pyridazine-1,4(4aH)-dione: To a suspension of Pd/C (0.20 g, 10% purity) in MeOH (40.0 mL) was added (4aR,7aS)-6-benzylhexahydro-1H-pyrrolo[3,4-d]pyridazine-1,4(4aH)-dione (1.00 g, 1.0 equiv.) under N.sub.2 atmosphere. The suspension was degassed and purged with H.sub.2 for 3 times. The mixture was stirred under H.sub.2 (15 Psi) at 25° C. for 1 hour. The mixture was filtered and concentrated to afford the title compound (0.23 g, 36% yield over two steps) as a yellow solid; .sup.1H NMR (400 MHz, DMSO-d4) δ 3.15-3.10 (m, 2H), 2.92-2.88 (m, 1H), 2.88-2.85 (m, 11H), 2.83-2.75 (m, 2H).

(1665) Step C. (4aR,7aS)-6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)hexahydro-1H-pyrrolo[3,4-d]pyridazine-1,4(4aH)-dione: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (150 mg, 1.0 equiv.) in DMF (2.0 mL) were added (4aR,7aS)-hexahydro-1H-pyrrolo[3,4-d]pyridazine-1,4(4aH)-dione (58.9 mg, 1.5 equiv.), DIEA (98.1 mg, 3.0 equiv.) and 4 Å molecular sieves (70 mg). The mixture was stirred at 40° C. for 16 hours. After reaction completion, the mixture was filtered and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 30%-50%, 9 min] and re-purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM ammonium hydroxide)-ACN, B %: 7%-37%, 9 min] to afford the title compound (16.9 mg, 10% yield, 99.2% purity) as a white solid; .sup.1H NMR (400 MHz, Methanol-d4) δ 9.29 (s, 1H), 7.68 (dd, J=5.6, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.07 (d, J=2.8 Hz, 1H), 5.43-5.21 (m, 1H), 4.51-4.21 (m, 6H), 3.58-3.46 (m, 2H), 3.31-3.16 (m, 3H), 3.03 (dt, J=5.6, 9.2 Hz, 1H), 2.55-2.42 (m, 1H), 2.42-2.11 (m, 4H), 2.07-1.85 (m, 3H), 0.81 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=648.4.

Example 364

(1666) ##STR00697##

7-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(1667) ##STR00698##

(1668) Step A. 4-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxyl-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol: To a mixture of 1-(1-(((7-chloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)cyclopropyl)-N,N-dimethylmethanamine (500 mg, 1.0 equiv.), 5-ethyl-6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-2-ol (464 mg, 1.2 equiv.), K.sub.3PO.sub.4 (1.5 M, 2.45 mL, 3.0 equiv.) in methoxycyclopentane (5 mL) was added CataCXium A Pd G3 (89.0 mg, 0.1 equiv.). The reaction mixture was stirred at 90° C. for 2 hours under N.sub.2 atmosphere. The reaction mixture was quenched by addition of water (5 mL), and extracted with ethyl acetate (15 mL). The combined organic layer was dried over anhydrous sodium sulfate, concentrated and purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to give title compound (333 mg, 47% yield) as a yellow solid; LCMS (ESI, M+1): m/z=563.3.

(1669) Step B. 7-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxyl-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a mixture of 4-(2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol (170 mg, 1.0 equiv.), 2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide (69.3 mg, 1.2 equiv.), 4 Å molecular sieves (20.0 mg) in DMF (2 mL) was added DIEA (195 mg, 5.0 equiv.). The reaction mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered. The filtrate was purified by prep-HPLC (column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 13%-43%, 10 min) and lyophilized to afford title compound (141 mg, 69% yield) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4): δ=9.12 (s, 1H), 8.52 (s, 1H), 7.68 (ddd, J=3.2, 5.6, 9.2 Hz, 1H), 7.31 (d, J=1.6 Hz, 1H), 7.28-7.22 (m, 1H), 7.03 (dd, J=2.4, 10.0 Hz, 1H), 4.59 (br t, J=12.0 Hz, 1H), 4.51-4.38 (m, 3H), 3.85-3.73 (m, 1H), 3.72-3.60 (m, 1H), 3.37 (dd, J=12.0, 14.4 Hz, 1H), 3.20 (d, J=12.0 Hz, 1H), 3.15-3.05 (m, 2H), 2.83 (s, 6H), 2.49 (ddd, J=6.4, 7.6, 14.4 Hz, 1H), 2.26-2.11 (m, 1H), 2.08-1.99 (m, 2H), 1.97-1.81 (m, 2H), 0.93 (s, 2H), 0.83-0.75 (m, 5H) LCMS(ESI, M+1): m/z=654.0.

Example 365

(1670) ##STR00699##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-thia-6-azaspiro[3.5]nonan-6-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1671) The title compound was synthesized according to the procedure described for example 330. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.15 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.07 (d, J=2.4 Hz, 1H), 5.52-5.32 (m, 1H), 4.63-4.41 (m, 3H), 4.30 (dd, J=13.2, 19.2 Hz, 1H), 4.18-4.04 (m, 1H), 3.93 (qd, J=6.4, 12.8 Hz, 1H), 3.64-3.40 (m, 3H), 3.24-3.11 (m, 3H), 2.95-2.78 (m, 2H), 2.55-2.30 (m, 3H), 2.28-2.09 (m, 4H), 2.06-1.87 (m, 3H), 1.77 (q, J=5.6 Hz, 2H), 0.80 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=636.4.

Example 366

(1672) ##STR00700##

(R)-1-(2-(3-(dimethylamino)-2,2-dimethylpropoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1673) The title compound was synthesized according to the procedure described for example 309. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.24 (d, J=2.0 Hz, 1H), 8.55-8.49 (m, 1H), 7.68 (dd, J=5.8, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (t, J=2.0 Hz, 1H), 4.34 (s, 2H), 4.33-4.28 (m, 1H), 3.69-3.59 (m, 1H), 3.55-3.38 (m, 2H), 3.02 (br s, 2H), 2.76 (br s, 6H),

2.52-2.42 (m, 1H), 2.24-2.13 (m, 2H), 1.89-1.77 (m, 3H), 1.30 (d, J=9.8 Hz, 3H), 1.18 (s, 6H), 0.81 (q, J=7.6 Hz, 3H); LCMS [ESI, M+1]: m/z=580.3.

Example 367

(1674) ##STR00701##

3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidine-2,5-dione

(1675) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, methanol-d₄) δ=9.21-9.18 (m, 1H), 8.47 (s, 1H), 7.69 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06-7.02 (m, 1H), 5.58-5.37 (m, 1H), 4.60-4.46 (m, 3H), 4.19-4.08 (m, 1H), 4.03-3.93 (m, 1H), 3.80-3.54 (m, 3H), 3.51-3.41 (m, 1H), 3.01-2.88 (m, 1H), 2.74 (dd, J=4.8, 18.2 Hz, 1H), 2.62-2.28 (m, 4H), 2.24-2.04 (m, 4H), 0.79 (t, J=7.2 Hz, 3H) LCMS [ESI, M+1]: m/z=621.3.

Example 368

(1676) ##STR00702##

3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one

(1677) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, MeOD-d₄) δ=9.15 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.42-5.22 (m, 1H), 4.33 (q, J=10.8 Hz, 2H), 4.04 (dd, J=6.0, 13.6 Hz, 1H), 3.83 (dd, J=8.0, 13.2 Hz, 1H), 3.46-3.33 (m, 3H), 3.27-3.11 (m, 3H), 3.06-2.91 (m, 2H), 2.54-.28 (m, 3H), 2.27-1.82 (m, 8H), 0.79 (t, J=7.2 Hz, 3H); ¹⁹F NMR (400 MHz, MeOD-d₄) δ=-121.18 (br s, 1F), -139.08 (br s, 1F), -173.55 (br t, J=14.2 Hz, 1F); LCMS (ESI, M+1): m/z=607.3.

Example 369

(1678) ##STR00703##

5-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)imidazolidine-2,4-dione

(1679) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, METHANOL-d₄) δ=9.19 (s, 1H), 8.49 (br s, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (s, 1H), 5.58-5.34 (m, 1H), 4.66-4.51 (m, 4H), 4.14 (td, J=4.4, 13.6 Hz, 1H), 4.05-3.94 (m, 1H), 3.75-3.50 (m, 3H), 2.61-2.27 (m, 4H), 2.24-1.98 (m, 4H), 0.78 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=622.3.

Example 370

(1680) ##STR00704##

4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)imidazolidin-2-one

(1681) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, methanol-d₄) δ=9.22-9.17 (m, 1H), 8.48 (br s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.26 (br t, J=9.2 Hz, 1H), 7.04 (br s, 1H), 5.56-5.40 (m, 1H), 4.64-4.48 (m, 3H), 4.24-4.10 (m, 1H), 4.05-3.89 (m, 1H), 3.81-3.58 (m, 4H), 3.44-3.35 (m, 1H), 2.58-2.09 (m, 9H), 0.78 (br t, J=6.0 Hz, 3H); LCMS (ESI, M+1): m/z=608.3.

Example 371

(1682) ##STR00705##

5-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)pyrrolidin-2-one

(1683) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, methanol-d₄) δ=9.22 (s, 1H), 8.45 (s, 1H), 7.71 (dd, J=6.0, 9.2 Hz, 1H), 7.34 (d, J=2.8 Hz, 1H), 7.28 (t, J=9.6 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 5.74-5.32 (m, 1H), 4.66-4.50 (m, 3H), 4.13 (br s, 1H), 3.96-3.66 (m, 5H), 2.70-2.45 (m, 4H), 2.43-2.33 (m, 3H), 2.31-2.20 (m, 2H), 2.19-1.98 (m, 3H), 0.81 (br t, J=7.2 Hz, 3H) LCMS [ESI, M+1]: m/z=607.3.

Example 372

(1684) ##STR00706##

1-(2-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl)imidazolidin-2-one

(1685) The title compound was synthesized according to the procedure described for example 330. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =10.13-9.76 (m, 1H), 9.24 (s, 1H), 9.12 (s, 1H), 7.76 (dd, J=6.0, 8.8 Hz, 1H), 7.41-7.29 (m, 2H), 7.00 (s, 1H), 6.34 (s, 1H), 5.45-5.16 (m, 1H), 4.22-4.12 (m, 1H), 4.12-4.03 (m, 1H), 3.82-3.70 (m, 1H), 3.69-3.58 (m, 1H), 3.55-3.45 (m, 2H), 3.41 (s, 1H), 3.28-3.21 (m, 2H), 3.16-3.06 (m, 2H), 3.05-2.98 (m, 1H), 2.89-2.78 (m, 1H), 2.44-2.25 (m, 2H), 2.18-1.97 (m, 4H), 1.89-1.75 (m, 3H), 0.72-0.68 (m, 3H); LCMS (ESI, M+1): m/z=622.1.

Example 373

(1686) ##STR00707##

4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)azetidin-2-one

Example 374

(1687) ##STR00708##

3-amino-4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino) butanoic Acid

(1688) ##STR00709##

(1689) Step A. 4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)azetidin-2-one and 3-amino-4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)butanoic acid: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (60.0 mg, 1.0 equiv.) in N,N-dimethylformamide (1.00 mL) was added 4 Å molecular sieves (30.0 mg), diisopropylethylamine (39.3 mg, 3.0 equiv.) and 4-(aminomethyl)azetidin-2-one (20.3 mg, 2.0 equiv.). The mixture was stirred at 40° C. for 16 hours. The mixture was filtered. The filtrate was purified by prep-HPLC [column: 3_Phenomenex Luna C18 75×30 mm×3 µm; mobile phase: water (0.1% HCl)-acetonitrile; B %: 23%-43%, 7 minutes] and prep-HPLC [column: water s Xbridge 150×25 mm×5 µm; mobile phase: water (ammonium bicarbonate)-acetonitrile; B %: 34%-64%, 9 min] to afford 4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)azetidin-2-one (7.14 mg, 11% yield) as a white solid (second eluting peak); .sup.1H NMR (400 MHz, MeOD-d.sub.4) δ =9.15 (s, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.40-5.20 (m, 2H), 4.70-4.51 (m, 2H), 4.39-4.24 (m, 2H), 4.07 (br d, J=1.6 Hz, 1H), 4.02-3.85 (m, 2H), 3.27-3.17 (m, 3H), 3.17-3.09 (m, 1H), 3.07-2.97 (m, 1H), 2.87-2.79 (m, 1H), 2.55-2.29 (m, 2H), 2.29-2.08 (m, 4H), 2.04-1.84 (m, 4H), 0.79 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=593.3; and 3-amino-4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)butanoic acid (11.8 mg, 17.5% yield) as white solid (first eluting peak); .sup.1H NMR (400 MHz, MeOD-d.sub.4) δ =9.17 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.44-5.22 (m, 1H), 4.42-4.31 (m, 2H), 4.00-3.88 (m, 2H), 3.84-3.73 (m, 1H), 3.42-3.33 (m, 1H), 3.26 (br d, J=7.6 Hz, 1H), 3.12-3.00 (m, 1H), 2.66-2.56 (m, 1H), 2.53-2.35 (m, 3H), 2.33-2.10 (m, 4H), 2.09-1.88 (m, 4H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=611.1.

Example 375

(1690) ##STR00710##

3-(2-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-

1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl)pyrrolidin-2-one
(1691) The title compound was synthesized according to the procedure described for example 330.
.sup.1H NMR (400 MHz, MeOD-d.sub.4) δ =9.18 (s, 1H), 7.68 (dd, J=6.0, 8.8 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.05 (d, J=2.4 Hz, 1H), 5.51-5.30 (m, 1H), 4.55-4.39 (m, 2H), 3.93-3.69 (m, 2H), 3.67-3.43 (m, 3H), 3.42-3.34 (m, 2H), 3.26-3.13 (m, 1H), 2.64-2.51 (m, 1H), 2.51-2.38 (m, 3H), 2.38-2.19 (m, 3H), 2.19-2.08 (m, 3H), 2.07-1.78 (m, 3H), 1.40-1.35 (m, 11H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=621.1.

Example 376

(1692) ##STR00711##

4-(2-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl)pyrrolidin-2-one
(1693) The title compound was synthesized according to the procedure described for example 330.
.sup.1H NMR (400 MHz, MeOD) δ =9.18 (s, 1H), 8.45 (s, 1H), 7.68 (dd, J=5.6, 9.2 Hz, 1H), 7.34-7.30 (m, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.05 (s, 1H), 5.68-5.38 (m, 1H), 4.64-4.52 (m, 3H), 3.90-3.66 (m, 5H), 3.61 (dd, J=8.0, 9.6 Hz, 1H), 3.15 (dd, J=6.8, 9.2 Hz, 1H), 2.73-2.31 (m, 6H), 2.30-2.03 (m, 5H), 2.02-1.87 (m, 2H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=621.3.

Example 377

(1694) ##STR00712##

5-(2-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl)pyrrolidin-2-one
(1695) The title compound was synthesized according to the procedure described for example 330.
.sup.1H NMR (400 MHz, MeOD) δ =9.12 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25-7.22 (m, 1H), 7.03 (d, J=2.4 Hz, 1H), 5.44-5.21 (m, 1H), 4.60 (s, 2H), 4.41-4.26 (m, 2H), 3.90-3.70 (m, 3H), 3.23 (s, 1H), 3.04-3.03 (m, 1H), 2.53-2.21 (m, 6H), 2.20-2.09 (m, 2H), 2.06-1.86 (m, 6H), 0.80-0.76 (m, 3H); LCMS (ESI, M+1): m/z=621.3.

Example 378

(1696) ##STR00713##

1-(2-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl)imidazolidine-2,4-dione
(1697) The title compound was synthesized according to the procedure described for example 330.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.11 (s, 1H), 8.48 (s, 1H), 7.69-7.66 (m, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.27-7.22 (m, 11H), 7.04 (d, J=2.8 Hz, 11H), 5.54-5.40 (m, 1H), 4.57-4.55 (m, 11H), 4.50-4.49 (m, 1H), 4.18 (t, J=5.6 Hz, 11H), 3.83-3.70 (m, 2H), 3.62-3.27 (m, 5H), 3.31-3.27 (m, 1H), 2.46-2.45 (m, 1H), 2.41-2.18 (m, 6H), 2.16-2.05 (m, 11H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=636.4.

Example 379

(1698) ##STR00714##

3-(2-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl)imidazolidine-2,4-dione
(1699) The title compound was synthesized according to the procedure described for example 330.
.sup.1H NMR (400 MHz, MeOD) δ =9.07 (s, 1H), 8.50 (s, 1H), 7.68 (dd, J=9.2, 6.0 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.0 Hz, 1H), 5.56-5.34 (m, 1H), 4.59-4.48 (m, 2H), 3.99-3.84 (m, 6H), 3.72 (s, 1H), 3.57-3.47 (m, 2H), 3.26-3.18 (m, 1H), 2.64-2.24 (m, 4H), 2.21-2.00 (m, 4H), 0.78 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=636.3.

Example 380

(1700) ##STR00715##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)hexahydroimidazo[1,5-a]pyrazin-3(2H)-one
(1701) The title compound was synthesized according to the procedure described for example 330.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.13 (s, 11H), 7.70 (dd, J=5.9, 9.0 Hz, 11H),

7.33 (d, J=2.4 Hz, 1H), 7.27 (t, J=9.2 Hz, 1H), 7.07 (t, J=2.4 Hz, 1H), 5.49-5.24 (m, 1H), 4.77-4.58 (m, 3H), 4.46-4.29 (m, 2H), 4.17-4.05 (m, 1H), 3.95 (br d, J=12.4 Hz, 1H), 3.73-3.61 (m, 1H), 3.52-3.39 (m, 3H), 3.31-3.19 (m, 3H), 3.16-3.04 (m, 1H), 2.57-2.46 (m, 1H), 2.45-2.26 (m, 2H), 2.24-2.13 (m, 2H), 2.11-2.01 (m, 2H), 2.00-1.89 (m, 1H), 0.81 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=634.3.

Example 381

(1702) ##STR00716##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl) tetrahydroimidazo[1,5-a]pyrazine-1,3(2H,5H)-dione

(1703) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, MeOD-d₄) δ=9.15 (s, 1H), 8.50 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.53-5.24 (m, 1H), 4.63 (d, J=12.4 Hz, 1H), 4.53-4.37 (m, 3H), 4.16 (d, J=13.2 Hz, 1H), 3.68-3.52 (m, 2H), 3.52-3.46 (m, 1H), 3.43-3.35 (m, 4H), 3.19-3.08 (m, 1H), 2.57-2.28 (m, 3H), 2.27-2.15 (m, 2H), 2.13-2.03 (m, 2H), 2.03-1.94 (m, 1H), 0.80 (dt, J=2.4, 7.2 Hz, 3H); LCMS (ESI, M+1): m/z=648.2.

Example 382

(1704) ##STR00717##

4-(2-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl) imidazolidin-2-one

(1705) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, METHANOL-d₄) δ=9.15 (s, 1H), 8.47 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.8 Hz, 1H), 7.25 (t, J=10.0 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.59-5.35 (m, 1H), 4.64-4.47 (m, 2H), 3.98-3.88 (m, 1H), 3.86-3.77 (m, 2H), 3.76-3.53 (m, 4H), 3.26-3.20 (m, 1H), 2.66-2.52 (m, 1H), 2.50-2.40 (m, 2H), 2.38-2.28 (m, 1H), 2.26-2.13 (m, 3H), 2.12-2.05 (m, 1H), 2.04-1.94 (m, 2H), 0.78 (t, J=7.2 Hz, 3H). LCMS (ESI, M+1): m/z=622.1.

Example 383

(1706) ##STR00718##

5-(2-((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)ethyl) imidazolidine-2,4-dione

(1707) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, METHANOL-d₄) δ=9.13 (d, J=4.0 Hz, 1H), 8.46 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.03 (t, J=3.2 Hz, 1H), 5.63-5.40 (m, 1H), 4.69-4.48 (m, 3H), 4.31-4.24 (m, 1H), 4.15-3.92 (m, 1H), 3.89-3.80 (m, 1H), 3.77-3.62 (m, 3H), 2.71-2.54 (m, 1H), 2.49-2.35 (m, 3H), 2.31-2.19 (m, 4H), 2.18-2.08 (m, 2H), 0.79 (q, J=6.8 Hz, 3H); LCMS (ESI, M+1): m/z=636.3

Example 384

(1708) ##STR00719##

4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)methyl)amino)methyl)azetidin-2-one

(1709) The title compound was synthesized according to the procedure described for example 330.

¹H NMR (400 MHz, DMSO-d₆) δ=10.21-9.90 (m, 1H), 9.28 (s, 1H), 8.18 (br d, J=4.8 Hz, 1H), 7.80-7.72 (m, 1H), 7.40-7.29 (m, 2H), 7.01 (d, J=2.4 Hz, 1H), 5.38-5.17 (m, 1H), 4.17-4.01 (m, 5H), 3.62 (s, 3H), 3.14-2.97 (m, 4H), 2.84-2.74 (m, 2H), 2.30-1.92 (m, 4H), 1.93-1.90 (m, 1H), 1.89-1.74 (m, 3H), 0.73 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=607.4.

Example 385

(1710) ##STR00720##

5-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1-methyl-1H-pyrazole-3-carboxamide

(1711) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-*d*₆) δ=9.92 (s, 1H), 9.50 (m, 1H), 9.34 (s, 1H), 8.13 (s, 1H), 7.76 (m, 1H), 7.41 (s, 1H), 7.38-7.29 (m, 2H), 7.20-7.12 (m, 1H), 6.99 (d, J=2.4 Hz, 1H), 6.66 (s, 1H), 5.42-5.20 (m, 1H), 4.93-4.74 (m, 2H), 4.24-4.07 (m, 2H), 3.98 (s, 3H), 3.17-3.05 (m, 3H), 2.91-2.84 (m, 1H), 2.36-2.30 (m, 1H), 2.20-1.99 (m, 4H), 1.89-1.74 (m, 3H), 0.71 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=647.1

Example 386

(1712) ##STR00721##

(R)-4-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1713) ##STR00722##

(1714) Step A. 4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of 6-methyl-1,4-oxazepan-6-ol (117 mg, 1.3 equiv.), 4 Å molecular sieves (30 mg) and DIEA (353 mg, 4.0 equiv.) in DMF (3 mL) was stirred at 15° C. for 0.5 hour. Then 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (300 mg, 1.0 equiv.) was added into the above mixture and the mixture was stirred at 40° C. for 4 hours. After reaction completion, the mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃ and concentrated in vacuum to remove acetonitrile. The aqueous phase was extracted with ethyl acetate (2×15 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous Na₂SO₄, concentrated in vacuum to afford the title compound (210 mg, 62% yield) as a white solid; ¹H NMR (400 MHz, DMSO-*d*₆) δ=9.31 (s, 1H), 5.27 (br d, J=53.6 Hz, 1H), 5.15 (s, 1H), 4.37-4.26 (m, 1H), 4.20-4.11 (m, 2H), 4.08-3.96 (m, 3H), 3.95-3.89 (m, 1H), 3.82-3.73 (m, 1H), 3.57-3.48 (m, 2H), 3.11-2.99 (m, 3H), 2.87-2.78 (m, 1H), 2.13-2.03 (m, 2H), 2.01-1.94 (m, 1H), 1.88-1.72 (m, 3H), 1.13 (s, 3H); LCMS [ESI, M+1]: 470.1.

(1715) Step B. 4-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To a mixture of 4-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (190 mg, 1.0 equiv.), ((2-fluoro-6-(methoxymethoxy)-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (311 mg, 1.5 equiv.) and K₃PO₄ (1.5 M in water, 809 µL, 3.0 equiv.) in THF (2.4 mL) was added CataCXium A Math.Pd G3 (29.5 mg, 0.1 equiv.) under N₂. The mixture was degassed and stirred at 60° C. for 2.5 hours under N₂. After completion, the mixture was diluted with water (5 mL) and extracted with ethyl acetate (2×6 mL). The combined organic layers were concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO₃, concentrated in vacuum to remove acetonitrile. The aqueous phase was extracted with ethyl acetate (25 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous Na₂SO₄, concentrated in vacuum to afford the title compound (316 mg, 89% yield) as a yellow solid; LCMS [ESI, M+1, M/2+1]: 820.3, 410.9.

(1716) Step C. 4-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: A mixture of 4-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (320 mg, 1.0 equiv.) and CsF (889 mg, 15 equiv.) in DMF (1.6 mL) was stirred at 25° C. for 2 hours. After reaction

completion, the mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected, neutralized with solid NaHCO₃ and concentrated in vacuum to remove acetonitrile. The mixture was extracted with ethyl acetate (20 mL). The organic phase was washed with brine (20 mL), dried over anhydrous Na₂SO₄, concentrated in vacuum to afford the title compound (242 mg, 92% yield) as a yellow solid; ¹H NMR (400 MHz, DMSO-d₆) δ=9.39 (d, J=47.2 Hz, 1H), 8.10 (dd, J=6.0, 9.2 Hz, 1H), 7.74 (d, J=2.4 Hz, 1H), 7.55 (t, J=8.8 Hz, 1H), 7.39 (dd, J=1.6, 9.2 Hz, 1H), 5.38 (s, 2H), 5.36-5.08 (m, 2H), 4.42-4.31 (m, 1H), 4.20-4.05 (m, 4H), 4.02-3.93 (m, 2H), 3.90-3.79 (m, 1H), 3.63-3.49 (m, 2H), 3.44 (s, 3H), 3.30 (br s, 1H), 3.15-3.00 (m, 3H), 2.87-2.79 (m, 1H), 2.15-2.03 (m, 2H), 1.99 (s, 1H), 1.88-1.73 (m, 3H), 1.19-1.16 (m, 3H); ¹⁹F NMR (376 MHz, DMSO-d₆) δ=-108.987, -140.166, -172.173; LCMS [ESI, M+1, M/2+1]: 664.3, 332.7.

(1717) Step D. 4-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To a mixture of 4-(7-(8-ethynyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (260 mg, 1.0 equiv.) in dichloromethane (1 mL) was added TFA (1.54 g, 34 equiv.) at 0° C. and the mixture was stirred at 0-15° C. for 2 hours. After reaction completion, the mixture was concentrated in vacuum. The residue was neutralized with saturated NaHCO₃ solution and extracted with ethyl acetate (2×8 mL). The combined organic layers were washed with brine (10 mL), dried over anhydrous Na₂SO₄ and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with NaHCO₃ solid, concentrated in vacuum to remove acetonitrile. The mixture was extracted with ethyl acetate (2×20 mL) and the combined organic phase was washed with brine (25 mL), dried over anhydrous Na₂SO₄. The mixture was concentrated in vacuum to afford the title compound (177 mg, 70% yield) as a yellow solid; LCMS [ESI, M+1]: 620.2.

(1718) Step E. (R)-4-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: Stereoisomers of 4-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol (177 mg) were separated by SFC (column: DAICEL CHIRALPAK AD (250 mm×30 mm, 10 μm); mobile phase: A: [0.1% NH₃.H₂O-IPA], B: CO₂, B %: 35%-35%, 6.8 min over 90 min). Both peaks were further re-purified by prep-HPLC [column: Water s Xbridge 150×25 mm×5 μm; mobile phase: water (10 mM NH₄HCO₃)-ACN; B %: 30%-60%, 8 min] followed by another prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 μm; mobile phase: water (0.05% ammonia hydroxide v/v)-ACN, B %: 25%-55%, 7 min] to afford the title compound (peak 2) (35.1 mg, 19% yield) as an orange solid; SFC: 97.4%: Chiralpak IE-3 50×4.6 mm I.D., 3 μm; A: Heptane (0.05% DEA); B: 25% EtOH (0.05% DEA); 1 mL/min; 220 nm, t_R: 6.204 min; ¹H NMR (400 MHz, methanol-d₄) δ=9.42 (d, J=43.2 Hz, 1H), 7.88-7.81 (m, 1H), 7.36-7.29 (m, 2H), 7.22 (dd, J=2.4, 10.4 Hz, 1H), 5.30 (br d, J=54.0 Hz, 1H), 4.64-4.43 (m, 2H), 4.35-4.15 (m, 3H), 4.07-3.83 (m, 3H), 3.75-3.56 (m, 2H), 3.48-3.41 (m, 1H), 3.29-3.13 (m, 3H), 3.05-2.97 (m, 1H), 2.38-2.18 (m, 2H), 2.16-2.08 (m, 1H), 2.04-1.95 (m, 2H), 1.94-1.83 (m, 1H), 1.27 (d, J=13.6 Hz, 3H); LCMS (ESI, M+1): m/z=620.3.

Example 387

(1719) ##STR00723##

1-(2-(2-(dimethylamino)ethoxy)-7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol

(1720) ##STR00724##

(1721) Step A. 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol: A solution of

2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.), DIEA (512 mg, 5.0 equiv.) and 4 Å molecular sieves (20 mg) in DCM (4 mL) at 18° C. under N.sub.2 was de-gassed and piperidin-3-ol was added dropwise at -40° C. (218 mg, 2.0 equiv., HCl) under N.sub.2. The reaction was stirred at -40° C. for 15 minutes. The mixture was quenched with water (20 mL) and extracted with ethyl acetate (2×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated in vacuum to give a residue and the residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (193 mg, 77% yield) as a yellow solid. LCMS (ESI, M+1): m/z=316.9.

(1722) Step B. 1-(7-chloro-2-(2-(dimethylamino)ethoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol: A stirred mixture of 1-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol (129 mg, 1.0 equiv.), DIEA (157 mg, 3.0 equiv.) and 2-(dimethylamino) ethanol (366 mg, 10 equiv.) was heated to 90° C. for 3 hours. The mixture was diluted with ethyl acetate (10 mL) and water (20 mL). The aqueous phase was extracted with ethyl acetate (20 mL), the combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered, concentrated and the residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (126 mg, 84% yield) as a yellow solid. LCMS (ESI, M+1): m/z=370.1.

(1723) Step C. 1-(2-(2-(dimethylamino)ethoxy)-8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol: A mixture of ((6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)ethynyl)naphthalen-2-yl)oxy)triisopropylsilane (243 mg, 1.2 equiv.), 1-(7-chloro-2-(2-(dimethylamino)ethoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol (120 mg, 1.0 equiv.), Cs.sub.2CO.sub.3 (1.5 M in water, 3.0 equiv.) and CataCXium A Pd G3 (24 mg, 0.1 equiv.) in methoxycyclopentane (4 mL) was de-gassed and attired at 90° C. for 2.5 hours under N.sub.2. The mixture was diluted with ethyl acetate (10 mL) and water (20 mL). The aqueous phase was extracted with ethyl acetate (10 mL), the combined organic phase was washed with brine (15 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue and the residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (115 mg, 42% yield) as a yellow solid; LCMS (ESI, M+1): m/z=832.4.

(1724) Step D. 1-(2-(2-(dimethylamino)ethoxy)-7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol: A mixture of 1-(2-(2-(dimethylamino)ethoxy)-8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-ol (60 mg, 1.0 equiv.) and CsF (164 mg, 15 equiv.) in DMF (0.6 mL) was stirred at 18° C. for 12 hours. After reaction completion, the mixture was filtered and the filtrate was purified by prep-HPLC (column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 24%-54%, 7 min) to afford the title compound (5.3 mg, 14% yield) as a white solid; .sup.1H NMR (400 MHz, CD3OD, 298 K) δ (ppm)=9.12-9.02 (m, 1H), 7.85 (dd, J=5.6, 8.8 Hz, 1H), 7.36-7.28 (m, 2H), 7.22 (dd, J=2.4, 10.4 Hz, 1H), 4.66-4.59 (m, 2H), 4.29-4.17 (m, 1H), 4.14-3.76 (m, 4H), 3.50-3.38 (m, 1H), 2.84 (t, J=5.6 Hz, 2H), 2.37 (s, 6H), 2.18-2.01 (m, 2H), 1.84-1.68 (m, 2H) LCMS [ESI, M+1]: m/z=520.2.

Example 388

(1725) ##STR00725##

(1R,5R,6R)-3-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol

(1726) ##STR00726##

(1727) Step A. 8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-

((((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (130 mg, 1.0 equiv.), ((6-fluoro-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)ethynyl)naphthalen-2-yl)oxy)triisopropylsilane (204 mg, 1.1 equiv.), Cs.sub.2CO.sub.3 (1.5 M in water, 3.0 equiv.) and CataCXium A Pd G3 (21.6 mg, 0.1 equiv.) in methoxycyclopentane (2 mL) was de-gassed and stirred at 100° C. for 2.5 hours under N.sub.2. The reaction mixture was diluted with ethyl acetate (15 mL) and water (30 mL). The aqueous phase was extracted with ethyl acetate (10 mL), the combined organic phase was washed with brine (15 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO.sub.3, concentrated in vacuum to afford the title compound (260 mg, 97% yield) as a yellow oil; LCMS (ESI, M+1): m/z=901.3.

(1728) Step B. 3-(8-fluoro-7-(7-fluoro-3-hydroxy-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol: A solution of 8-fluoro-7-(7-fluoro-8-((triisopropylsilyl)ethynyl)-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (260 mg, 1.0 equiv.), (1R,5R,6R)-3-azabicyclo[3,2,1]octan-6-ol (47.7 mg, 1.3 equiv.), DIEA (112 mg, 3.0 equiv.) and 4 Å molecular sieves (20 mg) in DMF (3 mL) was stirred at 40° C. for 12 hrs. Additional amounts of (1R,5R,6R)-3-azabicyclo[3,2,1]octan-6-ol (25.7 mg, 0.7 equiv.) and DIEA (37.3 mg, 1.0 equiv.) were introduced into the reaction mixture and the resulting was stirred at 40° C. for 12 hrs. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO.sub.3, concentrated in vacuum to remove ACN. The aqueous phase was extracted with ethyl acetate (10 mL×2). The combined organic phase was dried with anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuum to afford the title compound (110 mg, 44% yield) as a white solid; LCMS (ESI, M+1): m/z=772.3.

(1729) Step C. (1R,5R,6R)-3-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol: To a solution of 3-(8-fluoro-7-(7-fluoro-3-hydroxy-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol (100 mg, 1.0 equiv.) in DMF (1 mL) was added CsF (295 mg, 15 equiv.). The reaction was stirred at 20° C. for 12 hours. The mixture was filtered and the filtrate was purified by prep-HPLC [column: Water s Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 38%-68%, 9 min] and lyophilized to afford the title compound (33.8 mg, 42% yield) as a yellow solid; .sup.1H NMR (400 MHz, CD3SOCD3, 298 K) δ (ppm)=10.58-9.93 (m, 1H), 9.36-9.01 (m, 1H), 8.00-7.91 (m, 1H), 7.49-7.34 (m, 2H), 7.26-7.13 (m, 1H), 5.37-5.17 (m, 1H), 5.10-4.54 (m, 2H), 4.51-4.05 (m, 3H), 4.01-3.94 (m, 1H), 3.93-3.86 (m, 1H), 3.77-3.63 (m, 1H), 3.23 (br d, J=12.8 Hz, 1H), 3.14-2.97 (m, 3H), 2.86-2.77 (m, 1H), 2.32 (s, 1H), 2.18-1.92 (m, 5H), 1.92-1.58 (m, 5H), 1.33-1.20 (m, 1H); LCMS (ESI, M+1): m/z=616.2

Example 389

(1730) ##STR00727##

5-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1731) The title compound was synthesized according to the procedure described for example 388. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ 9.23 (s, 1H), 7.90-7.83 (m, 1H), 7.38-7.28 (m, 2H), 7.23 (d, J=2.4 Hz, 1H), 5.55-5.36 (m, 1H), 4.76-4.73 (m, 1H), 4.65-4.61 (m, 1H), 4.56-4.46 (m, 2H), 4.41-4.33 (m, 1H), 4.31-4.22 (m, 1H), 3.79-3.71 (m, 2H), 3.71-3.52 (m, 3H), 3.36-3.33 (m, 1H), 3.26-3.24 (m, 1H), 2.60-2.37 (m, 2H), 2.34-2.26 (m, 1H), 2.24-2.14 (m, 2H), 2.11-1.99 (m,

1H); LCMS (ESI, M+1): m/z=629.3.

Example 390

(1732) ##STR00728##

6-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3,2,1]octan-3-ol

(1733) ##STR00729##

(1734) Step A. 6-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol: To a solution of 5-ethynyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (100 mg, 1.0 equiv.) and 4 Å molecular sieves (20.0 mg) in DMF (1 mL) was added DIEA (371 mg, 0.50 mL, 16.9 equiv.) and 6-azabicyclo[3,2,1]octan-3-ol (123 mg, 3.0 equiv., TFA). The mixture was stirred at 40° C. for 12 hours. The mixture was filtered and the filtrate was purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 12%-42%, 10.5 min) to afford the title compound (15.6 mg, 14% yield) as a yellow solid; .sup.1H NMR (400 MHz, METHANOL-d) δ=9.27-9.10 (m, 1H), 7.84 (dd, J=5.6, 8.4 Hz, 1H), 7.36-7.27 (m, 2H), 7.27-7.17 (m, 1H), 5.40-5.21 (m, 1H), 4.97 (br s, 1H), 4.40-4.18 (m, 4H), 4.16-4.05 (m, 1H), 3.27-3.12 (m, 3H), 3.07-2.95 (m, 1H), 2.82-2.53 (m, 2H), 2.40-2.27 (m, 1H), 2.26-2.11 (m, 3H), 2.07-1.95 (m, 4H), 1.95-1.85 (m, 2H), 1.84-1.62 (m, 2H); LCMS (ESI, M+1): m/z=616.5.

Example 391

(1735) ##STR00730##

5-ethynyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1736) The title compound was synthesized according to the procedure described for example 390. 1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.19 (d, J=14.4 Hz, 1H), 8.51 (br s, 11H), 7.89 (dd, J=5.6, 9.2 Hz, 1H), 7.38 (d, J=2.4 Hz, 1H), 7.35 (t, J=9.2 Hz, 1H), 7.24 (dd, J=2.4, 10.8 Hz, 1H), 5.55-5.34 (m, 1H), 4.74-4.54 (m, 2H), 4.53-4.48 (m, 1H), 4.48-4.30 (m, 2H), 4.20-4.10 (m, 2H), 4.09 (br s, 2H), 3.94-3.77 (m, 1H), 3.70-3.61 (m, 3H), 3.60-3.48 (m, 3H), 3.29-3.20 (m, 1H), 2.59-2.34 (m, 3H), 2.32-2.22 (m, 1H), 2.21-2.11 (m, 2H), 2.10-1.97. (m, 1H); LCMS (ESI, M+1): m/z=620.3.

Example 392

(1737) ##STR00731##

6-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,6-diazaspiro[3.5]nonan-1-one

(1738) The title compound was synthesized according to the procedure described for example 390. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.01 (d, J=8.0 Hz, 1H), 7.85 (dd, J=5.6, 9.2 Hz, 1H), 7.36-7.27 (m, 2H), 7.23 (dd, J=2.4, 6.4 Hz, 1H), 5.40-5.20 (m, 1H), 4.43-4.27 (m, 3H), 4.26-4.19 (m, 1H), 4.13-3.98 (m, 1H), 3.85-3.60 (m, 1H), 3.42-3.33 (m, 1H), 3.29-3.12 (m, 5H), 3.00 (dt, J=5.6, 9.2 Hz, 1H), 2.39-2.26 (m, 1H), 2.24-2.02 (m, 5H), 2.02-1.82 (m, 4H); LCMS (ESI, M+1): m/z=629.5.

Example 393

(1739) ##STR00732##

6-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1740) The title compound was synthesized according to the procedure described for example 390. 1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.05-8.96 (m, 1H), 7.86 (dd, J=5.6, 9.2 Hz, 1H), 7.43-7.28 (m, 1H), 7.39-7.26 (m, 1H), 7.24-7.18 (m, 1H), 5.45-5.15 (m, 1H), 4.36-4.30 (m, 1H), 4.39-4.17 (m, 1H), 4.16-4.09 (m, 1H), 4.06 (br s, 1H), 4.02-3.95 (m, 1H), 3.95-3.84 (m, 1H), 3.47-3.35 (m, 1H), 3.28-3.12 (m, 3H), 3.06-2.98 (m, 1H), 2.39-2.32 (m, 1H), 2.32-2.26 (m, 1H), 2.25-

2.11 (m, 3H), 2.05-1.96 (m, 2H), 1.96-1.67 (m, 5H); LCMS (ESI, M+1): m/z=630.4.

Example 394

(1741) ##STR00733##

4-(4-(2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethynyl-6-fluoronaphthalen-2-ol

(1742) The title compound was synthesized according to the procedure described for example 390. ¹H NMR (400 MHz, METHANOL-d.sub.4) δ=9.19 (s, 1H), 8.47 (s, 1H), 7.87 (dd, J=5.6, 9.2 Hz, 1H), 7.36 (d, J=2.4 Hz, 1H), 7.36-7.30 (m, 1H), 7.22 (d, J=2.4 Hz, 1H), 5.79 (s, 1H), 5.53-5.34 (m, 1H), 5.18-5.03 (m, 2H), 4.54-4.41 (m, 2H), 4.38 (br d, J=4.8 Hz, 2H), 4.25 (br dd, J=4.0, 5.6 Hz, 2H), 3.70-3.47 (m, 3H), 3.40 (d, J=2.4 Hz, 1H), 3.24 (dt, J=6.0, 9.6 Hz, 1H), 2.55-2.43 (m, 1H), 2.39 (br d, J=4.0 Hz, 3H), 2.32-2.23 (m, 1H), 2.23-2.12 (m, 2H), 2.09-1.98 (m, 1H); LCMS (ESI, M+1): m/z=641.2

Example 395

(1743) ##STR00734##

7-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1744) The title compound was synthesized according to the procedure described for example 390. ¹H NMR (400 MHz, METHANOL-d.sub.4) δ=9.06 (s, 1H), 8.51 (br s, 1H), 7.86 (ddd, J=1.6, 5.6, 9.2 Hz, 1H), 7.39-7.26 (m, 2H), 7.22 (dd, J=2.4, 4.4 Hz, 1H), 5.52-5.29 (m, 1H), 4.52-4.36 (m, 2H), 4.36-4.28 (m, 1H), 4.25-3.98 (m, 2H), 3.97-3.78 (m, 1H), 3.63-3.49 (m, 1H), 3.49-3.40 (m, 3H), 3.40-3.36 (m, 1H), 3.28 (br d, J=2.0 Hz, 1H), 3.21-3.11 (m, 1H), 2.53-2.30 (m, 2H), 2.30-2.18 (m, 1H), 2.16-1.94 (m, 6H), 1.92-1.82 (m, 1H). LCMS (ESI, M+1): m/z=644.2.

Example 396

(1745) ##STR00735##

7-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(1746) The title compound was synthesized according to the procedure described for example 390. ¹H NMR (400 MHz, METHANOL-d.sub.4) δ=9.19 (d, J=14.4 Hz, 1H), 8.51 (br s, 1H), 7.89 (dd, J=5.6, 9.2 Hz, 1H), 7.38 (d, J=2.4 Hz, 1H), 7.35 (t, J=9.2 Hz, 1H), 7.24 (dd, J=2.4, 10.8 Hz, 1H), 5.55-5.34 (m, 1H), 4.74-4.54 (m, 2H), 4.53-4.48 (m, 1H), 4.48-4.30 (m, 2H), 4.20-4.10 (m, 2H), 4.09 (br s, 2H), 3.94-3.77 (m, 1H), 3.70-3.61 (m, 3H), 3.60-3.48 (m, 3H), 3.29-3.20 (m, 1H), 2.59-2.34 (m, 3H), 2.32-2.22 (m, 1H), 2.21-2.11 (m, 2H), 2.10-1.97 (m, 1H); LCMS (ESI, M+1): m/z=680.2.

Example 397

(1747) ##STR00736##

7-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1748) The title compound was synthesized according to the procedure described for example 390. ¹H NMR (400 MHz, METHANOL-d.sub.4) δ 9.05 (d, J=6.4 Hz, 1H), 7.86 (dd, J=6.0, 9.2 Hz, 1H), 7.37-7.29 (m, 2H), 7.23 (dd, J=2.4, 4.8 Hz, 1H), 5.38-5.22 (m, 1H), 4.64-4.58 (m, 1H), 4.56-4.49 (m, 1H), 4.45-4.13 (m, 3H), 3.97-3.56 (m, 3H), 3.23-3.12 (m, 2H), 3.05-2.96 (m, 1H), 2.37-2.23 (m, 2H), 2.23-2.13 (m, 2H), 2.12-2.03 (m, 2H), 2.03-1.91 (m, 4H). LCMS (ESI, M+1): m/z=658.4.

Example 398

(1749) ##STR00737##

5-(7-(8-ethynyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1750) The title compound was synthesized according to the procedure described for example 390. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.14 (s, 1H), 7.88 (dd, J=5.6, 9.2 Hz, 1H), 7.38-7.31 (m, 2H), 7.23 (d, J=2.4 Hz, 1H), 6.79 (s, 1H), 5.42-5.19 (m, 3H), 4.61-4.55 (m, 2H), 4.46 (br t, J=5.6 Hz, 2H), 4.35-4.21 (m, 2H), 3.43 (d, J=4.8 Hz, 1H), 3.36 (s, 2H), 3.28-3.14 (m, 3H), 3.10 (s, 3H), 3.06-2.98 (m, 1H), 2.46 (br s, 2H), 2.07-1.84 (m, 4H), 2.37-1.84 (m, 3H); LCMS (ESI, M+1): m/z=697.3.

Example 399

(1751) ##STR00738##

(R)-1-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1752) ##STR00739##

(1753) Step A. (S)-1-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 7-(8-bromo-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.) in DMF (0.5 mL) were added DIEA (129 mg, 8.0 equiv.), 4 Å molecular sieves (5 mg, 1.00 equiv.) and (3R)-3-methylpiperidin-3-ol (28.4 mg, 1.97 equiv.). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and the filtrate was purified by prep-HPLC (column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 16%-46%, 8 min) and lyophilized to afford the title compound (13.2 mg, 16% yield) as a brown solid; ¹H NMR (400 MHz, METHANOL-d₃) δ=9.23 (d, J=11.2 Hz, 1H), 8.61-8.38 (m, 1H), 7.91-7.82 (m, 1H), 7.41-7.33 (m, 2H), 7.24 (dd, J=2.8, 4.4 Hz, 1H), 5.52-5.35 (m, 1H), 4.59 (br d, J=13.2 Hz, 1H), 4.54-4.40 (m, 2H), 4.33 (br dd, J=5.6, 13.2 Hz, 1H), 3.73-3.59 (m, 2H), 3.59-3.51 (m, 2H), 3.49-3.41 (m, 1H), 3.28-3.19 (m, 1H), 2.60-2.33 (m, 2H), 2.32-2.24 (m, 1H), 2.23-2.09 (m, 3H), 2.07-1.98 (m, 1H), 1.90-1.73 (m, 3H), 1.30 (d, J=6.4 Hz, 3H); LCMS (ESI, M+1): m/z=658.2.

Example 400

(1754) ##STR00740##

4-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1755) The title compound was synthesized according to the procedure described for example 399. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.54 (dd, J=2.8, 12.0 Hz, 1H), 8.51 (s, 1H), 7.86 (ddd, J=1.2, 5.6, 9.2 Hz, 1H), 7.43-7.32 (m, 2H), 7.24 (dd, J=2.4, 14.4 Hz, 1H), 5.48-5.28 (m, 1H), 4.59-4.49 (m, 2H), 4.48-4.33 (m, 2H), 4.23-4.14 (m, 1H), 4.06-3.87 (m, 3H), 3.75-3.63 (m, 2H), 3.55-3.36 (m, 3H), 3.21-3.09 (m, 1H), 2.49-2.28 (m, 2H), 2.26-2.18 (m, 1H), 2.14-2.04 (m, 2H), 2.03-1.92 (m, 1H), 1.28 (s, 3H). LCMS (ESI, M+1): m/z=674.2.

Example 401

(1756) ##STR00741##

5-bromo-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1757) The title compound was synthesized according to the procedure described for example 399. ¹H NMR (400 MHz, METHANOL-d₃) δ=9.34-9.16 (m, 1H), 8.59-8.46 (m, 1H), 7.88 (dd, J=5.6, 9.2 Hz, 1H), 7.46-7.34 (m, 2H), 7.27 (dd, J=2.4, 11.2 Hz, 1H), 5.54-5.35 (m, 1H), 4.67 (dt, J=4.4, 9.2 Hz, 1H), 4.60-4.39 (m, 3H), 4.21-3.94 (m, 4H), 3.92-3.78 (m, 1H), 3.69-3.44 (m, 6H), 3.28-3.17 (m, 1H), 2.35 (br s, 3H), 2.31-1.98 (m, 4H); LCMS (ESI, M+1): m/z=674.2.

Example 402

(1758) ##STR00742##

3-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3.2.1]octan-6-ol

(1759) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.32-9.20 (m, 1H), 8.59-8.47 (m, 1H), 7.88 (ddd, J=1.2, 5.6, 9.2 Hz, 1H), 7.41 (d, J=2.4 Hz, 1H), 7.40-7.35 (m, 1H), 7.26 (dd, J=2.4, 16.8 Hz, 1H), 5.53-5.35 (m, 1H), 5.03 (br d, J=9.6 Hz, 1H), 4.80 (br s, 1H), 4.54-4.47 (m, 1H), 4.46-4.38 (m, 1H), 4.38-4.30 (m, 1H), 3.91-3.74 (m, 1H), 3.71-3.43 (m, 4H), 3.28-3.18 (m, 1H), 2.59-2.37 (m, 3H), 2.36-2.21 (m, 3H), 2.21-2.11 (m, 2H), 2.09-1.92 (m, 2H), 1.89-1.78 (m, 1H), 1.45-1.35 (m, 1H); LCMS (ESI, M+1): m/z=670.1.

Example 403

(1760) ##STR00743##

6-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol

(1761) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.30 (br d, J=7.2 Hz, 1H), 8.50 (s, 1H), 7.86 (dd, J=5.6, 9.2 Hz, 1H), 7.34 (s, 2H), 7.29-7.19 (m, 1H), 5.53-5.35 (m, 1H), 5.00 (br s, 1H), 4.55-4.34 (m, 3H), 4.31-4.24 (m, 1H), 4.14 (br s, 1H), 3.74-3.52 (m, 3H), 3.27-3.21 (m, 1H), 2.84-2.58 (m, 2H), 2.55 (br s, 1H), 2.45-2.26 (m, 2H), 2.17 (br dd, J=5.2, 9.2 Hz, 3H), 2.10-1.98 (m, 3H), 1.94-1.76 (m, 2H). LCMS (ESI, M+1): m/z=670.2.

Example 404

(1762) ##STR00744##

5-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1763) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.26 (s, 1H), 8.51 (br s, 1H), 7.86 (dd, J=5.6, 9.2 Hz, 1H), 7.40-7.33 (m, 2H), 7.24 (d, J=2.4 Hz, 1H), 5.47-5.29 (m, 1H), 4.64 (br t, J=12.0 Hz, 2H), 4.47-4.42 (m, 1H), 4.41-4.30 (m, 3H), 3.81-3.72 (m, 2H), 3.57-3.45 (m, 1H), 3.45-3.35 (m, 2H), 3.20-3.10 (m, 1H), 2.51-2.30 (m, 2H), 2.29-2.17 (m, 1H), 2.15-2.05 (m, 2H), 2.04-1.93 (m, 1H); LCMS (ESI, M+1): m/z=682.9.

Example 405

(1764) ##STR00745##

6-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1765) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.11 (s, 1H), 8.45 (br s, 1H), 7.87 (br dd, J=5.8, 8.0 Hz, 1H), 7.43-7.33 (m, 2H), 7.25 (br s, 1H), 5.58-5.34 (m, 1H), 4.63-4.26 (m, 4H), 4.09-3.96 (m, 1H), 3.92-3.54 (m, 4H), 3.29-3.22 (m, 1H), 2.99-2.72 (m, 2H), 2.62-2.25 (m, 3H), 2.23-1.92 (m, 7H) LCMS (ESI, M+1): m/z=683.2.

Example 406

(1766) ##STR00746##

6-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,6-diazaspiro[3.5]nonan-1-one

(1767) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.09 (d, J=2.0 Hz, 1H), 8.49 (br d, J=2.4 Hz, 1H), 7.86 (dd, J=5.6, 9.2 Hz, 1H), 7.42-7.33 (m, 2H), 7.25 (t, J=3.2 Hz, 1H), 5.57-5.28 (m, 1H), 4.54-4.27 (m, 4H), 4.26-3.91 (m, 2H), 3.70-3.45 (m, 3H), 3.29-3.17 (m, 3H), 2.57-2.35 (m, 2H), 2.32-2.08 (m, 6H), 2.07-1.89 (m, 2H); LCMS (ESI, M+1): m/z=683.2.

Example 407

(1768) ##STR00747##

6-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1769) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.11-9.05 (m, 11H), 8.54-8.43 (m, 1H), 7.91-7.82 (m, 1H), 7.43-7.31 (m, 2H), 7.30-7.22 (m, 1H), 5.52-5.36 (m, 1H), 4.56-4.44 (m, 2H), 4.32-4.24 (m, 1H), 4.15-3.95 (m, 4H), 3.69-3.50 (m, 3H), 3.27-3.19 (m, 1H), 2.53-2.15 (m, 7H), 2.09-2.00 (m, 11H), 1.88-1.72 (m, 6H); LCMS (ESI, M+1): m/z=684.2.

Example 408

(1770) ##STR00748##

4-(4-(2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-bromo-6-fluoronaphthalen-2-ol

(1771) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.20 (s, 1H), 7.87-7.83 (m, 1H), 7.42-7.33 (m, 2H), 7.25 (d, J=2.0 Hz, 1H), 5.78 (s, 1H), 5.51-5.33 (m, 1H), 5.19-5.01 (m, 2H), 4.51-4.39 (m, 2H), 4.38-4.32 (m, 2H), 4.27-4.18 (m, 2H), 3.65-3.42 (m, 3H), 3.25-3.15 (m, 11H), 2.54-2.41 (m, 1H), 2.40-2.30 (m, 3H), 2.30-2.21 (m, 1H), 2.19-2.09 (m, 2H), 2.07-1.96 (m, 1H); LCMS (ESI, M+1): m/z=695.2.

Example 409

(1772) ##STR00749##

7-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1773) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.09 (s, 1H), 7.91-7.79 (m, 1H), 7.46-7.31 (m, 2H), 7.24 (s, 11H), 5.44-5.19 (m, 11H), 4.37-4.23 (m, 2H), 4.19-4.06 (m, 2H), 4.05-3.89 (m, 2H), 3.52-3.34 (m, 2H), 3.27-3.18 (m, 3H), 3.05-2.96 (m, 1H), 2.39-2.12 (m, 3H), 2.05-1.84 (m, 7H); LCMS (ESI, M+1): m/z=698.1.

Example 410

(1774) ##STR00750##

7-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1775) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.09 (s, 1H), 7.85 (dd, J=6.0, 8.8 Hz, 1H), 7.40-7.20 (m, 3H), 5.45-5.15 (m, 11H), 4.54-4.40 (m, 2H), 4.31-4.18 (m, 2H), 3.86-3.70 (m, 2H), 3.23-3.16 (m, 2H), 3.13-2.93 (m, 2H), 2.39-2.17 (m, 4H), 2.16-2.08 (m, 2H), 2.03-1.98 (m, 2H), 1.95-1.88 (m, 2H). LCMS (ESI, M+1): m/z=712.0.

Example 411

(1776) ##STR00751##

7-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(1777) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.11 (d, J=7.2 Hz, 1H), 8.54-8.43 (m, 1H), 7.86 (td, J=5.2, 9.2 Hz, 1H), 7.41-7.31 (m, 2H), 7.23 (t, J=2.0 Hz, 1H), 5.57-5.34 (m, 1H), 4.65 (br dd, J=4.0, 14.4 Hz, 1H), 4.62-4.53 (m, 1H), 4.52-4.41 (m, 2H), 3.85-3.76 (m, 1H), 3.74-3.50 (m, 4H), 3.45-3.38 (m, 1H), 3.28-3.18 (m, 2H), 2.62-2.38 (m, 2H), 2.38-2.27 (m, 1H), 2.22-2.13 (m, 2H), 2.12-2.00 (m, 3H), 1.99-1.83 (m, 2H); LCMS (ESI, M+1): m/z=734.1.

Example 412

(1778) ##STR00752##

5-(7-(8-bromo-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1779) The title compound was synthesized according to the procedure described for example 399.

.sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.19 (s, 1H), 8.52 (br s, 1H), 7.86 (dd, J=5.4, 8.8 Hz, 1H), 7.43-7.33 (m, 2H), 7.24 (d, J=2.4 Hz, 1H), 6.76 (s, 1H), 5.46-5.19 (m, 3H), 4.60 (br s, 4H), 4.42-4.30 (m, 2H), 3.54-3.45 (m, 1H), 3.42-3.36 (m, 2H), 3.34 (s, 3H), 3.21-3.11 (m, 1H), 3.11-3.06 (m, 31H), 2.51-2.50 (m, 1H), 2.18-2.16 (m, 11H), 2.50-2.16 (m, 5H), 2.13-1.92 (m, 3H); LCMS (ESI, M+1): m/z=751.0.

Example 413

(1780) ##STR00753##

(R)-7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1781) ##STR00754##

(1782) Step A. (R)-7-(7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a mixture of 7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), (5R)-1,3,9-triazaspiro[4.5]decane-2,4-dione (78.9 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (1 mL) was added DIEA (100 mg, 5.0 equiv.). The reaction was stirred at 40° C. for 12 hours. The residue was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (110 mg, 92% yield) as a yellow solid; LCMS (ESI, M+1): m/z=712.3.

(1783) Step B. (R)-7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: To a solution of (R)-7-(7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione (50 mg, 1.0 equiv.) in DCM (2 mL) and MeOH (2 mL) was added TsOH (121 mg, 10 equiv.). The reaction was stirred at 10° C. for 12 hours. The residue was filtered and purified by prep-HPLC [column: Phenomenex C18 75×30 mm×3 µm; mobile phase: water (0.1% formic acid)/ACN] B %: 10%-40%, 7 min] and lyophilized to afford the title compound (20.0 mg, 41% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.15 (d, J=14.8 Hz, 1H), 7.83 (ddd, J=1.6, 5.6, 9.2 Hz, 1H), 7.52-7.29 (m, 2H), 7.27-7.17 (m, 1H), 5.64-5.43 (m, 1H), 4.77-4.28 (m, 6H), 3.95-3.67 (m, 4H), 3.44-3.32 (m, 1H), 2.72-2.52 (m, 1H), 2.51-2.29 (m, 2H), 2.29-2.05 (m, 4H), 2.05-1.87 (m, 2H); LCMS (ESI, M+1): m/z=668.2.

Example 414

(1784) ##STR00755##

1-(1-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1785) The title compound was synthesized according to the procedure described for example 413 except for HCl.dioxane in MeCN was added in step B. .sup.1H NMR (400 MHz, DMSO-d₆) δ =9.06 (d, J=2.0 Hz, 1H), 7.92 (dd, J=6.0, 9.2 Hz, 1H), 7.55 (t, J=8.8 Hz, 1H), 7.41 (d, J=2.4 Hz, 1H), 7.19 (s, 1H), 7.03-6.70 (m, 2H), 5.38-5.17 (m, 1H), 4.70-4.53 (m, 1H), 4.45-4.31 (m, 1H), 4.19-4.00 (m, 2H), 3.16-3.03 (m, 4H), 3.03-2.96 (m, 2H), 2.86-2.78 (m, 1H), 2.42-2.34 (m, 1H), 2.33 (m, 1H), 2.20-2.09 (m, 11H), 2.09-1.98 (m, 3H), 1.92-1.69 (m, 5H), 1.60-1.45 (m, 1H); LCMS (ESI, M+1): m/z=677.3.

Example 415

(1786) ##STR00756##

(S)-7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1787) The title compound was synthesized according to the procedure described for example 413. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.14 (d, J=7.2 Hz, 1H), 7.85-7.66 (m, 1H), 7.47-

7.31 (m, 2H), 7.25-7.18 (m, 1H), 5.65-5.33 (m, 1H), 4.63-4.54 (m, 2H), 4.39 (br t, J=13.2 Hz, 1H), 4.31-4.16 (m, 1H), 4.05-3.63 (m, 5H), 3.47-3.34 (m, 2H), 2.73-2.45 (m, 2H), 2.43 (br d, J=4.4 Hz, 5H), 2.09-1.76 (m, 4H); LCMS (ESI, M+1): m/z=654.2.

Example 416

(1788) ##STR00757##

(S)-6-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1789) ##STR00758##

(1790) Step A. (S)-6-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a solution of 2,4,7-trichloro-8-fluoropyrido[4,3-d]pyrimidine (500 mg, 1.0 equiv.), (S)-1,6-diazaspiro[3.5]nonan-2-one (138 mg, 0.5 equiv.) in DCM (5 mL) was added DIEA (767 mg, 1.03 mL, 3.0 equiv.). The mixture was stirred at -40° C. for 0.5 hour. After reaction completion, the mixture was concentrated under reduced pressure to give a residue. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (330 mg, 46% yield) as a white solid; LCMS [ESI, M+1]: m/z=356.1.

(1791) Step B. (S)-6-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxyl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a mixture of (S)-6-(2,7-dichloro-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (300 mg, 1.0 equiv.), ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (268 mg, 2.0 equiv.), 4 Å molecular sieves (10.0 mg, 1.0 equiv.) in dioxane (3 mL) was added DIEA (544 mg, 733 µL, 5.0 equiv.). The reaction was stirred at 90° C. for 12 hours. After completion, the reaction mixture was quenched by addition of water (3 mL), and extracted with Ethyl acetate (10 mL). The combined organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (210 mg, 52% yield) as a yellow solid; LCMS [ESI, M+1]: m/z=479.1.

(1792) Step C. (S)-6-(7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a solution of (S)-6-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (130 mg, 1.0 equiv.), (8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)trimethylstannane (328 mg, 3.0 equiv.) in toluene (3 mL) was added Pd(dppf)C12 (19.8 mg, 0.1 equiv.), BINAP (33.8 mg, 0.2 equiv.), CuI (15.5 mg, 0.3 equiv.) under N.sub.2. The reaction was stirred at 90° C. for 2 hours under N.sub.2. After reaction completion, the mixture was quenched by addition of water (3 mL), and then extracted with Ethyl acetate (6 mL). The organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (57.0 mg, 17% yield) as a green oil; LCMS [ESI, M+1]: m/z=683.4.

(1793) Step D. (S)-6-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a solution of (S)-6-(7-(8-chloro-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (52.0 mg, 1.0 equiv.) in DCM (0.5 mL) was added TFA (770 mg, 0.5 mL, 88 equiv.) at 0° C. The reaction was stirred at 0° C. for 0.5 hour. After reaction completion, the mixture was concentrated under reduced pressure to give a residue. The pH of the residue was adjusted to 9 with sat. aq. NaHCO.sub.3 and the resulting mixture was extracted with DCM (3×2 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 8%-38%,

10 min) and lyophilized to afford the title compound (26.3 mg, 49% yield, formic acid salt) as a white solid; ¹H NMR (400 MHz, methanol-d₄): δ=9.10 (d, J=3.2 Hz, 1H), 8.49 (br s, 1H), 7.82 (dd, J=5.6, 9.2 Hz, 1H), 7.43-7.37 (m, 2H), 7.22 (d, J=2.4 Hz, 1H), 5.49-5.32 (m, 1H), 4.47-4.37 (m, 3H), 4.36-4.26 (m, 1H), 3.99 (dd, J=2.8, 13.2 Hz, 1H), 3.86-3.75 (m, 1H), 3.62-3.43 (m, 3H), 3.18 (dt, J=5.6, 9.6 Hz, 1H), 2.91 (dd, J=6.8, 14.8 Hz, 1H), 2.79-2.73 (m, 1H), 2.53-2.33 (m, 2H), 2.31-2.21 (m, 1H), 2.17-2.08 (m, 3H), 2.06-1.90 (m, 4H); LCMS [ESI, M+1]: m/z=638.9.

Example 417

(1794) ##STR00759##

7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decan 2,2-dioxide

(1795) The title compound was synthesized according to the procedure described for example 414. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.19-9.05 (m, 1H), 7.81 (ddd, J=3.6, 5.6, 9.2 Hz, 1H), 7.43-7.35 (m, 2H), 7.24-7.19 (m, 1H), 5.58-5.33 (m, 1H), 4.73-4.55 (m, 2H), 4.55-4.25 (m, 3H), 3.90-3.67 (m, 2H), 3.66-3.45 (m, 3H), 3.44-3.33 (m, 1H), 3.29-3.19 (m, 2H), 2.61-2.37 (m, 2H), 2.36-2.25 (m, 1H), 2.22-2.12 (m, 2H), 2.11-1.99 (m, 3H), 1.95-1.79 (m, 2H); LCMS (ESI, M+1): m/z=690.2.

Example 418

(1796) ##STR00760##

(R)-1-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1797) The title compound was synthesized according to the procedure described for example 414. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.23 (s, 1H), 8.49 (br s, 1H), 7.85-7.79 (m, 1H), 7.44-7.35 (m, 2H), 7.22 (t, J=2.8 Hz, 1H), 5.57-5.29 (m, 1H), 4.60 (br s, 2H), 4.54-4.50 (m, 1H), 4.47-4.41 (m, 1H), 4.37-4.28 (m, 1H), 3.73-3.63 (m, 1H), 3.62-3.55 (m, 2H), 3.50-3.41 (m, 1H), 3.28-3.19 (m, 1H), 2.60-2.38 (m, 2H), 2.36-2.25 (m, 1H), 2.21-2.12 (m, 3H), 2.08-1.98 (m, 1H), 1.90-1.75 (m, 3H), 1.30 (d, J=8.4 Hz, 3H); LCMS (ESI, M+1): m/z=614.3.

Example 419

(1798) ##STR00761##

4-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(1799) ##STR00762##

(1800) Step A. 4-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol: To a mixture of 7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (126 mg, 1 equiv.) and 4 Å MS (15 mg) in DMF (0.5 mL) was added DIEA (172 mg, 8 equiv.) and 6-methyl-1,4-oxazepan-6-ol (83.9 mg, 3 equiv., HCl). The mixture was stirred at 40° C. for 12 hr. The reaction mixture was filtered and purified by prep-HPLC (column: 3_Phenomenex Luna C18 75×30 mm×3 μm; mobile phase: [water (0.1% formic acid)-ACN]; B %: 8%-38%, 10 min) to afford the title compound (28.7 mg, 25% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.74-9.39 (m, 1H), 8.57-8.42 (m, 1H), 8.01-7.66 (m, 1H), 7.50-7.30 (m, 2H), 7.29-7.12 (m, 1H), 5.50-5.31 (m, 1H), 4.57 (br s, 2H), 4.49-4.37 (m, 2H), 4.25-4.14 (m, 1H), 4.08-3.86 (m, 3H), 3.77-3.63 (m, 2H), 3.51-3.43 (m, 2H), 3.23-3.10 (m, 2H), 2.57-2.41 (m, 1H), 2.38-2.21 (m, 2H), 2.17-1.95 (m, 3H), 1.29 (br s, 3H); LCMS [ESI, M+1]: m/z=630.1.

Example 420

(1801) ##STR00763##

5-chloro-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1802) The title compound was synthesized according to the procedure described for example 413. ¹H NMR (400 MHz, methanol-d₄) δ=9.22 (d, J=6.8 Hz, 1H), 7.85-7.65 (m, 1H), 7.44-7.34 (m, 2H), 7.27-7.17 (m, 1H), 5.62-5.40 (m, 1H), 4.80-4.53 (m, 4H), 4.52-4.31 (m, 1H), 4.18-3.93 (m, 3H), 3.93-3.62 (m, 4H), 3.62-3.53 (m, 2H), 3.41-3.32 (m, 1H), 2.75-2.38 (m, 3H), 2.38-2.13 (m, 4H), 2.13-1.91 (m, 1H); LCMS (ESI, M+1): m/z=630.5.

Example 421

(1803) ##STR00764##

3-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-azabicyclo[3,2,1]octan-6-ol

(1804) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.25 (dd, J=1.6, 14.8 Hz, 1H), 7.83 (dd, J=5.6, 9.2 Hz, 1H), 7.45-7.36 (m, 2H), 7.23 (dd, J=2.4, 13.6 Hz, 1H), 5.51-5.30 (m, 11H), 4.98 (br s, 1H), 4.83 (br s, 1H), 4.49-4.40 (m, 1H), 4.39-4.29 (m, 2H), 3.85-3.74 (m, 1H), 3.59-3.50 (m, 1H), 3.37 (br s, 3H), 3.21-3.12 (m, 1H), 2.53-2.32 (m, 3H), 2.32-2.18 (m, 3H), 2.16-2.05 (m, 2H), 2.04-1.90 (m, 2H), 1.87-1.78 (m, 11H), 1.40 (br dd, J=2.0, 13.6 Hz, 11H); LCMS [ESI, M+1]: m/z=626.1.

Example 422

(1805) ##STR00765##

6-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azabicyclo[3.2.1]octan-3-ol

(1806) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.36-9.14 (m, 1H), 8.49 (br s, 1H), 7.82 (dd, J=5.6, 9.2 Hz, 1H), 7.44-7.34 (m, 2H), 7.24-7.16 (m, 11H), 5.57-5.38 (m, 11H), 5.00 (br s, 1H), 4.61-4.46 (m, 2H), 4.42-4.32 (m, 1H), 4.32-4.19 (m, 1H), 4.14 (br s, 1H), 3.72 (br s, 1H), 3.70-3.57 (m, 2H), 3.27 (br s, 11H), 2.86-2.62 (m, 2H), 2.60-2.48 (m, 11H), 2.45-2.32 (m, 1H), 2.31-2.17 (m, 3H), 2.17-2.10 (m, 1H), 2.10-1.96 (m, 3H), 1.95-1.74 (m, 2H). LCMS [ESI, M+1]: m/z=626.2.

Example 423

(1807) ##STR00766##

5-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1808) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.28 (s, 1H), 8.46 (br s, 1H), 7.82 (dd, J=5.6, 9.2 Hz, 1H), 7.43-7.36 (m, 2H), 7.22 (d, J=2.4 Hz, 11H), 5.54-5.36 (m, 1H), 4.64 (br d, J=13.2 Hz, 2H), 4.60-4.46 (m, 2H), 4.43-4.30 (m, 2H), 3.81-3.55 (m, 3H), 3.69-3.53 (m, 3H), 3.30-3.23 (m, 1H), 2.63-2.37 (m, 2H), 2.36-2.27 (m, 1H), 2.25-2.15 (m, 2H), 2.13-2.00 (m, 1H). LCMS [ESI, M+1]: m/z=639.1.

Example 424

(1809) ##STR00767##

6-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,6-diazaspiro[3.5]nonan-1-one

(1810) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.08 (s, 11H), 7.83-7.83 (m, 1H), 7.44-7.34 (m, 2H), 7.23 (s, 1H), 5.53-5.34 (m, 11H), 4.55-4.49 (m, 1H), 4.48-4.42 (m, 11H), 4.36-4.28 (m, 2H), 4.21-4.09 (m, 1H), 4.08-3.95 (m, 1H), 3.71-3.59 (m, 1H), 3.59-3.46 (m, 2H), 3.30-3.21 (m, 2H), 3.21-3.18 (m, 1H), 2.59-2.35 (m, 2H), 2.33-2.25 (m, 1H), 2.22-2.12 (m, 4H), 2.12-2.00 (m, 2H), 1.97-1.87 (m, 1H); LCMS [ESI, M+1]: 639.1.

Example 425

(1811) ##STR00768##

6-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1812) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-*d*.sub.4) δ=9.05 (d, J=7.6 Hz, 1H), 8.50 (br s, 1H), 7.80 (dd, J=5.2, 9.2 Hz, 1H), 7.45-7.32 (m, 2H), 7.22 (t, J=2.4 Hz, 1H), 5.55-5.27 (m, 1H), 4.50-4.35 (m, 2H), 4.32-4.22 (m, 1H), 4.12-3.90 (m, 4H), 3.65-3.36 (m, 3H), 3.18 (br d, J=4.5 Hz, 1H), 2.52-2.16 (m, 6H), 2.15-1.96 (m, 3H), 1.84-1.66 (m, 6H); LCMS [ESI, M+1]: m/z=640.2.

Example 426

(1813) ##STR00769##

4-(4-(2-amino-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepin-5(6H)-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-chloro-6-fluoronaphthalen-2-ol

(1814) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-*d*.sub.4) δ=9.19 (s, 1H), 7.83-7.79 (m, 1H), 7.43-7.34 (m, 2H), 7.22 (d, J=2.4 Hz, 1H), 5.78 (s, 1H), 5.48-5.30 (m, 1H), 5.17-5.01 (m, 2H), 4.45-4.40 (m, 1H), 4.39-4.32 (m, 3H), 4.27-4.21 (m, 2H), 3.56-3.36 (m, 3H), 3.18-3.10 (m, 1H), 2.48-2.29 (m, 4H), 2.28-2.17 (m, 1H), 2.15-2.05 (m, 2H), 2.05-1.92 (m, 1H); LCMS [ESI, M+1]: m/z<651.3.

Example 427

(1815) ##STR00770##

5-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(1816) The title compound was synthesized according to the procedure described for example 419. ¹H NMR (400 MHz, METHANOL-*d*.sub.4) δ=9.21 (s, 1H), 8.61-8.44 (m, 1H), 7.83 (dd, J=5.2, 9.2 Hz, 1H), 7.45-7.37 (m, 2H), 7.23 (d, J=2.4 Hz, 1H), 6.78 (s, 1H), 5.52-5.20 (m, 3H), 4.56 (br d, J=6.4 Hz, 2H), 4.51-4.38 (m, 4H), 3.66-3.43 (m, 3H), 3.35 (s, 3H), 3.25-3.17 (m, 1H), 3.10 (s, 3H), 2.55-2.26 (m, 5H), 2.22-1.98 (m, 3H); LCMS [ESI, M+1]: 707.2.

Example 428

(1817) ##STR00771##

7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(1818) ##STR00772##

(1819) Step A. 1-(1-(((7-chloro-8-fluoro-4(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)cyclopropyl)-N,N-dimethylmethanamine: To a solution of 2,7-dichloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (6 g, 1.0 equiv.) and 1-((dimethylamino)methyl)cyclopropyl)methanol (2.21 g, 0.9 equiv.) in THF (60 mL) was added DIEA (7.36 g, 3 equiv.) and 4 Å molecular sieves (1 g) at -40° C., The mixture was stirred at -40° C. for 0.5 hour. The mixture was filtered, diluted with water (60 mL) and extracted with ethyl acetate (3×60 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)—ACN] to afford the title compound (3.0 g, 35% yield) as a yellow solid; LCMS (ESI, M+1): m/z=409.1.

(1820) Step B. 1-(1-(((7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)cyclopropyl)-N,N-dimethylmethanamine: A mixture of 1-(1-(((7-chloro-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)cyclopropyl)-N,N-dimethylmethanamine (500 mg, 1.0 equiv.), ((5-chloro-6-fluoro-4-(trimethylstannyl)naphthalen-2-yl)oxy)triisopropylsilane (1.26 g, 2.0 equiv.), CuI (23.3 mg, 0.1 equiv.), BINAP (152 mg, 0.2 equiv.) and Pd(dppf)Cl.sub.2 (89.5 mg, 0.1 equiv.) in toluene (8 mL) was degassed and stirred at 90° C. for 6 hours under N.sub.2 atmosphere. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed

phase flash chromatography [C18, water (0.1% formic acid)—ACN] to afford the title compound (0.5 g, 48% yield) as a yellow oil; LCMS (ESI, M+1): m/z=725.3.

(1821) Step C. 7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of 1-(1-(((7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)cyclopropyl)-N,N-dimethylmethanamine (157 mg, 1.0 equiv.) and 2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide (58.0 mg, 1.1 equiv.) in DMF (1 mL) were added 4 Å molecular sieves (40 mg) and DIEA (107 mg, 3 equiv.). The mixture was stirred at 60° C. for 12 hours. The mixture was filtered, diluted with water (5 mL) and extracted with ethyl acetate (3×5 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] and prep-HPLC [column: Waters Xbridge 150×25 mm×5 pin; mobile phase: water (NH.sub.4HCO.sub.3)-ACN; B %: 30%-60%, 10 min] to afford the title compound (16.3 mg, 9% yield) as an off-white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.07 (d, J=2.4 Hz, 1H), 7.82-7.76 (m, 1H), 7.42-7.33 (m, 2H), 7.24-7.16 (m, 1H), 4.50-4.27 (m, 4H), 3.92 (d, J=13.3 Hz, 1H), 3.72 (d, J=13.2 Hz, 1H), 3.41 (d, J=11.6 Hz, 1H), 3.20 (dd, J=3.2, 12.0 Hz, 1H), 2.60-2.54 (m, 1H), 2.42-2.35 (m, 1H), 2.33 (s, 6H), 2.12-1.76 (m, 4H), 0.79-0.70 (m, 2H), 0.58-0.49 (m, 2H). LCMS (ESI, M+1): m/z=660.0.

Example 429

(1822) ##STR00773##

1-(8-fluoro-7-(2-fluoro-5-hydroxyphenyl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1823) ##STR00774##

(1824) Step A. J-(8-fluoro-7-(2-fluoro-5-hydroxyphenyl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50 mg, 1.0 equiv.), 4-fluoro-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol (55 mg, 2 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 229 µL, 3 equiv.) in THF (0.5 mL) was added cataCXium-A-Pd-G3 (8.4 mg, 0.1 equiv.). The mixture was degassed and stirred at 60° C. for 2 hours. Upon reaction completion, the mixture was diluted with water (5 mL) and extracted with ethyl acetate (3×5 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: water (0.225% formic acid)-ACN; B: 5%-35%, 11.5 min)] to afford the title compound (37.1 mg, 66.5 µmol, 56.0% yield, formic acid salt) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.26 (s, 1H), 8.50 (br s, 1H), 7.12-7.06 (m, 1H), 7.04 (dd, J=3.0, 5.6 Hz, 1H), 6.93 (td, J=3.6, 8.8 Hz, 1H), 4.64 (s, 2H), 4.60 (br d, J=13.1 Hz, 1H), 4.31 (br d, J=13.3 Hz, 1H), 3.74-3.66 (m, 2H), 3.62 (d, J=13.4 Hz, 1H), 3.44-3.36 (m, 1H), 3.30-3.24 (m, 2H), 2.32 (ddd, J=3.1, 6.8, 12.3 Hz, 2H), 2.27-2.05 (m, 7H), 1.90-1.71 (m, 3H), 1.28 (s, 3H); LCMS(M+H): 512.3.

Example 430

(1825) ##STR00775##

1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(5-methyl-1H-indazol-4-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1826) ##STR00776##

(1827) Step A. 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(5-methyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (30 mg, 1 equiv.) in n-BuOH (0.6 mL) was added 5-

methyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indazole (35.3 mg, 1.5 equiv.), K.sub.3PO.sub.4 (1.5 M in water, 138 µL, 3 equiv.) and X-Phos-Pd-G4 (5.92 mg, 0.1 equiv.). The mixture was stirred at 60° C. for 2 hours under N.sub.2 atmosphere. The mixture was poured into water (5 mL) and extracted with ethyl acetate (3×5 mL). The combined organic layers were washed with brine 5 mL, dried over Na.sub.2SO.sub.4, concentrated and purified by reverse phase flash chromatography (water (0.225% formic acid)) to afford the title compound (30 mg, 70.5% yield) as a yellow oil; LCMS (ESI, M+1): m/z=616.4.

(1828) Step B. 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(5-methyl-1H-indazol-4-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of 1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(5-methyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (30 mg, 1 equiv.) in MeOH (1 mL) was added HCl.dioxane (4 M, 0.5 mL, 41 equiv.). The mixture was stirred at 25° C. for 2 hours under N.sub.2 atmosphere. The mixture was dried with a stream of N.sub.2 and purified by prep-HPLC [column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: water (0.225% formic acid)-ACN; B %: 8%-38%, 10 minutes] to afford the title compound (6.31 mg, 22% yield) as an off-white solid; .sup.1HNMR (400 MHz, METHANOL-d.sub.4) δ=9.34 (s, 1H), 7.71 (s, 1H), 7.61 (d, J=8.4 Hz, 1H), 7.43 (d, J=8.8 Hz, 1H), 4.64 (s, 3H), 4.36 (m, 1H), 3.71-3.58 (m, 3H), 3.44 (br s, 1H), 3.28-3.20 (m, 2H), 2.38-2.27 (m, 5H), 2.25-2.12 (m, 5H), 2.08 (m, 2H), 1.80 (br s, 3H), 1.30 (s, 3H); LCMS (ESI, M+1): m/z=532.3.

Example 431

(1829) ##STR00777##

1-(8-fluoro-7-(6-methyl-1H-indazol-4-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1830) The title compound was synthesized according to the procedure described for example 430. .sup.1HNMR (400 MHz, METHANOL-d.sub.4) δ=9.32 (s, 1H), 8.21 (s, 1H), 7.51 (d, J=13.6 Hz, 2H), 4.58 (s, 3H), 4.36-4.28 (m, 1H), 3.66-3.53 (m, 3H), 3.48-3.37 (m, 1H), 3.22-3.11 (m, 2H), 2.58 (s, 3H), 2.35-2.24 (m, 2H), 2.23-2.08 (m, 5H), 2.04 (s, 2H), 1.91-1.72 (m, 3H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z=532.3.

Example 432

(1831) ##STR00778##

1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(5-hydroxy-2-methylphenyl)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1832) The title compound was synthesized according to the procedure described for example 429. .sup.1HNMR (400 MHz, METHANOL-d.sub.4) δ=9.24 (s, 1H), 7.17 (d, J=8.4 Hz, 1H), 6.89-6.73 (m, 2H), 4.65 (s, 2H), 4.63-4.56 (m, 1H), 4.35-4.28 (m, 1H), 3.74-3.66 (m, 2H), 3.62 (d, J=13.3 Hz, 1H), 3.45-3.36 (m, 1H), 3.30-3.24 (m, 2H), 2.38-2.28 (m, 2H), 2.26-2.16 (m, 4H), 2.14 (s, 3H), 2.13-2.06 (m, 3H), 1.88-1.74 (m, 3H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z=508.3.

Example 433

(1833) ##STR00779##

6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1834) ##STR00780##

(1835) Step A. 8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(R-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (150 mg, 1 equiv.), 4,4,5,5-tetramethyl-2-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-1,3,2-dioxaborolane (131 mg, 1 equiv.) and Cs.sub.2CO.sub.3 (1.5 M in water, 713 µL, 3 equiv.) in methoxycyclopentane (2.5 mL) was added cataCXium A Pd G3 (26.0 mg, 0.1 equiv.). The mixture was stirred at 90° C. for 3 hours under N.sub.2. The mixture was poured into water (10 mL) and extracted with ethyl acetate (2×10 mL).

The combined organic phase was dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography (water (0.225% formic acid)) to afford the title compound (80 mg, 35.8% yield) as a yellow oil; LCMS (ESI, M+1): m/z=627.3

(1836) Step B. 6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: A mixture of 8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (80 mg, 1 equiv.) and 1,6-diazaspiro[3.5]nonan-2-one (21.5 mg, 1.2 equiv.) and DIEA (82.5 mg, 5 equiv.) in DMF (1 mL) was stirred at 40° C. for 2 hours. The mixture was filtered and purified by reversed phase flash chromatography (water (0.225% formic acid)) to afford the title compound (60 mg, 70.5% yield) as a yellow oil; LCMS (ESI, M+1): m/z=667.5.

(1837) Step C. 6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a solution of 6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (30 mg, 1 equiv.) in EtOH (0.5 mL) was added TsOH.Math.H.sub.2O (9.41 mg, 1.1 equiv.). The mixture was stirred at 25° C. for 16 hours. The mixture was poured into water (5 mL). The pH of the mixture was adjusted to 8 with saturated aqueous NaHCO.sub.3. The resulting solution was extracted with ethyl acetate (2×5 mL). The combined organic phase was dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 µm; mobile phase: water (0.225% formic acid)-ACN; B %: 10%-40%, 10 minutes] to afford the title compound (5.75 mg, 21% yield) as a white solid; .sup.1H NMR (400 MHz, MeOD-d.sub.4) δ=9.20-9.07 (m, 1H), 8.09 (d, J=7.6 Hz, 1H), 8.02-7.93 (m, 1H), 7.73 (d, J=6.8 Hz, 1H), 7.65-7.54 (m, 2H), 7.48 (d, J=6.8 Hz, 1H), 4.64-4.54 (m, 2H), 4.46-4.22 (m, 4H), 4.05 (m, 1H), 3.98-3.82 (m, 1H), 3.56-3.55 (m, 2H), 3.19-3.13 (m, 2H), 2.97-2.73 (m, 2H), 2.29-2.00 (m, 12H); LCMS (ESI, M+1): m/z=583.5.

Example 434

(1838) ##STR00781##

2-(8-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-oxo-1,6-diazaspiro[3.5]nonan-6-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-1-yl)acetonitrile

(1839) ##STR00782##

(1840) Step A. 6-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a solution of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.), 1,8-diazaspiro[3.5]nonan-2-one (43.3 mg, 1.3 equiv.) in DMF (1.0 mL) was added DIEA (92.1 mg, 3.0 equiv.) and 4 Å molecular sieves (40 mg). The mixture was stirred at 40° C. for 12 hours. The mixture was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile]. The desired fractions were collected and neutralized with solid NaHCO.sub.3, concentrated in vacuum to remove ACN. The aqueous layer was lyophilized to afford title compound (73 mg, 65% yield) as a yellow solid; LCMS (ESI, M+1): m/z=461.1.

(1841) Step B. 2-(8-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-oxo-1,6-diazaspiro[3.5]nonan-6-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-1-yl)acetonitrile: A mixture of 6-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (50 mg, 1.0 equiv.), 2-(8-trimethylstannyl-1-naphthyl)acetonitrile (71.6 mg, 2.0 equiv.) and cataCXium A Pd G3 (7.90 mg, 0.1 equiv.) in DMAC (0.5 mL) was degassed stirred at 90° C. for 3 hours under N.sub.2 atmosphere. The mixture was directly purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN]. The desired fractions were re-purified by prep-HPLC [Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 34%-64%, 10 min] and lyophilized to afford title compound (16.6 mg, 25% yield) as a white solid; HPLC:>99% cc,

Chiralcel OJ-3 50×4.6 mm I.D., 3 µm column A: 60% MeOH+40% ACN (w/0.05% DEA), B: CO.sub.2, 3 mL/min, 220 nm, t.sub.R: 1.827 min; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.07 (s, 1H), 8.56 (s, 1H), 8.18 (d, J=7.2 Hz, 1H), 8.11 (d, J=7.2 Hz, 1H), 7.71-7.65 (m, 2H), 7.65-7.58 (m, 2H), 4.27-4.15 (m, 2H), 4.12-4.04 (m, 2H), 3.96-3.70 (m, 2H), 3.68v3.57 (m, 2H), 2.98-2.89 (m, 2H), 2.86-2.79 (m, 1H), 2.67-2.62 (m, 1H), 2.58-2.53 (m, 2H), 2.03-1.95 (m, 1H), 1.94-1.85 (m, 5H), 1.84-1.79 (m, 4H), 1.62-1.53 (m, 2H); LCMS (ESI, M+1): m/z=592.2.

Example 435

(1842) ##STR00783##

(R)-1-(7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1843) ##STR00784##

(1844) Step A. (R)-1-(7-(6-(bis(4-methoxybenzyl)amino)-4-methylpyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(7-chloro-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (1.80 g, 1.0 equiv.) in DMAC (20 mL) was added cataCXium A Pd G3 (320 mg, 0.1 equiv.) and N,N-bis[(4-methoxyphenyl)methyl]-4-methyl-6-tributylstannyl-pyridin-2-amine (4.20 g, 1.50 equiv.). The mixture was stirred at 90° C. for 12 hours. The reaction mixture was filtered, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (2.20 g, 69% yield) as a yellow solid; LCMS (ESI, M+1): m/z=722.4.

(1845) Step B. (R)-1-(7-(6-(bis(4-methoxybenzyl)amino)-3-iodo-4-methylpyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (R)-1-(7-(6-(bis(4-methoxybenzyl)amino)-4-methylpyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (1.0 g, 1.0 equiv.) in AcOH (20 mL) was added dropwise solution of NIS (467 mg, 1.5 equiv.) in AcOH (5 mL). The mixture was stirred at 25° C. for 2 hours. The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (800 mg, 64% yield) as a yellow solid; LCMS (ESI, M+1): m/z=848.3.

(1846) Step C. (R)-1-(7-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of CuI (225 mg, 5.0 equiv.) and KF (68.5 mg, 5.0 equiv.) were heated in vacuum with gentle shaking until a greenish color was obtained. To the mixture was added a solution of (R)-1-(7-(6-(bis(4-methoxybenzyl)amino)-3-iodo-4-methylpyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 1.0 equiv.) and trimethyl(trifluoromethyl)silane (168 mg, 5.0 equiv.) in DMF (3 mL). The mixture was stirred at 20° C. for 3 hours. The reaction mixture was diluted with water (10 mL) and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (3×5 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (60.0 mg, 32% yield) as a yellow solid. LCMS (ESI, M+1): m/z=790.4.

(1847) Step D. (R)-1-(7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A solution of (R)-1-(7-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-8-fluoro-2-(((S)-1-methylpyrrolidin-2-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (60.0 mg, 1.0 equiv.) in TFA (1 mL) was stirred at 25° C. for 2 hours. The reaction mixture was concentrated. The residue was purified by reversed-phase HPLC [C18, water (0.1% formic acid)/ACN] and further re-purified by prep-HPLC [column: Phenomenex Gemini-NX C18 75×30 mm×3 µm; mobile phase: water (0.225% formic acid)-ACN; B %: 12%-42%, 7 min] and lyophilized to afford the title compound (19.3 mg, 46% yield, formic acid salt) as a light yellow

solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.21 (s, 1H), 6.63 (s, 1H), 4.80-4.73 (m, 1H), 4.66-4.59 (m, 1H), 4.58-4.50 (m, 1H), 4.29 (br d, J=13.2 Hz, 1H), 3.66-3.50 (m, 3H), 3.47-4.37 (m, 1H), 3.08-2.96 (m, 1H), 2.93 (s, 3H), 2.46 (d, J=1.6 Hz, 3H), 2.38-2.27 (m, 1H), 2.18-1.94 (m, 4H), 1.89-1.72 (m, 3H), 1.28 (s, 3H) LCMS (ESI, M+1): m/z=550.2.

Example 436

(1848) ##STR00785##

(1-(8-fluoro-7-(naphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)pyridin-3-yl)methanesulfonamide

(1849) ##STR00786##

(1850) Step A. 8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxyl-7-(naphthalen-1-yl)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (300 mg, 1.0 equiv.), Cs.sub.2CO.sub.3 (1.5 M, 3.0 equiv.) and naphthalen-1-ylboronic acid (184 mg, 1.5 equiv.) in methoxycyclopentane (5 mL) was added CataCXium A Pd G3 (51.9 mg, 0.1 equiv.) under N.sub.2. The reaction was stirred at 90° C. for 2 hours under N.sub.2. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (2×10 mL). The combined organic layers were washed with brine (15 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (250 mg, 68% yield) as a yellow solid; LCMS (ESI, M+1): m/z=513.2.

(1851) Step B. (1-(8-fluoro-7-(naphthalen-1-yl)-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a solution of 8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(naphthalen-1-yl)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (0.10 g, 1.0 equiv.), piperidin-3-ylmethanesulfonamide (69.6 mg, 2.0 equiv.) and DIEA (75.7 mg, 3.0 equiv.) in DMF (0.5 mL) was added 4 Å molecular sieves (10 mg). The reaction was stirred at 40° C. for 12 hours. The mixture was filtered and purified by prep-HPLC [Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 33%-63%, 2 min] and lyophilized to afford the title compound (66.9 mg, 58% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.15 (s, 1H), 8.08-8.02 (m, 1H), 7.99 (d, J=8.4 Hz, 1H), 7.71-7.61 (m, 3H), 7.57-7.46 (m, 2H), 4.93 (br d, J=13.6 Hz, 1H), 4.57 (br d, J=12.8 Hz, 1H), 4.37-4.27 (m, 2H), 3.64-3.55 (m, 11H), 3.37 (dd, J=10.4, 13.2 Hz, 1H), 3.23-3.16 (m, 1H), 3.15-3.07 (m, 3H), 2.75-2.67 (m, 2H), 2.57-2.46 (m, 1H), 2.18-2.06 (m, 3H), 1.97-1.85 (m, 5H), 1.84-1.70 (m, 3H), 1.68-1.56 (m, 1H); .sup.19F NMR (400 MHz, methanol-d.sub.4) δ =-139.847; LCMS [ESI, M+1, M/2+1]: 591.3, 296.2.

Example 437

(1852) ##STR00787##

1-(1-(7-(8-ethynynaphthalene-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1853) ##STR00788##

(1854) Step A. 8-fluoro-24(hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)-7-(8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (250 mg, 1.0 equiv.), triisopropyl((8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)silane (310 mg, 1.2 equiv.), CataCXium A Pd G3 (43.3 mg, 0.1 equiv.) and Cs.sub.2CO.sub.3 (1.5 M in water, 1.19 mL, 3.0 equiv.) in methoxycyclopentane (3.5 mL) was degassed and then heated to 90° C. for 3 hours under N.sub.2. The mixture was diluted with EtOAc (50 mL) and water (60 mL). The aqueous layer was extracted with EtOAc (2×40 mL), the combined organic phase was washed with brine (40 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (158 mg, 36% yield, 93% purity) as a yellow oil; LCMS (ESI, M+1): m/z=693.3.

(1855) Step B. 1-(1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: A mixture of piperidin-3-ylmethanesulfonamide (84.9 mg, 2.0 equiv.), DIEA (154 mg, 5.0 equiv.), 8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)-7-(8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidine (165 mg, 1.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was stirred at 40° C. for 36 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (94 mg, 46% yield). Yellow solid; LCMS (ESI, M+1): m/z=771.3.

(1856) Step C. 1-(1-(7-(8-ethynynaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: A solution of 1-(1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide (95 mg, 123 µmol, 1.0 equiv.) and CsF (281 mg, 15 equiv.) in DMF (0.5 mL) was stirred at 18° C. for 2 hours. The mixture was filtered and the filtrate was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] and re-purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH₄sub.4HCO₃sub.3)-ACN; B %: 30%-60%, 8 min] to afford the title compound (12.3 mg, 16% yield, 98% purity) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.03 (s, 1H), 8.07 (dd, J=8.0, 18.4 Hz, 2H), 7.77-7.72 (m, 1H), 7.67 (m, 1H), 7.59 (m, 1H), 7.52 (t, J=7.6 Hz, 1H), 4.98-4.90 (m, 1H), 4.61-4.48 (m, 1H), 4.37-4.27 (m, 2H), 3.69-3.51 (m, 1H), 3.50-3.36 (m, 1H), 3.23-3.06 (m, 5H), 2.73 (m, 2H), 2.58-2.43 (m, 1H), 2.18-2.05 (m, 3H), 1.98-1.82 (m, 6H), 1.77-1.72 (m, 2H), 1.69-1.56 (m, 1H); LCMS (ESI, M+1): m/z=615.2

Example 438

(1857) ##STR00789##

1-(1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-methylnaphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1858) The title compound was synthesized according to the procedure described for example 436. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.08 (s, 1H), 8.04 (d, J=8.0 Hz, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.60-7.53 (m, 1H), 7.48-7.39 (m, 2H), 7.31 (d, J=6.8 Hz, 1H), 5.00-4.89 (m, 1H), 4.57 (dd, J=11.2, 7.6 Hz, 1H), 4.38-4.26 (m, 2H), 3.63-3.53 (m, 1H), 3.43-3.33 (m, 1H), 3.22-3.09 (m, 4H), 2.77-2.69 (m, 2H), 2.56-2.46 (m, 1H), 2.11 (dt, J=12.0, 5.6 Hz, 3H), 2.04 (d, J=5.2 Hz, 3H), 1.92 (td, J=12.0, 6.0 Hz, 5H), 1.82-1.70 (m, 3H), 1.67-1.58 (m, 1H); LCMS (ESI, M+1): m/z 605.3.

Example 439

(1859) ##STR00790##

1-(1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1860) The title compound was synthesized according to the procedure described for example 433. ¹H NMR (400 MHz, methanol-d₄) δ ppm=9.08 (s, 1H), 8.08 (d, J=8.0 Hz, 1H), 7.95 (d, J=8.2 Hz, 1H), 7.75 (d, J=7.2 Hz, 1H), 7.62-7.53 (m, 2H), 7.44-7.51 (m, 1H), 4.98-4.96 (m, 1H), 4.64-4.52 (m, 1H), 4.42-4.28 (m, 4H), 3.69-3.53 (m, 1H), 3.42-3.34 (m, 1H), 3.25-3.08 (m, 4H), 2.85-2.68 (m, 2H), 2.61-2.44 (m, 1H), 2.15-2.11 (m, 3H), 1.99-1.88 (m, 5H), 1.83-1.73 (m, 3H), 1.68-1.58 (m, 1H); LCMS (ESI, M+1): m/z=621.2.

Example 440

(1861) ##STR00791##

1-(7-(8-cyclopropylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1862) ##STR00792##

(1863) Step A. 1-bromo-8-cyclopropylnaphthalene: To a solution of 1,8-dibromonaphthalene (5.00

g, 1.0 equiv.) in dioxane (45.0 mL) and water (15.0 mL) was added cyclopropylboronic acid (2.25 g, 1.5 equiv.), K₂CO₃ (13.4 g, 3.6 equiv.) and Pd(dppf)Cl₂ (640 mg, 0.05 equiv.). The mixture was degassed and stirred at 100° C. for 6 hours. After reaction completion, the mixture was quenched by water (30 mL), and extracted with ethyl acetate (3×50 mL). The combined organic layers were concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (2.20 g, 14% yield, 82% purity) as a yellow solid; ¹H NMR (400 MHz, CDCl₃) δ=7.93 (dd, J=1.2, 7.6 Hz, 1H), 7.83 (dd, J=1.2, 8.4 Hz, 1H), 7.74 (d, J=8.0 Hz, 1H), 7.55-7.49 (m, 1H), 7.44-7.37 (m, 1H), 7.33-7.25 (m, 1H), 3.09-2.99 (m, 1H), 1.20-1.13 (m, 2H), 0.96-0.88 (m, 2H).

(1864) Step B. (8-cyclopropylnaphthalen-1-yl)trimethylstannane: To a solution of 1-bromo-8-cyclopropylnaphthalene (4.40 g, 1.0 equiv.) in toluene (45.0 mL) were added trimethyl(trimethylstannyl)stannane (31.6 g, 96.6 mmol, 20.0 mL, 5.0 equiv.) and Pd(PPh₃)₄ (2.06 g, 0.1 equiv.) under N₂ atmosphere. The mixture was stirred at 110° C. for 18 h. After reaction completion, the mixture was concentrated under reduced pressure at 40° C. The crude product was purified by column chromatography [Silica gel, petroleum ether] and reversed-phase flash chromatography C18, water (0.1% NH₃·H₂O)/ACN] to afford the title compound (2.46 g, 42% yield) as a yellow oil; ¹H NMR (400 MHz, CDCl₃) δ=7.87-7.77 (m, 2H), 7.70 (br dd, J=0.8, 8.0 Hz, 1H), 7.45-7.34 (m, 2H), 7.15 (d, J=7.2 Hz, 1H), 2.67-2.43 (m, 1H), 1.16-1.09 (m, 2H), 0.95-0.86 (m, 2H), 0.47-0.32 (m, 9H).

(1865) Step C. 7-(8-cyclopropylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a reaction mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (500 mg, 1.19 mmol, 1.0 equiv.) in toluene (10.0 mL) was added (8-cyclopropylnaphthalen-1-yl)trimethylstannane (1.18 g, 3.56 mmol, 3.0 equiv.), CuI (67.9 mg, 356 μmol, 0.3 equiv.), BINAP (148 mg, 238 μmol, 0.2 equiv.) and Pd(dppf)Cl₂ (86.9 mg, 119 μmol, 0.1 equiv.). The reaction mixture was stirred at 110° C. for 10 hours under N₂ atmosphere. After reaction completion, the mixture was diluted with water (10 mL), then extracted with ethyl acetate (3×30 mL). The combined organic layers were concentrated under reduced pressure at 40° C. The residue was purified by reversed-phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (160 mg, 23% yield) as a yellow solid; ¹H NMR (400 MHz, CDCl₃) δ=9.23 (s, 1H), 7.99 (d, J=8.0 Hz, 1H), 7.83 (t, J=4.8 Hz, 1H), 7.60-7.53 (m, 1H), 7.52-7.48 (m, 1H), 7.44 (d, J=4.8 Hz, 2H), 5.07 (q, J=8.2 Hz, 2H), 4.43 (br s, 2H), 3.27 (br s, 2H), 2.72 (br d, J=6.4 Hz, 2H), 2.12 (br dd, J=5.6, 11.3 Hz, 2H), 1.94 (br s, 4H), 1.81-1.68 (m, 2H), 1.66-1.58 (m, 1H), 0.58 (qd, J=4.9, 9.5 Hz, 1H), 0.51-0.40 (m, 1H), 0.37-0.27 (m, 1H), 0.04-0.09 (m, 1H);

(1866) Step D. 1-(7-(8-cyclopropylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 7-(8-cyclopropylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (140 mg, 1.0 equiv.) in DMF (2.0 mL) were added DIEA (164 mg, 5.0 equiv.) and 3-methylpiperidin-3-ol (58.7 mg, 2.0 equiv.). The mixture was stirred at 40° C. for 16 hours. After reaction completion, the mixture was quenched with water (10 mL), and extracted with ethyl acetate (3×30 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered, concentrated and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: water (10 mM NH₄HCO₃)-ACN, B %: 52%-82%, 2 min] to afford the title compound (47.5 mg, 32% yield, 98.5% purity) as a white solid; ¹H NMR (400 MHz, DMSO-d₆) δ=9.18 (d, J=9.6 Hz, 1H), 8.08 (d, J=8.0 Hz, 1H), 7.92 (d, J=8.0 Hz, 1H), 7.65-7.57 (m, 1H), 7.54-7.43 (m, 2H), 7.38 (d, J=7.2 Hz, 1H), 4.72 (d, J=11.6 Hz, 1H), 4.36-4.21 (m, 1H), 4.11-3.93 (m, 3H), 3.68-3.45 (m, 1H), 3.01-2.86 (m, 2H), 2.57-2.52 (m, 2H), 2.05-1.95 (m, 1H), 1.92-1.51 (m, 13H), 1.15 (d, J=9.6 Hz, 3H), 0.56 (dt, J=4.4, 8.8 Hz, 1H), 0.39-0.21 (m, 2H), 0.08-0.10 (m, 1H); LCMS (ESI, M+1): m/z=568.3.

Example 441

(1867) ##STR00793##

trans-6-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

Example 442

(1868) ##STR00794##

cis-6-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol

(1869) ##STR00795##

(1870) Step A. 6-(7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol: To a solution of 6-azaspiro[3.5]nonan-2-ol (317 mg, 5.0 equiv., HCl) in DMF (1.0 mL) was added 4 Å molecular sieves (20 mg) and DIEA (461 mg, 10 equiv.). The mixture was stirred at 60° C. for 30 minutes before 7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (240 mg, 1.0 equiv., formic acid salt) was added. The mixture was stirred at 60° C. for 12 hours. The reaction mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (100 mg, 40% yield) as a yellow solid; LCMS (ESI, M+1): m/z=668.4.

(1871) Step B. 6-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol: To a mixture of 6-(7-(7,8-difluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-azaspiro[3.5]nonan-2-ol (90.0 mg, 1.0 equiv.) in DCM (0.5 mL) was added TFA (308 mg, 20 equiv.) in one portion at 0° C. under N.sub.2. The mixture was stirred at 0° C. for 1. hour. The reaction mixture was diluted with water (3.0 mL), the aqueous phase was extracted with DCM (3×5 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography (C18, water (0.1% formic acid)-ACN) and re-purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 40%-70%, 10 min] and lyophilized to afford the first eluting peak—Example 441 (6.28 mg, 7.0% yield) as a white solid, .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.05 (s, 1H), 7.68-7.55 (m, 1H), 7.45-7.36 (m, 1H), 7.33 (t, J=2.4 Hz, 1H), 7.25 (d, J=2.4 Hz, 1H), 5.47-5.18 (m, 1H), 4.39-4.24 (m, 3H), 4.10-3.91 (m, 4H), 3.30-3.18 (m, 3H), 3.10-2.96 (m, 1H), 2.42-2.11 (m, 5H), 2.07-1.90 (m, 3H), 1.85-1.79 (m, 4H), 1.75-1.64 (m, 2H); LCMS (ESI, M+1): m/z=624.2;

(1872) and the second eluting peak—Example 442 (8.91 mg, 10% yield) as a white solid. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.07 (s, 1H), 7.65-7.57 (m, 1H), 7.45-7.36 (m, 1H), 7.33 (t, J=2.0 Hz, 1H), 7.26-7.22 (m, 1H), 5.48-5.19 (m, 1H), 4.45-4.19 (m, 3H), 4.11-3.93 (m, 4H), 3.45-3.34 (m, 1H), 3.30-3.21 (m, 2H), 3.14-2.97 (m, 1H), 2.46-2.12 (m, 5H), 2.10-1.92 (m, 3H), 1.89-1.72 (m, 6H); LCMS (ESI, M+1): m/z=624.2; LCMS (ESI, M+1): m/z=624.2.

Example 443

(1873) ##STR00796##

3-(((8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide

(1874) ##STR00797##

(1875) Step A: 3-(((7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide: To a solution of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) in DMF (2 mL) was added 3-(aminomethyl)thietane 1,1-dioxide (77.1 mg, 1.2 equiv.) and DIEA (307 mg, 5.0 equiv.). The mixture was stirred at 40° C. for 2 hours. The pH of the

mixture was adjusted to 7 with 1N HCl and the resulting mixture was filtered. The filter cake was dried in vacuum to give the title compound (140 mg, 65% yield) as a white solid. LCMS (ESI, M+1): 456.2.

(1876) Step B: 3-(((8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide: To a solution of 3-(((7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide (50 mg, 1.0 equiv.) and 4,4,5,5-tetramethyl-2-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)-1,3,2-dioxaborolane (60.6 mg, 1.5 equiv.) in n-BuOH (1.5 mL) was added K.sub.3PO.sub.4 (1.5 M, 3.0 equiv.) and XPhos Pd G3 (9.44 mg, 0.1 equiv.). The mixture was stirred at 60° C. for 2 hours. The mixture was filtered through a pad of celite and concentrated in vacuum. The residue was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] and Prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 µm; mobile phase: water (0.225% formic acid)-CAN, B %: 9%-39%, 10 min] to give the title compound (10 mg, 13.8% yield) as a yellow solid. LCMS [ESI, M+1]: 662.0.

(1877) Step C: 3-(((8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide: To a solution of 3-(((8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(((tetrahydro-2H-pyran-2-yl)oxy)methyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)thietane 1,1-dioxide (10 mg, 1.0 equiv.) in MeOH (0.5 mL) was added HCl.dioxane (4 M, 20 equiv.). The mixture was stirred at 25° C. for 1 hour. The mixture was concentrated in vacuum. The residue was purified by Prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 µm; mobile phase: water (0.225% formic acid)-CAN, B %: 9%-39%, 10 min] to give the title compound (2.39 mg, 27% yield) as an off-white solid. .sup.1H NMR (400 MHz, MeOD-d4) δ=9.18 (s, 1H), 8.13-8.05 (m, 1H), 7.96 (d, J=7.6 Hz, 1H), 7.73 (d, J=6.4 Hz, 1H), 7.64-7.54 (m, 2H), 7.46 (dd, J=1.2, 7.2 Hz, 1H), 4.72 (s, 2H), 4.36 (m, 2H), 4.29-4.20 (m, 2H), 4.17-4.03 (m, 4H), 3.72-3.63 (m, 2H), 3.28-3.26 (m, 2H), 3.11-2.99 (m, 1H), 2.38-2.07 (m, 8H). LCMS [ESI, M+1]: 578.3.

Example 444

(1878) ##STR00798##

7-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1879) ##STR00799##

(1880) Step A. 7-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione: A mixture of 5,6-difluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (60 mg, 1.0 equiv.), 1,3,9-triazaspiro[4.5]decane-2,4-dione (74 mg, 4.25 equiv.), DIEA (40.07 mg, 54 µL, 3.01 equiv.), 4 Å molecular sieves (20 mg) in DMF (1 mL) was degassed and stirred at 40° C. for 18 hours. The mixture was filtered. The filtrate was purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to give a crude product. The crude product was re-purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water (0.225% formic acid)-ACN]; B %: 11%-41%, 10 min) and lyophilized to give the title compound (34.1 mg, 50% yield) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.20 (s, 1H), 8.50 (s, 1H), 7.62 (dd, J=4.4, 9.2 Hz, 1H), 7.45-7.36 (m, 1H), 7.34 (t, J=2.0 Hz, 1H), 5.50-5.33 (m, 1H), 4.66 (br d, J=13.2 Hz, 1H), 4.56-4.35 (m, 3H), 3.83-3.49 (m, 5H), 3.25-3.14 (m, 1H), 2.56-1.87 (m, 10H); LCMS [ESI, M+1]: 652.3.

Example 445

(1881) ##STR00800##

7-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1882) The title compound was synthesized according to the procedure described for example 203. ¹H NMR (400 MHz, DMSO-d₆) δ=9.11 (s, 1H), 7.78-7.64 (m, 1H), 7.62-7.45 (m, 1H), 7.37 (s, 1H), 7.24 (s, 1H), 7.00 (br s, 1H), 6.30 (s, 1H), 5.41-5.15 (m, 1H), 4.20-3.91 (m, 4H), 3.78-3.63 (m, 2H), 3.24-2.94 (m, 6H), 2.88-2.75 (m, 1H), 2.20-1.97 (m, 4H), 1.88-1.70 (m, 6H); LCMS (ESI, M+1): m/z=638.2.

Example 446

(1883) ##STR00801##

7-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2,7-diazaspiro[4.5]decan-3-one

(1884) The title compound was synthesized according to the procedure described for example 203. ¹H NMR (400 MHz, methanol-d₄) δ=9.07 (s, 1H), 7.67-7.57 (m, 1H), 7.44-7.35 (m, 1H), 7.35-7.31 (m, 1H), 7.26-7.21 (m, 1H), 5.43-5.18 (m, 1H), 4.36-4.19 (m, 2H), 4.17-3.95 (m, 4H), 3.39-3.33 (m, 2H), 3.28-3.16 (m, 3H), 3.06-2.95 (m, 1H), 2.43-2.10 (m, 5H), 2.05-1.82 (m, 7H); LCMS (ESI, M+1): m/z=637.3;

Example 447

(1885) ##STR00802##

7-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1886) The title compound was synthesized according to the procedure described for example 433. ¹H NMR (400 MHz, methanol-d₄) δ=9.13-9.07 (m, 1H), 8.08 (dd, J=1.2, 8.0 Hz, 1H), 7.95 (d, J=7.6 Hz, 1H), 7.75 (br d, J=7.2 Hz, 1H), 7.58 (m, 2H), 7.48 (d, J=7.2 Hz, 1H), 4.63 (br d, J=6.8 Hz, 1H), 4.54-4.40 (m, 1H), 4.36-4.22 (m, 4H), 3.85-3.67 (m, 2H), 3.17-3.06 (m, 2H), 2.73 (m, 2H), 2.30-2.19 (m, 1H), 2.14-1.99 (m, 4H), 1.98-1.83 (m, 5H), 1.80-1.71 (m, 2H); LCMS (ESI, M+1): m/z=612.2.

Example 448

(1887) ##STR00803##

7-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-methylnaphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1888) The title compound was synthesized according to the procedure described for example 436. ¹H NMR (400 MHz, methanol-d₄) δ=9.17-9.06 (m, 1H), 8.04 (d, J=8.0 Hz, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.57 (t, J=7.6 Hz, 1H), 7.48-7.40 (m, 2H), 7.35-7.25 (m, 1H), 4.63 (br dd, J=5.6, 13.6 Hz, 1H), 4.52-4.37 (m, 1H), 4.35-4.23 (m, 2H), 3.86-3.64 (m, 2H), 3.17-3.06 (m, 2H), 2.73 (td, J=6.4, 10.4 Hz, 2H), 2.30-2.17 (m, 1H), 2.14-2.02 (m, 5H), 2.02-1.94 (m, 3H), 1.94-1.84 (m, 4H), 1.81-1.72 (m, 2H); LCMS (ESI, M+1): m/z=596.1.

Example 449

(1889) ##STR00804##

8-(4-(2,4-dioxo-1,3,7-triazaspiro[4.5]decan-7-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-1-naphthonitrile

(1890) The title compound was synthesized according to the procedure described for example 436. ¹H NMR (400 MHz, methanol-d₄) δ=9.14 (s, 1H), 8.41-8.35 (m, 1H), 8.27-8.20 (m, 1H), 8.07-8.02 (m, 1H), 7.85-7.78 (m, 2H), 7.73-7.67 (m, 1H), 4.65 (br d, J=13.2 Hz, 1H), 4.55-4.44 (m, 1H), 4.38-4.26 (m, 2H), 3.88-3.66 (m, 2H), 3.22-3.11 (m, 2H), 2.78 (td, J=6.4, 10.4 Hz, 2H), 2.31-2.20 (m, 1H), 2.16-2.02 (m, 4H), 1.99-1.87 (m, 5H), 1.84-1.74 (m, 2H); LCMS (ESI, M+1): m/z=607.2.

Example 450

(1891) ##STR00805##

6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-methylnaphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1892) The title compound was synthesized according to the procedure described for example 436. ¹H NMR (400 MHz, methanol-d₄) δ=9.13 (d, J=2.4 Hz, 1H), 8.05 (dd, J=1.2, 8.4 Hz,

1H), 7.87 (d, J=8.0 Hz, 1H), 7.63-7.53 (m, 1H), 7.48-7.41 (m, 2H), 7.32 (br d, J=7.2 Hz, 1H), 4.67-4.58 (m, 2H), 4.42 (d, J=13.2 Hz, 1H), 4.37-4.28 (m, 1H), 4.02 (dd, J=4.4, 13.2 Hz, 1H), 3.91-3.82 (m, 1H), 3.70-3.60 (m, 2H), 3.28-3.18 (m, 2H), 2.96-2.87 (m, 11H), 2.82-2.74 (m, 1H), 2.35-2.27 (m, 2H), 2.26-2.14 (m, 4H), 2.13-2.08 (m, 2H), 2.07-2.01 (m, 5H), 1.97 (br dd, J=4.0, 7.6 Hz, 2H); LCMS (ESI, M+1): m/z=567.1.

Example 451

(1893) ##STR00806##

8-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-oxo-1,6-diazaspiro[3.5]nonan-6-yl)pyrido[4,3-d]pyrimidin-7-yl)-1-naphthonitrile

(1894) The title compound was synthesized according to the procedure described for example 436. ^{sup}.1H NMR (400 MHz, methanol-d_{sub}.4) δ=9.11 (s, 1H), 8.40-8.35 (m, 1H), 8.26-8.20 (m, 1H), 8.04 (dd, J=1.2, 7.2 Hz, 1H), 7.85-7.78 (m, 2H), 7.73-7.66 (m, 1H), 4.41-4.18 (m, 4H), 4.09-4.00 (m, 1H), 3.97-3.73 (m, 1H), 3.19-3.09 (m, 2H), 2.81-2.92 (m, 1H), 2.80-2.70 (m, 3H), 2.16-2.03 (m, 4H), 2.01-1.87 (m, 6H), 1.83-1.73 (m, 2H); LCMS (ESI, M+1): m/z=578.3.

Example 452

(1895) ##STR00807##

6-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1896) The title compound was synthesized according to the procedure described for example 444. ^{sup}.1H NMR (400 MHz, methanol-d_{sub}.4) δ=9.09 (s, 1H), 7.65-7.57 (m, 1H), 7.44-7.35 (m, 1H), 7.33 (t, J=2.0 Hz, 1H), 7.26-7.22 (m, 1H), 5.45-5.17 (m, 1H), 4.49-4.20 (m, 4H), 4.03-3.90 (m, 1H), 3.83-3.68 (m, 1H), 3.30-3.14 (m, 3H), 3.06-2.96 (m, 1H), 2.95-2.86 (m, 1H), 2.80-2.71 (m, 1H), 2.40-2.07 (m, 4H), 2.06-1.83 (m, 6H); LCMS (ESI, M+1): m/z=623.2.

Example 453

(1897) ##STR00808##

7-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(hydroxymethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1898) The title compound was synthesized according to the procedure described for example 433. ^{sup}.1H NMR (400 MHz, methanol-d_{sub}.4) δ=9.10 (br d, J=6.4 Hz, 1H), 8.08 (br d, J=6.8 Hz, 1H), 7.95 (br d, J=7.6 Hz, 1H), 7.75 (br s, 1H), 7.63-7.53 (m, 2H), 7.49 (br s, 1H), 4.36-4.22 (m, 4H), 4.15-3.87 (m, 4H), 3.48-3.40 (m, 1H), 3.28 (br s, 1H), 3.16-3.05 (m, 2H), 2.78-2.67 (m, 2H), 2.13-2.02 (m, 2H), 2.01-1.82 (m, 8H), 1.81-1.71 (m, 2H); ^{sup}.19F NMR (376 MHz, methanol-d_{sub}.4) δ=-139.118; LCMS (ESI, M+1): m/z=598.3.

Example 454

(1899) ##STR00809##

7-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-methylnaphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decan-2-one

(1900) The title compound was synthesized according to the procedure described for example 436. ^{sup}.1H NMR (400 MHz, methanol-d_{sub}.4) δ=9.16 (d, J=2.4 Hz, 1H), 8.06 (dd, J=0.8, 8.4 Hz, 1H), 7.87 (d, J=8.0 Hz, 1H), 7.58 (t, J=7.6 Hz, 1H), 7.50-7.40 (m, 2H), 7.32 (dd, J=3.2, 6.8 Hz, 1H), 4.70-4.65 (m, 1H), 4.61-4.55 (m, 1H), 4.41 (dd, J=8.8, 12.8 Hz, 1H), 4.25 (br dd, J=4.0, 13.2 Hz, 1H), 4.06-3.93 (m, 1H), 3.88 (dd, J=10.4, 12.8 Hz, 1H), 3.74-3.62 (m, 2H), 3.43 (dd, J=1.6, 9.6 Hz, 1H), 3.30-3.21 (m, 3H), 2.37-2.27 (m, 2H), 2.26-2.14 (m, 4H), 2.09 (dd, J=6.4, 12.8 Hz, 2H), 2.03 (d, J=8.8 Hz, 3H), 1.98 (br s, 3H), 1.92-1.84 (m, 1H); LCMS (ESI, M+1): m/z=582.1.

Example 455

(1901) ##STR00810##

4-(3-(1H-1,2,4-triazol-3-yl)piperidin-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-methylnaphthalen-1-yl)pyrido[4,3-d]pyrimidine

(1902) The title compound was synthesized according to the procedure described for example 436. ^{sup}.1H NMR (400 MHz, methanol-d_{sub}.4) δ=9.09 (d, J=2.0 Hz, 1H), 8.22 (d, J=4.4 Hz, 1H),

8.08-7.99 (m, 1H), 7.86 (d, J=8.0 Hz, 1H), 7.57 (t, J=7.6 Hz, 1H), 7.48-7.39 (m, 2H), 7.31 (d, J=7.2 Hz, 1H), 4.65-4.50 (m, 1H), 4.35-4.19 (m, 2H), 3.78 (ddd, J=10.4, 13.2, 20.0 Hz, 1H), 3.73-3.64 (m, 1H), 3.42-3.32 (m, 2H), 3.11-3.02 (m, 2H), 2.70 (td, J=6.8, 10.4 Hz, 2H), 2.36-2.27 (m, 1H), 2.15-2.05 (m, 3H), 2.04 (s, 4H), 2.00-1.91 (m, 2H), 1.91-1.86 (m, 2H), 1.86-1.79 (m, 1H), 1.78-1.68 (m, 2H); LCMS (ESI, M+1): m/z=579.1.

Example 456

(1903) ##STR00811##

8-(4-(3-(1H-1,2,4-triazol-3-yl)piperidin-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-1-naphthonitrile

(1904) The title compound was synthesized according to the procedure described for example 436. ^{sup}.1H NMR (400 MHz, methanol-d.sub.4) δ=9.12 (s, 1H), 8.38-8.36 (m, 1H), 8.29-8.17 (m, 2H), 8.05-8.02 (m, 1H), 7.85-7.77 (m, 2H), 7.71-7.66 (m, 1H), 4.65-4.54 (m, 1H), 4.37-4.26 (m, 2H), 3.79-3.61 (m, 2H), 3.41-3.33 (m, 1H), 3.17-3.06 (m, 2H), 2.77-2.70 (m, 2H), 2.36-2.26 (m, 1H), 2.14-1.82 (m, 10H), 1.81-1.72 (m, 2H); LCMS (ESI, M+1): m/z=590.3.

Example 457

(1905) ##STR00812##

5,6-difluoro-4-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(6-(hydroxymethyl)-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1906) The title compound was synthesized according to the procedure described for example 203. ^{sup}.1H NMR (400 MHz, methanol-d.sub.4) δ=9.32-9.20 (m, 1H), 7.63 (br dd, J=4.8, 8.4 Hz, 1H), 7.46-7.39 (m, 1H), 7.38-7.33 (m, 1H), 7.32-7.22 (m, 1H), 4.81-4.64 (m, 3H), 4.52-4.35 (m, 1H), 4.18-4.03 (m, 2H), 4.02-3.94 (m, 1H), 3.88-3.79 (m, 1H), 3.75-3.67 (m, 2H), 3.61-3.53 (m, 2H), 3.35-3.31 (m, 1H), 3.30-3.20 (m, 2H), 3.12-2.97 (m, 1H), 2.40-2.28 (m, 2H), 2.27-2.16 (m, 4H), 2.15-1.99 (m, 3H); LCMS (ESI, M+1): m/z=596.3.

Example 458

(1907) ##STR00813##

(3aR,6aS)-5-(7-(8-ethynylnaphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1908) The title compound was synthesized according to the procedure described for example 437. ^{sup}.1H NMR (400 MHz, CD3OD) δ=9.24 (s, 1H), 8.58-8.53 (m, 1H), 8.16-8.11 (m, 1H), 8.10-8.05 (m, 1H), 7.78-7.74 (m, 1H), 7.73-7.66 (m, 1H), 7.66-7.60 (m, 1H), 7.58-7.50 (m, 1H), 4.81-4.74 (m, 1H), 4.70-4.60 (m, 1H), 4.42-4.31 (m, 1H), 4.27 (br dd, J=9.2, 12.4 Hz, 1H), 3.81-3.69 (m, 2H), 3.55-3.46 (m, 2H), 3.16-3.05 (m, 3H), 2.34-1.91 (m, 10H); LCMS (ESI, M+1): m/z=577.2.

Example 459

(1909) ##STR00814##

1-(7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1910) The title compound was synthesized according to the procedure described for example 435 using intermediate 48 instead of intermediate 46 in step A. ^{sup}.1H NMR (400 MHz, DMSO-d.sub.6) δ=9.13 (s, 1H), 6.78 (s, 2H), 6.50 (s, 1H), 5.40-5.16 (m, 1H), 4.78-4.64 (m, 1H), 4.37-4.25 (m, 1H), 4.15-4.07 (m, 1H), 4.05-3.96 (m, 2H), 3.56 (br d, J=13.2 Hz, 1H), 3.29-3.27 (m, 1H), 3.14-2.99 (m, 3H), 2.87-2.78 (m, 1H), 2.39-2.33 (m, 3H), 2.17-2.09 (m, 11H), 2.06-1.93 (m, 3H), 1.90-1.59 (m, 6H), 1.15 (s, 3H); LCMS (ESI, M+1): m/z=594.2.

Example 460

(1911) ##STR00815##

1-(1-(8-fluoro-7-(8-(fluoromethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1912) ##STR00816##

(1913) Step A. 1-bromo-8-(fluoromethyl)naphthalene: To a solution of 1-bromo-8-

(bromomethyl)naphthalene (2.0 g, 1.0 equiv.) in DMSO (20 mL) was added tetrabutylammonium fluoride trihydrate (10.5 g, 5.0 equiv.). The reaction was stirred at 120° C. for 2 hours. The mixture was diluted with ethyl acetate (30 mL) and water (50 mL) then separated. The aqueous phase was extracted with ethyl acetate (30 mL). The organic layer was washed with brine (3×20 mL) and dried over Na₂SO₄, concentrated and purified by column chromatography [silica gel, petroleum ether] to afford the title compound (1.30 g, 78% yield) as a yellow solid; ¹H NMR (400 MHz, dimethylsulfoxide-d₆) 5-8.10-8.02 (m, 2H), 7.96 (dd, J=1.2, 7.2 Hz, 1H), 7.79 (br d, J=7.2 Hz, 1H), 7.64-7.58 (m, 1H), 7.43 (t, J=7.6 Hz, 1H), 6.32-6.19 (d, J=47.6 Hz, 2H); ¹⁹F NMR (376 MHz, dimethylsulfoxide-d₆) δ=195.731.

(1914) Step B. 2-(8-(fluoromethyl)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: To a solution of 1-bromo-8-(fluoromethyl)naphthalene (1.20 g, 1.0 equiv.), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (1.91 g, 1.50 equiv.) and KOAc (1.48 g, 3.0 equiv.) in dioxane (20 mL) was added Pd(dppf)Cl₂ (367 mg, 0.1 equiv.) under N₂ atm. The reaction was stirred at 90° C. for 1.5 hours. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (2×15 mL). The combined organic layers were washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by prep-TLC [Silica gel, petroleum ether/ethyl acetate 100:1] to afford the title compound (1.08 g, 75% yield) as a yellow solid; ¹H NMR (400 MHz, chloroform-d) 6-7.93 (dd, J=1.2, 8.0 Hz, 1H), 7.88-7.82 (m, 2H), 7.55-7.43 (m, 3H), 6.15-6.02 (d, J=47.2 Hz, 2H), 1.46 (s, 12H); ¹⁹F NMR (376 MHz, chloroform-d) δ=-204.598.

(1915) Step C. 8-fluoro-7-(8-(fluoromethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (1.20 g, 1.0 equiv.), 2-(8-(fluoromethyl)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (979 mg, 1.20 equiv.) and Cs₂CO₃ (1.5 M, 3.0 equiv.) in methoxycyclopentane (15 mL) was added CataCXium A Pd G3 (208 mg, 0.1 equiv.) under N₂. The reaction was stirred at 90° C. for 2 hours under N₂. The mixture was diluted with water (10 mL) and extracted with ethyl acetate (2×15 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (0.3 g, 18% yield) as a yellow solid; ¹H NMR (400 MHz, chloroform-d) δ 9.26 (s, 1H), 8.05 (dd, J=1.2, 8.0 Hz, 1H), 7.98 (d, J=8.0 Hz, 1H), 7.65-7.59 (m, 2H), 7.59-7.50 (m, 2H), 5.15-5.06 (m, 3H), 5.02 (s, 1H), 4.45 (br s, 2H), 3.40-3.16 (m, 2H), 2.82-2.67 (m, 2H), 2.18-2.08 (m, 2H), 2.02-1.91 (m, 4H), 1.80-1.74 (m, 2H); ¹⁹F NMR (376 MHz, chloroform-d) δ=-73.277, -136.576; LCMS (ESI, M+1): m/z=545.3.

(1916) Step D. 1-(1-(8-fluoro-7-(8-(fluoromethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a solution of 8-fluoro-7-(8-(fluoromethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 1.0 equiv.), piperidin-3-ylmethanesulfonamide (24.6 mg, 1.5 equiv.) and DIEA (35.6 mg, 3.0 equiv.) in DMF (0.5 mL) was added 4 Å molecular sieves (8 mg). The reaction was stirred at 40° C. for 12 hours. The mixture was filtered and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: water (10 mM NH₄HCO₃)-ACN, B %: 31%-61%, 8 min]. The desired fraction was collected and concentrated lyophilized to afford the title compound (30.7 mg, 53% yield) as a yellow solid; ¹H NMR (400 MHz, methanol-d₄) δ=9.10 (s, 1H), 8.12 (dd, J=0.8, 8.0 Hz, 1H), 8.06 (d, J=8.0 Hz, 1H), 7.67-7.62 (m, 2H), 7.61-7.52 (m, 2H), 5.16-5.01 (m, 2H), 4.98 (br d, J=13.6 Hz, 1H), 4.58 (br d, J=12.4 Hz, 1H), 4.44-4.30 (m, 2H), 3.64-3.54 (m, 1H), 3.41-3.33 (m, 1H), 3.23-3.09 (m, 4H), 2.83-2.73 (m, 2H), 2.58-2.46 (m, 1H), 2.19-2.07 (m, 3H), 2.00-1.88 (m, 5H), 1.86-1.73 (m, 3H), 1.68-1.56 (m, 1H); ¹⁹F NMR (376 MHz, methanol-d₄) δ=-138.958, -204.127; LCMS (ESI, M+1): m/z=623.3.

Example 461

(1917) ##STR00817##

1-(1-(8-fluoro-7-(7-fluoro-3-hydroxy-8-methylnaphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide (1918) ##STR00818## ##STR00819##

(1919) Step A. 6-fluoro-5-methyl-1,4-dihydro-1,4-epoxynaphthalene: To a solution of 1-bromo-2,4-difluoro-3-methylbenzene (80.0 g, 1.0 equiv.) and furan (52.6 g, 2.0 equiv.) in toluene (1000 mL) was added n-BuLi (2.5 M, 185 mL, 1.2equiv.) at -40°C . The mixture was stirred at 15°C . for 16 hours. The mixture was quenched with saturated aqueous NH_4Cl (800 mL) and extracted with ethyl acetate (800 mL). The organic phase was concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 1:0 to 10:1] to afford the title compound (30.0 g, crude) as a yellow oil.

(1920) Step B. 7-fluoro-8-methylnaphthalen-1-ol: To a solution of 6-fluoro-5-methyl-1,4-dihydro-1,4-epoxynaphthalene (30.0 g, 1.0 equiv.) in EtOH (150 mL) was added HCl (150 mL, 36%, 8.9 equiv.). The reaction mixture was stirred at 80°C . for 3 hours. The mixture was concentrated and quenched with saturated aqueous NaHCO_3 (300 mL) and extracted with ethyl acetate (2×100 mL). The combined organic phase was washed with brine (150 mL), dried, concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 100:1 to 10:1] to afford the title compound (10.0 g, 15% yield over two steps) as a yellow oil; ^1H NMR (400 MHz, CHCl_3 -d) δ =7.59 (dd, J =5.6, 8.8 Hz, 1H), 7.37 (d, J =8.0 Hz, 1H), 7.24-7.16 (m, 2H), 6.76 (d, J =7.6 Hz, 1H), 5.67 (s, 1H), 2.84 (d, J =2.8 Hz, 3H); LCMS (ESI, $M+1$): m/z =177.2.

(1921) Step C. 7-fluoro-8-methylnaphthalen-1-yl pivalate: To a solution of 7-fluoro-8-methylnaphthalen-1-ol (10.0 g, 1.0 equiv.), DIEA (22.0 g, 3.0 equiv.) in DCM (100 mL) was added 2,2-dimethylpropanoyl chloride (9.58 g, 1.4 equiv.) at 0°C . The reaction mixture was stirred at 0°C . for 1 hour. The mixture was quenched with water (50 mL) and extracted with ethyl acetate (100 mL). The combined organic phase was washed with brine (100 mL), dried over anhydrous Na_2SO_4 , concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 50:1 to 20:1] to afford the title compound (10.8 g, 73% yield) as a yellow oil; ^1H NMR (400 MHz, CHCl_3 -d) δ =7.74-7.65 (m, 2H), 7.39 (t, J =7.6 Hz, 1H), 7.27-7.21 (m, 1H), 7.03 (d, J =7.6 Hz, 1H), 2.64 (d, J =2.4 Hz, 3H), 1.46 (s, 9H).

(1922) Step D. 7-fluoro-8-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl pivalate: To a mixture of 7-fluoro-8-methylnaphthalen-1-yl pivalate (7.00 g, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (6.83 g, 1.0 equiv.), 4-tert-butyl-2-(4-tert-butyl-2-pyridyl)pyridine (433 mg, 0.06 equiv.) in hexane (240 mL) was added (1,5-Cyclooctadiene)(methoxy)iridium(1) dimer (891 mg, 1.34 mmol, 0.05 equiv.) under N_2 . The mixture was stirred at 60°C . for 3 hours. The mixture was concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 100:1 to 15:1] to afford the title compound (6.00 g, crude) as a white solid.

(1923) Step E. 7-fluoro-3-hydroxy-8-methylnaphthalen-1-yl pivalate: To a mixture of 7-fluoro-8-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl pivalate (6.00 g, 1.0 equiv.), THF (45 mL) and water (21 mL) were added H_2O_2 (13.4 g, 30% purity, 7.6 equiv.) and AcOH (46.6 g, 50 equiv.) at 10°C . The mixture was stirred at 25°C . for 1 hour. The mixture was quenched with saturated aqueous Na_2SO_3 (400 mL) and extracted with ethyl acetate (2×100 mL). The combined organic phase was washed with brine (200 mL), dried over anhydrous sodium sulfate, concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 20:1 to 5:1] and reversed phase flash chromatography [C18, water (0.1% NH_3 · H_2O)-ACN] to afford the title compound (1.70 g, 23% yield over two steps) as an off-white solid; ^1H NMR (400 MHz, CHCl_3 -d) δ =7.41 (dd, J =5.6, 8.8 Hz, 1H), 7.16 (t, J =9.2 Hz, 1H), 6.92 (d, J =2.4 Hz, 1H), 6.65 (d, J =2.4 Hz, 1H), 2.58 (d, J =2.0 Hz, 3H), 1.45 (s, 9H).

(1924) Step F. 7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl pivalate: To a solution of 7-fluoro-3-hydroxy-8-methylnaphthalen-1-yl pivalate (2.30 g, 1.0 equiv.), DIEA (3.23 g, 3.0 equiv.)

in DCM (30 mL) was added chloro(methoxy)methane (889 mg, 1.3 equiv.) at 0° C. The mixture was stirred at 20° C. for 1 hour. The mixture was quenched with saturated aqueous NaHCO₃ (30 mL) and extracted with ethyl acetate (2×30 mL). The combined organic layer was concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 20:1 to 5:1] to afford the title compound (2.60 g, 97% yield) as a white solid; ¹H NMR (400 MHz, chloroform-d) δ=7.57 (dd, J=5.2, 9.0 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.20 (t, J=9.2 Hz, 1H), 6.79 (d, J=2.4 Hz, 1H), 5.27 (s, 2H), 3.52 (s, 3H), 2.59 (d, J=2.4 Hz, 3H), 1.45 (s, 9H); LCMS (ESI, M+1): m/z=321.2.

(1925) Step G. 7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-ol: To a solution of 7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl pivalate (2.60 g, 1.0 equiv.) in MeOH (26 mL) was added KOH (1.37 g, 3.0 equiv.). The reaction mixture was stirred at 20° C. for 0.5 hour. The pH of the mixture was adjusted to 5 with 0.1 N HCl and extracted with ethyl acetate (2×80 mL). The organic phase was concentrated to afford the title compound (1.60 g, 83% yield) as a brown solid; ¹H NMR (400 MHz, chloroform-d) δ=7.46 (dd, J=5.6, 8.8 Hz, 1H), 7.16 (t, J=9.2 Hz, 1H), 6.96 (d, J=2.4 Hz, 1H), 6.55 (d, J=2.0 Hz, 1H), 5.25 (s, 2H), 3.52 (s, 3H), 2.78 (d, J=2.6 Hz, 3H).

(1926) Step H. 7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl trifluoromethanesulfonate: To a solution of 7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-ol (1.60 g, 1.0 equiv.), DIEA (2.63 g, 3.0 equiv.) in DCM (25 mL) was added Tf₂O (2.48 g, 1.3 equiv.) at -40° C. The mixture was stirred at -40° C. for 0.5 hour. The reaction mixture was diluted with DCM (50 mL) and water (15 mL). The organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 20:1 to 6:1] to afford the title compound (2.30 g, 89% yield) as a yellow oil; ¹H NMR (400 MHz, chloroform-d) δ=7.61 (dd, J=5.2, 9.2 Hz, 1H), 7.43 (d, J=2.4 Hz, 1H), 7.32 (d, J=2.0 Hz, 1H), 7.31-7.28 (m, 1H), 7.26 (s, 1H), 5.29 (s, 2H), 3.53 (s, 3H), 2.73 (d, J=2.4 Hz, 3H).

(1927) Step I. 2-(7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: To a mixture of 7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl trifluoromethanesulfonate (1.00 g, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (1.38 g, 2.0 equiv.), AcOK (933 mg, 3.5 equiv.) in dioxane (10 mL) was added Pd(dppf)Cl₂ (199 mg, 0.1 equiv.) under N₂. The reaction was stirred at 100° C. for 1.5 hours. The mixture was diluted with water (30 mL) and extracted with ethyl acetate (30 mL). The organic phase was washed with brine (20 mL), dried over anhydrous Na₂SO₄, concentrated and purified by column chromatography [Silica gel, petroleum ether/ethyl acetate 50:1 to 15:1] to afford the title compound (500 mg, 53% yield) as a colorless oil; ¹H NMR (400 MHz, chloroform-d) δ=7.57 (dd, J=5.6, 8.8 Hz, 1H), 7.45-7.39 (m, 2H), 7.20 (t, J=9.2 Hz, 1H), 5.32-5.23 (m, 2H), 3.51 (s, 3H), 2.63 (d, J=2.4 Hz, 3H), 1.44 (s, 12H).

(1928) Step J. 1-(1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: A mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (60.0 mg, 1.0 equiv.), 3-piperidylmethanesulfonamide (26.8 mg, 1.1 equiv.), and DIEA (53.0 mg, 3.0 equiv.) in DMF (1 mL) was stirred at 20° C. for 1 hour. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (60.0 mg, 82% yield) as a white solid; LCMS (ESI, M+1): m/z=517.3.

(1929) Step K. 1-(1-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a mixture of 1-(1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide (60.0 mg, 1.0 equiv.), 2-(7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (48.2 mg, 1.2 equiv.),

K.sub.3PO.sub.4 (1.5 M in water, 232 μ L, 3.0 equiv.) in methoxycyclopentane (1 mL) was added cataCXium A Pd G3 (8.45 mg, 0.1 equiv.) under N.sub.2 atmosphere. The reaction mixture was stirred at 90° C. for 2 hours. The reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (2 $\times$ 15 mL). The combined organic phase was washed with brine (5 mL), dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (42.0 mg, 51% yield) as a yellow solid; LCMS (ESI, M+1): m/z=701.2.

(1930) Step L. 1-(1-(8-fluoro-7-(7-fluoro-3-hydroxy-8-methylnaphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a solution of (1-(8-fluoro-7-(7-fluoro-3-(methoxymethoxy)-8-methylnaphthalen-1-yl)-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide (39.0 mg, 1.0 equiv.) in ACN (0.5 mL) was added HCl-dioxane (4 M, 1 mL, 72 equiv.) at 0° C. The mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was concentrated at room temperature. The residue was dissolved in ethyl acetate (20 mL) and water (3 mL). The pH of the mixture was adjusted to 8 with solid NaHCO.sub.3 while cooling the mixture with ice bath. The mixture was extracted with ethyl acetate (2 $\times$ 15 mL), dried over anhydrous Na.sub.2SO.sub.4, concentrated and purified by prep-HPLC [column: Phenomenex Synergi C18 150 $\times$ 25 mm $\times$ 10 μ m; mobile phase: water (0.1% formic acid)-ACN; B %: 10%-40%, 10 minutes] to afford the title compound (22.3 mg, 57% yield, 0.64 formic acid salt) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.13 (d, J=1.2 Hz, 1H), 7.68 (dd, J=5.6, 8.8 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.11 (dd, J=2.8, 8.8 Hz, 1H), 5.57-5.34 (m, 1H), 5.20-5.03 (m, 1H), 4.60-4.44 (m, 3H), 3.76-3.46 (m, 4H), 3.29-3.11 (m, 3H), 2.65-2.30 (m, 4H), 2.23-2.00 (m, 4H), 1.97-1.88 (m, 1H), 1.85-1.71 (m, 4H), 1.70-1.56 (m, 1H); LCMS (ESI, M+1): m/z=657.3.

Example 462

(1931) ##STR00820##

7-(8-fluoro-7-(8-(fluoromethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1932) The title compound was synthesized according to the procedure described for example 460. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.17 (s, 1H), 8.19-8.05 (m, 2H), 7.74-7.50 (m, 4H), 5.12 (d, J=47.6 Hz, 2H), 4.68-4.64 (m, 2H), 4.59-4.45 (m, 2H), 3.95-3.80 (m, 2H), 3.76-3.61 (m, 2H), 3.28-3.17 (m, 2H), 2.35-1.96 (m, 12H); LCMS (ESI, M+1): m/z=614.4.

Example 463

(1933) ##STR00821##

1-(1-(7-(8-(difluoromethyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1934) ##STR00822##

(1935) Step A. 8-bromo-1-naphthaldehyde: To a solution of 1,8-dibromonaphthalene (10.0 g, 1.0 equiv.) in THF (200 mL) was added n-BuLi (2.5 M, 18.2 mL, 1.3 equiv.) at -60° C. dropwise under N.sub.2 atm. After stirring for 5 minutes at -60° C., DMF (25.6 g, 26.9 mL, 10 equiv.) was added dropwise to the mixture. The reaction was warmed up to 25° C. and stirred for another 25 minutes. The mixture was quenched with water (150 mL) and concentrated to remove THF. The residue was extracted with ethyl acetate (2 $\times$ 150 mL). The organic layer was dried over Na.sub.2SO.sub.4, concentrated in vacuum. The residue was purified by silica gel chromatography (petroleum ether/ethyl acetate 100:1 to 50:1) to afford the title compound (5.0 g, 61% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ =11.44 (s, 1H), 8.03-7.98 (in, 1H), 7.93-7.87 (m, 3H), 7.57 (t, J=8.0 Hz, 1H), 7.39 (t, J=8.0 Hz, 1H).

(1936) Step B. 1-bromo-8-(difluoromethyl)naphthalene: To a solution of 8-bromo-1-naphthaldehyde (2.50 g, 1.0 equiv.) in DCM (30 mL) was added a solution of DAST (6.86 g, 4.0 equiv.) in DCM (10 mL) slowly at -40° C. The reaction was stirred at 25° C. for 12 hours. The

mixture was neutralized with saturated aqueous ice-cold NaHCO₃ solution (300 mL) and layers were separated. The aqueous phase was extracted with ethyl acetate (3×100 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by silica gel chromatography [petroleum ether/ethyl acetate 1:0 to 100:1] to afford the title compound (1.40 g, 51% yield) as a white solid; ¹H NMR (400 MHz, chloroform-d) δ=8.56-8.56 (m, 1H), 8.12 (d, J=7.2 Hz, 1H), 8.00-7.92 (m, 2H), 7.91-7.86 (m, 1H), 7.61-7.56 (m, 1H), 7.33 (t, J=7.6 Hz, 1H).

(1937) Step C. 2-(8-(difluoromethyl)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane: A mixture of 1-bromo-8-(difluoromethyl)naphthalene (1.50 g, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (2.22 g, 1.5 equiv.), KOAc (1.72 g, 3.0 equiv.) and Pd(dppf)Cl₂ (427 mg, 0.1 equiv.) in dioxane (30 mL) was degassed and stirred at 90° C. for 2 hours under N₂ atmosphere. The mixture was diluted with water (4.0 mL) and extracted with ethyl acetate (3×5.0 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by silica gel chromatography [petroleum ether/ethyl acetate 1:0 to 50:1] to afford the title compound (750 mg, 42% yield) as a white solid; ¹H NMR (400 MHz, chloroform-d) δ=8.08-7.77 (m, 5H), 7.54-7.46 (m, 2H), 1.46 (s, 12H).

(1938) Step D. 7-(8-(difluoromethyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.), 2-(8-(difluoromethyl)naphthalen-1-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (159 mg, 1.1 equiv.), CataCXium A Pd G3 (34.6 mg, 0.1 equiv.) and K₃PO₄ (1.5 M in water, 951 μL, 3.0 equiv.) in methoxycyclopentane (4.0 mL) was degassed and stirred at 90° C. for 3 hours under N₂ atmosphere. The mixture was diluted with water (3.0 mL) and extracted with ethyl acetate (3×3.0 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (70 mg, 26% yield) as a yellow solid; ¹H NMR (400 MHz, chloroform-d) δ=9.26 (s, 1H), 8.13-8.05 (m, 2H), 7.95 (d, J=7.2 Hz, 1H), 7.66-7.54 (m, 2H), 7.58-7.54 (m, 1H), 6.58-6.25 (m, 1H), 5.13-5.05 (m, 2H), 4.36 (s, 2H), 3.19-3.09 (m, 2H), 2.68 (td, J=6.8, 10.0 Hz, 2H), 2.11-2.06 (m, 1H), 2.05-2.00 (m, 1H), 1.90 (q, J=6.4 Hz, 4H), 1.74-1.69 (m, 2H).

(1939) Step E. 1-(1-(7-(8-(difluoromethyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a solution of 7-(8-(difluoromethyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (30 mg, 1.0 equiv.) and piperidin-3-ylmethanesulfonamide (14.3 mg, 1.50 equiv.) in DMF (0.5 mL) were added DIEA (20.7 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg). The mixture was stirred at 40° C. for 12 hours. The mixture was filtered and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 μm; mobile phase: water (10 mM NH₄HCO₃)-ACN, B %: 40%-70%, 10 min] and re-purified by prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 μm; mobile phase: water (0.225% formic acid)-ACN, B %: 15%-45%, 10 min] and lyophilized to afford the title compound (11.7 mg, 34% yield) as a white solid; ¹H NMR (400 MHz, methanol-d₄) δ 9.17 (s, 1H), 8.24-8.15 (m, 2H), 7.94 (d, J=7.2 Hz, 1H), 7.68 (q, J=8.0 Hz, 2H), 7.63-7.57 (m, 1H), 6.59-6.27 (m, 1H), 5.14 (br d, J=12.0 Hz, 1H), 4.71-4.65 (m, 1H), 4.64-4.54 (m, 2H), 3.76-3.67 (m, 1H), 3.61-3.50 (m, 2H), 3.25-3.10 (m, 4H), 2.61-2.47 (m, 1H), 2.33 (td, J=6.4, 12.8 Hz, 2H), 2.21-2.11 (m, 4H), 2.11-1.88 (m, 5H), 1.85-1.71 (m, 1H), 1.68-1.56 (m, 1H); LCMS (ESI, M+1): m/z=641.4.

Example 464

(1940) ##STR00823##

7-(7-(8-(difluoromethyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,3,7-triazaspiro[4.5]decane-2,4-dione

(1941) The title compound was synthesized according to the procedure described for example 463.

.sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.13 (s, 1H), 8.23-8.14 (m, 2H), 7.95 (d, J=7.2 Hz, 1H), 7.73-7.65 (m, 2H), 7.63-7.58 (m, 1H), 6.60-6.26 (m, 1H), 4.69-4.65 (m, 1H), 4.54-4.45 (m, 1H), 4.38-4.26 (m, 2H), 3.84-3.70 (m, 2H), 3.21-3.11 (m, 2H), 2.81-2.72 (m, 2H), 2.30-2.20 (m, 1H), 2.14-2.02 (m, 4H), 1.99-1.87 (m, 5H), 1.84-1.74 (m, 2H); LCMS (ESI, M+1): m/z=632.4.

Example 465

(1942) ##STR00824##

1-(1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(trifluoromethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide

(1943) ##STR00825##

(1944) Step A. 1-(1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.), piperidin-3-ylmethanesulfonamide (102 mg, 1.2 equiv.), 4 Å molecular sieves (10.0 mg) in DMF (2 mL) was added DIEA (184 mg, 248 μ L, 3.0 equiv.). The reaction was stirred at 40° C. for 4 hours. After reaction completion, the mixture was filtered to give a solution. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) to afford the title compound (135 mg, 57% yield) as a yellow solid; LCMS [ESI, M+1]: m/z=499.2.

(1945) Step B. 1-(1-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(trifluoromethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide: To a mixture of (1-(1-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)piperidin-3-yl)methanesulfonamide (135 mg, 1.0 equiv.), 4,4,5,5-tetramethyl-2-(8-(trifluoromethyl)naphthalen-1-yl)-1,3,2-dioxaborolane (113 mg, 1.3 equiv.), K.sub.3PO.sub.4 (1.5 M, 541 μ L, 3.0 equiv.) in methoxycyclopentane (1 mL) was added CataCXium A Pd G3 (19.7 mg, 0.1 equiv.) under N.sub.2. The reaction was stirred at 90° C. for 2 hours. After completion, the reaction mixture was quenched by addition of water (1 mL) and extracted with ethyl acetate (6 mL). The organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give a residue. The crude product was purified by reversed-phase flash chromatography (water (0.1% formic acid)-ACN) and prep-HPLC (column: Waters Xbridge 150×25 mm×5 μ m; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN, B %: 36%-66%, 10 min) and lyophilized to afford the title compound (3.53 mg, 2% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4): δ =9.06 (d, J=2.4 Hz, 1H), 8.30 (d, J=8.0 Hz, 1H), 8.20 (d, J=8.4 Hz, 1H), 8.05 (d, J=7.2 Hz, 1H), 7.78-7.72 (m, 11H), 7.71-7.63 (m, 2H), 4.98 (br s, 1H), 4.60-4.48 (m, 1H), 4.40-4.27 (m, 2H), 3.69-3.54 (m, 11H), 3.40-3.34 (m, 1H), 3.26-3.07 (m, 4H), 2.79-2.67 (m, 2H), 2.53 (br dd, J=4.0, 7.2 Hz, 1H), 2.11 (qd, J=5.6, 12.0 Hz, 3H), 1.98-1.88 (m, 5H), 1.98-1.71 (m, 3H), 1.68-1.57 (m, 1H); LCMS [ESI, M+1]: m/z=659.5.

Example 466

(1946) ##STR00826##

6-(7-(8-(difluoromethyl)naphthalen-1-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1947) The title compound was synthesized according to the procedure described for example 463. H NMR (400 MHz, methanol-d.sub.4) δ =9.14 (s, 1H), 8.25-8.13 (m, 2H), 7.95 (d, J=7.2 Hz, 1H), 7.72-7.65 (m, 2H), 7.62-7.62 (m, 1H), 6.59-6.26 (m, 1H), 4.53-4.39 (m, 3H), 4.37-4.27 (m, 11H), 4.07-3.98 (m, 1H), 3.91-3.78 (m, 1H), 3.42-3.34 (m, 2H), 3.05-2.95 (m, 2H), 2.94-2.74 (m, 2H), 2.24-2.15 (m, 2H), 2.15-2.07 (m, 2H), 2.06-1.96 (m, 6H), 1.96-1.87 (m, 2H); LCMS (ESI, M+1): m/z=603.4.

Example 467

(1948) ##STR00827##

6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(trifluoromethyl)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1949) The title compound was synthesized according to the procedure described for example 465.
.sup.1H NMR (400 MHz, methanol-d.sub.4): δ =9.06 (d, J=3.6 Hz, 1H), 8.31 (d, J=8.0 Hz, 1H), 8.21 (dd, J=1.2, 8.4 Hz, 1H), 8.06 (d, J=7.6 Hz, 1H), 7.80-7.60 (m, 3H), 4.43-4.20 (m, 4H), 4.00 (dd, J=5.6, 13.2 Hz, 1H), 3.88-3.71 (m, 1H), 3.17-3.07 (m, 2H), 2.94-2.84 (m, 1H), 2.80-2.70 (m, 3H), 2.08 (ddd, J=6.0, 13.2, 19.2 Hz, 4H), 2.01-1.87 (m, 6H), 1.82-1.73 (m, 3H); LCMS [ESI, M+1]: m/z=621.4.

Example 468

(1950) ##STR00828##

6-(8-fluoro-7-(8-(fluoromethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1951) The title compound was synthesized according to the procedure described for example 460.
.sup.1H NMR (400 MHz, methanol-d.sub.4): δ =9.14 (s, 1H), 8.51 (s, 1H), 8.16-8.04 (m, 2H), 7.69-7.51 (m, 4H), 5.15-4.98 (m, 2H), 4.70-4.57 (m, 2H), 4.42 (br d, J=13.2 Hz, 1H), 4.37-4.27 (m, 11H), 4.00 (d, J=13.2 Hz, 1H), 3.90-3.80 (m, 11H), 3.69-3.58 (m, 2H), 3.26-3.17 (m, 2H), 2.95-2.88 (m, 1H), 2.81-2.73 (m, 1H), 2.35-2.25 (m, 2H), 2.24-2.01 (m, 8H), 2.00-1.92 (m, 2H); LCMS [ESI, M+1]: m/z=585.3.

Example 469

(1952) ##STR00829##

6-(8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)-7-(8-(methylthio)naphthalen-1-yl)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1953) The title compound was synthesized according to the procedure described for example 436.
.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.08 (d, J=4.8 Hz, 1H), 8.06 (dd, J=1.2, 8.0 Hz, 1H), 7.86-7.784 (m, 1H), 7.68-7.58 (m, 1H), 7.55-7.49 (m, 3H), 4.58-4.48 (m, 2H), 4.45-4.18 (m, 2H), 4.07-4.01 (m, 1H), 3.96-3.73 (m, 1H), 3.54-3.41 (m, 2H), 3.13-3.01 (m, 2H), 2.93-2.71 (m, 2H), 2.25 (s, 3H), 2.24-2.19 (m, 1H), 2.17-2.02 (m, 6H), 2.00-1.96 (m, 4H) LCMS (ESI, M+1): m/z=599.3.

Example 470

(1954) ##STR00830##

6-(8-fluoro-7-(8-(2-fluoroethyl)naphthalen-1-yl)-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(1955) The title compound was synthesized according to the procedure described for example 434;
.sup.1H NMR (400 MHz, methanol-d.sub.4) δ =9.15 (d, J=2.0 Hz, 1H), 8.09 (dd, J=1.2, 8.0 Hz, 1H), 7.95 (d, J=7.2 Hz, 1H), 7.66-7.41 (m, 4H), 4.70-4.60 (m, 2H), 4.47-4.19 (m, 4H), 4.10-3.79 (m, 2H), 3.72-3.61 (m, 2H), 3.30-3.20 (m, 2H), 2.94-2.84 (m, 1H), 2.83-2.65 (m, 3H), 2.37-2.27 (m, 2H), 2.26-2.14 (m, 4H), 2.13-1.93 (m, 6H); LCMS (ESI, M+1): m/z=599.3.

Example 471

(1956) ##STR00831##

1-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-yl)methylpiperidin-3-ol

(1957) ##STR00832##

(1958) Step A. 7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.), 5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indazole (406 mg, 2.5 equiv.), cataCXium A Pd G3 (33.2 mg, 0.1 equiv.) and K.sub.3PO.sub.4 (1.5 M, 912 μ L, 3.0 equiv.) in THF (4.0 mL) was degassed and stirred at 60° C. for 2 hours under N.sub.2 atmosphere. The mixture was diluted with water (5.0 mL) and extracted with ethyl acetate (3 $\times$ 5.0 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford

title compound (176 mg, 60% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ =9.35 (d, J=4.4 Hz, 1H), 7.63-7.57 (m, 1H), 7.54 (d, J=16.9 Hz, 11H), 5.78-5.68 (m, 11H), 5.46-5.23 (m, 11H), 5.15-5.01 (m, 2H), 4.54-4.34 (m, 2H), 4.10-3.98 (m, 1H), 3.82-3.72 (m, 11H), 3.54-3.22 ((m, 3H), 3.11-2.99 (m, 1H), 2.64-2.53 (m, 1H), 2.51 (s, 3H), 2.40-2.26 (m, 2H), 2.20 (s, 3H), 2.17-2.13 (m, 1H), 2.11-2.07 (m, 1H), 2.04-1.99 (m, 2H), 1.85-1.60 (m, 5H); LCMS (ESI, M+1): m/z=633.4.

(1959) Step B. 1-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (200 mg, 1.0 equiv.) in DMF (2.0 mL) was added 3-methylpiperidin-3-ol (95.9 mg, 2.0 equiv., HCl), DIEA (204 mg, 5.0 equiv.) and 4 Å molecular sieves (50 mg). The reaction was stirred at 40° C. for 12 hours. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford title compound (173 mg, 84% yield) as a yellow solid; .sup.1H NMR (400 MHz, chloroform-d) δ =9.28-9.19 (m, 1H), 7.69-7.62 (m, 1H), 7.51 (br d, J=10.4 Hz, 1H), 5.77-5.68 (m, 1H), 5.39-5.15 (m, 1H), 4.52-4.37 (m, 2H), 4.35-4.18 (m, 2H), 4.09-3.96 (m, 1H), 3.83-3.69 (m, 1H), 3.54-3.37 (m, 1H), 3.36-3.22 (m, 3H), 3.20-3.10 ((m, 11H), 3.02-2.92 (m, 1H), 2.63-2.52 (m, 1H), 2.50 (s, 3H), 2.30-2.24 (m, 1H), 2.22 (s, 3H), 2.20-2.10 (m, 3H), 2.08 (br s, 1H), 1.99-1.84 (m, 5H), 1.78-1.62 (m, 5H), 1.35 (s, 3H); LCMS (ESI, M+1): m/z=648.6.

(1960) Step C. 1-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of 1-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (50 mg, 1.0 equiv.) in ACN (0.15 mL) was added HCl.dioxane (4 M, 289 μ L, 15 equiv.). The reaction was stirred at 0° C. for 1 hour. The mixture was concentrated and purified by prep-HPLC [column: Waters Xbridge 150 $\times$ 25 mm $\times$ 5 μ m; mobile phase: water (10 mM NH₄HCO₃), B: ACN, B %: 34%-64%, 9 min] and lyophilized to afford the title compound (27.2 mg, 62% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d₄) δ 9.28 (dd, J=1.0, 7.2 Hz, 11H), 7.62 (d, J=2.2 Hz, 1H), 7.50 (s, 11H), 5.40-5.19 (m, 11H), 4.68-4.51 (m, 2H), 4.37-4.23 (m, 3H), 3.69-3.61 (m, 11H), 3.51-3.37 (m, 1H), 3.24 (br s, 11H), 3.19 (br s, 11H), 3.07-2.96 (m, 11H), 2.51 (s, 3H), 2.41-2.26 (m, 1H), 2.26-2.19 (m, 4H), 2.19-2.10 (m, 2H), 2.05-1.95 (m, 2H), 1.94-1.83 (m, 2H), 1.82-1.72 (m, 2H), 1.29 (d, J=1.2 Hz, 3H); LCMS (ESI, M+1): m/z=564.4.

Example 472

(1961) ##STR00833##

(3R)-1-(7-(8-ethyl-2-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1962) ##STR00834##

(1963) Step A. (3R)-1-(7-(3-((benzyloxy)methoxy)-2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a mixture of (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (2.00 g, 1.0 equiv.), ((6-((benzyloxy)methoxy)-7-fluoro-8-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)naphthalen-1-yl)ethynyl)triisopropylsilane (3.89 g, 1.5 equiv.), K₃PO₄ (1.5 M in water, 8.81 mL, 3.0 equiv.) in methoxycyclopentane (30 mL) was added CataCXium A Pd G3 (321 mg, 0.1 equiv.) under N₂. The mixture was stirred at 90° C. for 8 hours. The reaction mixture was diluted with water (10 mL) and extracted with ethyl acetate (2 $\times$ 15 mL). The combined organic phase was washed with brine (20 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title

compound (1.20 g, 31% yield) as a brown oil; LCMS (ESI, M+1): m/z=880.5.

(1964) Step B. (3R)-1-(7-(3-((benzyloxy)methoxy)-8-ethynyl-2-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To the solution of (3R)-1-(7-(3-((benzyloxy)methoxy)-2-fluoro-8-((triisopropylsilyl)ethynyl)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (1.10 g, 1.0 equiv.) in DMF (5 mL) was added CsF (1.90 g, 10 equiv.). The mixture was stirred at 25° C. for 1 hour. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)/ACN] to afford the title compound (820 mg, 89% yield) as a yellow oil; LCMS (ESI, M+1): m/z=724.4.

(1965) Step C. (3R)-1-(7-(8-ethyl-2-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (3R)-1-(7-(3-((benzyloxy)methoxy)-8-ethynyl-2-fluoronaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (200 mg, 1.0 equiv.) in MeOH (3 mL) was added Pd/C (40.0 mg, 10% purity) under N.sub.2 atmosphere. The mixture was degassed, purged with H.sub.2 and stirred at 25° C. for 12 hours under H.sub.2 (15 psi) atmosphere. The mixture was filtered, concentrated and purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: water (0.25% formic acid)-ACN; B %: 14%-44%, 10 minutes] to afford the title compound (37.8 mg, 21% yield, 0.5% FORMIC ACID) as a yellow solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.28 (d, J=3.2 Hz, 1H), 7.64 (d, J=7.6 Hz, 1H), 7.46 (d, J=9.6 Hz, 1H), 7.35 (t, J=7.6 Hz, 1H), 7.20 (d, J=7.2 Hz, 1H), 5.56-5.33 (m, 1H), 4.64-4.40 (m, 3H), 4.37-4.26 (m, 1H), 3.75-3.38 (m, 5H), 3.28-3.18 (m, 1H), 2.61-2.33 (m, 2H), 2.33-2.23 (m, 2H), 2.23-2.10 (m, 4H), 2.09-1.97 (m, 1H), 1.91-1.71 (m, 3H), 1.29 (d, J=9.2 Hz, 3H), 0.89 (td, J=7.5, 9.7 Hz, 3H); LCMS (ESI, M+1): m/z=608.4.

Example 473

(1966) ##STR00835##

(3R)-1-(7-(5-ethyl-1H-benzof[*f*]indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(1967) ##STR00836## ##STR00837##

(1968) Step A. 4-bromo-2-(bromomethyl)-1-chlorobenzene: A mixture of 4-bromo-1-chloro-2-methylbenzene (50 g, 1.0 equiv.), NBS (47.6 g, 1.1 equiv.), AIBN (799 mg, 0.02 equiv.) in MeCN (600 mL) was stirred at 90° C. for 12 hours. The mixture was concentrated in vacuum and the residue was purified by column chromatography (Silica gel, ethyl acetate/petroleum ether 10:1) to afford the title compound (60 g, 87% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃-d) δ 7.61 (d, J=2.4 Hz, 1H), 7.40 (dd, J=2.4, 8.4 Hz, 1H), 7.30-7.26 (in, 1H), 4.54 (s, 2H).

(1969) Step B. ethyl 1-(N,N-dimethylsulfamoyl-1H-pyrazole-4-carboxylate: To a mixture of ethyl 1H-pyrazole-4-carboxylate (10 g, 1.0 equiv.) and DABCO (8.81 g, 1.1 equiv.) in MeCN (100 mL) was added N,N-dimethylsulfamoyl chloride (11.3 g, 8.41 mL, 1.1 equiv.). The mixture was stirred at 20° C. for 1 hour before being concentrated in vacuum and the residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 3:1) to afford the title compound (15 g, 85% yield) as a white solid. LCMS (ESI, M+1): m/z=248.0.

(1970) Step C. ethyl 5-(5-bromo-2-chlorobenzyl-1-(N,N-dimethylsulfamoyl-1H-pyrazole-4-carboxylate: To a solution of ethyl 1-(N,N-dimethylsulfamoyl)-1H-pyrazole-4-carboxylate (6 g, 1.0 equiv.) in THF (60 mL) was added LDA (2 M, 14.6 mL, 1.2 equiv.) and HMPA (5.22 g, 5.12 mL, 1.2 equiv.) at -78° C. After stirring at -78° C. for 1 hour, 4-bromo-2-(bromomethyl)-1-chlorobenzene (8.28 g, 1.2 equiv.) was added into the mixture. The mixture was stirred at -78° C. for 2 hours and at 20° C. for 1 hour before being diluted with ethyl acetate (100 mL), washed with water (100 mL) and brine (100 mL), dried over Na₂SO₄ and concentrated in vacuum. The residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 3:1). The

desired fractions were collected and concentrated in vacuum to give a residue and the residue was triturated with methanol (30 mL), dried in vacuum to afford the title compound (3 g, 27% yield) as a white solid. LCMS (ESI, M+1): m/z=452.1.

(1971) Step D. 5-(5-bromo-2-chlorobenzyl)-1-(N,N-dimethylsulfamoyl)-1H-pyrazole-4-carboxylic acid: A mixture of ethyl 5-(5-bromo-2-chlorobenzyl)-1-(N,N-dimethylsulfamoyl)-1H-pyrazole-4-carboxylate (20 g, 1.0 equiv.) and NaOH (44.4 g, 25 equiv.) in dioxane (120 mL) and water (120 mL) was stirred at 90° C. for 2 hours. The mixture was extracted with ethyl acetate (300 mL), washed with brine (200 mL), dried over Na₂SO₄ and concentrated in vacuum to afford the title compound (20.5 g, crude) as a white solid and used into next step without further purification.

(1972) Step E. 5-bromo-8-chloro-1H-benzo[f]indazol-4(9H)-one: A mixture of 5-(5-bromo-2-chlorobenzyl)-1-(N,N-dimethylsulfamoyl)-1H-pyrazole-4-carboxylic acid (22 g, 1.0 equiv.) in CF₃SO₃H (220 mL) was stirred at 90° C. for 3 hours. The mixture was poured into ice water (500 mL) and filtered. The filter cake was diluted with ethyl acetate (500 mL) and filtered, the filtrate was washed with saturated sodium bicarbonate (400 mL), brine (400 mL), dried over Na₂SO₄ and concentrated in vacuum to afford the title compound (7.5 g, 48% yield) that was used in the next step without further purification as a yellow solid. LCMS (ESI, M+1): m/z=298.8.

(1973) Step F. 5-bromo-8-chloro-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4(9H)-one: To a mixture of 5-bromo-8-chloro-1H-benzo[f]indazol-4(9H)-one (6.5 g, 1.0 equiv.) and TsOH (376 mg, 0.1 equiv.) in THF (60 mL) was added DHP (5.51 g, 5.99 mL, 3.0 equiv.) at 15° C. The mixture was stirred at 15° C. for 15 mins before being concentrated in vacuum. The residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (3.7 g, 40% yield) as a yellow oil. LCMS (ESI, M-83): m/z=298.8.

(1974) Step G. 8-chloro-1-(tetrahydro-2H-pyran-2-yl)-5-vinyl-1H-benzo[f]indazol-4(9H)-one: A mixture of 5-bromo-8-chloro-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4(9H)-one (550 mg, 1.0 equiv.), potassium trifluoro(vinyl)boranuide (579 mg, 3 equiv.), Pd(dppf)Cl₂ (105 mg, 0.1 equiv.) and Na₂CO₃ (458 mg, 3.0 equiv.) in dioxane (5 mL) and water (1.5 mL) was stirred at 90° C. for 3 hours. The mixture was concentrated in vacuum and the residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (220 mg, 46% yield) as a yellow solid. LCMS (ESI, M+1): m/z=328.9.

(1975) Step H. 5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4(9H)-one: A mixture of 8-chloro-1-(tetrahydro-2H-pyran-2-yl)-5-vinyl-1H-benzo[f]indazol-4(9H)-one (300 mg, 1.0 equiv.), Pd/C (10 mg, 10%, 1 equiv.) and NaHCO₃ (76.6 mg, 1.0 equiv.) in MeOH (10 mL) was stirred at 20° C. for 1 hours under H₂ atmosphere (15 psi). The mixture was filtered and concentrated in vacuum and the residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (110 mg, 41% yield) as a yellow solid. LCMS (ESI, M+1): m/z=297.2.

(1976) Step I. 5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4-yl trifluoromethanesulfonate: To a solution of 5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4(9H)-one (50 mg, 1.0 equiv.) and DIEA (87.2 mg, 4.0 equiv.) in DCM (1.0 mL) was added Tf₂O (95.2 mg, 2.0 equiv.) at -40° C. The mixture was stirred at -40° C. for 15 mins. After completion, the residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 5:1) to afford the title compound (45 mg, 51% yield) as a yellow solid. LCMS (ESI, M+1): m/z=429.1.

(1977) Step J. 5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-benzo[f]indazole: A mixture of 5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4-yl trifluoromethanesulfonate (30 mg, 1.0 equiv.), 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (35.9 mg, 40.6 μL, 4.0 equiv.), TEA (28.3 mg, 4.0 equiv.) and Pd(dppf)Cl₂ (5.12 mg, 0.1 equiv.) in MeCN (1 mL) was stirred at 80° C. for 5 hours under N₂ atm. The mixture was concentrated

in vacuum and the residue was purified by column chromatography (Silica gel, petroleum ether/ethyl acetate 5:1) to afford the title compound (24 mg, 71% yield) as a yellow oil. LCMS (ESI, M+1): m/z=407.2.

(1978) Step K. (3R)-1-(7-(5-ethyl-)-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of 5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-benzo[f]indazole (15 mg, 1.0 equiv.), (R)-1-(7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (33.5 mg, 2.0 equiv.), CataCXium A Pd G3 (2.69 mg, 0.1 equiv.) and K.sub.3PO.sub.4 (1.5 M in water, 3.0 equiv.) in THF (1.0 mL) was stirred at 60° C. for 12 hours under N.sub.2. The mixture was concentrated in vacuum to give a residue and the residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] to afford the title compound (18 mg, 50% yield) as a yellow oil. LCMS (ESI, M+1): m/z=698.4.

(1979) Step L. (3R)-1-(7-(5-ethyl-1H-benzo[f]indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (3R)-1-(7-(5-ethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-benzo[f]indazol-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (16 mg, 1.0 equiv.) in TFA (1.23 g, 471 equiv.) and DCM (0.5 mL) was stirred at 15° C. for 0.5 hours. After completion, the mixture was concentrated in vacuum and the residue was purified by reversed phase flash chromatography [water (0.1% formic acid)/acetonitrile] followed by prep-HPLC (column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: water (0.225% formic acid)-ACN, B %: 16%-36%, 10 min) to afford the title compound (2.51 mg, 17% yield) as a yellow solid. .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.34 (d, J=10.4 Hz, 1H), 8.22 (s, 1H), 7.97 (d, J=8.4 Hz, 1H), 7.77 (d, J=18.4 Hz, 1H), 7.43 (dd, J=6.8, 8.4 Hz, 1H), 5.49-5.31 (m, 1H), 4.61 (br s, 1H), 4.48-4.33 (m, 3H), 3.75-3.60 (m, 1H), 3.55-3.37 (m, 4H), 3.19-3.09 (m, 1H), 2.63-2.30 (m, 4H), 2.29-1.95 (m, 5H), 1.94-1.76 (m, 3H), 1.33 (d, J=10.0 Hz, 3H), 1.00 (q, J=7.6 Hz, 3H). LCMS (ESI, M+1): m/z=614.4

Example 474

(1980) ##STR00838##

7-(8-chloronaphthalen-1-yl)-N-(3-(difluoromethoxy)propyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-amine

(1981) The title compound was synthesized according to the procedure described for example 101. LCMS (ESI, M+1): m/z=572.2

Example 475

(1982) ##STR00839##

3-((7-(8-chloronaphthalen-1-yl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)-2-hydroxypropanamide

(1983) The title compound was synthesized according to the procedure described for example 101. LCMS (ESI, M+1): m/z=551.1

Example 476

(1984) ##STR00840##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2-thia-7-azaspiro[4.5]decan-7-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1985) The title compound was synthesized according to the procedure described for example 135. .sup.1H NMR (400 MHz, Acetic) δ=9.50 (s, 1H), 8.16-8.09 (m, 1H), 7.71 (dd, J=6.0, 8.8 Hz, 1H), 7.42 (d, J=2.4 Hz, 1H), 7.29 (t, J=9.6 Hz, 1H), 7.22 (t, J=2.8 Hz, 1H), 5.69-5.49 (m, 1H), 5.02-4.73 (m, 2H), 4.72-4.52 (m, 1H), 4.51-4.26 (m, 1H), 4.21-4.09 (m, 2H), 4.07-3.63 (m, 3H), 3.49-3.36 (m, 1H), 3.07-2.74 (m, 4H), 2.72-2.46 (m, 4H), 2.39-2.25 (m, 4H), 2.08 (br s, 1H), 2.02-1.78 (m, 5H), 0.86 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=650.2.

Example 477

(1986) ##STR00841##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepin-2-yl)(4-methylpiperazin-1-yl)methanone

(1987) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.23 (s, 1H), 7.69 (dd, J=6.0, 9.0 Hz, 1H), 7.32 (d, J=2.4 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 6.87 (s, 1H), 5.67-5.51 (m, 1H), 5.41-5.20 (m, 3H), 5.01-4.90 (m, 2H), 4.82 (br s, 2H), 4.75-4.36 (m, 7H), 4.11-3.80 (m, 3H), 3.46 (td, J=5.6, 10.7 Hz, 3H), 2.95 (s, 3H), 2.81-2.03 (m, 11H), 0.78 (br t, J=7.2 Hz, 3H), LCMS (ESI, M+1): m/z=756.5.

Example 478

(1988) ##STR00842##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-oxa-1,7-diazaspiro[4.5]decan-2-one

(1989) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.10 (s, 1H), 7.71-7.65 (m, 1H), 7.31 (d, J=2.6 Hz, 1H), 7.25 (t, J=9.4 Hz, 1H), 7.05 (d, J=2.6 Hz, 1H), 5.45-5.26 (m, 1H), 4.44-4.28 (m, 4H), 4.16 (d, J=9.0 Hz, 2H), 4.00-3.82 (m, 2H), 3.51-3.38 (m, 1H), 3.14-3.05 (m, 1H), 2.62-2.11 (m, 6H), 2.09-1.96 (m, 6H), 1.94-1.79 (m, 2H), 0.86-0.73 (m, 3H); LCMS (ESI, M+1): m/z=649.2.

Example 479

(1990) ##STR00843##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-isopropyl-5,6,7,8-tetrahydropyrazolo[4,3-c]azepine-2(4H)-carboxamide

(1991) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=10.16-9.72 (m, 1H), 9.15 (s, 1H), 8.25 (s, 1H), 8.22 (s, 1H), 8.05 (d, J=8.6 Hz, 1H), 7.76 (dd, J=6.0, 9.1 Hz, 1H), 7.40-7.29 (m, 2H), 7.01 (d, J=2.4 Hz, 1H), 5.38-5.18 (m, 1H), 5.11-4.97 (m, 2H), 4.31 (br s, 2H), 4.17-4.05 (m, 2H), 4.00-3.89 (m, 1H), 3.12-3.05 (m, 2H), 3.01 (br s, 1H), 2.99-2.92 (m, 2H), 2.86-2.79 (m, 1H), 2.41-2.35 (m, 1H), 2.21-1.98 (m, 6H), 1.88-1.72 (m, 3H), 1.17 (d, J=6.8 Hz, 6H), 0.70 (t, J=7.2 Hz, 3H), LCMS (ESI, M+1): m/z=715.5.

Example 480

(1992) ##STR00844##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,6-dioxa-9-azaspiro[3.6]decan-9-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(1993) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.93 (s, 1H), 9.22 (s, 1H), 7.83-7.70 (m, 1H), 7.42-7.28 (m, 2H), 7.01 (d, J=2.4 Hz, 1H), 5.28 (d, J=64.0 Hz, 1H), 4.62-4.48 (m, 3H), 4.36 (t, J=6.8 Hz, 3H), 4.18-4.12 (m, 1H), 4.07 (d, J=1.2 Hz, 1H), 4.02-3.92 (m, 4H), 3.89 (d, J=4.4 Hz, 2H), 3.09 (d, J=10.0 Hz, 2H), 3.01 (s, 1H), 2.86-2.79 (m, 1H), 2.41-2.34 (m, 1H), 2.21-2.08 (m, 2H), 2.02 (d, J=10.4 Hz, 2H), 1.90-1.71 (m, 3H), 0.73 (t, J=7.2 Hz, 3H); LCMS [M+1]: m/z=636.2.

Example 481

(1994) ##STR00845##

6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-6-azaspiro[3.5]nonane 2,2-dioxide

(1995) The title compound was synthesized according to the procedure described for example 135. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.12 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.31 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.5-5.32 (m, 1H), 4.52 (s, 2H), 4.46-4.30 (m, 2H), 4.22-4.06 (m, 4H), 4.05-3.97 (m, 2H), 3.70-3.55 (m, 1H), 3.55-3.42 (m, 2H), 3.26-3.16 (m, 1H), 2.61-2.42 (m, 2H), 2.40-2.26 (m, 2H), 2.25-2.10 (m, 3H), 2.10-2.02 (m, 3H),

1.82 (br s, 2H), 0.80 (dt, J=1.2, 7.2 Hz, 3H); LCMS (ESI, M+1): m/z=668.4.

Example 482

(1996) ##STR00846##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3a-methyltetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(1997) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=11.85-10.93 (m, 1H), 9.26 (s, 1H), 8.18 (s, 1H), 7.77 (dd, J=6.0, 9.2 Hz, 1H), 7.42-7.29 (m, 2H), 7.02 (d, J=2.4 Hz, 1H), 5.41-5.17 (m, 1H), 4.55-4.41 (m, 2H), 4.40-4.30 (m, 1H), 4.20-4.12 (m, 1H), 4.10-4.04 (m, 1H), 3.95 (d, J=12.0 Hz, 1H), 3.44 (s, 1H), 3.10 (d, J=11.2 Hz, 2H), 3.02 (s, 1H), 2.87-2.81 (m, 1H), 2.36-2.31 (m, 1H), 2.17-1.97 (m, 4H), 1.89-1.74 (m, 3H), 1.50 (s, 3H), 0.80-0.66 (m, 3H); LCMS [M+1]⁺: m/z=647.2.

Example 483

(1998) ##STR00847##

5-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-N,N,1-trimethyl-1H-pyrazole-3-carboxamide

(1999) ##STR00848##

(2000) Step A. methyl 5-[(tert-butoxycarbonylamino)methyl]-1-methyl-pyrazole-3-carboxylate: A mixture of methyl 5-bromo-1-methyl-pyrazole-3-carboxylate (0.95 g, 1.0 equiv.), potassium (((tert-butoxycarbonyl)amino)methyl)trifluoroborate (1.2 g, 1.2 equiv.), Pd(PPh₃)₂Cl₂ (152 mg, 0.05 equiv.), RuPhos (405 mg, 0.2 equiv.) and Na₂CO₃ (1.4 g, 3.0 equiv.) in dioxane (20 mL) and H₂O (2 mL) was degassed and stirred at 120° C. for 12 hours. The residue was diluted with H₂O (30 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (3×30 mL), dried over Na₂SO₄, filtered and concentrated to give a residue, which was purified by flash silica gel chromatography (Ethyl acetate in Petroleum ether 42~46%) to afford the title compound (0.82 g, 70% yield) as a yellow gum; ¹H NMR (400 MHz, CDCl₃) δ=6.71 (s, 1H), 4.82 (s, 1H), 4.37 (d, J=5.6 Hz, 2H), 3.92 (m, 6H), 1.46 (s, 9H). (2001) Step B. 5-[(tert-butoxycarbonylamino)methyl]-1-methyl-pyrazole-3-carboxylic acid: To a solution of methyl 5-[(tert-butoxycarbonylamino)methyl]-1-methyl-pyrazole-3-carboxylate (700 mg, 1.0 equiv.) in THF (10 mL) and H₂O (10 mL) was added LiOH·H₂O (218 mg, 2.0 equiv.). The reaction was stirred at 25° C. for 2 hours. The pH of the mixture was adjusted to 2~3 with 1 N HCl. The resulting mixture was diluted with H₂O (150 mL) and extracted with DCM (3×150 mL), dried over Na₂SO₄, filtered and concentrated under reduced pressure to afford the title compound (560 mg, 2.19 mmol, 84% yield) as a yellow gum. ¹H NMR (400 MHz, CDCl₃) δ=6.76 (s, 1H), 4.89 (br s, 1H), 4.39 (d, J=4.4 Hz, 2H), 3.95 (s, 3H), 1.46 (s, 9H).

(2002) Step C. tert-butyl N-[[5-(dimethylcarbamoyl)-2-methyl-pyrazol-3-yl]methyl]carbamate: To a solution of 5-[(tert-butoxycarbonylamino)methyl]-1-methyl-pyrazole-3-carboxylic acid (550 mg, 1 equiv.) and dimethylamine (351 mg, 2.0 equiv, HCl) in DMF (10 mL) was added HATU (1.64 g, 2.0 equiv.) and DIEA (1.39 g, 5.0 equiv.). The reaction was stirred at 25° C. for 2 hours. The reaction mixture was diluted with water (30 mL) and extracted with ethyl acetate (3×30 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered and concentrated to give a residue. The residue was purified by flash silica gel chromatography (Ethyl acetate in Petroleum ether 72~80%) to afford the title compound (600 mg, 96% yield, 97% purity) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ=6.54 (s, 1H), 4.96 (s, 1H), 4.36 (d, J=5.6 Hz, 2H), 3.87 (s, 3H), 3.32 (s, 3H), 3.09 (s, 3H), 1.45 (s, 9H).

(2003) Step D. 5-(aminomethyl)-N,N,1-trimethyl-pyrazole-3-carboxamide: To a solution of tert-butyl N-[[5-(dimethylcarbamoyl)-2-methyl-pyrazol-3-yl]methyl]carbamate (300 mg, 1.0 equiv.) in DCM (3 mL) was added HCl.dioxane (4 M, 0.1 equiv.). The reaction was stirred at 25° C. for 12

hours. The reaction mixture was concentrated under reduced pressure to afford the title compound (228 mg, 98% yield, HCl) as a white solid.

(2004) Step E. 5-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-N,N,1-trimethyl-1H-pyrazole-3-carboxamide: To a solution of 5-(aminomethyl)-N,N,1-trimethylpyrazole-3-carboxamide (129 mg, 5.0 equiv, HCl) in DMSO (1.5 mL) were added K.sub.2CO.sub.3 (49.0 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg). The reaction was stirred at 25° C. for 0.5 hour before 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (70 mg, 1 equiv.) was added and the resulting mixture was stirred at 40° C. for 2 hours. The mixture was concentrated and the residue was purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: [water(0.1% formic acid)-ACN]; B %: 13%-43%, 9 min) to afford the title compound (12.3 mg, 13% yield, 93.1% purity, formic acid salt) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.4) δ=9.49 (br t, J=5.2 Hz, 1H), 9.34 (s, 1H), 8.18 (s, 1H), 7.76 (dd, J=6.0, 9.2 Hz, 1H), 7.40-7.28 (m, 2H), 7.00 (d, J=2.4 Hz, 1H), 6.59 (s, 1H), 5.42-5.15 (m, 1H), 4.93-4.75 (m, 2H), 4.19-4.05 (m, 2H), 3.98 (s, 3H), 3.28 (s, 31H), 3.09 (d, J=10.4 Hz, 2H), 3.01 (s, 1H), 2.95 (s, 3H), 2.87-2.78 (m, 1H), 2.38-2.28 (m, 11H), 2.18-1.95 (m, 4H), 1.88-1.72 (m, 3H), 0.71 (t, J=7.2 Hz, 3H). LCMS [M+1].sup.+ : m/z=675.3

Example 484

(2005) ##STR00849##

6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-6-azaspiro[3.5]nonane 2-oxide

(2006) ##STR00850##

(2007) Step A. 6-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-6-azaspiro[3.5]nonane 2-oxide: To a solution of NaIO.sub.4 (7.07 mg, 1.05 equiv.) in H.sub.2O (0.5 mL) was added 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2-thia-6-azaspiro[3.5]nonan-6-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (20.0 mg, 1.0 equiv.) at 0° C. Then MeOH (0.4 mL) and dioxane (0.3 mL) was added to the mixture. The reaction mixture was stirred at 0° C. for 12 hours. The residue was filtered, washed with MeOH (1 mL), and the filtrate was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH.sub.4HCO.sub.3)-ACN; B %: 32%-62%, 8 min] and lyophilized to afford the title compound (4.36 mg, 21% yield) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.06 (s, 1H), 7.68 (dd, J=6.0, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.42-5.17 (m, 1H), 4.43-4.19 (m, 4H), 4.08-3.96 (m, 2H), 3.73-3.62 (m, 2H), 3.27-3.12 (m, 5H), 3.06-2.97 (m, 1H), 2.55-2.43 (m, 1H), 2.41-2.20 (m, 2H), 2.20-2.12 (m, 2H), 2.04-1.83 (m, 7H), 0.80 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=652.4.

Example 485

(2008) ##STR00851##

3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)azetidin-2-one

(2009) ##STR00852##

(2010) Step A. 1-[tert-butyl(dimethyl)silyl]azetidin-2-one: azetidin-2-one (2.00 g, 1.0 equiv.) and tert-butylchlorodimethylsilane (4.88 g, 1.1 equiv.) were dissolved in dichloromethane (25 mL) at 20° C. under nitrogen atmosphere. A solution of diisopropylethylamine (5.46 g, 1.5 equiv.) in dichloromethane (15 mL) was added dropwise. The resulting mixture was stirred at 25° C. for 20 hours. The mixture was concentrated. The residue was diluted with petroleum ether/ethyl acetate (1:1, 100 mL) and the mixture was filtered. The filtrate was concentrated. The residue was purified by flash silica gel chromatography (ethyl acetate/petroleum ether 1:4) to afford the title compound (4.90 g, 94% yield) as a colorless oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ=3.21-3.17 (m, 2H),

3.09-3.04 (m, 2H), 0.95 (d, J=0.8 Hz, 9H), 0.23 (d, J=1.2 Hz, 6H); LCMS (ESI, M+1): m/z=186.2.

(2011) Step B. 3-(benzyloxymethyl)-1-[tert-butyl(dimethyl)silyl]azetidin-2-one: To a solution of ((chloromethoxy)methyl)benzene (1.86 g, 1.1 equiv.) in tetrahydrofuran (40 mL) was added lithium diisopropylamide (2.0 M in THF, 1.2 equiv.) at -70° C. under nitrogen atmosphere. After stirring at -70° C. for 0.5 hour, a solution of 1-[tert-butyl(dimethyl)silyl]azetidin-2-one (2.00 g, 1.0 equiv.) in tetrahydrofuran (8 mL) was added to above mixture at -70° C. The resulting mixture was stirred at -70° C. for 1 hour and then warmed up to 25° C. and stirred at 25° C. for 16 hours. The mixture was quenched with saturated ammonium chloride solution (150 mL) at 0° C. under nitrogen atmosphere and extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with brine, dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. The residue was purified by flash silica gel chromatography (petroleum ether/ethyl acetate 0 to 100%) to afford the title compound (1.8 g, 39% yield) as a colorless oil; LCMS (ESI, M+1): m/z=306.2.

(2012) Step C. 1-[tert-butyl(dimethyl)silyl]-3-(hydroxymethyl)azetidin-2-one: To a solution of 3-(benzyloxymethyl)-1-[tert-butyl(dimethyl)silyl]azetidin-2-one (2.06 g, 1.0 equiv.) in methanol (40 mL) was added palladium/carbon (2.00 g, 10% purity, wet) under nitrogen atmosphere. The suspension was degassed under vacuum and purged with hydrogen. The mixture was stirred under hydrogen atmosphere (15.0 psi) at 40° C. for 16 hours. The reaction mixture was filtered. The filter cake was washed with methanol (2×20 mL). The combined filtrate was concentrated in vacuum to afford the title compound (1.20 g, 66% yield) as a yellow oil; LCMS (ESI, M+1): m/z=216.1.

(2013) Step D. [1-[tert-butyl(dimethyl)silyl]-2-oxo-azetidin-3-yl]methyl 4-methylbenzenesulfonate: To a solution of 1-[tert-butyl(dimethyl)silyl]-3-(hydroxymethyl)azetidin-2-one (0.20 g, 1.0 equiv.) in dichloromethane (2 mL) was added 4-methylbenzenesulfonyl chloride (195 mg, 1.1 equiv.) and triethylamine (188 mg, 2.0 equiv.) at 0° C. The mixture was stirred at 25° C. for 16 hours. The mixture was poured into water (10 mL) and extracted with dichloromethane (3×10 mL). The combined organic phase was washed with brine (2×10 mL), dried over anhydrous sodium sulfate, filtered and concentrated in vacuum. Then the crude was purified by flash silica gel chromatography (petroleum ether/ethyl acetate 0 to 100%) to afford the title compound (130 mg, 37% yield) as a colorless oil; ¹H NMR (400 MHz, DMSO-d₆) δ=7.78 (m, 2H), 7.50 (m, 2H), 4.23 (m, 2H), 3.67-3.59 (m, 1H), 3.26 (m, 1H), 2.93 (m, 1H), 2.43 (s, 3H), 0.87 (s, 9H), 0.17 (s, 3H), 0.13 (s, 3H); LCMS (ESI, M+1): m/z=370.2.

(2014) Step E. 3-(azidomethyl)-1-(tert-butyl(dimethyl)silyl)azetidin-2-one: To a solution of [1-[tert-butyl(dimethyl)silyl]-2-oxo-azetidin-3-yl]methyl 4-methylbenzenesulfonate (130 mg, 1.0 equiv.) in dimethylformamide (2 mL) was added sodium azide (68.6 mg, 3.0 equiv.). The mixture was stirred at 60° C. for 16 hours. The mixture was poured into water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with brine (3×10 mL), dried over anhydrous sodium sulfate, filtered and concentrated to afford the title compound (80 mg, crude) as a colorless liquid; LCMS (ESI, M+1): m/z=240.1.

(2015) Step F. 3-(aminomethyl)-1-[tert-butyl(dimethyl)silyl]azetidin-2-one: To a solution of 3-(azidomethyl)-1-[tert-butyl(dimethyl)silyl]azetidin-2-one (80.0 mg, 1.0 equiv.) in methanol (1 mL) was added palladium/carbon (40.0 mg, 10% purity, wet) under nitrogen atmosphere. The suspension was degassed under vacuum and purged with hydrogen. The mixture was stirred under hydrogen atmosphere (15.0 psi) at 25° C. for 16 hours. The reaction mixture was filtered. The filter cake was washed with methanol (2×5 mL). The filtrate was concentrated in vacuum to afford the title compound (60 mg, crude in 1 mL N,N-dimethylformamide) as a white liquid; LCMS (ESI, M+1): m/z=215.1.

(2016) Step G. 3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)azetidin-2-one: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (30.0 mg, 1.0 equiv.) and 3-(aminomethyl)-1-[tert-

butyl(dimethyl)silyl]azetidin-2-one (21.7 mg, 2.0 equiv.) in N,N-dimethylformamide (0.5 mL) was added diisopropylethylamine (19.6 mg, 3.0 equiv.) and 4 Å molecular sieves (30 mg) at 25° C. The mixture was stirred at 40° C. for 2 hours. The reaction mixture was filtered and the filtrate was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM ammonium bicarbonate)/acetonitrile. B %: 34%-64%, 8 min] to afford the title compound (2.33 mg, 7.7% yield) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=9.95 (s, 1H), 9.32 (s, 1H), 9.20 (m, 1H), 7.92 (s, 1H), 7.77 (dd, J=6.0, 9.2 Hz, 1H), 7.40-7.31 (m, 2H), 7.00 (d, J=2.4 Hz, 1H), 5.46-5.24 (m, 1H), 4.25-4.10 (m, 2H), 4.03-3.94 (m, 1H), 3.78 (m, 1H), 3.60 (m, 1H), 3.14 (m, 2H), 2.94-2.75 (m, 2H), 2.44-2.23 (m, 2H), 2.22-2.04 (m, 4H), 1.91-1.77 (m, 3H), 1.24 (br s, 1H), 0.72 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=593.4.

Example 486

(2017) ##STR00853##

(1 R,5R,6S,7R)-3-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-(hydroxymethyl)-3-azabicyclo[3.2.1]octan-6-ol

(2018) ##STR00854## ##STR00855##

(2019) Step A. dimethyl 2-(5-methoxycarbonyl)pyridin-3-yl)malonate: A mixture of methyl 5-bromopyridine-3-carboxylate (25 g, 1.0 equiv.), dimethyl malonate (45.9 g, 3.0 equiv.), dicyclohexyl-[2-(2,4,6-triisopropylphenyl)phenyl]phosphane (11.0 g, 0.2 equiv.) and K.sub.3PO.sub.4 (73.7 g, 3.0 equiv.) in toluene (500 mL) was degassed and Pd.sub.2(dba).sub.3 (10.6 g, 11.57 mmol, 0.1 equiv.) was added. The mixture was stirred at 100° C. for 12 hours under N.sub.2 atmosphere. The mixture was diluted with water (100 mL), extracted with ethyl acetate (4×100 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash by reversed phase flash [C18, water 0 (0.1% formic acid)-ACN] to afford the title compound (25 g, 79% yield) as a yellow solid; LCMS (ESI, M+1): m/z=268.1.

(2020) Step B. methyl 5-(2-methoxy-2-oxoethyl)nicotinate: To a solution of dimethyl 2-(5-methoxycarbonyl-3-pyridyl)propanedioate (25 g, 1.0 equiv.) in DMSO (250 mL) was added LiCl (7.93 g, 2.0 equiv.) and H.sub.2O (1.69 g, 1.0 equiv.). The mixture was stirred at 120° C. for 12 hours. The mixture was diluted with water (50 mL), extracted with ethyl acetate (4×50 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by column chromatography (SiO.sub.2, Petroleum ether/Ethyl acetate 1:1) to afford the title compound (7 g, 34% yield) as a yellow solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=8.98 (d, J=1.6 Hz, 1H), 8.65 (d, J=2.0 Hz, 1H), 8.31-8.30 (m, 1H), 3.94 (s, 3H), 3.82 (s, 2H), 3.71 (s, 3H); LCMS (ESI, M+1): m/z=210.1.

(2021) Step C. 1-benzyl-3-(2-methoxy-2-oxoethyl)-5-(methoxycarbonyl)pyridin-1-ium: A mixture of methyl 5-(2-methoxy-2-oxoethyl)nicotinate (800 mg, 1.0 equiv.), TEA (1.06 g, 3.0 equiv.) in DCM (8 mL) was degassed and purged with N.sub.2 for 3 times, BnCl (1.79 g, 3.0 equiv.) was added into the mixture at -40° C. Then the mixture was stirred at -40° C. for 0.5 hour under N.sub.2 atmosphere. The mixture was diluted with water (20 mL), extracted with DCM (4×15 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated to afford the title compound (1 g, crude) as a yellow oil.

(2022) Step D. (3R, 5S)-methyl 1-benzyl-5-(2-methoxy-2-oxoethyl) piperidine-3-carboxylate: To a solution of methyl 1-benzyl-5-(2-methoxy-2-oxo-ethyl) pyridine-3-carboxylate (4 g, 1.0 equiv.) in MeOH (40 mL) was added PtO.sub.2 (1.19 g, 0.4 equiv.) under N.sub.2 atmosphere. The suspension was degassed and purged with H.sub.2. The mixture was stirred under H.sub.2 (50 Psi) at 40° C. for 12 hours. The reaction mixture was filtered and purified by prep-HPLC [column: Kromasil Eternity XT 250×80 mm×10 µm; mobile phase: water (0.1% ammonia hydroxide)-ACN; B %: 45° %6-75%, 20 min) and lyophilized to afford the title compound (1.2 g, 22% yield) as a yellow oil; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=7.33-7.23 (m, 5H), 3.65-3.60 (m,

6H), 3.60-3.49 (m, 2H), 3.12-3.04 (m, 1H), 2.93-2.91 (m, 1H), 2.69-2.59 (m, 1H), 2.08-2.07 (m, 2H), 2.05-2.04 (m, 2H), 1.95-1.63 (m, 1H), 1.68-1.63 (m, 1H), 1.14-1.01 (m, 1H); LCMS (ESI, M+1): m/z=306.0.

(2023) Step E. methyl 3-benzyl-7-oxo-3-azabicyclo [3.2.1] octane-6-carboxylate: To a solution of (3R, 5S)-methyl 1-benzyl-5-(2-methoxy-2-oxo-ethyl) piperidine-3-carboxylate (1.0 g, 1.0 equiv.) in THF (90 mL) was added t-BuOK (1.0 M, 9.82 mL, 3.0 equiv.). The mixture was stirred at 25° C. for 12 hours. The mixture was diluted with water (50 mL), extracted with ethyl acetate (4×30 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by column chromatography (SiO₂, Petroleum ether/Ethyl acetate 15:1 to 10:1) to afford the title compound (350 mg, 24% yield) as a yellow oil; LCMS (ESI, M+1): m/z=274.0.

(2024) Step F. (6S,7R)-3-benzyl-7-(hydroxymethyl-3-azabicyclo [3.2.1] octan-6-ol: To a solution of methyl 3-benzyl-7-oxo-3-azabicyclo [3,2,1]octane-6-carboxylate (350 mg, 1.0 equiv.) in THF (1 mL) was added LiAlH₄ (146 mg, 3.0 equiv.). The mixture was stirred at 0° C. for 2 hours. The mixture was diluted with water (20 mL), extracted with ethyl acetate (4×20 mL). The combined organic layers were dried over anhydrous sodium sulfate, concentrated and purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (0.1% ammonia hydroxide)-ACN; B %: 23%-53%, 9 min] to afford the title compound (200 mg, 63% yield) as a yellow oil; LCMS (ESI, M+1): m/z=248.0.

(2025) Step G. (6S,7R)-7-(hydroxymethyl)-3-azabicyclo[3.2.1]octan-6-ol: To a solution of 3-benzyl-7-(hydroxymethyl)-3-azabicyclo[3,2,1]octan-6-ol (200 mg, 1.0 equiv.) in MeOH (3 mL) was added Pd/C (50 mg, 1.0 equiv.) under N₂ atmosphere. The suspension was degassed and purged with H₂. The mixture was stirred under H₂ (15 psi) at 25° C. for 12 hours before being filtered and concentrated to afford the title compound (100 mg, 79% yield) as a white oil.

(2026) Step H. (1R,5R,6S,7R)-3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-(hydroxymethyl)-3-azabicyclo[3.2.1]octan-6-ol: To a solution of 7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (100 mg, 1.0 equiv.) and (1R,5R,6S,7R)-7-(hydroxymethyl)-3-azabicyclo[3,2,1]octan-6-ol (44.4 mg, 1.8 equiv.) in DMF (0.5 mL) was added DIEA (60.9 mg, 3.0 equiv.) and 4 Å molecular sieves (20 mg). The mixture was stirred at 40° C. for 12 hours. The reaction mixture was filtered and purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: water (0.1% formic acid), /ACN; B %: 15%-45%, 2 min] to afford the title compound (50 mg, 46% yield) as white solid.

(2027) Step I (1R,5R,6S,7R)-3-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-(hydroxymethyl)-3-azabicyclo[3.2.1]octan-6-ol: To a solution of (1R,5R,6S,7R)-3-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-(hydroxymethyl)-3-azabicyclo[3,2,1]octan-6-ol (50 mg, 1.0 equiv.) in MeOH (2 mL) was added HCl-MeOH (4 M, 2 mL, 111 equiv.). The mixture was stirred at 0° C. for 1 hour. The residue was filtered and the filtrate was purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (0.1% ammonia hydroxide)/ACN; B %: 28%-58%, 9 min] and lyophilized to afford the title compound (37.8 mg, 80% yield) as a yellow solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.31-9.18 (m, 1H), 7.68-7.65 (m, 1H), 7.29-7.26 (m, 1H), 7.24-7.22 (m, 1H), 7.07-7.04 (m, 1H), 5.40-5.21 (m, 1H), 5.07-4.95 (m, 1H), 4.83-4.75 (m, 1H), 4.39-4.15 (m, 2H), 3.90-3.68 (m, 2H), 3.68-3.58 (m, 1H), 3.57-3.41 (m, 2H), 3.29-3.11 (m, 3H), 3.19-2.99 (m, 1H), 2.51-2.30 (m, 1H), 2.38-2.07 (m, 6H), 2.05-1.74 (m, 6H), 0.86-0.72 (m, 3H); LCMS (ESI, M+1): m/z=650.4.

Example 487

(2028) ##STR00856##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-

pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)hexahydro-1H-pyrrolo[3,4-c]pyridine-1,3(2H)-dione

(2029) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.22 (s, 1H), 8.57 (s, 1H), 7.69 (dd, J=5.6, 9.2 Hz, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.08 (t, J=2.8 Hz, 1H), 5.43-5.21 (m, 1H), 4.75 (br dd, J=3.6, 13.6 Hz, 1H), 4.41-4.24 (m, 2H), 4.20-4.04 (m, 2H), 4.04-3.92 (m, 1H), 3.36 (br d, J=3.6 Hz, 1H), 3.32-3.23 (m, 3H), 3.15 (br d, J=1.6 Hz, 1H), 3.05-2.99 (m, 1H), 2.52-2.39 (m, 2H), 2.38-2.33 (m, 1H), 2.33-2.25 (m, 1H), 2.24-2.10 (m, 3H), 2.04-1.91 (m, 3H), 1.28-1.20 (m, 1H), 0.81 (q, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=647.

Example 488

(2030) ##STR00857##

(3R,3aR,6aS)-5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-(hydroxymethyl)hexahydropyrrolo[3,4-c]pyrrol-1(2H)-one

(2031) ##STR00858##

(2032) Step A. (S)-5-(((tert-butyldimethylsilyl)oxy)methyl)pyrrolidin-2-one: To a solution of (5S)-5-(hydroxymethyl)pyrrolidin-2-one (10.0 g, 1.0 equiv.) and imidazole (8.80 g, 1.5 equiv.) in dichloromethane (100 mL) was added tert-butyldimethylsilyl chloride (15.7 g, 1.2 equiv.) portion wise at 0-5° C. The resulting mixture was allowed to warm up to 25-30° C. and stirred for 12 hours. The mixture was diluted with water (150 mL) and the layers were separated. The aqueous phase was extracted with dichloromethane (2×50 mL). The combined organic layers were washed with brine, dried over sodium sulfate, filtered, and concentrated in vacuum. The residue was purified by flash silica gel chromatography [Ethyl acetate/Petroleum ether 50 to 100%] to afford the title compound (17.5 g, 83% yield) as a colorless liquid; ¹H NMR (400 MHz, CDCl₃-d₃) δ=5.98 (br s, 1H), 3.81-3.71 (m, 1H), 3.62 (dd, J=4.0, 10.0 Hz, 1H), 3.44 (dd, J=7.6, 10.0 Hz, 1H), 2.40-2.30 (m, 2H), 2.22-2.11 (m, 1H), 1.81-1.69 (m, 1H), 0.88 (s, 9H), 0.06 (s, 6H); LCMS (ESI, M+1): m/z=230.2.

(2033) Step B. tert-butyl (S)-2-(((tert-butyldimethylsilyl)oxy)methyl)-5-oxopyrrolidine-1-carboxylate: To a solution of (5S)-5-[[tert-butyl(dimethyl)silyl]oxymethyl]pyrrolidin-2-one (2.0 g, 1.0 equiv.) in dichloromethane (20 mL) was added tert-butyl dicarbonate (2.8 g, 1.5 equiv.), triethylamine (1.7 g, 2.0 equiv.) and 4-dimethylaminopyridine (106 mg, 0.1 equiv.). The resulting mixture was stirred at 25° C. for 12 hours. The mixture was concentrated in vacuum. The residue was purified by flash silica gel chromatography [Ethyl acetate/Petroleum ether 1:4] to afford the title compound (2.25 g, 78% yield) as a colorless oil; ¹H NMR (400 MHz, CDCl₃-d₃) δ=4.22-4.13 (m, 1H), 3.92 (dd, J=4.0, 10.4 Hz, 1H), 3.69 (dd, J=2.0, 10.4 Hz, 1H), 2.78-2.64 (m, 1H), 2.38 (ddd, J=2.0, 9.6, 17.6 Hz, 1H), 2.16-1.96 (m, 2H), 1.54 (s, 9H), 0.88 (s, 9H), 0.04 (d, J=5.2 Hz, 6H); LCMS (ESI, M-99): m/z=230.2.

(2034) Step C. tert-butyl (5S)-5-(((tert-butyldimethylsilyl)oxy methyl)-2-oxo-3-(phenylselanyl)pyrrolidine-1-carboxylate: To a solution of tert-butyl (2S)-2-[[tert-butyl(dimethyl)silyl]oxymethyl]-5-oxo-pyrrolidine-1-carboxylate (1.0 g, 1.0 equiv.) in tetrahydrofuran (25 mL) was added lithium hexamethyldisilazide (1 M, 3.3 mL, 1.1 equiv.) dropwise at -60° C. under nitrogen atmosphere. The solution was stirred at -60° C. for 0.5 hour before a solution of phenyl selenohypochlorite (1.0 g, 1.75 equiv.) in tetrahydrofuran (5.0 mL) was added at -60° C. The resulting mixture was stirred at -60 for additional 1 hour and then warmed to 25° C. and stirred at 25° C. for 12 hours. The mixture was quenched with saturated ammonium chloride (40 mL) at 0-5° C. under nitrogen, and allowed to warm to 25° C. and stirred for 0.5 hour. The mixture was extracted with ethyl acetate (3×25 mL). The combined organic layers were washed with brine, dried over sodium sulfate, filtered and concentrated in vacuum. The residue was purified by flash silica gel chromatography [Ethyl acetate/Petroleum ether 10 to 15%] to afford the title compound (740 mg, 50% yield) as a yellow oil; LCMS (ESI, M-99): m/z=386.2.

(2035) Step D. tert-butyl (S)-2-(((tert-butyldimethylsilyl)oxy)methyl)-5-oxo-2,5-dihydro-1H-pyrrole-1-carboxylate: To a solution of tert-butyl (5S)-5-[[tert-butyl(dimethyl)silyl]oxymethyl]-2-oxo-3-phenylselanyl-pyrrolidine-1-carboxylate (740 mg, 1.0 equiv.) in dichloromethane (10.0 mL) was added pyridine (362 mg, 3.0 equiv.) at -70°C ., followed by slow addition of hydrogen peroxide (606 mg, 30% in water, 3.5 equiv.). The resulting mixture was allowed to warm up to 25°C . and stirred at 25°C . for 12 hours. The mixture was diluted with water (20 mL) and extracted with dichloromethane (2×10 mL). The combined organic layers were washed with saturated sodium sulfite and brine, dried over sodium sulfate, filtered and concentrated in vacuum. The residue was purified by flash silica gel chromatography [Ethyl acetate/Petroleum ether 10 to 15%] to afford the title compound (230 mg, 34% yield) as a colorless oil; LCMS (ESI, M-99): $m/z=228.1$.

(2036) Step E. tert-butyl (1S,3aS,6aR)-5-benzyl-1-(((tert-butyldimethylsilyl)oxy)methyl)-3-oxohexahydropyrrolo[3,4-c]pyrrole-2(1H)-carboxylate: To a solution of tert-butyl (2S)-2-[[tert-butyl(dimethyl)silyl]oxymethyl]-5-oxo-2H-pyrrole-1-carboxylate (180 mg, 1.0 equiv.) and N-(methoxymethyl)-1-phenyl-N-(trimethylsilylmethyl)methanamine (391 mg, 3.0 equiv.) in dichloromethane (5.0 mL) was added TFA (25 mg, 0.4 equiv.) at 0°C . After addition, the resulting mixture was stirred at 25°C . for 12 hours. The mixture was concentrated in vacuum. The residue was purified by prep-HPLC [column: Phenomenex Luna C18 $75\times 30\text{ mm}\times 3\text{ }\mu\text{m}$; mobile phase: water (0.1% HCl)—CH₂sub.3CN, B %: 39%-59%, 6 min] to afford the title compound (102 mg, 37% yield) as a yellow solid; LCMS (ESI, M+1): $m/z=461.4$.

(2037) Step F. (3S,3aR,6aS)-5-benzyl-3-(hydroxymethyl)hexahydropyrrolo[3,4-c]pyrrol-1(2H)-one: To a solution of tert-butyl (3aS,6S,6aR)-2-benzyl-6-[[tert-butyl(dimethyl)silyl]oxymethyl]-4-oxo-3,3a,6,6a-tetrahydro-1H-pyrrolo[3,4-c]pyrrole-5-carboxylate (102 mg, 1 equiv.) in methanol (1.0 mL) was added HCl-MeOH (1.0 mL, 4M). The resulting mixture was stirred at 25°C . for 2 hours. The mixture was concentrated in vacuum to afford the title compound (65 mg, crude, HCl) as a white solid. LCMS (ESI, M+1): $m/z=247.1$.

(2038) Step G. (3S,3aR,6aS)-5-benzyl-3-(hydroxymethyl)hexahydropyrrolo[3,4-c]pyrrol-1(2H)-one: To a flask charged with Pd/C (20 mg, 10% purity) was added a solution of (1S,3aS,6aR)-5-benzyl-1-(hydroxymethyl)-1,2,3a,4,6,6a-hexahydropyrrolo[3,4-c]pyrrol-3-one (65 mg, 1.0 equiv, hydrochloride) in methanol (2.0 mL). The mixture was degassed and stirred at 70°C . under H₂ (50 psi) for 24 hours. The mixture was filtered and the filtrate was concentrated. The residue was diluted with acetonitrile/water (v/v, 1/2) and lyophilized directly to give the title compound (45 mg, crude, hydrochloride) as a white gum; LCMS (ESI, M+1): $m/z=157.2$.

(2039) Step H. (3R,3aR,6aS)-5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-(hydroxymethyl)hexahydropyrrolo[3,4-c]pyrrol-1(2H)-one: To a solution of (3aS,6S,6aR)-6-(hydroxymethyl)-2,3,3a,5,6,6a-hexahydro-1H-pyrrolo[3,4-c]pyrrol-4-one (32.5 mg, 2.5 equiv, hydrochloride) and 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (40 mg, 1.0 equiv.) in N,N-Dimethylformamide (3.0 mL) was added N,N-Diisopropylethylamine (34.9 mg, 4.0 equiv.) and 4 Å molecular sieves (80 mg). The resulting mixture was stirred at 40°C . for 12 hours. The filtrate was purified by prep-HPLC [column: Phenomenex Luna C18 $75\times 30\text{ mm}\times 3\text{ }\mu\text{m}$; mobile phase: water (0.1% HCl)—CH₂sub.3CN; B %: 19%-39%, 6 min] to afford the title compound (13.8 mg, 29% yield, hydrochloride) as yellow solid; ¹H NMR (400 MHz, MeOD-d₄) $\delta=9.48$ (br s, 1H), 7.73 (br dd, J=5.6, 8.8 Hz, 1H), 7.40 (s, 1H), 7.31 (br t, J=9.2 Hz, 1H), 7.17 (br s, 1H), 5.73-5.47 (m, 1H), 4.55 (br s, 6H), 4.30-3.81 (m, 4H), 3.70 (br s, 3H), 3.48 (br s, 2H), 2.88-2.11 (m, 8H), 0.86 (br s, 3H); LCMS (ESI, M+1): $m/z=649.4$.

Example 489

(2040) ##STR00859##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-

pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3,3-dimethylhexahydroimidazo[1,5-a]pyrazin-1(5H)-one

(2041) ##STR00860##

(2042) Step A. 3,3-dimethylhexahydroimidazo[1,5-a]pyrazin-1(5H)-one: To a solution of piperazine-2-carboxamide (300 mg, 1.0 equiv.) in acetone (3.0 mL) was added 2-methoxyethanol (265 mg, 1.50 equiv.). The mixture was stirred at 40° C. for 2 hours. After completion, the reaction mixture was cooled to 25° C., filtered and the filtrate was concentrated under reduced pressure to afford the title compound (170 mg, 43% yield) as a white solid; LCMS (ESI, M+1): m/z=170.1.

(2043) Step B. 7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3,3-dimethylhexahydroimidazo[1,5-a]pyrazin-1(5H)-one: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (60.0 mg, 1.0 equiv.) and 3,3-dimethyl-2,5,6,7,8,8a-hexahydroimidazo[1,5-a]pyrazin-1-one (85.7 mg, 5.0 equiv.) in DMF (0.50 mL) was added 4 Å molecular sieves (10 mg) and diisopropylethylamine (39.3 mg, 3.0 equiv.). The mixture was stirred at 60° C. for 12 hours. The reaction mixture was cooled to 25° C., filtered and the filtrate was partitioned between ethyl acetate (10 mL) and water (10 mL). The organic phase was separated, washed with saturated brine (2×5 mL), dried over sodium sulfate, filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: A: water (0.1% formic acid)-ACN; B %: 14%-44%, 10 min) to afford the title compound (2.88 mg, 4.23% yield) as an off-white solid; .sup.1H NMR (400 MHz, CD3OD) δ=9.16 (s, 1H), 7.72-7.65 (m, 1H), 7.32 (d, J=2.8 Hz, 1H), 7.26 (t, J=9.6 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.53-5.36 (m, 1H), 4.58 (br s, 3H), 4.55-4.44 (m, 2H), 3.70-3.64 (m, 1H), 3.58-3.50 (m, 2H), 3.46 (br s, 2H), 3.17 (br d, J=11.2 Hz, 1H), 2.85-2.76 (m, 1H), 2.59-1.99 (m, 9H), 1.46 (s, 3H), 1.31 (s, 3H), 0.83-0.76 (m, 3H); LCMS (ESI, M+1): m/z=662.4.

Example 490

(2044) ##STR00861##

(3R,6R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-3,6-diol

(2045) ##STR00862##

(2046) Step A. (3R,6R)-tert-butyl 3,6-dihydroazepine-1-carboxylate: To a solution of tert-butyl 3,6-dihydroxy-2,3,6,7-tetrahydroazepine-1-carboxylate (500 mg, 2.18 mmol, 1.0 equiv.) in MeOH (8 mL) was added Pd/C (100 mg, 10% purity) under Ar. The suspension was degassed and purged with H.sub.2. The mixture was stirred under H.sub.2 (50 psi) at 30° C. for 16 hours. The mixture was filtered and the filtrate was concentrated under vacuum to afford the title compound (500 mg, crude) as a yellow oil; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=3.84 (br s, 2H), 3.63 (br dd, J=4.8, 14.4 Hz, 1H), 3.53 (br dd, J=4.8, 14.2 Hz, 1H), 3.35 (s, 1H), 3.14 (br dd, J=7.6, 14.0 Hz, 1H), 2.02-1.89 (m, 2H), 1.74-1.59 (m, 1H), 1.47 (s, 9H), 1.45-1.38 (m, 1H).

(2047) Step B. (3R,6R)-azepane-3,6-diol: To a solution of tert-butyl 3,6-dihydroazepine-1-carboxylate (500 mg, 2.16 mmol, 1.0 equiv.) in MeCN (3 mL) was added HCl.Math.dioxane (4 M, 6 mL) at 0° C. The reaction mixture was stirred at 0° C. for 1 hour. The mixture was concentrated in vacuum to afford the title compound (283 mg, crude, HCl) as a yellow oil.

(2048) Step C. (3R,6R)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-3,6-diol: A suspension of azepane-3,6-diol (111 mg, 660 µmol, 3.9 equiv, HCl), DIEA (109 mg, 844 µmol, 5.0 equiv.) and 4 Å molecular sieves (10 mg) in DMF (0.5 mL) was stirred at 25° C. for 30 minutes. Then 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (100 mg, 169 µmol, 1.0 equiv.) was added into the mixture and the mixture was stirred at 40° C. for 72 hours.

The reaction mixture was diluted with ethyl acetate (5 mL) and water (10 mL). The mixture was extracted with ethyl acetate (5 mL) and the combined organic phase was washed with brine (5 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by column chromatography (SiO₂, petroleum ether/ethyl acetate 3:1 then Dichloromethane: Methanol 5:1). Then the mixture was re-purified by prep-HPLC [column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH₄HCO₃)-ACN, B %: 34%-54%, 8 min] to afford the title compound (5.8 mg, 5.47% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.57-9.43 (m, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.05 (s, 1H), 5.41-5.20 (m, 1H), 4.38-4.19 (m, 6H), 3.94-3.83 (m, 2H), 3.29-3.16 (m, 3H), 3.07-2.97 (m, 1H), 2.56-2.42 (m, 1H), 2.40-2.06 (m, 6H), 2.05-1.86 (m, 3H), 1.70-1.58 (m, 2H), 0.84-0.76 (m, 3H); 1.53-1.31 (m, 1H); LCMS (ESI, M+1): m/z=624.4.

Example 491

(2049) ##STR00863##

(3R,6S)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-3,6-diol

(2050) ##STR00864##

(2051) Step A. (3R,6S)-tert-butyl 3-((tert-butyldiphenylsilyl)oxy)-6-hydroxyazepane-1-carboxylate: To a mixture of Pd/C (20 mg, 10% purity) in MeOH (5 mL) was added (3R,6S)-tert-butyl 3-((tert-butyldiphenylsilyl)oxy)-6-hydroxy-2,3,6,7-tetrahydro-1H-azepine-1-carboxylate (0.20 g, 1.0 equiv.). The mixture was degassed and purged with H₂ before being stirred at 25° C. for 16 hours (50 Psi). The mixture was filtered and concentrated to afford the title compound (165 mg, 82% yield) as a white solid. LCMS (ESI, M+1): m/z=470.2.

(2052) Step B. (3R,6S)-azepane-3,6-diol: To a solution of (3R,6S)-tert-butyl 3-((tert-butyldiphenylsilyl)oxy)-6-hydroxyazepane-1-carboxylate (160 mg, 1.0 equiv.) in ACN (2 mL) was added HCl-dioxane (4 M, 2.0 mL) at 0° C. The solution was stirred at 0° C. for 0.5 hour. The mixture was filtered and triturated with ethyl acetate (2 mL) at 25° C. for 0.5 hour to afford the title compound (40 mg, 70% yield) as a white solid.

(2053) Step C. (3R,6S)-1-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)azepane-3,6-diol: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (150 mg, 1.0 equiv.) and (3R,6S)-azepane-3,6-diol (39.8 mg, 1.2 equiv.) in DMF (0.3 mL) were added DIEA (98.1 mg, 132 µL, 3.0 equiv.) and 4 Å molecular sieves (20 mg). The mixture was stirred at 60° C. for 12 hours. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (15.8 mg, 10% yield) as a white solid. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.37 (s, 1H), 7.67 (dd, J=5.6, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.06 (d, J=2.4 Hz, 1H), 5.30 (d, J=54.4 Hz, 1H), 4.56-4.40 (m, 2H), 4.37-4.13 (m, 4H), 3.92-3.83 (m, 2H), 3.29-3.16 (m, 3H), 3.05-2.88 (m, 1H), 2.56-2.42 (m, 1H), 2.39-2.10 (m, 4H), 2.07-1.88 (m, 5H), 1.86-1.74 (m, 2H), 0.80 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=624.1.

Example 492

(2054) ##STR00865##

(3S,4S)-4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)tetrahydrofuran-3-ol

Example 493

(2055) ##STR00866##

(3R,4R)-4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-

yl)amino)methyl)tetrahydrofuran-3-ol

(2056) ##STR00867##

(2057) Step A. 4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)tetrahydrofuran-3-ol: To a solution of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (200 mg, 337 μ mol, 1 equiv.) in DMF (4 mL) was added 4 Å molecular sieves (50 mg), cis-4-(aminomethyl)tetrahydrofuran-3-ol (79.1 mg, 2.0 equiv.) and DIPEA (175 mg, 1.35 mmol, 4.0 equiv.). The mixture was stirred at 40° C. for 2 hours and concentrated. The residue was purified by prep-HPLC (column: Phenomenex luna C18 150×25 mm×10 μ m; mobile phase: A: water (0.1% formic acid), -ACN; B %: 12/0-42%, 9 min) to afford the title compound (120 mg, 51% yield) as a white solid; LCMS(ESI, M+1): m/z=610.2.

(2058) Step B. (3S,4S)-4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)tetrahydrofuran-3-ol: The mixture of 4-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)tetrahydrofuran-3-ol (120 mg, 196.84 μ mol, 1.0 equiv.) was separated by chiral SFC (column: DAICEL CHIRALCEL OJ (250 mm×30 mm, 10 μ m); mobile phase: A: [0.1% NH₃.sub.3.Math.H.sub.2O, B: IPA; B %: 25%-25%, 10.6; 159 min) to afford Example 492, second eluting peak (33.4 mg, 24% yield, HCOOH) as a yellow solid; ¹H NMR (400 MHz, DMSO-d₆) δ =9.94 (s, 1H), 9.31 (s, 1H), 8.98-8.94 (m, 1H), 7.78-7.74 (m, 1H), 7.37-7.32 (m, 2H), 6.99 (s, 1H), 5.35-5.16 (m, 2H), 4.31 (s, 1H), 4.16 (d, J=10.4 Hz, 1H), 4.07 (d, J=10.4 Hz, 1H), 3.91-3.74 (m, 3H), 3.64-3.54 (m, 3H), 3.14-3.08 (m, 2H), 3.02 (s, 1H), 2.85-2.79 (m, 1H), 2.59-2.54 (m, 1H), 2.37-2.32 (m, 1H), 2.19-2.06 (m, 3H), 2.04-1.91 (m, 2H), 1.85-1.77 (m, 3H), 1.74 (s, 1H), 0.71 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=610.2, and Example 493, first eluting peak (35.7 mg, 26% yield, HCOOH) as a yellow solid; ¹H NMR (400 MHz, DMSO-d₆) δ =9.96 (s, 1H), 9.31 (s, 1H), 8.96 (d, J=4.4 Hz, 1H), 8.16 (s, 1H), 7.76 (t, J=6.8 Hz, 1H), 7.36-7.32 (m, 2H), 6.99 (s, 1H), 5.34-5.16 (m, 2H), 4.31 (s, 1H), 4.15 (d, J=10.4 Hz, 1H), 4.07 (d, J=10.4 Hz, 1H), 3.91-3.83 (m, 2H), 3.80-3.74 (m, 1H), 3.64-3.55 (m, 4H), 3.14-3.02 (m, 1H), 2.83-2.82 (m, 2H), 2.33 (s, 1H), 2.19-1.94 (m, 4H), 1.90-1.77 (m, 3H), 0.71 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=610.2.

Example 494

(2059) ##STR00868##

3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1,2,5-thiadiazolidine 1,1-dioxide

(2060) ##STR00869##

(2061) Step A. tert-butyl (2-amino-2-cyanoethyl)carbamate: To a solution of tert-butyl (2-oxoethyl)carbamate (1.00 g, 1.0 equiv.) in methanol (10 mL) were added Ti(i-PrO)₄ (1.79 g, 1.0 equiv.) and NH₃.sub.3 in methanol (7.00 M, 2.0 equiv.). TMSCN (1.25 g, 2.0 equiv.) was added dropwise to the mixture at 0° C. The mixture was stirred at 25° C. for 16 hours. The mixture was diluted with water (50 mL) and filtered. The filtrate was extracted with ethyl acetate (3×100 mL). The combined organic layers were dried over Na₂SO₄ and concentrated to afford the title compound (1.14 g, 83% yield) as a yellow oil; ¹H NMR (400 MHz, dimethylsulfoxide-d₆) δ =7.14 (br t, J=5.6 Hz, 1H), 3.75 (br s, 1H), 3.21-3.06 (m, 2H), 2.36 (br s, 2H), 1.38 (s, 9H).

(2062) Step B. tert-butyl (2,3-diaminopropyl)carbamate: A mixture of tert-butyl (2-amino-2-cyanoethyl)carbamate (500 mg, 1.0 equiv.) and Raney Ni (694 mg, 3.0 equiv.) in MeOH (5 mL) and NH₃.sub.3 in methanol (7.00 M, 1 mL) was degassed and purged with H₂ for 3 times. The mixture was stirred at 25° C. for 16 hours under H₂ atmosphere (15 psi). The reaction mixture

was filtered and the filtrate was concentrated to afford the title compound (500 mg, crude) as a yellow oil; ¹H NMR (400 MHz, dimethylsulfoxide-*d*.sub.6) δ=6.71 (br s, 1H), 2.94-2.84 (m, 1H), 2.82-2.73 (m, 1H), 2.57 (br s, 1H), 2.45 (br d, J=6.2 Hz, 1H), 2.35-2.23 (m, 1H), 1.37 (s, 9H). (2063) Step C. tert-butyl ((1,1-dioxido-1,2,5-thiadiazolidin-3-yl)methyl)carbamate: To a solution of tert-butyl (2,3-diaminopropyl)carbamate (400 mg, 1.0 equiv.) in pyridine (4 mL) was added dropwise sulfamide (203 mg, 1.0 equiv.) at 25° C. The resulting mixture was stirred at 115° C. for 20 hours. The reaction mixture was filtered and the filtrate was concentrated. The residue was purified by prep-TLC (dichloromethane/methanol 10:1) to afford the title compound (100 mg, 16% yield over two steps) as a yellow oil; ¹H NMR (400 MHz, dimethylsulfoxide-*d*.sub.4) δ=6.96-6.85 (m, 3H), 3.65-3.55 (m, 1H), 3.33-3.30 (m, 1H), 3.09-2.97 (m, 3H), 1.38 (s, 9H).

(2064) Step D. 3-(aminomethyl)-1,2,5-thiadiazolidine 1,1-dioxide: To a solution of tert-butyl ((1,1-dioxido-1,2,5-thiadiazolidin-3-yl)methyl)carbamate (41.0 mg, 1.0 equiv.) in DCM (0.50 mL) was added HCl-dioxane (4 M, 12.6 equiv.) at 0° C. The mixture was stirred at 25° C. for 1 hour. The reaction mixture was concentrated in vacuum to afford the title compound (30.0 mg, 98% yield, HCl) as a brown solid.

(2065) Step E. 3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)amino)methyl)-1,2,5-thiadiazolidine 1,1-dioxide: To a mixture of 3-(aminomethyl)-1,2,5-thiadiazolidine 1,1-dioxide (30.0 mg, 1.0 equiv, HCl), 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (47.4 mg, 1.0 equiv.) in DMF (0.5 mL) were added DIEA (31.0 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg). The mixture was stirred at 60° C. for 24 hours. The mixture was filtered and purified by prep-HPLC [column: Waters xbridge 150×25 mm×10 µm; mobile phase: water (10 mM NH₄HCO₃), B: ACN, B %: 29%-59%, 8 min] to afford the title compound (10.4 mg, 19% yield) as a white solid; ¹H NMR (400 MHz, dimethylsulfoxide-*d*.sub.4) δ=9.94 (s, 1H), 9.33 (s, 1H), 9.13 (br d, J=2.4 Hz, 1H), 7.77 (dd, J=6.4, 9.2 Hz, 1H), 7.39-7.28 (m, 2H), 7.26-7.16 (m, 1H), 7.05 (br t, J=7.6 Hz, 1H), 6.99 (d, J=2.4 Hz, 1H), 5.48-5.18 (m, 1H), 4.31-3.97 (m, 3H), 3.76-3.59 (m, 2H), 3.57-3.45 (m, 1H), 3.22 (br dd, J=5.2, 11.2 Hz, 5H), 2.97-2.84 (m, 1H), 2.27-2.17 (m, 1H), 2.16-2.00 (m, 3H), 1.97-1.72 (m, 3H), 0.71 (br t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=644.0.

Example 495

(2066) ##STR00870##

3-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)methyl)amino)methyl)-1,2,5-thiadiazolidine 1,1-dioxide

(2067) The title compound was synthesized according to the procedure described for example 494 using tert-butyl methyl(2-oxoethyl)carbamate instead of tert-butyl (2-oxoethyl)carbamate in step A. ¹H NMR (400 MHz, dimethylsulfoxide-*d*.sub.4) δ=9.95-9.87 (m, 1H), 9.35-9.24 (m, 1H), 7.81-7.70 (m, 1H), 7.40-7.27 (m, 2H), 7.25-7.15 (m, 1H), 7.08-7.02 (m, 1H), 7.02-6.98 (m, 1H), 5.42-5.17 (m, 1H), 4.30-3.99 (m, 4H), 3.98-3.82 (m, 1H), 3.67-3.58 (m, 3H), 3.57-3.45 (m, 1H), 3.23-2.96 (m, 5H), 2.94-2.77 (m, 1H), 2.19-2.02 (m, 4H), 1.91-1.75 (m, 3H), 0.78-0.70 (m, 3H); LCMS (ESI, M+1): m/z=658.2.

Example 496

(2068) ##STR00871##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(6-(hydroxymethyl)-6-methyl-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(2069) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-*d*.sub.4) δ=9.19 (d, J=5.3 Hz, 1H), 7.67 (dd, J=6.0, 8.8 Hz, 1H), 7.30 (d, J=2.8 Hz, 1H), 7.24 (t, J=9.2 Hz, 1H), 7.06 (s, 1H), 5.43-5.25 (m, 1H), 4.49-4.23 (m, 4H), 4.22-3.98 (m, 4H), 3.77-3.65 (m, 1H), 3.59-3.45 (m, 2H), 3.44-3.34 (m, 2H), 3.29-3.19

(m, 1H), 3.11-3.02 (m, 1H), 2.58-2.34 (m, 2H), 2.34-2.09 (m, 4H), 2.08-1.98 (m, 2H), 1.97-1.87 (m, 1H), 0.98 (br d, J=12.4 Hz, 3H), 0.84-0.74 (m, 3H); LCMS (ESI, M+1): m/z=638.2

Example 497

(2070) ##STR00872##

(2R,6R)-4-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-(hydroxymethyl)-6-methyl-1,4-oxazepan-6-ol

(2071) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, CDCl₃) δ=9.67 (d, J=5.6 Hz, 1H), 7.69-7.66 (m, 1H), 7.30 (t, J=2.4 Hz, 1H), 7.25 (t, J=9.6 Hz, 1H), 7.08-7.01 (m, 1H), 5.49-5.35 (m, 1H), 4.54 (d, J=11.2 Hz, 1H), 4.44 (d, J=11.2 Hz, 1H), 4.40-4.35 (m, 1H), 4.32 (d, J=14.8 Hz, 1H), 4.18-4.02 (m, 3H), 3.86-3.82 (m, 1H), 3.74-3.64 (m, 3H), 3.55-3.49 (m, 2H), 3.25-3.18 (m, 1H), 2.56-2.11 (m, 8H), 2.06-2.00 (m, 1H), 1.25 (d, J=4.0 Hz, 3H), 0.84-0.75 (m, 3H); LCMS (ESI, M+1): m/z=654.4.

Example 498

(2072) ##STR00873##

4-(4-((S)-1,1-difluoro-5-oxa-8-azaspiro[2.6]nonan-8-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(2073) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.95 (s, 1H), 9.15 (d, J=6.4 Hz, 1H), 8.22 (s, 1H), 7.77 (m, 1H), 7.40-7.26 (m, 2H), 7.05-6.93 (m, 1H), 5.41-5.17 (m, 1H), 4.40-4.28 (m, 2H), 4.26-4.18 (m, 1H), 4.18-4.10 (m, 1H), 4.10-4.02 (m, 2H), 4.02-3.91 (m, 2H), 3.86-3.73 (m, 2H), 3.15-3.03 (m, 2H), 3.01 (s, 1H), 2.88-2.77 (m, 1H), 2.40-2.34 (m, 1H), 2.21-2.09 (m, 2H), 2.05 (m, 1H), 1.99 (m, 1H), 1.90-1.68 (m, 5H), 0.73 (m, 3H); LCMS (ESI, M+1): m/z=656.1.

Example 499

(2074) ##STR00874##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(6-methoxy-1,4-oxazepan-4-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(2075) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=10.04-9.84 (m, 1H), 9.23 (d, J=2.4 Hz, 1H), 8.15 (s, 1H), 7.77 (dd, J=6.0, 9.2 Hz, 1H), 7.42-7.25 (m, 2H), 7.02 (dd, J=2.0, 5.6 Hz, 1H), 5.40-5.16 (m, 1H), 4.50-4.32 (m, 1H), 4.24-3.98 (m, 6H), 3.97-3.88 (m, 1H), 3.87-3.76 (m, 2H), 3.76-3.64 (m, 1H), 3.38 (d, J=2.0 Hz, 3H), 3.14-3.05 (m, 2H), 3.02 (s, 1H), 2.87-2.78 (m, 1H), 2.21-1.93 (m, 5H), 1.90-1.75 (m, 3H), 0.73 (d, J=3.2 Hz, 3H); LCMS (ESI, M+1): m/z=624.2.

Example 500

(2076) ##STR00875##

5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(3,8-dioxa-11-azaspiro[5.6]dodecan-11-yl)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol

(2077) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.94 (s, 1H), 9.19 (s, 1H), 8.15 (s, 1H), 7.76 (m, 1H), 7.48-7.18 (m, 2H), 7.03 (d, J=2.4 Hz, 1H), 5.29 (d, J=54.0 Hz, 1H), 4.55 (d, J=14.0 Hz, 1H), 4.26-4.12 (m, 2H), 4.11-4.02 (m, 4H), 3.68-3.47 (m, 6H), 3.18-3.00 (m, 3H), 2.86-2.80 (m, 1H), 2.54 (s, 1H), 2.41-2.34 (m, 1H), 2.20-1.94 (m, 4H), 1.89-1.89 (m, 3H), 1.54-1.27 (m, 4H), 0.72 (t, J=7.38 Hz, 3H); LCMS (ESI, M+1): m/z=664.3.

Example 501

(2078) ##STR00876##

4-(4-(2,2-dimethyl-1,4-oxazepan-4-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(2079) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=10.05-9.87 (m, 1H), 9.10 (s, 1H), 8.24 (s, 1H), 7.76 (dd, J=5.2, 8.4 Hz, 1H), 7.39-7.29 (m, 2H), 7.03 (d, J=2.0 Hz, 1H), 5.21 (s, 1H), 4.45-4.31 (m, 1H),

4.19-4.00 (m, 4H), 4.00-3.92 (m, 1H), 3.88-3.77 (m, 2H), 3.08 (d, J=6.8 Hz, 2H), 3.00 (s, 1H), 2.86-2.79 (m, 1H), 2.67 (s, 1H), 2.33 (s, 2H), 2.17-1.96 (m, 7H), 1.89-1.75 (m, 3H), 1.21-1.08 (m, 6H), 0.73 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=622.2.

Example 502

(2080) ##STR00877##

4-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1-oxa-9-thia-4-azaspiro[5.5]undecane 9,9-dioxide

(2081) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=9.16 (s, 1H), 8.18 (s, 1H), 7.79-7.76 (m, 1H), 7.37 (d, J=9.6 Hz, 1H), 7.34 (d, J=2.4 Hz, 1H), 7.02 (d, J=2.4 Hz, 1H), 5.36-5.22 (m, 1H), 4.18-4.15 (m, 1H), 4.10-4.07 (m, 1H), 4.02-4.00 (m, 2H), 3.96-3.90 (m, 4H), 3.17-3.14 (m, 2H), 3.11 (s, 1H), 3.07 (t, J=3.2 Hz, 1H), 3.02 (s, 1H), 2.86-2.81 (m, 1H), 2.54 (s, 1H), 2.53-2.52 (m, 1H), 2.37-2.33 (m, 3H), 2.15-2.01 (m, 6H), 1.87-1.77 (m, 3H), 0.73 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=698.3.

Example 503

(2082) ##STR00878##

4-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-9-imino-1-oxa-9-thia-4-azaspiro[5.5]undecane 9-oxide

(2083) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, DMSO-d₆) δ=10.08-9.84 (s, 1H), 9.22-9.06 (s, 1H), 7.79-7.75 (m, 1H), 7.40-7.30 (m, 2H), 7.01 (d, J=2.4 Hz, 1H), 5.28 (d, J=64.0 Hz, 1H), 4.21-4.12 (m, 1H), 4.10-4.05 (m, 1H), 4.04-3.95 (m, 2H), 3.95-3.84 (m, 4H), 3.53 (s, 1H), 3.11-3.00 (m, 5H), 2.95 (d, J=12.8 Hz, 2H), 2.86-2.78 (m, 1H), 2.26 (d, J=12.4 Hz, 2H), 2.18-2.10 (m, 2H), 2.10-1.94 (m, 5H), 1.87-1.74 (m, 3H), 0.72 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): m/z=697.7.

Example 504

(2084) ##STR00879##

3-chloro-5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(2085) ##STR00880##

(2086) Step A. tert-butyl 3-chloro-2-(dimethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate: To a solution of tert-butyl 2-(dimethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (1.2 g, 1.0 equiv.) in DMF (12 mL) was added NCS (779 mg, 1.5 equiv.) at 0° C. The mixture was stirred at 55° C. for 1 hour. The mixture was diluted with H₂O (30 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with brine (3×10 mL), dried over anhydrous Na₂SO₄, concentrated and purified by column chromatography [SiO₂, petroleum ether/ethyl acetate 10:1 to 1:1] to afford the title compound (740 mg, 51% yield) as a light yellow oil; ¹H NMR (400 MHz, CDCl₃) δ=4.50 (s, 2H), 4.45-4.35 (m, 2H), 3.75 (s, 2H), 3.10 (s, 6H), 1.97 (s, 2H), 1.44 (s, 9H); LCMS (ESI, M+1): m/z=343.4.

(2087) Step B. 3-chloro-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: A mixture of tert-butyl 3-chloro-2-(dimethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (100 mg, 292 μmol, 1.0 equiv.) in HCl-dioxane (1.0 mL) was stirred at 25° C. for 1 hour. The mixture concentrated to afford the title compound (45 mg, 50% yield, HCl) as a white solid, LCMS (ESI, M+1): m/z=243.0.

(2088) Step C: 3-chloro-547-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: A mixture of 3-chloro-N,N-

dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (45 mg, 3.5 equiv, HCl), 4 Å molecular sieve (100 mg) and DIEA (16.4 mg, 3.0 equiv.) in DMF (0.05 mL) was stirred at 25° C. for 30 minutes. 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (25 mg, 1.0 equiv.) was added and the mixture was stirred at 40° C. for 36 hours. The mixture was filtered and the filtrate was purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: A: water (0.1% formic acid)-ACN; B %: 15%-45%, 9 minutes] to afford the title compound (8.52 mg, 24% yield, formic acid salt) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=9.93 (s, 1H), 9.16 (s, 1H), 8.13 (s, 1H), 7.77 (dd, J=9.2 Hz, 6.0 Hz, 1H), 7.39-7.30 (m, 2H), 7.00 (d, J=2.4 Hz, 1H), 5.45-5.35 (m, 1H), 5.31-5.09 (m, 3H), 4.47 (d, J=6.8 Hz, 2H), 4.40-4.30 (m, 2H), 4.25-4.06 (m, 2H), 3.21-3.04 (m, 3H), 2.99 (d, J=15.2 Hz, 6H), 2.96-2.72 (m, 2H), 2.42-2.37 (m, 2H), 2.17-1.77 (m, 8H), 0.72 (t, J=7.6 Hz, 3H); LCMS (ESI, M+1): m/z=735.2.

Example 505

(2089) ##STR00881##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-fluoro-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(2090) ##STR00882##

(2091) Step A. tert-butyl 2-(dimethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (1.0 g, 1.0 equiv.) in Me.sub.2NH (2 M in THF, 8.89 mL, 5.0 equiv.) was added DIEA (689 mg, 1.5 equiv.) and HATU (2.03 g, 1.5 equiv.). The mixture was stirred at 25° C. for 12 hours. The mixture was diluted with H.sub.2O (4 mL) and filtered. The filtrate was extracted with ethyl acetate (3×5 mL). The combined organic layer was washed with brine (5 mL), dried over anhydrous sodium sulfate, concentrated and purified by reversed phase flash [water (0.1% formic acid)/acetonitrile] to afford the title compound (981 mg, 89% yield) as a yellow oil; .sup.1H NMR (400 MHz, CDCl.sub.3) δ=6.67-6.39 (m, 1H), 4.61-4.31 (m, 4H), 3.80-3.60 (m, 2H), 3.45-3.20 (m, 3H), 3.18-2.93 (m, 3H), 1.99-1.88 (m, 2H), 1.41 (s, 9H); LCMS (ESI, M+1): m/z=308.9.

(2092) Step B. tert-butyl 2-(dimethylcarbamoyl)-3-fluoro-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate: To a solution of tert-butyl 2-(dimethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (760 mg, 1.0 equiv.) in ACN (10 mL) was added 1-chloromethyl-4-fluoro-1,4-diazoniabicyclo[2.2.2]octane bis(tetrafluoroborate) (4.37 g, 5.0 equiv.). The mixture was stirred at 25° C. for 16 hours. The mixture was concentrated and purified by prep-HPLC [column: Phenomenex luna C18 200×40 mm×10 µm; mobile phase: A: water (0.1% formic acid)-MeCN; B %: 25%-55%, 10 minutes] to afford the title compound (107 mg, 13% yield) as a yellow oil; LCMS (ESI, M+1): m/z=326.9.

(2093) Step C. 3-fluoro-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of tert-butyl 2-(dimethylcarbamoyl)-3-fluoro-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (56 mg, 1.0 equiv.) in DCM (0.25 mL) and TFA (0.25 mL) was stirred at 25° C. for 1 hour. The mixture was concentrated to afford the title compound (91 mg, crude, TFA) as a yellow oil; LCMS (ESI, M+1): 226.9.

(2094) Step D: 5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-fluoro-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: A mixture of 5-ethyl-6-fluoro-4-(8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidin-7-yl)naphthalen-2-ol (31.7 mg, 1.0 equiv.), 3-fluoro-N,N-dimethyl-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (91 mg, crude), 4 Å molecular sieve (10 mg) and K.sub.2CO.sub.3 (37.0 mg, 5.0 equiv.) in DMSO (0.5 mL) was stirred

at 40° C. for 2 hours under N.sub.2 atmosphere. The mixture was filtered and the filtrate was purified by prep-HPLC [column: Phenomenex luna C18 100×30 mm×5 µm; mobile phase: A: water (0.1% formic acid)-MeCN; B %: 10%-40%, 10 minutes] to afford the title compound (8.89 mg, 20% yield, formic acid salt) as a white solid; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ=9.94 (s, 1H), 9.18 (s, 1H), 8.15-8.12 (m, 1H), 7.77 (m, 1H), 7.40-7.29 (m, 2H), 7.00 (d, J=2.4 Hz, 1H), 5.47-5.26 (m, 1H), 5.26-5.19 (m, 1H), 5.18-5.09 (m, 1H), 4.46 (d, J=6.0 Hz, 4H), 4.29-3.95 (m, 2H), 3.14-3.09 (m, 3H), 2.98-2.91 (m, 3H), 2.42-2.28 (m, 4H), 2.27-1.53 (m, 9H), 0.71 (t, J=7.2 Hz, 3H); LCMS (ESI, M+1): 719.1.

Example 506

(2095) ##STR00883##

5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide

(2096) ##STR00884## ##STR00885##

(2097) Step A. tert-butyl 2-((2-hydroxyethyl)carbamoyl)-7,8-dihydro-4H-pyrazolo[1,5-a][1,4]diazepine-5(6H)-carboxylate: To a solution of 5-tert-butoxycarbonyl-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxylic acid (200 mg, 1.0 equiv.) and TEA (108 mg, 1.5 equiv.) in DMF (2 mL) was added HATU (405 mg, 1.5 equiv.) and 2-aminoethanol (174 mg, 4.0 equiv.). The reaction was stirred at 20° C. for 0.5 hour. The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-TLC (SiO.sub.2, Petroleum ether/Ethyl acetate 1:1) to afford the title compound (160 mg, 69% yield) as a colorless oil; LCMS (ESI, M+1): m/z=325.1

(2098) Step B. N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: A solution of tert-butyl 2-(2-hydroxyethylcarbamoyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-5-carboxylate (160 mg, 1.0 equiv.) in HCl.dioxane (2 M, 2 mL, 8.1 equiv.) was stirred at 20° C. for 0.5 hour. The reaction mixture filtered, concentrated under reduced pressure to give a residue. The residue was purified by prep-H PLC (column: Waters Xbridge 150×25 mm×5 µm; mobile phase: A: water (10 mM NH.sub.4HCO.sub.3)-ACN; B %: 1%-10%, 8 minutes) to afford the title compound (70.0 mg, 63% yield) as a white solid; LCMS (ESI, M+1): m/z=225.2.

(2099) Step C. 5-(2-chloro-7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (70.0 mg, 1.0 equiv.) in DMF (1 mL) were added 2,4-dichloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidine (141 mg, 1.0 equiv.) and DIEA (121 mg, 3.0 equiv.), and then the mixture was stirred at -40° C. for 10 minutes. The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The crude product was purified by reversed phase flash chromatography (C18, water (0.1% formic acid)-ACN) to afford the title compound (60.0 mg, 30% yield) as a white solid; LCMS (ESI, M+1): m/z=638.3.

(2100) Step D. 5-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of 5-[2-chloro-7-[8-ethyl-7-fluoro-3-(methoxymethoxy)-1-naphthyl]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-N-(2-hydroxyethyl)-4,6,7,8-tetrahydropyrazolo[1,5-a][1,4]diazepine-2-carboxamide (30.0 mg, 1.0 equiv.) in THF (1.0 mL) were added ((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methanol (11.0 mg, 1.5 equiv.), DIEA (18.0 mg, 3.0 equiv.) and 4 Å molecular sieves (10 mg). The mixture was stirred at 50° C. for 3 hours. The reaction mixture was concentrated under reduced pressure to afford the title compound as a colorless oil. LCMS (ESI, M+1): m/z=761.5.

(2101) Step E. 5-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-

fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide: To a solution of 5-(7-(8-ethyl-7-fluoro-3-(methoxymethoxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-N-(2-hydroxyethyl)-5,6,7,8-tetrahydro-4H-pyrazolo[1,5-a][1,4]diazepine-2-carboxamide (25.0 mg, 1.0 equiv.) in MeOH (1 mL) was added HCl-MeOH (4 mol/L, 1.0 mL, 122 equiv.). The mixture was stirred at 0° C. for 0.5 hour. The reaction mixture was filtered and concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (column: Waters Xbridge 150×25 mm×5 µm; mobile phase: water (10 mM NH₄sub.4HCO₃)-ACN]; B %: 35%-55%, 8 minutes) to afford the title compound (16.0 mg, 68% yield) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.19 (d, J=2.8 Hz, 1H), 7.81-7.57 (m, 1H), 7.36-7.21 (m, 2H), 7.05 (d, J=2.4 Hz, 1H), 6.84 (s, 1H), 5.43-5.15 (m, 3H), 4.68-4.16 (m, 9H), 3.67 (t, J=5.6 Hz, 2H), 3.49-3.44 (m, 2H), 3.14-3.04 (m, 1H), 2.53-2.32 (m, 4H), 2.33-2.15 (m, 3H), 2.12-1.91 (m, 3H), 0.78 (br t, J=6.8 Hz, 3H); LCMS (ESI, M+1): m/z=717.5.

Example 507

(2102) ##STR00886##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-7-azaspiro[4.5]decane 2,2-dioxide (2103) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, Methanol-d₄) δ 9.08 (s, 1H), 7.66 (dd, J=5.6, 9.0 Hz, 1H), 7.28 (d, J=2.8 Hz, 1H), 7.23 (t, J=9.6 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 5.42-5.18 (m, 1H), 4.49-4.31 (m, 2H), 4.30-4.14 (m, 2H), 4.06-3.83 (m, 2H), 3.44-3.33 (m, 1H), 3.29-3.12 (m, 5H), 3.08-2.96 (m, 2H), 2.58-2.42 (m, 1H), 0.85-0.73 (m, 3H); LCMS [ESI, M+1]: m/z=682.4.

Example 508

(2104) ##STR00887##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-imino-2-thia-7-azaspiro[4.5]decane 2-oxide

(2105) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.16-9.04 (m, 11H), 7.74-7.63 (m, 1H), 7.37-7.20 (m, 2H), 7.12-6.99 (m, 1H), 5.40-5.38 (m, 1H), 5.28-5.25 (m, 1H), 4.52-4.14 (m, 4H), 4.09-3.88 (m, 2H), 3.34 (br s, 4H), 3.26-2.98 (m, 4H), 2.49 (br s, 6H), 2.12-1.84 (m, 8H), 0.85-0.74 (m, 3H); LCMS[ESI, M+1]: m/z=681.3.

Example 509

(2106) ##STR00888##

4-(4-(2,2-difluoro-7-azaspiro[4.5]decan-7-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-7-yl)-5-ethyl-6-fluoronaphthalen-2-ol

(2107) The title compound was synthesized according to the procedure described for example 330. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.10 (s, 1H), 8.58-8.46 (m, 1H), 7.70 (dd, J=6.0, 8.8 Hz, 11H), 7.33 (d, J=2.4 Hz, 11H), 7.27 (t, J=9.6 Hz, 11H), 7.08 (s, 11H), 5.53-5.31 (m, 1H), 4.53-4.36 (m, 2H), 4.33-4.17 (m, 1H), 4.16-3.95 (m, 2H), 3.95-3.79 (m, 1H), 3.63-3.39 (m, 3H), 3.27-3.12 (m, 1H), 2.57-2.34 (m, 3H), 2.28-2.09 (m, 7H), 2.05-1.94 (m, 2H), 1.93-1.66 (m, 6H), 0.87-0.77 (m, 3H); ¹⁹F NMR (377 MHz, METHANOL-d₄) δ=-76.89 (br s, 1F), -88.50-94.55 (m, 1F), -121.07 (br d, J=5.7 Hz, 1F), -137.84-141.32 (m, 1F), -173.71 (br s, 1F); LCMS (ESI, M+1): m/z=668.5.

Example 510

(2108) ##STR00889##

7-(7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-7-azaspiro[4.5]decan-2-ol

(2109) The title compound was synthesized according to the procedure described for example 330.

.sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =0.69-0.47 (m, 3H) 1.15 (s, 1H) 1.30-1.64 (m, 7H) 1.71-1.86 (m, 6H) 2.00-2.15 (m, 4H) 2.78-2.86 (m, 1H) 3.01 (s, 1H) 3.08 (d, J=11.6 Hz, 2H) 3.76 (m, 2H) 3.93 (s, 1H) 4.03-4.19 (m, 4H) 4.42-4.56 (m, 1H) 5.17-5.38 (m, 1H) 7.03 (s, 1H) 7.28-7.40 (m, 2H) 7.76 (s, 1H) 8.23 (s, 1H) 9.12 (s, 1H) 9.93 (s, 1H); LCMS [M+1].sup.+ : m/z=648.3.

Example 511

(2110) ##STR00890##

((3S,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl morpholine-4-carboxylate

(2111) The title compound was synthesized according to the procedure described for example 313.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.21 (d, J=3.6 Hz, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.30 (d, J=2.4 Hz, 1H), 7.25 (t, J=9.2 Hz, 1H), 7.07 (t, J=3.2 Hz, 1H), 4.71-4.41 (m, 2H), 4.34-4.19 (m, 3H), 4.16-4.06 (m, 1H), 4.06-3.92 (m, 1H), 3.68-3.57 (m, 5H), 3.49-3.42 (m, 4H), 3.10-2.97 (m, 2H), 2.91-2.70 (m, 1H), 2.55-2.40 (m, 1H), 2.25-2.09 (m, 3H), 2.07-1.99 (m, 1H), 1.98-1.88 (m, 3H), 1.87-1.72 (m, 5H), 1.72-1.62 (m, 1H), 1.29 (d, J=10.4 Hz, 3H), 0.88-0.74 (m, 3H); LCMS (ESI, M+1): m/z=733.4.

Example 512

(2112) ##STR00891##

((3R,7aS)-7a-(((7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-4-((R)-3-hydroxy-3-methylpiperidin-1-yl)pyrido[4,3-d]pyrimidin-2-yl)oxy)methyl)hexahydro-1H-pyrrolizin-3-yl)methyl dimethylcarbamate

(2113) The title compound was synthesized according to the procedure described for example 362.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.21 (d, J=4.0 Hz, 1H), 7.70-7.66 (m, 1H), 7.31-7.23 (m, 2H), 7.07-7.06 (m, 1H), 4.59-4.49 (m, 2H), 4.39-4.19 (m, 3H), 4.18-3.92 (m, 2H), 3.71-3.55 (m, 1H), 3.52-3.40 (m, 1H), 3.22-3.02 (m, 2H), 2.93-2.88 (m, 6H), 2.53-2.39 (m, 1H), 2.30-2.10 (m, 3H), 2.09-2.02 (m, 1H), 2.00-1.89 (m, 3H), 1.88-1.63 (m, 6H), 1.30-1.28 (m, 3H), 0.84-0.78 (m, 3H); LCMS (ESI, M+1): m/z=691.4.

Example 513

(2114) ##STR00892##

6-(2-(((1-((dimethylamino)methyl)cyclopropyl)methoxy)-7-(8-ethyl-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(2115) The title compound was synthesized according to the procedure described for example 364.

.sup.1H NMR (400 MHz, Methanol-d.sub.4) δ 9.07 (s, 1H), 7.67 (dd, J=5.6, 9.2 Hz, 1H), 7.29 (d, J=2.4 Hz, 1H), 7.24 (t, J=9.6 Hz, 1H), 7.04 (t, J=2.8 Hz, 1H), 4.42-4.21 (m, 4H), 3.96 (dd, J=13.2, 18.4 Hz, 1H), 3.84-3.64 (m, 1H), 2.88 (dd, J=6.0, 14.8 Hz, 1H), 2.78-2.68 (m, 1H), 2.50-2.36 (m, 3H), 2.29 (s, 6H), 2.21-2.04 (m, 2H), 2.04-1.86 (m, 3H), 0.79 (dt, J=2.8, 7.2 Hz, 3H), 0.75-0.71 (m, 2H), 0.56-0.50 (m, 2H); LCMS [ESI, M+1]: m/z=603.4.

Example 514

(2116) ##STR00893##

(R)-1-(7-(7,8-difluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(2117) The title compound was synthesized according to the procedure described for example 133.

.sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ =9.20 (d, J=7.2 Hz, 1H), 7.64-7.56 (m, 1H), 7.44-7.35 (m, 1H), 7.32 (t, J=2.0 Hz, 1H), 7.24 (dd, J=2.0, 7.6 Hz, 1H), 5.30 (d, J=54.4 Hz, 1H), 4.53 (br d, J=12.4 Hz, 1H), 4.37-4.21 (m, 3H), 3.63 (dd, J=2.4, 13.2 Hz, 1H), 3.45 (br t, J=12.0 Hz, 1H), 3.30-3.12 (m, 3H), 3.04-2.96 (m, 1H), 2.40-2.09 (m, 4H), 2.05-1.71 (m, 6H), 1.29 (d, J=5.2 Hz, 3H); LCMS (ESI, M+1): m/z=598.2.

Example 515

(2118) ##STR00894##

(R)-7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-

1H-pyrrolizin-7a(SB)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(2119) ##STR00895##

(2120) Step A. 7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: A mixture of 7-chloro-8-fluoro-2-(((2R,7aS)-2-fluorohexahydro-1H-pyrrolizin-7a-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (800 mg, 1.0 equiv.), ((5-chloro-6-fluoro-4-(trimethylstannyl)naphthalen-2-yl)oxy)triisopropylsilane (1.40 g, 1.49 equiv.), Pd(dppf)Cl.sub.2 (133 mg, 0.1 equiv.), BINAP (227 mg, 0.2 equiv.) and CuI (104 mg, 0.3 equiv.) in toluene (30 mL) was degassed and stirred at 90° C. for 6 hours under N.sub.2 atmosphere. The mixture was filtered and the filtrate was concentrated under reduced pressure to give a residue. The residue was purified by column chromatography by prep-TLC (SiO.sub.2, Petroleum ether/Ethyl acetate 1:1) to give the title compound (230 mg, 14% yield) as a yellow oil. LCMS (ESI, M+1): m/z=754.9.

(2121) Step B. (R)-7-(7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of 7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (230 mg, 1.0 equiv.), (R)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide (87.3 mg, 1.5 equiv.) in DMF (2 mL) was added DIEA (196 mg, 5.0 equiv.) and 4 Å molecular sieves (300 mg). The mixture was stirred at 60° C. for 12 hours. The mixture was poured into water (20 mL) and extracted with EtOAc (2×30 mL). The combined organic layer was washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered and the filtrate was concentrated to afford the title compound (262 mg, crude) as a yellow oil.

(2122) Step C. (R)-7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of (R)-7-(7-(8-chloro-7-fluoro-3-((triisopropylsilyl)oxy)naphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide (260 mg, 1.0 equiv.) in DMF (2.0 mL) was added CsF (46.6 mg, 1.0 equiv.). The mixture was stirred at 25° C. for 0.5 hour. The mixture was poured into water (30 mL) and extracted with EtOAc (2×30 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, then the mixture was filtered and the filtrate was concentrated. The residue was purified by prep-HPLC (column: Phenomenex C18 75×30 mm×3 µm; mobile phase: water (0.1% formic acid)-ACN; B %: 20%-30%, 7 min) to afford the title compound (108 mg, 50.4% yield) as a white solid; .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.11 (d, J=1.8 Hz, 1H), 8.48 (s, 1H), 7.81 (ddd, J=3.4, 5.5, 9.0 Hz, 1H), 7.46-7.34 (m, 2H), 7.21 (t, J=2.8 Hz, 1H), 5.59-5.34 (m, 1H), 4.76-4.37 (m, 5H), 3.86-3.54 (m, 5H), 3.42 (dd, J=2.4, 11.9 Hz, 1H), 3.22 (dd, J=3.2, 11.9 Hz, 1H), 2.65-2.37 (m, 2H), 2.37-2.29 (m, 1H), 2.27-2.15 (m, 2H), 2.13-1.99 (m, 3H), 1.98-1.93 (m, 1H), 1.92-1.83 (m, 1H); LCMS (ESI, M+1): m/z=690.2.

Example 516

(2123) ##STR00896##

(S)-7-(7-(8-chloro-7-fluoro-3-hydroxynaphthalen-1-yl)-8-fluoro-2-(((2R,7aS)-2-fluorotetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(2124) The title compound was synthesized according to the procedure described for example 515. .sup.1H NMR (400 MHz, methanol-d.sub.4) δ=9.12 (d, J=3.3 Hz, 1H), 8.47 (s, 1H), 7.81 (ddd, J=3.5, 5.5, 9.1 Hz, 1H), 7.47-7.34 (m, 2H), 7.21 (dd, J=2.5, 4.2 Hz, 1H), 5.60-5.37 (m, 1H), 4.77-4.32 (m, 5H), 3.89-3.53 (m, 5H), 3.41 (dd, J=3.1, 12.0 Hz, 1H), 3.22 (dd, J=3.4, 11.9 Hz, 1H), 2.65-2.40 (m, 2H), 2.38-2.29 (m, 1H), 2.27-2.16 (m, 2H), 2.13-2.00 (m, 3H), 1.96 (br d, J=11.0 Hz,

1H), 1.92-1.83 (m, 1H); LCMS (ESI, M+1): m/z=690.2.

Example 517

(2125) ##STR00897##

(R)-1-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(2126) ##STR00898##

(2127) Step A. (R)-1-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (R)-1-(7-chloro-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol (100 mg, 1.0 equiv.), 3-chloro-4-cyclopropyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol (104 mg, 1.5 equiv.), cataCXium® A Pd G3 (34.4 mg, 0.2 equiv.), Cs.sub.2CO.sub.3 (1.0 M, 3.0 equiv.) in methoxycyclopentane (1.0 mL) was degassed and stirred at 80° C. for 3 hours under N.sub.2 atmosphere. The mixture was diluted with water (5 mL), extracted with ethyl acetate (3×5 mL), the combined organic layers were dried over Na.sub.2SO.sub.4 and concentrated. The crude product was purified by prep-HPLC [column: Phenomenex luna C18 150×25 mm×10 µm; mobile phase: water (0.1% formic acid)-ACN; B %: 13%-43%, 10 min] and lyophilized to afford the title compound (53.4 mg, 39% yield) as a yellow solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.24 (s, 1H), 6.97 (d, J=2.8 Hz, 1H), 6.79 (d, J=2.8 Hz, 1H), 4.61-4.52 (m, 1H), 4.42 (s, 2H), 4.31-4.26 (m, 1H), 3.60 (d, J=13.2 Hz, 1H), 3.49-3.37 (m, 1H), 3.14-2.99 (m, 2H), 2.79 (s, 6H), 2.22-2.08 (m, 1H), 1.91-1.72 (m, 4H), 1.29 (s, 3H), 0.94-0.88 (m, 2H), 0.79-0.76 (m, 2H), 0.67-0.61 (m, 2H), 0.10-0.07 (m, 2H); LCMS (ESI, M+1): m/z=556.2.

Example 518

(2128) ##STR00899##

6-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(2129) ##STR00900##

(2130) Step A. 6-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one: To a solution of 6-(7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one (50.0 mg, 1.0 equiv.) in CPME (1 mL) was added CataCXium A Pd G3 (15.8 mg, 0.2 equiv.), 3-chloro-4-cyclopropyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol (47.9 mg, 1.5 equiv.) and Cs.sub.2CO.sub.3 (1.5 M, 3.0 equiv.). The mixture was stirred at 100° C. for 2 hours under nitrogen atmosphere. The reaction mixture was poured into saturated aqueous NH.sub.4Cl (5 mL) and extracted with EtOAc (3×2 mL). The combined organic layers were washed with brine (5 mL), dried over anhydrous sodium sulfate, filtered and concentrated. The residue was diluted with MeOH (1 mL). The residue was purified by prep-HPLC [column: Phenomenex Luna C18 150×25 mm×10 µm; mobile phase: water (0.1% formic acid)-ACN. B %: 12%-42%, 10 min] to afford the title compound (11.8 mg, 17% yield, formic acid salt) as a yellow solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.13 (s, 1H), 8.51 (br s, 1H), 6.98 (d, J=2.4 Hz, 1H), 6.82-6.79 (m, 1H), 4.69-4.58 (m, 2H), 4.44-4.28 (m, 2H), 4.01 (br dd, J=2.9, 13.0 Hz, 1H), 3.88-3.81 (m, 1H), 3.73-3.60 (m, 2H), 3.26 (td, J=6.0, 11.6 Hz, 2H), 2.94-2.87 (m, 1H), 2.81-2.74 (m, 1H), 2.36-2.28 (m, 2H), 2.25-2.15 (m, 4H), 2.08 (td, J=6.5, 12.6 Hz, 4H), 1.97 (br s, 2H), 1.89-1.81 (m, 1H), 0.67-0.47 (m, 2H), 0.13-0.02 (m, 2H); LCMS (ESI, M+1): m/z=593.3.

Example 519

(2131) ##STR00901##

5-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione

(2132) ##STR00902##

(2133) Step A. 7-(3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine: To a solution of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (600 mg, 1.0 equiv.) in THF (12 mL) was added 2-[3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (579 mg, 1.2 equiv.), K.sub.3PO.sub.4 (1.5 M, 3.0 equiv.) and CataCXium A Pd G3 (104 mg, 0.1 equiv.). The mixture was stirred at 60° C. for 3 hours under nitrogen atmosphere. The mixture was diluted with water (200 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were washed with brine (100 mL), dried over anhydrous sodium sulfate, filtered and concentrated. The residue was purified by column chromatography [SiO.sub.2, petroleum ether/ethyl acetate 10:1 to 0:1] to afford the title compound (400 mg, 41% yield) as a light yellow solid; LCMS (ESI, M+1): m/z=597.3.

(2134) Step B. 5-(7-(3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione: To a solution of 7-(3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (50.0 mg, 1.0 equiv.) and 2,3,3a,6a-tetrahydro-1H-pyrrolo[3,4-c]pyrrole-4,6-dione (14.1 mg, 1.2 equiv.) in DMF (0.25 mL) and ACN (0.25 mL) was added K.sub.3PO.sub.4 (53.3 mg, 3.0 equiv.). The mixture was stirred at 40° C. for 16 hours. The mixture was diluted with water (20 mL) and extracted with EtOAc (2×10 mL). The organic layer was washed with brine (20 mL), dried over anhydrous sodium sulfate, filtered and concentrated to afford the title compound (50.0 mg, crude) as a light yellow oil; LCMS (ESI, M+1): m/z=637.4.

(2135) Step C. 5-(7-(3-chloro-2-cyclopropyl-5-hydroxyphenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione: To a solution of 5-(7-(3-chloro-2-cyclopropyl-5-(methoxymethoxy)phenyl)-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)tetrahydropyrrolo[3,4-c]pyrrole-1,3(2H,3aH)-dione (50.0 mg, 1.0 equiv.) in ACN (1.0 mL) was added HCl-dioxane (4 M, 102 equiv.). The mixture was stirred at 20° C. for 1 hour. The mixture was concentrated. The residue was purified by prep-HPLC [column: Phenomenex Synergi C18 150×25 mm×10 µm; mobile phase: water (0.1% formic acid)-ACN. B %: 8/0-38%, 10 min] to afford the title compound (19.5 mg, 36% yield, FORMIC ACID) as a white solid; .sup.1H NMR (400 MHz, METHANOL-d.sub.4) δ=9.29 (s, 1H), 8.53 (s, 1H), 6.98 (d, J=2.5 Hz, 1H), 6.81 (d, J=2.5 Hz, 1H), 4.64 (br d, J=13.9 Hz, 2H), 4.60 (s, 2H), 4.41-4.29 (m, 2H), 3.82-3.73 (m, 2H), 3.65-3.55 (m, 2H), 3.26-3.14 (m, 2H), 2.35-2.24 (m, 2H), 2.15 (tt, J=6.6, 13.3 Hz, 4H), 2.09-1.99 (m, 2H), 1.90-1.79 (m, 1H), 0.69-0.53 (m, 2H), 0.06 (br s, 2H); LCMS (ESI, M+1): m/z=593.2.

Example 520

(2136) ##STR00903##

(3R)-1-(7-(5,6-dimethyl-1H-indazol-4-yl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol

(2137) ##STR00904##

(2138) Step A. (3R)-1-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: A mixture of (3R)-1-[7-chloro-2-[[1-((dimethylamino)methyl)cyclopropyl]methoxy]-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-3-methylpiperidin-3-ol (120 mg, 1.0 equiv.), 5,6-dimethyl-1-tetrahydropyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indazole (111 mg, 1.10 equiv.), K.sub.3PO.sub.4 (1.5 M in water, 0.5 mL, 2.70 equiv.), and CataCXium A Pd G3 (20.6 mg, 0.1 equiv.) in methoxycyclopentane (2 mL) was degassed and stirred at 90° C. for 4 hours under N.sub.2 atmosphere. The reaction mixture was diluted with ethyl acetate (5 mL) and water (10 mL) and the aqueous phase was extracted with

ethyl acetate (5 mL). The combined organic phase was washed with brine (10 mL), dried over anhydrous sodium sulfate, filtered and concentrated to give a residue. The residue was purified by reversed phase flash [water (0.1% formic acid)-ACN] to afford the title compound (135 mg, 74% yield) as a white solid; LCMS (ESI, M+1): m/z=618.5.

(2139) Step B. (3R)-1-(7-(5,6-dimethyl-1H-indazol-4-yl)-2-((1-((dimethylamino)methyl)cyclopropyl)methoxy)-8-fluoropyrido[4,3-d]pyrimidin-4-yl)-3-methylpiperidin-3-ol: To a solution of (3R)-1-[2-[[1-((dimethylamino)methyl)cyclopropyl]methoxy]-7-(5,6-dimethyl-1-tetrahydropyran-2-yl-indazol-4-yl)-8-fluoro-pyrido[4,3-d]pyrimidin-4-yl]-3-methyl-piperidin-3-ol (120 mg, 194 μ mol, 1.0 equiv.) in DCM (2.0 mL) was added TFA (3.08 g, 139 equiv.). The reaction mixture was stirred at 20° C. for 1 hour. The mixture was poured into saturated NaHCO₃ solution (60 mL) and the pH was adjusted to 8. The mixture was extracted with ethyl acetate (2×20 mL), the organic layer was washed with saturated brine (20 mL). The combined organic layers were dried over anhydrous sodium sulfate and concentrated under vacuum to give a residue. The residue was purified by prep-HPLC [column: Unisil 3-100 C18 Ultra 150×50 mm×3 μ m; mobile phase: water (0.1% formic acid)-ACN, B %: 3%-33% over 10 min] to afford the title compound (45.8 mg, 44% yield) as a white solid. ¹H NMR (400 MHz, METHANOL-d₄) δ =9.30 (d, J=10.4 Hz, 1H), 8.53 (s, 1H), 7.62 (d, J=2.4 Hz, 1H), 7.51 (s, 1H), 4.67-4.53 (m, 1H), 4.49-4.39 (m, 2H), 4.32 (br d, J=1.2 Hz, 1H), 3.62 (dd, J=4.4, 13.2 Hz, 1H), 3.49-3.34 (m, 1H), 3.17-3.02 (m, 2H), 2.82 (s, 6H), 2.51 (s, 3H), 2.23 (d, J=3.2 Hz, 3H), 2.20-2.08 (m, 1H), 1.91-1.72 (m, 3H), 1.29 (d, J=2.4 Hz, 3H), 0.96-0.88 (m, 2H), 0.83-0.75 (m, 2H); LCMS (ESI, M+1): m/z=534.4.

Example 521

(2140) ##STR00905##

7-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide

(2141) ##STR00906##

(2142) Step A. 7-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: A mixture of 7-chloro-8-fluoro-2-((tetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-4-(2,2,2-trifluoroethoxy)pyrido[4,3-d]pyrimidine (0.4 g, 1 equiv.), 2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide (200 mg, 1.1 equiv.), DIPEA (246 mg, 2 equiv.) and 4 Å molecular sieves (50 mg) in DMF (2.5 mL) was stirred at 40° C. for 13 hours under N₂ atmosphere. The mixture was filtered and purified by reversed phase flash chromatography [C18, water (0.1% formic acid)-ACN] to afford the title compound (0.3 g, 59% yield) as a yellow solid; LCMS (ESI, M+1): m/z=512.2.

(2143) Step B. 7-(7-(5,6-dimethyl-1-tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a mixture of 7-(7-chloro-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide (140 mg, 1.0 equiv.), 5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indazole (100 mg, 1.0 equiv.) and K₃PO₄ (1.5 M, 547 μ L, 3.0 equiv.) in methoxycyclopentane (2.5 mL) was added CataCXium A Pd G3 (20 mg, 0.1 equiv.) and the mixture was degassed and stirred at 90° C. for 2 hours under N₂ atmosphere. The mixture was diluted with H₂O (1 mL) and extracted with ethyl acetate (4×2 mL). The combined organic layers were dried over anhydrous Na₂SO₄, concentrated and purified by reversed phase flash [C18, water (0.1% formic acid)-ACN] to afford the title compound (74 mg, 36% yield) as a light yellow solid; LCMS (ESI, M+1): m/z=706.4.

(2144) Step C. 7-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-triazaspiro[4.5]decane 2,2-dioxide: To a solution of 7-(7-(5,6-dimethyl-1-(tetrahydro-2H-pyran-2-yl)-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-2-thia-1,3,7-

triazaspiro[4.5]decane 2,2-dioxide (85 mg, 1.0 equiv.) in DCM (1.0 mL) was added TFA (1.2 mL) dropwise at 0° C. The mixture was stirred at 0° C. for 0.75 hour. The mixture was poured into DCM (10 mL) and saturated NaHCO₃ aqueous (20 mL) at 0° C. The pH of the mixture was adjusted to 9 with solid Na₂CO₃ while maintaining the temperature below 10° C. The mixture was extracted with DCM (4×5 mL). The combined organic layers were dried over anhydrous Na₂SO₄ and concentrated. The residue was purified by prep-HPLC [column: Phenomenex Luna C18 150×25 mm×10 μm; mobile phase: water (0.1% formic acid)-ACN; B %: 7/-37%, 10 minutes] to afford the title compound (66.6 mg, 83% yield, 0.6 formic acid salt) as a white solid; ¹H NMR (400 MHz, METHANOL-d₄) δ=9.21 (s, 1H), 7.63 (br d, J=17.2 Hz, 1H), 7.52 (br s, 1H), 4.75 (br d, J=12.0 Hz, 2H), 4.64-4.54 (m, 2H), 3.82-3.67 (m, 4H), 3.45-3.36 (m, 1H), 3.27-3.22 (m, 3H), 2.51 (s, 3H), 2.33 (dt, J=7.2, 12.0 Hz, 2H), 2.27-2.13 (m, 7H), 2.10-1.87 (m, 6H); LCMS (ESI, M+1): m/z=622.4.

Example 522

(2145) ##STR00907##

4-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-6-methyl-1,4-oxazepan-6-ol

(2146) 26. The title compound was synthesized according to the procedure described for example 521. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.67 (br d, J=6.8 Hz, 1H), 7.60 (d, J=7.2 Hz, 1H), 7.52 (s, 1H), 4.67-4.51 (m, 4H), 4.27-4.15 (m, 11H), 4.09-3.85 (m, 3H), 3.78-3.66 (m, 4H), 3.30-3.26 (m, 2H), 2.51 (s, 3H), 2.38-2.27 (m, 2H), 2.26-2.09 (m, 9H), 1.29 (s, 3H); LCMS (ESI, M+1): m/z=562.4.

Example 523

(2147) ##STR00908##

6-(7-(5,6-dimethyl-1H-indazol-4-yl)-8-fluoro-2-((hexahydro-1H-pyrrolizin-7a-yl)methoxy)pyrido[4,3-d]pyrimidin-4-yl)-1,6-diazaspiro[3.5]nonan-2-one

(2148) The title compound was synthesized according to the procedure described for example 521. ¹H NMR (400 MHz, METHANOL-d₄) δ=9.21 (s, 1H), 7.60 (s, 1H), 7.52 (3, 1H), 4.69-4.58 (m, 2H), 4.41-4.39 (m, 1H), 4.38-4.29 (m, 1H), 4.03-3.93 (m, 1H), 3.92-3.79 (m, 11H), 3.72-3.61 (m, 2H), 3.27-3.21 (m, 2H), 2.95-2.90 (m, 1H), 2.81-2.73 (m, 1H), 2.51 (s, 3H), 2.35-2.27 (m, 2H), 2.25-2.20 (m, 4H), 2.20-2.11 (m, 4H), 2.11-2.02 (m, 3H), 2.07-1.95 (m, 2H); LCMS (ESI, M+1): m/z=571.4.

Example A: KRas Binding Assay

(2149) This Example illustrates that exemplary compounds of the present invention bind to KRas and are capable of displacing a labeled tracer ligand occupying the KRas binding site.

KRas^{WT}, KRas^{G12A}, KRas^{G12C}, KRas^{G12D}, KRas^{G12R}, KRas^{G12S}, KRas^{G12V}, KRas^{G13D}, or KRas^{Q61H} was used in the assay.

(2150) The ability of a compound to bind to KRas was measured using a TR-FRET displacement assay. Biotinylated KRas (corresponding to amino acids 1-169, produced at Accelegan Inc.) was incubated with custom made Cy5 labelled tracer, terbium streptavidin (Cisbio Inc.) and compound (1% DMSO final) in buffer (50 mM HEPES, pH 7.5, 5 mM MgCl₂, 0.005% Tween-20 and 1 mM DTT). After a 60-minute incubation at room temperature, the reaction was measured using a BMG LABTECH CLARIO star Plus via TR-FRET. 100 percent of control (POC) is determined by using a DMSO control and 0 POC is determined using a concentration of control compound that completely inhibits binding of the tracer to KRas. The POC values were fit to a 4-parameter IC₅₀ equation and the IC₅₀ value reported.

(2151) TABLE-US-00003 TABLE 2 Binding to KRas (IC₅₀ nM) by Exemplary Compounds of Formula (I)

Ex. No.	G12D	G12V	G12R	G12A	G12S	G12C	WT	G13D	Q61H
1	330	565	866	2632	1006	2179	38181	11740	20400
4	43	21	31	23	5	≤2	≤2	≤2	≤2
5	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2
6	29	19	14	16	20	18	22	21	23
7	≤2	≤2	≤2	≤2	≤2	≤2	3	3	3
8	2242	2626	5406	9456	333	354	10	≤2	≤2
9	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2
10	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2
11	15510	20810	40130	12	3	3	4	13	≤2
12	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2
13	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2	≤2
14	29	52	109	58	58	51	51	61	80
15	24	42	81						

38 42 37 41 42 52 26 39 60 29 31 29 32 33 40 17 746 1488 2227 961 921 1188 956 925 1861
18 23950 4446 6776 7540 4981 8795 6809 9372 13160 19 ≥ 100000 64510 84320 20 2395 3398
7389 2779 2982 3178 2967 21 14850 9727 10720 12570 18800 12710 11430 12570 19430 22
9970 12130 14880 62060 23 5275 4301 6176 4737 24 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 25$ 433 633 1312 679
26 845 1261 2317 27 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 28$ 3 ≤ 2 3 $\leq 2 \leq 2 \leq 2$ 3 3 3 29 20540 1965 4382
6198 3114 4319 5842 8190 13860 30 209 800 1271 1758 1452 1287 1743 1999 2900 31 .14040
13200 12810 32 98600 ≥ 100000 86270 33 6373 6612 10050 34 11090 14960 ≥ 100000 48730
53460 34950 28140 35 12110 2522 1611 1659 4013 36 9 20 24 37 41 57 93 52 63 56 68 88 88 38
 $\leq 2 \leq 2 \leq 2$ 39 12520 25480 19750 40 54950 66350 76920 41 6477 3407 6391 42 1861 1614 3088 43
3782 1301 2270 44 3229 2987 6238 45 89 189 362 46 14070 24030 22490 8793 18460 47 63390
70100 ≥ 100000 48 18360 20350 24070 16010 19890 49 69970 27940 18940 50 39000 22590
22670 51 21 35 69 36 38 35 32 36 50 52 235 25 39 53 66380 49380 90250 54 162 321 528 55 139
212 410 295 56 2614 3757 8341 57 $\leq 2 \leq 2 \leq 2$ 3 ≤ 2 58 6 $\leq 2 \leq 2$ 59 $\leq 2 \leq 2 \leq 2$ 60 $\leq 2 \leq 2 \leq 2$ 61 $\leq 2 \leq 2$
 ≤ 2 62 3 $\leq 2 \leq 2$ 3 ≤ 2 63 3 ≤ 2 4 5 3 64 568 1356 2551 1303 1149 1059 997 1359 1547 65 14090
14990 45130 21030 66 26920 44540 67780 67 10370 11520 16100 14240 68 11610 16950 27560
69 2089 2733 4168 70 14070 397 215 403 418 224 2857 11660 10840 71 3 ≤ 2 6 72 14050 10770
14020 73 5 $\leq 2 \leq 2$ 74 $\leq 2 \leq 2$ 3 75 63100 2393 977 1835 76 15490 3148 8872 77 12820 1350 2205
1065 1172 1640 1410 1208 1907 78 677 1532 1759 2198 1242 1263 2676 3797 3297 79 15460
539 650 80 29790 1595 1475 1576 2566 1253 6843 6313 10670 81 3564 802 1039 759 531 891
864 860 1168 82 $\leq 2 \leq 2 \leq 2$ 83 392 498 992 735 703 509 1195 1885 1992 84 12740 1864 2499 1647
2073 863 4288 5071 5864 85 44020 3704 2118 2845 4641 2153 4518 4656 10870 86 6901 3098
8490 2245 2391 3078 2408 3001 4775 87 3363 3657 11910 7119 10850 5938 6086 5257 9503 88
30710 2177 4367 1821 4239 2509 7175 9308 14660 89 2283 1021 2600 1252 1548 1061 1399
1811 3272 90 40220 3466 4190 5138 6483 4182 7236 8214 19550 91 ≥ 100000 12660 2197 12700
17910 11350 14140 36630 31360 92 20760 1718 1976 6549 6867 3411 8804 8557 17330 93 83570
2935 3386 7842 10190 4053 14880 20410 21820 94 4510 146 115 240 368 183 1395 1246 1846
95 54980 4991 3061 4996 4122 3619 3994 7183 10350 96 62940 1944 3191 2497 3229 1947 5709
10730 22110 97 175 189 348 227 272 192 333 297 412 98 1889 3009 6415 2908 4014 2841 3356
3207 4765 99 ≥ 100000 3967 16650 13590 21750 8334 32380 59770 98550 100 11810 17290
 ≥ 100000 50370 28810 39930 40320 49290 42010 101 15310 966 615 994 1343 679 3032 4233
2578 102 2289 365 479 634 191 399 1168 769 1178 103 15520 1834 1503 2086 2452 1891 3946
4212 4944 104 7613 2868 2819 1742 2018 2300 2333 3638 3208 105 9051 2806 5925 6703 5754
3511 8005 13840 13810 106 3631 612 1507 722 369 1125 2098 2335 2340 107 10750 3414 4619
6518 6848 3519 8525 8792 8776 108 19720 8102 4192 109 10160 7426 4113 4275 7389 4250
3991 4340 4944 110 26900 3974 2482 9285 4365 5222 8814 9330 10080 111 2716 2012 3717
2416 2696 1742 2324 3624 3445 112 37540 25940 28120 25320 30990 27470 36620 13260 20740
113 14250 12000 16010 9476 10570 7110 8467 12060 12230 114 4380 2856 3747 2645 2806 2475
2705 29890 43270 115 30890 5804 4636 10530 10860 6558 11600 21170 22290 116 3309 4089
6267 2743 3593 4411 3449 2698 5199 117 31970 11110 10080 9152 11010 6570 9856 5679 8239
118 16100 7531 11350 8142 6237 7451 8554 8813 13960 119 31940 18040 15560 14400 14580
11600 15680 17210 27610 120 ≥ 100000 ≥ 100000 ≥ 100000 33300 85290 19770 34070 ≥ 100000
 ≥ 100000 121 42820 6637 12320 15460 18580 9100 12990 17330 25370 122 ≥ 100000 6909 12450
14410 21750 3471 25970 86370 88200 123 4436 3192 7503 3181 2760 2341 3499 3484 4979 124
12620 1507 2190 2068 3869 1483 7022 7821 7135 125 60420 3923 4394 5517 7425 4250 13310
29060 39020 126 67250 ≥ 100000 18430 13800 3967 9288 5967 7293 7502 127 25080 1698 2578
2681 1715 2192 4220 8699 11080 128 48980 8793 4142 11460 7803 9569 12810 13400 18430 129
2255 2795 4434 2687 2795 2531 3113 2496 4359 130 8361 10850 18260 10510 14240 9076 11430
25770 21930 131 14640 8300 2793 4463 9294 3453 8212 6561 7694 132 9 5 6 133 $\leq 2 \leq 2 \leq 2$ 134
17 $\leq 2 \leq 2$ 135 31 3 3 136 3 $\leq 2 \leq 2$ 137 $\leq 2 \leq 2$ 3 138 12 5 5 139 8 $\leq 2 \leq 2$ 140 $\leq 2 \leq 2 \leq 2$ 141 $\leq 2 \leq 2 \leq 2$
142 14 $\leq 2 \leq 2$ 143 4 $\leq 2 \leq 2$ 144 43 48 96 71 65 77 74 60 82 145 9 6 3 146 $\leq 2 \leq 2$ 3 147 4 3 3 148 3

≤2 ≤2 149 4 ≤2 3 150 ≤2 ≤2 ≤2 151 12 8 9 152 175 8 10 153 6 5 10 7 7 7 6 8 7 154 33 28 66 42 43
43 54 34 53 155 118 4 3 156 ≤2 ≤2 ≤2 157 12 ≤2 ≤2 158 152 78 81 85 69 41 103 83 102 159 39 3
5 160 15 ≤2 ≤2 161 68 7 9 1.2 15 7 20 16 17 162 31 ≤2 3 163 95 14 17 164 28 3 3 165 51 7 13 17
17 12 13 13 14 166 7 ≤2 ≤2 167 16 ≤2 ≤2 168 5 ≤2 ≤2 169 4 ≤2 ≤2 170 ≤2 ≤2 ≤2 171 89 3 9 172 9
≤2 ≤2 173 6 4 6 174 89 4 13 175 38 ≤2 ≤2 176 108 20 30 177 3 ≤2 ≤2 178 18 7 14 179 12 4 3 180
4 ≤2 4 181 29 17 10 182 9 ≤2 ≤2 183 4 4 5 184 ≤2 ≤2 3 185 11 9 9 186 21 9 10 187 12 5 6 188 48
19 23 189 32 18 18 190 3 ≤2 ≤2 191 9 4 4 192 ≤2 ≤2 ≤2 193 ≤2 ≤2 ≤2 194 ≤2 ≤2 ≤2 195 16600
4909 1714 196 ≤2 ≤2 ≤2 197 ≤2 ≤2 ≤2 198 86 9 8 199 3 ≤2 ≤2 200 ≤2 ≤2 ≤2 201 ≤2 ≤2 ≤2 202 ≤2
≤2 ≤2 ≤2 203 ≤2 ≤2 ≤2 204 1287 43 21 38 51 20 233 317 283 205 307 8 7 13 16 10 80 69 95 206
≤2 ≤2 ≤2 207 17270 6166 13080 18100 46210 11750 26720 30390 46370 208 17450 824 494 1297
2263 677 4011 6246 7154 209 63350 2345 1615 3699 10080 2517 4957 13260 12720 210
≥100000 2940 37770 66550 ≥100000 43260 ≥100000 ≥100000 ≥100000 211 ≥100000 14680 8034
12240 18720 10400 16940 20220 29830 212 15590 1349 880 2831 6449 3124 8356 10070 17370
213 ≥100000 5573 4252 6194 7252 4198 16840 20540 17690 214 ≥100000 12550 16350 28800
34560 9588 60030 95150 81100 215 ≥100000 17000 8948 26070 23480 17750 33970 47090
57470 216 ≥100000 17890 2571 13750 41050 6302 30890 59030 83190 217 64450 5159 7568
8986 11330 9878 13980 11610 19530 218 65930 2844 6534 14830 23180 6879 28910 ≥100000
84810 219 65290 8833 4777 16210 13130 5652 15020 18130 20180 220 10820 6067 8672 6749
6309 8246 5570 6699 7473 221 586 556 191 271 313 498 284 308 231 222 ≥100000 18230 31250
17790 26890 8984 35880 70090 52900 223 21110 930 665 1357 1801 1189 5601 6870 8654 224
64760 8476 4018 8455 7111 11710 6285 4834 9675 225 21660 6241 9389 14900 9898 8121 16590
23510 26140 226 6070 14070 51930 227 1242 2430 3056 2718 3386 228 147 148 224 1212 202
1019 2698 465 822 229 85340 3121 2117 4724 36850 230 38750 7987 9849 11260 19660 10640
28440 35510 23350 231 97900 24560 8681 52100 31620 29610 49370 61490 62320 232 11920
7781 9993 17340 19370 13600 19190 17970 25900 233 6699 4635 5563 2647 7451 6154 4303
7383 5348 234 2022 1603 1846 2302 1764 1622 2640 1979 2587 235 2660 1700 2870 2152 2011
1650 2653 2714 3356 236 2331 623 3352 3395 897 2043 3497 2462 4439 237 9560 7612 9743
5337 8916 7594 4362 8292 6382 238 ≥100000 19350 9182 239 8653 4934 4812 1978 2212 1608
1830 1640 2673 240 1138 1550 777 586 492 597 478 350 498 241 17150 13610 8624 242 8032
7632 12950 8192 6226 6956 11010 7859 13620 243 25360 2187 1056 1049 1808 1146 3320 2963
3585 244 18210 3574 3095 1848 3061 2823 5138 4856 5077 245 2991 11710 9831 246 6414
22140 11260 247 429 1827 2555 2108 2152 2682 1585 1182 2058 248 6807 9778 10530 249 7856
1287 424 981 742 979 1061 1280 808 250 7315 3408 3992 3927 3799 6839 3579 3513 5313 251
6831 3870 14580 6125 5753 6401 3841 3784 6246 252 11 17 27 5 8 17 7 6 9 253 67330 4353
4082 3032 3589 2408 6002 14090 7718 254 2839 10830 7908 255 6227 11010 15130 256 2416
4536 8144 4627 5308 4048 4031 3796 6017 257 2432 3793 6914 5493 5219 4004 4026 3957 6028
258 320 858 1183 2066 1014 1381 3249 4383 4614 259 158 380 769 1521 649 825 2208 3299
3181 260 61 204 455 294 419 591 864 1173 1062 261 1606 3984 2606 5743 2317 2392 7099 9609
10650 262 960 1790 1663 1534

(2152) TABLE-US-00004 Ex. No. G12D G12V G12R G12A G12S G12C WT G13D Q61H 263
1981 4033 3038 4936 3554 4408 9086 10960 12120 264 8160 13570 9856 265 5309 2735 1923
3019 2505 3311 2458 2384 3188 266 6061 5255 13860 267 4369 1712 1719 2496 2074 2253 1565
1367 2430 268 11350 6442 8884 269 1909 1000 1056 1052 725 608 1063 1494 1826 270 2627
2095 7593 2834 4194 271 2429 2444 7693 3001 272 194 130 120 138 132 141 116 134 144 273
5150 2309 5202 3068 274 375 867 2762 14700 275 1656 9053 7165 24340 276 212 303 1215
3219 277 765 1801 4911 5287 278 801 1985 1860 3673 279 34330 3388 3624 280 ≥100000
≥100000 ≥100000 ≥100000 ≥100000 9859 ≥100000 ≥100000 ≥100000 281 213 170 81 59 67 75
61 125 51 282 30 15 18 16 15 16 15 18 17 283 97 90 129 97 91 77 113 157 142 284 38 41 58 46
45 44 46 54 50 285 976 864 350 367 359 426 323 447 293 286 313 521 674 370 407 410 608 498
555 287 8169 225 225 648 590 321 838 423 1158 288 53 67 78 95 82 68 89 71 113 289 5610 807

754 607 120 407 818 960 1071 290 582 171 176 183 147 207 276 236 290 291 140 277 439 229
192 161 269 209 218 292 21 61 180 97 26 88 68 55 77 293 10 4 5 5 6 6 6 7 5 294 9 9 15 13 12 13
14 16 14 295 2454 948 593 1801 2385 1896 2851 2873 3259 296 4 $\leq 2 \leq 2$ 297 54 ≤ 2 9 298 1516
133 99 299 3 $\leq 2 \leq 2$ 300 $\leq 2 \leq 2 \leq 2$ 301 $\leq 2 \leq 2 \leq 2$ 302 $\leq 2 \leq 2 \leq 2$ 303 $\leq 2 \leq 2 \leq 2$ 304 $\leq 2 \leq 2 \leq 2$ 305 25 6
14 306 7 4 4 307 $\leq 2 \leq 2 \leq 2$ 308 6 4 10 309 4 ≤ 2 3 310 8 $\leq 2 \leq 2$ 311 $\leq 2 \leq 2 \leq 2$ 312 388 254 203 156
172 212 202 242 280 313 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 315 $\leq 2 \leq 2$
 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 3 ≤ 2 316 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 318 $\leq 2 \leq 2$
 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 319 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 321 $\leq 2 \leq 2$
 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 322 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 324 $\leq 2 \leq 2 \leq 2 \leq 2$
 $\leq 2 \leq 2 \leq 2 \leq 2$ 325 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 327 $\leq 2 \leq 2 \leq 2$ 328
9 6 18 9 9 329 6602 1784 3379 2618 1029 330 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 331 $\leq 2 \leq 2 \leq 2$ 332 $\leq 2 \leq 2$ 3 333 ≤ 2
 ≤ 2 5 334 1480 699 738 335 7827 ≥ 100000 84490 336 $\leq 2 \leq 2 \leq 2$ 337 3 3 5 338 ≤ 2 3 4 339 $\leq 2 \leq 2 \leq 2$
340 $\leq 2 \leq 2 \leq 2$ 341 ≤ 2 3 ≤ 2 342 3 $\leq 2 \leq 2$ 343 $\leq 2 \leq 2 \leq 2$ 344 138 83 94 345 5 3 4 346 $\leq 2 \leq 2 \leq 2$ 347 3
3 3 348 6 3 ≤ 2 349 20 4 7 350 30 8 8 351 13 15 11 352 $\leq 2 \leq 2 \leq 2$ 353 $\leq 2 \leq 2 \leq 2$ 354 $\leq 2 \leq 2 \leq 2$ 355
 $\leq 2 \leq 2 \leq 2$ 356 $\leq 2 \leq 2 \leq 2$ 357 $\leq 2 \leq 2 \leq 2$ 358 $\leq 2 \leq 2 \leq 2$ 359 $\leq 2 \leq 2 \leq 2$ 360 5 $\leq 2 \leq 2$ 361 $\leq 2 \leq 2 \leq 2$ 362 3
3 4 363 201 57 10 364 $\leq 2 \leq 2 \leq 2$ 365 $\leq 2 \leq 2 \leq 2$ 366 3 3 9 367 $\leq 2 \leq 2 \leq 2$ 368 4 24 11 369 5 3 4 370
11 ≤ 2 5 371 7 ≤ 2 5 372 56 5 13 373 5 $\leq 2 \leq 2$ 374 ≤ 2 9 15 375 44 8 9 376 16 3 4 377 ≤ 2 3 12 378
104 14 25 379 111 14 34 21 26 380 275 51 71 54 18 381 23 11 21 16 5 382 9 ≤ 2 3 383 13 3 8 384
38 12 32 385 $\leq 2 \leq 2 \leq 2$ 386 7 5 6 387 124 130 125 128 113 96 106 130 153 388 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2$
 $\leq 2 \leq 2 \leq 2$ 389 $\leq 2 \leq 2$ 3 $\leq 2 \leq 2 \leq 2 \leq 2 \leq 2$ 390 $\leq 2 \leq 2 \leq 2$ 391 $\leq 2 \leq 2 \leq 2$ 392 ≤ 2 3 3 393 $\leq 2 \leq 2 \leq 2$ 394
 $\leq 2 \leq 2 \leq 2$ 395 $\leq 2 \leq 2 \leq 2$ 396 ≤ 2 3 3 397 $\leq 2 \leq 2 \leq 2$ 398 $\leq 2 \leq 2 \leq 2$ 399 $\leq 2 \leq 2 \leq 2$ 400 $\leq 2 \leq 2 \leq 2$ 401 ≤ 2
 $\leq 2 \leq 2$ 402 $\leq 2 \leq 2 \leq 2$ 403 6 $\leq 2 \leq 2 \leq 2$ 404 $\leq 2 \leq 2 \leq 2$ 405 $\leq 2 \leq 2 \leq 2$ 406 4 5 11 407 $\leq 2 \leq 2 \leq 2$ 408 $\leq 2 \leq 2$
 ≤ 2 409 $\leq 2 \leq 2 \leq 2$ 410 $\leq 2 \leq 2 \leq 2$ 411 $\leq 2 \leq 2 \leq 2$ 412 $\leq 2 \leq 2 \leq 2$ 413 $\leq 2 \leq 2$ 3 414 $\leq 2 \leq 2$ 4 415 ≤ 2 40 3
416 $\leq 2 \leq 2 \leq 2$ 417 $\leq 2 \leq 2 \leq 2 \leq 2$ 418 $\leq 2 \leq 2$ 3 419 3 3 8 420 3 ≤ 2 4 421 3 $\leq 2 \leq 2$ 422 7 ≤ 2 5 423
 $\leq 2 \leq 2 \leq 2$ 424 3 4 8 425 $\leq 2 \leq 2 \leq 2$ 426 $\leq 2 \leq 2 \leq 2$ 427 $\leq 2 \leq 2 \leq 2$ 428 $\leq 2 \leq 2 \leq 2$ 429 2456 4438 8253
430 6452 4764 8774 6824 7717 7825 8049 5493 8858 431 9624 15880 34430 19620 19670 17850
19720 13300 22220 432 1139 1930 3036 2267 2405 2413 2476 1675 2808 433 8780 10030 15370
434 2203 3209 4510 5619 3523 3480 5130 6601 9266 435 3886 5440 6857 4911 5636 4839 4452
2062 6096 436 798 652 331 204 217 271 193 212 173 437 1432 2681 2764 1359 904 1401 715
754 897 438 238 239 99 109 106 130 95 117 87 439 7966 5785 2833 2145 2258 2528 2035 2435
1905 440 510 844 1722 758 690 550 714 767 940 441 3 $\leq 2 \leq 2$ 442 $\leq 2 \leq 2 \leq 2$ 443 53010 4206 6379
8474 6309 8200 18790 22220 32610 444 5 $\leq 2 \leq 2$ 445 4 $\leq 2 \leq 2$ 446 4 ≤ 2 4 447 1100 744 964 665
587 608 557 578 752 448 87 54 61 59 49 53 53 62 60 449 8394 3659 4382 5620 4163 3604 4072
5528 6403 450 250 238 344 356 276 235 335 481 388 451 2577 2856 4173 3123 2061 1902 2271
1994 3814 452 4 $\leq 2 \leq 2$ 453 274 569 1119 559 459 484 467 503 679 454 105 191 421 176 167 157
163 200 213 455 6352 5153 4506 5987 8171 8469 11720 13070 11620 456 48800 10680 6371 457
31 6 10 458 15 39 172 97 28 75 53 46 67 459 537 766 1006 815 821 822 688 586 793 460 2322
1469 548 760 759 916 693 763 664 461 $\leq 2 \leq 2 \leq 2$ 462 880 350 418 321 311 430 493 393 567 463
6494 3064 1362 1549 1401 2145 2402 1908 1685 464 986 469 513 507 355 459 661 506 717 465
15640 6097 10060 466 2323 1656 2255 3312 2135 2313 2684 3579 3508 467 915 651 1015 968
1136 1110 1477 1800 1648 468 2876 2311 2902 2807 2482 2406 3018 4191 5109 469 127 105 149
180 146 135 196 256 231 470 406 355 503 347 341 348 475 565 663 471 309 427 1072 332 391
459 362 319 451 472 6 16 18 473 11 8 14 8 9 9 10 11 11 474 8598 1751 3628 2953 3452 2386
6166 5974 6350 475 8546 1455 2621 3676 2748 2749 4600 7516 8141 476 $\leq 2 \leq 2$ 4 477 $\leq 2 \leq 2 \leq 2$
478 $\leq 2 \leq 2$ 5 479 5 4 12 480 38 7 6 481 4 4 7 482 $\leq 2 \leq 2$ 3 483 $\leq 2 \leq 2 \leq 2$ 484 23 28 48 485 9 ≤ 2 3
486 336 170 186 487 14 5 6 488 22 27 51 489 221 151 310 490 $\leq 2 \leq 2$ 3 491 4 4 6 492 84 15 14
493 130 11 9 494 3 $\leq 2 \leq 2$ 495 8 ≤ 2 7 496 24 11 6 497 13 3 ≤ 2 498 5 3 3 499 27 8 6 500 235 104
103 501 606 169 176 502 638 457 607 503 305 82 124 504 $\leq 2 \leq 2 \leq 2$ 505 $\leq 2 \leq 2 \leq 2$ 506 $\leq 2 \leq 2 \leq 2$
507 $\leq 2 \leq 2$ 4 508 5 12 14 509 6 6 12 510 10 8 9 511 $\leq 2 \leq 2 \leq 2$ 512 5 5 8 513 $\leq 2 \leq 2$ 3 514 $\leq 2 \leq 2 \leq 2$
515 $\leq 2 \leq 2 \leq 2$ 516 517 ≤ 2 3 6 518 $\leq 2 \leq 2 \leq 2$ 519 $\leq 2 \leq 2 \leq 2$ 520 692 614 1749 521 4 ≤ 2 3 522 292

Example B: Inhibition of KRas Phosphorylation of ERK by Exemplary Compounds of Formula (I) (2153) This Example illustrates that exemplary compounds of the present invention inhibit the phosphorylation of ERK downstream of KRas WT, G12C, G12D, G12R, G12S, G12V, G13D, Q61H.

(2154) AsPC-1 (G12D, ATCC CRL-1682), A549 (G12S, ATCC CCL-185), HCT116 (G13D, ATCC CCL-247) cells were grown in DMEM medium supplemented with 10% fetal bovine serum and Penicillin/Streptomycin. NCI-H358 (G12C, ATCC CRL-5807), NCI-H460 (Q61H, ATCC HTB-117), NCI-H727 (G12V, ATCC CRL-5815), MKN1 (WTdep, JCRB JCRB0252), PSN-1 (G12R, ATCC CRM-CRL-3211) cells were grown in RPMI medium supplemented with 10% fetal bovine serum, 10 mM HEPES, 10 mM Sodium Pyruvate, and Penicillin/Streptomycin. Cells were plated in black well clear bottom tissue culture treated 96 well plates (Corning, 3904) at a density of 20,000 cells/well and allowed to attach for 12-14 hours. Diluted compounds were then added in a final concentration of 0.5% DMSO. After 3 hours, 50 μ L of 4.0% formaldehyde was added and the plates incubated at room temperature for 20 minutes. The plates were then dumped and permeabilized with 150 μ L of ice cold 100% methanol for 10 minutes. Non-specific antibody binding to the plates was blocked using 100 μ L Odyssey blocking buffer (LI-COR Biosciences, 927-60010) for 1 hour at room temperature.

(2155) The amount of phospho-ERK was determined using an antibody specific for the phosphorylated form of ERK and compared to the amount of GAPDH. Primary antibodies used for the detection were added as follows: Phospho-ERK(Cell Signaling CS-9101) diluted 1:500 and GAPDH (Millipore MAB374) diluted 1:5000 in Odyssey blocking buffer+0.05% Tween 20. The plates were incubated overnight at 4° C. The plates were washed 3 $\times$ with 150 μ L PBS+0.1% Tween 20.

(2156) Secondary antibodies used to visualize primary antibodies were added as follows: Goat Anti-Rabbit-800 (LI-COR, 926-32211) and Goat Anti-Mouse-680 (LI-COR, 926-68070) diluted 1:800 both in Odyssey blocking buffer+0.05% Tween 20, and were incubated for 1 hour at room temperature. The plates were washed 3 $\times$ with 150 μ L PBS+0.1% Tween 20. Plates were imaged dry on a Li-COR Odyssey CLX plate reader.

(2157) The phospho-ERK (Thr202/Tyr204) signal was normalized to the GAPDH signal for each well and percent of DMSO control values were calculated. IC50 values were generated using a 4-parameter fit of the dose response curve.

(2158) TABLE-US-00005 TABLE 3 Inhibition (IC_{sub}.50, nM) of KRas-mediated Phosphorylation of ERK, Exemplary Compounds of Formula (I) Ex. No. AsPC-1 H727 MKN1 PSN1 A549 H460 HCT116 H358 4 519 5 9 29 3 681 11 12 23 6 7 802 530 32 3592 1094 416 627 199 10 297 12 497 13 18 44 10 773 140 13 42 11 14 847 4585 1383 ≥ 10000 ≥ 10000 ≥ 10000 1594 725 15 731 ≥ 10000 1255 ≥ 10000 3635 ≥ 10000 5157 495 16 654 ≥ 10000 802 ≥ 10000 1758 113 2176 1179 24 1684 4725 406 3687 ≥ 10000 2825 1368 454 27 45 232 20 2924 300 33 88 22 28 78 308 24 ≥ 10000 523 54 153 60 36 1544 3387 1181 ≥ 10000 2151 557 1610 1127 37 2213 ≥ 10000 1674 ≥ 10000 8947 ≥ 10000 4358 1719 38 966 1187 143 7617 3274 573 814 324 51 586 7226 873 ≥ 10000 2638 ≥ 10000 3719 715 57 82 115 22 4143 668 119 1075 50 58 24 242 23 1312 455 111 1457 24 59 577 769 107 ≥ 10000 3562 296 1272 176 60 3892 1265 230 ≥ 10000 5533 1031 3600 389 61 306 1109 46 ≥ 10000 699 550 518 162 62. 591 178 7 ≥ 10000 87 154 410 24 63 15 56 5 1043 139 38 44 27 71 4734 2237 140 ≥ 10000 ≥ 10000 2170 1947 361 73 3494 756 23 ≥ 10000 191 176 128 98 74 72 355 8 ≥ 10000 63 55 662 10 82 18 65 10 1741 25 180 216 3 132 104 ≥ 10000 27 ≥ 10000 ≥ 10000 206 ≥ 10000 131 133 156 612 171 ≥ 10000 55 49 650 41 134 ≥ 10000 ≥ 10000 178 ≥ 10000 1489 666 ≥ 10000 48 135 2061 ≥ 10000 53 ≥ 10000 2458 647 3272 37 136 351 181 7 ≥ 10000 19 75 631 4 137 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 2035 138 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 139 ≥ 10000 ≥ 10000 307 ≥ 10000 ≥ 10000 7514 ≥ 10000 115 140 27 28 2 1 9 17 11 141 210 ≥ 10000 282 ≥ 10000 875 1108 ≥ 10000 88 142 2866 ≥ 10000 45

≥10000 ≥10000 212 6959 78 143 ≥10000 ≥10000 300 ≥10000 ≥10000 ≥10000 527 145
≥10000 ≥10000 258 ≥10000 ≥10000 847 ≥10000 379 146 22 ≥10000 22 2526 1057 204 1003 15
147 ≥10000 ≥10000 65 ≥10000 1951 383 ≥10000 79 148 105 260 76 3785 550 68 ≥10000 15 149
9 206 16 1758 83 71 292 10 150 435 626 13 ≥10000 45 362 996 4 152 ≥10000 ≥10000 6290
≥10000 ≥10000 ≥10000 ≥10000 ≥10000 153 233 500 306 154 ≥10000 ≥10000 8447 1455 840 155
1724 2666 196 ≥10000 ≥10000 399 ≥10000 329 156 335 189 14 ≥10000 33 141 323 10 157 751
4725 31 ≥10000 448 585 893 31 159 ≥10000 1997 133 ≥10000 4070 1508 2112 139 160 5520 727
146 ≥10000 974 471 1597 149 162 4526 4166 6769 164 ≥10000 ≥10000 207 ≥10000 ≥10000 6825
≥10000 602 165 ≥10000 ≥10000 412 ≥10000 ≥10000 2923 ≥10000 701 166 ≥10000 ≥10000
≥10000 ≥10000 328 167 6763 1252 623 168 1805 ≥10000 ≥10000 821 156 169 2892 1807 931
170 3395 ≥10000 ≥10000 2353 470 171 ≥10000 ≥10000 ≥10000 ≥10000 ≥10000 172 481 192 79
173 180 304 145 174 ≥10000 ≥10000 4952 ≥10000 1288 175 1725 756 675 176 ≥10000 3954 701
≥10000 ≥10000 3511 ≥10000 1254 177 ≥10000 ≥10000 764 1699 54 178 606 2110 728 179 785
675 274 180 205 623 113 181 1543 1613 584 182 547 355 157 183 399 720 137 184 66 201 30
185 306 481 424 186 1593 1062 592 187 470 550 133 188 2067 2136 478 189 1241 1426 377 190
189 272 62 191 665 714 230 192 90 191 51 193 94 133 4 12 44 63 41 194 92 89 7 65 196 361 452
48 197 144 841 11 293 198 2686 1754 803 199 1318 798 314 200 47 67 3 3 22 21 13 201 91 110 2
100 202 60 159 3 74 203 8014 1266 758 206 6961 3330 3384 296 2195 2085 619 297 3430 450
660 298 ≥10000 ≥10000 2712 ≥10000 ≥10000 ≥10000 5008 299 126 ≥10000 108 333 29
300 251 372 37 71 301 216 520 15 416 302 152 360 5 193 303 3288 3687 4115 304 1369 2576
804 305 ≥10000 ≥10000 ≥10000 306 405 461 306 307 117 215 11 308 489 627 372 309 176 272
59 310 ≥10000 3372 3793 311 738 2603 501 386 1185 1024 299 388 16 16 441 119 139 70 442
134 234 124 444 2438 1736 656 445 1615 2934 1189 446 3669 2500 2194 452 182 793 154 457
3954 9409 459 9936 ≥10000 ≥10000 473 320 325

Example C

(2159) Inhibition of KRas Phosphorylation of ERK (HTRF) by Exemplary Compounds of Formula (I)

(2160) Cisbio HTRF Advanced pERK Assay Catalog #64AERPEH

(2161) Cells: MKN1, PSN1 Procedure: Day 1: Seed 6,000 cells/well –25 µl/well in 384-well white solid bottom plate; RPMI_10% FBS. Incubate overnight at 37° C./5% CO.sub.2. Day 2: Echo transfer 25 nl of 10 mM compound 10 point dilution at 1:3 (Cf=10 uM) and incubate for 3 hour at 37° C./5% CO.sub.2. Add 8.5 µl/well of 4× Lysis Buffer/25× Blocking reagent (do not dump media) and incubate for 30 min at room temperature on shaker. Add conjugate mixture of 4.25 ul/well 1×-pERK-D2 and 1×-pERK-K diluted in Detection Buffer for a total of 8.5 µl/well. Incubate for 4 hours at room temperature covered. Read HTRF using ClarioStar Cells: ASPC1, H727, A549, H460, HCT116, H358 Culture/Assay media: RPMI-1640+10% h FBS Procedure: Cell seeding 1. To harvest cells from flask using 0.05% Trypsin/EDTA solution. Add 10 mL of media to stop trypsinizing. Pipette the cells into a conical bottom 50 mL centrifuge tube and centrifuge 5 min×1000 rpm. 2. Re-suspend the cell pellet in media, take a cell count, and then adjust the cell density using fresh media. 3. Seed 6,000 cells into cell culture plate with 50 µL media. The 4. Incubate cell plate overnight in a 37° C., 5% CO.sub.2 incubator. Compound titrations 1. Use Tecan to complete the compound addition. Compounds start from 10 uM top, 3-fold dilution, and 10 doses. The final DMSO concentration is 0.8%. Dispensed 0.2 uM Trametinib as Min control. 2. Incubate cell plate for 3 hrs in the incubator. Detection with cisbio pERK HTRF kit 1. Dilute 1 volume of 4× lysis buffer with 3 volumes of deionized water. Then, add 100× the blocking reagent. Keep lysis buffer on the ice. 2. At the end of the compound treatment, flick-off the media. 3. Add 35 µL of lysis buffer per well using a Multidrop Combi. Then place on a plate agitator shaking at 300 rpm at 4° C. for 40 mins. 4. Make up the HTRF antibody buffer. For each assay plate, mix 50 µL of d2-conjugate antibody with 950 µL of detection buffer. Similarly, mix 50 µL of Cryptate antibody with 950 µL of detection buffer. Then mix the two diluted antibodies

together. 5. Dispense 3.4 μ L the antibody buffer to wells of an empty assay plate. Seal the plate and centrifuge plate 30 sec $\times$ 1000 rpm. 6. At the end of the 4° C. lysis, centrifuge the lysate plates 3 mins $\times$ 1500 rpm. 7. Use the Bravo to transfer 13.6 μ L of lysate from cell culture plate to assay plate.

(2162) Then incubate assay plate for 2 hrs at room temperature. 8. At the end of incubation, read plate on the Envision after centrifuging plate 30 sec $\times$ 1000 rpm.

Table 4

(2163) TABLE-US-00006 Inhibition (HTRF IC.sub.50 nM) of KRas-mediated Phosphorylation of ERK by Exemplary Compounds of Formula (I) Ex. No. AsPC-1 H727 MKN1 PSN1 A549 H460 HCT116 H358 4 943 113 ≥ 10000 5 2 6 1 364 2 5 10 2 10 13 3341 12 26 ≥ 10000 13 4 13 5 1090 43 27 33 16 275 ≥ 10000 25 ≥ 10000 ≥ 10000 26 ≥ 10000 ≥ 10000 28 20 159 21 7246 156 101 155 57 32 55 175 119 2011 58 75 45 69 179 61 126 250 336 1644 63 4 28 19 21 63 71 2103 1924 74 65 84 151 336 82 15 16 6 2325 15 22 106 10 132 623 912 133 144 246 360 136 318 52 83 243 140 7 4 1 922 2 4 7 141 641 2134 146 92 400 327 607 148 59 110 132 2209 149 25 111 290 180 276 150 66 12 167 115 156 125 72 143 349 162 187 ≥ 10000 167 3 ≥ 10000 169 100 ≥ 10000 172 96 26 41 ≥ 10000 32 170 6 173 56 263 42 5542 176 454 95 175 39 ≥ 10000 178 49 ≥ 10000 179 10 6955 180 10 6905 181 139 6701 182 5 6896 183 2 4234 184 7 90 7 1228 44 39 91 15 185 17 4837 186 108 ≥ 10000 187 38 ≥ 10000 188 77 ≥ 10000 189 29 5543 190 11 4990 191 15 5847 192 4 7340 193 8 36 7 3789 44 42 82 194 20 34 3 1346 63 55 686 4 196 97 167 39 314 33 114 31 197 57 348 23 ≥ 10000 183 620 112 198 183 ≥ 10000 199 7 ≥ 10000 200 3 6 2 1010 5 5 11 201 23 42 15 2388 223 118 187 12 202 7 33 5 2493 9 28 47 4 203 35 3711 206 34 ≥ 10000 296 53 ≥ 10000 297 154 ≥ 10000 300 38 89 7 1378 14 301 54 199 14 3928 216 456 46 302 31 82 3 2533 65 120 179 26 303 358 ≥ 10000 304 29 ≥ 10000 306 57 6803 307 58 85 5 106 12 33 15 308 155 278 40 ≥ 10000 284 309 54 37 2 8328 113 57 26 26 311 15 ≥ 10000 313 13 28 7 2120 38 38 42 4 314 9 21 3 407 19 14 21 2 315 20 20 5 1959 28 34 24 5 316 8 32 1 2040 51 69 30 9 317 40 46 20 3931 197 287 98 985 318 133 69 13 877 65 60 319 433 808 120 ≥ 10000 6358 320 115 82 10 5182 100 65 321 16 30 3 7757 99 71 39 7 322 484 255 32 9090 300 323 8 8 2 600 19 11 22 2 324 35 22 3 1442 95 99 61 14 325 49 45 17 5152 61 56 164 326 10 26 8 ≥ 10000 138 253 274 18 327 23 98 14 6687 238 206 381 10 328 1719 271 ≥ 10000 1278 329 ≥ 20000 ≥ 10000 ≥ 10000 ≥ 20000 330 29 61 8 305 84 52 57 11 331 4 35 1 6 16 27 3 332 6 31 2 710 13 22 48 3 333 603 2108 425 ≥ 10000 9189 336 32 129 4 4120 75 162 208 59 337 514 395 49 1843 438 338 695 377 24 2878 487 339 48 72 12 ≥ 10000 557 991 826 95 340 39 149 26 ≥ 10000 353 373 999 55 341 472 1036 86 ≥ 10000 1320 342 23 46 4 2245 167 203 470 62 343 60 24 4 5928 91 128 128 7 344 3407 4815 796 ≥ 10000 345 83 138 13 8785 82 296 20 346 99 113 11 7392 166 133 57 347 118 73 98 4458 127 195 348 275 273 7 ≥ 10000 245 349 475 1856 149 ≥ 10000 2251 350 241 334 209 ≥ 10000 797 351 228 551 48 7176 185 182 346 352 948 1267 46 ≥ 10000 353 7 34 24 4807 80 149 396 19 354 14 48 14 ≥ 10000 65 304 623 17 355 8 30 17 6773 140 270 471 16 356 16 63 40 9301 232 458 767 20 357 13 63 71 7478 576 805 1354 27 358 12 15 4 9767 44 131 167 3 359 345 782 60 4095 576 360 49 62 2 9186 54 140 19 361 8 49 3 6817 122 85 231 17 362 73 313 37 5130 629 404 763 113 363 ≥ 10000 ≥ 10000 ≥ 10000 ≥ 10000 364 121 578 26 ≥ 10000 219 117 645 365 17 24 10 2856 11 17 77 6 366 60 185 11 5531 108 321 61 367 505 244 58 ≥ 10000 375 368 4683 617 191 ≥ 10000 369 1341 471 294 ≥ 10000 370 2985 516 330 ≥ 10000 371 ≥ 20000 387 ≥ 10000 9341 372 660 ≥ 10000 373 4066 103 ≥ 10000 3841 374 9382 ≥ 10000 376 ≥ 10000 6491 377 ≥ 10000 7783 2298 ≥ 10000 378 ≥ 10000 ≥ 10000 379 ≥ 20000 ≥ 10000 ≥ 10000 ≥ 20000 380 ≥ 20000 1103 ≥ 10000 3215 381 ≥ 20000 282 ≥ 10000 741 382 ≥ 10000 5899 781 ≥ 10000 383 ≥ 10000 ≥ 10000 384 ≥ 10000 ≥ 10000 385 ≥ 10000 4648 421 ≥ 10000 386 44 2877 388 14 5 8 409 18 21 9 1 389 268 970 95 ≥ 10000 1090 390 195 16 8 748 114 175 190 391 61 74 12 3987 143 563 22 392 1184 1371 336 6019 2250 393 68 24 8 449 133 110 68 20 394 158 368 46 ≥ 10000 567 1072 1715 395 477 1377 299 ≥ 10000 396 3075 7481 525 ≥ 10000 397 903 1000 1460 398 25 26 7 6694 102 387 348 10 399 3 16 1 357 14 18 23 4 400 27 87 7 3538 62 41 113 9 401 162 253 82 9961 1146 402 12 14 3 514 34 51 46 4 403 426 79 20 3815 96 182 487 404 136 822 55

≥10000 1375 405 238 114 45 6734 247 155 2214 406 880 748 306 ≥10000 407 27 15 15 1333 27
 26 41 3 408 721 570 351 ≥10000 409 398 593 94 ≥10000 1176 410 365 603 157 3146 602 411 63
 83 40 146 48 92 9 412 19 51 34 5981 242 413 586 11 413 479 408 33 2530 484 414 1334 269 32
 3475 514 415 548 903 46 ≥10000 2557 416 141 83 3 1305 283 2194 417 182 57 20 110 63 79 418
 25 32 3 596 74 52 60 11 419 71 154 14 3481 182 319 58 420 859 366 56 ≥10000 1313 421 61 26 4
 1498 39 73 70 16 422 430 112 26 1813 303 539 423 556 2218 110 ≥10000 424 817 1300 95 2815
 425 42 37 5 921 54 99 73 12 426 652 1220 264 ≥10000 427 32 59 84 7684 379 878 916 52 428
 339 425 28 3280 183 896 441 6 2884 442 9 2847 444 1690 1161 37 ≥10000 734 1559 1761 445
 8199 ≥10000 446 786 ≥10000 452 204 165 41 4084 224 639 1512 461 1148 239 18 3759 172 472
 89 406 80 9295 414 1072 231 476 36 148 14 3592 99 275 35 477 61 456 67 ≥10000 865 1571
 2835 180 478 192 871 80 ≥10000 1247 713 5434 479 203 115 328 ≥10000 987 480 240 100 6
 ≥10000 215 47 111 481 387 643 49 5411 827 3150 482 546 128 26 ≥10000 390 124 483 890 339
 83 ≥10000 1589 484 4859 3034 1059 ≥10000 485 6623 1221 486 ≥10000 ≥10000 487 ≥10000
 4993 488 ≥10000 ≥10000 ≥10000 490 395 205 41 ≥10000 298 305 368 491 1705 1532 182
 ≥10000 492 3519 254 ≥10000 493 ≥10000 622 ≥10000 494 354 795 135 ≥10000 982 2169 495
 1350 1821 496 95 407 68 8288 261 245 525 117 497 987 503 43 ≥10000 498 47 64 13 ≥10000 48
 66 168 21 499 109 100 13 ≥10000 145 162 474 500 1103 1898 239 ≥10000 501 2064 3234 225
 ≥10000 502 2102 6458 1938 ≥10000 503 ≥10000 ≥10000 ≥10000 ≥10000 504 4 6 8 4834 11 53 81
 2 505 14 50 31 ≥10000 327 16 506 302 2221 703 ≥10000 8909 ≥10000 507 25 262 7 1952 345 179
 341 40 508 1552 2191 509 144 338 89 6139 211 206 314 510 186 388 73 5103 515 397 954 511 8
 25 2 1041 23 27 45 6 512 55 80 14 2827 68 130 291 53 513 1313 869 221 ≥10000 514 15 19 5 859
 20 60 30 11 515 35 19 36 19 24 7 516 9316 5686 517 203 321 42 ≥10000 518 ≥10000 9212 519
 ≥10000 ≥10000 520 ≥10000 ≥10000 521 ≥10000 ≥10000 522 ≥10000 ≥10000 523 ≥10000 ≥10000
 Example D: Anti-Proliferative Activity of Pan-KRas Inhibitors Against Mutations that Confer

Resistance to Adagrasib

(2164) To test the anti-proliferative activity of a prototype pan KRas inhibitor against mutations that confer resistance to adagrasib, mouse 3T3 fibroblasts were transduced with retroviruses that expressed various engineered human KRas mutant constructs. Cells were selected with puromycin to select for cells that were successfully transduced by the retrovirus and plated in ultra-low attachment plates where cells grew as 3 dimensional cultures. Cells were treated with a serial dilution of MRTX849 or Example 5 and 50% inhibitory concentration (IC50) values were calculated (Table 5). Example 5 demonstrated activity against numerous codon 12 mutations including the G12W mutation predicted to result from a single nucleotide substitution from the cysteine 12 codon.

(2165) TABLE-US-00007 TABLE 5 IC50 Values of the KRas G12C Inhibitor MRTX849 and the Pan KRas Inhibitor Example 5 in a 5-day Viability Assay in 3T3 Cells Engineered to Express MRTX849 Resistance Mutations
 MRTX849 (nM) Example 5 (nM) G12A >3000 32 G12C 16.62 28.1 G12D >3000 20.25 G12R >3000 1742 G12V >3000 94 G12W >3000 50 G13D >3000 610 Q61H >3000 58

(2166) While the invention has been described in connection with specific embodiments thereof, it will be understood that it is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure as come within known or customary practice within the art to which the invention pertains and as may be applied to the essential features hereinbefore set forth, and as follows in the scope of the appended claims.

Claims

1. A compound of Formula (I): ##STR00909## or a pharmaceutically acceptable salt thereof, wherein: W is: ##STR00910## A is naphthyl optionally substituted with 1-4 R.sup.1; B is:

##STR00911## Y.sup.1 and Y.sup.2 join to form: ##STR00912## where X is selected from: a bond, —S—, —O—, —N< bound to a fused ring, —CH.sub.2—, —CH.sub.2—N—, —CH.sub.2—N—CH.sub.2—, —CH.sub.2—CH.sub.2—CH.sub.2—, —CH.sub.2—CH.sub.2—, —O—CH.sub.2— and —S—CH.sub.2—; each R.sup.1 is independently halogen, cyano, hydroxy, C.sub.1-C.sub.4 alkyl, —S—C1-C3 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C2-C4 hydroxyalkynyl, C1-C3 cyanoalkyl, triazolyl, C1-C3 haloalkyl, —O—C1-C3 haloalkyl, —S—C1-C3 haloalkyl, C1-C3 alkoxy, hydroxyC1-C3 alkyl, —CH.sub.2C(=O)N(R.sup.5).sub.2, —C3-C4 alkynyl(NR.sup.5).sub.2, —N(R.sup.5).sub.2, deuterioC2-C4 alkynyl, (C1-C3 alkoxy)haloC1-C3 alkyl-, or C3-C6 cycloalkyl wherein said C3-C6 cycloalkyl is optionally substituted with halogen or C1-C3 alkyl; each R.sup.2 is independently hydrogen, hydroxy, halogen, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R.sup.5).sub.2, —CO.sub.2R.sup.5, or —CO.sub.2N(R.sup.5).sub.2; each R.sup.3 is independently hydrogen, hydroxy, halogen, C1-C3 alkyl, C1-C3 cyanoalkyl, C1-C3 hydroxyalkyl, HC(=O)—, -L-OC(O)N(R.sup.5).sub.2, —CO.sub.2R.sup.5, or —CO.sub.2N(R.sup.5).sub.2; each R.sup.4 is independently hydrogen, halogen or C1-C3 alkyl; each R.sup.5 is independently hydrogen or C1-C3 alkyl; each R.sup.6 is independently hydrogen, hydroxy, C1-C4 hydroxyalkyl or heteroaryl, or two R.sup.6 join to form C3-C6 cycloalkyl or heterocycle; each R.sup.7 is independently hydrogen, C1-C3 alkyl, hydroxy, halogen, C1-C3 haloalkyl, -L-NH.sub.2, —NH(C1-C3 alkyl), —N(C1-C3 alkyl).sub.2, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)-OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R.sup.10).sub.2, —CN, aryl, —(CH.sub.2)S(O).sub.2N(R.sup.10).sub.2, or heteroaryl optionally independently substituted with 1-2 substituents independently selected from C1-C3 alkyl, —CN and C(O)NH.sub.2, two R.sup.7 on the same atom optionally join to form a spirocyclic ring selected from C3-C6 cycloalkyl and heterocycle, where said spirocyclic ring is optionally substituted with 1-4 substituents independently selected from oxo (=O), halogen, hydroxy, C1-C3 alkyl and —O—(C1-C3 alkyl), two R.sup.7 on adjacent atoms optionally join to form a bond or a fused ring selected from C3-C6 cycloalkyl optionally substituted with 1-4 R.sup.8, heteroaryl optionally substituted with 1-4 R.sup.8, aryl optionally substituted with 1-4 R.sup.8, and heterocycle optionally substituted with 1-4 R.sup.8, and two R.sup.7 on non-adjacent atoms optionally join to form a 1-2 carbon bridge; each R.sup.8 is independently C1-C3 alkyl, hydroxy, halogen, oxo (=O), —O—(C1-C3 alkyl), —(C1-C3 alkyl)-OH, —C(O)OH, —C(O)O(C1-C3 alkyl), —C(O)N(R10).sub.2, or —CN; each R.sup.10 is hydrogen L is a bond each n is 0-3; and p is 1-8.

2. The compound or salt of claim 1, wherein: X is selected from: —CH.sub.2—, —CH.sub.2—CH.sub.2— and —O—CH.sub.2.
3. The compound or salt of claim 2, wherein each R.sup.1 is independently selected from halogen, hydroxy, C1-C3 alkoxy and C1-C4 alkyl.
4. The compound or salt of claim 2, wherein each R.sup.2, if present, is selected from hydrogen and halogen, and wherein each R.sup.3, if present, is selected from hydrogen and halogen.
5. The compound or salt of claim 2, wherein each R.sup.7 is independently selected from hydrogen, C1-C3alkyl, hydroxy, C1-C3 alkoxy, and wherein two R.sup.7 on non-adjacent atoms optionally join to form a 1-2 carbon bridge.
6. The compound or salt of claim 2, wherein each R.sup.6 is independently hydrogen or hydroxy.
7. The compound or salt of claim 1, wherein at least one R.sup.1 is C1-C4 alkyl.
8. The compound or salt of claim 1, wherein at least one R.sup.1 is halogen.
9. The compound or salt of claim 8, wherein said halogen is a fluorine.
10. The compound or salt of claim 1, wherein at least one R.sup.1 is hydroxy.
11. The compound or salt of claim 1, wherein at least one R.sup.2 is halogen.
12. The compound or salt of claim 11, wherein said halogen is a fluorine.
13. The compound or salt of claim 1, wherein at least one R.sup.3 is C1-C4 alkyl.
14. The compound or salt of claim 1, wherein at least one R.sup.3 is halogen.

15. The compound or salt of claim 14, wherein said halogen is fluorine.
16. The compound or salt of claim 1, wherein R.sup.4 is halogen.
17. The compound or salt of claim 16, wherein said halogen is fluorine.
18. The compound or salt of claim 1, wherein at least one R.sup.5 is C1-C4 alkyl.
19. The compound or salt of claim 1, wherein at least one R.sup.5 is hydrogen.
20. The compound or salt of claim 1, wherein one or both R.sup.6 are hydrogen.
21. The compound or salt of claim 1, wherein Y.sup.1 and Y.sup.2 join to form piperidine, azepane, azocane, thiazepine, diazepane, oxazepane, pyrrolidine, piperazine bound to a fused ring via nitrogen or thiomorpholine.
22. The compound or salt of claim 1, wherein two R.sup.7 on the same atom join to form a spirocyclic ring selected from C3-C6 cycloalkyl and heterocycle, where said spirocyclic ring is optionally substituted with one or more substituents selected from oxo (=O), halogen, hydroxy, C1-C3 alkyl and —O—(C1-C3 alkyl).
23. The compound or salt of claim 1, wherein two R.sup.7 on adjacent atoms join to form a bond or a fused ring selected from C3-C6 cycloalkyl optionally substituted with 1-4 R.sup.8 heteroaryl optionally substituted with 1-4 R.sup.8; aryl optionally substituted with 1-4 R.sup.8, and heterocycle optionally substituted with 1-4 R.sup.8.
24. The compound or salt of claim 1, wherein two R.sup.7 on nonadjacent atoms join to form a 1-2 carbon bridge.
25. A compound selected from: ##STR00913## ##STR00914## ##STR00915## ##STR00916## ##STR00917## ##STR00918## ##STR00919## ##STR00920## ##STR00921## ##STR00922## ##STR00923## ##STR00924## ##STR00925## ##STR00926## ##STR00927## ##STR00928## ##STR00929## ##STR00930## ##STR00931## ##STR00932## ##STR00933## ##STR00934## ##STR00935## ##STR00936## ##STR00937## ##STR00938## ##STR00939## ##STR00940## ##STR00941## ##STR00942## ##STR00943## ##STR00944## ##STR00945## ##STR00946## ##STR00947## ##STR00948## ##STR00949## ##STR00950## ##STR00951## ##STR00952## ##STR00953## ##STR00954## ##STR00955## ##STR00956## ##STR00957## ##STR00958## ##STR00959## ##STR00960## ##STR00961## ##STR00962## ##STR00963## ##STR00964## ##STR00965## ##STR00966## ##STR00967## ##STR00968## ##STR00969## ##STR00970## ##STR00971## ##STR00972## ##STR00973## ##STR00974## ##STR00975## ##STR00976## ##STR00977## ##STR00978## ##STR00979## ##STR00980## ##STR00981## ##STR00982## ##STR00983## ##STR00984## ##STR00985## ##STR00986## ##STR00987## ##STR00988## ##STR00989## ##STR00990## ##STR00991## ##STR00992## ##STR00993## ##STR00994## ##STR00995## ##STR00996## ##STR00997## ##STR00998## ##STR00999## ##STR01000## ##STR01001## ##STR01002## ##STR01003## ##STR01004## ##STR01005## ##STR01006## ##STR01007## ##STR01008## ##STR01009## ##STR01010## ##STR01011## ##STR01012## ##STR01013## ##STR01014## ##STR01015## and pharmaceutically acceptable salts thereof.
26. A pharmaceutical composition, comprising a therapeutically effective amount of a compound of claim 1 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient.
27. A method for inhibiting wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H activity in a cell, comprising contacting a cell in which inhibition of KRas activity is desired with an effective amount of a compound of according to claim 1 or a pharmaceutically acceptable salt thereof, or the pharmaceutical composition of claim 26.
28. A method for treating cancer comprising administering to a patient having cancer a therapeutically effective amount of a compound according to claim 1 or a pharmaceutically acceptable salt thereof, or the pharmaceutical composition of claim 26.
29. The method of claim 28, wherein the therapeutically effective amount of the compound is between about 0.01 to 100 mg/kg per day.
30. The method of claim 29, wherein the therapeutically effective amount of the compound is

between about 0.1 to 50 mg/kg per day.

31. The method of claim 28, wherein the cancer is selected from the group consisting of Cardiac: sarcoma, myxoma, rhabdomyoma, fibroma, lipoma and teratoma; Lung: bronchogenic carcinoma, alveolar carcinoma, bronchial adenoma, sarcoma, lymphoma, chondromatous hamartoma, mesothelioma; Gastrointestinal: esophagus, stomach, pancreas, small bowel, large bowel; Genitourinary tract: kidney, bladder and urethra, prostate, testis; Liver: hepatoma, cholangiocarcinoma, hepatoblastoma, angiosarcoma, hepatocellular adenoma, hemangioma, Biliary tract: gall bladder carcinoma, ampullary carcinoma, cholangiocarcinoma; Bone: osteogenic sarcoma, fibrosarcoma, malignant fibrous histiocytoma, chondrosarcoma, Ewing's sarcoma, malignant lymphoma, multiple myeloma, malignant giant cell tumor chordoma, osteochondroma, benign chondroma, chondroblastoma, chondromyxofibroma, osteoid osteoma and giant cell tumors; Nervous system: skull, meninges, brain; Gynecological: uterus, vulva, vagina; Hematologic: blood, Hodgkin's disease, non-Hodgkin's lymphoma; Skin: malignant melanoma, basal cell carcinoma, squamous cell carcinoma, Kaposi's sarcoma, moles dysplastic nevi, lipoma, angioma, dermatofibroma, keloids, psoriasis; and Adrenal glands: neuroblastoma neuroblastoma.

32. The method of claim 28, wherein the cancer is a KRas G12A-associated cancer.

33. The method of claim 28, wherein the cancer is a KRas G12C-associated cancer.

34. The method of claim 28, wherein the cancer is a KRas G12D-associated cancer.

35. The method of claim 28, wherein the cancer is a KRas G12R-associated cancer.

36. The method of claim 28, wherein the cancer is a KRas G12S-associated cancer.

37. The method of claim 28, wherein the cancer is a KRas G12V-associated cancer.

38. The method of claim 28, wherein the cancer is a KRas G13D-associated cancer.

39. The method of claim 28, wherein the cancer is a KRas Q61H-associated cancer.

40. The method of claim 28, wherein the cancer is a KRas G12A-associated cancer.

41. The method of claim 28, wherein the cancer is associated with at least one of wild type KRas, KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H.

42. The method of claim 28, wherein the cancer is non-small cell lung cancer, small cell lung cancer, colorectal cancer, rectal cancer or pancreatic cancer.

43. A method for treating cancer in a patient in need thereof, the method comprising (a) determining that the cancer is associated with wild type KRas or a KRas G12A, KRas G12C, KRas G12D, KRas G12R, KRas G12S, KRas G12V, KRas G13D or KRas Q61H mutation; and (b) administering to the patient a therapeutically effective amount of a compound according to claim 1 or a pharmaceutically acceptable salt thereof, or the pharmaceutical composition of claim 26.

44. The method of claim 28, wherein the administering is done via a route selected from the group consisting of parenteral, intraperitoneal, intradermal, intracardiac, intraventricular, intracranial, intracerebrospinal, intrasynovial, intrathecal administration, intramuscular injection, intravitreal injection, intravenous injection, intra-arterial injection, oral, buccal, sublingual, transdermal, topical, intratracheal, intrarectal, subcutaneous, and topical administration.

45. The method of claim 44, wherein the administration route is oral.

46. The method of claim 44, wherein the administration is intravenous injection.

47. The method of claim 44, wherein the administration route is intramuscular injection.

48. The method of claim 44, wherein the administration route utilizes a delivery device.

49. The method of claim 44, wherein administration is done in a hospital setting.
